

Transitioning: Evaluating approaches to flow ramp-down in a complex floodplain system

Alex A Stacy¹, Colden V Baxter¹, Ben T Crosby¹, Clark F Johnson²

1. Idaho State University, Pocatello, ID, United States

2. Wyoming Game and Fish Department, Jackson, WY, United States

Abrupt changes in flow regimes, particularly those associated with dam operations, can have significant ecological consequences. While many environmental flow alterations work toward endpoints such as maintaining base flows, less work has been done on the transition between high and low flows in manipulated systems. These impacts are especially pronounced in systems with complex floodplain habitats, where abrupt flow reductions can disrupt connectivity and ecological function. The Upper Snake River (Wyoming, USA), below Jackson Lake Dam, is reduced annually from high summer flows to winter base flows over 4-5 days, restricting available aquatic habitat and stranding fish and macroinvertebrates in side channels. Water and resource managers seek recommendations for fall ramp-down protocols that result in less stranding and loss of native fish (e.g., cutthroat trout) and macroinvertebrates. We address this complex problem by using a cross-disciplinary approach that utilizes both hydrological modeling and ecological investigations to assess the impacts associated with two different years of scheduled ramp-down events. Are there rates of change or ranges of flows at which more mortality and habitat loss occur? During the first and second year of the study, we observed substantial stranding of non-salmonid larval fishes and macroinvertebrates, on the order of thousands and hundreds of thousands of individuals, respectively. While no major direct effects on cutthroat trout were observed during either ramp-down, we anticipate and are assessing indirect impacts due to the loss of larval fishes and macroinvertebrates as critical food sources. Preliminary analyses suggest a disproportionate concentration of stranded fishes and macroinvertebrates in side channel habitats. Our analyses of potential differences in responses between the two years are ongoing. By improving flow management practices, we aim to enhance the resilience and sustainability of the Upper Snake River's aquatic ecosystems.

How are changes in stream water N₂:Ar and CH₄:Ar saturation ratios linked to estimates of ecosystem metabolism?

Robert S. Brown¹, Jonathan R. Behrens¹, Scott C. Brooks¹, Natalie A. Griffiths¹, Erin R. Hotchkiss², Marie J. Kurz¹

1. Oak Ridge National Laboratory, Oak Ridge, Tennessee, United States

2. Virginia Tech, Blacksburg, Virginia, United States

Traditional reach-scale O₂-derived measures of stream metabolism, gross primary production (GPP) and aerobic ecosystem respiration (ER), overlook anaerobic processes (e.g., denitrification and methanogenesis). The inclusion of anaerobic processes alongside estimates of stream metabolism can improve predictions and understanding of watershed biogeochemical dynamics. Daily estimates of reach-scale GPP and ER are facilitated by high frequency O₂ sensors, but simultaneous daily estimates of reach-scale denitrification and methanogenesis remain challenging to conduct across multiple sites in a stream network. To explore linkages between GPP, ER, net denitrification, and net methane flux, we used membrane inlet mass spectrometry to measure nitrogen N₂:Ar and CH₄:Ar saturation ratios (hereafter N₂sat and CH₄sat) in water samples along the mainstem and four tributaries of a 4th order stream draining a watershed with heterogeneous landcover. These ratios served as proxies for the direction and magnitude of net denitrification or methane flux where N₂sat or CH₄sat > 1 indicate net production and likely flux of either N₂ or CH₄ from the stream to the atmosphere. We collected concurrent water samples for nitrate and dissolved organic carbon analyses to assess potential variation in substrate availability for denitrification and methanogenesis. Additionally, we estimated daily GPP and ER from a time series of dissolved oxygen and temperature data collected at the same locations. N₂sat ranged from 0.999 to 1.046 and CH₄sat ranged from 1.2 to 8.3. CH₄sat was highest in summer and positively related to water temperature. N₂sat had apparent peaks in summer and winter and was unrelated to water temperature. Preliminary analyses alongside antecedent GPP and ER (mean values for preceding 7 days) suggest CH₄sat was negatively correlated with GPP and positively correlated with ER in colder months. We continue to assess N₂sat and CH₄sat as functions of antecedent GPP, ER, temperature, precipitation, and water chemistry throughout the year. This research will facilitate generation of hypotheses that shed light on relationships between anaerobic as well as aerobic carbon and nitrogen cycling.

Predicting biodiversity patterns and the importance of connectivity, stochasticity, and environmental variation in river ecosystems

Adrian Ortiz-Velez¹, Kurt Anderson², Bryan Brown³, Victor Saito⁴, Tadeu Siqueira⁵, Chris Swan⁶

1. UCR and SDSU, San Diego, CA, United States

2. Biology, University of California, Riverside, Riverside, CA, United States

3. Biology, Virginia Tech University, Blacksburg, VA, United States

4. Biology, Federal University of São Carlos, São Carlos, State of São Paulo, Brazil

5. University of Canterbury, Canterbury, New Zealand

6. University of Maryland, Baltimore, Maryland, United States

Differences within and between regional metacommunities could be driven by differences in abiotic and biotic environmental conditions, dispersal connectivity, and ecological drift. Drift—stochastic variation in population dynamics—has recently received attention but is still much less understood relative to environmental and dispersal effects. This is especially the case in freshwater ecosystems despite evidence of the importance of ecological drift on river diversity patterns. Here, we present a stochastic modeling framework where we examine how ecological drift links with abiotic and biotic environmental drivers, intrinsic survival,

fecundity, and dispersal connectivity to generate measurable variation in metacommunity assembly. We considered a range of population dynamic parameters underlying demographic stochasticity, forms of environmental stochasticity, dispersal stochasticity, maximum survival rate, and species interactions in order to represent the breadth of conditions encountered among different stream types, locations and communities. Model output generated predicted patterns of local (alpha) diversity, changes in diversity over space (beta), and time (beta and synchrony). We found that increased negative environmental effects on survival rates reduce local population fluctuations over time due to ecological drift. However, the increased negative environmental effect on survival rate reverses the theory-predicted patterns of community size, drift, and temporal beta diversity. Our model predicts that as headwaters become more similar to one another or more variable over time, the metacommunity's average spatial beta decreases, while temporal beta values and their spread increase.

id #135427

Do Geophysical Habitat Types and Their Network-Scale Diversity Predict Fish Species Richness Across Freshwater Ecoregions?

Abhiram Pamula¹, Ryan McManamay¹

1. Baylor University, Waco, TX, United States

Geophysical diversity can structure freshwater biodiversity by shaping habitat heterogeneity through hydrology, geomorphology, and thermal regimes. Conservation efforts to preserve biodiversity might prioritize preserving geophysical diversity and connectivity of habitats, operating under the assumption that habitat diversity within topological networks necessarily begets aquatic biodiversity. However, this hypothesis has not been adequately tested at a variety of spatial scales, levels of community organization (, , and diversity), and transferability amongst divergent ecoregions, yet its predictive value is often unclear when models are transferred across broad ecological regions. We modeled reach-scale fish species richness based on testing a series of alternative hypotheses about what structures biodiversity: 1) universal predictors of biodiversity, such as Ecoregion, system size (i.e., discharge, drainage area), and terrestrial productivity (NPP), 2) geophysical habitat types (e.g., hydrological types, stream thermal regimes), and 3) diversity in geophysical habitat types. Alternative models were constructed using a Bayesian negative binomial generalized linear mixed models (NB-GLMM) with random intercepts for freshwater ecoregions.. We evaluate predictive skill both within ecoregions and under transfer to entirely new ecoregions, revealing strong within-region performance but markedly reduced transferability, consistent with region-specific structure and/or missing transferable gradients. Building on this baseline, we are extending analyses to quantify habitat diversity explicitly at multiple spatial scales (local "alpha" vs watershed "gamma"; e.g., HUC8 summaries) and to compare hierarchical models with machine-learning approaches that can capture nonlinearities and interactions while retaining interpretability. More flexible machine-learning (ML) models (e.g., gradient-boosted trees, random forests, or neural networks) are currently tested that can complement NB-GLMMs by capturing nonlinearities, high-order interactions, and thresholds in geodiversity–biodiversity relationships, while modern interpretation tools (e.g., SHAP/partial dependence) can retain ecological insight. Comparing ML and hierarchical statistical models under the same "new-ecoregion" validation can also identify which predictor sets and model structures generalize best, supporting transferable biodiversity mapping and conservation prioritization.

id #135683

A hybrid approach to rating curve development: refining hydraulic models with field data

Andrew N Chu¹, Christina Bryant¹

1. City of Austin, Austin, TEXAS, United States

The City of Austin's flagship environmental monitoring program, the Environmental Integrity Index (EII), is a tool used to assess the hydrology, water quality, geomorphology and biology of all streams within the City's jurisdiction. Hydrology is the mechanism for sediment and constituent transport within streams; therefore, accurate records of continuous discharge are necessary to compute stream pollutant loads and assess long-term biologic integrity. Traditional rating curve development often requires months or years of discharge measurements to establish site-specific relationships between continuously monitored stage and computed discharge. High-flow events further complicate data collection due to safety constraints and specialized equipment. Alternatively, theoretical rating curves can be developed using hydraulic modeling, but these models represent fixed conditions and may not capture current or dynamic site conditions. To accelerate rating curve development, we implemented a hybrid strategy that integrates hydraulic modeling with field data. This effort was initiated after observing issues in our original models, such as hydraulic control structures at modeled elevations lower than the gage pools they create. We developed provisional rating curves in HEC-RAS 1D, a river hydraulics modeling software, by updating terrain-based floodplain models with surveyed cross-sections and structural measurements. Overall, the revised cross sections were consistently deeper than the original digital elevation map, and the slopes varied along the surveyed stream sections. These hydraulic models will be further refined over time using flow rates from wading measurements to update model coefficients and ensure models respond to seasonal site conditions. This approach enables us to immediately estimate flow and associated uncertainty while establishing a protocol for iterative refinement of rating curves. Thus, pollutant loading can also be more accurately estimated.

id #136195

How invasion can create evolutionary mismatches that decouple positive interactions in a freshwater cleaning symbiosis: consequences for crayfish plague

Bryan Brown¹, Robert P Creed², Lindsey Albertson³, James Skelton⁴, Christopher M Swan⁵

1. Department of Biological Sciences, Virginia Tech, Richmond, VT, USA

2. Department of Biology, Appalachian State University, Boone, NC, USA

3. Department of Ecology, Montana State University, Bozeman, MT, USA

4. *Department of Plant Pathology, University of Florida, Gainesville, FL, USA*

5. *Department of Geography and Environmental Science, University of Maryland, Baltimore, MD, USA*

Positive interactions with symbionts can provide protection from disease in host organisms. However, these positive interactions often rely on relationships built through shared co-evolutionary history between hosts and symbionts. Invasions by non-native hosts can disrupt or decouple those positive interactions because invasive hosts do not share an evolutionary history with native symbionts, effectively decoupling disease-preventing positive interactions. We demonstrate how such symbiotic decoupling may be occurring in the freshwater cleaning symbiosis between crayfish and their branchiobdellidan symbionts, resulting in an increased intensity of crayfish plague. Recent evidence suggests that this cleaning symbiosis can reduce the effects of plague in host crayfish. However, because non-native crayfish are poor hosts for symbionts, symbiont abundance and diversity decline dramatically as the density of non-native hosts increases relative to native hosts. We combine evidence from field studies, laboratory experiments, and host immune gene expression to show how decoupling in this common cleaning symbiosis may be exacerbating the effects of crayfish plague, and to hypothesize how these effects project at regional and even continental scales.

id #135428

Embracing the zoogeochemical framework in freshwaters to understand and predict the ecosystem-level consequences of conservation (Joint talk with Taylor Michael)

Jonathan W Lopez^{1,2}, Taylor C Michael¹, Carla L Atkinson¹, Matthew B Lodato^{1,3}, Lauren M Morris¹, Stephen Plont^{1,4}, Chelsea R Smith¹, Frank C Akamagwuna¹, Lindsey K Albertson⁵, D. Kyle Breault¹, Jacob D Dorris¹, Samuel F Fritz⁵, Jennifer Haase⁶, Tori A Hebert¹, Adam C Hensley¹, Garrett W Hopper⁶, Madison P Kuczek⁶, Julia N Morin⁶, Hayley C Oakland⁵, Adam C Siders¹, Augustine Sitati¹, Benjamin B Tumolo⁷

1. *University of Alabama, Tuscaloosa, UNITED STATES, United States*

2. *University of Mississippi, Oxford, UNITED STATES OF AMERICA, United States*

3. *Texas Tech University, Lubbock, TX, United States*

4. *University of Georgia, Athens, Georgia, United States*

5. *Montana State University, Bozeman, Montana, United States*

6. *Louisiana State University, Baton Rouge, Louisiana, United States*

7. *Northern Michigan University, Marquette, Michigan, United States*

Freshwater ecosystems are under immense pressure from global environmental change. These stressors have led to biogeochemical alterations and rapid rates of defaunation and losses of associated animal-driven ecosystem functions. Zoogeochemistry is an ideal framework for trying to meet the dual challenges of global change and defaunation in freshwater ecosystems, but zoogeochemical frameworks that explicitly link conservation biology with biogeochemistry have not yet been fully embraced by freshwater ecologists. We conducted a systematic review and meta-analysis of the mechanistic roles that animals play in freshwater biogeochemistry, with the aim of advancing the application of zoogeochemistry in freshwaters. We reviewed 14 major mechanisms of freshwater zoogeochemistry. We grouped mechanisms into three primary categories: (1) trophic interactions that mediate the movement and transformation of elements through food webs, (2) metabolic processes, including excretion and egestion, that directly alter nutrient availability, and (3) physical ecosystem engineering, such as bioturbation and sediment resuspension, which modify redox conditions and enhance benthic–pelagic coupling. Our review shows that a diversity of taxa impact ecosystem-level elemental pools and fluxes of dozens of elements including the traditional macronutrients, C, N, and P, and a range of micronutrients, both within and across freshwater ecosystem boundaries. However, these effects are not always placed in context at the ecosystem level, which limited the scope of our meta-analysis. Zoogeochemical impacts tended to be stronger in wetlands and lakes than streams and freshwater estuaries. Consumption, excretion, and respiration had the strongest direct impacts on elemental pools and fluxes; bioturbation, consumption, and egestion had the strongest indirect impacts. These mechanisms often interacted, generating complex influences on elemental pools and fluxes. However, creative experimental designs can be used to isolate individual mechanisms, allowing complex effects with multiple mechanisms to be incorporated into ecosystem models. Anthropogenic pressures such as biodiversity loss, altered species interactions, habitat fragmentation, and climate-driven shifts in movement and phenology, are disrupting animal-mediated biogeochemical processes at local and landscape scales. Quantifying and modelling zoogeochemical effects in freshwater ecosystems will allow ecologists and managers to make informed decisions that balance biodiversity conservation with healthy ecosystem function.

id #134405

Integrated hydrological and biological monitoring reveal ecological patterns in intermittent streams

Daren M Carlisle¹, Jason T May²

1. *US Geological Survey, Lawrence, KS, United States*

2. *US Geological Survey, Sacramento, CA, United States*

Integrated hydrological and biological data are crucial to understanding the ecological dynamics of streamflow intermittency. Our objectives were to determine the hydrologic characteristics that drive biological community structure along an intermittency gradient, and to identify biological signatures of flow intermittency. We sampled macroinvertebrate assemblages in 21 southern California streams that varied primarily in hydrologic permanence. Sampling sites were located at stream gages in basins with minimal human influence. Mean annual zero-flow days ranged from 0–200 based on daily flow records ranging from 11–100 years in length. Biological assemblage composition was evaluated for associations with several hundred streamflow metrics calculated over long- (decades) and short-term (antecedent 365 days) temporal windows. We identify several streamflow characteristics that were the most influential predictors of benthic assemblage structure. We also report the taxonomic and trait-based biological indicators representative of various degrees of intermittency.

Beyond the daily average: Why high temporal resolution streamflow data matters in urban watersheds

Kristina Hopkins¹, Malia Scott¹, Kristin Jaeger¹

1. U.S. Geological Survey, Tacoma, WA, USA

Daily mean streamflow measurements are widely used to compute streamflow metrics due to the viability and completeness of daily streamflow data. In small, urban watersheds with flashy streamflow, characterizing hydrological dynamics may necessitate the use of hourly, 15-minute, or even 5-minute streamflow data. This presentation will explore differences between streamflow metrics calculated from daily mean streamflow compared to those calculated from 15-minute data and highlight considerations for selecting the temporal resolution for hydrologic analyses. We analyzed data from twelve small watersheds (<50 square kilometers) located in the Puget Sound Lowlands within western Washington, including metropolitan Seattle. Streamflow metrics were calculated at 12 sites from streamflow data collected by the King County Hydrologic Monitoring Program for water years 2021 through 2024. Selected watersheds spanned a gradient of urban development from 9% to 44% impervious cover. We tested different approaches to define the start and end of flow events from both the daily mean and 15-minute streamflow time series, then compared the resultant statistics that describe flow frequency, magnitude, duration, timing, and rate of change. When possible, metrics were normalized based on drainage area and precipitation amount obtained from three King County Hydrologic Monitoring Program stations. Lastly, we compared the predictive power of common urban land cover metrics to explain variability among daily and sub-daily streamflow metrics. Urban land cover metrics included impervious cover at high resolution (1-meter), impervious cover at low resolution (30-meter), road density, population density, and housing density. Evaluating the sensitivity of streamflow metric values to data resolution can inform monitoring strategies by identifying the metrics and watershed conditions where high-frequency data provide advantages over daily mean streamflow.

Ecosystem Processes in the Age of Antibiotics

Michael Strickland¹

1. University of Idaho, Moscow, IDAHO, United States

In 1928, Alexander Fleming identified the antibiotic properties of penicillin. Since this discovery antibiotics have been a boon for human health as well as for agricultural livestock production. In fact, the latter utilizes 80% of the antibiotics produced in the United States, a staggering 33 million pounds a year. Most of these antibiotics and antibiotic derivatives ultimately enter the environment, leading to the assertion that no environment on earth is free from the influence of agricultural antibiotics. Despite this, little research has examined the environmental influence agricultural antibiotics are likely to have, for example, on microbial communities and the ecosystem processes they mediate. Here we demonstrate through a series of field- and laboratory-based studies, leveraging next-generation sequencing and stable isotope tracing, that antibiotics can dramatically alter microbial community composition and function. For instance, we show that antibiotics lead to increased abundance of antibiotic resistance genes and shifts in microbial community composition. Additionally, we find that microbial carbon-use efficiency is negatively impacted by exposure to antibiotics. This also leads to decreased ecosystem efficiency (i.e., more carbon is lost via respiration than is reattained within the system) and shifts in the entire food web (i.e., arthropod communities as well as microbial communities). Furthermore, the impact of antibiotics on microbial communities can be exacerbated by other stressors, such as increasing temperature. This leads to more drastic changes in microbial community composition and network structure, and exposure to antibiotics may fundamentally shift the response of microbial communities to global change. In conclusion, antibiotics are an often under-appreciated and diffuse environmental contaminant, but antibiotics have the potential, even at low concentrations, to fundamentally alter microbial communities and ecosystem processes.

Decomposition rate and microplastic flux in urban environment: two contrasting watersheds in central kenya

Erasmus Uyizeye¹, Timothy Hoellein², George Ndiritu³, Krista Capps⁴, Scott Tiegs⁵

1. Brevard College, Brevard, NC, U.S

2. Loyola University Chicago, Chicago, IL, U.S

3. Karatina University, Karatina, Kenya

4. University of Georgia, Athens, GA, U.S

5. Oakland University, Rochester, MI, U.S

The effects of urbanization and watershed land-use patterns on the abundance and composition of microplastics, and their relationship with key drivers of organic matter decomposition is not well documented in tropical African river systems. We assessed microplastic contamination and quantified organic matter decomposition rates in the Nairobi River and two major tributaries draining a highly urbanized watershed. To provide a contrasting land-use context, we also examined the Ragati River within the Karatina town watershed, which is less urbanized but predominantly agricultural. Microplastic concentrations in the Nairobi River increased along an urbanization gradient, with significantly higher concentrations at downstream sites (135.2 items L⁻¹) compared to upstream locations and the Ragati River ($p < 0.05$). Within the Nairobi watershed, the Mathare River, whose drainage is predominantly residential, contributed disproportionately high microplastic loads (77.3 items L⁻¹) relative to other tributaries and upstream cumulative concentrations in the Nairobi River mainstem (31.2 items L⁻¹), which drains a mix of commercial, industrial, institutional and residential land uses. Microplastic composition differed between watersheds: fibers dominated in the Nairobi system, indicating strong inputs from domestic wastewater and textile-related sources, followed by foams associated with packaging and construction materials. In contrast, fibers and fragments were most prevalent in the Ragati

River, likely reflecting the degradation of plastic consumer products such as bottles and containers. Organic matter decomposition rates did not follow a clear urbanization gradient but instead showed strong site-specific variability. Tensile loss rates were highest in the Mathare River (4.955 Kd d^{-1}) and at the nearest downstream site on the Nairobi River mainstem, exceeding upstream rates ($0.127 \pm 0.01 \text{ Kd d}^{-1}$) and higher than the previously predicted regional rate (0.015 Kd d^{-1}). Decomposition rates in the Ragati River were comparatively similar between sites ($0.188 \pm 0.03 \text{ Kd d}^{-1}$ upstream; $0.143 \pm 0.182 \text{ Kd d}^{-1}$ downstream). These findings highlight the influence of urbanization and associated anthropogenic pressures on microplastic pollution, while emphasizing the role of local physico-chemical conditions in regulating organic matter processing. Multi-season monitoring across broader spatial scales is needed to better capture variability and inform targeted mitigation strategies of pollution in rapidly urbanizing tropical river systems.

id #135430

Seeing the Unseen: Serendipity, Perception, and the Science Hidden in Plain Sight

Amy Burgin¹

1. *Iowa State University, Ames, IA, United States*

Scientific discovery is often portrayed as the outcome of careful planning and tightly reasoned hypotheses, yet much of what we learn emerges when our expectations fail. In this talk, I explore how two concepts —serendipity and figure-ground perception— have shaped my scientific trajectory and reframed my understanding of “failure” as a productive force. Serendipity, the experience of finding something unforeseen, but valuable while looking for something else, can seem at odds with rigorous science. But serendipity only becomes possible when we attend to the unexpected and stay open to alternative interpretations of our data. Likewise, figure-ground perception—a neurological principle describing how the brain shifts between different interpretations of the same visual stimulus—offers a powerful metaphor for scientific thinking. By reconsidering which part of a problem is the “figure” and which is the “background,” we can surface overlooked processes, challenge group-think, and identify novel knowledge gaps.

Across my career, from undergraduate research to professor, serendipity has repeatedly redirected my scientific path. My dissertation emerged from a pattern in nitrogen and sulfur coupling that did not conform to predicted behavior, prompting a deeper inquiry that ultimately defined my research agenda. Since then, “accidental experiments” and unexpected field conditions have repeatedly provided opportunities for insight—revealing patterns and mechanisms I would not have found through planned hypotheses alone. Furthermore, careful and intentional shifts in perspective have also guided my work identifying underexplored conceptual gaps, leading to several synthesis efforts, including papers on nitrate fate in aquatic ecosystems, the biogeochemical economy, and the redox paradox. Each grew from asking what becomes visible when we shift our focus to what others have ignored. By embracing serendipity and adopting flexible perception mindset, we can transform unexpected results from setbacks into invitations—opening creative pathways to understanding complex ecological systems and strengthening our resilience as scientists.

id #135687

Science Communication and Outreach in the Age of Film: How to Expand Audiences by Turning Science into Stories with Documentary Film

Brandon Gaesser¹

1. *Media Arts, University of North Texas, Denton, Texas, United States*

Did video kill the journal publication? No, it certainly did not. But video is now one of the best tools to help scientists communicate their research and reach broader audiences beyond their respective fields. To be clear, this is not about TikTok videos, Instagram posts, and the proverbial ‘like and subscribe’ dyad. Instead, this is about science-based documentary filmmaking with interviews and observational footage both in the laboratory and in the field. These films can range from 5-minute videos embedded in your journal publications to feature-length films screened at science-themed film festivals and broadcast on the Public Broadcasting Service (PBS, USA). Films can highlight just one aspect or stage of a research project or document the entire scientific narrative. Why use film to communicate research findings? A study in *Nature* found that video as supplementary material led to increased citations, increased number of views, and increased social attention, while Wiley Online Library found an 80% increase in full text views. Does filmmaking have to be tackled by scientists? No. Scientists are increasingly collaborating with filmmakers to work with and communicate directly with the public. In this presentation, current project examples will include how film and science collaboration is shedding light on the dangers oil and gas drilling waste pose to freshwater sources and ecosystems. Additionally, past examples of films addressing freshwater quality and soil health will also be used. These science films have been published in *Nature*, screened in environmental film festivals, and broadcast on PBS, and will demonstrate how collaborations among scientists, filmmakers, and community partners broaden communication and outreach and expand educational resources for increased public participation, engagement, and representation.

id #135943

Applying remote sensing and GIS to assess hydrologic connectivity and potential riparian habitat development at a historic bentonite mining site

Patrick K Kozak^{2,1}, Jennifer N Walker^{2,1}, Michael Beason³, Lisa A Kunza^{2,1}

1. *Center for Sustainable Solutions, South Dakota School of Mines, Rapid City, SD, USA*

2. *Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines, Rapid City, South Dakota, USA*

3. *South Dakota Wing, Civil Air Patrol, Rapid City, South Dakota, USA*

Mining of bentonitic clays in the Cretaceous Belle Fourche Shale northwest of Belle Fourche, South Dakota, began during the 1920s. Areas where mining predated regulations requiring site reclamation are characterized by a unique landscape of unvegetated spoil mounds interspersed with water-filled pits. Topsoil in the disturbed areas is largely lost, and vegetation is absent or sparse. The post-mining landscape changes represent a permanent departure from pre-disturbance conditions, making a return to a historical reference state unattainable as a restoration goal. We evaluate the potential for restoration of hydrologic connectivity to promote the development of vegetation communities and riparian habitats. To assess hydrologic connectivity and riparian habitat, we initially analyzed half-meter USDA National Agricultural Imagery Program (NAIP) visible and near infrared (VNIR) imagery to classify open water, vegetation type, and bare ground. The initial classification appropriately separated riparian from other vegetation types and the percentage of bare ground. Initial difficulty in classifying open water pixels due to suspended sediments being identified as bare ground. We revised the initial training data and post-classification interpretation of imagery. From our initial classification, we subset the area and, in July of 2025, flew a Wingtra Gen II fixed-wing drone with a Micasense RedEdge-P multispectral camera at 7 cm pixel resolution over 2 weeks, covering approximately 15 square kilometers, producing a higher-resolution multispectral dataset and a photogrammetric digital elevation model for analysis. We applied the same classification methodology to the Drone dataset results were similar. However, the higher-resolution drone data also identified regions along gullies and rills where vegetation was observed and field-checked, indicating soil development and moisture retention. Finally, we used the drone-identified riparian vegetation locations, the slope derived from the drone-based DEM, and field observations to assess riparian vegetation development as a function of slope, hydrologic connectivity, and shoreline structure. We found that slope, shoreline structure, including hydrologic connectivity, and prevalent wind direction influenced riparian vegetation development. An understanding of how slope and hydrologic connectivity influence vegetation (riparian, mesic, and upland) will allow us to suggest cost-effective approaches for long-term remediation of the bentonite mining area and potentially other similar locations regionally and globally.

id #134152

Moving beyond novel ecosystems: Biogeochemical cycling in de novo aquatic ecosystems

Jacqueline Gerson¹, Anika Bratt², Audrey Goeckner³, Meredith Holgerson¹, AJ Reisinger³

1. Cornell University, NY, United States

2. Macalester, Saint Paul, Minnesota

3. University of Florida, Gainesville, FL

The concept of novel ecosystems was first proposed in 2006 and has attracted much attention due to the proliferation of new ecosystems resulting from environmental change. Over the past two decades, the novel ecosystem framework has been used to evaluate when ecosystems have reached a tipping point from merely a hybrid system to one that is unique from its previous state. We propose that this framework leaves out an important nuance, what we call *de novo ecosystems* – completely human-made ecosystems that mimic natural ecosystems in some way. A critical component of de novo ecosystems is that they do not exist prior to human interference – differentiating them from novel ecosystems – yet still allowing them to be subject to some of the same considerations as novel ecosystems. De novo aquatic ecosystems are becoming increasingly common globally, often as a result of resource extractive industries and for management of water in urban and agricultural environments; for example, the construction of mining ponds on land that was previously forested and the construction of stormwater retention ponds on land that was previously grassland. There is therefore a need to consider how these de novo ecosystems behave in comparison to natural and novel ecosystems. In this presentation, we argue for the consideration of de novo ecosystems as unique but important aspects of the landscape that are critically understudied, particularly in regard to biogeochemical processes. We begin by comparing the biogeochemical cycling of nutrients and contaminants in de novo aquatic ecosystems to constructed vs. naturally formed landscapes. We continue by examining how three prominent de novo aquatic ecosystems – abandoned mining ponds, stormwater retention ponds, and agricultural ponds – impact biogeochemical cycling of elements and the implications for ecological function. We conclude with future research directions into de novo ecosystems and how addressing these research gaps could better inform management efforts.

id #134409

Environmental conditions induce growth rate and stoichiometric trait plasticity in captivity reared freshwater mussels

Garrett Hopper¹, Irene Sánchez González², Jamie Bucholz³, Jefferey Lozier⁴, Carla Atkinson⁴, Paul Johnson⁵

1. Louisiana State University, LA, United States

2. University of Georgia, Athens, Georgia

3. University of Rhode Island, Rhode Island

4. The University of Alabama, Tuscaloosa, Alabama

5. Alabama Department of Natural Resources, Marion, Alabama, United States

Conservation programs for imperiled freshwater taxa often include a propagation and reintroduction component designed to enhance existing population sizes but may also be used to recover lost ecosystem functions resulting from population declines. Freshwater mussels are imperiled organisms that enhance habitat and control aquatic food webs through bottom-up subsidies via their nutrient recycling traits. Improving efforts to recover mussel populations and subsequent mussel-mediated nutrient cycling requires information about how species' performance and functional traits are influenced by conditions of the reintroduction environment. We addressed how environmental conditions influence the growth and stoichiometric traits of species by conducting a 16-week reciprocal transplant experiment with broods of captivity-raised mussels in two rivers within the Mobile River Basin (MRB; Cahaba and Sipsey) and two rivers within the Tennessee River Basin (TRB; Duck and Paint Rock). The reciprocal transplant occurred during week eight. We expected growth rates to differ among rivers due to underlying environmental conditions, and nutrient excretion rates and stoichiometric ratios would differ among individuals stocked into different rivers because tissue stoichiometry should remain stable regardless of water column and food nutrient stoichiometry as animals have evolved post-assimilatory mechanisms to balance tissue demands with resource supply to maintain

homeostasis. MRB mussels grew most in the Cahaba, or when transplanted from the Sipsey to the Cahaba. TRB mussels grew most in the Paint Rock, or when transplanted from the Paint Rock into the Duck. TRB mussel tissue C:N, C:P, N:P, and P excretion were highest in the Duck, but transplanting had no effect. Transplanting mussels between rivers led to opposing excretion N:P ratios between transplanted and non-transplanted MRB mussels. Mussel growth, tissue stoichiometry, and nutrient excretion rates were strongly shaped by river-specific environmental conditions, with clear evidence that environmental context creates lasting physiological legacies that influence mussel performance and ecosystem nutrient cycling following transplantation. The persistence of growth and stoichiometric effects following transplantation suggests that conservation efforts involving mussel translocations should consider source populations and recipient river conditions. Relocating mussels to seemingly favorable habitats may not immediately yield expected growth or ecosystem benefits if individuals are physiologically constrained by prior conditions.

id #135177

Environmental DNA metabarcoding detection dynamics from evolving reference libraries: a decade of change

Christopher L Jerde¹, Andrew R Mahon²

1. *University of California, Santa Barbara, Santa Barbara, CALIFORNIA, United States*

2. *Department of Biology, Central Michigan University, Mount Pleasant, MICHIGAN, United States*

DNA (eDNA) metabarcoding enables broad biodiversity monitoring; however, its accuracy depends on the completeness of the reference databases. Early studies often underestimated or misclassified diversity due to the absence of reference sequences. We revisited an eDNA dataset first analyzed in 2014 (Simmons et al., 2016), which surveyed fish communities in the Muskingum River, a part of the Mississippi River Basin (MRB). Original samples (collected in 2013, amplified with 16S primers, and sequenced on an Illumina MiSeq) were reanalyzed against GenBank Actinopterygii 16S reference libraries, which were updated annually from 2014 to 2024. During this period, the database nearly doubled in size. Species richness estimates remained relatively stable (46–54 taxa), but species identifications improved: non-MRB detections declined while MRB taxa increased, reducing spurious matches. The ratio of non-MRB to MRB detections dropped from 0.2 (2014–2019) to 0.06 (2024). Metrics of overlap, gain, loss, and turnover show that new reference sequences primarily drove species gains, while losses often reflected correction of errant matches. These dynamics highlight how database growth reshapes biodiversity inferences and reinforce the value of archived eDNA as evolving digital records for conservation monitoring.

id #135689

Dissolved organic matter primarily determines the potential microbial nutrient acquisition strategy in stormwater ponds in Florida, USA

Shuo Chen¹, AJ Reisinger¹

1. *University of Florida, GAINESVILLE, FL, United States*

More than 76,000 stormwater ponds (SWPs) have been constructed in Florida to control organic and inorganic nutrient loadings from the surrounding landscape. Human decisions on SWP design and management control external nutrient inputs, potentially altering the form and magnitude of nutrient availability and subsequently influencing microbial metabolism within SWPs and downstream habitats. Microbes can excrete extracellular enzymes to acquire metabolic organic substances when there is a relative limitation of bioavailable carbon (C) or inorganic nitrogen (N) and phosphorus (P) for their metabolism. Therefore, extracellular enzyme activities (EEA) can reflect the microbial demand for, and environmental availability of, organic nutrient subsidies. However, the quantity and quality of dissolved organic matter (DOM) that control microbial nutrient (C, N and P) demand that may predict nutrient removal by SWPs are poorly understood. Furthermore, seasonal shifts in DOM composition and inorganic nutrient availability, driven by the imbalance between terrestrial inputs and autochthonous production, may alter microbial nutrient acquisition strategies, although this remains unclear. We sampled 19 SWPs from two master-planned communities (GNV and LWR) throughout 2025–2026 to understand how microbial nutrient acquisition strategies responded to the amount, sources, chemical composition and bioavailability of DOM and inorganic nutrient availability across wet and dry seasons. We incorporated DOM characterization and extracellular enzyme assays to better understand microbial reliance on organic substrates. We hypothesized that, spatially, SWPs with algal blooms will have a larger fraction of bioreactive DOM but an inorganic nutrient deficit, leading to high N-, P- and labile C-acquiring enzyme activities. Seasonally, we expect that N-, P- and labile C-acquiring enzyme activities will be positively correlated with N- and P-containing DOM compounds during the summer wet season. Preliminary results revealed strong co-limitation (C and N) for microbial uptake in GNV, whereas P and C were co-limiting in LWR, suggesting that different pond management can alter microbial metabolism. In the hot, wet season, the N- and P-acquiring enzyme activities were constrained by the inorganic nutrient deficiency and terrestrial organic carbon inputs. This study will improve our understanding of microbial transformations of organic carbon, potential nutrient limitation and microbial function from human-made aquatic ecosystems.

id #135946

Process-Based Ecological Restoration Involving the Local Community of the 65 Square Mile Tenmile Creek Watershed in Mendocino County, Northern California, USA

Patrick T Higgins¹

1. Eel River Recovery Project, McKinleyville, CA, United States

The Eel River Recovery Project is a non-profit 501c3 organization operating in the 3600 square mile Eel River watershed in northwestern California. Our mission is "To empower communities to collaborate in monitoring the ecological conditions of the Eel River, to share information about the health of the watershed, and to work together to formulate and implement a restoration strategy." Since 2018, ERRP has been focused on the ecological restoration of the 65 square mile Tenmile Creek watershed in northern Mendocino County, which is a major branch of the South Fork Eel River. According to the National Marine Fisheries Service *Southern Oregon/Northern California Coho Recovery Plan*, the upper South Fork Eel River has one of the last viable meta-populations of coho in the region. While NMFS models suggest that Tenmile Creek had higher intrinsic potential than the upper SF Eel, coho were at low levels or absent in Tenmile Creek because of limited flow and excess sediment. Monitoring has included water temperature, fish community structure, sediment in pools, flow, and riparian air temperatures in key tributary Cahto Creek. Data allowed definition of limiting factors and targeted restoration projects. These include bioengineering and large wood structure installation to stop bank erosion and enhance riparian and fish habitat, increasing water storage to allow forbearance from water withdrawal during the dry season in Streeter Creek that serves as a steelhead refugia, road related erosion control to stop sediment in a Tenmile Creek reach with high Chinook salmon spawning, healing gullies in meadow and restoring native grasses there, and carrying out widespread forest health thinning to help reduce tree evapotranspiration in order to improve stream base-flow and prevent catastrophic wildfire. Coho are experiencing a resurgence regionally and juveniles have been present and even abundant in most recent years since 2022. Chinook salmon and steelhead are also rebounding as wet years, good ocean conditions and almost no ocean fishing align. New committees are forming aimed at invasive species removal targeting bullfrogs and feral pigs, and native grass restoration at the Tenmile Creek watershed scale. "Like it or not, we are in charge of evolution" – Julian Huxley.

id #134155

Assessing environmental, spatial, and phylogenetic drivers in crayfish assemblages of the Ogeechee River

Reginald Turner¹, Jose C. Colon Gaud¹, Nicholas Macias²

1. Georgia Southern University, GA, United States

2. University of Georgia Marine Institute, Sapelo Island, GA, USA

The southeastern United States harbors the greatest diversity of crayfish in North America, yet this biodiversity is increasingly threatened by land-use change and habitat alteration. Many species exhibit extreme endemism, often restricted to single river systems, making them especially vulnerable to urbanization, forestry, and hydrologic shifts. Despite this, few studies have examined the ecological drivers shaping crayfish assemblages in inter-ecosystem rivers. We investigated crayfish assemblage structures in the Ogeechee River Basin, Georgia, using land cover, hydrological, spatial, and phylogenetic predictors. Gradient-forest analysis identified key environmental variables, which were tested with Threshold Indicator Taxa ANalysis (TITAN) and incorporated into a Hierarchical Model of Species Communities (HMSC). Our results show that hydrologic subunits (HUC8), wetland cover, and upstream channel length are the strongest predictors of crayfish assemblage composition, contributing to a cross-validated model with a mean AUC of 0.7. Wetland cover was particularly influential, suggesting that wetlands act as critical refugia for some species and, for a select few, as a barrier to dispersal across the basin. Hydrologic subunits structured assemblages along the river continuum, while channel length reflected connectivity and habitat heterogeneity. Some species showed significant positive or negative associations with the model's variables, suggesting that some species are specialists for those factors and can act as indicators of changes in those variables across the basin. We also found no significant biotic interactions among crayfish species within the basin, after accounting for variation due to the covariates used in the model. The model supports the existence of four major metacommunities across the basin, as illustrated by a community turnover plot generated from the first three PC axes of the model's predicted presence/absence values. These findings demonstrate that crayfish assemblages in the Ogeechee are strongly shaped by a complex interplay of hydrologic and landscape features, underscoring the importance of conserving wetlands and maintaining river connectivity. By linking community composition to environmental drivers, this study provides a framework for predicting crayfish responses to ongoing land-use change and informs conservation strategies for endemic species in southeastern rivers.

id #135435

Harnessing artificial intelligence to automate environmental predictions

Avni Malhotra¹, Brienne Forbes¹, Stefan Gary², Amy Goldman¹, Breanna Rivera Waterman³, Vanessa Garayburu-Caruso¹, Etienne Fluet-Chouinard¹, Sushant Mehan⁴, Michael Bruen⁵, Marcelo Ardon⁶, Bayani Cardenas⁷, Walter Dodds⁸, Christian Lonborg⁹, William H. McDowell¹⁰, Moussa Moustapha¹¹, Allison Myers-Pigg¹, Peter Regier¹, Tod Rubin¹², Hyun Song¹³, Ryan Stewart¹⁴, Jorge Villa¹⁵, Nicholas Ward¹, Tim Scheibe¹, James Stegen¹

1. Pacific Northwest National Laboratory, Richland, WA, United States

2. Parallel Works, Chicago

3. University of Kansas, Lawrence

4. South Dakota State University, Brookings

5. University College Dublin, Dublin

6. North Carolina State University, Raleigh

7. The University of Texas at Austin, Austin

8. Kansas State University, Manhattan

9. Aarhus University, Aarhus
10. The University of New Hampshire, Durham
11. The University of Ebolowa, Ebolowa
12. The Watershed Project, Richmond
13. The University of Nebraska–Lincoln, Lincoln
14. Virginia Tech, Blacksburg
15. The University of Louisiana at Lafayette, Lafayette

Predicting heterogeneous and non-linear processes remains a fundamental challenge in Earth sciences. Here, we present an artificial intelligence (AI)-guided, closed-loop workflow that iteratively combines predictive modeling with targeted field sampling to rapidly improve environmental predictions. We demonstrate our workflow by predicting oxygen consumption, a key process in rivers, across the contiguous United States (CONUS). Our approach consists of multiple iterative loops of measurements and models, combining distributed participatory field sampling, lab analysis, automated machine learning (ML) predictions, and error and distinctiveness analyses to autonomously guide the next sampling at optimal site locations. Specifically, we used an ensemble of stacked regressors to both maximize predictive performance and quantify predictive uncertainty, which then informed the data acquisition policy. Over 18 iterative loops, we increased the predictive power of sediment oxygen consumption across CONUS rivers by over fifteenfold between the first and last iteration. Relative to our last sampling iteration, our first sampling missed rivers with outlier values and underestimated median oxygen consumption rates by 68%. In addition to identifying areas of high oxygen consumption rates, iterations enabled refinement of laboratory and data handling methods, and engagement with a broad community of field researchers. We conclude that AI-guided iterative loops between targeted sampling and predictive modeling are a powerful and efficient approach for improving predictions of heterogeneous environmental processes. The architecture of our workflow balances model flexibility with interpretability and can be readily applied to other domains where uncertainty- and distinctiveness-aware site selection is crucial.

id #135948

Facilitation of aquatic bryophytes by baldcypress extends oxic periods and offsets carbon emissions in blackwater swamps

Andrew S Mehring¹, Dustin W Kemp²

1. University of Louisville, Louisville, KY, United States
2. Department of Biology, University of Alabama at Birmingham, Birmingham, Alabama, USA

We used laboratory and field studies to determine whether forest composition influences aquatic bryophyte primary productivity, oxygen release, and carbon dioxide (CO₂) uptake in forested blackwater swamps. To model total bryophyte oxygen release and CO₂ uptake, we measured *Porella pinnata* biomass within a range of heights supporting bryophyte growth on tree trunks and knees. We used photosynthesis-irradiance (P-I) curves to estimate oxygen production and CO₂ uptake across varying temperatures and light levels. These estimates informed four modeling scenarios of riverine oxygen dynamics and bryophyte-mediated CO₂ flux: 1) current forest composition, with baldcypress (*Taxodium distichum*) knee surface area included, 2) current composition with knee surface area removed, and 3) forests composed entirely of gum/tupelo (*Nyssa* spp) or 4) entirely of baldcypress. At moderate discharge and temperature, aquatic bryophytes may exert strong influence on ecosystem metabolism, counteracting ~10-25 percent of the oxygen demand and CO₂ released from decomposing leaf litter. Modelling results suggest that forest composition influences aquatic bryophyte primary production: buttressed trunks and abundant knees in baldcypress-dominated swamps produce greater surface area for bryophyte growth, which delays seasonal onset of hypoxia. This study demonstrates the importance of aquatic primary production in forested blackwater rivers and shows how baldcypress, a foundation tree species, can significantly modify aquatic ecosystem function.

id #135182

From Projects to Resilient Watersheds: Core Outcome Themes from the EU MERLIN Project

Sebastian Birk¹, Gerardo Anzaldúa², Annette Baattrup-Pedersen³, Anna Bérczi-Siket⁴, Kirsty Blackstock⁵, Tom Buijse⁶, Laurence Carvalho⁷, Nadine Gerner⁸, Nicolas Grondard⁹, Tomasz Okruszko¹⁰, Ellis Penning⁶, Josselin Rouillard², Astrid Schmidt-Kloiber¹¹, Daniel Hering¹

1. University of Duisburg-Essen, Duisburg-Essen, Germany
2. Ecologic Institute, Berlin, Germany
3. Aarhus University, Aarhus, Denmark
4. WWF Hungary, Budapest, Hungary
5. James Hutton Institute, Aberdeen, Scotland
6. Deltares, Delft, the Netherlands
7. Norwegian Institute for Water Research, Oslo, Norway
8. Emschergerossenschaft/Lippeverband, Essen, Germany
9. Wageningen University, Wageningen, the Netherlands
10. Warsaw University of Life Sciences, Warsaw, Poland
11. University of Natural Resources and Life Sciences, Vienna, Austria

Freshwater restoration is increasingly expected to deliver resilience at the scale of watersheds and the communities that depend on them. The EU Horizon 2020 project MERLIN (Mainstreaming Ecological Restoration of freshwater-related ecosystems in a Landscape context: INnovation, upscaling and transformation) offers a rare opportunity to reflect on how such ambitions can be operationalised across diverse socio-ecological contexts. MERLIN worked with 18 large-scale restoration case studies across

Europe, spanning small streams and their basins, large rivers and their floodplains, and peatlands and other wetlands. This presentation synthesises MERLIN's core outcome themes, structured around five interlinked domains that proved critical for building resilient freshwater landscapes. First, place-based learning highlighted how restoration trajectories are shaped by local histories, turning points, and adaptations beyond initial project designs. Second, upscaling and landscape planning showed the importance of embedding site-level measures within long-term watershed visions and regional coordination frameworks. Third, mainstreaming across sectors revealed both opportunities and limits for integrating freshwater restoration into agriculture, climate adaptation, and spatial planning. Fourth, resource management and financing underscored the need to move beyond short-term project funding toward diversified and blended finance models supported by institutional capacity. Finally, capacity building and knowledge transfer, including practitioner learning and training, emerged as a prerequisite for sustaining restoration outcomes over time. By integrating these outcome themes, the talk positions MERLIN not as a collection of cases, but as a structured learning space for understanding how freshwater restoration can contribute to resilient watersheds and communities. The synthesis is intended to stimulate dialogue with North American experiences and support cross-continental learning on the governance and practice of large-scale freshwater restoration.

id #135694

Explicit representation of hyporheic hydrology in whole-ecosystem metabolism models

Hayley Oakland¹, Samuel Fritz¹, Anna French¹, Lindsey Albertson¹, Geoffrey Poole¹

1. Montana State University, BOZEMAN, MONTANA, United States

Dynamic water exchange between stream channels and hyporheic zones plays a crucial role in ecosystem processing and function, yet adequately representing these dynamics in whole-stream metabolism models remains challenging. We demonstrate an approach for explicitly incorporating hyporheic hydrology into ecosystem process models by testing how foundational mathematical relationships between residence time, hyporheic exchange, and subsurface storage affect metabolite delivery and thus influence estimates of ecosystem metabolism. Our analysis employs a flexible, user-defined functional form for hyporheic water age distributions, which avoids inappropriate assumptions inherent to simpler models, such as well-mixed hyporheic zones and/or infinite water residence times. This framework allows us to more realistically represent the dynamic nature of hyporheic zone effects on whole ecosystem metabolism and, critically, to distinguish between the effects of hydrologic delivery and biotic activity. We evaluate how this integrated approach affects our ability to estimate ecosystem metabolism parameters and discuss implications for modeling coupled hydrologic-biotic processes in real stream systems.

id #135183

Home range size and movement of giant salamanders, *Amphiuma means* and *Siren lacertina*, in seasonally inundated, isolated wetlands

Madeline N Zickgraf^{1,2}, Carla L Atkinson¹, Alexander D Huryn¹, Lora L Smith²

1. Biology, University of Alabama, Tuscaloosa, AL, USA

2. Herpetology, The Jones Center at Ichauway, Newton, GA, USA

In wetlands across the Southeastern Coastal Plain, two large aquatic salamanders, Two-toed amphiuma (*Amphiuma means*) and Greater siren (*Siren lacertina*), can be abundant. As both predators and prey, *A. means* and *S. lacertina* likely exert influence on wetland community structure and biogeochemistry; especially in isolated wetlands where their biomass dominates the upper trophic levels as fish are generally absent and apex predators such as, American alligators (*Alligator mississippiensis*) and large wading birds, are typically transient. Despite their potential impact in wetlands, basic aspects of their ecology such as habitat use and seasonal activity remain unclear. To understand salamander movement and estimate home range sizes, we conducted a radiotelemetry study. We surgically implanted ten individuals of both species with radio-transmitters and collected locations 2-3 times per week from March through August 2025. From these locations we generated minimum convex polygons from individuals with ≥ 5 unique locations. *S. lacertina* exhibited the largest home range (82.4 m²) while *A. means* exhibited the smallest (2.1 m²). Mean home range size was 28.1 m² for *S. lacertina* (n = 7) and 4.2 m² for *A. means* (n = 9) though high variation in *S. lacertina* ranges resulted in no significant difference. However, the variances in home range size of the two species were significantly different. Distance moved varied significantly between species; mean distance moved was 4.5 m for *A. means* (n = 10) and 10.1 m for *S. lacertina* (n = 8). Movement frequency did not vary across time and was marginally significant between species. *A. means* had a higher predicted probability of movement than *S. lacertina* throughout the study period. By examining the space use and activity patterns of these two abundant aquatic mesopredators we can begin to understand their role and potential impacts within these vulnerable aquatic systems.

id #135695

Cool tools for warming streams: A new thermal stressor index helps managers in King County, WA interpret trends in macroinvertebrate communities

Kate Macneale¹, Beka Stiling¹, Diane Yeh¹, Jeremy Walls¹

1. King County - Water and Land Resources, WA, United States

Biological indices indicate stream conditions in King County, WA have improved over the last twenty years, despite increased development and warming stream temperatures. To help explain these surprising yet encouraging trends, we used a new thermal stressor index and improved estimates of thermal preferences that Shannon Hubler and others developed for freshwater macroinvertebrates in the Pacific Northwest. We found macroinvertebrate taxa that are increasing in density across lowland streams are more tolerant of warmer water than those that are not increasing or spreading. In addition to providing insights on how macroinvertebrate communities may be responding to warming streams, the thermal stressor index and improved taxon-specific thermal preferences can be used to help identify stressors and prioritize streams for targeted restoration actions. The

thermal stressor index can also be used to biologically validate modeled stream temperatures, including providing biological evidence that temperatures may be too warm for certain macroinvertebrate taxa and salmonids.

id #135951

La ruta de pesca: Fishers' knowledge provides insight to ecology, conservation, and management of migratory catfishes in the Amazon

Elizabeth Anderson¹, Fiorella Briceño¹, Vanessa Rodriguez²

1. Florida International University, Miami, FLORIDA, United States

2. Independent, Lima, Peru

Large migratory catfishes of the Amazon are iconic species that connect distant habitats with their migrations, thereby influencing ecological conditions throughout much of the basin. Amazon migratory fishes can be loosely grouped based on the direction (lateral, longitudinal) and scale (local, regional, continental) of their migrations, which often occur for feeding or reproduction. Beyond their ecological importance, large migratory catfishes sustain freshwater fisheries throughout much of the lowland Amazon, providing a major source of protein and income for many Amazonian people. Concerns over perceived and reported declines in their populations caused by overfishing and hydropower dams have motivated a recent proposal to include large migratory catfishes as a conservation target under the Convention on Migratory Species. However, much information on the species' ecological needs and existing management approaches is still lacking. Large migratory catfishes are hard to study, in part because they move over continental-scale distances. Only in the past decade have detailed scientific syntheses been published about their migratory routes and movement ecology. These studies offer a basin-scale perspective on migration as an ecological process, but less published information exists about the ecology and habitat use of migratory catfishes in specific localities, or about the fishing communities that depend on them. These fishing communities are likely the best source of knowledge on large migratory catfish, yet in few instances has this knowledge been recorded. Within this context, we aimed to document the experience, knowledge, and management of large migratory catfishes among fishers along the Amazon River in Peru. We had four lines of inquiry: (1) important species; (2) techniques and fishing gear; (3) local management of the fishery; and (4) change over time. We conducted our research in collaboration with seven communities in Loreto, Peru, during 2024-26. Here we describe the research process, fishers' perspectives, and preliminary results to date. We also share how fishers' knowledge from Peru and other parts of the Amazon has strengthened the case for their inclusion in the Convention on Migratory Species.

id #135440

Do Beaver Dam Analogs Mimic Beaver Dams?: Effects on Food Webs, Structure, Function, and Water Quality in a Restored Stream

Miles Milbrath^{1, 2, 3}, Lusha Tronstad^{1, 2, 3}

1. University of Wyoming, Laramie, WYOMING, United States

2. Wyoming Natural Diversity Database, Laramie, Wyoming, USA

3. Department of Zoology and Physiology, Laramie, Wyoming, USA

Beavers are ecosystem engineers whose dams increase habitat complexity, water retention, and ecological resilience in stream ecosystems; however, beavers have been extirpated from 95% of the streams they once occupied. Beaver dam analogs (BDAs) are increasingly used in the western USA to replicate these functions in degraded ecosystems, yet their effects on food web structure remain poorly understood. We evaluated how BDAs and natural beaver dams influence fish and macroinvertebrate abundance and biomass, food web structure, and nutrient dynamics in Muddy Creek, a semi-arid stream in southern Wyoming, USA, across four reach types: reference (no structure), natural beaver dams, BDAs, and BDAs enhanced by beaver activity. A before-after-control-impact framework was applied at the BDA reach, where sampling occurred before and after installing BDAs. Stable isotope ratios of carbon ($\delta^{13}\text{C}$), nitrogen ($\delta^{15}\text{N}$), and sulfur ($\delta^{34}\text{S}$) were measured from basal resources and fishes to evaluate shifts in energy pathways and trophic structure. Following BDA installation, fish abundance doubled at the BDA reach relative to pre-installation, while the reference reach showed no comparable increase. Notably, Colorado River Cutthroat Trout (*Oncorhynchus clarkii pleuriticus*), a native species that is the focus of restoration efforts, colonized the reach after installing BDAs. Preliminary isotope results suggest that fishes were enriched in $\delta^{15}\text{N}$ in BDA and beaver-influenced reaches relative to the reference reach, indicating more complex food chains. Carbon isotope values of some fishes were more depleted in the BDA reach, suggesting increased reliance on benthic production. Additionally, sulfur isotope signatures were more enriched in the BDA reach, consistent with enhanced wetland or groundwater connectivity. Patterns in isotope composition and fish responses in BDA and beaver-influenced reaches were more similar to each other than to the reference reach. We are currently analyzing chlorophyll a, nutrients, water quality, and macroinvertebrate density and biomass, which will be presented. Our findings suggest that while BDAs do not fully replace beavers, they rapidly initiate ecological conditions that converge toward those created by natural beaver dams. These results support the use of BDAs as an effective, beaver-mediated restoration tool for rebuilding food web complexity and freshwater resilience.

id #135696

Temporal and spatial changes in nutrient concentrations in the War Eagle Creek Watershed, Arkansas

Brandy N Everett¹, Brian E Haggard^{1,2}

1. *Arkansas Water Resources Center, Fayetteville, AR, United States*

2. *Biological and Agricultural Engineering, University of Arkansas, Fayetteville, Arkansas, United States*

The War Eagle Creek Watershed (WEC), a major tributary to Beaver Lake, the primary drinking water source for Northwest Arkansas, was identified as a hotspot for nutrient and sediment transport and a priority for conservation in the EPA approved watershed management plan. To evaluate nutrient patterns within WEC, water samples were collected from May 2025 through 2026 and analyzed for total nitrogen (TN), total phosphorus (TP), nitrite, soluble reactive phosphorus (SRP), and conductivity. Recent findings were compared to historical data from 2003–04 to assess nutrient changes over time. Additionally, land cover at the watershed and site catchment scales was calculated to identify potential sources of nutrient inputs.

Spatial variability in nutrient concentrations was observed across WEC, with higher concentrations generally occurring at downstream sites. Geometric mean TN and TP ranged from 0.09 to 4.66 mg/L and from <0.01 to 0.38 mg/L, respectively. These patterns are likely due to increased pasture land use and inputs from a wastewater treatment plant.

Comparison with historical data revealed temporal variability as well. Most sites showed increased conductivity and P concentrations over time, while N trends were more varied. TN increased at 54% of sites between 2003–04 and 2025–26, but the median TN geomean across sites decreased by 34%. In contrast, TP and conductivity increased at 63% and 72% of sites, respectively, with median site-level concentrations increasing by 2% for TP and 19% for conductivity. This indicates that temporal changes in nutrient concentrations are not uniform across WEC and vary in magnitude across sites.

Catchment land cover changed slightly between 2003 to 2024, including increases in urban (4.5% to 5.5%) and pasture cover (37% to 39%), and decreases in forest cover (58% to 55%). Site catchments experiencing increases in pasture and urban cover, along with declining forest cover, corresponded with higher observed nutrient concentrations in 2025–26 relative to 2003–04, highlighting land use change as a likely driver of nutrient inputs. Additionally, results will be compared with priority areas identified in the 2025 Beaver Lake Watershed Protection Strategy to assess patterns between observed nutrients and modeled management priorities.

id #135952

Stressor-response relationships linking human activities to Pacific salmon in freshwater ecosystems

Marta Ulaski¹, Jonathan Moore¹

1. *Earth to Ocean Research Group, Simon Fraser University, Burnaby, BC, Canada*

Understanding how Pacific salmon respond to multiple human activities is essential for predicting vulnerability and informing management in increasingly altered freshwater ecosystems. Stressor-response relationships quantify changes in biological responses as exposure to environmental stressors varies. Estimating these relationships has become increasingly important for predicting the cumulative effects of activities on species, including salmon. Salmon ecology literature is vast, yet existing evidence describing stressor-response relationships for Pacific salmon are fragmented across stressors, activities, and species.

We conducted a comprehensive synthesis and quantitative reanalysis of stressor-response relationships linking human activities to Pacific salmon responses in watersheds of the Pacific Northwest. We first developed a pathway-of-effects framework linking major human activities (e.g., forestry, agriculture, urban development, mining, hydroelectric infrastructure) to freshwater stressors (e.g., temperature, flow alteration, sediment) and salmon responses (e.g., survival, recruitment, fecundity). This framework informed search terms for a systematic literature review. Studies were screened using predefined inclusion criteria, including restriction to stressors in fresh water and studies conducted in the Pacific Northwest (e.g., Alaska, British Columbia, Washington, Oregon, and Idaho).

For studies reporting underlying data, we extracted or digitized observations and fit stressor-response curves across studies and species using generalized linear mixed models. Models included species-level random intercepts and slopes and study-level random intercepts to account for hierarchical structure and cross-study variability. We quantified where stressor-response curves changed most rapidly and compared the strength of responses at those locations across stressors using standardized metrics.

Across synthesized stressors, Pacific salmon responses frequently exhibited nonlinear dynamics, with pronounced changes in response occurring over relatively narrow ranges of stressor exposure. The magnitude and location of these changes varied among stressors and species, with many relationships exhibiting weak, linear responses. Many stressor-response relationships lacked sufficient data for robust estimation and studies testing for interactive effects among multiple stressors were scarce. This synthesis integrates pathway-based review and quantitative reanalysis to provide stressor-response curves that can be used in cumulative effects frameworks and help prioritize future research and management actions for Pacific salmon in freshwater ecosystems.

id #135697

Duckweed (*Lemna* and *Wolffia* spp.) coverage does not alter organic matter decomposition rates in small ponds

McKenzie E Goodwyn¹, Nathan O Earl¹, Andrew S Mehring¹, Joshua Snipes¹

1. *University of Louisville, Louisville, KY, United States*

Duckweeds (subfamily *Lemnoideae*) are among the smallest and fastest-growing vascular plants on Earth. These free-floating species can rapidly cover the surface of a pond, reducing light and creating hypoxic conditions in the water column, and we predicted these conditions would result in slower organic matter breakdown. To explore this, plastic mesh litter bags containing sugar maple (*Acer saccharum*) wood or leaf litter were deployed in six small peri-urban ponds near Louisville, Kentucky—three with persistent duckweed cover and three without. Four bags of wood and two bags of leaf litter were retrieved on six dates over a five-month incubation period, during which time dissolved oxygen, temperature, pH, and floating duckweed biomass were monitored at each site. Decomposition was quantified as the change in the percentage of dry mass remaining over time. A generalized linear mixed-effects modeling approach was used to identify environmental and structural factors influencing

decomposition. Leaf litter decomposed more rapidly than wood in both pond types ($p = 2.2e-16$), litter material did not decompose significantly between groups, despite significantly lower dissolved oxygen ($p = 8.1e-5$) and moderately lower temperatures in duckweed covered ponds. Temperature had a significant effect on decomposition ($p = 0.038$) while DO and pH did not. This work improves our understanding of how floating vegetation influences organic matter processing and carbon cycling in small wetland ecosystems.

id #138257

Alternate pathways to restoring fish passage at high head dams

Conor Giorgi¹

1. *Spokane Tribe of Indians, Spokane, WA, United States*

Installation of new fish passage facilities is often the result of hydroelectric dam relicensing processes or driven by required compliance with Endangered Species Act mandates. In the absence of those regulatory hooks, what compels federal agencies and other stakeholders to support new fish passage installation and salmon reintroduction efforts? In the upper Columbia River watershed there are five dams blocking anadromous fish from previously occupied habitats. The Upper Columbia United Tribes' Phase 2 Implementation Plan ("P2IP") addresses those blockages through systematic feasibility studies of fish behavior, survival, and passage systems. In this talk I'll review regional history for context, discuss the phased approach to reintroduction, and how the combination of persistence, hope, and transparency led to broad support and commitment to the tribally led effort.

id #134162

Assessing Spatial and Temporal Variation in the Size Spectra of Benthic Macroinvertebrates in the Urban Santa Ana River

Ria Ghosh¹, Kurt Anderson¹

1. *University of California, Riverside, Riverside, CALIFORNIA, United States*

Urbanization generates novel spatial and temporal heterogeneity, altering stream ecosystem structure and function. This includes benthic macroinvertebrates (BMIs), widely used as bioindicators of freshwater health, that respond quickly to changing environmental conditions at multiple scales. However, BMI taxonomic responses are often inconsistent, making it difficult to generalize urban impacts (Gjoni et al., 2024; Collyer et al., 2023). A useful tool for evaluating ecosystem dynamics is the size spectrum (SS) which describes the relationship between organism body size and biomass or abundance. Urban stressors like wastewater treatment plant (WWTP) effluent can alter stream thermal regimes, along with flow stability and water chemistry, all of which influence organismal metabolic rates and size distributions. Metabolic theory predicts that warmer environments produce steeper slopes with more small-bodied organisms, whereas colder environments should produce shallower slopes and large-bodied ones (Brown et al., 2004). However, recent studies show that SS patterns often deviate from these predictions, suggesting additional urban stressors beyond temperature influence SS slopes.

To better understand ecosystem effects in urban areas, the study investigates spatial and temporal variation in BMI size spectra in the urbanized Santa Ana River, California, where 85% baseflow is from WWTP discharge. We ask how abiotic drivers—temperature, substrate, depth, pH, conductivity, dissolved oxygen, velocity, and discharge—shape SS patterns across eight sites over one year. BMIs were identified and converted to biomass (Benke et al., 2011). SS slopes are estimated using the MLEbin method to evaluate environmental drivers of spatial and temporal variation (Edwards et al., 2017).

Preliminary results show body-size distributions with shallower slopes at effluent-influenced sites near WWTPs than at downstream non-effluent sites. Seasonal SS variation is evident across all sites, but does not consistently distinguish effluent and non-effluent reaches. Depth is the main driver of slope variation, with weak interactions among substrate, discharge, conductivity, and velocity. By improving understanding of SS variability under complex urban pressures, this work informs urban ecosystem resilience and freshwater management.

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id #135698

Assessing patterns of freshwater fish extirpations in the southeastern United States

Eric Walther¹, Mary Freeman¹, Seth Wenger¹

1. River Basin Center, University of Georgia, Athens, GA, United States

The southeastern United States is a global hotspot for freshwater biodiversity. There are at least 600 freshwater fish species found in river systems across the southeastern United States, of which about half are locally endemic with limited geographic range. However, following nearly two centuries of habitat modification, fragmentation, and land use change, many southeastern freshwater fish populations are now vulnerable to local extinction. We compiled stream fish sampling data for over 18,000 sites across the southeastern United States and generated lists of candidate species extirpations for watersheds with sufficient monitoring data at the HUC-8 watershed scale. During a workshop held in 2023, the candidate lists were reviewed by a group of experts from across the Southeast with deep knowledge of freshwater fish distributions and assigned an extirpation status to candidate species in 134 watersheds. Over half of the watersheds assessed ($n = 78$) are hypothesized to have lost at least one species (range: 1 – 17 species extirpated) and 113 species are presumed extirpated from at least one HUC-8 watershed (range: 1 – 10 watersheds). We fit hurdle models to examine how land use change, habitat fragmentation, and watershed characteristics are associated with the variation in the presence of any extirpations and the number of expected extirpations in HUC-8 watersheds. This analysis can be used to identify landscape features associated with “extirpation hotspots” and watersheds where additional extirpations are likely absent appropriate conservation actions. For example, our preliminary modeling identified fifteen watersheds that may have at least one extirpated species but currently have no documented extirpations. The ability to infer where there may be undetected extirpations and the compilation of a publicly available database of documented fish extirpations can help target conservation, e.g., to watersheds that support species that have been extirpated elsewhere and locations that could benefit from reintroduction efforts.

id #135954

Landscape Drivers of Lake Thermal Structure in Southwest Alaska’s National Parks

Matthew Dunkle¹, Paul W.C. Gabriel^{2, 3}, Sadie R. Textor³, Krista K. Bartz³, Jeffrey D. Muehlbauer^{1, 4}

1. Institute of Arctic Biology, University of Alaska, Fairbanks, Alaska, USA

2. Division of Water, Alaska Department of Environmental Conservation, Anchorage, Alaska, USA

3. Southwest Alaska Inventory and Monitoring Research Network, National Park Service, Anchorage, Alaska, USA

4. Alaska Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Fairbanks, Alaska, USA

Lakes are sentinels of climate change because their physical and chemical conditions integrate regional climate, local weather, and watershed-scale land cover and hydrology. In southwest Alaska, these lakes provide critical habitat for iconic populations of sockeye salmon and other species, underpinning regional culture, ecology, and economy. To evaluate how long-term changes in climate and watershed conditions influence lake temperature and water quality, we analyzed 15 years of vertical profile data collected by the National Park Service Inventory and Monitoring Program in lakes of Katmai and Lake Clark National Parks and Preserves. Using in situ measurements, downscaled climate data, and remotely sensed lake and watershed characteristics, we developed separate models for mixed and stratified lakes. For stratified systems, we identified drivers of epilimnetic and hypolimnetic temperatures as well as thermal gradients between layers. We then applied future climate projections to assess potential trajectories of lake thermal structure under continued warming and shifting precipitation regimes. Our results indicate that lake thermal conditions respond not only to contemporaneous summer weather, but also to legacy effects of temperature, precipitation, and wind from preceding fall and winter seasons. Lake size and morphometry, glacier influence, and surrounding land cover further mediate these responses. Together, these findings highlight the importance of cross-seasonal and landscape-level controls on subarctic lake thermal dynamics and provide a foundation for salmon habitat modeling, long-term monitoring, and climate-adaptive management in Alaska’s national parks.

id #138258

Alternate pathways to restoring fish passage at high head dams

Desiree Tullos¹

1. Oregon State University, CORVALLIS, OR, United States

Dam removal is a growing practice to address fish passage barriers, often occurring in convergence with other infrastructure objectives, such as reducing safety and financial liabilities for dam owners and permitting processes that require reducing negative environmental impacts, among others. Thus, the pathways to dam removal can be as varied as the dams and the landscapes on which they occur, but they tend to require a common set of catalysts and actors to reach. In this presentation, I will summarize some of those key catalysts and actors that drive decisions to remove a dam, in particular summarizing the pathway that led to the Klamath River dam removals and how that pathway intersected with dam removal design. I will conclude with a short summary of some of the early responses to dam removal on the Klamath River to illustrate the full life cycle of dam removal decisions, design, and outcomes.

id #135699

Chironomid community composition in alpine ponds

Mason Chow¹, Mia Ashby¹, Kurt Anderson¹

1. University of California, Riverside, Diamond Bar, CALIFORNIA, United States

Aquatic invertebrate community composition is a standard measure of freshwater ecosystem health. Monitoring of invertebrate communities usually focuses on taxa such as Trichoptera, Plecoptera, or Ephemeroptera. Collectively referred to as EPTs, these taxa are widely studied owing to their nutrient cycling capabilities and water quality sensitivity (Lenat, 1988). However, their unreliable presence in some unstable and remote environments such as ephemeral alpine pools, creates a challenge in assessing a full range of ecological conditions. Chironomids represent a widespread and abundant taxon that remain understudied due to the technical challenges of their identification, especially in remote areas where invertebrate sampling rarely occurs (Ward et al. 2002). This research aims to investigate how Chironomidae subfamily diversity and abundance reflects biotic and abiotic conditions of ephemeral aquatic habitats in the alpine Green Lakes Valley. These bodies of water represent a model system due to reduced direct anthropogenic influence. We aim to quantify and categorize the benthic invertebrate taxa using invertebrate and environmental samples collected from 2021 to 2024 and compared with historical data collected in 1987-88 (Mihuc and Toetz, 1996). All specimens were identified using light microscopy, and chironomids were slide-mounted and identified to subfamily and genus when possible. The changes in chironomidae subfamily composition may reflect the availability of genera-specific trophic conditions due to the presence or absence of prey. Our findings represent a step towards developing Chironomids as an indicator taxa for alpine aquatic research in the face of volatile future climates.

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id #135955

Integrating biodiversity and allometric theory to explain macroecological patterns of community biomass in streams

James R Junker¹, Vojsava Gjoni², Justin Pomeranz³, Jeff Wesner⁴

1. *Dept. of Biological Sciences, University of North Texas, Denton, TX, United States*
2. *Institute for Marine Biological Resources and Biotechnology, Italian National Research Council, Messina, Italy*
3. *Dept. of Biological Sciences, Colorado Mesa University, Grand Junction, Colorado, U.S.A.*
4. *Dept. of Biology, University of South Dakota, Vermillion, South Dakota, U.S.A.*

One of the most pressing aims in ecology is discerning the interrelationships among the environment, species diversity, and ecosystem functions. A wealth of theoretical and experimental research has shown that more diverse communities support higher standing biomass and energy fluxes in ecosystems (biodiversity–ecosystem function; BEF); however, BEF relationships in natural communities show much more complex and idiosyncratic patterns. This discrepancy is especially apparent in multi-trophic communities, where the relative importance of bottom-up and top-down control leads to diverse relationships between diversity and ecosystem functions such as total community biomass and energy flux. Recent work on the BEF relationships in multi-trophic communities has highlighted that complementary approaches, particularly allometric theory (AT), may help explain variability in BEF relationships across ecosystems. Here, we develop a framework leveraging a widely observed macroecological rule from AT—that organism abundance (N) is negatively related to organism body size (M) within and across ecosystems—to explore the environmental and community drivers of community standing stock biomass in the continental-scale stream gradient of the National Ecological Observatory Network (NEON). Using two size-abundance relationships: 1) the individual size distribution (ISD), which describes the relative frequency of body sizes within a community and represents a proxy of food web energetic efficiency, and 2) the cross-community scaling relationship (CCSR), which describes covariation in community abundance and body size across communities, we integrate BEF and AT concepts to explain patterns in community biomass. Streams across NEON vary $>20^{\circ}\text{C}$ in mean temperature, orders of magnitude in basal resource availability (i.e., gross primary production and organic matter standing stock) and, accordingly, communities of macroinvertebrate and fish vary multiple orders of magnitude in community abundance (10^4 – 10^6 individuals/ m^2), biomass (10^1 – 10^3 g/ m^2), mean body size (10^1 – 10^3 mg/individual), and >3 -fold in species richness (30–115 species). We find weak bivariate relationships between diversity and community biomass, but stronger congruence between predictions of AT and community biomass patterns. We further incorporate determinants of ecosystem energetics such as basal resource availability and temperature, with BEF and AT concepts towards developing a richer understanding of macroecological drivers of community biomass and ecosystem function.

id #134420

Storm hydrologic connectivity in non-perennial streams and its influence on dissolved organic carbon

Erin VanderJeugd¹, Kenneth Lowe¹, Xin Gu¹, Xiangping Yin¹, Elizabeth Herndon¹

1. *Oak Ridge National Laboratory, Oak Ridge, TN, United States*

Non-perennial streams flow intermittently but are otherwise comprised of disconnected pools and dried stream beds. Storms increase hydrologic connectivity around the stream channel, facilitating dissolved organic carbon (DOC) transport from surrounding uplands to perennial stream networks. Organic matter lability, meaning the ease of which organic matter is broken down by microorganisms, is dependent in part on its chemical properties, which vary with its source. **We aim to understand how surface and subsurface flow paths connect and influence carbon lability and transport before, during, and after storm events.** Our study took place in two tributaries draining catchments with contrasting land cover (one urban, one forested) in the East Fork Poplar Creek watershed in Oak Ridge, TN. We collected stream and groundwater from wells installed in perennial and non-perennial reaches of both areas during baseflow and storm flow in Fall 2024 and Winter 2025. All samples were analyzed for major anions and cations, DOC, nutrients, and trace elements while baseflow and storm groundwater samples were also

analyzed for dissolved inorganic carbon, sulfides, and Fe(II). Carbon lability was assessed spectroscopically by measuring ultraviolet absorbance at 254 nm (SUVA₂₅₄), a measurement positively correlated with aromatic (not labile) carbon. During baseflow conditions, DOC concentrations were slightly higher in the urban (3.1 ± 0.6 mg/L) than the forested tributary (1.9 ± 0.3 mg/L) but were comparable between streams and groundwater within each tributary. SUVA₂₅₄ values, however, were higher in streams (6.2 ± 0.6 L/(mg*m)) than groundwater (3.8 ± 0.3 L/(mg*m)) at baseflow, regardless of land cover. During the storm event, DOC concentrations started high in the stream (22.7 mg/L) and decreased through the storm (10.3 mg/L) but remained low in groundwater (0.88 ± 0.06 mg/L). SUVA₂₅₄ values in the stream were low relative to baseflow but increased throughout the storm (2.9 - 5.6 L/(mg*m)), while SUVA₂₅₄ values in groundwater were elevated yet variable (2.9 ± 0.3 L/(mg*m)). While low SUVA₂₅₄ values in the stream early in the storm indicate inputs of labile carbon from groundwater, elevated DOC concentrations and SUVA₂₅₄ values in the stream later in the storm imply that runoff from surface soils is contributing recalcitrant carbon.

id #135188

Monitoring long-term water quality benefits of urban stream restoration

Kira D Burnett¹, Shannon L Speir¹

1. *University of Arkansas, Fayetteville, ARKANSAS, United States*

Urbanization without conscious environmental planning can result in the degradation of stream ecosystems. This tends to result in urban stream syndrome, which is often characterized by increased sediment and nutrient loading, decreased biodiversity, elevated flashiness, and overall water quality degradation. As such, stream restoration practices are often implemented to address urban stream syndrome. Commonly used restoration practices include applying natural channel design to alter a stream's geomorphology and stabilize eroding channels and establishing diverse riparian vegetation to filter pollutant inputs. However, the benefits of restoration are not often quantified due to a lack of post-restoration monitoring, making it challenging to determine what restoration success looks like. Therefore, we monitored Mullins Creek, a partially restored urban stream in Fayetteville, Arkansas, for physical and chemical parameters. We sampled five sites in the watershed - two sites upstream of the unrestored reach, one site within the restored reach, and two downstream sites - for dissolved oxygen, nitrate and nitrite (NO₃-N + NO₂-N), ammonium (NH₃-N), specific conductivity, soluble reactive phosphorus, temperature, and turbidity. Importantly, there has been no active maintenance of the restoration since its establishment. Ten years post-restoration, we found that all parameters met Clean Water Act standards in 2024, including turbidity which exceeded limits in the previous year. We did document substantial variation in nutrient concentrations between years, with phosphorus and nitrate showing fewer extreme values in 2024 compared to 2023. The year-to-year variability indicates that the restoration may no longer be resilient and able to maintain water quality under typical interannual hydrologic variation. This year-to-year variation in water quality in Mullins Creek suggests that initial restoration benefits may degrade without continued management, highlighting the need for adaptive maintenance strategies to sustain water quality improvements in urban streams.

id #135700

Seasonality and Land Use Influence Organic Matter Processing Rates in Tropical Watersheds

Krista Capps¹, Irene Sanchez Gonzalez¹, Maria Mercedes Castillo², Erick Antonio Chacón Hartleven², Aaron Jarquin Sanchez², Arutro Marcial Álvarez Merino²

1. *University of Georgia, GA, United States*

2. *El Colegio de la Frontera Sur, Tabasco, Mexico*

Rivers receive and process large quantities of terrestrially derived organic matter. Global studies using standardized cotton strip assays show that organic matter processing varies widely across biomes, is fastest at low latitudes, and is consistently higher in rivers than in adjacent riparian zones, reflecting differences in temperature sensitivity and moisture limitation. Despite the importance of tropical rivers to global runoff and carbon emissions, organic matter processing in tropical stream and riparian systems remains understudied, particularly across seasons and land-cover types. We examined seasonal shifts in organic matter processing across forested, pasture (intermittent and perennial), and urban streams and adjacent riparian zones in southern Mexico using a standardized cotton strip assay deployed during the wet and dry seasons of 2023. We predicted higher processing rates in streams than in riparian zones, strong seasonal effects, and elevated instream decomposition in urban systems, particularly during the dry season. We also expected higher processing rates in perennial than in intermittent pasture sites. Instream processing rates were high relative to other tropical sites, especially in urban streams during the dry season under hotter, drier conditions. As expected, processing rates were consistently higher in streams than in riparian zones, which exhibited uniformly low rates and weak responses to season and land cover. Seasonality was the dominant driver of organic matter processing, with substantially higher decomposition during the wet season across land cover types. Urban land cover exerted a strong but season-dependent effect, with the highest instream decomposition rates overall. Contrary to expectations, intermittent pasture streams did not show reduced instream processing during the dry season relative to perennial streams, despite flow intermittency. Together, these results indicate that seasonal hydrology is the primary regulator of organic matter processing in tropical stream and riparian systems. Additionally, urbanization amplifies instream decomposition relative to forested and pasture systems, with differences especially pronounced during the dry season. Because stream research in the tropics is often conducted during the dry season, our findings suggest that seasonal sampling bias may lead to systematic underestimation of organic matter processing in tropical watersheds and their contribution to global organic matter processing.

id #135189

Post-fire suspended-sediment and phosphorus attenuation in a central Montana headwater catchment.

Gregory D Clark¹, Michael W Suplee², Haylie A Brown², Molly A Moloney¹, Rodney R Caldwell¹

1. US Geological Survey, Helena, MONTANA, United States
2. Montana Department of Environmental Quality, Helena, Montana

Eutrophication and nuisance filamentous algal blooms (i.e. *Cladophora*) are increasingly common occurrences as increased nutrient loading is coupled with warmer and drier regional climates. Wildfire may be contributing to the frequency and magnitude of localized algal blooms through excess sediment and phosphorus loading. This study evaluates how the 2021 Woods Creek Fire affected sediment concentrations and phosphorus (total and bioavailable) partitioning in Camas Creek (Montana, USA), where *Cladophora* are now consistently reaching nuisance levels. Both before and after the fire, we collected water quality samples during snowmelt events as well as routine interval sampling using streamgages instrumented with a continuous water quality monitor and an automatic peristaltic pump sampler. The asynchronous relationship between turbidity and streamflow was evaluated to interpret sediment sourcing and connectivity. Water samples were analyzed for total phosphorus (TP), sediment-bound bioavailable phosphorus (S-BioP), soluble reactive phosphorus (SRP), and suspended sediment concentrations. Hysteresis analysis of 91 delineated events indicated turbidity often reached its maximum before streamflow during snowmelt events. This may indicate a lower hillslope to channel connectivity, with the major sediment and total phosphorus supply originating from the channel and/or riparian areas. Water quality results indicate that the Woods Creek Fire did not significantly influence TP and S-BioP in Camas Creek. However, there was a significant increase in SRP and turbidity in both postfire years (2022 and 2023). Hydroclimatic conditions likely played a role in the observed response as both postfire years were markedly different in terms of snowpack and streamflow magnitude. While a lack of postfire response in TP and S-BioP is contrary to many other studies, our findings highlight the role climate and event behavior play in attenuating a disturbance effect.

id #135701

RAD-seq generated single nucleotide polymorphisms resolve patterns of genetic diversity and structure of the freshwater mussel *Ptychobranchus fasciolaris* across its distribution

Nichelle M VanTassel^{1,2}, Brant E Fisher³, Wendell R Haag⁴, Trevor L Hewitt⁵, Todd J Morris⁶, Nicholas M Sard⁷, David T Zanatta², Nichelle M VanTassel^{1,2}

1. South Carolina Department of Natural Resources, Charleston, SOUTH CAROLINA, United States
2. Biology, Central Michigan University, Mount Pleasant, Michigan, United States
3. Division of Fish & Wildlife, Indiana Department of Natural Resources, Edinburgh, IN, USA
4. Southern Research Station, US Forest Service, Frankfort, KY, USA
5. Department of Ecology and Evolutionary Biology and Museum of Zoology, University of Michigan, Ann Arbor, Michigan, USA
6. Fisheries and Oceans Canada, Burlington, Ontario, Canada
7. Biological Sciences Department, State University of New York-Oswego, Oswego, New York, USA

Resolving geographic patterns of genetic diversity is necessary for understanding the evolution and biogeography of freshwater biodiversity and has important implications for its conservation and recovery. The Kidneyshell (*Ptychobranchus fasciolaris*), a freshwater mussel (Bivalvia: Unionidae) distributed throughout the Ohio River watershed and southern portions of the Laurentian Great Lakes watershed. *Ptychobranchus fasciolaris* is considered secure globally, but it is considered a species of conservation concern in portions of its distribution (e.g., Endangered in Canada). Restriction-site associated DNA sequencing (RADseq) was used to develop a genome-wide dataset of single nucleotide polymorphisms (SNPs) to examine patterns genetic diversity and structure across the range of *P. fasciolaris* and further used to make inferences about its natural history and biogeography and recommendations for conservation and restoration of populations. We identified two major populations corresponding to the Great Lakes and Ohio River watersheds, indicating that each should be considered a separate management unit. Patterns of genetic variation supported post-Pleistocene colonization of the Great Lakes drainage via probable stream capture of the St. Joseph River from the Wabash River (Ohio River drainage) by the Maumee River (Great Lakes drainage). The lower genetic diversity metrics and effective population size estimates in the Great Lakes population compared to the Ohio River watershed population are indicative of a founder effect and even more evidence of post-Pleistocene colonization of the Great Lakes basin. The lack of clear structure within the Great Lakes and most of the Ohio River watershed indicates that there is, as yet, little evidence of the effects of recent fragmentation on stream systems by dams and habitat degradation.

id #135957

Where do fish get their gut bacteria?

Cedar Mackaness¹, Theodore Kennedy², Claire Couch³, Anna Jolles¹, Isaac Schuman¹, David Lytle¹

1. Oregon State University, Corvallis, OR, United States
2. Southwest Biological Science Center, US Geological Survey, Flagstaff, AZ, USA
3. Columbia River Research Laboratory, USGS Western Fisheries Research Center, Cook, WA, USA

Freshwater fish populations are under threat in a rapidly changing environment. Many of these impacts on individual fish health will be indirect and mediated by the microbial landscape. Upon hatching, fish start off with a naïve gut—a blank slate which quickly becomes colonized by microbes present in the environment. The community of microbes that establish is a result of a mix of deterministic processes such as host genetic and immune factors, diet, and physiology and stochastic processes like priority effects, ecological drift, and dispersal limitations of the microbes. To better understand the connection between the microbial landscape and fish health, we used 16S metabarcoding to describe the gut microbiomes of 3 fish species in the Colorado River (Rainbow Trout, Humpback Chub, and Flannelmouth Sucker). A total of 250 fish fecal samples were collected across 24 sites in the Colorado River, Grand Canyon. After strictly filtering the sequencing data to only analyze sequence variants with a relative abundance greater than 0.05% of a sample, we found an average of 179 sequence variants representing seven major phyla across fish fecal samples. However, the total diversity across samples was 2,229 sequence variants. The discrepancy is driven by high turnover in community composition. Distance based redundancy analysis showed spatial distance constrained a greater proportion of the variance in community composition than host related variables such as fish species, size class, or condition factor, or a suite of environmental variables related to water chemistry. These results indicate that stochastic processes dominate

the community composition of fish gut microbiomes in the Colorado River, and further analyses are needed to link differences in fish gut microbiomes to changing environmental factors.

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id #134422

Assessing Stream Habitat Conditions across Alaska to Inform Future Restoration Efforts

Audrey Huff¹, Erin Larson¹, Paul Gabriel², Amber Crawford²

1. *University of Alaska Anchorage, Anchorage, ALASKA, United States*

2. *Alaska Department of Environmental Conservation, Anchorage, Alaska, USA*

Pacific salmon returns, including Chinook, chum, and coho, are of enormous cultural, economic, and recreational value for Alaskan communities. In recent years, however, the number of returning salmon in the Arctic-Yukon-Kuskokwim (AYK) region has declined dramatically. Warming water temperatures, mining, and other climactic and human impacts have altered freshwater habitats throughout Alaska, with past land uses, including in-stream mining, disproportionately impacting headwater streams - critical habitat for juvenile salmon spawning and rearing. Understanding, conserving, and restoring these and other freshwater habitats will support efforts to increase the number of salmon returning to the region, but comprehensive habitat data throughout the region is currently lacking. This data gap impairs our understanding of current conditions, which limits our ability to identify restoration needs and have informed restoration targets. By expanding the Bureau of Land Management's Assessment, Inventory, and Monitoring (AIM) program, researchers at the Alaska Center for Conservation Science and the Alaska Department of Environmental Conservation are providing quantitative baseline data on stream chemical, biological, and physical conditions across the AYK. This data will be used to inform future restoration efforts, support adaptive management decisions, and improve the resiliency of Pacific salmon populations. Additionally, data from this effort will be publicly available through BLM's public data portal and data analysis tools.

id #135447

Towards transparent, robust use of LLMs in writing peer reviewed publications

James Stegen¹, Yunxiang Chen¹

1. *Pacific Northwest National Laboratory, WA, United States*

The release of ChatGPT as an app marked a turning point in how researchers use AI. Although many scientists are comfortable using AI for technical tasks, there remains unease with AI in scientific writing. That discomfort matters: writing is a primary interface between science and the community that supports it. If AI can help researchers communicate more clearly and efficiently—without compromising rigor—avoiding it may be more harmful than helpful.

Here we describe the use of large language models (LLMs) to accelerate manuscript drafting while preserving human ownership of the science and accountability. Our case study is a multidisciplinary paper currently under review that presents an AI system for automated sediment grain size analysis from streambeds. We did not prompt the LLM to write the paper from scratch. Instead, human authors first completed the research, generated figures, and produced initial text. We then iteratively engaged an LLM to draft topical sentences to convey a compelling story based on scientific outcomes and draft paragraph content informed by initial human text.

A safeguard was human verification and editing as part of iterations. Figures with captions were provided to help the LLM understand the technical science and improve paragraph content; a domain expert reviewed and corrected LLM interpretations and refined prompts, and the LLM then regenerated text. We also provided human identified keywords to constrain the LLM-generated storyline. Paragraphs were drafted one at a time. The LLM was told to not modify figures or tables and to not suggest references to avoid hallucinations. Use of the LLM was discontinued once a full draft was developed. Final revisions, references, and compliance sections were human-completed.

This human-LLM loop produced a high-quality initial draft in roughly eight hours, dramatically faster than our experiences in human-only writing. Critically, the LLM was constrained by human scientific insights and human generated figures/data; the LLM helped generate text constrained by human technical outcomes and expertise. We argue that openly sharing workflows and experiences turns individual LLM-engaged writing efforts into replicates of a community-wide experiment—accelerating convergence on robust norms as LLMs evolve.

id #135961

Hybridization in the Toxic Bloom-Forming Microalga *Prymnesium parvum*

Nathan Watervoort¹

1. *Washington State University, Pullman, WASHINGTON, United States*

Hybridization is a source of genetic diversity whose frequency and ecological importance is increasingly clear. However, the importance of hybridization in microbial eukaryotes is relatively unknown due to issues such as lack of sampling or difficulty in detection. *Prymnesium parvum* (Haptophyta) is a cryptic species complex of toxic microalgae with member species that form ecosystem disruptive blooms around the world. Two closely related clades within the species complex have been reported to form hybrids, but this observation has been disputed. We screened 27 strains of *P. parvum* for signatures of hybridization. A SNP based evolutionary analysis identified 4 clades including a hybrid clade containing 13 strains found in inland Texas and along the eastern coast of the United States. Single cell re-isolates of two strains in this hybrid clade further confirmed that this signature is due to contamination and not contamination. Identified hybrids form five distinct subclades that can be differentiated based on unique patterns of parental SNP retention. Additionally, a new clade of *P. parvum* was discovered in Japanese waters. This supports the existence of a hybrid genotype in North American waters and a sexual stage to the *P. parvum* lifecycle, and expands the cryptic diversity present in the *P. parvum* species complex.

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id #135962

Carbon availability mediates nutrient removal in high-arctic streams in NE Greenland.

Jennifer L. Tank¹, Caroline G.T. Anscombe¹, Ursula H. Mahl¹, Tenna Riis²

1. *University of Notre Dame, Notre Dame, IN, United States*

2. *Department of Biology, Aarhus University, Aarhus, Denmark*

While high-Arctic landscapes are at the frontlines of climate change, there remain knowledge gaps about how climate-induced warming will modify coupled nutrient and carbon (C) dynamics in headwater streams. Thermokarst processes, including retrogressive thaw slumping (RTS) along Arctic stream banks, can mobilize permafrost C, fundamentally altering stream biogeochemistry. To understand the impact of RTS-mobilized C on nitrogen (N) and phosphorus (P) cycling, we conducted short-term solute additions in paired reaches (RTS-impacted reach vs. upstream reference) of two streams in the Zackenberg Basin (74°N) in NE Greenland. Both Kaerelv (Q=11-16 L/s) and Graenselv (Q=19-20 L/s) have been shown previously to be co-limited by N+P, but have recently (<2 yrs) exhibited RTS. We hypothesized that RTS-mobilized organic C could stimulate heterotrophic N and P uptake. We conducted ammonium (NH₄⁺-N), nitrate (NO₃⁻-N), and soluble reactive phosphorus (SRP) releases, with and without acetate, allowing us to quantify nutrient spiraling metrics, before and after the addition of labile C, in reaches with and without RTS. For NH₄⁺-N, uptake velocities (V_f) were consistently higher in the RTS reach (10-11.5 mm/min), compared to upstream (5.8-7.9), but NH₄⁺-N+C consistently decreased NH₄⁺-N V_f (7.4-7.6). Surprisingly, this suggests that C-priming may stimulate mineralization of benthic NH₄⁺-N and decrease overall N demand. For NO₃⁻-N, V_f was slightly higher in RTS reaches (5.9-6.1 mm/min) compared to upstream (4.7-5.1), but co-addition of acetate decreased NO₃⁻-N V_f in all but one reach. In general, NO₃⁻-N demand was lower than NH₄⁺-N. Finally, SRP V_f was higher in the RTS reach for Kaerelv (13.5 mm/min) than upstream (5.6), and the co-addition of acetate decreased SRP V_f in both reaches (4.2-4.7). In both streams, the ambient chemistry in RTS vs. upstream reaches gave no indication of RTS-related differences. In contrast, short-term solute additions suggest that very small increases in availability can stimulate nutrient demand, and that C, possibly supplemented by RTS events, can alter nutrient demand in surprising ways, stimulating mineralization and decreasing nutrient demand from the water column. These results emphasize the need to understand solute-specific patterns of nutrient uptake, especially given the implications for nutrient-limited coastal waters.

id #135707

Biogeochemical drivers of mercury cycling and speciation in temperate ponds

Samantha Davies¹, Kaci Zarek¹, Meredith Holgerson¹, Jacqueline Gerson¹

1. *Cornell University, Ithaca, NY, United States*

Methylmercury (MeHg) is a bioavailable neurotoxin known to biomagnify in the aquatic food web. MeHg is produced by sulfate reducing bacteria, iron reducing bacteria, and methanogenic archaea in the anoxic water column and sediments. Anoxic freshwater environments are thus sources of MeHg production, though this process is predominantly studied in wetlands and lakes. MeHg cycling has rarely been studied in ponds, yet ponds are globally abundant, biodiverse, and support ideal conditions for MeHg production. This research aims to unveil the drivers of mercury cycling in 16 temperate ponds (ranging from 2-6 m deep) across two ecoregions of upstate New York. We sampled each pond twice during summer 2025 (June and August). We took vertical temperature and oxygen profiles to quantify stratification strength at each sampling event and used temperature loggers to record thermal stratification throughout July. At each sampling event, we collected sediment from the pond's deepest point and collected surface and bottom water. Water was analyzed for total mercury, MeHg, major cations and anions, dissolved organic carbon (DOC), SUVA₂₅₄, trace elements, and iron speciation. Sediment was analyzed for total mercury, MeHg, and loss on ignition (a proxy for organic matter content). All ponds were summer stratified, with most (12/16) already exhibiting bottom water anoxia in June and all exhibiting bottom water anoxia in August. In June, total mercury concentrations did not differ significantly between surface and bottom waters. By contrast, August bottom waters exhibited significantly higher total mercury concentrations than surface waters, indicating that length of stratification influences mercury cycling in ponds. Across sampling events, aqueous total mercury concentrations were positively correlated with DOC, suggesting allochthonous loading of carbon-bound mercury. Our next steps include analyzing sediment and water for MeHg. We expect to see high variability in bottom water

and sediment %MeHg between ponds, and higher %MeHg in August than in June due to increased stratification strength. Our results will provide new insight into mercury cycling in ponds, allowing for a greater understanding of the threat MeHg poses to organisms in these ecosystems.

id #134428

Disentangling seasonal and interannual drivers of macroinvertebrate community responses to flow and specific conductance in Central Appalachian streams.

Amaryllis Adey¹, Daniel McLaughlin², Erin Hotchkiss³, Ashleigh Kirker², Caleigh Meehan³, Lisa Tabor¹, Megan Underwood¹, Sally Entekin¹

1. Department of Entomology, Virginia Tech, Blacksburg, Virginia, United States

2. Department of Forest Resources and Environmental Conservation, Virginia Tech, Blacksburg, Virginia, United States

3. Department of Biological Sciences, Virginia Tech, Blacksburg, Virginia, United States

Macroinvertebrate Indices of Biotic Integrity (IBIs) in Central Appalachian streams often exhibit substantial variation between fall and spring sampling periods, yet mechanisms driving these seasonal differences remain poorly understood. In streams where mountaintop-mining has altered land-water linkages, these seasonal differences likely reflect both direct flow disturbances and variation in the strength of terrestrial-aquatic connections, as flow regimes regulate the magnitude and timing of material fluxes, thus altering the specific conductance (SC) signals of salt transported from mined landscapes. Previous studies suggest that SC affects pollution-sensitive taxa, while sample timing influences IBI assessment outcomes. However, the relative importance of flow disturbance versus elevated SC, and whether this relationship is consistent across space and time, has not yet been evaluated. We used a 12-year dataset (2011-2023) from 23 streams in the coalfields of southern West Virginia and southwest Virginia (including nine streams with continuous water level and three streams with estimated flow) to test whether seasonal differences in flow regimes and SC explain overall patterns in macroinvertebrate communities. We hypothesized that: 1. flow disturbances differentially affect communities between seasons due to variation in flow magnitude and community composition, 2. SC impacts at low flows vary seasonally as flow-SC relationships change, and 3. flow and SC effects are consistent both temporally (across years within each season) and spatially (across streams within each year). Using Bayesian hierarchical models with random effects for stream and year, and random slopes to assess spatial and temporal consistency, we evaluated IBI metrics as functions of season, flow (or water level), SC, and their interactions. Preliminary analyses across 12 years of paired macroinvertebrate and SC data reveal distinct seasonal patterns in IBI scores. The full model results will quantify roles of flow, SC, and their interactions to improve our understanding of how mining-altered terrestrial-aquatic connections influence biological impairment through seasonal variation in contaminant transport pathways. These findings will inform appropriate timing for bioassessments in mining-impacted watersheds and clarify whether management should prioritize flow stabilization, SC reduction, or other seasonal considerations to restore the strength and resilience of aquatic-terrestrial linkages and mitigate biological impairment from mining activities.

id #135196

An eDNA Approach to Mussel Diet Analysis: A preference for protists

Bruce Cahoon¹, Tim W Lane², Isabella Maggard³, Raegan Cox¹

1. The University of Virginia's College at Wise, Wise, Virginia, United States

2. Cumberland River Aquatic Center, Tennessee Wildlife Resource Agency, Nashville, Tennessee, USA

3. University of Tennessee, Knoxville, TN, USA

Freshwater mussels (Mollusca: Unionidae) play a crucial role in freshwater river environments where they live in multi-species aggregations and often serve as long-lived benthic ecosystem engineers. Many of these species are imperiled and it is imperative that we understand their basic needs to aid in the reestablishment and maintenance of mussel beds in rivers. To expand our knowledge of the diet of these organisms, ten species of mussel were introduced into enclosed systems in three experiments. In the first, five species of relatively common mussels were incubated in water from the Clinch River (Virginia, USA) and in the second, water from a manmade pond at the Commonwealth of Virginia's Aquatic Wildlife Conservation Center in Marion, VA. In the third experiment, five endangered species were incubated in water from the AWCC pond. Quantitative PCR and eDNA metabarcoding were used to determine which planktonic microbes were present before and after the introduction of mussels into each experimental system. It was found that all ten species preferentially consumed microeukaryotes over bacteria. Most microeukaryotic taxa, including Stramenopiles and Alveolates were quickly consumed by all ten mussel species. We also found that they consumed fungi but not as quickly as the microalgae, and that only two species of mussels, *Ortmanniana pectorosa* and *Medionidus conradicus*, may have consumed bacteria. Our results provide evidence that siphon feeding Unionid mussels can select preferred microbes from mixed plankton, and mussel species may exhibit dietary niche differentiation.

id #135708

Temporal dynamics of invasive dreissenid mussels in nearshore Lake Michigan

Rachel Smith¹, Harvey Bootsma¹, Brenda M Lafrancois², Ben Turschak³

1. University of Wisconsin-Milwaukee, Milwaukee, WI, United States

2. National Park Service, Ashland, WI

3. Michigan Department of Natural Resources, Charlevoix, MI

Invasive dreissenid mussels, introduced to the Great Lakes in the 1980s, dominate the benthic community of Lake Michigan. Quagga mussels (*dreissena rostriformis bugensis*) are present in the lake with populations as dense as ~10,000 individuals/m². These mussels have dramatically altered nutrient dynamics and food web structure within the Great Lakes. Dreissenids are sessile filter feeders that serve as integrators of local water quality conditions. Therefore, they can be useful as indicators of long-term change. Using a 14-year dataset from a 10-meter site near Sleeping Bear Dunes National Lakeshore in northeastern Lake Michigan, we examined trends in nearshore dreissenid density and condition. Although mussel density has remained relatively stable over time, we observed a significant decline in the mussel weight:length ratio, suggesting a decrease in mussel condition. The decline is particularly pronounced in small to midsize mussels. A potential driver of this trend is a change in the nutritional quality of phytoplankton in the study region, as revealed in measurements of seston nutrient stoichiometry. This may in turn have consequences for much of the Lake Michigan food web. We will discuss the implications for nearshore dreissenid populations and nutrient cycling in Lake Michigan.

id #135964

Vertebrate Biodiversity via eDNA at the air-water interface

Aden Ip¹, Pedro FP Brandão-dias¹, Gledis Guri¹, Elizabeth A Allan¹, Ryan P Kelly¹, Pedro Brandao Dias F Pinto¹

1. University of Washington, Seattle, WA, United States

Although aquatic, aerial, and terrestrial habitats are often treated as separate ecological systems, these environments exist along a continuum of connectivity: flows of biomass and energy routinely create linkages across water and air, blurring traditional boundaries. Here, we use eDNA sequencing to illustrate and quantify the movement of trace genetic information between water and air. We collected 27 paired air-and-water samples from two urban-wildland interface sites using passive air sampling and active water filtering. Metabarcoding with the MiFish-U 12S marker recovered 35 vertebrate taxa, 40% of which were detected in both water and air, ranging from the strictly aquatic salmon to wholly terrestrial cottontail rabbit. This reciprocal relationship suggests that eDNA pools form within each partition (water or air), with the probability of transfer governed by DNA concentration. In this view, detecting aquatic eDNA in the air is not contamination or stochastic noise but an expected, repeatable phenomenon. Logistic models suggest that higher abundance in one medium predicts spillover into the other. For instance, peaks in Coho and Chinook salmon eDNA in water and air align within 24 hours, demonstrating that passive air sampling reflects the temporal abundance trends of the most common aquatic species to some degree. In contrast, low-read-abundance taxa appear only sporadically, reflecting their expected stochastic behavior at low abundance. Routine detection of such rare species would require intensified sampling in their primary habitat. Together, our findings bridge the conventional separation of aquatic and airborne eDNA and establish a unified, non-invasive framework for holistic vertebrate monitoring at the land-water interface. This approach offers transformative potential for conservation, invasive-species early warning, and One Health surveillance.

id #135965

Sediment conditions may support benthic cyanobacteria bloom formation in an oligotrophic river reach

John Buster¹, Sarah Roley¹

1. Washington State University, WA, United States

Benthic cyanobacterial blooms (cyanoHABs) have become more common in oligotrophic systems, but their cause is unclear. Higher temperatures and higher pH levels may increase benthic cyanoHAB formation in systems lacking sufficient surface water phosphorus (P) because high temperatures decrease gas solubility and increase microbial respiration. Resultant anoxic conditions can also stimulate P release by reducing iron and manganese, which then releases sorbed P. Benthic photosynthesis can increase pH in pore water and stimulate the release of aluminum-bound P, particularly along shallow river margins with abundant solar radiation.

To test the hypothesis that sediment P release drives benthic cyanoHABs in oligotrophic rivers, we measured sediment P on the Hanford Reach of the Columbia River, where recent benthic cyanoHABs killed dogs and sickened humans. We collected pore water and surface water from several sites along the river reach. We also collected littoral sediment cores and performed sequential P extractions to determine the conditions that could lead to P release. These P fractions included loosely bound, redox-sensitive, pH-sensitive, organically bound, and calcium bound P. Our aim was to identify dominant pools of sediment P to determine if higher temperatures and pH might cause elevated P flux to pore water, potentially providing a nutrient source for benthic cyanobacteria.

Surface water P concentrations were generally low (mean = 0.018 mgP/L, range = 0.002 - 0.125 mgP/L), whereas pore water P concentrations were much higher (mean = 0.484 mgP/L, range = 0.012 - 4.943 mgP/L), often exceeding EPA safe water guidelines (0.100 mgP/L). Releasable P fractions (by percentage of totals) in the Hanford Reach varied by site, but in general, pH-sensitive and organically bound P dominated (redox-sensitive range = 8% - 28%, pH-sensitive range = 31% - 65%, organically bound range = 18% - 61%). Our fractionation data suggests that a high proportion of sediment P in the Hanford Reach is releasable under increased temperature and pH, potentially causing elevated pore water P concentrations. Our results also suggest that low-P surface water can overlie high-P pore water, with pore water P potentially fueling benthic cyanobacteria blooms in oligotrophic surface waters.

id #135198

Evaluating the contribution of wetland ecosystems to carbon storage in a dryland landscape using a data fusion model

Erik Nati-Johnson¹, Jodi Brandt¹, Trevor Caughlin¹

1. Boise State University, Boise, ID, United States

Interior wetlands supported by snowmelt, such as those in the arid and semi-arid Intermountain American West, serve critical functions for the wild and human communities of dryland regions, and sequester carbon in quantities that could meaningfully contribute to climate mitigation. Evaluations of individual reaches have shown higher carbon stocks in floodplains than neighboring uplands, but there are currently no maps that estimate the overall carbon storage benefits of mesic ecosystems in the American West, and mapping them is challenging due to their small size and dispersion across the landscape. While existing models of biomass and soil organic carbon (SOC) at regional-and-greater extents are well suited to represent larger, more homogenous upland areas, the spatial resolution of these models are too coarse to capture many small interior wetlands. To fill this gap, I have developed a mapped product that highlights the carbon stock contributions of mesic ecosystems in the American West. Underlying this product is a LiDAR-multispectral data fusion model constrained to wetland vegetation extent that predicts canopy height at a 10-meter spatial resolution with lower-than-typical error metrics for models of this kind. By leveraging these height predictions and spectral differences in phenology, as well as publicly available data on height/biomass relationships, I can estimate carbon at every pixel according to vegetation type and size. The final product is a landscape carbon map that overlays our high-resolution wetland carbon map onto coarser existing products and highlights the contribution of mesic areas relative to their extent, allowing for broader understanding of the spatial distribution of natural carbon stores in dryland regions. Preliminary results indicate a highly disproportionate amount of carbon stock found in mesic ecosystems, but substantial uncertainties remain, particularly with SOC estimates, and efforts that improve our ability to predict floodplain SOC heterogeneity will be key to future improvements in locally-specific estimates.

id #135454

Modeling Nitrate Attenuation in Tidal Riparian Zones Using a 1D Boussinesq-Based Monte Carlo Framework

Chuanhui Gu¹

1. Duke Kunshan University, Suzhou Shi, JIANGSU SHENG, China

Tidal riparian zones are increasingly recognized as biogeochemical filters that mitigate nutrient loading to estuaries, yet the mechanisms controlling nitrate removal under oscillatory flow remain poorly quantified. Here we integrate one-dimensional Boussinesq-based flow modeling, reactive transport simulations, and Monte Carlo experiments to evaluate nitrate attenuation in tidal riverbanks. Results show that water table oscillations drive short but frequent infiltration–exfiltration events that create temporally discrete windows of reactivity. The near-bank margin (0–8 m) functions as a hotspot of nitrate transformation, with removal strongly coupled to tidal exchange. Sensitivity and Sobol analyses identify first-order degradation rate (λ) as the dominant control, with hydraulic conductivity and tidal amplitude modulating its effectiveness. A Damköhler framework reveals threshold behavior distinguishing transport-limited from reaction-limited regimes, while Buckingham Pi–based regression achieves high predictive skill ($R^2 > 0.94$) across diverse conditions. Monte Carlo ensembles highlight strong variability in removal potential, underscoring the need for probabilistic assessment. Collectively, this work demonstrates that tidal riparian buffers can substantially attenuate nitrate when reactivity aligns with hydrologic exchange, and it provides transferable, dimensionless tools to predict performance across coastal systems. These insights inform management strategies aimed at enhancing hydrologic connectivity and sustaining ecosystem services under changing climate and tidal regimes.

id #134943

Wetland reconnection enhances food resource availability in the Kootenai River

Brooke Long-Fox^{1,2}, Shelby Therrian³, Kurt M Chowanski^{1,2}, Brandon Diller³, Lisa A Kunza^{1,2}

1. Center for Sustainable Solutions, South Dakota School of Mines and Technology, Rapid City, SD, United States

2. Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines and Technology, Rapid City, SD, United States

3. Fish & Wildlife Department, Kootenai Tribe of Idaho, Bonners Ferry, ID, United States

Floodplain wetlands are among the most productive components of large river ecosystems, yet regulation, leveeing, and nutrient retention in reservoirs have left many rivers nutrient-poor and hydrologically disconnected from their floodplains. In the lower Kootenai River (Idaho, USA), an oligotrophic, regulated system managed in part through nutrient supplementation, reestablishing seasonal connectivity to off-channel habitats is a key restoration strategy for providing food resources to higher trophic levels. We evaluated how reconnection projects influence water quality and lower-trophic level standing stock by comparing river (mainstem), connector-channels, and interior wetland/pond habitats across seasons and hydrologic conditions from 2018–2024. During pre-connection periods and during hydrologic connection windows, we quantified water chemistry (N, P, C), chlorophyll-a, phytoplankton density, and zooplankton density. The wetland/pond had elevated nutrient concentrations, chlorophyll-a, and plankton densities. During connection windows, connector channels exhibited intermediate to enriched conditions relative to the mainstem river, indicating the mobilization and export of wetland-derived nutrients and plankton to the mainstem during reconnection. These patterns support a process-based mechanism in which reconnected floodplain wetlands accumulate biomass and nutrients under baseflow and export high-quality subsidies to connector channels and the mainstem river during inundation and drawdown. Our results demonstrate that targeted floodplain reconnection can restore productive wetland habitats that act both as biodiversity and productivity hotspots and as dynamic sources of nutrient and plankton subsidies to nutrient-limited mainstem rivers. Wetland reconnection can complement nutrient-addition programs to enhance food resource availability and resilience in large, oligotrophic rivers. These productive wetlands may also provide high-quality food resources for higher consumers, including waterfowl, linking river restoration to broader floodplain and wildlife benefits.

id #136223

Two Decades of Nutrient Sampling in the Klamath River, CA, Reveal Impacts of Dams and Agricultural Pollution across Time and Space

John R. Oberholzer Dent¹

1. University of North Dakota, Happy Camp, CA, United States

Nutrients regulate algal growth, drive microbial processes, and determine food web structure. Despite major alterations to nutrient dynamics, temporal and spatial patterns of nutrient retention are poorly understood in large rivers in part due to limited sample records. On the dammed, nutrient-polluted Klamath River (CA), the Karuk and Yurok Tribes have collected over two decades of biweekly nutrient (N and P) samples, allowing us to model daily nutrient concentration and loads from 2005-2023 and then calculate monthly, reach-scale (25-65 km) nutrient retention and release. P was high compared to N in all reaches of the Klamath River with N:P < 16:1, and nutrient concentrations generally decreased while loads increased in a downriver direction due to tributary accretions. Seasonal variation was high, spanning two orders of magnitude, suggesting that baseflow metrics will severely underestimate annual retention. Nutrients also varied longitudinally, with dams and major tributary confluences shifting the relationship between discharge and nutrient retention. In stable upriver reaches, higher discharge suppressed retention, while in larger downriver reaches, it enhanced retention via delivery of the limiting nutrient N. Other differences aligned with diverse hydrology, geology, and autotrophic assemblages. Nutrient dynamics in the Klamath River help us understand more broadly how dams and nutrient pollution lead to nuances in eutrophication patterns across time and space, and how large-scale dam removal may impact ecosystems from the bottom up. This thorough investigation into nutrient dynamics in a large, 7th order river was only possible due to long-term, place-based Tribal datasets, which offer insight into how nutrient pollution and dams impact the aquatic ecology of large rivers more broadly.

id #134176

Modeling the spread of invasive species and emerging infectious diseases in heterogeneous environments

John M Drake¹

1. University of Georgia, Athens, GA, United States

The spread of invasive organisms and emerging pathogens often unfolds across spatially heterogeneous landscapes where environmental variation, connectivity, and ecological context shape transmission or establishment dynamics. A longstanding approach to modeling this process has relied on "gravity-type" formulations, in which the attractiveness of locations and the flux among them are expressed parametrically, often in proportion to site size or distance. While widely used, gravity models lack an explicit mechanistic basis. In particular, it is unclear whether standard parameterizations capture the true ecological drivers of spread or whether covariates added to improve fit adequately reflect causal pathways. As a result, predictions may be difficult to generalize beyond the specific system or data set to which the model was calibrated.

In this talk, I propose an alternative framework for modeling biological invasions and disease spread that explicitly decomposes the invasion process into three distinct sub-mechanisms: (i) production of propagules at occupied sites, (ii) entrainment of those propagules into physical, ecological, or anthropogenic redistribution pathways, and (iii) successful establishment at uninvaded locations. The key advance is that each component process can be parameterized independently, thereby enabling a fully mechanistic or partially mechanistic formulation. This modular structure naturally accommodates environmental covariates—for example, hydrological connectivity influencing transport, local habitat suitability affecting establishment, or economic activity modulating anthropogenic movement—without ad hoc modifications to the model's core structure.

I further show that, under reasonable assumptions, the resulting process-based formulation yields a likelihood function for observed invasion times, enabling maximum-likelihood estimation of parameters governing propagule production, transport, and establishment. This allows formal statistical inference, hypothesis testing, and model comparison.

To illustrate the framework, I present a reanalysis of classical data on the spread of zebra mussels (*Dreissena polymorpha*) through North America. By fitting the model to historical records of first detection, we infer system-specific dispersal pressures, quantify how environmental heterogeneity mediates spread, and demonstrate improved interpretability relative to traditional gravity models. More broadly, this approach provides a principled foundation for forecasting the spread of invasive species and infectious diseases, designing surveillance systems, and evaluating intervention strategies in complex, spatially structured environments.

id #135712

Seasonal hydrology effects on CO₂ and CH₄ emissions from a geographically isolated cypress swamp in the Southeastern United States

Chloe Hall¹, Jamie Rogers², Jewell Johnson², Dustin Benton², Amanda DelVecchia¹, Nicholas Marzolf²

1. University of North Carolina at Chapel Hill, Chapel Hill, NC, United States

2. The Jones Center at Ichauway, Newton, GA, United States

The variable hydrology of freshwater wetlands fuels a dynamic biogeochemical regime, in which the degree of inundation influences oxygen availability within the water column and sediments. These hydrologically mediated oxygen conditions govern aerobic and anaerobic microbial pathways that directly influence the production and emission of carbon dioxide (CO₂) and methane (CH₄). Although hydrology, vegetation, and organic matter availability are well-established drivers of wetland greenhouse gas emissions, the seasonal variability of these controls and their effects on CO₂ and CH₄ fluxes remain understudied. We hypothesized that wetlands with longer periods of inundation are often dominated by anoxia which can cause increases in methane (CH₄) production, while drier wetlands are often characterized by increased carbon dioxide (CO₂) emissions due to increased aerobic respiration from higher oxygen availability. This study quantified CO₂ and CH₄ emissions from a geographically isolated wetland during peak inundation through complete drying (March–June) as water levels transitioned from inundated to exposed sediments. Our study wetland was an 18-

acre cypress swamp located within a managed longleaf pine forest at The Jones Center at Ichauway in southwest Georgia. We collected in-situ emission data from water and sediment surfaces using floating and soil chambers placed throughout our study wetland along transects from the dry wetland perimeter to the deepest point of inundation in the wetland. Total CO₂ and CH₄ emission rates were calculated based on chamber gas accumulation rates and averaged for each site along the transect. Results showed higher mean CH₄ emissions at the deepest water levels consistently throughout the three months of this study. CO₂ emissions were highest at our perimeter sites where we measured soil gas accumulation. Our results highlight the seasonal variability of GHG emissions in southeastern wetlands and demonstrate the influence of hydrologic variation on these emissions.

id #135968

CONDITION OF STREAMS ACROSS BUREAU OF LAND MANAGEMENT LANDS: ASSOCIATED STRESSORS AND TRENDS

Jennifer L Courtwright¹, Charles P Hawkins¹

1. Department of Watershed Science, Utah State University, National Aquatic Monitoring Center, Logan, UT, United States

The lack of standardized data collected has limited our ability to quantify aquatic resource conditions and determine stressors of degraded conditions across large scales. Additional challenges, particularly in dryland regions, have included high natural spatial and temporal variability of condition metrics and dispersed land uses that are poorly represented in national geospatial datasets. We conducted two large-scale probabilistic surveys of perennial streams on Bureau of Land Management (BLM) lands in the continental western U.S. from 2013-2015 and 2019-2022 to determine aquatic resource conditions, trends, and stressors. We assessed the condition of 13 biological, chemical, and physical habitat metrics. To account for natural spatial and temporal variability when determining conditions, we used recent models that use field and remote sensing datasets to predict site-specific reference conditions. We used relative risk and random forest modeling to assess stressor-response relationships and sources of stressors. Stressors and their associated sources were assessed both with field data and national and agency specific geospatial datasets. We were able to infer the conditions of a total of 18,423 and 17,324 perennial stream km on BLM from 2013-2015 and 2019-2022, respectively. Chemical (total nitrogen, total phosphorous, and salinity) and biological (macroinvertebrate assemblage composition) metrics had the greatest extent of poor conditions (26-37% of stream km). Reaches with excessive total nitrogen were twice as likely to have poor macroinvertebrate assemblage conditions from 2013-2015. For both total nitrogen and macroinvertebrate assemblages, the extent of stream km in poor condition decreased by 10% between survey one and two. However, salinity, percent fine sediment, percent shade, and temperature conditions posed high risks to macroinvertebrates in 2019-2022. Sources of excessive nutrients were primarily agriculture and livestock or wild horse and burro grazing. Sources of excessive salinity were primarily oil and gas wells and agriculture. Additionally, excess nutrient concentrations and salinity and poor macroinvertebrate conditions were associated with streams with a high likelihood of hydrologic alteration. These results will assist the BLM with prioritizing management actions and achieving their goal of sustaining the health, diversity, and productivity of public lands for the use and enjoyment of present and future generations

id #135713

A virtuous cycle of innovations in science, policy, and markets to manage invasions

David M Lodge¹

1. Cornell University, NY, -

Successful management of biological invasions requires the virtuous cycle of research innovation, policy, and commercialization to spin much faster. During the 42 years of my research career on invasive species, multiple research innovations from many researchers have produced a sophisticated policy-relevant understanding of all stages of biological invasions, and many technological innovations relevant to species management. However, policy innovation to move research innovations along the technology readiness scale into the marketplace has often stalled. Investments to develop practical tools to identify, interdict, eradicate, or slow-the-spread of species have lagged well behind research innovation, and remain grossly insufficient to support species management at the frequency and geographic scales required. Thus the toolbox for managing biological invasions remains poorly stocked. This is especially true for freshwater and marine ecosystems. Using environmental DNA (eDNA) and other recent technological advances, I will describe policy changes that could quickly produce great societal benefit via effective and efficient biosecurity measures, including early detection and rapid responses.

id #135969

Headwaters are fish habitat: Fish diversity and protection status of headwater streams in the United States

Jared A Ross¹, Dana Infante¹, Susan Colvin², Keith Gido³, Patrick Shirey⁴, Jill Silver⁵, Robert Vadas⁶, Megan Malish¹, Arthur Cooper¹, Robert Hughes⁷

1. Michigan State University, East Lansing, MICHIGAN, United States

2. Minnesota State University, Mankato

3. Kansas State University, Manhattan

4. University of Pittsburgh, Pittsburgh

5. 10,000 Years Institute, Forks

6. Washington Department of Fish and Wildlife, Olympia

7. Amnis Opes Institute, Corvallis

Headwater streams are the smallest, yet most abundant reaches in any river network and comprise most of the stream length; as a result, they drain the vast majority of terrestrial landscapes across the globe. These small but numerous ecosystems are essential components in the structure and function of larger streams and rivers, wetlands, and receiving lentic water bodies including lakes as well as marine systems. Despite the prominence of headwaters, their role as permanent or temporary fish habitat remains understudied, including the degree to which they're used by freshwater fishes, one of the most threatened taxa groups globally. While many fishes are described in the literature as accessing headwaters for portions of their life cycles, a large-scale assessment documenting species presences in headwaters has not occurred. To meet this need, we characterize the extent to which freshwater fishes occur in headwater streams across the United States. First, using an existing database characterizing presence of species from nearly 60,000 headwater stream reaches, we document the frequency with which species occur in these habitats, and we include results summarized for endemic, rare, and threatened species. Additionally, we summarize and map the degree to which headwater habitats supporting these species are currently protected based on the Protected Areas Database of the United States, thus highlighting the vast number of species relying on headwater habitats and the mismatch in the protection of these catchments. Finally, we summarize literature describing assumed headwater dependency of hundreds of fishes identified from our analyses. These results help to amplify awareness of the elevated vulnerability of many freshwater fishes stemming from recent legislative changes in the Clean Water Act, which removes many headwaters from federal protection and from changes in how the Endangered Species Act treats protection of habitat supporting imperiled species. Collectively, our results show the overall importance of headwater habitats to freshwater fishes and underscores the need to conserve these critical habitats for our most vulnerable species.

id #134690

Direct emissions and bioturbation from the invasive Asian clams (*Corbicula* spp.) drive CO₂, CH₄, and N₂O cycling in streams

Nguyen Tien Anh Quach¹, Yuseung Shin¹, Nicholas S Marzolf², Amanda G DelVecchia¹

1. University of North Carolina at Chapel Hill, NC, United States

2. The Jones Center at Ichauway, Newton, GA, United States

Greenhouse gas (GHG; CO₂, CH₄, N₂O) emissions from running waters are supplied from instream processes (e.g., metabolism, weathering) and external sources (e.g., soils, groundwater). However, little information is known about the contribution of macroinvertebrates to lotic GHG emissions. To address this knowledge gap, we asked how the invasive filter-feeding and burrowing Asian clams (*Corbicula* spp.), whose population has exploded in North American streams, affect stream GHG cycling. We predicted that (1) clams directly emit CO₂, CH₄, and N₂O, (2) direct CH₄ and N₂O emissions from clams are higher in agricultural streams, and (3) bioturbation from clams leads to higher total sediment to water gas fluxes. In urban streams in North Carolina (n=5) and agricultural streams in Georgia (n=4), we conducted field microcosm incubations using three treatments: clam vs. no clam, natural vs. sterile sediment, and light vs. dark treatments. For each treatment, net GHG production was calculated as the difference in GHG concentrations between the start and end of the incubations. We found that, regardless of light and sediment treatment, CO₂ concentrations in clam treatments doubled, while stabilizing and even decreasing in control treatments. The concentrations of N₂O generally increased in clam treatments from two to four times, except for two study sites where the concentrations decreased up to three times. Notably, the N₂O concentration in clam treatment was higher in natural sediment than in sterilized sediment, implying an interactive effect between the Asian clams and the sediment microbial communities. Contrary to our initial prediction, CH₄ concentration across all treatments declined, which could be due to clam bioturbation causing sediment oxygenation and CH₄ oxidation. These results suggest that at high densities, the Asian clams can play a substantial role in affecting reach scale GHG flux. Our findings highlight the need for future studies and ultimately, for management of U.S. streams, to explicitly incorporate macroinvertebrates into models of lotic carbon cycling.

id #135970

Enhanced greenhouse gas emissions with floating and emergent plants in wetlands

Nathan O Earl¹, McKenzie E Goodwyn¹, Andrew S Mehring¹

1. University of Louisville, Louisville, KY, United States

Wetlands are often important carbon sinks, but their carbon storage efficacy may be offset by greenhouse gas (GHG) emissions, especially of methane (CH₄), which may be altered by plant community structure. Floating plants such as duckweed (*Lemna*, *Wolffia* spp.) augment methanogenesis by inducing anoxia through rapid turnover of biomass and the creation of physical barriers to oxygen diffusion. Much of the CH₄ in ponds can be retained by sediments and oxidized via methanotrophy. However, cattails (genus *Typha*), despite being effective at sequestering carbon dioxide (CO₂), provide a conduit through their aerenchyma for CH₄ to bypass methanotrophic bacteria in sediments and water. To determine whether the copresence of these two plants synergistically enhances CH₄ emissions in small wetlands, we sampled CH₄ and CO₂ fluxes from ponds characterized by the presence or absence of duckweed or cattails in a 2 x 2 full factorial design. Fluxes of GHGs were analyzed with linear mixed effects models as a function of vegetation type and other environmental parameters, and model selection was performed with Akaike's Information Criterion. The copresence of these two plants synergistically enhanced CH₄ emissions, and cattail-dominated systems featured the greatest rates of carbon uptake. These results suggest that plant community assemblages are important in determining a wetland's carbon source-sink status, that wetlands with emergent vegetation may be most effective for carbon capture, while also generating greater tradeoffs in CH₄ emissions. These results provide context for considering plant community structure while managing wetlands as carbon sinks and nature-based solutions to climate change.

id #135459

Assessing the Impact of Anthropogenically Modified Land Uses on Wetland Health

Stanley M.S Liphadzi¹, Sylvester N.S Mpandeli¹, Luxon L Nhamo¹

1. Water Research Commission, South Africa, Pretoria, GAUTENG, South Africa

Wetlands are critical ecological infrastructures that improve water quality, serve as habitat for fish and other aquatic life, accumulate floodwaters, and maintain surface water flow during dry periods. However, the health of wetlands has been compromised by anthropogenic activities that affect the constant supply of ecosystem services. This study assessed the impact of anthropogenically modified land use on wetland health in the Witbank Dam Catchment in South Africa, whose land use has been severely modified for agriculture and mining purposes. The study developed a model linking surface runoff generated in the catchment with land use and wetland typology to comprehend diffuse pollution from pollution-source land uses. Runoff data and related wetland spatial information were processed and analysed in a Geographic Information System (GIS) to estimate pollutants (agricultural nutrients and acid mine drainage) from runoff detained and released by wetlands. The analysis facilitated the assessment of the value of wetlands in enhancing water quality, as well as human and environmental health. The runoff volume from pollution-source land uses (urban areas, farmlands, and mining) was used to evaluate annual pollution levels. Wetland types are ranked according to their efficiency levels to filter pollutants. The assumption is that the difference between filtered and unfiltered runoff is the quantity of polluted runoff water discharged into the river system. The analysis has shown that 85% of polluted runoff generated in the catchment ends up in the river system. An important observation is that although wetlands have a substantial ability to absorb excess pollutants, they have finite boundaries. Once they reach their full holding capacity, they can no longer absorb any further pollutants. The excess is discharged into the river system, risking human and environmental health. This explains why the Limpopo River is heavily polluted resulting in the death of fish, crocodiles and other aquatic life.

id #135971

Estimating Organism Abundance Using Within-Sample Haplotype Frequencies of eDNA Data

Pedro FP Brandão-Dias¹, Gledis Guri¹, Megan R Shaffer¹, Elizabeth A Allan¹, Ryan P Kelly¹, Pedro Brandao Dias F Pinto¹

1. University of Washington, Seattle, WA, United States

Environmental DNA (eDNA) provides powerful insights into species presence and community composition but remains limited in its capacity to infer species abundance or population structure. Here, we show that the deviation between within-sample haplotype frequencies and the overall population-level haplotype frequencies can be used to estimate the number of individual contributors to a given sample. We first establish the theoretical framework for approximating population haplotype frequencies directly from eDNA data, enabling application even in the absence of tissue-derived references. Building on this foundation, we introduce a maximum likelihood estimator to infer the number of contributors and assess its performance through simulations spanning a range of haplotype frequency distributions and noise scenarios. These approaches assume that all samples are drawn from a single, panmictic population. We find that accurate estimates are attainable when haplotypes are sufficiently variable, population frequencies are well-characterized, and samples are large enough to capture frequency deviations. By bridging population genetic theory and eDNA, our method complements existing molecular approaches and offers a novel path toward quantifying abundance from eDNA metabarcoding data.

id #136227

‘River, keep on rolling. Carry my soul to the sea.’

David Merritt¹

1. Functional River Ecology, Fort Collins, CO, Australia

For centuries, composers, artists and poets have been inspired by rivers; rivers of ever-shifting flow and form. The changing tones, percussion, legato of cobble bed movement as river stage declines. The cacophony and staccato of turbulence and standing waves during a flood. The surging swell of shedding vortices-flow separation. All rhythms and melodies that lull us to sleep and inspire conductors to bring symphony orchestras to crescendo. The figurative music of rivers is understood by those who know them, the literal music of rivers can only be expressed second hand. What if the flow regime of rivers were the pen-to-paper on a musical score. Sonification of data is an age-old endeavor. Direct translation of measured river discharge into musical notes, harmonies, and articulations holds expression unique to each. The gap between wavelet analysis, the Fast Fourier transformation, the position in multidimensional ordination space and resonance with the public on the beauty, importance, evolutionary role, and imminence of conservation action for river protection lies in translation. In this work, the rivers play the band and their expressions allow us to aesthetically distinguish between natural flow regimes and how their expressed music varies by region and is altered in response to climate change and flow alteration. This talk attempts to bridge science with art to compel us to maintain, enhance and restore natural flow regimes in rivers.

id #135460

Wastewater Regulatory Challenges and Strategies in Austin, Texas

Kaela Champlin¹

1. City of Austin Watershed Protection Department, Austin, TEXAS, United States

With continued population growth and development in Central Texas, the number of applications for municipal domestic wastewater permits regulated by the Texas Commission on Environmental Quality (TCEQ) has increased. TCEQ has no nutrient criteria for streams, and applicants are required to provide limited modeling of nutrient impacts. Wastewater permits can affect water quality, threatening the health of receiving waters, and treatment levels are often not set to prevent nutrient concentrations that cause undesirable effects on aquatic life and algae blooms. Discharging effluent may also degrade nearby springs that recharge aquifers and provide habitat for endangered species, such as the Barton Springs and Austin Blind salamanders. While the City of Austin Watershed Protection Department has no regulatory authority over these permits, it has

worked to develop a policy for reviewing them, sharing the results with regional partners, and prioritizing wastewater permits of concern, with the goal of nudging applicants and state regulators toward more protective permits. By developing specific review criteria, our department has streamlined its internal review process for permits of concern. And by collaborating with regional stakeholders, we have augmented our resources and inputs into the decision-making process. The presentation will summarize the strategies the Austin Watershed Protection Department has used to address permits of concern, along with the outcomes, ongoing challenges, and future strategies being explored to address wastewater regulatory challenges in the Austin area.

id #135972

Ecosystem metabolism across a large oligotrophic river network

Kurt Chowanski^{1,2}, Brandon Diller³, Genny Hoyle⁴, Mark Elliston³, Greg Hoffman⁵, Lisa Kunza^{1,2}

1. Center for Sustainable Solutions, South Dakota School of Mines and Technology, Rapid City, SD, USA

2. Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines and Technology, Rapid City, SD, USA

3. Fish and Wildlife Department, Kootenai Tribe of Idaho, Bonners Ferry, ID, USA

4. Environmental Department, Kootenai Tribe of Idaho, Bonners Ferry, ID, USA

5. United States Army Corps of Engineers, Libby Dam, Libby, MT, USA

Anthropogenic modifications have altered aquatic communities, food web structures, and energetic pathways along rivers and within river networks. While algae and aquatic plants support lotic food webs with in-situ primary production, current understanding of spatial variability in ecosystem metabolism along a river or river network remains limited. We used dissolved oxygen measurements to estimate gross primary production (GPP) and ecosystem respiration (ER) in the oligotrophic Kootenai River and tributaries. Mainstem sites spanned 158 river km across Montana and Idaho with reaches characterized by different nutrient concentrations and channel morphology including a narrow canyon reach with limited floodplain, a braided reach with wider and shallower channel, a deeper slow moving meander reach, and a reach downstream of phosphorus addition. Tributary sites were situated along 70 mainstem river km and were characterized by different land use and channel depth. Higher GPP and ER were associated with reaches characterized by greater agricultural land use, phosphorus addition, and shallow river depths. Our results suggest that adjacent land use, river management, river morphology, and nutrient concentrations affect the timing and magnitude of GPP and ER. Understanding how ecosystem metabolism varies within a river network may provide a baseline for evaluating the effectiveness of future river management decisions intended to improve food resources for fish.

id #134949

The Invasive Crayfish Collaborative: translating science into regional coordination and action

Natalia Szklaruk¹, Katherine O'Reilly¹

1. Illinois-Indiana Sea Grant, Prospect Heights, ILLINOIS, United States

Invasive crayfish research has advanced rapidly, yet translating this science into coordinated management remains a persistent challenge, particularly across the North American Great Lakes region where jurisdictions, capacity, and monitoring approaches vary widely. The Invasive Crayfish Collaborative (ICC) was formed in 2017 to bridge this gap by connecting researchers, managers, educators, and other stakeholders together. Facilitated collaboration among these partners enables crayfish science to inform practice while management needs shape research priorities. This presentation describes the ICC as a model for translational science, highlighting three projects: (1) a five-year strategic plan co-developed with regional partners to align priorities, (2) interjurisdictional workshops conducted in four Great Lakes states to train professionals in crayfish identification, monitoring, and reporting, and (3) a crayfish curriculum for grades 2–12 designed to build scientific literacy. Together, these efforts improve consistency in data collection, enhance early detection and reporting, and strengthen cross-sector collaboration. This case study demonstrates how structured collaboration can accelerate the translation of crayfish research into effective regional action and advance the management of invasive crayfish.

id #135717

Flow reductions shape invertebrate communities in dryland river systems

Mack White¹, Chloe E Lyles², Lindsey A Bruckerhoff¹, Charles B Yackulic³, Phaedra Budy^{4,5}, Casey A Pennock¹

1. Department of Evolution, Ecology, and Organismal Biology, The Ohio State University, Columbus, OH, United States

2. New Mexico Department of Wildlife, Santa Fe, NM

3. Southwest Biological Science Center, U.S. Geological Survey, Flagstaff, AZ

4. U.S. Geological Survey, Utah Cooperative Fish and Wildlife Research Unit, Utah State University, Logan, UT

5. Department of Watershed Sciences and the Ecology Center, Utah State University, Logan, UT

The natural flow regime of rivers is being reshaped by numerous global change stressors, such as increasing drought and human water use. Aquatic invertebrates are particularly susceptible to these changes due to their limited mobility and reliance on multiple components of the flow regime for their life history. While the effects of hydrologic alteration on invertebrate communities are commonly viewed through changes in taxonomic identity, linking traits to environmental gradients associated with hydrologic alteration – such as flow reduction – can provide mechanistic insight to community assembly rules in the wake of global change. This may be particularly relevant for regions experiencing severe aridification resulting from consumptive water demands, warming temperatures, and decreased run-off, such as the upper Colorado River basin (UCRB). However, little is known about invertebrate assemblages in the UCRB, including the effect of widespread flow reductions in shaping the taxonomic identity and trait distribution of assemblages. Here, we use a series of univariate and multivariate approaches to investigate the role of flow reductions in shaping the taxonomic structure and trait composition of assemblages across the UCRB. Invertebrate communities

were less diverse in rivers that have undergone large flow reductions and characterized by intolerant taxa such as Simuliidae and Ceratopogonidae. Spatial patterns in biodiversity were accompanied by differences in the taxonomic structure of communities and correlated with multiple abiotic factors including flow reduction, flow skewness, baseflow, conductivity, gross primary production, and temperature. However, only a subset of these variables (flow reduction, conductivity, and baseflow) explained variation in trait distributions. In rivers with large flow reductions, invertebrate assemblages were characterized by collector-filterer trophic roles, synchronized emergence strategies, and strong adult flying strengths. We show that flow reductions shape the structure and function of invertebrate communities across the UCRB. As warming temperatures and consumptive water needs continue to intensify, understanding the impact of water scarcity across multiple levels of ecological organization will be needed to conserve freshwater ecosystems in the wake of a hotter, drier future.

id #136229

Evaluating environmental flow performance in California's Central Valley under alternative management scenarios

Sooyeon Yi¹

1. California State University, Chico, Chico, United States

Rivers in California's Central Valley have experienced substantial alteration of their natural flow regimes due to dams, water conveyance systems, and land-use change, resulting in degraded aquatic ecosystems and declining native fish populations. Environmental flows provide a framework for identifying the flow conditions necessary to sustain ecological processes; however, incorporating these requirements into existing water management systems remains challenging, particularly under competing human demands and climate change. As part of the publicly funded COLaboratory for EQUity in Water Allocations (COEQWAL), this study evaluates how alternative water management strategies and future climate conditions affect the ability to meet environmental flow objectives, using the Functional Flows framework as an ecological benchmark. Functional Flows characterize key seasonal components of the natural flow regime that support ecosystem function. We assess monthly environmental water budgets across basins and water-year types to quantify performance under multiple operational scenarios. Results indicate that, under select management approaches, Functional Flow targets can be met in specific basins and hydrologic conditions, though performance varies spatially and seasonally. We further examine the implications of projected warming on flow regime outcomes. Overall, this work highlights trade-offs and opportunities for integrating environmental flow objectives into Central Valley water operations and supports more ecologically grounded and equitable water management decisions.

id #134182

Combined effects of wildfire and historical mining on stream insect communities in Idaho

Maricela N Alaniz¹, Jeremy M Brooks¹, Johanna M Kraus², William H Clements¹, Daniel L Preston¹

1. Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, CO, United States

2. US Geological Survey, Columbia, MO, United States

Wildfire and mining are two disturbances that often occur together, but their combined effects on stream ecosystems are not well understood. In historically mined watersheds, wildfire can mobilize sediment and toxic metals through erosion, which may impact aquatic communities. The Napias Creek Watershed in the Leesburg Mining District of Lemhi County, Idaho, has a long history of gold mining dating back to the late 1860s, with ongoing small-scale placer and lode operations. From July to October 2022, the Moose Fire burned approximately 141,766 acres (57,370 ha) across multiple watersheds in the region. We conducted a field survey to assess the combined effects of wildfire and mining on periphyton, benthic macroinvertebrates, and emergent insects in natural streams. Twelve stream sites were selected across unburned/unmined, unburned/mined, burned/unmined, and burned/mined conditions with similar geological characteristics. Chlorophyll-a biomass was measured using a Benthotorch, benthic macroinvertebrates were collected with a Surber sampler, and emergent insects were captured using emergence traps. Overall, we found no significant differences in periphyton chlorophyll-a biomass between stream types, though there was weak evidence for lower biomass in mined sites. Total macroinvertebrate abundance was reduced at both burned and mined sites relative to controls, with substantial variation among taxa. There was some evidence of an earlier pulse of emergence at fire-only sites, with no consistent patterns observed among the other sites. These results indicate that mining and wildfire influence benthic macroinvertebrate communities, potentially driving changes in both short-term community structure and longer-term food web dynamics. The limited effects on periphyton and insect emergence, however, may reflect additional factors such as fire severity, elevation, and soil metal concentrations, highlighting the complex interactions of wildfire and mining in natural streams.

id #134694

From fundamental ecology, to restoration experimentation, and on onward to policy-relevant restoration.

Margaret Palmer¹

1. University of Maryland, MD, -

For many freshwater scientists, research starts with a focus on biodiversity, species interactions, and/or some fundamental community or ecosystem question. Such research is important and has advanced our understanding of ecological dynamics in running and lentic waters enormously. But today many or most freshwater researchers also undertake research that has the potential to influence decisions that enhance the health of freshwaters and those people they support. Having been first influenced by LeRoy Poff's research mindset as early as 1990 when he was at the University of Maryland, I have followed his work and his impact with great interest. I describe our early restoration ecology work, how the field has advanced, and my recent work that reflects his scientific impact.

Ecosystem metabolism of tropical rivers: exploring river ecology and people perceptions

Maria M Castillo^{2,1}, Krista A Capps³, Dora Ramos^{2,1}, Azahara Mesa², Amber J Ulseth⁴, Diana Blanco²

1. *El Colegio de la Frontera Sur, Villahermosa, Tabasco, México*

2. *El Colegio de la Frontera Sur, -, Mexico*

3. *Odum School of Ecology, University of Georgia, Athens, GA, United States*

4. *Natural Resources Research Institute, University of Minnesota, Duluth, MN, United States*

Tropical rivers face increasing human pressures that fundamentally reshape human-river relationships. We estimated ecosystem metabolism in non-wadeable rivers in southern México with varying levels of human disturbance. In addition, we used the ecosystem metabolism concept to explore how communities perceive the condition of rivers and relate to these ecosystems. The smaller, less disturbed rivers draining forested catchments showed reduced gross primary production, ecosystem respiration, and net ecosystem production approaching autotrophy relative to larger and more disturbed catchments. The smaller rivers exhibited greater and sustained GPP during periods of low discharge and a more constrained metabolic fingerprint compared to larger and more disturbed rivers. We used an analogy with human metabolism to introduce the concept of ecosystem metabolism. Seasonal patterns in dissolved oxygen and ecosystem metabolism were shared with local communities in workshops to explore how local communities perceive river conditions and their connections to these ecosystems. Co-designed infographics, developed with community input, integrated ecological data with local knowledge, perceptions, and observations of how human activities affect river health. Interestingly, people can identify longitudinal zonation in minimally disturbed rivers based on water color and depth, while in the more disturbed sites, spatial zonation was not perceived. Historical changes in water quality and flow were also reported. Local people engaged in recreation, subsistence fishing and tourism at the forest rivers. In the more disturbed rivers, pollution restricts water use and fish consumption, while the increased abundance of crocodiles further limits recreational activities. We concluded that human pressures are causing profound changes in the ecology of tropical rivers. People can perceive these transformations, which may progressively undermine the resilience of human-river relationships.

Discovering in reverse: using isotopes and omics to reveal ecological interactions in microbiomes in stream ecosystems.

Jane Marks¹

1. *Northern Arizona University, Flagstaff, ARIZONA, United States*

Since Haeckel coined the term, *ecology* has advanced by studying mechanisms and feedbacks that govern how organisms interact with their environment and each other. From individuals to populations to communities to ecosystems, ecologists have designed reductionist experiments for strong inference and developed theory from first principles. Inference in microbiome science often runs in reverse, starting with gene surveys or meta-genomics of microbiomes, and making inferences from informatics, with little direct knowledge of the organisms. Many microbial sequences remain classified as biological "dark matter", known only by the order of their nucleotides, their connection to organisms inferred from bioinformatics, and their roles in nature predicted from genes of organisms known in culture. While the 'omics revolution holds great promise, it lags in connecting microbiome 'omic descriptions with the ecological theories that predict organismal responses to environmental change. Bridging 'omics insights and ecological theory has been hampered by three key problems: First, 'omics techniques are only partially quantitative--insofar as they estimate relative abundances or proportions. Most 'omic techniques were not designed to be capable of estimating process rates of a specific microbiome population, such as growth, mortality, and resource assimilation. Second, microbiomes are metabolically complex, using an array of electron donors and acceptors and associated reactions that far outnumber the basic biochemical reactions that support macrosystems. Third, living microbes are far more difficult to observe in their native environments than are macro-biota, challenging hypothesis-driven discovery of the species interactions that structure biological assemblages. Here, we present a new approach to detect and quantify microbial interactions, an approach that draws on ecological theory and concepts, with tools and techniques from isotope geochemistry, biochemistry, genomics, and machine learning.

Current state of ballast water management in the Laurentian Great Lakes and challenges with microbial discharge standards

Noah W Grode¹, Sophie J Husby¹, Christine N Polkinghorne¹, Matthew C TenEyck¹

1. *University of Wisconsin, Superior - Lake Superior Research Institute, Superior, WI, USA*

Ballast water serves as a prevalent vector for aquatic invasive species transport in the Laurentian Great Lakes region. The International Maritime Organization (IMO) currently requires ballast water exchange for cargo ships entering the St. Lawrence Seaway, which serves to limit introduction of species from outside of the Great Lakes. Regulations within the Great Lakes, however, are inconsistent with Canada requiring that vessels built before 2009 must install ballast water management systems (BWMS) by 2030, while vessels built after 2009 must install BWMS by 2024. In the United States, there is currently no BWMS regulation for vessels trading exclusively in the Great Lakes. Even with no regulation, BWMS are currently in use in the Great Lakes region, and while not required, do use the numeric IMO discharge standards, designed for saltwater, as a target for treatment. These discharge standards use size class as their definition of organisms, with less than 10 µm being the microorganism size class. Fecal indicators such as *Escherichia coli*, *Enterococci spp.* and *Vibrio cholerae* are used to assess whether BWMS treatments meet these discharge standards. BWMS treatments are in general successful in disinfecting this size class, however there is often not sufficient concentration of fecal indicators in uptake water to challenge the system and make

meaningful conclusions about the treatment. The Lake Superior Research Institute has been researching BWMS since 1998 and in 2020 have established a research and development plan to inform regulators of whether market ready BWMS are able to effectively achieve discharge standards when operated in the Great Lakes region. Here we describe the current state of BWMS in the Laurentian Great Lakes, along with challenges with microbial discharge standards.

id #135463

Embracing the zoogeochemical framework in freshwaters to understand and predict the ecosystem-level consequences of conservation (Joint Talk with Jonathan Lopez)

Taylor C Michael^{1,2}, Jonathan W Lopez^{1,3}, Carla L Atkinson¹, Matthew B Lodato^{1,4}, Lauren M Morris¹, Stephen Plont^{1,5}, Chelsea R Smith¹, Frank C Akamagwuna¹, Lindsey K Albertson⁶, David K Breault¹, Jacob D Dorris¹, Samuel F Fritz⁶, Jennifer Haase⁷, Tori A Hebert¹, Adam C Hensley¹, Garrett W Hopper⁷, Madison P Kuczek⁷, Julia N Morin⁷, Hayley C Oakland⁶, Adam C Siders¹, Augustine Sitati¹, Benjamin B Tumolo⁸

1. University of Alabama, Tuscaloosa, AL, United States

2. Kent State University, Kent, OH, United States

3. University of Mississippi, Oxford, MS, United States

4. Texas Tech University, Lubbock, TX, United States

5. University of Georgia, Athens, GA, United States

6. Montana State University, Bozeman, MT, United States

7. Louisiana State University, Baton Rouge, LA, United States

8. Northern Michigan University, Marquette, MI, United States

Freshwater ecosystems are under immense pressure from global environmental change. These stressors have led to biogeochemical alterations and rapid rates of defaunation and losses of associated animal-driven ecosystem functions. Zoogeochemistry is an ideal framework for trying to meet the dual challenges of global change and defaunation in freshwater ecosystems, but zoogeochemical frameworks that explicitly link conservation biology with biogeochemistry have not yet been fully embraced by freshwater ecologists. We conducted a systematic review and meta-analysis of the mechanistic roles that animals play in freshwater biogeochemistry, with the aim of advancing the application of zoogeochemistry in freshwaters. We reviewed 14 major mechanisms of freshwater zoogeochemistry. We grouped mechanisms into three primary categories: (1) trophic interactions that mediate the movement and transformation of elements through food webs, (2) metabolic processes, including excretion and egestion, that directly alter nutrient availability, and (3) physical ecosystem engineering, such as bioturbation and sediment resuspension, which modify redox conditions and enhance benthic–pelagic coupling. Our review shows that a diversity of taxa impact ecosystem-level elemental pools and fluxes of dozens of elements including the traditional macronutrients, C, N, and P, and a range of micronutrients, both within and across freshwater ecosystem boundaries. However, these effects are not always placed in context at the ecosystem level, which limited the scope of our meta-analysis. Zoogeochemical impacts tended to be stronger in wetlands and lakes than streams and freshwater estuaries. Consumption, excretion, and respiration had the strongest direct impacts on elemental pools and fluxes; bioturbation, consumption, and egestion had the strongest indirect impacts. These mechanisms often interacted, generating complex influences on elemental pools and fluxes. However, creative experimental designs can be used to isolate individual mechanisms, allowing complex effects with multiple mechanisms to be incorporated into ecosystem models. Anthropogenic pressures such as biodiversity loss, altered species interactions, habitat fragmentation, and climate-driven shifts in movement and phenology, are disrupting animal-mediated biogeochemical processes at local and landscape scales. Quantifying and modelling zoogeochemical effects in freshwater ecosystems will allow ecologists and managers to make informed decisions that balance biodiversity conservation with healthy ecosystem function.

id #135719

A 44-year study of amphibian-mediated zoogeochemical connectivity in Carolina bays

Irene Sanchez¹, David E Scott², Stacey L Lance², Krista Capps¹

1. Odum School of Ecology, University of Georgia, Athens, GA, United States

2. Savannah River Ecology Laboratory, University of Georgia, Aiken, South Carolina, United States

Consumer-mediated transfers of energy and nutrients across ecosystem boundaries can strongly influence ecosystem structure and function. Mobile consumers link habitats by transporting biomass and nutrients during reproduction and dispersal, creating cross-ecosystem subsidies. Carolina bays are shallow, isolated depression wetlands distributed across the Atlantic Coastal Plain of the southeastern United States. Although individually small, these wetlands occur at high densities and collectively can influence biodiversity and nutrient cycling at landscape scales. Carolina bays support diverse amphibian communities and exhibit pronounced interannual variability in hydroperiod, creating conditions for episodic exchanges of biomass and nutrients across aquatic and terrestrial boundaries. Amphibians are well-suited to mediate these exchanges because of their biphasic life cycles. Adults transport nutrients into wetlands during breeding, while juveniles export biomass and nutrients to surrounding terrestrial habitats upon emergence. We tested three hypotheses: (1) amphibian-mediated fluxes constitute a meaningful component of cross-ecosystem nutrient transport, (2) hydroperiod length regulates the magnitude of nutrient and biomass export to terrestrial systems, and (3) amphibian life history traits mediate community-level zoogeochemical outcomes under variable hydrologic conditions. We analyzed a long-term amphibian dataset from a single Carolina bay in South Carolina, where amphibian movements have been monitored continuously using a drift fence and pitfall array from 1979 to 2023. Records of amphibian abundance, community composition, body size, egg production, juvenile mass, and adult breeding mortality were integrated with annual hydroperiod data. Species-specific stoichiometric information was used to estimate annual dry mass and elemental fluxes associated with amphibian movements across the aquatic and terrestrial boundary. Amphibian-mediated fluxes exhibited substantial interannual variability, with total export to the terrestrial environment associated with wetland inundation duration. Longer hydroperiods supported greater amphibian production and higher biomass and nutrient export, whereas shorter

hydroperiods reduced export and shifted species contributions. Life history traits such as larval development time and body size influenced species-level fluxes, while demographic processes, including breeding mortality, further shaped net outcomes. Overall, Carolina bays function as temporally dynamic zoogeochemical hotspots, where hydroperiod variability governs the strength of animal-mediated nutrient transport. Changes in hydroperiod regimes therefore have the potential to alter both amphibian communities and freshwater–terrestrial connectivity.

id #135975

Occurrence and spatial variability of PFAS in surface waters across the Northern Great Plains

Brooke Long-Fox^{1,2}, Jenny Kozak², Kurt M Chowanski^{1,2}, Lisa A Kunza^{1,2}

1. Center for Sustainable Solutions, South Dakota School of Mines and Technology, Rapid City, SD, United States

2. Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines and Technology, Rapid City, SD, United States

Per- and polyfluoroalkyl substances (PFAS) are persistent, mobile contaminants increasingly detected in freshwater systems, yet their spatial distribution and controlling factors across river networks remain poorly resolved. We assessed PFAS occurrence in surface waters across four Northern Great Plains watersheds: Boxelder Creek (SD), Rapid Creek (SD), the Big Sioux River (SD), and the Missouri River (NE). These systems represent small to large watersheds influenced by varying degrees of urban and agricultural land use. Surface water samples were collected during a synoptic sampling event in June 2025 from upstream, urban, tributary, and downstream reaches and analyzed for 40–43 PFAS using U.S. EPA Method 1633 or EPA Method 537.1, depending on site. PFAS occurrence and summed concentrations varied markedly among watersheds. Boxelder Creek exhibited the highest PFAS signal, although elevated concentrations were confined to a small number of downstream sites, with most locations showing substantially lower levels. The Big Sioux River showed intermediate PFAS occurrence, with substantial spatial heterogeneity and higher concentrations at downstream sites relative to upstream locations. Rapid Creek exhibited fewer detections and lower summed concentrations, while the Missouri River showed minimal PFAS occurrence, with only a single PFAS compound detected per site and consistently low summed concentrations. Across all systems, most targeted PFAS were not detected, indicating that observed PFAS patterns were dominated by a limited subset of compounds rather than broad detection across the analytical suite. We used the EPA National Primary Drinking Water Regulations (NPDWR) benchmarks, including individual Maximum Contaminant Levels for six PFAS and the Hazard Index framework for PFAS mixtures, as health-based reference values to support interpretation of surface water data. While the NPDWR does not apply to surface waters, several urban-influenced sites exhibited concentrations or mixture profiles exceeding drinking-water reference values, highlighting areas where continued monitoring may be warranted. Together, these results demonstrate the value of comparative, watershed-scale monitoring for characterizing PFAS occurrence in river systems. This work provides baseline data to support future investigations of seasonal dynamics, mixture behavior, and potential source differentiation in freshwater environments.

id #135208

Understanding thermal mixing dynamics at tributary confluences in a large managed river to enhance cold-water refuges.

Francine H Mejia^{2,1}, Eric K Berntsen³, Christian E Torgersen¹, Sylvia R Gholson¹, Matthew I Barker⁴, Andy Johnsen³, Daniel S Auerbach⁵, Alexander K Fremier⁵, Francine Mejia^{2,1}

1. Forest and Rangeland Ecosystems Center, Seattle, WA, USA

2. Natural Resources, Salish Kootenai College, Pablo, Montana, USA

3. Kalispel Tribe Natural Resources Department, Usk, Washington, USA

4. USGS Washington Water Science Center, Tacoma, WA, USA

5. Washington State University, Pullman, WA, USA

Dam operations restrict or eliminate connectivity longitudinally, laterally, vertically, and temporally and can affect thermal heterogeneity, particularly mixing of the water column. We examined thermal heterogeneity across space and time in two tributary confluences before and after instream structures were constructed (i.e., large woody debris, gravel spits) in a dam-influenced river in the inland Pacific Northwest (USA) during summer months. Structures were placed to reduce mixing of cold water from tributaries with warm water from the mainstem and promote scouring to create pools for coldwater fishes. To describe thermal and hydrologic patterns, we used thermal infrared (TIR) imagery from an uncrewed aerial vehicle (UAV), high-resolution in-situ thermographs, and flow data from an upstream gage. Preliminary results revealed that cold water ($<16^{\circ}\text{C}$) was mostly maintained near the substratum within the confluence areas throughout July and August after structures were installed. Mainstem flows were more important than air temperature in influencing the magnitude of stratification and can dissipate the cold-water plumes rapidly once the plumes move past the gravel spits. However, flows below $140\text{ m}^3\text{s}^{-1}$ in the mainstem reduced the amount of usable area for fish behind the gravel spits due to insufficient water depth, suggesting that minimum instream flows may be needed to maximize the effectiveness of instream structures designed to enhance cold-water refuges. Preliminary findings also suggest that using multiple tools can help identify the extent to which surface-release flows influence the persistence and size of cold-water refuges.

id #135720

Aquatic macroinvertebrates at the continental scale: Using NEON data to detect community patterns

Stephanie M. Parker¹

1. National Ecological Observatory Network (NEON), CO, United States

Aquatic macroinvertebrate data have long been used as core indicators of ecosystem health and productivity. The National Ecological Observatory Network (NEON) is a research platform designed to assess the effects of climate, land-use change, and invasive species across North America. NEON's aquatic program consists of 24 streams, 7 lakes, and 3 rivers across 19 eco-climatic domains. NEON's macroinvertebrate data collection began in 2014 with samples collected in the two most common habitats at each site. Samples are collected three times per year, roughly in spring, summer, and fall, then identified by expert taxonomists to the lowest practical taxonomic level and measured to the nearest mm. These data enable analyses of diversity and community metrics, organismal abundance, and biomass estimates over space and time. Within NEON macroinvertebrate data, alpha diversity is highest in streams with seasonal snowmelt pulses like Blacktail Deer Creek (BLDE), MT (50.6) and Teakettle Creek (TECR), CA (46.8), and lowest in streams with seasonal drying like Kings Creek (KING), KS (18.7) and Sycamore Creek (SYCA), AZ (19.3). Macroinvertebrate density is also highest in snowmelt-dominated streams such as West St. Louis Creek (WLOU), CO (44,530 organisms m⁻²) and TECR (34250 organisms m⁻²), however, abundance at the sites with seasonal drying falls in the middle of the range (KING = 11037 and SYCA = 21817 organisms m⁻²), with low macroinvertebrate abundance occurring in the northern latitude streams Oksrukuyik Creek (OKSR) and Caribou (CARI) Creek, AK (2421 and 1735 organisms m⁻², respectively). Patterns in primary producers will also be explored through this analysis. These data leverage multiple NEON data products, including Macroinvertebrate Collection (DP1.20120.001), Periphyton, seston, and phytoplankton chemical properties (DP1.20163.001), and Continuous Discharge (DP4.00130.001) to explore patterns in the stream macroinvertebrate community across space and time.

id #135976

Wet wipes as potential carriers of pathogens and antibiotic resistance genes in urban streams

David Calvo-Mora¹, Benoit Cournoyer², Miquel Ribot¹, David Pineda-Morante¹, Sara Castelar¹, Baptiste Luton², Ana Luisa Mejía-Camacho³, Natàlia Corcoll³, Eugenia Martí¹, Helena Guasch¹

1. *Integrative Freshwater Ecology Group, Center for Advanced Studies of Blanes, Spanish National Research Council (CEAB-CSIC), Blanes, Spain*

2. *"Bacterial Opportunistic Pathogens and Environment" (BPOE) Research Group, CNRS 5557, INRAE 1418, VetAgro Sup, Université Claude Bernard Lyon 1, UMR Ecologie Microbienne – Lyon (LEM), Marcy l'Etoile, France*

3. *Department of Biological and Environmental Sciences, University of Gothenburg, Gothenburg, Sweden*

Wet wipe pollution is increasing in aquatic ecosystems. In urban catchments, wipes can become dominant plastic substrata for microbial colonization. It has been described that plastic can be colonized by pathogens and antibiotic-resistant bacteria, with growing evidence that plastic surfaces may also facilitate their occurrence, representing potential health risk spots. Effluents from wastewater treatment plants are known to be sources of several antibiotic-resistant pathogenic bacteria from the ESKAPEE group. We hypothesize that wet wipes may provide novel substrata in aquatic ecosystems that facilitate the microbial establishment given their fibrous plastic structure. The combination of a high organic matter retention capacity and the hydrophobicity of plastic fibers may additionally promote denser or compositionally distinct biofilm assemblages. These features may enhance the potential for wet wipes to act as physical supports for pathogenic cell lines. To address this hypothesis, a biofilm colonization experiment was conducted in a flume mesocosm facility and in three urban rivers impacted by wastewater treatment plant effluents, all located near Barcelona (Spain). We compared the 4-week colonization of two wet wipes with different plastic compositions to that of two selected inorganic and organic control substrata. DNA extractions of biofilms on the different substrata were performed, and targeted DNA qPCR assays for five pathogens (*Aeromonas caviae*, *Clostridioides difficile*, *Clostridium perfringens*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*) and four antibiotic resistance genes (*tetA*, *blaCTX-M*, *sul1*, and *ermB*) were conducted. To contextualize these results, occurrences of the main functional microbial groups within such periphytic-like assemblages were assessed by qPCR assays targeting phototrophs, microeukaryotes, and fungi. The metabolic activity (respiration rates) of the biofilms was also performed. Preliminary results show a high variability among stream sites and experimental substrata in the relative abundances of pathogen DNA targets. Statistical analyses of these data are currently underway to examine the drivers of this variability and elucidate whether biofilms in wipes hold pathogenically distinct microbial communities. These results will be shown in the presentation.

id #134441

Chasing Salt: Utilizing Community Science to Understand How Salt Stresses Riparian-Stream Ecosystems

Lisa M. Tabor¹, Tara Muenz², Amaryllis K. Adey¹, David Costello³, Natalie Clay⁴, Scott Tiegs⁵, Michelle Evans-White⁶, Megan Underwood¹, Sally Entekin¹

1. *Entomology, Virginia Tech, Blacksburg, Virginia, United States*

2. *Education, Stroud Water Research Center, Avondale, Pennsylvania, United States*

3. *Biological Sciences, Kent University, Kent, Ohio, United States*

4. *Entomology and Plant Pathology, University of Arkansas, Fayetteville, Arkansas, United States*

5. *Biological Sciences, Oakland University, Rochester, Michigan, United States*

6. *Biological Sciences, University of Arkansas, Fayetteville, Arkansas, United States*

We tested how excess sodium (Na) from road salt and agriculture alters carbon exchange across terrestrial-aquatic boundaries, by evaluating organic-matter decomposition. The Na ecosystem-respiration hypothesis (SER) posits that more Na in riparian forests will increase leaf decomposition rates; we extend this hypothesis to streams, adding that Na will slow rates past a threshold concentration (subsidy-stress hypothesis). We predicted that 1. streams with more urban and agricultural land cover will have more Na in riparian soil and stream water and that organic matter will decompose faster at saltier sites 2. urban sites will have more Na than agricultural in autumn and winter from road salt addition and 3. more precipitation will accelerate decomposition in streams and riparia from greater microbial activity. We recruited 56 community partners from the central Appalachians to the western edge of the Great Plains and provided standardized monitoring kits, protocols, and other resources for them to use in

streams selected by participants. Community study sites span 14 ecoregions within the U.S, across a gradient of land cover from either 60% forested, 10% urban and 25% cropland, pasture, or a combination of both. To measure decomposition rates, we used standardized cotton-strip assays, deployed within a 50-meter reach in both the riparian zone and stream in late November and retrieved them one month later. Results from this community science project will be shared publicly through an open-access database to support local environmental stewardship for mitigating salt impacts to ecosystems. Our data will be part of a global decomposition study using consistent methods and are framed to evaluate how salt influences riparian–stream connections through carbon transformation.

id #134953

A Partnership for Ecohydrologic Modeling of Headwater Habitat and Refugia for Stream-breeding Amphibian Species in Western Washington and Oregon

Nicholas A Sutfin¹, Lindsey L Thurman², Tristan Weiss³, Reed Ojala-Barbour³, Aimee McIntyre³

1. *Washington Water Science Center, USGS, Tacoma, WA*

2. *Northwest Climate Adaptation Science Center, USGS, Corvallis, WA, United States*

3. *Washington Department of Fish and Wildlife, Olympia, WA*

Pacific Northwest forests have undergone substantial changes in the past century as a result of extensive logging, changes in forest management methods, development, and climate change. These forests contain a dense network of small headwater streams that supply cold, clean water and support rich amphibian diversity. The USGS Washington Water Science Center, USGS Northwest Climate Adaptation Science Center, and the Washington Department of Fish and Wildlife are developing models and maps of amphibian habitat in headwater stream ecosystems. We are collaborating with academic, private, tribal, state, and federal partners to ensure our project will produce useful data products for natural resource managers. We facilitate virtual triannual meetings of a Technical Advisory Committee (TAC) through which we provide project updates and receive helpful input in a structured discussion. We seek guidance from the TAC on habitat field survey protocol, the use of spatial datasets to develop proxies for important characteristics of amphibian habitat, and the modeling approach for six target amphibian species in western Washington and northwestern Oregon west of the Cascade Range crest. Spatial datasets include precipitation, stream temperature, streamflow permanence, land cover, land use, and lidar-derived elevation. We are hydrologically conditioning 3-m resolution lidar-derived digital elevation models for accurate hydrologic routing through the landscape and drainage area delineation. We will use estimates of streamflow, stream power, and valley confinement combined with pre-existing datasets for canopy cover and lithology to derive proxies of channel substrate grain size and presence of large instream wood. Random sampling was stratified by EPA Ecoregion to select 10-digit hydrologic unit code watersheds (HUC10s) from the USGS National Hydrography Dataset. In summer 2025, we conducted field surveys of 122 streams across eleven HUC10s, which will be used to validate the habitat proxies derived from spatial datasets. The TAC has indicated that multi-species occupancy and joint species distribution models that include both current and potential habitat refugia for the six species of interest will yield the most useful output for natural resource managers. Through consistent dialogue with the TAC, modeled outputs will be tailored to the needs of regional natural resource managers.

id #135721

Urban stream syndrome at the river network scale in a Mediterranean watershed.

Wilfred Wollheim¹, Eugenia Martí², Sara Castelar², Susana Bernal², Joan Blavia², Berta Bonet², David Calvo Mora², Nuria Catalan², DELFINA CORNEJO², Maria Espinosa², Esperanca Gacia², Xavier Galceran², Helena Guasch², Joan Chavarria-Hierro², Carolina Jativa², Jonas Lower², Anna Lupon², Rafel Marce², Stephanie Merbt², Maria Niswonger³, Nicki Novak², David Pineda-Morante², Miquel Ribot², Krati Sharma²

1. *University of New Hampshire, Durham, NH, United States*

2. *Centre d'Estudis Avançats de Blanes, Blanes, Catalunya, Spain*

3. *York County Community College, Wells, Maine, USA*

The urban stream syndrome describes the multiple changes that co-occur when a stream becomes urbanized, affecting flow, biogeochemistry, microbiota, and biodiversity. The multiple symptoms of the urban stream syndrome are attenuated as water flows downstream either by dilution or via active transformations by aquatic processes which are affected by flow conditions. We addressed the question of how a suite of changes caused by urbanization are transferred through an entire river network under both high and low flow conditions. We conducted synoptic surveys throughout the river network of the Tordera River watershed, Catalonia, Spain, during March (high flow) and July (low flow) 2025. We measured flow, specific conductance, pH, dissolved oxygen, temperature, cations, anions, inorganic and organic nutrients, organic carbon, greenhouse gases, microplastics, and microbial communities from headwaters of different land use through tributaries and mainstem, as well as inputs from wastewater treatment plants. We found that specific conductance, a conservative tracer, increased steadily along the Tordera continuum under both high and low flow conditions, consistent with increasing population and wastewater treatment plant effluent moving downstream. Surprisingly, overall specific conductance levels were only slightly higher under low flow compared to high flow indicating minimal dilution at high flows. Nitrate increased along the Tordera continuum similar to those for conductance during high flow, but during low flow, nitrate did not increase despite considerable wastewater and diffuse N sources. This result suggests that in stream processes are able to mitigate bioreactive compounds (e.g. nitrate) entering the river network under certain conditions, attenuating a portion of the urban stream syndrome from being expressed further downstream.

id #135977

Indigenous student engagement in freshwater field education: lessons, challenges, and opportunities

Carolyn Cummins¹, Kaitlin Mattos^{1,2}, Colten Ashley^{3,1}, Elizabeth Waring⁴

1. *Four Corners Water Center, Fort Lewis College, Durango, CO, USA*

2. *Department of Environment and Sustainability, Fort Lewis College, Durango, CO, USA*

3. *Tribal Media Center, KSUT Public Radio, Ignacio, CO, USA*

4. *Department of Natural Sciences, Northeastern State University of Oklahoma, Tahlequah, OK, USA*

Climate change, hydrologic alterations, non-native species, and water quality concerns threaten freshwater ecosystems worldwide (Dudgeon and Strayer 2025). At the same time, opportunities for college students to engage in field-based experiential learning are declining, though these opportunities are critical for cultivating the next generation of freshwater scientists, stewards, and storytellers (Soga and Gaston 2025). Field-based learning comes with barriers to participation, including financial, time, social, and cultural constraints (Carlin et al. 2023). However, these experiences can be transformative for students, fostering personal growth, leadership, and science identity (Soga and Gaston 2025). To ensure that field-based experiential learning is successful, programs must be intentionally designed to be safe, welcoming, and inclusive—especially for Indigenous students. In working to bridge Indigenous knowledge and perspectives with Western science in field settings, instructors have the essential opportunity to build courses that honor cultural perspectives, acknowledge the complex history of environmental science, and enhance ecological understanding. In this presentation, we will share examples of place-based, experiential, freshwater field education at two Native American-Serving, non-Tribal institutions in the United States. We will discuss program successes, challenges, and lessons learned, as well as actionable steps and best practices informed by these programs that can help make field-based learning more accessible and inclusive. We will highlight Fort Lewis College's Tribal Water Media Fellowship program, in which students learn about, then create independent media projects on, Tribal and Indigenous water topics. In addition, we will discuss professional development and curricular opportunities such as the nationwide River Studies and Leadership Certificate (RSLC) program, a program administered by the River Management Society with the goal of empowering students to pursue river careers. Overall, our programs highlight the need to work collaboratively with freshwater science students, faculty, and practitioners to improve inclusion in field education and build the next generation of diverse water leaders and storytellers.

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id #135210

Effects of climate stressors on positive outcomes of ecosystem engineers: Current patterns and future expectations

Lindsey K Albertson¹, Samuel F Fritz¹, Hayley C Oakland¹, Lydia JL Bushey¹, Anna C French¹, Nathaniel M Heili¹, Johnathan Reyes de Merkle¹, Stacy Schmidt¹, Hamish S Greig², Matthew F Johnson³, Gemma L Harvey⁴

1. *Montana State University, Bozeman, MT, United States*

2. *University of Maine, Orono*

3. *University of Nottingham, Nottingham, U.K.*

4. *Queen Mary University London, London, U.K.*

Ecosystem engineers that create and maintain the physical environment epitomize the interconnectedness of the global crises of climate change, biodiversity loss, and pollution. Ecosystem engineer species and their activities are impacted by changing hydrology, temperature, and extreme weather. But these same species also influence how climate change drivers affect ecosystems by buffering and amplifying climate conditions. These feedbacks have often been overlooked in freshwater ecosystems compared to terrestrial ecosystems. To understand how climate affects the ability of ecosystem engineers to support other species, we conducted a quantitative literature review. We found that temperature was studied 2.5x more often as a climate driver of ecosystem engineering potential in terrestrial ecosystems when compared to freshwater. In freshwater ecosystems, high flow was more studied than low flow or temperature; mammal and plant species were studied over 5x more than the next most common group insects; and most studies to date were from locations with climates projected to get wetter and warmer, leading to gaps in our knowledge about drier locations. Our findings highlight the need to better understand how positive interactions imparted by ecosystem engineer species may become more or less ecologically meaningful when beneficiaries experience stressors such as those related to climate change.

id #135722

Using macroinvertebrate colonization of artificial *Podostemum ceratophyllum* (a swiftwater macrophyte) substrates to quantify restoration impacts

James Wood¹, Elizabeth Cika², Zackary Graham²

1. *Tennessee Tech University, Cookeville, TN, United States*

2. *West Liberty University, West Liberty, WV, United States*

Podostemum ceratophyllum "Hornleaf riverweed", henceforth referred to as *Podostemum*, is a foundation species in fast-flowing Appalachian rivers. *Podostemum* is a benthic epilithic macrophyte known to provide habitat for macroinvertebrates and fish, and previous studies have found significant reductions in invertebrate abundance when *Podostemum* has been removed or when stem length is reduced. Similarly, multiple species of darters and mad toms have been positively associated with *Podostemum*. While the plant appears to have declined substantially from its historic abundance, drivers of loss are unclear, which poses a major restoration challenge. Additionally, concerns over moving live plants between watersheds exist due to the possibility of transferring diseases or nuisance species. In order to better understand the effects of restoring *Podostemum* to rivers where the plant has been extirpated, we created artificial *Podostemum* substrates (APS) and deployed them into four rivers in the Appalachian Plateau of WV and PA. At each site, we deployed the APS for approximately six weeks, then collected all macroinvertebrates on the APS. Inverts were identified to family, assigned functional feeding groups (FFGs), and biomass was estimated. We found significant increases in macroinvertebrate abundance and biomass with increasing stem length of our APS. We also found increases in Collector-Gatherer FFG in APS compared to bare rock controls. Our results indicate that restoration of *Podostemum* could have positive impacts on macroinvertebrate abundance and biomass, and suggest that restoration of *Podostemum* into rivers where the plant has been extirpated may benefit the broader benthic community.

id #135978

Assessing Morphological, Developmental, and Genetic Responses of Hydropsychid Caddisflies to Cry1Ab Exposure

Ethan P Bull¹, Scott P Egan¹, Jennifer L Tank², Elise Snyder³, Yufei Qi¹, Amanda Potts¹, Maryam Rajabi Faghihi², Pedro FP Brandao-Dias^{4,1}, Pedro Brandao Dias F Pinto^{4,1}

1. Department of BioSciences, Rice University, Houston, Texas, USA
2. Department of Biological Sciences, University of Notre Dame, Notre Dame, Indiana, USA
3. Illinois Natural History Survey, Champaign, Illinois, USA
4. University of Washington, Seattle, WA, United States

Transgenic crops expressing Cry proteins are widely cultivated to reduce crop damage from insect pests. However, Cry proteins can leach from transgenic crop detritus into nearby aquatic ecosystems, potentially affecting non-target organisms. Here, we assess the impacts of exposure to a commonly used Cry protein, Cry1Ab, on net-spinning caddisflies (Hydropsychidae), a common and ecologically important group of stream insects. We collected and genetically barcoded 1862 larval caddisflies from 25 streams across the Midwestern United States, comparing populations from Cry1Ab-positive and Cry1Ab-negative streams. We evaluated potential impacts associated with body size, developmental stage, species assemblages, and mitochondrial haplotype diversity. Overall, we observed limited effects of Cry1Ab exposure, including most comparisons showing no difference between Cry1Ab-positive and Cry1Ab-negative streams with a few variable exceptions. Second instar *Hydropsyche betteni* showed significantly reduced body size at Cry1Ab-positive sites, while two other species showed size increases. Mean developmental stage within species and species assemblages showed no consistent association with Cry1Ab. In contrast, environmental variables such as water temperature, stream width, and watershed land use were stronger predictors of body size, instar progression, and species assemblages. Haplotype networks revealed strong geographic structure but no genetic patterns related to Cry1Ab exposure. These results indicate that Bt-Cry exposure is not a major driver of morphological, developmental, or genetic variation in Hydropsychid caddisflies under natural field conditions. Environmental gradients remain the dominant structuring force, although further work is needed to assess potential context-dependent or cumulative effects of transgenic crop byproducts in aquatic ecosystems.

id #134955

Trophic highways and dead ends: Algal identity regulates nutrient flux and energy transfer in stream food webs

Michael C Zampini¹, Jane C Marks², Steve Thomas³, Michael Wulf², Saeed Kariunga², Augustine Sitati³, Mary Power¹

1. University of California Berkeley, Berkeley, CALIFORNIA, United States
2. Northern Arizona University, Flagstaff, ARIZONA, United States
3. Biological Sciences, University of Alabama, Tuscaloosa, AL, United States

Primary producers form the energetic base of river food webs, yet the fate of fixed carbon (C) and nitrogen (N) depends on how these nutrients are packaged within producer biomass and accessed by consumers. In nitrogen-limited rivers, biological N₂ fixation can be an important N source, but it does not uniformly support higher trophic levels. Here, we test how algal identity regulates nutrient flux and trophic transfer in the South Fork Eel River, California. Using stable isotope tracers (¹³C and ¹⁵N), we measured nutrient acquisition across common benthic producers and evaluated their contribution to the river food base by tracking assimilation by the grazing caddisfly *Gumaga nigricula*.

Although several algal types exhibited high rates of N₂ fixation, trophic outcomes differed sharply. Assemblages dominated by epiphytic diatoms of the genus *Epithemia*, which host N-fixing cyanobacterial endosymbionts, consistently supported the highest grazer assimilation and trophic transfer efficiency for both C and N. Epiphytized *Cladophora* and epilithic periphyton functioned as trophic highways, efficiently routing newly fixed nutrients to consumers. In contrast, free-living cyanobacteria (e.g., *Nostoc* and *Anabaena*) and mucilage-rich filamentous green algae were highly productive but transferred little of their production to grazers, forming trophic dead ends. High fixation rates alone were a poor predictor of food-web support.

These contrasting outcomes reflect differences in food quality and accessibility. *Epithemia* diatoms provide highly palatable, nutrient-dense biomass with favorable stoichiometry and essential biochemicals, while cyanobacteria and filamentous green algae are limited by physical structure, chemical defenses, or poor nutritional composition. This study quantifies large variation in the flow of newly fixed C and N from the atmosphere into algal biomass and onward to primary consumers, and demonstrates that trophic transfer efficiency is governed by algal functional traits. Together, these results show that shifts from diatom-dominated assemblages toward cyanobacterial or green algal dominance alter not only elemental fluxes but also the molecular

pathways of nutrient assimilation, decoupling primary production from consumer support and creating trophic dead ends in river food webs.

id #135723

Instream flow standards support amphidromous macrofauna in tropical island freshwater communities

Ayron M Strauch¹, Cody L Chacon¹, Jody Kimmel², Dean D Uyeno¹

1. Commission on Water Resource Management, Honolulu, HI, United States

2. Hawaii Coral Reef Initiative, University of Hawaii, Honolulu, Hawaii, United States

Tropical islands support unique communities of endemic amphidromous species, with life cycles necessitating connectivity between freshwater habitats and the ocean. Simultaneously, surface water diverted at low-head dams is a critical source of municipal and agricultural water supply in resource limited islands. Water withdrawals at low-head dams can dewater downstream habitat and limit connectivity to upstream habitat, with negative consequences for the recruitment and survival of amphidromous species. After approximately seven years of full flow restoration (2016-2023), water withdrawals restarted in some East Maui streams, providing a unique opportunity to study the consequences of partial flow restoration. We conducted ecological surveys at two elevations: in 2025 at lower elevation reaches to test the hypothesis that there is no statistical difference in habitat availability, the density and abundance of species, or community composition of aquatic macrofauna (e.g., fish, crustaceans, mollusks) among streams with partial flow restoration compared to streams with full flow restoration; and in 2024 at upper elevation reaches to test the hypothesis that recruitment of an amphidromous mountain shrimp, *Atyoida bisulcata*, is not affected by low-head dams with partial flow restoration compared to full flow restoration.

Despite differences in streamflow characteristics, in lower elevation reaches we found no significant differences in habitat availability, the density or abundance of species, or community composition among streams differing in flow restoration. We observed greater densities of the transient freshwater *Kuhlia* species in streams with open estuaries compared to estuaries closed by wave action, independent of streamflow. At upper elevation reaches, populations of *A. bisulcata* were not significantly different in streams with partial flow restoration compared to full flow restoration. We conclude that partial flow restoration, in combination with frequent, naturally occurring runoff events that overtop low-head dams, drive recruitment and reproduction, substrate reorganization, and habitat heterogeneity, all supporting endemic freshwater species.

id #135979

No snow? Do years with lower snowpack have warmer summer stream temperatures and lower streamflow?

Sherri L Johnson^{1,2}, Garrett Pignotti³, Steve Wondzell¹, John Mallard⁴, Becky Fasth², Gordon Grant^{1,2}

1. US Forest Service, Pacific Northwest Research Station, Corvallis, OR, United States

2. Oregon State University, Corvallis, OR, USA

3. Interstate Commission on the Potomac River Basin, Rockville, MD, United States

4. Tetra Tech, Raleigh, NC, United States

Streams that receive snowmelt contributions have been hypothesized to remain cooler and have more streamflow during critical periods in the summer. Headwater streams in the Pacific Northwest are anticipated to experience major declines of snowpack in the future. To evaluate the potential effects of reduced future snowpacks on summer stream temperatures and streamflow, we explored 20 years of hydro-climatic data from gaged watersheds and weather stations in the H.J. Andrews Experimental Forest, Oregon, USA. The distribution of large and small sites across a range of elevations (400–950 m) allowed us to further explore spring and summer temperature and hydrologic responses to snowpack. We found that years with low snowpack corresponded to higher stream temperatures primarily in spring and early summer, but not in late summer. Other hydrologic and climatic variables, including timing of precipitation, were stronger predictors for late summer stream temperatures and streamflow. We also noted that the strength and persistence of snowpack influences on stream temperature were greater in the larger watersheds. Our findings suggest that for Mediterranean climates in the Pacific Northwest, snowpack variability has a limited relationship to summer stream temperatures and late summer streamflow. Instead, late-summer stream conditions appeared to be most influenced by multiple factors, including precipitation after snowmelt has ceased. We hypothesize that the timing of new water into these watersheds may be a critical predictor for late summer stream dynamics and when streams reach baseflow, rather than the form of water (rain or snowmelt) or its temperature.

id #134956

Ecological Structure and Function of River Confluences

Amy R Chance¹, Christopher M Swan¹

1. GES, UMBC, Baltimore, MD

River confluences are hydromorphologically distinct zones within fluvial networks. These junctions interrupt longitudinal continuity and create localized areas of physical and chemical heterogeneity that can influence ecosystem structure and function. Despite their ubiquity and geomorphic significance, the ecology of confluences remains poorly understood compared to other freshwater

habitats. This study examines patterns of macroinvertebrate biodiversity and organic matter decomposition within and around confluences. Field surveys were conducted at 13 tributary–mainstem junctions in Spring 2025, spanning a range of environmental conditions. Leaf litter decomposition was assessed utilizing a standard leaf litter pack design, and quantitative macroinvertebrate sampling was performed to evaluate functional and structural responses in the confluence mixing zone compared to the upstream reaches of both the mainstem and tributary channels. Due to the increased heterogeneity inherent at confluences, it was predicted that both macroinvertebrate diversity and leaf litter decomposition rates would be higher at confluences than in the mainstem river. Preliminary results indicate increased macroinvertebrate diversity at confluences compared to surrounding reaches, whereas decomposition rates varied across locations. These findings suggest that confluences may represent ecologically important nodes of riverine systems but still remain understudied.

id #135724

Identifying ecologically relevant host fishes across the Atlantic Slope

Stefanie J Farrington^{1,2}, Allison H Roy³, David Perkins², Graziella V DiRenzo³, John G Gibbons⁴, Christina A Murphy⁵, Timothy Warren²

1. *Organismic and Evolutionary Biology Program, Massachusetts Cooperative Fish and Wildlife Research Unit, University of Massachusetts, University of Massachusetts Amherst, Amherst, MA, USA*

2. *Richard Cronin Aquatic Resource Center, U.S. Fish & Wildlife Service, Sunderland, MA, USA*

3. *Department of Environmental Conservation, University of Massachusetts, U.S. Geological Survey, Massachusetts Cooperative Fish and Wildlife Research Unit, Amherst, MA, USA*

4. *Department of Food Science, University of Massachusetts Amherst, Amherst, MA, USA*

5. *Department of Wildlife, Fisheries, and Conservation Biology, University of Maine, U.S. Geological Survey, Maine Cooperative Fish and Wildlife Research Unit, Orono, Maine, USA*

Freshwater mussels (Unionids) rely on access to suitable host fishes to reproduce, making understanding of this obligate-affiliate relationship vital to developing effective conservation and restoration plans for mussels. Nearly all host fish assessments have occurred in a single location with a single population of mussels and fishes, although it is unlikely that host fishes are equally suitable and available across a mussel's distribution. Furthermore, knowledge of host fishes for freshwater mussel species is largely based on laboratory studies, despite an inherent disconnect between physiological host observations and in situ host interactions. The goal of this study was to assess host fish use by Unionids in six watersheds across the Atlantic Slope. Working in collaboration with state, federal, and academic organizations conducting fish collections in our target areas, we collected twelve species of fishes in the Percidae, Centrarchidae, and Moronidae families, and we examined their gills for attached glochidia. Of the 518 individuals collected, 45 fishes had at least one glochidium present (232 total glochidia). We ran a Principal Component Analysis (PCA) to compare our samples to a reference database for morphometric measurements of glochidia (shell height, length, hinge length, and overall shape), compiled using existing literature. We mapped where in multivariate space our unknown glochidia samples fell within the 95% confidence ellipses for each species in the PCA: 42 glochidia samples fell within an ellipse and could be identified to species or genus. We identified five novel mussel-host relationships with no previous evidence of transformation or infection in the literature, highlighting the utility of natural infection as a means of host fish identification. Future work for this project will expand the reference data used to identify candidate species and will incorporate DNA sequencing of the glochidia samples to validate the morphometric assessments.

id #135980

Seasonal Variability in Phytoplankton Growth and Microzooplankton Grazing Rates in the Lower Columbia River

Jacie Fabela¹, Gretchen Rollwagen-Bollens¹, Stephen M Bollens¹

1. *School of the Environment, Washington State University Vancouver, Vancouver, WA, United States*

Phytoplankton growth and microzooplankton grazing rates are well studied in marine systems and lakes; however, much less is known about these dynamics within fully freshwater reaches of large rivers. The Columbia River (CR) discharges the largest amount of freshwater into the northeastern Pacific Ocean and is located in one of the most extensively managed river basins in the world (Columbia River Basin). Our study is the first experimental assessment of phytoplankton growth rates and microzooplankton grazing rates in the lower CR and provides novel insights into the dynamics of the CR planktonic food web. We undertook a two-water-year (2023-2025) study to measure phytoplankton standing stock in the field and simultaneously measure intrinsic phytoplankton growth rates and microzooplankton grazing rates within the lower CR in Vancouver, Washington, USA by employing monthly "dilution experiments" (Landry and Hassett 1982). Surface water chlorophyll measurements have been taken from this site monthly since 2005. We found the seasonal pattern of chlorophyll concentration to be similar between both time periods (2005-2023 and 2023-2025), with peaks in spring and lower concentrations in the winter months. However, the average annual chlorophyll concentration was significantly lower in 2023-2025 (3.10 µg/L) compared to the long-term dataset (5.57 µg/L). We found that average monthly chlorophyll concentration is typically positively related to increases and decreases in the phytoplankton intrinsic growth rate. From October 2023 to September 2025, the phytoplankton intrinsic growth rate ranged from -0.26 to 0.83 d⁻¹ and microzooplankton grazing rates ranged from -0.690 to 0.729 d⁻¹. However, we found that in most months, microzooplankton grazing rates on phytoplankton were very low and not significantly different than zero. These low and sometimes negative grazing rates could have impacts on higher trophic levels, as energy from primary production may not be transferred to larger grazers. Our findings help to advance our understanding of the planktonic food web and possible higher trophic level impacts on larger grazers by limited grazing by microzooplankton in the lower CR, as well as show the advantage of multi-year sampling programs to illustrate seasonal patterns of phytoplankton growth and microzooplankton grazing rates within a large river system.

id #134957

Beaver augmentation of BDA structures increases wetland characteristics

Susan Washko^{1,2}, Laura Croll¹, Aydah Daniel^{2,3}, Rosalie Hinke^{2,4}

1. *Western Colorado University, CO, United States*

2. *Rocky Mountain Biological Laboratory, Gothic, CO*

3. *Sewanee: University of the South, Sewanee, TN*

4. *University of Richmond, Richmond, VA*

Installing beaver dam analogues (BDAs) in degraded streams is an increasingly common restoration technique in western North America. In some locations, such as Trail Creek near Gunnison, Colorado, the goal of these installations is to create wetland areas within the stream corridor. Using a reach-scale approach, we sampled aquatic invertebrate community composition, water velocity, and substrate grain size in three 500m reaches: a beaver pond reach (with breached dams), a BDA-treated reach, and an untreated reach. Our objective was to assess whether the BDA-treated reach exhibited physical and biological characteristics consistent with wetland habitats. We found that the BDA reach supported a more homogenous aquatic invertebrate community composition compared to the other reaches. The BDA reach had lower flow and deeper depths than the beaver reach and the untreated reach, and the untreated reach had larger median grain size than the other two reaches.

The next year, the beaver ponds had been abandoned and were no longer intact; however, the BDA structures had been colonized and augmented by beavers. Taking a habitat-scale approach, we then compared aquatic invertebrate community composition, temperature, substrate size, pond depth, and pond area between untreated stream pools, BDA ponds, and beaver-augmented BDA ponds. Community composition differed among habitat types. The BDA ponds had significantly lower daily high temperatures compared to stream pools, and the beaver-augmented ponds had significantly lower daily highs than the BDA ponds. Beaver-augmentation also resulted in lengthening the dams and deepening the ponds. Beaver-augmented BDA ponds were rounder in shape than BDA ponds, which often were just deeper versions of the existing stream channel shape. Our results have implications for how BDA installation spacing should be designed, such as by potentially leaving room between some structures for more riffle-like flow to avoid loss of habitat diversity. Overall, these results support that BDAs change habitat towards wetland status, but beavers enhance wetland outcomes.

id #135213

Spatial and seasonal variation in occupancy of native and invasive fishes in an aridland spring ecosystem

Joey Perez¹, Matthew Troia¹, Robert Mollenhauer², Monica McGarrity³, Matthew Troia¹

1. *University of Texas at San Antonio, TX, United States*

2. *Coast Ecology and Life Cycle Monitoring Unit, Washington Department of Fish and Wildlife, Olympia, WA, United States*

3. *Inland Fisheries Division – Science Program, Texas Parks and Wildlife Department, Austin, TX, United States*

Springs and spring-influenced streams (SSIS) in aridland riverscapes support endemic faunas, yet these ecosystems are threatened by groundwater extraction, climate disruption, and invasive species. San Felipe Creek in South Texas, USA, is fed by the third largest spring complex in the state, harbors several endemic and threatened fishes, and has been invaded by suckermouth armored catfish, *Hypostomus species*. Our objective was to use this prototypical aridland SSIS to characterize the roles spring influence (e.g., longitudinal distance from springs, temperature stability) versus traditional mesohabitat characteristics (e.g., depth, velocity, substrate, vegetation cover) play on the occupancy of eight fishes among seasons. These focal species included six native fishes, one temperate invasive fish, and one tropical invasive fish. To quantify these alternate habitat associations, we established 36 ten-meter-long transects representing pool, run, and riffle habitats and distributed along a 5-km longitudinal continuum of spring proximity. Next, we performed snorkel counts of fishes at these transects on four consecutive days and measured habitat variables and thermal regimes in January, March, August, and November of 2024. We then used multi-season occupancy models and information theoretic model selection to quantify the influence of environmental covariates on detection and occupancy. Across the eight focal species, detection probability (p) ranged from 0.18 for *Notropis megalops* to 0.61 for *Astyanex mexicanus* and occupancy (ψ) from 0.23 for *Notropis megalops* to 0.50 for *Micropterus salmoides*. Occupancy was more strongly affected by spring proximity than mesohabitat factors for three of the eight species whereas season had a lesser effect on occupancy. Notably, occupancy of the endemic *Etheostoma grahami* decreased with distance from springs. These findings provide insight into the environmental filtering mechanisms driving fish community assembly in aridland SSIS and offer a quantitative baseline for future monitoring and conservation planning in the face of anthropogenic environmental changes.

id #135725

Grab some attention! Twelve minutes to dramatically enhance your poster or slides . . . in PowerPoint

Daniel McGarvey¹

1. *Virginia Commonwealth University, VA, United States*

From time to time, we come across a poster or slide deck with outstanding aesthetic qualities. We know when we see it because we tend to stop and admire it, perhaps ponder how it was built. Through 10+ years of teaching graphic design classes and workshops for scientists, I've developed a basic workflow to create strong visuals and help presenters stand out in a crowd. In this 12-minute presentation, I will demonstrate a rapid process to create posters and slides that really pop, even when compared

with Canva or AI generated content. Quick lessons/demonstrations on font and color selection, page layout, importing graphs, and judicious use of supporting images will be covered. And we'll do it all in PowerPoint. Come give me 12 minutes of your attention. I will give you the tools to nail your next presentation.

id #134702

Developing an assessment framework for nature-based recreation, an overlooked ecosystem service in urban settings that depends on ecological health

Raphael D Mazor¹, Kris Taniguchi-Quan¹, Lara Jansen¹, Abel Santana¹

1. *Southern California Coastal Water Research Project, Costa Mesa, CA, United States*

Nature-based recreation (NBR; e.g., bird watching, fishing, hiking) is vital for a community's physical and mental health. However, watershed managers need tools to determine how well their waterbodies support NBR, and thus they struggle to communicate the benefits of their decisions to the public. We used community-science datasets of natural history observations (e.g., iNaturalist, eBirds) to determine where NBR occurs within Southern California's coastal watersheds. We identified "hot spots" with high NBR and "cold spots" where NBR is less than expected. Regionally, NBR was not strongly related to ecological health, as the highest levels of NBR occurred in urban areas where ecological quality was poorest. However, within urban areas, NBR was higher near parks with natural areas and good public access points. The next phase of this study is to quantitatively evaluate factors that drive support for NBR and develop models to predict increases in NBR from restoration activities or water quality improvements. We will establish thresholds to identify waterbodies that adequately support NBR at levels appropriate for the surrounding land use and population density. Ultimately, this assessment framework will be integrated into decision-support systems that help managers evaluate multiple (sometimes competing) benefits of projects, such as stream restorations.

id #135470

Distribution and point-source contributions of pre-production plastic pellets in an urban river

Olivia Schaul¹, Timothy Hoellein¹

1. *Biology, Loyola University Chicago, Chicago, IL, United States*

Pre-production plastic pellets (nurdles) are a pervasive but understudied component of freshwater plastic pollution. Whereas most plastic pollution has diffuse sources, nurdles can originate from point sources associated with plastic manufacturing, transport, and storage. However, few studies have quantified spatial variation in nurdle abundance or evaluated how distribution patterns reflect proximity to potential sources. We investigated the distribution of nurdles relative to a suspected point source along a 5.3 km reach of the Middle Fork of the North Branch of the Chicago River (Illinois, USA), an urban stream with a suspected industrial input. Sampling sites were established upstream, at, and downstream of a drainage pipe connected to a plastic manufacturing facility ($N = 9$). At each site, we quantified nurdle density within riparian and in-stream habitats and measured channel characteristics and discharge. Nurdles in transport were sampled using surface nets deployed over short (10 min) and extended (24 h) intervals. Laboratory analyses included quantification of nurdle density (count and mass per area) and polymer identification. Results showed that nurdles were absent at upstream sites and present at all downstream sites, consistent with a localized point-source input. Preliminary spatial patterns suggest that nurdle density did not peak immediately downstream of the source, but instead farther downstream, potentially reflecting variation in transport and retention influenced by habitat structure and pulsed nurdle release. Polymer composition was dominated by polyethylene. Ongoing analyses will further evaluate spatial patterns in nurdle density, mass, and composition across habitats and sampling periods. Results from this study contribute to understanding point-source plastic pollution in urban rivers and inform targeted mitigation strategies.

id #135726

Warming alters biofilm nutrient cycling recovery after drought

Jacqueline E Todd¹, Ariel J Shogren¹, Nina K Keck¹, Annabel R Schreiber¹, Jonathan P Benstead¹, Mia Berwick², Wyatt F Cross³, Alexander D Huryn¹, Kieran Khamis², Mark E Ledger², Steve A Thomas¹

1. *The University of Alabama, Tuscaloosa, AL, United States*

2. *The University of Birmingham, Birmingham, United Kingdom*

3. *Montana State University, Bozeman, MT, United States*

In a changing climate, freshwater networks are experiencing more frequent and intense droughts that co-occur with rising temperatures. Intensifying drought events are thought to destabilize stream ecosystems and disrupt nutrient cycling, but emerging hypotheses suggest that warming may enhance ecosystem resilience to drought by triggering thermal compensatory responses in stream biofilms. However, to what degree warming influences biofilm-mediated nutrient cycling in response to drought remains poorly understood. We conducted an experiment in 8 ambient ($\sim 20^\circ\text{C}$) and 8 warmed ($+5^\circ\text{C}$) outdoor stream flumes and applied a one-month experimental drought in all flumes after one year of biofilm establishment. Before and shortly after the drought, we conducted short-term nutrient additions of nitrate (NO_3^- -N), ammonium (NH_4^+ -N), and soluble reactive phosphorus (SRP) to quantify uptake velocity (v_i , mm min^{-1}) and volumetric nutrient uptake (U_{vol} , $\mu\text{g N or SRP cm}^{-3} \text{ hr}^{-1}$), as well as incubations to quantify net N_2 flux rates (F_{net} , $\mu\text{mol N m}^{-2} \text{ hr}^{-1}$) of the stream biofilms in chambers within the stream. We characterized biofilms using ash-free dry mass, chlorophyll-*a*, and community composition with FlowCam microscopy. To quantify recovery, we calculated recovery ratios of post- to pre-drought measurements, with a ratio greater than 1 indicating recovery. Preliminary results suggested variable responses between N and P v_i and U_{vol} in warmed versus ambient flumes. Recovery ratios for NO_3^- -N uptake metrics indicated that flumes at both temperature treatments did not exhibit full recovery after drought, while recovery ratios for NH_4^+ -N uptake metrics indicated full recovery. Recovery ratios for SRP uptake metrics indicated full recovery in the ambient mesocosms but not in the warmed mesocosms. Early F_{net} results revealed greater N_2 uptake in the ambient mesocosms pre- and post-drought. Our study's findings inform predictions about the response of nutrient cycling to the interactive effects of

rising temperatures and drought in freshwater ecosystems and suggest that warming does not necessarily confer resilience of stream biofilm nutrient cycling to flow cessation and drying.

id #135982

Arkansas's calm before the storm: Sub-seasonal weather whiplash intensifies following a multi-decadal decline

Shannon L Speir¹, Admin Husic², Kathleen J Cutting¹

1. *University of Arkansas, Fayetteville, AR, United States*

2. *Virginia Tech, Blackburg, Virginia*

Hydroclimatic extremes, such as floods and droughts, challenge water resources management, and the severity, duration, and frequency of these events are expected to increase with future climate change. These "weather whiplash" events have historically been characterized on an annual or decadal basis (i.e., shifts from drought to wet years); however, there have been recent calls to understand sub-annual weather whiplash events and their impacts on both hydrology and water quality. Here, we analyzed data from 86 USGS gages across Arkansas (1950-2024) to detect sub-seasonal whiplash events, defined as transitions from drought to flood conditions within 30 days. We classified events as moderate (e.g., shifts from 10th percentile of flow to 90th percentile) and severe (e.g., shifts from 5th percentile of flow to 95th percentile) and also characterized their magnitude and transition speed. We also adjusted for the number of active gages each year. Over a 70-year period, the severity of whiplash events declined steadily from 20% of events in the 1950s to a historic minimum of 6% in the 2010s. However, Arkansas experienced a statewide, unprecedented increase in severity in the 2020's. The state recorded 26 weather whiplash events in 2024, representing a 500-year return interval ($\sigma=2.9$). Moreover, 31% of 2024 events were categorized as severe, a 5x increase compared to the 2010s. The shift in the 2020s coincided with the warmest 5-year period on record in Arkansas, and 2024 was the second hottest year in the 74-year record. The contrast between long-term decreasing severity and the 2024 surge suggests that managing water resources for gradual change may not be sufficient under future climate conditions. Looking to the future, effective water resources management requires infrastructure and planning frameworks capable of handling extreme sub-seasonal variability that may significantly exceed historical precedent, particularly as warming patterns continue.

id #135471

Teaching taxonomy to undergraduate students

Fredric R Govedich¹, Bonnie A Bain¹, Samuel A Wells¹

1. *Southern Utah University, UT, United States*

Taxonomy is often taught as a side note in general biology classes with students having limited taxonomic exposure in upper division courses such as entomology, mammalogy, ornithology, herpetology, and invertebrate zoology. Many biology students often do not understand the opportunities related to taxonomy and conservation biology and may only be focused on the fields of healthcare or biotechnology. Students who are given opportunities to explore the taxonomy of groups which they are interested in gain a stronger understanding of the importance of evolutionary biology, conservation, and ecology. We have developed an upper division taxonomic workshop for undergraduates to learn the history and application of taxonomy to all living organisms. Skills and taxonomic concepts include: differences in species definitions, descriptions and diagnoses of species, type specimens, taxonomic keys, biogeography, taxonomic literature, naming (etymology) and synonymizing species, taxonomic revisions, familiarization with international Codes of taxonomic nomenclature, and basic cladistic methodology. Students are encouraged to present their findings at university, regional, national, and international venues and are also given an opportunity to engage with and visit regional museums and collections as well as visit with professional taxonomists.

id #135727

Growing climate is more important than induced defenses for predicting chemistry and consumption of riparian tree leaves

Claire W Asmussen¹, Olivia A Steinmetz¹, Richard L Lindroth², Thomas G Whitham¹, Gerard J Allan¹, Rebecca J Best¹

1. *Northern Arizona University, Flagstaff, AZ, United States*

2. *Department of Entomology, University of Wisconsin, Madison, WI, United States*

Changing climates threaten to disrupt the function of riparian areas as biodiversity hotspots and providers of important ecosystem services. In the American Southwest, riparian trees, such as the foundation species *Populus fremontii*, cycle terrestrial nutrients into support for aquatic species, linking these ecosystems. Changes in the trait characteristics of *Populus fremontii* leaf litter can therefore have consequences for aquatic functioning and foodwebs. Here, we investigated how interactive genetic and environmental effects alter the leaf traits of these trees, and subsequently how this shapes the shredding rate of an aquatic invertebrate shredder, the caddisfly larvae *Phylloicus mexicanus*. We used an experimental design of two common gardens spanning a temperature gradient on the warmer end of the *Populus fremontii* range, each containing six clonal genotypes originating from across the species' climatic range. On some clonal replicates of each genotype, we applied a simulated herbivory treatment to simulate the increased insect activity predicted to accompany warming climates. We found that genetics and growing environment shaped both leaf traits and shredding rates more than simulated herbivory. Though simulated herbivory did have some small impacts on defensive leaf chemistry, this did not translate into effects on shredding. Interestingly, the highest shredding rates were observed when the growing climate was closer to the climate at the genotype's origin. Variation in shredding rates was best explained by a combination of genetics, condensed tannins, carbon to nitrogen ratio, and specific leaf area, with most of these traits themselves influenced by garden/environment. Our findings confirm that genetic and environmental variation in this foundation tree species directly influence changing nutrient flows to aquatic systems by altering trait-based species interactions.

Mind the gap: Where Clean Water Act rulemaking fails to meet the science and legislative intent in defining waters of the United States and the biodiversity left unguarded.

Susan Colvin¹, Robert Keast², Leanne Roulson³, Nayeli Sanchez², Patrick Shirey⁴

1. *Minnesota State University - Mankato, MN, United States*

2. *Clemson University, Clemson, SC, United States*

3. *Hydrosolutions, Bozeman, Montana, United States*

4. *Department of Geology and Environmental Science, University of Pittsburgh, Pittsburgh, Pennsylvania, United States*

The intent of the U.S. Clean Water Act (CWA) is to restore and maintain the chemical, physical, and biological integrity of the nation's waters and achieve fishable and swimmable lakes, rivers, and streams. To accomplish this, the CWA aimed to eliminate pollutant discharges and restrict the dredging and filling of waters, including wetlands, to protect water quality for fish, wildlife, recreation, and human consumption. Yet, Congress did not specify which of the millions of miles of streams, rivers, and lakes would be the Waters of the United States (WOTUS), or those waters that required federal protection to achieve these goals. This definition was left to the executive branch and thereby the changing administrations to compose. In the past decade, WOTUS definitions have changed every several years, over time largely eroding to protections less than those of the mid-1980's, ultimately creating a "Gap" between protections and CWA goals. Wetlands contain approximately 1/4 of all vertebrate animals yet are being lost three times faster than forests. Half of the world's Groundwater Dependent Ecosystems (GDE's) exist in areas that show declining groundwater levels. In the U.S. there are 36 mostly freshwater fish families, 9 exclusively found in headwaters, most highly endemic, and 9 families with >50% headwater associates. We review the biodiversity of unprotected waters, the intent of the U.S. CWA and international protections, the scientific consensus on needed protections, and the path towards achieving clean and diverse waters.

Naturalness shapes dry-phase refuges and invertebrate community recovery in headwater streams

Kieran Gething¹, Judy England², Emma Fox², Ro Sarremejane¹, Rachel Stubbing¹

1. *Nottingham Trent University, Nottingham, NOTTINGHAMSHIRE, United Kingdom*

2. *Environment Agency, England*

When non-perennial stream reaches, dry-phase refuges retain some water, or at least remain damp, enabling aquatic organisms including macroinvertebrates to survive. However, human pressures including physical habitat modification alter refuge accessibility, diversity, extent and quality. Whereas homogeneous channels may have few accessible, high-quality refuges, damming can hold up water, potentially creating artificial refuges. It is thus unclear how anthropogenic habitat modification alters refuge quality and thus community recovery after water returns. We addressed this knowledge gap in the non-perennial, seasonally intermittent headwater reaches of three chalk streams in south England, one with higher, intermediate and lower levels of anthropogenic modification. We predicted a negative relationship between modification and macroinvertebrate community recovery. We conducted physical habitat surveys, measured water quality variables and sampled communities four times at two-month intervals starting in the month of lowest flows, at four sites on each river, from downstream perennial reaches to upstream intermittent headwater reaches. We calculated metrics to summarize community characteristics then quantified differences between the 'most recovered' communities (i.e. those at sites which had dried then had most time to recover) and other communities. Supporting our prediction, recovery was greater on the least-impacted river and lower on the most-impacted river, and dissolved oxygen concentrations and sediment composition influenced recovery. In addition, the least-impacted river was the only one on which we recorded a nationally rare non-perennial-specialist mayfly. However, the most highly modified site retained ponded surface water upstream of a weir, potentially promoting upstream survival but also preventing colonization of upstream from downstream reaches. As river drying increases even in cool, wet countries such as England, maintaining and restoring natural habitats that include a range of dry-phase refuges is key to supporting the resilience of aquatic communities.

From Land to Lake: A Continuous Flow-Path Framework for Land-Use Impacts on Water Quality

Mikaela Sako¹, Erich Seamon¹, Thad Scott²

1. *Environmental Science, Baylor University, Waco, TX, United States*

2. *Biology, Baylor University, Waco, TX, United States*

Land-use composition within a watershed is a critical driver of variation in water chemistry across waterbodies at a landscape scale. However, the magnitude of land-use effects may vary across climate gradients and along hydrologic flow paths. Here, we quantify land-use influences on lake water quality across Texas lakes using a continuous, flow-path based distance metric within hydrologically connected subwatersheds. Land-use data from the National Land Cover Database (NLCD) are integrated with routed flow paths to weight upstream land-use contributions by hydrologic distance from each lake. Statistical models are used to relate these metrics to observed nutrient and chlorophyll-a concentrations from Texas lakes monitored by the Texas

Commission on Environmental Quality (TCEQ). Generalized additive and mixed model approaches are used to quantify nonlinear land-use-water-quality relationships and examine how these relationships vary with climate conditions, spatial scale, and hydrologic context. The aim of this research is to improve inference on the land-use composition and water quality correlations by accounting for variation in spatial scale and hydrologic context.

id #135728

Effect of dams and dam removals on freshwater mussels: a global meta-analysis

Alexa Hershberger¹, Allison Roy²

1. *Organismic and Evolutionary Biology Program, University of Massachusetts Amherst, Massachusetts Cooperative Fish and Wildlife Research Unit, Amherst*

2. *U.S. Geological Survey, Department of Environmental Conservation, University of Massachusetts Amherst, Massachusetts Cooperative Fish and Wildlife Research Unit, Amherst*

Historically, dams were built to power mills, provide energy (e.g. hydro power), and for recreation, yet, many dams are no longer needed for these purposes, leaving them abandoned and unregulated. Dam removal has become an attractive stream restoration strategy; however, some dam removals may harm surrounding biotic communities due to sediment mobilization, high flows, and rapid water level declines during the removal, especially impacting species that are relatively sessile like freshwater mussels. Studies documenting mussel responses to dam removal are often limited to one site and have conflicting results. The uniqueness of each dam (e.g., dam height, dam purpose), site conditions (e.g., channel flow, stream width, surrounding landscape), removal process (e.g., timing, phased, breached), and species-specific tolerances may account for the variability on how dams and dam removals affect freshwater mussel assemblages, but make it challenging for practitioners to predict the outcome of a planned dam removal and address essential mitigation strategies. Here, we conducted a global meta-analysis to synthesize information across studies and inform expected impacts of dams and dam removals on freshwater mussels based on site-specific conditions. We extracted information from peer-reviewed and grey literature on freshwater mussel survey methods and response, habitat, dam characteristics, and mitigation practices. We hypothesized that older dams and run-of-river dams will have higher mussel abundances downstream compared to newer, larger dams due to the longer recolonization periods and the semi-consistent flow creating habitat refugia. We expect opportunistic species to recover quicker and the immediate impact to be less stressful from a phased dam removal as compared to all-at-once dam breaching. Information gained from this study will directly inform managers and researchers on how to mitigate any impacts of dams/dam removals based on their site-specific conditions.

id #135984

Surface mining increases the contribution of chlorophyll-a to aquatic insect food webs in central Appalachia.

Kelley Sinning¹, Amaryllis Adey¹, Stephen Schoenholtz¹, Teresa Brown², Erin Hotchkiss¹, Lisa Tabor¹, Daniel McLaughlin¹, Caleigh Meehan¹, Greg Pond³, Rachel Reid¹, Megan Underwood¹, Carl Zipper¹, Sally A Entekin¹

1. *Virginia Polytechnic Institute and State University, Blacksburg, VA, United States*

2. *Department of Natural Sciences, University of Virginia at Wise, Norton, Virginia, United States*

3. *U.S. Environmental Protection Agency, Wheeling, West Virginia, United States*

Central Appalachian streams with drainage from surface mining sites have increased Cl⁻, Ca²⁺, and SO₄²⁻ ion concentrations, resulting in elevated specific conductivity (SC) gradients (18-1,185 µS/cm). Elevated SC decreases mostly mayfly genera that are scrapers but this loss does not change total macroinvertebrate secondary production despite increased chlorophyll-a concentration. Instead, salt-tolerant shredder, collector, and predator production increased with increasing periphyton chlorophyll-a. We are now investigating how the contribution of basal food resources changes with SC among scrapers and shredders. We predicted that shredders and collectors are supported by more a suite of ion subsidized primary production in high-SC streams where the remaining macroinvertebrates experience ionic and osmoregulatory stress. To test this, we quantified the trophic basis of insect production using natural abundance of nitrogen (N) and carbon (C) stable isotopes and their presumed food resources of coarse benthic organic matter (CBOM), fine benthic organic matter (FBOM), and periphyton. All basal resources had higher annual average δ¹³C and δ¹⁵N values along an increasing SC gradient with some seasonal variation, particularly CBOM that was lower in spring and autumn. Coarse Benthic Organic Matter δ¹³C and δ¹⁵N were different from FBOM and periphyton but FBOM and periphyton displayed isotopic overlap. Still, scrapers and shredders had a greater % contribution to biomass from periphyton and FBOM in streams with greater SC. We will present the contribution of resources to dominant scraper and shredder production and explore these relationships with streamwater ion concentrations to characterize the role of salts in resource transfer to macroinvertebrates.

id #134961

From reservoirs to rivers: documenting ecosystem recovery and project success after two years of the Klamath River Renewal Project restoration journey.

Caitlin Boise¹

1. *Resources Environmental Solutions, CA, United States*

In January 2024, the largest dam removal and restoration project in history in the United States began with the drawdown of three reservoirs. Ten months later, the Klamath River was free-flowing through the former hydroelectric reach, where four dams once stood, for the first time in over a century. During that time and in the two years since, RES and partners have focused on implementing and monitoring ecologically sound restoration throughout the former reservoir footprints and high priority tributaries. This presentation will look back on the restoration strategies, implementation, successes, and challenges from the

ongoing Klamath River Renewal Project restoration. The talk will also highlight the robust and multi-faceted monitoring program that is utilizing novel approaches and technologies to aid in real-time decision making and document Project outcomes.

id #135729

The dynamic colors of the Klamath River and what they tell about the source and transport of sediments.

Nicholas Tufillaro¹

1. Aqualytics, CORVALLIS, OR, United States

I examine changes in water color along the Klamath River from 2016 to the present using a combination of satellite-based remote sensing (Sentinel-2) and in situ water gauging. The Klamath River has nominally blue water in flows from the Trinity, Salmon, Shasta, and Scott Rivers that alter the water's composition and color. In addition, storm runoff also carries significant sediment loads, giving the water a brownish or reddish hue, particularly in the region affected by the 2022 McKinney fire. Lastly, significant phytoplankton populations from the reservoirs above the Keno Dam color the water green. We develop spectral fingerprints for these different source waters based on above-water hyperspectral measurements and use these ground-based measurements to aid in identifying and tracking sediment and other water constituent transport in the Klamath River from Keno Dam to its entrance into the Pacific Ocean at Requa. A spatially dense view of the river is presented with imagery from the Sentinel-2 satellite (nominally 10-meter resolution every 2-3 days), and temporally dense water-quality data (every 15 minutes) is also available from a few sites thanks to the extensive monitoring efforts of the [Karuk](#) and [Yurok](#) Tribes, as well as from the USGS. An initial attempt is described to track the journey of different source waters and sediments down the Klamath River by combining remote sensing and in situ gauging.

id #134962

Aquatic insect drift and emergence in response to metals exposure: accounting for emigration using stream mesocosms

Christopher J Kotalik¹, Pete Cadmus², William H Clements³, David M Walters¹

1. U.S. Geological Survey, Columbia Environmental Research Center, COLUMBIA, MISSOURI, United States

2. Colorado Parks and Wildlife, Fort Collins, CO

3. Department of Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, CO

Aquatic insect drift and emergence are essential "movements" for dispersal and colonization in streams and rivers; however, contaminant exposure can alter these ecologically important processes. We exposed naturally occurring stream benthic communities to two trace metal (Cu, Zn) exposure regimes: a continuous exposure and a pulse exposure implemented intermittently over the 10-day experiment. We measured aquatic insect drift and emergence daily and collected benthic invertebrates at the end of the experiments. We observed significant changes in the timing of emergence, with both stimulated and delayed emergence, which were taxon-dependent and varied between exposure regimes. Larval drift increased with metal exposure, with greater abundance under continuous exposure than under pulse exposure. Assessment of benthic community composition at the end of the experiments showed differences among treatments, including significantly reduced abundance of mayflies and caddisflies due to emigration. Altered drift and emergence have important implications for invertebrate population dynamics and for the animals (e.g., fish, birds, spiders) that depend on invertebrate prey resources. By accounting for emigration, we can more accurately estimate how aquatic invertebrates and their linked food webs are affected by contaminant exposure.

id #135986

Beaver restoration as a nature-based solution for sustaining perennial streamflow during drought in Sierra Nevada headwater streams

Chloe Faehndrich¹, Albert Ruhi Vidal¹, Stefano Casirati¹, John Stella², Jessie Moravek³, Manuela Girotto¹

1. Environmental Science, Policy, and Management, UC Berkeley, Berkeley, CA, United States

2. Sustainable Resources Management, SUNY ESF, Syracuse, New York, United States

3. Geography, Environment & Society, University of Minnesota, Minneapolis, Minnesota, United States

Beavers are ecosystem engineers that modify watersheds and river flow regimes. By building dams, they restore channel-floodplain connectivity, stabilizing flows and increasing groundwater storage. Notably, beavers may be considered a Nature-based Solution to water scarcity in the dry U.S. West, where snow droughts are reducing snowpack and advancing snowmelt in free-flowing mountain streams - often inducing hydrologic regime shifts from perennial to intermittent hydrographs. In California, the Department of Fish and Wildlife is working to restore beavers to Sierra Nevada watersheds where they could help capture and store snowmelt water to sustain flows during drought. However, despite promising ecological and hydrological implications, the consequences of this restoration on California's watersheds and freshwater resources remain largely unknown. This work seeks to address this knowledge gap by adapting the Soil and Water Assessment Tool Plus (SWAT+), a process-based hydrologic model, to simulate the effects of beaver dams on water storage and river flows. The modeling approach is developed and evaluated in two hydrologically similar headwater catchments in the Lake Tahoe Basin. Ward Creek and Blackwood Creek watersheds are adjacent, have very similar drainage areas (25 and 29 km², respectively), as well as aspect, soil, land use, and climate. Additionally, each catchment has a USGS in-situ stream gauge measuring daily discharge where the creeks drain to Lake Tahoe. No reservoirs are present in either catchment, and beaver dams are only observed in Blackwood Creek throughout the duration of the streamflow records. Despite these watersheds being physical "twins", they experience different flow regimes, with Ward Creek becoming increasingly intermittent over time and Blackwood Creek maintaining perennial flows. This study deploys a hydrologic model to assess whether this difference in flow regime can be attributed to beaver presence on the

landscape. With the potential for transferability across the Sierra Nevada, this tool can help inform restoration planning by identifying where beavers are most likely to counteract the ongoing and projected effects of climate change by preserving perennial flow in free-flowing headwater streams.

id #135475

The Urbanization Affect: How Biodiversity is Impacted Along an Urbanization Gradient

Cameron S Lockett¹, Christopher M Swan¹

1. *Geography & Environmental Systems, University of Maryland, Baltimore County, Catonsville, MARYLAND, United States*

Biodiversity is a foundational aspect of ecology. However, anthropogenic effects have impacted biodiversity across the globe. A type of anthropogenic effects that can disrupt biodiversity urbanization. As urbanization increases, less adaptable species may decrease. Additionally urbanization may cause behavioral changes in some organisms, and both results may lead to changes in community composition and species interactions. To further understand the effects of urbanization on species interactions we will look at the relationship between crayfish and branchiobdellidan worms. Crayfish and branchiobdellidan worms work well as a model system, because they have an established context-dependent relationship and are abundant in freshwater ecosystems. The objective behind this study was to see if there are any changes in the community structure of crayfish and branchiobdellidan worms depending on the amount of urbanization.

To complete the objective, crayfish and branchiobdellidan worms were sampled at different sites along Baltimore City, Baltimore County, and Howard County Maryland. Previous research has shown that as stream order increases so does the average amount of impervious surface cover. We sampled 24 sites with the sampling split up evenly between first, second, and third order streams. Crayfish and branchiobdellidan worms were sampled by using dip nets and the one person hour method. Along with crayfish and branchiobdellidan worms environmental samples were assessed as well. Results suggest an overwhelming negative effect on native crayfish abundance as well as the presence of worms due to non-native species. Additionally non-native crayfish increased in abundance as stream order increased. The implications of this work will provide researchers with further understanding on how species interactions within communities can be affected by urbanization.

id #135731

Refining fish-dragonfly mercury relationships to inform resource managers of potential risks to local anglers

James J. Willacker¹, Collin A. Eagles-Smith¹, Carson Lillard Van Delden²

1. *U.S. Geological Survey, Corvallis, OR, United States*

2. *Congaree National Park, Columbia, SC, USA*

Environmental mercury contamination is a widespread, global threat to aquatic resources and human health. Fish can accumulate high levels of mercury, especially the more toxic methylmercury, and represent the primary vector of exposure to human consumers. Many communities relying on subsistence fishing as a primary protein source are particularly at risk, especially in areas with limited monitoring activity where the necessary information to communicate risk is unavailable. Widespread fish monitoring programs are costly to implement and maintain, commonly resulting in a limited focus on areas with known contamination or the greatest recreational fishing pressure. Other biosentinels, such as dragonfly larvae, can mitigate many of these issues and may be effective proxies for fish sampling; however, the relationships between biosentinel and fish mercury concentrations are poorly resolved. We assessed mercury impairment in 20 waterbodies from Congaree National Park, SC, USA, which is known to have elevated mercury and supports significant subsistence fishing activity but currently has limited information for communicating risk to local anglers. We co-collected dragonfly larvae and fishes to better understand the relationships between these metrics. This presentation will provide refined fish-dragonfly mercury relationships for four taxonomic groups of fish frequently consumed by local anglers: Largemouth Bass (*Micropterus nigricans/ salmoides*), sunfishes (*Lepomis* spp.), catfishes (Ictaluridae), and Bowfin (*Amia calva*). Such taxa-specific relationships provide resource managers the details necessary to effectively evaluate mercury exposure risk to local anglers while minimizing the financial and logistical burden of data collection.

id #135987

Transport of 6ppd-quinone in water and sediment from small streams in the Puget Sound lowlands during fall storms

Rich Sheibley¹, Andrew Spanjer¹, Rachael Lane², Katie Marx¹, Julia David¹

1. *Washington Water Science Center, U.S. Geological Survey, Tacoma, WA, United States*

2. *Central Plains Water Science Center, U.S. Geological Survey, Lawrence, KS, United States*

Coho salmon (*Oncorhynchus kisutch*) have been experiencing pre-spawn mortality events in small urban streams in the Seattle area for years. Recent research has shown that a tire additive, 6PPD-quinone (6PPDQ), that is highly toxic to aquatic life, is responsible for these mortality events. Since the identification of 6PPDQ, there has been increased research into which other aquatic species are affected by this chemical, as well as water quality sampling to measure concentrations and assess the extent of this chemical in local streams and rivers. However, little is known about concentrations of 6PPDQ in stream sediments or how it is transported in the environment during fall storm events. For this study, we sampled 33 small streams for sediment-bound 6PPDQ during the summer of 2024 to establish baseline conditions in small streams in Puget Sound. Sites spanned a range of total impervious area (TIA%) in their basins, which included forested to highly urban stream sites. At 15 of these streams, a

passive water sampler was installed for approximately 2-3 months during the fall of 2024 to understand the movement of 6PPDQ in the water column during early-season storms. During retrieval of the passive samplers, a repeat sediment sample was collected to determine the extent of 6PPDQ transport in the sediment during this time. Preliminary data showed that baseline sediment-bound 6PPDQ was variable and ranged from non-detects to 2.0 ug/kg of sediment and concentrations were greater in the more urban sites. Similar results were shown in both water and sediment collected after several early season storms. The more urban sites had higher concentrations in both water and sediment. At the sites where there were paired summer and fall sediment samples, changes in concentrations were variable. Results from this study document some of the first sediment-bound 6PPDQ data from small streams and showed that even during summer baseflow, this toxic chemical is present in these systems. More work is needed to determine if and how sediment-bound 6PPDQ is released into the water column.

id #134196

Is the rusty crayfish (*Faxonius rusticus*) a transient invader?

Eric R. Larson¹

1. *University of Illinois, IL, United States*

The rusty crayfish (*Faxonius rusticus*) is among the best-studied freshwater invasive species in North America, as invasions of this crayfish into lakes of the Midwest United States with long-term monitoring programs allowed clear quantification of its impacts for decades by researchers including David Lodge, John Magnuson, Roy Stein, and Jake Vander Zanden. However, the past decade of rusty crayfish monitoring and research finds impacts of this invader are attenuating over time due to a combination of declines in both populations and body size. Here I review and update the latest on rusty crayfish invasions in northern Wisconsin, documenting the extent and magnitude of population and body size declines for this invasive crayfish, as well as patterns of community and ecosystem recovery to pre-invasion states within study lakes. Causes of rusty crayfish declines remain poorly known, and I outline a research agenda to use this uncommonly well-documented case study to investigate mechanisms of invasive species population decline and phenotypic change. Finally, if rusty crayfish are a transient invader in these ecosystems, then opportunities for subsequent serial or over-invasion by other crayfish species may be likely. What lessons have we learned from the rusty crayfish to guide risk analysis, surveillance, and rapid response or management of future crayfish invasions in north temperate lakes?

id #134964

Oops...we did it again! Introduced fishes deliver both nutrient and contaminant subsidies to freshwater ecosystems.

Gary Lambert¹, **Alison Zachritz**¹, **Daniele Miranda**¹, **Dominic Chaloner**¹

1. *University of Notre Dame, Notre Dame, IN, United States*

Fishes are among the most intentionally introduced organisms on planet Earth to support sportfishing, aquaculture, commercial harvest, and biological control. The proliferation of introduced fish that have annual spawning migrations, such as salmonids and clupeids, delivers millions of kilograms of labile resources to streams, rivers, wetlands, and lakes globally. However, these subsidies carry with them a milieu of toxic chemicals, such as metals and synthetic organic compounds, that can be spread 100s of kilometers from their source. The Laurentian Great Lakes provide a classic example of intentional and unintentional fish introductions that have transformed the watershed in dramatic and unanticipated ways. Some 40 species of non-indigenous fishes have been introduced into the Great Lakes, resulting in novel phenomena, such as extensive spawning migrations into naïve tributaries, and the reshaping of the native food web. The delivery and cycling of macro- and micro-nutrients from the excreta, gametes, and decomposing bodies of introduced fishes, such as Pacific salmon (*Oncorhynchus* spp.), is less important in the Great Lakes than in their native, oligotrophic habitats (e.g., Alaska). However, our research shows that adult salmon also transport organic contaminants in their bodies that are subsequently transferred to stream food webs of the Great Lakes watershed and, potentially, human consumers. We demonstrate that PFAS, or 'forever chemicals', acquired in the lakes are concentrated in the bodies and gametes of salmon and then released into spawning streams and subsequently accumulated by aquatic and riparian biota. This 'dark web' of salmon subsidies has serious implications for both wildlife and humans that consume these fish, particularly for individuals reliant on this protein source. The synergistic nexus of fish introductions, chemical pollution, and fish behavior can result in unintended consequences for ecosystems already challenged with emerging contaminants and other environmental changes.

id #135476

Cobalt as an Emerging Contaminant: Ecosystem-Scale Responses to Metal Mixtures in Freshwater Mesocosms

Travis S Schmid¹, **Christopher A Mebane**², **Laurie S Balistrieri**³, **Janet L Miller**⁴

1. *U.S. Geological Survey, Helena, MT*

2. *U.S. Geological Survey, Boise, ID*

3. *U.S. Geological Survey, Grafton, WI*

4. *National Operations Center, Bureau of Land Management, Denver, CO*

Cobalt (Co), copper (Cu), and nickel (Ni) are essential trace elements to life but can cause toxicity when elevated in freshwater. These elements are important commodities to global technological and energy economies. However, Co and Ni have been declared critical minerals for the economic and national security of the United States (U.S.) due to vulnerabilities in their supply chains. Mineral deposits that contain economically recoverable Co also contain either Ni, or Cu and Ni. To sustain economic and national security, previously unexploited domestic Co deposits are likely to be explored and mined. While Cu and Ni are commonly

observed in freshwater ecosystems and have aquatic life standards, no standard exist for Co. To address this knowledge gap, we conducted a 30-day mesocosm experiment exposing naïve benthic invertebrate communities to unary, binary, and ternary mixtures of Co, Cu, and Ni. Effect Concentration causing 50% change (EC50) to total mayfly abundance for each unary exposure series were: 11 ug/L for Cu, 18 ug/L for Co, and 38 ug/L for Ni. In biofilms, metal accumulation suggests that inclusion of Cu in mixtures enhanced accumulation of Co and Ni relative to unary exposures, yet Cu accumulation was unaffected by the inclusion of Co or Ni in metal mixture exposures. In filter feeding caddisflies, metal accumulation generally decreased in metal mixture exposures relative to that observed in unary exposures. We employed the response addition model to determine if metal mixtures elicited effects more/less than expected under the assumption of additivity. In all mixtures, invertebrate community responses measured as total mayflies or Bray-Curtis Similarity, suggest that mixtures cause effects at lower concentrations than unary metal exposures caused. While at higher concentrations metal mixtures cause less toxicity than expected. This suggests mixture complexity leads to unexpected toxicity suggesting that Co can be toxic to aquatic life and that mixtures including Cu and Ni do not follow the standard assumption of additivity in all situations.

id #135732

Evolving metal-related risks and sources in an ecosystem impacted by a century of mining

Madison J Foster¹, Johanna M Blake¹, Travis S Schmidt¹, Meryl B Storb¹, Nathan A Cook²

1. *USGS Wyoming Montana Water Science Center, MT, United States*

2. *Montana Natural Resource Damages Program, Missoula, MT*

A legacy of hard rock mining has historically impaired the ecological health of the upper Clark Fork Basin (Montana, USA). Since the Superfund designation in the 1980s, large stretches of the upper Clark Fork River and its tributaries have been restored, leading to improvements in water quality and fish populations. However, in the last decade, nearly all three-year-old or younger brown trout have been absent from the most upstream reach of the Clark Fork River, from below Warm Springs to Galen. To better understand metal-related risks to aquatic life, we evaluated metal(loid) (arsenic, cadmium, copper, lead, zinc) concentrations and loads. Assessments looked across four sites in the upper Clark Fork Basin with varying degrees of restoration, exposure to active and historical mine wastes, and fish population histories. Flow-normalized trends in metal concentrations and loads were estimated using the Weighted Regressions on Time, Discharge, and Season model. Annual average concentrations of most metal(loid)s have declined over nearly 30 years, with the largest declines at restored sites and smallest declines at unrestored sites. For example, from 1997 to 2024, total copper concentrations declined by 91% at Silver Bow at Opportunity (restored) yet only 16% at Clark Fork at Deer Lodge (unrestored). Despite declines in metal(loid) concentrations, all sites still commonly exceed chronic water quality criteria that are protective of aquatic life. Concentration-discharge analysis and differences in loads among sites suggest that restoration is effective at reducing the source of copper available for transport to the river. However, upstream improvements due to restoration are not translated to downstream unrestored sites because the restored sites contributed only 12% of the total copper load at the most downstream site (Clark Fork at Deer Lodge) between 2019 and 2024. These results suggest arsenic, cadmium, copper, lead, and zinc concentrations may not be related to the decline in fish populations, but are a persistent stressor to aquatic life, especially in reaches that have yet to be restored.

id #135733

Nutrient Addition to Address Low Productivity in the Kootenai River

Brandon Diller¹

1. *Kootenai Tribe of Idaho, BONNERS FERRY, ID, United States*

Over 100 years of land use changes and the construction and operation of Libby dam has significantly altered the Kootenai River's aquatic food chain. Decreased productivity in the lower end of the food chain has led to dramatic declines in important Tribal fisheries such as Kootenai River white sturgeon, burbot and native trout species, which are scarce relative to historical accounts. The Kootenai River White sturgeon are currently listed as endangered and burbot are a species of concern. The impoundment of the Kootenai River by Libby dam has cut-off key nutrients. Very low levels of phosphorus have been found downstream of Libby Dam resulting in nutrient-poor conditions in most river zones. Decreased biological productivity in the Kootenai River ecosystem has cascaded across all trophic levels. In order to mitigate for this loss of nutrients the Kootenai Tribe of Idaho is implementing and maintaining a large-scale nutrient addition effort to stimulate production in the Kootenai River's depleted food web and reverse downward trends in native fish population abundance and ecosystem health.

id #135989

Effects of tropical forest restoration on stream invertebrate and periphyton communities.

Paula dos Reis Oliveira¹, Joao Osti², Antonio Camargo³, Solange Filoso⁴, Silvio Ferraz¹

1. *University of Sao Paulo - ESALQ, -, Brazil*

2. *UNG, Guarulhos*

3. *UNESP, Rio Claro*

4. *University of Maryland Center for Environmental Science, Maryland*

Catchment-scale forest restoration is expected to promote the recovery of tropical stream ecosystems by improving habitat conditions, channel structure, and, ultimately, biodiversity and ecosystem functioning. However, empirical evidence linking forest restoration to stream ecosystem benefits in the tropics remains limited. Here, we evaluated how stream habitat, water quality, and aquatic communities respond along a gradient of forest restoration in catchments previously dominated by pasturelands.

We sampled 30 independent headwater streams (150 m reaches) during the driest season (July 2023), representing six forest cover stages (N = 5 per stage), ranging from active pasturelands to mature forests. In each stream, we assessed physical habitat structure and water quality. Benthic invertebrates were sampled using a 0.25 mm mesh D-net, with each composite sample consisting of six habitat-proportional subsamples (1 m × 0.25 m each). Periphyton biomass, chlorophyll-a concentration, and community composition were quantified using acetate sheets.

Streams draining remnant and older restored forests exhibited lower water temperatures and nutrient concentrations, alongside higher habitat heterogeneity, compared to streams in pasture-dominated catchments. Invertebrate richness and the abundance of sensitive taxa (Ephemeroptera, Plecoptera, and Trichoptera – EPT) increased consistently with forest cover and restoration age. In contrast, periphyton chlorophyll-a, biomass, and taxonomic richness were highest in pasture streams and declined toward more conserved catchments. Functional composition of both invertebrate and periphyton communities also shifted along the restoration gradient, indicating changes in resource availability and ecosystem processes.

Overall, our results reveal a clear gradient of stream ecosystem recovery associated with forest restoration, encompassing improvements in water quality, habitat complexity, and biological communities. While structural attributes and water quality responded relatively rapidly, changes in invertebrate and periphyton communities suggest a more gradual reorganization. These findings highlight both the benefits and temporal limitations of tropical forest restoration for stream health and reinforce the importance of long-term, catchment-scale approaches for freshwater ecosystem recovery.

id #134966

Salinity affects carbon cycling in streams and the carbon use efficiency of sediment microbes

Allyson Marrs¹, Michelle Baker¹

1. Utah State University, Logan, UTAH, United States

Rivers and streams contribute greatly to global carbon dioxide (CO₂) emissions, but there are large uncertainties associated with global fluxes from these systems and the drivers of CO₂ fluxes in streams appear to be system dependent. Here, we aim to investigate differences in CO₂ dynamics between natural and artificial waterways by comparing the magnitude and drivers of CO₂ concentrations and fluxes from the Logan River and manmade canals in northern UT. CO₂ flux samples were collected using the floating chamber method, and linear mixed models were used to assess relationships between concentrations/fluxes and environmental variables. We found that, on average, the canals had higher fluxes of CO₂, but there was no difference in CO₂ concentrations. We also found that the environmental factors associated with CO₂ concentrations and fluxes differed between the artificial and natural sites, providing evidence that these waterbody types should be considered separately in GHG models and budgets.

We then conducted a secondary study to investigate the effects of salinity, a growing threat to freshwater ecosystems, on the carbon use efficiency (CUE) of sediment microbes, which are key players in GHG production in streams. CUE refers to the portion of C assimilated by microbes that is incorporated in biomass, rather than being respired as CO₂. Higher microbial CUE could enhance the C sink capacity of stream ecosystems, potentially mitigating rising atmospheric CO₂ levels. We investigated the effects of freshwater salinity on microbial CUE by incubating sediments from the Logan River at environmentally relevant salinities, ranging from ambient to 100 uS/cm above ambient. After 24 hours, we processed the sediments and estimated microbial CUE using two methods: stoichiometric modeling and ¹³C glucose tracing. Our results from the ¹³C tracing experiment surprisingly suggest that salinity has a positive effect on CUE, indicating that microbial growth may be stimulated under moderate osmotic stress. We found no significant relationship between salinity and CUE using the stoichiometric modeling technique. This study illustrates important differences in CUE estimation methods and provides evidence that salinity may increase the C storage capacity of the Logan River.

id #135734

An Evaluation of the Mechanisms Driving Dissolved Oxygen Downstream of the Klamath Dam Removal

Kristine Alford¹, Desiree Tullos¹

1. Oregon State University, Corvallis, OR, United States

As the world's largest dam removal began in the Klamath River in January 2024, fisheries managers were concerned that dissolved oxygen would fall below lethal levels during and after the drawdown of the associated reservoirs. The initiation of drawdown at the two farthest dams downstream, Copco II and Iron Gate Dams, resulted in a substantial amount of suspended sediment, rich with organic carbon from a century of cyanobacteria blooms, released downstream into the mainstem Klamath River. Associated with these releases were pulses of anoxic hypolimnetic water from the reservoirs. These events were accompanied by two sags in dissolved oxygen to levels considered fatal to the at-risk fish, such as Pacific salmon (*Oncorhynchus* spp.) and Green sturgeon (*Acipenser medirostris*), that are native to the Klamath River. The primary objectives of this study were to measure the magnitude and duration of dissolved oxygen sags during the dam removal, identify the mechanisms that contributed most to these sags, and assess whether these dynamics changed with distance downstream. Dissolved oxygen, streamflow, turbidity, and water temperature were monitored for two months prior to and four months after initiation of drawdown at three sites located downstream of the lowest dam. Sediment and water samples were collected weekly at these three sites to evaluate the relationships between particulate and dissolved organic carbon and biochemical oxygen demand in a controlled laboratory setting. The study documented four distinct dissolved oxygen sags during the drawdown process - two of which were to lethal levels. These sags were associated with rises in turbidity, particulate organic matter, and oxygen consumption rate. Data from closed incubations suggest at least two distinct processes consuming dissolved oxygen at different time scales during the sediment pulses. These data provide important insights into mechanisms driving dissolved oxygen dynamics that can inform future river restoration efforts.

id #135990

Temporal dynamics of biosolids-derived PFAS across the terrestrial-aquatic continuum of agroecosystems

Diana Oviedo Vargas¹, Melinda Daniels¹, Jimmy Murillo Gelvez², Jessica Anton²

1. Stroud Water Research Center, Avondale, PA, United States

2. Center for PFAS Solutions, New Castle, DE, United States

Per- and polyfluoroalkyl substances (PFAS) are a highly diverse class of contaminants of emerging concern produced intentionally and unintentionally over the past 80 years, with an ever-expanding inventory of thousands of unique structures. Some PFAS have been linked to adverse human health effects, and many are known to bioaccumulate and biomagnify in terrestrial and aquatic species. The application of biosolids (sludge from wastewater treatment facilities) as soil amendments on farmland is a common practice worldwide. While beneficial for soil health and sustainable waste management, this practice represents an understudied pathway for PFAS contamination of soil and water. Here we investigate PFAS fate and transport in a working farm in the northeastern US with 24 years of biosolids use (ceased permanently in 2023). Since 2023, we have monitored PFAS concentrations in surface soils (0-15 cm), stream benthic compartments, and stream water using USEPA Method 1633. In soils, long-chain (>7 carbons) compounds PFOA, PFOS, PFNA, PFDA, PFDS, PFUnA, and PFDoA have been consistently above the limit of quantification (>LOQ) in monthly samples over the last 20 months, with no evidence of significant temporal change (total PFAS mean $\pm$ sd = 126 $\pm$ 25 ng g⁻¹). Compounds consistently >LOQ in the stream water at baseflow conditions were mostly short-chain and included PFBA, PFBS, PFPeA, PFHxA, PFHxS, PFHpA, PFOA, PFOS, and PFNA (total PFAS mean $\pm$ sd = 124 $\pm$ 69 ng L⁻¹). Water samples collected at high temporal resolution during two storm events showed a dilution behavior for most compounds (stormflow<baseflow), except for PFNA and PFOS, whose concentrations increased during stormflow. In addition, precursor compounds NEtFOSAA and NMeFOSAA were only >LOQ in storm samples. Our results suggest that long-chain PFAS are persistent in the soil over months/years, and that short-chain PFAS are more readily mobilized from the surface soil to the stream water. Furthermore, storm events can be hot moments for the downstream transport of certain PFAS compounds. Results from this work improve our understanding of biosolids-derived PFAS dynamics in agroecosystems and the potential implications for human and environmental health.

id #134711

Browning or Clearing: Does recovery of acidification consistently alter dissolved organic carbon exports from small watersheds?

Jonathan P. Gilman¹, Heili E. Lowman¹, Michael J. Vlah¹, Nic Gubbins², Matthew R.V. Ross², Emily S. Bernhardt¹

1. Duke University, Durham, North Carolina, United States

2. Colorado State University, Fort Collins, Colorado, United States

Long-term changes in dissolved organic carbon (DOC) concentrations in streams are widely documented, and increasing DOC ("browning") is often attributed to watershed recovery from acid deposition. Recovery from acid rain is expected to reduce sulfate deposition, raise pH, and weaken metal complexation in soils and streams, increasing DOC solubility and mobility. It remains unclear whether DOC responses to acidification recovery are consistent across systems, or whether rising pH reliably predicts increasing DOC export at continental scales, given heterogeneity in buffering capacity, soil chemistry, and hydrologic setting. MacroSheds, a continental-scale dataset of long-term watershed observations, encompasses sites with varying degrees of historical acid deposition and buffering capacity under contemporary climate change, enabling the separation of chemical recovery signals from climate-driven variability. We analyzed multi-decadal records of DOC, pH, ionic strength, atmospheric sulfate deposition, and selected climate variables across minimally disturbed watersheds in the MacroSheds database. Among sites with sufficient data for long-term trend analyses, atmospheric sulfate deposition declined at all sites (n=166), indicating a consistent, continental-scale recovery forcing. Stream pH increased significantly at 48% of qualifying sites, demonstrating widespread chemical recovery. DOC responses were heterogeneous: ~6% of qualifying sites exhibited significant DOC increases and ~15% exhibited significant DOC decreases, indicating that significant DOC trends are confined to a subset of chemically responsive systems. Sites with significant DOC trends include cases of both increasing and decreasing DOC despite rising pH, suggesting that chemical recovery alone does not guarantee browning. Together, these results show that the potential for acidification recovery is widespread, yet DOC responses can decouple from recovery trajectories. This decoupling indicates that recovery forcing alone cannot predict browning and that watershed biogeochemical and hydrologic context likely mediates whether recovery translates into long-term changes in DOC export. As a result, projections of terrestrial-to-aquatic carbon export under continued recovery and ongoing climate change are likely to require watershed-specific controls rather than relying on pH or deposition trends alone.

id #135479

Multiple stressors and biomagnifying contaminants jeopardize restoration of mining and industrially-impacted ecosystems: the Upper Clark Fork River case.

Chris Frissell¹

1. Salish Kootenai College, Polson, MONTANA, United States

The headwaters of the Clark Fork River in western Montana have been the focus of four decades of extensive and expensive remediation and restoration effort to reverse impacts of a century of mining, mineral transportation and processing. Historic mining spread dozens of contaminants via air and water, but federal and state agencies chose to focus cleanup on directly reducing exposures of a handful of the most acutely toxic heavy metals and arsenic. After three decades of positive response to remediation, trout populations have experienced precipitous, unexplained declines. Following convergent lines of evidence, I posit that bioaccumulation and biomagnification of selenium is the most likely explanation. 1) Early surveys showed Se present at elevated levels in smelter emissions, and other leachate sources are known, but monitoring of Se was largely suspended by

the 1980s. 2) Fish declines are clearly associated with recruitment failure, and selenium toxicity is primarily manifest through embryo and larval mortality, also not directly monitored. 2) Literature review confirms that application of lime and phosphate fertilizer, treatments widely applied to immobilize heavy metals in soils, have the opposing effect of mobilizing Se. 3) Reducing heavy metals successfully facilitated the emergence of complex food webs in streams, which likely created the opportunity for bioaccumulation and biomagnification of Se, something not possible in past highly simplified food webs. 4) Recent riparian bird studies have substantiated nesting failure and elevated Se concentrations in body tissues. Because Se has not been consistently monitored in fish, soils, or water, direct evidence to confirm this hypothesis is lacking. The Se hypothesis impugns a prevailing narrative of restoration success, further contributing to the reluctance of managers to consider it. Divergent biogeochemistry and pathways of toxicity among multiple contaminants can fundamentally undermine or entirely negate restoration success in mining and industrially-contaminated ecosystems. Planning and adaptive management of restoration in mining and industry-impacted ecosystems must pay special heed to biomagnifying toxins like Se, mercury, and some persistent hydrocarbons, even if environmental concentrations appear relatively low, because restoration of food complex webs can set the stage for their toxic biomagnification.

id #135991

A framework for assessing how urban stormwater systems shape water quality in streams in King County, Washington

Malia Scott¹, Kristina Hopkins¹, Kristin Jaeger¹

1. U.S. Geological Survey, Tacoma, WASHINGTON, United States

Urban development alters hydrology and can degrade water quality, with impacts being pronounced in areas with a high density of impervious surfaces. It is not just the extent of impervious surfaces that affects stream health, but also the connectivity of impervious surfaces to streams. To address this connectivity, we created reproducible workflows using geospatial stormwater system data and high-resolution (1-meter) elevation and land cover data to delineate 1) topo-watersheds, watersheds based predominantly on topography, and 2) storm-watersheds, watersheds that were hydro-enforced. We calculated and compared common urban metrics using the boundaries from both the topo-watersheds and storm-watersheds. Urban metrics included percentage of impervious surfaces at high resolution (1-meter), percentage of impervious surfaces at low resolution (30-meter), road density, population density, and housing density. We evaluated differences in correlations between urban metrics and water quality metrics as a function of topo-watershed or storm-watershed delineations. The study area included 25 small watersheds (<75 square kilometers) in the Puget Sound lowlands in King County, Washington, which includes the Seattle metropolitan area. Water quality metrics were based on monthly grab samples for water years 2019 through 2024 for constituents of nitrogen and phosphorus, specific conductance, *E. coli*, and fecal coliform. Initial correlation results between urban metrics and water quality metrics by topo-watersheds indicated the strongest significant ($p < 0.05$) Spearman rank correlations between road density and median specific conductance ($p = 0.65$), low-resolution impervious cover and median ammonia concentration ($p = 0.53$), high-resolution impervious and median orthophosphate concentration ($p = 0.61$) and median total phosphorus ($p = 0.50$), and housing density and median *E. coli* ($p = 0.72$) and fecal coliform counts ($p = 0.69$). There were no significant ($p < 0.05$) positive correlations between urban metrics and median nitrate-nitrite concentration or median total nitrogen concentration. Future work will evaluate correlations between urban metrics and water quality metrics based on storm-watershed boundaries. The findings from this study will not only help inform more efficient implementation of best management practices, but the reproducible workflows and analysis frameworks produced from this study can also be used by other researchers and water quality managers to conduct similar studies in other regions.

id #134200

An alliance for cleaner waters: Source tracking fecal pollution in Poverty Bay, WA

Jeremy Walls¹, Cameron Chapman¹, Diane Yeh¹

1. Department of Natural Resources and Parks - Water and Land Resources Division, King County, Seattle, Washington, United States

Urban watersheds draining to Poverty Bay (King County, Washington, USA) have experienced persistent fecal pollution, resulting in closure of 825 of the area's 1,100 acres of commercial shellfish growing areas. Multiple sources, including aging side sewers, onsite sewage systems (OSS), and diffuse nonpoint inputs typical of urban basins, make these watersheds difficult to diagnose and restore. The King County Pollution Identification and Correction (PIC) program addresses this challenge through a collaborative, science-based approach that pairs targeted monitoring with coordinated corrective action across agencies and disciplines. From 2023–2025, the PIC team conducted a source-tracking study in Massey Creek, one of five urban subbasins discharging to the shellfish beds. We implemented a phased workflow designed to isolate likely sources, narrow down hotspots within the stream and stormwater conveyance network, and coordinate in parallel with outreach specialists to address both point and nonpoint contributions. The field science team conducted wet- and dry-season sampling of fecal indicator bacteria across the creek and stormwater system, then pursued hotspots using targeted DNA biomarkers and conductivity loggers to identify potential sewage signals. Field observations and sanitary surveys informed outreach priorities for pet waste management and OSS/side sewer maintenance. Across the study, we collected over 800 samples for fecal indicator bacteria analysis. This iterative approach led to investigation and correction of a failing septic system discharging to the Massey Creek mainstem. Following cleanup and connection to sewer, fecal bacteria concentrations declined by over 30%. Water quality improvements were bolstered by the collaborative nature of this project, including pairing financial assistance and public outreach work with field sampling efforts. The Massey Creek study provides a transferable framework for managing fecal pollution in complex urban watersheds and highlights the necessity and benefits of collaborative freshwater management and science.

id #134968

Overview of emerging research needs on the Spokane Valley Rathdrum Prairie Aquifer

Meg Wolf¹

1. Idaho Water Resources Research Institute, Coeur D' Alene, IDAHO, United States

Through a coordinated, statewide effort to identify priority water research needs in Idaho, the Spokane Valley–Rathdrum Prairie (SVRP) Aquifer emerged as a primary area of concern requiring additional research investment. Due to its unique geologic attributes, the SVRP Aquifer is a highly transmissive, fast-flowing system and is designated as a Sole Source Aquifer, providing drinking water for the Coeur d'Alene–Spokane region. These same characteristics that make the aquifer productive also increase its vulnerability to contamination from human activities. While existing regulatory designations help protect this critical resource, additional research is needed to better understand its hydrogeologic behavior, contaminant transport pathways, and susceptibility to future stressors. Three state water research priority projects have been identified for the 2025–2026 research team at the Idaho Water Resources Research Institute. This presentation will review ongoing research efforts on the SVRP Aquifer and highlight how these projects advance understanding of this groundwater system.

id #135480

Leaf litter processing rates shift in response to human-modified land cover in freshwater ecosystems

Anna Mothersole¹, Christopher M Swan¹

1. Geography and Environmental Systems, University of Maryland, Baltimore County, Baltimore, Maryland, USA

Land use transformation is a significant driver of habitat degradation across diverse ecosystems, leading to biodiversity loss and disrupting essential ecological processes. This study is a part of the global LandComp initiative, which investigates how changing land use impacts biological communities and processes in freshwater ecosystems worldwide through modifying flow patterns, temperature regimes, and sediment dynamics. Furthermore, LandComp aims to determine whether shifts in leaf litter breakdown vary with the quality of the litter and the exclusion of consumers.

This work specifically examines the relationship between leaf litter breakdown in freshwater streams, surrounding land use patterns, and macroinvertebrate community composition. Using the standardized leaf litter bag methodology, we measured breakdown rates in representative streams within three distinct landscape categories (urban, forested, and agricultural) in central Maryland, USA. Three leaf types with distinct foliar chemistries (*Quercus rubra*, *Acer rubrum*, and *Alnus spp.*) were used, and consumer exclusion bags were also utilized. The instream breakdown of senesced leaves—a process mediated by foliar chemistry, consumers, and environmental conditions—serves as a sensitive indicator of ecosystem condition.

Preliminary results from global LandComp collaborator locations running the same study design have revealed that both agricultural and urban land cover types significantly reduce breakdown rates compared with forested sites. This aligns with our work at our Baltimore, USA location, which shows similar trends. These findings provide valuable insight into the cascading effects of land use change on critical ecosystem processes and biodiversity maintenance in freshwater environments.

id #135736

Kern River in Crisis: Lessons learned while exploring how excessive water loss transforms the ecology of a river ecosystem

Rae E McNeish¹, Andrew Alba¹

1. California State University, Bakersfield, Bakersfield, CA, United States

Anthropogenic activities can result in the conversion of perennial rivers and streams into long-term, non-perennial systems, ultimately causing devastating effects on freshwater and riparian habitats, and wildlife. The Lower Kern River (LKR) historically functioned as a perennial-like river system; however, the river currently ecologically functions as an ephemeral river due to a series of weirs and canals that often divert 100% of flows away from the river. We present findings and lessons learned on how an unexpected, rapid dewatering event impacted the LKR ecosystem and fish community. We deployed temperature loggers ($n=1$ logger/site; $n=5$ sites) and measured stream physiochemical parameters (e.g., salinity, pH, etc.) before and during rapid water loss in the Kern River during summer and fall 2024. Underwater cameras, eDNA, and field surveys were conducted to survey the fish populations pre- and post-dewatering of the river. Preliminary results suggested that 1) water temperatures exceeded recommended thresholds for aquatic wildlife ($>30^{\circ}\text{C}$) throughout the river during summer 2024, and 2) rapid water loss resulted in multiple mass fish deaths ($>3,000$ fish) compared to the upstream reference site in early fall 2024. Important lessons learned included 1) adapting to drastic, unexpected changes when launching a new research project linked with a lawsuit, and 2) recognizing the importance of exploring the regional water politics that impact our local river to better inform research efforts. While data collection is still ongoing, these preliminary results highlight that ecosystem downgrading happens when perennial rivers are routinely dewatered, and thus, ecologically function as long-term ephemeral rivers, which can potentially have long-lasting effects on long-term ecosystem health.

id #135481

Continental-scale monitoring to identify high-nutrient, low-chlorophyll (HNLC) regions in streams and rivers

David Costello¹

1. Kent State University, Kent, OH, USA

In marine ecosystems, high-nutrient, low-chlorophyll (HNLC) regions are defined areas of the ocean where primary producer biomass is low despite high concentrations of nitrogen, phosphorus, and silica. Intensive study of these regions for the past 50

years has shown that primary production in HNLCs is limited by trace metals (iron and zinc) and to a lesser degree grazing. My lab group has been studying trace-metal limitation of primary producers in streams and found that metal limitation of primary producers is surprisingly common. Experimental nutrient and metal enrichments in 41 study streams demonstrated iron and zinc limitation in 50 and 33% of streams, respectively, which is comparable to the frequency of oceanic metal limitation. Predictive models suggest that iron and zinc limitation is more likely in streams with relatively high surface-water phosphorus concentrations ($>10 \mu\text{g/L}$). Both the experimental results and predictive models suggest that HNLC watersheds may be common and that, similar to oceanic HNLCs, low primary producer biomass is a result of metal limitation. I used datasets from the US EPA 2018-2019 Rivers and Streams assessment to identify streams in the continental US that have HNLC conditions. I then compare those data to a newly compiled dataset of metal concentrations to determine spatial overlap between HNLC and low concentrations of iron and zinc. Overall, the identification of HNLCs in freshwaters can be useful for locating areas of relatively inefficient carbon and nutrient storage in streams. Furthermore, many trace metals that can limit primary producer biomass are also of economic importance (i.e., critical minerals) and discovering the locations where metals may limit primary production can inform risk assessment as we enter an era of greater exploitation of critical minerals.

id #135993

Describing the diversity of *Cirolanides* spp. (Isopoda) in Texas, USA.

Benjamin F Schwartz¹, Ramon R Villarreal², Ben T Hutchins³, Kathryn E Perez²

1. Texas State University, San Marcos, TX, United States

2. School of Integrative Biological & Chemical Sciences, University of Texas Rio Grande Valley, Edinburg, TX, USA

3. Edwards Aquifer Research and Data Center, Texas State University, San Marcos, TX, USA

The genus *Cirolanides* was erected 130 years ago with description of *Cirolanides texensis* Benedict 1896: one of the first groundwater-obligate invertebrates from Texas. Since then, the species has been documented across a wide range of karst aquifers in central and Trans-Pecos Texas, USA, and northern Mexico, spanning over 145,000 km². Later work by Krejca suggested that additional diversity existed, but revisionary taxonomic work did not begin until 2019 with description of a second species; *Cirolanides wassenichae* Schwartz, Hutchins, Schwartz, Hess & Bonett, 2019. Here, we present preliminary results describing two additional species and redescribing the genus and type species.

A combination of morphological and molecular data (mitochondrial COI and nuclear 28S) supports the new taxa and confirms that additional undescribed diversity exists. What was once thought to be a single widespread species is clearly a species complex, with co-occurrence of at least three species at some sites. The most recent work shows that *Cirolanides texensis* s. str. has a relatively small range and primarily occurs in deeper confined aquifer settings, with other species having different habitats and ranges. Our findings bring the number of *Cirolanides* spp. confirmed from Texas to four, and the combination of molecular and morphological data provide a firm foundation for clearly delineating and describing more diversity in the future.

After 130 years of investigation, groundwater diversity in Texas remains under studied and under sampled. The description of two new *Cirolanides* spp. contributes to the knowledge base but revealed even greater diversity than was previously known. Despite the rate of taxonomic descriptions lagging the rate at which new groundwater species are discovered in Texas, more species are being described now than at any time in the past. This, in combination with new data on species occurrences, should be used to better manage and conserve groundwater fauna and their aquifer habitats in Texas.

id #135995

Testing an AI-Assisted Approach to Freshwater Macroinvertebrate Biomass Estimation

Ezekiel Peterson¹, Courtney Hendrickson¹, Tiffany Garcia¹

1. Oregon State University, Corvallis, OR, United States

Macroinvertebrate biomass estimation is essential for answering ecological questions in aquatic ecosystems. Length-mass regression is the most common method used, as it is precise and fast compared to other methods (e.g., destructively drying specimens and measuring mass). However, these regressions are data depauperate because regression production is challenged by several factors. The traditional method of hand measuring individuals is time-intensive and costly, and precise measurements of individual length are essential to accurately predict mass. Additionally, the specific coefficients of regressions vary widely across taxon, and having regression data derived from specific taxa is crucial for getting accurate measurements. As such, the use of semi-automatic image analysis techniques is a promising avenue for increasing the efficiency and accuracy of aquatic macroinvertebrate biomass data. One such technique is the ZooScan measurement tool, which uses an industrial waterproof scanner to image organisms and artificial intelligence software to separate and measure individuals semi-automatically. The ZooScan has been used extensively for biomass estimates of marine zooplankton to the effect of improved efficiency and accuracy; freshwater macroinvertebrates present an underutilized application with similar implications. One potential benefit is the availability of novel metrics such as body area, which may be a better predictor of biomass as it captures morphological traits that body length may not. This project aimed to test the ZooScan tool as a novel image analysis technique against traditional hand measurement techniques by producing length-mass regressions for three underrepresented aquatic invertebrate families. The relationship between dry mass and body metric was assessed by fitting a power function to the data, and regression coefficients were compiled. Published compilations of length-mass regressions for aquatic macroinvertebrates exist but are spatially limited. There is evidence to indicate that published regressions do not hold up as distributions of genera shift within families across spatial gradients as well, demonstrating a need for regionally targeted length-mass regression compilations. Regional compilations of regression data, while feasible with traditional hand-measurement techniques, are made significantly more achievable with the onset and availability of automatic measurement techniques improving the efficiency and accuracy of biomass estimation of invertebrates in aquatic systems.

id #134716

The effects of microplastic uptake on amphibian growth and development

Andre Felton¹, Daniela Cortez², Oscar Hernandez-Reyes², Cristina Mendez², Meredith Slimp², Brianna Zamarripa, Jeffrey Hutchinson¹

1. *Biology, Health, and the Environment, University of Texas at San Antonio, San Antonio, TX, USA*

2. *University of Texas at San Antonio, TX, United States*

Global declines of amphibian populations have primarily focused on classic chemical contaminants and compounds (i.e., pesticides, fertilizers, salts, etc.), but there is a paucity in research addressing the role of emerging pollutants such as microplastics and nanoplastics on amphibian health and contributory role to global declines. Microplastic pollution has been shown to be ubiquitous across freshwater systems yet our understanding of exposure routes and interactions within ephemeral fluvial systems that exhibit high spatiotemporal variability in MP concentrations on co-occurring amphibians that utilize ephemeral pools for reproductive and development purposes is limited. In this study, the effect of chronic microplastic exposure on amphibian growth and development throughout larval and metamorphic stages was examined using *Xenopus laevis*. Tadpoles across four treatment concentrations were exposed to secondary generated polyethylene microplastic fragments throughout pre- and pro-metamorphic development. Results regarding impacts on growth and developmental stages are currently being processed and analyses will be reported during the poster presentation. Amphibians are among the most threatened taxa globally and have served as effective models for other ecotoxicological studies, yet less than 1% of all studies investigating the effects of MP on biota have used amphibians. As global plastic production is expected to continue and increase, understanding MP exposure routes and effects of chronic exposure on amphibian health can provide insight to effective mitigative practices in conservation efforts.

id #134972

Caddisflies of Florida: advancing larval taxonomy by leveraging a species-level inventory based on faunal survey of adults

Andrew Rasmussen¹, Alexander Orfinger², Dana Denson³, Steven Harris⁴

1. *Center for Water Resources, Florida A&M University, Tallahassee, Florida, United States*

2. *Department of Life Science, Dalton State College, Dalton, Georgia, United States*

3. *Retired, Macon, Georgia, United States*

4. *Retired, Knox, Pennsylvania, United States*

A benthologist's focus on the larval life-stage of aquatic macroinvertebrates can lead to an, "I don't do adults" mentality. This is especially true in regards to primary aquatic insect orders such as Trichoptera in which genus and species descriptions are originally based on adult features (e.g., wing venation and adult genitalia) that are completely different compared to morphological features of the larvae. The outcome of larvae-only studies can be uncertain and low-resolution (family/genus level) taxonomic identifications in groups such as Trichoptera where larval taxonomy is poorly known. To address this issue, species-level knowledge of Trichoptera based on adults can be leveraged to facilitate larval-adult associations through traditional approaches such as larval/adult rearings, the metamorphotype method, and newer approaches using DNA barcoding. Additionally, we believe that comprehensive faunal surveys of Trichoptera adults by deploying collecting equipment (e.g., light traps, emergence and Malaise traps) within a prescribed area is a feasible way to develop species-level inventories that can be used to strengthen aquatic bioassessments and ecological studies. This approach is scalable, and by combining species distributional data of smaller-scale surveys, identification keys can be tailored to larger areas such as states or provinces. In Florida, we have now documented county-level distributions of 220 Trichoptera species representing 46 genera and 19 families within the state (Rasmussen et al. 2025). We are currently using these data as a foundation for developing a new identification guide to the caddisfly larvae of Florida that will include keys to families, genera, and species when possible, species distributional maps, and species notes on biology and ecology. As part of this process, the guide is being illustrated with many new line drawings and specimen photographs.

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id #134717

Community responses to environmental variation in alpine pond ecosystems

Mia Ashby¹, Kurt Anderson¹, Samuel Yevak², Mason Chow¹

1. *University of California, Riverside, Riverside, California, United States*

2. *University of Colorado, Boulder, Boulder*

Alpine freshwater communities are especially vulnerable to climate change due to their extreme environment and specialized biota and are already experiencing measurable impacts (Lamouille-Hébert et al., 2025). Alpine ponds are even more vulnerable, yet remain understudied due to their remoteness and limited accessibility (Čiamporová-Zaťovičová & Čiampor, 2017; Fehlinger et al., 2026; Freitas et al., 2024). As a result, little is known about what drives community dynamics at present and how alpine ponds are predicted to respond to climate change (Wissinger et al., 2016). This study examined the environmental drivers influencing macroinvertebrate community composition in alpine ponds. We hypothesized that ponds sharing similar substrate types and water sources would exhibit similar community composition, and that altitude and temperature would exert distinct but

comparable effects on community structure. Macroinvertebrate communities and environmental variables were sampled across 27 ponds in the Green Lakes Valley at the Niwot Ridge Long Term Ecological Research (LTER) site in Boulder County, Colorado, during three summer seasons (2021, 2022, and 2024). Macroinvertebrates were identified to the lowest taxonomic level possible, most commonly genus. Community responses to environmental drivers were assessed using non-metric multidimensional scaling (NMDS), redundancy analysis (RDA), and beta diversity metrics. Preliminary results indicate overall low to moderate macroinvertebrate diversity across ponds, with unexpectedly high diversity within the family *Chironomidae*. Community composition was generally similar within individual sampling years but varied in richness among years. Results have also shown a potential relationship between community composition of ponds with mixed and river water source and ponds dominated by groundwater and snowmelt sources. These results give some insight into alpine pond community dynamics, but warrant future studies in this sensitive, but poorly studied aquatic ecosystem.

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id #135741

Temporal dynamics of CO₂ and CH₄ concentrations and fluxes at 27 National Ecological Observatory Network (NEON) stream and river sites

Yuseung Shin¹, Emily Bernhardt², Amanda DelVecchia¹

1. Department of Geography and Environment, University of North Carolina at Chapel Hill, Chapel Hill, NC, United States

2. Department of Biology, Duke University, Durham, North Carolina, United States

In flowing waters, fluxes of greenhouse gases (GHGs), carbon dioxide (CO₂) and methane (CH₄), are products of dissolved gas concentrations and gas transfer velocity (k_{600}). Although spatial patterns in lotic GHG concentrations and fluxes have been widely studied, temporal dynamics remain relatively understudied due to limited long-term, regularly collected concentration data. In addition, CO₂ and CH₄ have rarely been analyzed simultaneously. Here, we synthesized temporal concentration- k_{600} -flux (C-k-F) patterns for CO₂ and CH₄ across 27 National Ecological Observatory Network (NEON) stream and river sites with monthly observations since 2016. We found that k_{600} was significantly negatively related to CO₂ concentrations at 15 sites and to CH₄ concentrations at 11 sites, suggesting inconsistent importance of k_{600} on the concentrations across sites. High fluxes were consistently associated with higher k_{600} for CO₂ and higher concentrations for CH₄. These results suggested that, temporally, CO₂ flux was primarily controlled by physical gas exchange whereas CH₄ flux was driven by supply across sites. This study provides the first synthesis of lotic GHG time-series data and offers a foundation for investigating site-specific and continental-scale temporal patterns and controls of lotic GHG concentrations and fluxes.

id #136509

Beyond the Baseline: The Natural Flow Regime in Complex Decision Making Processes

John Matthews¹, Deb Finn²

1. Alliance for Global Water Adaptation (AGWA), Corvallis, OR, United States

2. Missouri State University, Missouri, USA

The dominant paradigms for how we manage water resources for the past century have been derived from engineering (quantity and quality) and from economics (water use efficiency). At best, these paradigms resulted in a "balanced" sustainability framework for freshwater ecosystems. The natural flow regime (NFR) presented a challenge to these paradigms by mobilizing ecohydrological variables, particularly seasonality and variability, to link ecological and human outcomes. Updates to the NFR added elements of infrastructure design and operations, planning, and non-stationarity and climate adaptation. Concurrent with these trends has been a deepening awareness that freshwater resources are embedded across many sectors and that economic development is essential to how we align efforts around water policy, investment, and poverty alleviation.

A balanced approach to water management may actually be counterproductive today, however. Quickening shocks in political, trade, and technological systems are interacting with new climate regimes, which reinforce both the limitations of engineering and economic paradigms and the need to sustain the ecological and hydrological foundations of our economies despite climate-economic shocks and transformations. Our challenge now focuses on how to manage dynamic but data-scarce ecohydrological systems for uncertain futures, especially in emerging economies.

Traditionally, the NFR has served as a baseline and reference point; given the complexity of our current decision environment, we may be able to use the NFR to constrain and inform our ecological and economic futures. New tools are beginning to emerge that translate core NFR concepts into operational pathways, ranging from project scales to financial and national planning processes. This talk will describe several initiatives that reenvision the NFR for new domains and audiences, including infrastructure design, climate finance, measuring climate resilience, and water allocation systems spanning multiple economic sectors.

id #134462

Summer stream water specific-conductivity responses to a sodium-chloride salinization gradient and mode of delivery in experimental riparian-stream mesocosms.

Michelle A. Evans-White¹, Samuel Dias¹, Sally A. Entrekin¹, Connor Foley², Misty Goings³, Maggie Herrman², Clay Prater¹, Cale Prosen⁴, Natalie Clay²

1. Department of Biological Sciences, University of Arkansas, Fayetteville, AR
2. Department of Entomology and Plant Pathology, University of Arkansas, Fayetteville, AR, US
3. Department of Crop, Soil, and Environmental Sciences, University of Arkansas, Fayetteville, AR, USA
4. Department of Biological Sciences, College of St. Scholastica, Duluth, Minnesota, USA

Salinization of terrestrial soil and freshwater ecosystems is a global phenomenon. Despite being intimately connected, a mechanistic understanding of how salinization moves from terrestrial to aquatic ecosystems as a pulse or ramp input is not well understood and can have biological consequences. We constructed experimental paired riparian-stream mesocosms to study these connections across a low-level sodium chloride (NaCl) gradient [100, 200, 300, 400, 500 mg/L; n=5 riparian-stream mesocosms per treatment and ambient control (mean ~ 7 mg/L)] added as precipitation in the riparian mesocosm and dripped directly into the stream mesocosm starting in May 2025 in the Ozark Highlands ecoregion of Arkansas. We examined the effect of NaCl delivery method at 300-mg/L NaCl by adding two additional treatments [terrestrial precipitation only (TO) and stream drip only (SO); n= 5] to be compared with the 300 mg/L treatment receiving both. We measured conductivity and temperature and estimated specific conductivity (SC) every hour for the summer (June 21 to September 20) using Hobo U-24-001 loggers. The mean summer SC over time was estimated for each replicate mesocosm (n=40) and analyzed using a linear mixed effects model. Mean SC differed across NaCl treatment ($p < 0.001$) and increased across the NaCl gradient as expected from $210.6 \pm 16.5 \mu\text{S/cm}$ in the control to $1675.8 \pm 680.3 \mu\text{S/cm}$ in the 500 mg/L treatment. The 300-mg/L TO treatment had a lower mean SC ($380.7 \pm 177.6 \mu\text{S/cm}$) than the 300 mg/L treatment ($836.6 \pm 106.8 \mu\text{S/cm}$) receiving salty precipitation and drip. The mean SC in the 300-mg/L SO treatment ($803.4 \pm 125.6 \mu\text{S/cm}$) did not differ from the TO nor the 300-mg/L treatment receiving both, suggesting that the SO drip contributed more to streamwater SC than TO salty precipitation. While the strength of the connection between salty precipitation and mean streamwater SC was low, salty precipitation in the riparia did result in short-term spikes in stream water SC in the TO treatment and it may have more of a pulse than a ramp or press effect on stream water SC. We will examine the increase or decrease relative to baseflow SC after precipitation in each mesocosm to examine how these pulses might differ across treatments.

id #134718

Identifying tradeoffs between agro-economics and water resources to guide future management decisions under a drier climate

Breanna Rivera Waterman¹, Sam Zipper², Katherine S Nelson³, Denise Chavez⁴, Gaurav Talukdar⁵, Erin C Seybold², Ikenna Onyekwelu⁶, Vaishali Sharda⁷, Archie Howlader¹, Camden Hatley²

1. University of Kansas, Lawrence, KS, United States
2. Department of Geology, University of Kansas, Lawrence, KS, United States
3. School of Natural Resources, University of Missouri, Columbia, MO, United States
4. Kansas State University, Manhattan, KS, United States
5. Department of Civil & Environmental Engineering, Indian Institute of Technology Delhi, Delhi, India
6. Agriculture and Natural Resource Management, Schreiner University, Kerrville, TX, United States
7. Assistant Professor of Agriculture and Natural Resource Management, Kansas State University, Manhattan, KS, United States

Aridity is increasing across the United States Great Plains region, shifting the dividing line between the humid East and arid West, and intensifying pressure on already stressed water resources. Within the Great Plains, the Eastern Kansas River Basin spans this dividing line and provides an opportunity to evaluate how climatic shifts will impact outcomes for agriculture and water resources. Using data from 2006-2024, we developed and integrated process-guided statistical models of irrigation water use, crop yields, agricultural net returns, and river nitrate export. We then evaluated tradeoffs between agricultural productivity and profitability, irrigation water use, and nitrate export under eight alternative management, land use scenarios, and two future climate scenarios using the multi-objective optimization NSGA-II algorithm. Within the optimization, land use (defined using crop type and irrigated/rainfed management) served as the decision variable for identifying outcomes that balance competing objectives, such as maximizing net returns and minimizing nitrate flux and irrigation water use. We found that under historic climate and land use conditions, years that minimized agro-economic and water resource tradeoffs were drier and characterized by higher-than-average soybean and wheat shares and lower corn and sorghum shares. Preliminary scenario results suggest that both diversified expansion and rainfed management maintain net returns while reducing impacts to water quantity and quality, highlighting promising pathways for climate-resilient agricultural planning in the Eastern Kansas River Basin and similar Great Plains landscapes.

id #135742

Torrent salamanders and gene flow in an experimental watershed in the Pacific Northwest, USA.

Christopher Cousins¹, Deanna Olson², Sarah Emel³, Bryce Wade⁴, Marie Tosa⁵, Tiffany Garcia¹

1. Oregon State University, Philomath, OR, United States
2. United States Forest Service, Corvallis, OR, United States
3. Indiana University of Pennsylvania, Indiana, PA
4. University of Tennessee, Knoxville, TN, USA

5. Washington Department of Fish and Wildlife, Olympia, WA, USA

The torrent salamanders (Rhyacotritonidae) are a family of headwater-obligate species endemic to the Pacific Northwest. Cascade torrent salamanders (*Rhyacotriton cascadae*) are one of two species being considered as candidate species for listing under the Federal Endangered Species Act, due to ongoing and projected habitat loss from climate change and timber harvest. The cold, fast-flowing water of the often intermittent headwater streams that they inhabit is crucial for completion of their life cycle, making them at particular risk from decreased snowpack and rising water temperatures. To mitigate these threats, establishing riparian buffers along their streams has been proposed, but their fishless headwater streams are currently not afforded the same protections as fish bearing streams throughout most of their range. To investigate the landscape covariates that impact dispersal and gene flow for *R. cascadae*, we conducted a landscape genomics study in the H.J. Andrews Long-Term Experimental Research (LTER) site. We collected tissue samples from individuals across the 10km watershed, and used double digest reduced-representation sequencing (ddRADseq) to generate a dataset of single nucleotide polymorphisms (SNPs) from this population. Using our SNP dataset, we investigated the impact of multiple landscape features including topographic roughness, heat load index, and canopy cover on gene flow between sites. As *R. cascadae* is highly desiccation intolerant, they are thought to use streams as corridors for dispersal. We used our dataset to test this hypothesis, evaluating the relationship between distance to stream and dispersal. We found that, to *R. cascadae*, not all streams are created equal. The best performing models contained distance to stream, but only when accounting for other landscape features. Our results provide direct guidance to managers wishing to maintain corridors for gene flow, and subsequent population persistence, across the range of this threatened Pacific Northwest salamander.

id #135487

Going with the flow: morphological plasticity and performance of rusty crayfish (*Faxonius rusticus*) in different flow regimes.

William Perry¹

1. University of Minnesota - Duluth, Duluth, MINNESOTA, United States

Morphological plasticity allows organisms to adapt to varying environmental pressures and can be an important component of invasive species success. The rusty crayfish (*Faxonius rusticus*) is a prolific invader of North American lakes; however, its ability to expand through stream corridors may depend on its ability to tolerate hydraulic stresses. We utilized geometric morphometrics to examine shape variation in *F. rusticus* populations across three environments in Wisconsin and Michigan: lakes, low-velocity streams (<25 cm/s), and high-velocity streams (>40 cm/s) and then evaluate the ability of crayfish from different flow regimes to tolerate elevated water velocities in a recirculating flume.

Crayfish from high-velocity populations exhibited significantly more streamlined body forms and proportionally shorter, broader chelae compared to lentic populations. These adaptive traits likely reduce drag and facilitate effective flow deflection, allowing individuals to navigate high-velocity environments. When testing their ability to maintain position in the flume, we found that crayfish sourced from high-flow streams significantly higher velocities compared to those from low flow streams and lakes. Our results suggest that morphological plasticity and prior experience in flow may affect invasion of contrasting flow environments by optimizing both form and hydro-mechanical performance. These findings provide a framework for our current research into the plasticity and performance of *F. rusticus* and *F. propinquus* hybrids in Minnesota watersheds.

id #134976

Assessing Surface Water–Groundwater Connectivity Using Remote Sensing–Enhanced Recharge Modeling

Pooja P Preetha¹

1. Alabama A&M University, Madison, AL, AL, United States

This study investigates the potential of assimilating satellite-derived surface soil moisture to improve the representation of surface water–groundwater interactions through enhanced groundwater recharge estimates generated by the Soil and Water Assessment Tool (SWAT). The Tombigbee River Basin in the southeastern United States was used as a test case. A baseline SWAT model was first developed to simulate surface hydrologic processes and groundwater recharge, followed by an enhanced modeling framework that assimilated the Soil Moisture Active Passive (SMAP) Enhanced L3 Radiometer Global Daily 9 km soil moisture product (SPL3SMP_E). To assess the influence of spatial representation on surface water–groundwater connectivity, soil moisture assimilation was implemented using both subbasin-level and hydrologic response unit (HRU)-level discretization schemes. Model performance was evaluated against a spatially explicit 1 km monthly recharge dataset from the U.S. Geological Survey. Results show that the enhanced SWAT models substantially improved recharge estimates ($R^2 = 0.70$; $NSE = 0.63$; $p < 0.0001$), indicating a more realistic simulation of surface-to-subsurface fluxes. Variations in interannual recharge across HRU-based simulations underscore the importance of spatially explicit calibration when quantifying surface water–groundwater interactions. Overall, this framework strengthens the linkage between surface hydrologic states and groundwater processes, offering a transferable approach for integrated surface water–groundwater studies across multiple temporal scales.

id #135488

Seasonality and disturbance: three years of diatom assemblages post-Hurricane Ian

Hanna Innocent¹, Evelyn Gaiser¹

1. Florida International University, Miami

Hydrologic disturbances can alter algal communities, yet how these communities will respond to predicted increases in runoff associated with extreme precipitation events with climate change remains poorly understood. Precipitation has increased globally and is expected to continue changing in both frequency and intensity. Extreme precipitation events and associated flood pulses influence nutrient loading and water-column mixing in lakes which are key regulators of plankton succession, particularly in subtropical systems. However, it remains unclear how biological communities respond to these pulses and whether their responses are transient or persist over longer timescales. Understanding how hydrologic disturbance affects the temporal dynamics of algal communities is critical for predicting future lake conditions. We used three years of bimonthly data on benthic diatom composition from 14 sites across canal and pelagic zones to examine how a flow pulse associated with Hurricane Ian influenced benthic diatom assemblages in Lake Okeechobee, Florida. We assessed temporal changes in diatom communities in both zones in relation to flow, hypothesizing that assemblages would exhibit season patterns in species composition punctuated by flow pulse disturbances. Over the three-year period, diatom assemblages displayed clear seasonal cycles, with winter and summer communities differing consistently. Additionally, assemblage composition shifted over time after a large initial flood pulse of 40 m³/s that declined to a minimum of 4 m³/s. Flow emerged as a key driver of benthic diatom community structure in Lake Okeechobee, with observed compositional shifts likely reflecting lagged effects of Hurricane Ian that persist three years later. This research provides insight into algal community resilience and recovery following extreme precipitation and resultant flood pulse disturbances, which is essential for interpreting algal responses to other stressors, including nutrient enrichment, and for understanding implications for harmful algal bloom formation.

id #134465

Distribution and Diversity of Aquatic Macrophyte Taxa in the Carmel River Watershed

Audrey G Longan¹, John R Olson¹

1. *Applied Environmental Science, California State University, Monterey Bay, Seaside, CA, United States*

Macrophytes serve important roles in improving water quality, combating erosion, and nutrient cycling. Research on freshwater macrophytes largely focuses on those in lakes or larger water bodies with little flowing current, and there is little consensus on what factors determine aquatic macrophyte distributions. We surveyed macrophytes at nine sites in the Carmel River Watershed in Monterey County, CA, to determine what environmental predictors affect their distributions and diversity within rivers, as well as the conditions that support important species. At each site, 50-meter transects were set up on both banks, one meter out from shore, or in the middle of a stream if it was two meters wide or less. Macrophyte counts and total cover were recorded, as well as depth, velocity, substrate size, and stream flow conditions. Additional measurements of water quality and canopy cover were taken at the midpoint of each transect. Macrophytes were identified using field guides and the i-Naturalist app, a global biodiversity database. Across all sites, we identified 17 species, of which four are invasive species. With 1,230 observations across all sites, the invasive Water Speedwell (*Veronica anagallis-aquatica*) is the dominant species within the system, followed by the invasive Water Mint (*Mentha aquatica*), with 864 observations. On the other hand, five native species were determined to be rare, with five or fewer observations. To better understand how environmental conditions affect the diversity and distributions of ubiquitous (often invasive) and rare taxa, I will conduct correlation analyses and run generalized linear models. I also plan to run a random forest model to determine the most likely environment in which rare taxa may be found. Ultimately, this research will lead to a greater breadth of knowledge on aquatic plants and their distributions and can provide essential information for their conservation.

id #135489

Comparative vulnerability of riverine benthic and aquifer stoneflies in large alluvial river floodplains

Rachel L Malison¹, Art Woods², James Frakes³, Jack Stanford¹

1. *Flathead Lake Biological Station, The University of Montana, MT, United States*

2. *University of Montana, Missoula, MT, USA*

3. *The Salmonfly Project, Missoula, MT, USA*

Floodplains of gravel-bed rivers occur throughout the globe and are hotspots of biodiversity, maintained in part by their connectivity in longitudinal, lateral, and vertical dimensions. Below the surface, alluvial aquifers are vast and interconnected, supporting a unique community of both obligate and opportunistic fauna. Living in alluvial aquifers, aquifer stoneflies exhibit unique adaptations, including remarkable tolerance to hypoxia and anoxia. These adaptations allow them to exist as abundant consumers in aquifer food webs subsidized by ancient methane. However, living in alluvial aquifers, aquifer stoneflies are exposed to much more stable groundwater temperatures compared to benthic species in the river. The Climate Variability Hypothesis predicts that organisms exposed to more variable temperatures should evolve broader thermal tolerance limits. We predicted that benthic stoneflies would be less sensitive to warming temperatures because they are exposed to more variable river temperatures. We measured critical thermal maximum (CT_{MAX}) and conducted intermittent respirometry experiments on nymphs of five benthic and three aquifer stonefly species to test their sensitivity to warming temperatures. CT_{MAX} values of benthic taxa were ~ 5 °C higher than those of aquifer taxa and varied among genera. In contrast, the thermal sensitivities of metabolic rates were surprisingly indistinguishable between benthic and aquifer taxa. During respirometry experiments benthic and aquifer taxa stopped moving (an indicator of stress) at similarly high temperatures and survived at similar rates. Differences in CT_{MAX} suggest that benthic stoneflies may tolerate warming temperatures better than aquifer species, however metabolic responses were surprisingly similar for the two habitats. Regardless of having higher CT_{MAX} responses, benthic taxa will also experience greater near-term mid-summer and midday temperatures as the climate warms. Assessing which group is more vulnerable therefore likely requires understanding relative rates of warming in the two habitats, which currently is poorly known. We call for more work on patterns of temperature change in a floodplain ecosystem context.

id #135745

From Data to Decisions: Coordinating Freshwater Temperature Monitoring and Action in Alaska

Erin Larson¹, Sue Mauger², Marcus Geist¹, Jeff Walker³, Rebecca Shaftel^{1,4}, Ben Letcher⁵, Jennifer Fair⁶, Rebecca Bellmore⁷, Erik Schoen⁸, Krista Bartz⁹

1. University of Alaska Anchorage - Alaska Center for Conservation Science, AK, United States
2. Cook Inletkeeper, Homer, Alaska, United States
3. Walker Environmental Research, Brunswick, Maine
4. Bureau of Land Management, Anchorage, Alaska, USA
5. WestBrook DataViz, Montague, Massachusetts, USA
6. Eastern Ecological Science Center, USGS, Turners Falls, MA, USA
7. Southeast Alaska Watershed Coalition, Juneau, Alaska, United States
8. International Arctic Research Center, University of Alaska Fairbanks, Fairbanks, Alaska, United States
9. National Park Service, Anchorage, Alaska, United States

Temperature drives many biological, chemical, and physical processes in freshwater systems, as well as being an easy-to-measure indicator of water quality. In Alaska, freshwater temperature is of particular importance and concern because of its effects on culturally and economically important freshwater fisheries. Over the past several decades, partners across different types of organizations have worked together to collect more data to understand freshwater temperature patterns and trends. This increase in data collection led to the need for a centralized portal (AKTEMP) to view, upload, download, and manage stream and lake temperature across the state. Concurrently, a rise in interest in freshwater temperature created a need to update a statewide freshwater temperature action plan, to identify high priority actions to take to conserve salmon habitat under changing thermal conditions over the next decade. Both AKTEMP and the Freshwater Temperature Action Plan (2025-2034) were developed with a diverse group of stakeholders participating, including state and federal agencies, tribal organizations, non-profits, and academic institutions. This talk will share information about AKTEMP and the Freshwater Temperature Action Plan (2025-2034) as well as lessons learned from these collaborative efforts.

id #134466

Can Hydrologically Relevant Stormwater Management Regulations Improve In-stream Biological Conditions? A 20-year Case Study from Northern Kentucky.

Matt Wooten¹, Robert Hawley², Nora Korth², Elizabeth Fet³, Katie Macmannis²

1. Kenton County Soil and Water Conservation District, KY, United States
2. Sustainable Streams, LLC, Louisville, KY, USA
3. Sanitation District No.1 of Northern Kentucky, Ft. Wright, KY, USA

In 2006, the stormwater management utility in Northern Kentucky (NKY), Sanitation District No. 1 (SD1), initiated a comprehensive stream monitoring program that included biological, habitat, water quality, and geomorphological components with goals of better understanding baseline stream conditions and how streams might respond to modernized, science-based management decisions regarding stormwater runoff. The first decade of monitoring identified several key NKY stream characteristics, such as degraded biology, habitat loss, and accelerated erosion rates, and how they responded to development and the associated hydrologic changes. This led SD1, in October 2015, to require new developments to comply with a stormwater discharge rate aimed at limiting excess downstream impacts. Based on trends observed at an array of regional monitoring sites, this discharge rate is both geomorphically and biologically relevant – flows that mobilized the streambed not only contributed to chronic channel downcutting, degradation, and enlargement, but also created an unnatural disturbance frequency for benthic macroinvertebrates. The new " $Q_{critical}$ criteria" limited the discharge of the 2-year design storm to not exceed the regional critical discharge for streambed erosion. Since the adoption of $Q_{critical}$ criteria in 2015, several hundred new development sites have been permitted to incorporate stormwater control designs with the updated criteria. During this time SD1 has continued its monitoring across its 3-county service area, creating a dataset with nearly 20 years of macroinvertebrate data. Prior to the implementation of the $Q_{critical}$ criteria, the Kentucky Macroinvertebrate Bioassessment Index (MBI) and total impervious area in the sampling site's watershed (%Imp) had a strong negative relationship. Since the adoption of the $Q_{critical}$ criteria, the coefficient between MBI and %Imp has become sequentially less negative compared to prior years. This presentation will attempt to unpack whether the positive trend in macroinvertebrates can be reasonably attributable, at least in part, to the $Q_{critical}$ criteria.

id #134978

High severity wildfire elevated turbidity and suppressed riverine metabolism for 1.5 years following rain-on-fire triggered debris flows on the Klamath River, CA, USA

Laurel Genzoli¹, Robert O Hall², John R Oberholzer-Dent³, Grant Johnson³

1. Independent Researcher, Missoula, MT
2. Flathead Lake Biological Station, Polson, MT
3. Karuk Tribe Department of Natural Resources, Somes Bar, CA

Wildfires alter terrestrial subsidies to rivers, but the effects of wildfire subsidies on ecosystem metabolism (gross primary production, GPP and ecosystem respiration, ER) and dissolved O_2 are not well known due to a paucity of monitoring during and after wildfires. In July and August 2022, the McKinney Fire burned 240 km² in Northern California along the Klamath River and caused a series of debris flows when heavy rains fell on the active wildfire. We examined the impacts of this high-severity wildfire on GPP, ER, and dissolved O_2 by comparing the immediate (days) and mid-term (17 mo) post-fire metabolic fluxes to the 4.5 y preceding the wildfire. While the initial sediment pulse briefly turned the river anoxic, due in part to extremely high fluxes of ER

(< -60 g O₂ m⁻² d⁻¹), low dissolved O₂ concentrations lasted <1 d. Following the wildfire, turbidity remained elevated over background levels for the duration of the study, causing lower post-fire fluxes of GPP (mean GPP = 8.7 and 3.3 g O₂ m⁻² d⁻¹ before and after the fire, respectively). While net ecosystem production decreased following the fire (mean NEP = 1.7 and -0.4 g O₂ m⁻² d⁻¹ before and after the fire, respectively), ER continued to reflect GPP on most days, causing lower fluxes of ER after the initial debris flow. Despite more negative NEP, the coupled reductions in GPP and ER after the fire caused less daily variation in dissolved O₂, resulting in fewer days of minimum daily dissolved O₂ falling below the management target of 90% saturation. While wildfire associated debris flows can have short-term catastrophic effects on water quality, including anoxia leading to fish kills, moderate increases in turbidity may benefit dissolved O₂ where high fluxes of GPP cause water quality impairment. Long-term data that includes pre-disturbance baselines are needed to detect these changes to ecosystem function and water quality that could otherwise go unnoticed.

id #135746

Linking Patterns of Atmospheric Mercury Deposition with Bioaccumulation in Aquatic Ecosystems at a National Scale

Collin Eagles-Smith¹, James Willacker¹, Colleen Flanagan Pritz², Sarah Nelson³, Branden Johnson¹, Sarah Janssen⁴, David Walters⁵, Chris Kotalik⁵, Jennifer Wilkening⁶, Cailin Sinclair⁷, David Krabbenhoft⁴

1. US Geological Survey, Corvallis, OR, United States

2. National Park Service, Denver, Colorado, United States

3. Appalachian Mountain Club, Boston, MA

4. US Geological Survey, Madison, WI, United States

5. US Geological Survey, Columbia, MO, United States

6. US Fish and Wildlife Service, Fort Collins, CO, United States

7. Fisheries, Wildlife, and Conservation Science, Oregon State University, Corvallis, OR, United States

Despite decadal declines in atmospheric mercury emissions and deposition, clear linkages with similarly trending biotic concentrations in the environment remain elusive. We evaluated interannual temporal patterns in biosentinel dragonfly larvae mercury concentrations from more than 100 sites in protected lands (e.g. national parks, national wildlife refuges, and national forests) across the US, and also assessed the relationships between watershed level atmospheric deposition and dragonfly larvae mercury concentrations from more than 700 paired site-year measurements across the US. Preliminary findings suggested high variability in site-specific temporal trends of dragonfly mercury concentrations, with some sites decreasing, some increasing, and some with no directional trends. Additionally, at the site-scale there were no relationships between paired wet Hg deposition or precipitation Hg concentrations and mercury concentrations in dragonfly larvae, suggesting a decoupling between delivery of atmospheric inorganic mercury to a waterbody and the amount of mercury accumulating through the food web. A substantial proportion of this variability can be explained by site-specific biogeochemical characteristics that regulate mercury bioavailability and methylmercury production. However, when accounting for variability associated with individual sampling locations, we found that at a national scale, annual average dragonfly mercury concentrations declined by approximately 16% between 2011 and 2020, and were highly correlated with average annual Hg concentrations in precipitation, which similarly declined by 19% over the same time frame. Collectively, our preliminary findings indicate that despite considerable site-level variation in relationships between Hg deposition and bioaccumulation, when integrated at a national scale, temporal patterns in mercury bioaccumulation are consistent with patterns in the amount of mercury delivered in precipitation.

id #136004

Metal and Nitrogen Contamination in Poudre Lake and the Potential Impacts to Sediment Microbial Communities

Memphis J Hill¹, Jessica Z Buser Young¹, Daniel J Bain²

1. University of Alaska Anchorage, Anchorage, AK, United States

2. Geology and Environmental Science, University of Pittsburgh, Pittsburgh, PA, United States

Lake sediments provide records of metal and nitrogen deposition over time. In Rocky Mountain National Park (RMNP), studies have shown that upslope weather events transport nitrogen from the eastern Colorado plains to the park. Steps like the RMNP Early Warning System have been implemented on a voluntary basis to reduce nitrogen emissions from crop and livestock producers during such weather events. However, Poudre Lake in RMNP is less than 10 meters from Trail Ridge Road, and therefore likely receives substantial nitrogen and metal loadings from vehicle emissions and road run-off. This study uses sediment core analysis to reconstruct the past 300 years of metal contamination and nitrogen deposition in Poudre Lake. Copper, zinc, lead, cadmium, and total nitrogen fluxes increase dramatically after the construction of Trail Ridge Road and the accumulation of metals correlates positively with RMNP visitor count. Potential ecological impacts of sediment contamination are explored by characterizing the diversity and composition of sediment microbial communities and potential nitrogen cycling functions. Increases in nitrogen cycling genes *nirS* and *nosZ* start around 1930 CE and then drop back down in the 1970s. Sediment microbial communities play a key role in nitrogen removal, yet little is known about how combinations of metal and nitrogen contaminants influence microbial community structure and nitrogen cycling activities. Interestingly, bacteria and archaea found in relatively high abundance in Poudre Lake sediments are also commonly found in urban pond sediments and wastewater treatment facilities. Results indicate that roadside lakes in RMNP are impacted by metal and nitrogen contamination from vehicle exhaust and non-exhaust emissions, and that contamination may alter the nitrogen cycling potential of sediment microbial communities over time.

id #135749

Stream restoration in the Last Frontier: Restoring mined streams in Alaska

Jeffrey D Muehlbauer^{1,2}, Dakota E Keller³

1. *Alaska Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Fairbanks, Alaska, USA*

2. *Institute of Arctic Biology, Department of Fisheries, and Department of Biology and Wildlife, University of Alaska Fairbanks, Fairbanks, Alaska, USA*

3. *Institute of Arctic Biology and Department of Biology and Wildlife, University of Alaska Fairbanks, Fairbanks, Alaska, USA*

Stream restoration practice is burgeoning in the State of Alaska generally and in Interior Alaska and the Yukon River watershed specifically. The biological goals of these restorations, at least implicitly if not stated explicitly, are to improve conditions for Alaskan freshwater and anadromous fishes, most notably Pacific salmon. In the face of recent catastrophic declines in Yukon River Chinook salmon, restoration offers one of the few alternatives for not just slowing habitat degradation and population collapse, but in affirmatively increasing habitat availability and resilience for growing salmon populations. However, widespread restoration practice is relatively new to this region, which differs dramatically in climate, hydrology, geography, and biota from most conditions in the contiguous United States and temperate Europe where stream restoration now has a multi-decadal history of practice. Further, many restoration outcomes from those lower latitude regions have shown dispiriting results in terms of improving biotic in-stream conditions. Collectively this points to a need for assessment that explores the efficacy of stream restoration specific to Interior Alaska, and particularly for improving conditions such as habitat and food availability for salmonids. In this presentation I will synthesize ongoing research and preliminary results from my research lab in assessing restoration effectiveness within Yukon River watershed streams. I will focus on how habitat features included in each of these restorations influence macroinvertebrate community and other food web responses to restoration. I emphasize these resources because they represent the food base available for salmonids, including juvenile Chinook salmon (*Oncorhynchus tshawytscha*) and Arctic grayling (*Thymallus arcticus*). I will conclude by exploring how applied science may guide future restoration practice to improve conditions for Pacific salmon throughout the Yukon River watershed.

id #136005

Time series dynamics of remotely sensed mesic vegetation used to model shifts in macroinvertebrate communities across the Sagebrush Biome

Colin Martin¹, Charles Hawkins², Jodi Brandt¹, Trevor Caughlin¹

1. *Boise State University, Boise, IDAHO, United States*

2. *Utah State University, Logan, Utah*

Intact riverscape floodplains promote water security, increase water supply late in the growing season, mitigate flooding and wildfire, and serve as biological hotspots. These systems often have intact macroinvertebrate communities, an important pillar of riverscape function. However, many of the riverscapes in the Western U.S. are in poor ecological condition and have degraded macroinvertebrate communities. Remotely sensed time series information captures dense seasonal variability that would be nearly impossible to recreate with in-situ monitoring. We begin linking remote sensing to in-situ measurements in order to build a more complete picture of landscape dynamics, resulting in better management decisions and improved watershed health.

This study leverages remote sensing to address two critical knowledge gaps by: (1) improving the use of remote sensing indicators as proxies for field-based ecological monitoring, and (2) deepening our understanding of how macroinvertebrate assemblages are shaped by riverscape conditions. We ask: Can remotely sensed indicators of riverscape structure, when integrated with large-scale field sampling, improve our understanding of biota–environment relationships?

We analyze two major datasets: (1) **MRRMaid**, a Landsat-based tool that classifies mesic vegetation across the growing season and (2) **BLM AIM** benthic macroinvertebrate (BMI) data from 2164 sites across the Sagebrush Biome. In order to effectively use remote sensing data in our model, we develop distinct “signatures” of how mesic vegetation is expressed across the landscape at different spatiotemporal scales. Specifically, our signatures are: **seasonal minimums, maximums, and the coefficient of variation** of mesic vegetation across riparian buffers at three spatial scales. We use generalized linear mixed effects models to examine associations between these signatures and taxonomic shifts in over 1.6 million BMI's.

I hypothesize that in systems where all three signatures co-indicate a pronounced water regime [**Wet stable, Variable, Dry stable**] these signatures will be highly correlated with BMI assemblage. By integrating high-resolution remote sensing data with broad-scale field monitoring efforts, this study aims to refine our understanding of the drivers of BMI assemblage. This study offers a scalable, replicable workflow for combining remote sensing and field data, expanding the toolkit for ecological monitoring and informing effective riverscape management.

id #134470

Investigating the Impact of Behavioral Ecology on eDNA Particle Size Distribution for Two *Gambusia* Species

Sarah Baggett¹, Matthew A Barnes¹

1. *Natural Resources Management, Texas Tech University, Lubbock, TX, United States*

Environmental DNA (eDNA) analysis has become increasingly popular in freshwater ecological studies, but relating eDNA samples to source organism locations continues to pose a challenge in lotic systems due to downstream flow. We hypothesize that particle size distribution (PSD), the size range of eDNA fragments, may yield insight on this issue, since larger fragments are removed more quickly by deposition and degradation, while smaller fragments, produced by degradation of larger fragments, remain suspended in the water column for longer periods of time, and are therefore transported longer distances. However, previous studies have shown that PSD may vary by species, possibly due to behavioral factors such as proximity to benthic habitats or physiology. We compared PSDs in the water column between two species in the same genus: the benthic Pecos gambusia (*Gambusia nobilis*) and the pelagic Western mosquitofish (*Gambusia affinis*). We collected surface water samples at 7 sites along the Lower Pecos River, Texas, and segregated eDNA into small and large size classes

via sequential filtration. Next, we applied species-specific qPCR assays to quantify eDNA of our focal species. We then compared the quantity of DNA on each filter size to determine the ratios of large to small DNA fragments in the water column for each species. Preliminary findings show that PSD differs between sites and between species. Further analysis of PSD will improve our understanding of downstream eDNA transport in lotic systems and inform future research related to predicting source organism location based on the PSD of eDNA samples.

id #134982

Abundance and types of anthropogenic litter in southeast Pennsylvania headwater streams and effects of single-use plastic bans

Megan Fork¹, Victoria Moreira¹, Gianna Parrish¹

1. West Chester University Dept. of Biology, West Chester, PENNSYLVANIA, United States

Anthropogenic litter, particularly plastic, is ubiquitous in urban and suburban streams. In an effort to limit the amount of plastic waste in the environment, several nations and US states have enacted bans on distribution of single-use plastics (such as shopping bags and drinking straws). Other states, like Pennsylvania, do not have statewide bans, but do allow for municipal-scale bans on single-use plastics.

We conducted an 18-month study of nine headwater streams in southeast Pennsylvania to characterize the types and abundance of anthropogenic litter in stream beds and riparian zones. We categorized each piece of litter recovered according to the material of which it is composed and its use, in addition to recording the mass and size. In addition, we used a replicated before-after-control-impact study to ask how municipal-scale plastic bans affect the abundance and types of anthropogenic litter. Each of our sites had spatially-homogenous policy regarding single-use plastics within their watersheds, including three streams in municipalities without plastic bans, three streams that had plastic bans for at least one year before we began sampling, and three streams in which plastic bans were enacted partway through our sampling period. While we have shared preliminary results at past meetings, this presentation is the first to report on the full dataset from this study. We will describe the “community” of litter among sites, and describe whether and how the abundance and types of litter are affected by watershed characteristics and policy.

id #135750

Spatio-temporal changes in the physicochemical characteristics of Lake Atitlán Basin

Natalia Vargas López^{1,2}, Juan Estuardo Bocel^{1,3}, Monica Martínez¹, Krista Capps^{2,4}

1. Centro de Estudios de Atitlán, Universidad del Valle de Guatemala, Sololá, Guatemala

2. Odum School of Ecology, University of Georgia, Athens, GEORGIA, United States

3. Department of Natural Resources & Environmental Science, University of Reno, Reno, NEVADA, United States

4. River Basin Center, University of Georgia, Athens, GEORGIA, United States

Lake biogeochemistry is influenced by both internal processes, such as water-column stratification, and external pressures, including nutrient inputs from streams draining surrounding watersheds. In Central America, Lake Atitlán is a caldera lake supported by a long-term freshwater monitoring program that offers a valuable opportunity to examine biogeochemical dynamics in a tropical mountain watershed. In this study, we analyzed data from 2017-2024 to characterize spatiotemporal patterns in water chemistry across tributary rivers and sites in the lake, and (2) assess seasonal variability in water quality in both rivers and the lake. We present data from rivers that drain into Lake Atitlán, including San Francisco, Catarata, and San Buenaventura, as well as from two lake sites: San Buenaventura, a shallow nearshore site at 10 m depth, and Panajachel, a deep pelagic site at 100 m depth. Across rivers, concentrations of nitrate and nitrite (NO_3+NO_2), ammonium (NH_4^+), soluble reactive phosphorus (SRP), and total phosphorus (TP) showed no consistent differences between the dry and rainy seasons. NO_3+NO_2 concentrations were generally higher in the San Buenaventura and Catarata rivers than in San Francisco, whereas SRP and TP concentrations were highest in San Francisco. In contrast, conductivity and total dissolved solids were seasonal, with higher values during the dry season and consistently greater in San Francisco. Most analytes did not display trends; however, NO_3+NO_2 concentrations increased over time in San Buenaventura and San Francisco. Physicochemical patterns in the lake were heterogeneous. In Panajachel, the lake exhibited monomictic mixing, with stratification beginning in February, a fully developed thermocline by April, and complete water-column mixing each January. Dissolved oxygen varied with depth, with the highest values in the epilimnion and metalimnion and the lowest in the hypolimnion. In contrast, San Buenaventura displayed polymictic mixing, with dissolved oxygen more evenly distributed throughout the water column. Nutrients in Panajachel were typically lowest in the epilimnion and accumulated in the hypolimnion, while at San Buenaventura, nutrients were more uniformly distributed across depths. Insights into freshwater quality can inform management strategies that protect the lake's ecological integrity.

id #134983

Partitioning assimilatory and dissimilatory processes in ammonium uptake in response to light availability and biofilm colonization using experimental streams

Anna Vincent¹, Jennifer Tank², Robert Hall³

1. Northwestern University, Evanston, IL, United States

2. University of Notre Dame, South Bend, IN, United States

3. Flatheat Lake Biological Station, Polson, MT

Stream biogeochemical cycling, including both assimilatory and dissimilatory processes, can be altered via the alleviation of a light limitation in open canopy streams. However, few studies have examined ammonium (NH_4^+-N) removal via both assimilatory and dissimilatory processes, and assessed the combined effects of light availability and biofilm growth on removal in the same

experiment. Here, we quantified water column $\text{NH}_4^+\text{-N}$ and nitrate ($\text{NO}_3^-\text{-N}$) removal using $n=176$ short-term additions in four experimental streams over two years. We developed a Bayesian multilevel model to assess the influence of light availability (via experimental shading) and biofilm colonization on $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ removal rates, while also accounting for the simultaneous decline in $\text{NH}_4^+\text{-N}$ and increase in $\text{NO}_3^-\text{-N}$ via the dissimilatory process of nitrification at the reach-scale. Using this approach, we can describe the effects of biofilm and light availability on both $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ removal rates, while also providing estimates of reach-scale $\text{NO}_3^-\text{-N}$ production across a continuum of biofilm colonization and diel conditions that would inform the relative roles of these effects on inorganic N fluxes to downstream ecosystems. This method offers a new approach for quantifying nitrogen biogeochemical cycling in streams without the use of an isotopic ^{15}N tracer and informs the variability around removal rates.

id #135751

Spatial and temporal variation in occupancy of stream-associated amphibians.

Tiffany Garcia¹, Jake Verschuyt², AJ Kroll³

1. Oregon State University, Corvallis, OR, United States

2. National Council for Air and Stream Improvement, Anacortes, Washington

3. Rocinante Consulting, Corvallis, Oregon

Forest management regulations often require riparian buffer protections, with buffer length and width generally determined by fish presence. New forest practices rules in Oregon add buffers to streams without fish and increase buffer width on larger fish-bearing streams. Several species predicted to benefit from these added habitat protections are headwater stream amphibians such as the Columbia torrent salamander (*Rhyacotriton kezeri*), southern torrent salamander (*Rhyacotriton variegatus*); coastal giant salamander (*Dicamptodon tenebrosus*); Cope's giant salamander (*Dicamptodon copei*); and coastal tailed frog (*Ascaphus truei*). To inform management efforts, we estimated site-level occupancy for these species across the northern Oregon Coast Range during summer low flow to quantify species status as a function of forest management and streamflow conditions. We completed visual and environmental DNA sampling at 80 randomly selected 3rd order basins that vary in forest management intensity in 2026-2025. Sampling occurred on six 100-meter reaches (one 3rd-order, two 2nd-order, and three 1st-order) in each basin. We observed *Dicamptodon* species in 76/80 basins with detections throughout all stream orders; *Ascaphus* in 54/80 basins, also across all stream orders; and *Rhyacotriton* species most frequently in 1st order streams in 51/80 basins. We developed an eDNA metabarcoding assay to estimate occupancy of our target amphibian species and to compare across physical and eDNA sampling methodologies to maximize sampling efficiency. Preliminary results on 2024 sampling show that our metabarcoding assay detected all amphibian species observed using visual survey methods. Further, in samples without method agreement, eDNA surveys identified species that were not detected with visual surveys. This study establishes a baseline for assessing responses of headwater amphibians to forest management and streamflow conditions over time, explores species distributions within headwater stream reaches and across Oregon's northern coast range, and identifies appropriate survey methodologies as a function of environmental conditions.

id #134984

Examining Spatial and Temporal Dynamics of Nitrous Oxide in a Restored Wetland Receiving Elevated Nitrate Loads

Jonathan Norton¹, William Crumpton¹, David Green¹, Carlos Tenesaca¹, Steven Hall², Matthew Leeford²

1. EEOB, Iowa State University, Ames, IOWA, United States

2. Plant and Agroecosystem Sciences, University of Wisconsin - Madison, Madison, Wisconsin, United States

Surface water wetlands positioned to intercept nonpoint-source nutrient loads play a critical role in reducing nitrate (NO_3^-) exports from tile-drained agricultural watersheds but at the potential cost of increased nitrous oxide (N_2O) emissions. The magnitude of these emissions, and the underlying mechanisms controlling them, remain uncertain, largely due to historically infrequent monitoring that fails to consider the possibility of strong spatial and temporal variability in wetland N_2O dynamics, potentially biasing wetland-scale emission estimates. Addressing this uncertainty requires a whole-system approach that integrates close-interval monitoring of surface water N_2O mass imports and exports with detailed measurements of wetland pool N_2O flux rates. Here, we apply such an approach to characterize multi-season N_2O dynamics in a wetland restored to intercept nutrient loads from an intensively cropped, tile-drained catchment in central Iowa, USA. During spring through fall 2025, surface water N_2O imports and exports were estimated using continuous flow measurements paired with intensive dissolved N_2O sampling at the wetland inlet and outlet. Wetland N_2O emission rates were also surveyed weekly at 20 internal pool locations using a flow-through dynamic floating chamber coupled to a LiCOR 7820 trace gas analyzer, alongside measurements of dissolved N_2O , NO_3^- , water temperature, and vegetation cover. Dissolved N_2O concentrations at the inlet and outlet varied from 0.29 to 25 $\mu\text{g N}_2\text{O-N L}^{-1}$, while internal pool concentrations ranged from 0.11 to 75 $\mu\text{g N}_2\text{O-N L}^{-1}$. Wetland N_2O fluxes exhibited pronounced spatial and temporal variability, ranging from -0.010 to $1.56 \mu\text{g N}_2\text{O-N m}^{-2} \text{ s}^{-1}$, with a mean annual flux of $0.144 \mu\text{g N}_2\text{O-N m}^{-2} \text{ s}^{-1}$. These data provide the foundation for developing a coupled NO_3^- - N_2O mass balance model for this system that explicitly accounts for dissolved N_2O imports, exports, and surface emissions, allowing for the reasonable estimation of net internal N_2O production. This work promises to improve our understanding of how nutrient reduction wetlands contribute to NO_3^- reduction and N_2O emissions from agricultural landscapes.

id #134985

The influence of an invasive wetland plant on mercury dynamics

Caillin Sinclair¹, Tiffany Garcia¹, Branden Johnson², Christopher Kotalik³, Collin Eagles-Smith²

1. Oregon State University, Corvallis, OREGON, United States
 2. U.S. Geological Survey, Corvallis, OR, United States
 3. U.S. Geological Survey, Columbia, MO, United States
- Publish consent withheld

id #136009

Cryosphere change and alpine aquatic ecosystems in a high-elevation catchment of the Southern Rocky Mountains, Colorado, USA

Daniel L Preston¹, Scott Hotaling², Isabella M Errigo³, Paul B Frandsen⁴, Taylor L Price⁵, Trinity L Hamilton⁵, Kurt E Anderson⁶, Samuel E Yevak⁷, Jennifer E Morse⁷, Joshua Volmer²

1. Colorado State University, FORT COLLINS, CO, United States
2. Utah State University, Logan, Utah
3. Cornell University, Ithaca, NY
4. Brigham Young University, Provo, UT
5. University of Minnesota, St. Paul, MN
6. University of California, Riverside, CA
7. University of Colorado, Boulder, CO

Alpine aquatic ecosystems are sensitive to climate change due to the central role of snow and ice in shaping community and ecosystem characteristics. Long-term data on abiotic factors and biological communities remains relatively rare, however, making it challenging to understand shifts over time or space. Focusing on the Niwot Ridge Long-term Ecological Research Site in the Southern Colorado Rockies, we review evidence for long-term shifts in climate and cryosphere characteristics in relation to aquatic ecosystems. To understand possible future consequences of changing hydrology, we also document benthic stream communities using morphology and metabarcoding approaches across sites varying in source waters, including ephemeral and permanent streams dominated by glacier meltwater, snowmelt, lake outflows, and mixed source waters. Discharge, water temperature, conductivity, pH, total dissolved nitrogen, and total dissolved phosphorus differed between water source types. Macroinvertebrate composition was more strongly correlated with stream permanence and conductivity than with water temperature or nutrient availability. Lake outlets showed the highest beta diversity, with unique communities dominated by filter feeders. Alpha diversity was similar at the glacial, snowmelt, and lake outlet sites, but increased at the downstream mixed source sites which had proportionally more non-dipteran taxa. Taken together, these results help to understand and predict how ongoing and future changes in climate and hydrology will influence alpine aquatic communities.

id #136010

Modeling changes in temporal beta diversity across seasonal and non-seasonal temperature regimes.

Kurt E Anderson¹, Bryan L Brown², Tadeu Siqueira³, Victor S Saito⁴, Christopher M Swan⁵

1. University of California, Riverside, Riverside, CALIFORNIA, United States
2. Virginia Tech University, Blacksburg, Virginia, USA
3. Departamento de Ecologia, Universidade Estadual Paulista, Rio Claro, Brazil
4. Federal University of São Carlos, São Carlos, Brazil
5. University of Maryland, Baltimore County, Baltimore, Maryland, USA

Despite decades of climate change research, we still lack an understanding of how temperature changes will alter community dynamics. This is particularly the case when considering seasonal patterns of variation. Climate-induced changes in phenology have been well-documented, but there is still much to learn about the broader-scale community effects. One challenge is that changing temperatures can influence community dynamics through multiple pathways. Metabolic changes at the level of individuals scale to alter population dynamic rates and community interactions. This can lead to deterministic changes, but also stochastic ones, as relative abundances in small populations tend to exhibit greater ecological drift that generates spatial and temporal variation independent of environmental heterogeneity. Thus, changes in community composition may deviate from those predicted by seasonality. Here, we present a temperature-dependent, stochastic community model. Each simulated species is assigned a "body size" that determines how temperature sets key demographic processes such as reproduction, survival, and competition strengths. The model is parameterized with data from year-long spatial surveys of tropical and temperate stream macroinvertebrates, allowing us to explore how communities might respond to changes in different seasonal thermal regimes. Our analyses focus on temporal β -diversity to quantify compositional changes through time - a key indicator of community assembly processes. We have previously shown that simulated warmer communities tend to have smaller sizes compared to colder ones because warmer temperatures and accelerated metabolic rates lead to lower survival and carrying capacities. This in turn leads to greater compositional variability, reflected in higher temporal β -diversity. Here, we show that β -diversity in warmer temperature communities exhibits lower tracking of seasonal variation than in temperature ones. Preliminary analyses suggest that these differences arise from lower demographic responsiveness to temperature variation in combination with greater stochastic variation. Dispersal recruitment reduces stochastic variation by increasing community sizes across temperatures, also reducing differences in temperature tracking. Our results represent a step towards a framework for predicting changes in community composition in response to changing seasonal temperature regimes across ecosystems.

id #134987

Changes in macroinvertebrate communities in national parks of the southwestern United States across a ten-year period: patterns and challenges

Rocio J. Guzmán Ojeda¹, Michael T. Bogan¹

1. School of Natural Resources and the Environment, University of Arizona, Tucson, ARIZONA, United States

The increased variation in climate in the southwestern United States is leading to dramatic fluctuations in stream discharge, including atypical large floods and long-term droughts. These hydrological changes will have impacts on aquatic biota, including aquatic invertebrates. Examining these impacts can be challenging due to the presence of confounding factors. However, streams in national parks are ideal for study because they are protected from most confounding anthropogenic disturbances. We studied changes in macroinvertebrate community structure in five national parks using a 10-year biomonitoring dataset (2012 to 2022). Macroinvertebrate data were collected from riffles in perennial streams in the Sonoran Desert and Southern Plains ecoregions. We found a total of 193 invertebrate taxa, including a total of 73 EPT taxa. Most parks had distinct communities, except for Montezuma Castle National Monument (MOCC and MOWE) and Tuzigoot National Monument, which are in the same larger Verde River watershed and were compositionally similar. Average EPT taxonomic richness was highest at Pecos National Historical Park (PECO) and MOCC with 18 taxa, and lowest at Tumacácori National Historical Park (TUMM) with 3 taxa, which does receive treated wastewater inputs from an urban area 24 kilometers upstream. Across the time period, we found that all parks had weighted tolerance values (TV) generally ranging from 4 to 6 (representing very good to fair water quality, respectively), with the lowest tolerance values at PECO and the highest values at TUMM. TUMM had a TV peak of 7.3 in 2015, which was related to a large increase in density of snails in the genus *Physa* following an extraordinary flood event. The variation in TV throughout the 10-year period is part of the communities' responses to changes in flow and habitat. Further analyses, including IBIs and statistical modelling of species abundances, will clarify these community responses to flow changes.

id #135755

Beavers & People: Scaling Effective Stream Restoration & Rebuilding Watershed Resilience after Wildfire

Alexa Whipple¹

1. Methow Okanogan Beaver Project, a program of Methow Salmon Recovery Foundation, Oroville, WASHINGTON, United States

Restoring streams after wildfire and subsequent debris flows is possible and essential to our future. However, the scale of restoration must address the scale of degradation and simultaneously build in resilience to future disturbance. Humans are unlikely to achieve effective and resilient scaling of this magnitude and complexity alone. Partnering with native North American beavers, *Castor canadensis*, the original freshwater ecosystems engineer is well proven. However, many streams cannot support industrious and highly adaptive beavers following massive wildfire and debris flows on top of human legacy degradation and simplification. Many streams couldn't support beavers before recent large fires due to pervasive simplification of streams that rarely provide adequate beaver essential resources anymore such as: deep enough water to evade predation, dense enough deciduous vegetation to provide reliable year round food resources, other beavers nearby to fulfill social needs, and tolerance of their presence from humans. All of these challenges can be mitigated, most of the time... simply, affordably, and efficiently to expand beaver benefits and sustainable populations across our watersheds. Methow Okanogan Beaver Project (est. 2008) will share case studies showing effective beaver-based stream restoration and essential beaver coexistence strategies to accelerate and scale restoration of wildfire impacted streams and build resilience back into our watersheds in partnership with beavers and people. People and beavers working together will accomplish far more and much faster than humans can achieve alone. Our vast watersheds supporting so many diverse and threatened biological communities in the Pacific Northwest need us to scale up and accelerate restoration with built in resiliency as only beavers and people can do.

id #136011

River frequencies for river health

David A Lytle¹, Jonathan B Armstrong¹, Laurel Genzoli², Theodore A Kennedy³, David M Merritt⁴, Julian D Olden⁵, Albert Ruhi Vidal⁶, Jonathan D Tonkin⁷, Desiree D Tullios¹

1. Oregon State University, Corvallis, OR, United States

2. University of Montana, Missoula, MT, United States

3. US Geological Survey Grand Canyon Monitoring and Research Center, Flagstaff, AZ, USA

4. Functional River Ecology, Fort Collins, CO, United States

5. University of Washington, Seattle, WA, United States

6. University of California, Berkeley, CA, United States

7. University of Canterbury, Christchurch, New Zealand

The river flow regime is a defining feature of waterways, shaping physical, chemical, ecological, and evolutionary processes. While temporal variation in river flows is generally considered an essential component of river health, the frequency and context of variation can produce different outcomes, especially in the presence of streamflow modification. For example, high frequency flow signatures, caused by diel hydropedaking fluctuations, can stress river health via rapid inundation and desiccation of the varial

zone, imposing limits on recruitment for riparian plants and aquatic insects with complex life cycles. Flat signatures arising from flow homogenization produce a static varial zone prone to encroachment of rooted macrophytes and filamentous algae, which are of lower nutritional quality than diatom-based biofilms, and also increase the probability of toxic cyanobacterial blooms. Infrequent but large magnitude high flows can have positive effects on sediment dynamics and stimulate population growth but can sometimes favor invasive species. Here, we examine river frequency spectra across temporal scales and document river health outcomes in terms of physical, ecological, and evolutionary processes. Physical processes include hydrology, hydraulics, temperature regimes, and sediment flux. Ecological processes include population dynamics, biological invasions, species interactions, and cross-ecosystem linkages. Evolutionary processes include population genetics, trait evolution, and phylogenetic structure. We illustrate our findings with examples from rivers where management has altered river frequencies: the Colorado River below Glen Canyon dam (lowered diel flow frequencies that stimulated aquatic insect productivity) and the Klamath River basin (dam removal that altered sediment flux and macrophyte dynamics). We also examine the value of “breaking the brick” – implementing hydrographs that incorporate flow fluctuations with variable frequencies, amplitudes, and phases, rather than imposing seasonally-prescriptive flows or minimum flows that abruptly shift in response to management demands.

id #134732

Flow permanence alters size spectra of stream macroinvertebrate communities

Chelsea R Smith¹, Jonathan P Benstead¹, Steve W Golladay², Alexander D Huryn¹, Ariel J Shogren¹, Carla L Atkinson¹

1. *Biological Sciences, The University of Alabama, Tuscaloosa, AL, United States*

2. *The Jones Center at Ichauway, Newton, Georgia, United States*

Periodic cessation of stream flow results in changes to invertebrate community structure over time. Initially resistant taxa or resilient colonizers from nearby perennial streams are able to reestablish quickly when flow resumes followed by continued community recovery. Previous research has noted that taxa in non-perennial streams are often smaller-bodied with faster life cycles compared to those in perennial streams. Few studies, however, have examined how such alterations to community structure change food-web properties and energy transfer. Size-spectrum analysis examines body mass-abundance relationships to assess how they affect the efficiency of energy transfer and food-web structure. We examined how flow regimes, ranging from non-perennial to perennial, altered invertebrate size spectra across three nested reaches within multiple headwater streams in Alabama, USA. We collected riffle ($n=4$) and pool ($n=4$) samples from each reach four times from 2021 to 2022. We processed and identified samples to the lowest possible taxonomic resolution and measured lengths to quantify biomass. We collected continuous stage-data within each reach to characterize flow regime and physical stream attributes including width, substrate type, and depth, on each sampling occasion. We observed a reduction in invertebrate mean body size and abundance as flow declined within riffles but not pools. Future work will examine the shifts in functional feeding groups across sites, with the potential to examine shifts in energy transfer pathways. By examining both pool and riffle habitats in non-perennial sites relative to nearby perennial sites, we are able to compare the relative effects of flow cessation across habitats while accounting for available local diversity. Size-spectra analysis provides a more nuanced understanding of shifts in energy transfer as stream communities experience more frequent or longer periods of drying.

id #135757

Green ribbon in a blue-ribbon fishery: exploring synchrony between algal biomass and river metabolism

Cora M Steinbach¹, Robert O Hall¹

1. *University of Montana, Missoula, MONTANA, United States*

Assessing eutrophication in rivers is difficult compared to lakes and coastal waters, as nearly all algal production occurs on the riverbed. Surveying benthic algae is challenging, and variation in algal growth across space and time complicates reach-scale estimates of algal biomass. River metabolism integrates gross primary production (GPP) and ecosystem respiration (ER) to capture the processes that underlie algal growth at the reach scale. Yet, the role of algal biomass and growth form in structuring variation in GPP and ER remains poorly resolved. We explored the dynamics of reach-scale metabolism and algal biomass at six sites during the growing season in the productive Gallatin River (Montana). Sites were instrumented with dissolved oxygen sensors and selected for varying coverages of three algal growth forms—filamentous algae (*Cladophora glomerata*), diatomaceous mats (*Didymosphenia geminata*), and mixed epilithic biofilms. We surveyed transects bi-weekly to quantify percent cover and condition of each growth form and sampled all taxa to construct reach-scale biomass. Across sites, peak GPP varied in timing and magnitude but always occurred when epilithic biofilms dominated algal cover. At three of six reaches, low GPP ($> 3 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$) for nearly two months coincided with low early-season biomass. At some sites and times, *Cladophora* and *Didymosphenia* dominated total biomass, but did not appreciably increase GPP. We observed synchrony between algal biomass and GPP, in terms of total biomass accumulation and seasonal increase in GPP, but found dominant algal growth form alone did not control GPP and ER. By integrating algal dynamics with metabolism estimates, we can derive rich information on production and establish strong baselines to monitor eutrophication in rivers.

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id #136013

Effect of warming on solute fluxes from peatland streams: synthesizing results from a 10-year, whole-ecosystem experiment

Natalie Griffiths¹, Stephen Sebestyen², Keith Oleheiser¹

1. Oak Ridge National Laboratory, Oak Ridge, TENNESSEE, United States

2. Northern Research Station, USDA Forest Service, Grand Rapids, MN, United States

Increased temperatures are altering biogeochemical processes in wetland ecosystems. In the SPRUCE experiment, a whole-ecosystem warming and elevated atmospheric CO₂ manipulation in an ombrotrophic peatland in northern Minnesota, USA, we are examining how warming affects solute concentrations and fluxes from peatland streams. After 10 years, total organic carbon concentrations almost doubled with warming (up to +9°C). In contrast, there was a muted increase in nitrogen and phosphorus concentrations, likely due to biological uptake in this nutrient-limited ecosystem. While solute concentrations generally increased with warming, stream flow decreased, likely due to increased evapotranspiration, resulting in an overall decrease in total organic carbon and nutrient efflux (in the early years) or no change (in the later years) with warming. These results will be put into context of the peatland carbon budget to evaluate the extent to which warming influences the carbon balance of these vulnerable ecosystems.

id #135758

Mercury biomagnification across food webs with varying non-native fish presence: Implications for native fish conservation in the upper Colorado River Basin

Charles Wahl¹, Natalie Day², Travis Schmidt³, James Roberts⁴, Jessica Brandt⁵, Craig Stricker⁶

1. Colorado Water Science Center, U.S. Geological Survey, Lakewood, CO, USA

2. Colorado Water Science Center, U.S. Geological Survey, Grand Junction, CO, USA

3. Wyoming-Montana Water Science Center, U.S. Geological Survey, Helena, MT, USA

4. Great Lakes Science Center, Lake Erie Biological Station, U.S. Geological Survey, Huron, OH, USA

5. Department of Natural Resources and the Environment Center for Environmental Sciences and Engineering, University of Connecticut, Storrs, CT, USA

6. Fort Collins Science Center, U.S. Geological Survey, Lakewood, CO, USA

Environmental contamination and the introduction of non-native species are two anthropogenic disturbances that have profound implications for ecosystems. Mercury (Hg) transfers through food webs where it can bioaccumulate and biomagnify to toxic concentrations. Non-native introductions alter food webs and influence energy and Hg movement. This study examined Hg concentrations and stable isotope compositions in native and non-native fishes among four river subbasins of the upper Colorado River Basin (UCRB) to inform the combined influence of Hg exposure and non-native competition on the conservation status of native fishes. We assessed Hg biomagnification at the fish assemblage level, compared concentrations to risk thresholds for fish health, and evaluated niche overlap. Generalist-invertivores had the lowest Hg concentrations (0.001-0.191 ug g⁻¹ wet weight) and no individuals exceeding the U.S. Environmental Protection Agency aquatic life criterion (0.225 ug g⁻¹). Piscivores had the highest Hg concentrations (0.008-1.840) and 74% of individuals exceeded the criterion. Hg biomagnification was observed across the basin. However, Hg concentrations and trophic magnification varied across subbasins, with higher magnification factors in the Lower Green and White-Yampa subbasins. These subbasins also had the highest proportion of non-natives, suggesting subbasin-specific patterns in Hg cycling across the UCRB. Stable isotopes revealed niche overlap among native and non-native species and trophic position varied spatially within species. Non-native species have likely reshaped food webs and intensified the threat Hg poses to native fish conservation. Continued monitoring is needed to track contamination trends, particularly studies that focus on Hg sources entering the UCRB food web, vectors for accumulation, and competitive interactions among native and non-native species.

id #136014

Pinpointing drivers of covariance, persistence, and synchrony in stream metabolism across a heterogeneous watershed

Jonathan Behrens¹, Robert Brown¹, Natalie Griffiths¹, Scott Brooks¹, Erin Hotchkiss², Saubhagya Rathore¹, Marie Kurz¹

1. Oak Ridge National Laboratory, Oak Ridge, TN, United States

2. Virginia Tech, Blacksburg

Differences in light, temperature, and discharge lead to distinct rates of gross primary production (GPP) and ecosystem respiration (ER) in stream ecosystems. As such, within-day, magnitudes of metabolism and its drivers may covary across sites (spatial covariance) and within a site over time (temporal covariance). We examined how spatial and temporal covariance, spatial

persistence, and temporal synchrony of metabolism drivers (light, temperature, and discharge) align with GPP and ER in East Fork Poplar Creek in Oak Ridge, TN, a 4th order river (77 km²) that drains heterogeneous landcover. On a given day, an analyte has high spatial persistence when the ranked ordering of each site's magnitude of the analyte on that day is nearest to the median site ranked ordering calculated from the ranked ordering of each day with complete data (n = 100+ days). At a given site, an analyte has high temporal synchrony when the ranked ordering of each day's magnitude of the analyte at that site is nearest to the median ranked ordering of days across each individual site (n = 12 sites). We sampled across two years at 9 main channel sites and 3 tributaries. Seasonal shifts in light and temperature reduced spatial persistence of GPP and ER. As the canopy closed and light availability differed across sites, spatial covariance of light and GPP increased; this period had low spatial persistence relative to the full time series (i.e., site rank shifted). Temporal synchrony of light and GPP were lowest at the sites with little to no canopy cover due to low temporal covariance in GPP and light (and limited shifts in ordering of days). GPP and ER were not always correlated with one another, leading to differences in covariance, spatial persistence, and temporal synchrony. In the fall, consistent leaf inputs but variable light are associated with higher spatial covariance for GPP relative to ER. By investigating covariance, persistence, and synchrony independently, we can pinpoint specific sites or time periods for further analysis to explore the complex interplay of metabolic drivers or other drivers not commonly included in analysis (e.g., nutrients).

id #134735

Afromontane damselflies as indicators of freshwater ecosystem change

Lenin D Chari¹, Dorothy C Madamba², Kudzai Mafuwe³, Blessing M Mugaviri⁴, Gerald Chikowore^{5,6}, Sydney Moyo¹

1. Biological Sciences, Louisiana State University, Baton Rouge, 70803, Louisiana, United States

2. Entomology, Natural History Museum of Zimbabwe, Bulawayo, Zimbabwe

3. Entomology and Wildlife Ecology, University of Delaware, Newark, Delaware, United States

4. Nyanga Station, Zimbabwe National Parks, Nyanga, Manicaland, Zimbabwe

5. Zoology and Entomology, Rhodes University, Grahamstown, Eastern Cape, South Africa

6. Tsetse Control Services, Directorate of Veterinary Services, Ministry of Lands, Agriculture, Fisheries, Water and Rural Development, Harare, Zimbabwe

Afromontane freshwater ecosystems are global biodiversity hotspots characterized by high endemism and strong sensitivity to land-use change and climate variability. Yet for many threatened damselfly species (Odonata: Zygoptera), particularly those restricted to montane headwaters and seepage systems, baseline distribution data remain limited, constraining effective conservation planning. Because damselflies respond predictably to changes in riparian vegetation, hydrology, and water quality, they provide a powerful lens for assessing freshwater ecosystem condition in data-limited regions. In this study, we examine the distribution and habitat associations of threatened damselflies across the Eastern Highlands of Zimbabwe by integrating multiple data sources. We develop a historical database of species occurrences sourced from museum records, literature, and opportunistic surveys, in collaboration with regional partners. We complement these data with contemporary field surveys, habitat characterization, and environmental DNA (eDNA) sampling. Study sites span montane streams, seepages, and forested riparian habitats across gradients of elevation, land use, and disturbance. Using species occurrence data, including historical records, we examine emerging patterns in community composition and model the probability of damselfly presence as a function of local habitat features and broader landscape context. Analyses are ongoing, and this presentation focuses on consistent patterns rather than finalized trend estimates. Preliminary results indicate strong associations between threatened damselfly taxa and intact, shaded headwater systems, and highlight the value of combining historical records, contemporary surveys, eDNA, and spatial modeling to identify priority habitats. Together, this work demonstrates how damselflies can serve as sensitive indicators of freshwater ecosystem change and provides a framework for integrating heterogeneous data to support conservation planning in threatened Afromontane freshwater systems.

id #135247

Modeling Surface and Subsurface Transport of Land-Applied Treated Wastewater in Urban Watersheds

Harshita Mahaseth¹

1. City of Austin, Austin, TEXAS, United States

The Texas Land Application Permit (TLAP), issued by the Texas Commission on Environmental Quality (TCEQ), allows treated wastewater to be applied to land rather than discharged directly into rivers or streams. Within the City of Austin, approximately 50 water treatment and reclamation facilities operate under this framework, releasing treated water back into the environment through methods such as spray irrigation and drip application. These land-application practices rely on the assumption that evapotranspiration removes a significant portion of the applied water, thereby reducing the amount that reaches nearby streams or groundwater. However, that assumption has not been field verified in Austin. Furthermore, those 50 sites consist of varying site conditions and treatment characteristics, which can lead to a variety of outcomes for stream health. Thus, assessing their impact with observational data and knowing which factors play a role in their effectiveness is a key Austin Watershed Protection Department (AWP) goal.

To that end, we have developed a modeling approach that considers both surface and subsurface water movement. In this study, three nested models were run to evaluate water quality and pollutant behavior: an evapotranspiration model, a hydraulic

conductivity model, and a groundwater flow model based on the Boussinesq approximation. Together, these models provide a simplified but integrated way to examine how treated wastewater moves through the surrounding landscape. Through these models, we will be able to identify those facilities that are at risk of leaching treated wastewater to their receiving streams and prioritize for future water quality sampling

We estimated evapotranspiration using recent empirical data and water-quality information, such as pollutant concentrations, to estimate contaminants in the system. Next, we assessed subsurface movement by using hydraulic conductivity. Areas with higher hydraulic conductivity generally allow water and dissolved contaminants to move more rapidly. Finally, we further examined groundwater flow by using the Boussinesq model, which represents the subsurface contribution of land application on the stream. By combining these modeling approaches, this study provides a practical framework for evaluating the hydrologic and water-quality implications of land-applied treated wastewater, supporting improved planning and regulatory decision-making for urban watersheds.

id #135759

Extreme genome diversity, hybridization, and cryptic speciation in the toxic alga *Prymnesium parvum*

Jennifer Wisecaver¹

1. Washington State University, Pullman, WASHINGTON, United States

Harmful algal blooms (HABs) of the toxic haptophyte *Prymnesium parvum* are a recurrent problem around the world. Strains of *P. parvum* vary in the toxins they produce and in other physiological traits associated with HABs, but the genetic basis for this variation is unknown. To investigate genome diversity in this morphospecies, we generated genome assemblies for fifteen diverse strains of *P. parvum*. Comparative analysis revealed considerable DNA content variation between strains. Strains included haploids, diploids, and polyploids, but not all differences in DNA content were due to variation in genome copy number. Haploid genome size between strains of different chemotypes differed by as much as 243 Mbp. Syntenic and phylogenetic analyses indicate that the Texas strain UTEX 2797 is a hybrid that retains two phylogenetically distinct haplotypes. Investigation of gene families variably present across strains identified functional categories associated with metabolic and genome size variation in *P. parvum* including genes for biosynthesis of toxic metabolites and proliferation of transposable elements. Together, our results indicate that *P. parvum* is comprised of multiple cryptic species. Using these genome resources, we identified giant polyketide synthase enzymes (PKZILLAs) for prymnesin toxin biosynthesis and characterized variation at the PKZILLA loci responsible for differences in prymnesin structure. These genomes provide a robust phylogenetic and genomic framework for investigations into the eco-physiological consequences of the intra- and inter-specific genetic variation present in *P. parvum* blooms and demonstrate the need for similar resources for other HAB-forming morphospecies.

id #135760

Effects of biochar-modified poultry litter on water quality

Mahima Quazi¹, Shannon L Speir¹, Brian E Haggard¹

1. University of Arkansas, Fayetteville, ARKANSAS, United States

Arkansas is one of the leading poultry producers in the nation, resulting in a surplus of animal waste to be disposed of or land applied as an organic fertilizer source. However, poultry litter contains significant amounts of nitrogen (N) and phosphorus (P), and direct application of poultry litter to the landscape has been linked to the eutrophication of downstream surface waters. As a result, poultry litter amendments have become both a challenge and opportunity for farmers in waste and nutrient management. To address these concerns, biochar-amended poultry litter (BAPL) can be used as a plant-available fertilizer, while reducing nutrient runoff because of its absorbent nature and ability to improve soil structure. Yet, few studies have quantified the effectiveness of BAPL as a practice for protecting water quality across scales while maintaining its effectiveness as a fertilizer. Here, we compared nutrient leaching from poultry litter versus BAPL using both water extraction tests and a soil incubation study to understand the potential for BAPL to reduce the impact of traditional fertilizers on downstream water quality. Preliminary results BAPL releases significantly less water soluble nutrients compared to raw poultry litter in water extraction tests. In soil incubation studies, we also found that BAPL improved nutrient retention compared to raw litter, with biochar amendments reducing N and P leaching by >15%. These results suggest that BAPL amendments may effectively reduce nutrient release potential, displaying lower water-extractable N and P concentrations after simulated runoff events. Our findings demonstrate the potential for BAPL to serve as a conservation practice that can reduce nutrient loss from agricultural lands, while addressing the critical challenge of managing Arkansas's abundant poultry litter waste.

id #136016

The Puget Sound Stream Benthos database: A powerful regional tool and local application in Bellingham, Washington.

Mateo Scoggins¹, Kate Macneale², Beka Stiling², Sara Brooke Benjamin¹

1. City of Bellingham, Bellingham, WA, United States

2. Water and Land Resources, King County, Seattle, WA, United States

The Puget Sound Stream Benthos database (PSSB) and website was developed by local, regional and state partners to standardize and share freshwater benthic macroinvertebrate data and biological metrics at a meaningful scale. King County has been managing this powerful tool since 2007, and it currently holds data from over 11,000 samples collected by over 40 partners from 1994 through 2025. This tool, and the Benthic Index of Biotic Integrity (B-IBI) scores it calculates, are used widely by

waterway managers, researchers and private sector consultants. Although it initially was limited to the Puget Sound region, it now houses data from across Washington. PSSB users can quickly get index scores and make reliable and robust spatial and temporal comparisons among various groups' data, thanks to the requirement that data must be entered by certified professional taxonomists. The database also provides raw metrics and taxonomic data for all samples. .

The City of Bellingham Public Works department has been monitoring the creeks that run through the city since about 2000, starting with the Olympic Pipeline explosion that occurred in 1999. That event had far-reaching effects on how the city and nation managed public safety and the environment and set in motion development of a robust restoration and stewardship program focused on natural resources. There are four primary catchments in the City of Bellingham, Squalicum, Whatcom, Padden and Chuckanut, along with a handful of associated tributaries. Macroinvertebrate data was pulled from the PSSB and evaluated using the Index of Biotic Integrity as well as representative raw metrics to better understand temporal and spatial patterns among these primarily urban catchments. Results suggest that there is relatively high temporal variability among different dates, but that the streams are not indicating degradation over time, despite increasing development pressure, and that the benthic communities are responding to urban stressors consistent with relative levels of development among the catchments. Resource managers in Bellingham are using the PSSB effectively to both manage their data in a safe and effective manner but are also using it to make better decisions about how to manage local waterways.

id #135761

Nitrous oxide production and emissions in tailwater recovery systems in Eastern Arkansas over a year (2025).

Nellie I Little¹, Shannon L Speir¹, Auldyn L Faulk¹, Frank E Johnson II², Mike B Daniels³

1. *Department of Crop, Soil, and Environmental Sciences, University of Arkansas, Fayetteville, Arkansas, United States of America*

2. *U.S. Department of Agriculture, Agricultural Research Service, Oxford, Mississippi, United States of America*

3. *Department of Crop, Soil, and Environmental Sciences, Cooperative Extension Service, University of Arkansas Division of Agriculture, Little Rock, Arkansas, United States of America*

Eastern Arkansas is dominated by row-crop production; however, local aquifer declines have resulted in the implementation of on-farm water management systems to support continued irrigation needs. As such, many farms in Eastern Arkansas have introduced tailwater recovery systems (TWRS) that collect runoff from adjacent fields, storing it in ditches and reservoirs for recirculation and reuse. While downstream water quality benefits have been well documented, we lack understanding of the potential role of TWRS in releasing greenhouse gases, like nitrous oxide (N₂O), due to the accumulation of nutrients in runoff from cropland. Here, we sampled three Arkansas Discovery Farm sites with TWRS's in Eastern Arkansas, collecting samples to determine the production (e.g., dissolved gas samples) and surface emissions (e.g., via floating chambers) from TWRS. Preliminary results indicate that higher water column nitrate concentrations drive higher N₂O emissions from TWRS's. Surprisingly, peaks of N₂O emissions were observed in the winter, which did not align with the growing season when fertilizer application typically occurs. Overall, results suggest that TWRS predominately act as a net source of N₂O to the atmosphere. While TWRS provides benefits to water quality and water conservation, our results highlight a need to adapt TWRS to mitigate their production and emissions of N₂O hopefully leading to more resilient farms under a changing climate.

id #136017

Rainfall and temporal controls on inorganic nutrients in a salinization mesocosm experiment

Samuel AT Dias¹, Natalie A Clay¹, Sally A Entekin², Maggie Herrmann¹, Connor Foley¹, Clay Prater¹, Michelle A Evans-White¹

1. *University of Arkansas, AR, United States*

2. *Entomology, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, United States*

Salinization is a pervasive threat to aquatic and terrestrial ecosystems. Increased ionic loads can disrupt cation-anion exchange, biotic uptake, nutrient release, and fundamentally alter dissolved nutrient availability. While salinization alters ecosystem processes experimental studies often examine aquatic and terrestrial systems independently, limiting our understanding of how salinization alters nutrient dynamics across riparian-stream interfaces. In a greenhouse mesocosm experiment, we manipulated NaCl concentrations across an aquatic-terrestrial system by adding 100, 200, 300, 400, and 500 mg/L to soil and water simultaneously (n=5 per treatment). To isolate system-specific responses, 300 mg/L of NaCl was also added to either aquatic or terrestrial components alone. Aquatic systems received constant discharge while terrestrial systems experienced discrete rain events twice per week, simulating rainfall as either saline or freshwater. To assess how orthophosphate, nitrate + nitrite, and ammonia concentrations responded to rainfall through time we analyzed control mesocosms (no NaCl addition), single system controls (aquatic- or terrestrial-only additions), and coupled 300 mg/L NaCl treatment using linear mixed-effect models with rain event type (baseline, freshwater, saline) as a fixed effect and mesocosm as a random effect. Nutrient concentrations were primarily driven by rain event type rather than NaCl addition. Ammonia responded strongly to freshwater rain events exhibiting a parabolic shape with levels decreasing during the summer and subsequently increasing during the fall/winter with baseline and salt rain events had comparable concentrations. Nitrate + Nitrite concentrations were significantly associated with rain event and exhibited a rain event by time interaction resulting in an overall decline through time. In contrast, orthophosphate concentrations showed modest increases through time with limited impacts by rain event. Preliminary results indicate that rain event identity and time exert control over inorganic nutrient pools producing seasonal and nutrient-specific responses. However, these results should be tempered as they do not account for changes in the total nutrient pool (sodium, chloride, total/organic nitrogen and phosphorus) or responses across all experimental treatments.

id #134994

A one-two punch: fire and hydrological effects on stream communities in coastal southern California

Audrey Thellman¹, Scott D. Cooper¹, Sheila W. Wiseman¹, Conor McMahon¹, Kristie Klose²

1. University of California, Santa Barbara, Santa Barbara, CA, United States

2. United States Forest Service, Solvang, California, United States

As the climate warms, drought, intense storms, and wildfires are expected to increase in frequency, severity, and duration or extent. In coastal southern California, streams are experiencing pressures from wildfires, prolonged drought, and intense floods. Although wildfire impacts on stream abiotic conditions, algal abundance, leaf litter inputs, and macroinvertebrate communities depend on the extent of severe fire and whether riparian vegetation remains intact, the compound impacts of fire and drought or floods remain understudied. Since 2008, we have evaluated the effects of 7 fires on 23 stream reaches. In that time, two of the fires were followed by 5 years of drought and two were followed by intense rains and debris flows. We quantified the extent and severity of burns, and vegetation impacts and recovery, using remote sensing. We then compared stream community responses to several key factors: burn severity and extent (e.g., riparian zone intact or not), time since fire, hydrological conditions, and trout presence. Whether the riparian zone burned and a debris flow occurred were the most important factors affecting stream communities. Debris flows and riparian burning greatly reduced riparian vegetation, increasing light levels and, hence, algal abundance and reducing leaf litter inputs. Stream reaches disturbed by riparian burning or debris flows were dominated by algivores (baetid mayflies and hyliid tadpoles), whereas streams in burned basins with intact riparian vegetation recovered quickly to an undisturbed configuration dominated by detritivores (e.g., cased caddis larvae) and, where trout were absent, various invertebrate predators. Stream drying or intense floods after wildfire extirpated local trout populations, resulting in increases in invertebrate predator abundance. Rainfall and stream flow patterns influenced stream responses to wildfire, both by affecting the rapidity of vegetation recovery after fire and by directly disturbing streams through debris flows, floods, and drying.

id #136018

Drying out diversity: functional, but not taxonomic, shifts in aquatic insects under ephemeral conditions

Hayden Hays¹, Matthew Barnes²

1. East Central University, Ada, OK, United States

2. Texas Tech University, TX, United States

As hydrologic regimes shift under climate change, ephemeral freshwater ecosystems and their aquatic insect communities face increasing instability. Because aquatic insects are sensitive to environmental conditions and fill diverse ecological roles, they serve as useful bioindicators of ecosystem health. I conducted a mesocosm experiment to test whether taxonomic identity or functional trait composition respond more predictably to drying conditions. Using 24 mesocosms, I manipulated surface area and water permanence to simulate variable hydroperiods and sampled aquatic insects over time. I recorded 6,521 individuals from 24 morphospecies and 19 families, dominated by Odonata, Coleoptera, and Hemiptera. Multivariate analyses revealed contrasting patterns. Redundancy analysis showed hydrologic variables explained moderate variation in both taxonomic (adjusted $R^2 = 0.592$) and functional composition (adjusted $R^2 = 0.561$). Neither hydrologic treatment, nor the interaction factor had a significant effect on taxonomic and functional composition, yet they both changed over time. Overall, my study demonstrates that taxonomic and trait composition appear to follow successional processes in the assembly of aquatic insect communities. Further, despite the lack of a significant relationship between hydrology and functional composition, trait-based approaches can serve as an additional indicator to taxonomy for how communities work under rapidly changing conditions.

id #136019

Diet of Pygmy Whitefish *Prosopium coulterii* in Sullivan Lake, Pend Oreille County, Washington

Haley Bedell¹, Camille McNeely¹, Allan Scholz¹, Paul Spruell¹, Jason Connor²

1. Department of Biology, Eastern Washington University, Cheney, WA, United States

2. Kalispel Natural Resources Department, Kalispel Tribe of Indians, Cusick, WA, United States

Pygmy Whitefish were collected at Sullivan Lake on 6/30/2025 ($n = 25$) and 9/23/2015 ($n = 39$), in order to determine the composition of their diets and the potential for future cohabitation with Bull Trout (*Salvelinus confluentus*). Contents of each stomach were examined with a dissection scope, and count and length measurements were taken of individuals in each taxonomic group. Dry weights of prey items were determined using published length to weight regression equations. For each fish, we determined the number (and numerical percentage) and weight (and weight percentage) of each taxonomic group in the diet. The data from individual fish were combined to obtain a mean ($\pm$ SD) numerical percentage (%N) and weight percentage (%W), and frequency of occurrence (%FO) of items in the diet across the population. %FO represents the total percentage of stomachs in which a particular item occurred at least once. These parameters were combined to determine an Index of Relative Importance (IRI) for each food item [i.e., $IRI = (\%N + \%W) \times \%FO$]. Of the 64 stomachs examined, 9 (14.1%) were empty, each one remaining containing at least one food item. Of those that contained food, Chironomidae accounted for 90.7 (± 22.2)%N and 54.5% FO. The frequency of Chironomidae is unsurprising as other investigators determined that Pygmy Whitefish with counts

of < 15 gill rakers on the first arch feed almost exclusively on Chironomidae, whereas those with > 15 on the first arch eat comparatively more zooplankton. The mean number ($\pm$) SD of gill rakers counted from our samples was 13 ± 1.9 ($n = 55$).

id #134741

Landscape Watershed Condition and Drought Vulnerability Assessment in the Western U.S.

Jeffrey Falke¹, Thomas Dilts², Robert Al-Chokhachy³, Haley Glassic³, Nick Som³, Jonathan Burnett⁴, Jason Dunham⁵

1. Nevada Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Reno, Nevada, United States

2. Natural Resources and Environmental Science, University of Nevada, Reno, Reno, Nevada, United States

3. Northern Rocky Mountains Science Center, U.S. Geological Survey, Bozeman, Montana, United States

4. Oregon State University, Corvallis, Oregon, United States

5. Forest and Rangeland Ecosystem Science Center, U.S. Geological Survey, Corvallis, Oregon, United States

Climate variability and persistent drought have focused attention on declining water availability across western U.S. riverscapes. Although meteorological drought indicators are well established, relationships between climate and ecosystem responses are less recognized and need to reflect multiple drought "flavors" and complex mediating processes. A better understanding of drought sensitivity in an ecological context (i.e., how ecosystems change in response to water deficits) will provide essential information for adaptation planning and management. We developed a prototype watershed drought vulnerability assessment using a combination of remotely-sensed observations and results from recent modeling efforts focused on the Donner und Blitzen River basin in southeastern Oregon. We compiled spatially-explicit time series representing meteorological indicators, hydrologic responses (streamflow and permanence, soil moisture, spring zones), and vegetation productivity (upland, riparian) and summarized interannual variability across years (1980-2021). We assessed sensitivity by examining variability within and among components, testing spatial patterns across landform classifications, and evaluating temporal trends. Our preliminary results reveal substantial spatial and temporal heterogeneity among landform types with mean productivity differing substantially among valley bottoms, warm slopes, and cool slopes. Streamflow variance exhibited strong spatial patterns, with headwater reaches showing highest variability. Our next steps include expanding to additional watersheds that represent gradients in ecological condition and landscape intactness, developing integrated indicators through data reduction, and constructing relational models, probabilistic networks, and tools for adaptation planning. Ultimately, we hope to provide an understanding of drought sensitivity that incorporates social and ecological elements to inform broad- to local-scale decisions and facilitate identification of watersheds and landscape positions most sensitive to drought.

id #135766

LeRoy Poff's Role in Defining the Form, Function, and Future of Flow Regimes

Ted Grantham¹, Kris Taniguchi-Quan²

1. University of California, Berkeley, Berkeley, CA, United States

2. Southern California Coastal Water Research Project, Costa Mesa, CA, USA

Dr. LeRoy Poff's seminal paper on the Natural Flow Regime represented a paradigm shift in the way scientists and managers conceptualize and manage rivers. It also represented a cornerstone upon which my own career in freshwater science, policy, and management has been built. In this talk, I will highlight three of Poff's important contributions to the field: defining the form of river flow regimes, linking flow regime form and function, and charting a path for how to protect and manage flow regimes into the future. I will also describe how these contributions have created a foundation for much of my own work, seeking solutions to global water challenges that sustain the health of freshwater ecosystems.

id #135767

Vive la Résistance: Rock Glaciers Are Buffering Alpine Streams Against Climate Change

Scott Hotaling¹, Ashlesha Khatiwada², Dan McGrath², Lusha Tronstad³, Debra Finn⁴, Gordon Gianniny¹, Justin Pomeranz⁵, Simeon Caskey⁶

1. Utah State University, Logan, UT, United States

2. Colorado State University, Fort Collins, CO

3. University of Wyoming, Laramie, WY

4. Missouri State University, Springfield, MO

5. Colorado Mesa University, Grand Junction, CO

6. Grand Teton National Park, Moose, WY

Climate change is dramatically impacting mountain ecosystems around the world. Perhaps the most visceral of these impacts is the ongoing recession of glaciers and perennial snowfields. However, surface glaciers and snowfields are not the only perennial ice in mountain landscapes. Many forms of subsurface ice are also present and likely play an important, albeit understudied, role in water availability and aquatic ecosystem integrity in mountain systems. Rock glaciers—large masses of debris-covered ice that flow downhill—are the most common form of "alternative ice" alpine systems. Theory predicts that surface glaciers and snowfields that are exposed to ambient warming will recede faster than rock glacier that is insulated by debris cover. A limited amount of

evidence supports this expectation. Since 2015, we have been monitoring high-alpine streams in the Teton Range, USA fed by three different sources—surface glaciers, rock glaciers, and seasonal snowpack—to understand how different sources may yield differing rates of change amidst rapid climate warming. In 2014 and 2022, LiDAR data was also generated for the entire Teton Range. By pairing our 10+ years of aquatic monitoring data with the physical change inferred from LiDAR, we were able to gain rare insight into the links between physical change to ice sources and downstream ecosystems. Specifically, we found that rock glaciers in the Teton Range have been resistant to climate-induced ice loss while surface ice features have seen dramatic declines. However, these physical changes have not been mirrored in changes to the streams draining these features. At least not yet. For instance, streams fed by seasonal snowpack and small perennial ice features have warmed rapidly during the summer while streams fed by surface glaciers and rock glaciers have remained largely unchanged. Put together, our data provide new insight into the future dynamics of headwater ecosystems as climate change increasingly impacts them.

id #136023

A bridge not too far: Scaling PFOS toxicokinetics from lab studies to stream food webs

Alison Zachritz¹, Daniele Miranda¹, Gary Lamberti¹, Jeffery Steevens²

1. *University of Notre Dame, IN, United States*

2. *Columbia Environmental Research Center (USGS), Columbia, MO*

Per- and polyfluoroalkyl substances (PFAS) move through aquatic food webs, linking contaminated basal resources to invertebrates, fish, and human consumers. Yet predicting when and where PFAS propagate through food webs remains challenging because field exposure varies across resource types, abundance, and timing. Translating toxicokinetic rates from laboratory experiments to field conditions is particularly difficult because natural variability in exposure, chemical properties, and organismal traits can strongly influence observed concentrations and rate estimates. Our goal was to develop a quantitative bridge between laboratory toxicokinetics and field exposure by combining diet-based PFOS uptake and elimination rates with PFAS concentrations measured in basal food resources sampled opportunistically across Lake Michigan tributaries (biofilm, detritus, algae) as well as spawned salmon eggs which represent a seasonal, pulsed, resource subsidy that may alter PFAS exposures. Field concentrations were paired with laboratory-derived toxicokinetic parameters from dietarily exposed amphipods (*Hyalella azteca*). Amphipods were fed contaminated [~ 87 ng/g PFOS] benthic diatoms for 7 days followed by a 4-day elimination period with clean diatoms for food. In laboratory exposures, one-compartment models produced PFOS uptake rates of $0.095\text{--}0.150\text{ d}^{-1}$ and elimination rates of $0.165\text{--}0.389\text{ d}^{-1}$, yielding kinetics-based bioaccumulation factors ($\text{BAF}_{\text{kin}} = 0.39\text{--}0.58\text{ kg kg}^{-1}$) that indicate limited bioaccumulation under short-term diet-only exposure. We combined these empirically derived parameters with field-measured PFOS concentrations in basal and seasonal food resources to predict invertebrate body burdens from various diet types, including pulsed subsidy scenarios relevant to salmonid spawning events. This paired framework provides a novel quantitative link between field exposure measurements and food-web modeling to improve ecosystem-scale understanding of PFAS transfer.

id #135000

Relating Fish Ovary Selenium and Maturation Stage: Implications for Monitoring

Margot Grimmelmont¹, Travis Schmidt², Molly A Moloney², Vicki S Blazer³, Heather L Walsh³, James Dunnigan⁴, Noëlie Molbert¹, Brian Balmer⁵, Trevor Selch⁴, Jessica E Brandt¹

1. *Natural Resources and the Environment, University of Connecticut, Storrs, Connecticut, United States*

2. *Wyoming-Montana Water Science Center, U.S. Geological Survey, Helena, MT, USA*

3. *Eastern Ecological Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

4. *Montana Fish, Wildlife and Parks, Bozeman, MT, USA*

5. *U.S. Fish and Wildlife Service Montana Ecological Services Office, Helena, MT, USA*

Coal mining in the Elk River Valley (British Columbia) has been ongoing since the 1880s, elevating trace-element concentrations in surface and ground waters across the Elk River basin. Of particular concern is selenium (Se), which is transported downstream to the transboundary waters of Lake Koocanusa (Canada-USA). Se is an essential micronutrient that bioaccumulates in aquatic food webs and can cause toxicity at slightly elevated concentrations in egg-laying animals. Current management guidance assumes Se is transferred from the liver to the ovary during vitellogenesis, therefore, ovary samples collected early in development may underestimate ovary Se concentrations at spawning. We evaluated this assumption by quantifying how ovary Se changes with oocyte maturation, comparing species that develop eggs at the same time (synchronous spawner; northern pikeminnow, peamouth) versus species that develop eggs in multiple batches over time (asynchronous spawner; reidside shiner) in Lake Koocanusa. Across the three fish species, we paired ovary Se with gonado-somatic index (GSI) and histological staging to test whether GSI reflects gonadal maturation and whether ovary Se differs among maturation stages and spawning status. By clarifying how ovarian development affects ovary Se variability, results will inform the timing of monitoring Se in reproductive tissues.

id #136024

Stream warming and light availability mediate post-disturbance recovery of nutrient uptake rates and reach-scale metabolism.

Ryan T Meyer¹, Jennifer L Tank¹, Mitchell J Liddick¹, Emma M Thrift-Cahall¹, Ursula H Mahl¹

1. *University of Notre Dame, South Bend, IN, United States*

Stream ecosystems are increasingly shaped by both climate and land use change, yet the consequences of interacting stressors on stream ecosystem function remain poorly understood. We used experimental streams to test how temperature and light

influence recovery of stream ecosystem function after stream scouring. We hypothesized that during recolonization, nutrient uptake would be highest in warm-open canopy streams because warming stimulates metabolism of both autotrophs and heterotrophs, while light availability further stimulates algae and hence gross primary production (GPP). In contrast, nutrient uptake would be lowest in the cool-shaded treatment because cooling suppresses metabolic activity while a lack of light limits GPP. Finally, we predicted that ammonium ($\text{NH}_4^+\text{-N}$) uptake would be higher than that for nitrate ($\text{NO}_3^-\text{-N}$) uptake due to solute-specific preferential demand. During Summer 2025, we conducted short-term nutrient additions of $\text{NO}_3^-\text{-N}$, $\text{NH}_4^+\text{-N}$, and soluble reactive phosphorus (SRP) in replicated (N=4) streams at the Notre Dame Linked Experimental Ecosystem Facility using the Linked Experimental Temperature System (ND-LEEF-LETS), to warm (+2.5°C) and cool (-2.5°C) two open and two shaded (90% shade cloth) streams in a fully-crossed design. We conducted the short-term additions one day before, and on days 1, 7, 14, 21, and 28 after a scouring event. We found that nutrient demand, (expressed as Vf: mm min^{-1}) regardless of solute, was highest in the warm-open, followed by warm-shaded, cool-open, and cool-shaded treatments, indicating that warming is a key control on uptake velocity (ANOVA; $\text{NH}_4^+\text{-N}$; $F=6.9$, $p<0.001$, SRP; $F=4.5$, $p<0.001$, and $\text{NO}_3^-\text{-N}$; $F=2.7$, $p>0.05$). Only $\text{NH}_4^+\text{-N}$ and SRP in the cool-shaded treatment exceeded pre-scouring Vf . All streams were heterotrophic, with ecosystem respiration (ER) exceeding GPP ($\text{GPP:ER}<1$), although the warm-open treatment exhibited the highest GPP:ER ratio (0.51; ANOVA; $F=7.9$, $p<0.001$). Cumulative GPP was highest in the warm-open treatment, followed by the cool-open treatment, while both shaded streams had very low GPP (ANOVA; $F=22.4$, $p<0.001$). These results demonstrate that warming altered both autotrophic and heterotrophic processes, whereas light availability only mediated autotrophs. By partitioning these effects we show that temperature and light regulate different components of metabolism and nutrient demand after disturbance.

id #134746

Thermal ecology meets disease: stress responses to crayfish plague in *Faxonius virilis*

Stacy A Schmidt¹, Lindsey K Albertson², Lydia L.J. Bushey², Laura Martín-Torrijos³, David A Schmetterling⁴, Susan B Adams⁵

1. Montana Fish, Wildlife & Parks, Montana State University, Helena, MT, United States

2. Ecology, Montana State University, Bozeman, MT, USA

3. Department of Mycology, Real Jardín Botánico, Madrid, Spain

4. Fisheries Research, Montana Fish, Wildlife & Parks, Montana State University, Missoula, MT, USA

5. Southern Research Station, U.S.D.A. Forest Service, Oxford, MS, USA

Temperature drives physiological performance, disease expression and ecological interactions in freshwater ectotherms. For host-pathogen systems, warming can alter both host tolerance and pathogen dynamics, yet empirical tests of how thermal stress interacts with disease history at the population level remain unknown. Crayfish are keystone species and ecosystem engineers whose individual condition and performance can strongly influence community structure and ecosystem processes. In Montana, USA, lesion prevalence from *Aphanomyces astaci*, the oomycete responsible for crayfish plague, varies widely in *Faxonius virilis* populations.

We experimentally evaluated how warming temperature regulated physiological and behavioral stress responses in *F. virilis* originating from populations with contrasting plague expression histories (high versus low lesion prevalence). Individuals were acclimated to multiple temperature treatments and monitored for survival, molting success, lesion progression, growth and behavioral performance under thermal challenge. We found that short term survival was high across temperature treatments and did not differ strongly between crayfish from high and low lesion prevalence populations. Instead, temperature primarily influenced physiological performance, as acclimation history strongly shaped loss of righting response during thermal ramps, while lesion prevalence structured baseline lesion burden and recovery after exposure to thermal stress.

Considering that crayfish exert both strong top-down and bottom-up effects in freshwater ecosystems, temperature-disease interactions at the individual level may scale to influence community structure and ecosystem function under future climate scenarios.

id #135002

Effects of light availability on microbe-metal interactions in wetlands

Marilee C Hoyle¹, Raven Bier^{2,3}, Xiaoyu Xu^{3,4}, Emma Kelsick^{2,3}

1. Auburn University, Sweet Home, OR, United States

2. Odum School of Ecology, University of Georgia, Athens, Georgia, United States

3. Savannah River Ecology Laboratory, University of Georgia, Aiken, South Carolina, United States

4. Warnell School of Forestry and Natural Resources, University of Georgia, Athens, Georgia, United States

Wetlands are vital ecosystems vulnerable to threats such as metal contamination. Wetland microorganisms can mediate metal fate and bioavailability; however, the influence of environmental factors, particularly light exposure, on these interactions remains unclear. While previous studies suggest that light affects microbial communities, its consequences for metal dynamics are understudied. In this study, we conducted a microcosm experiment using two wetlands: a constructed wetland designed for metal retention and a natural depressional wetland at the Savannah River Site (Aiken, South Carolina USA). Microcosms were established from water and sediment collected from the two wetlands and incubated for eight days under natural light cycles or constant dark conditions, with copper spiked in at the experiment's midpoint. Analysis of the water microbial community revealed that light significantly altered microbial alpha diversity illustrating higher species richness and lower Pielou's evenness compared to communities under dark conditions. Compared to dark conditions, copper retention in the sediments was enhanced with natural

light under ambient conditions but decreased after the addition of a copper spike. With the use of predicted microbial functional analysis, this study found that light exposure, particularly under copper-amended conditions, significantly influenced the relative abundances of functional groups including those for sulfur respiration, photosynthesis, and methanogenesis, but there were no significant variations between wetland sources. These results highlight how abiotic factors, such as light, shape microbial processes, informing strategies for wetland conservation and management of metal contamination. More broadly, they emphasize the importance of incorporating landscape-scale features such as vegetation structure, foliage coverage, and light-mediated microbial responses into the design, management, and restoration of constructed wetlands

id #134491

Insect colonization alters resistant taxa population dynamics but not community composition in ephemeral wetland mesocosms

Christopher F Frazier^{1, 2}, Michelle H Busch³, Andrew T Karlin¹, James H Thorp^{1, 4}

1. *Kansas Biological Survey & Center for Ecological Research, University of Kansas, Lawrence, Kansas, USA*

2. *Nevada Cooperative Fish and Wildlife Research Unit, University of Nevada Reno, Reno, Nevada, USA*

3. *Michigan Natural Features Inventory, Michigan State University Extension, Lansing, Michigan, USA*

4. *Department of Ecology and Evolutionary Biology, University of Kansas, Lawrence, Kansas, USA*

Ephemeral wetlands support unique communities dominated by drought-resistant species that persist through dry periods and resilient taxa that recolonize from permanent refuges. Climate change is predicted to reduce the availability of aquatic refuges, likely reducing insect colonization and potentially shifting communities toward dominance by resistant taxa, notably large branchiopods (fairy, clam, and tadpole shrimp). To understand the impacts imposed by this shift, we conducted a mesocosm experiment using rehydrated ephemeral wetland sediment in which tanks were either open or closed to aerial colonization during different seasons. While colonization treatment significantly altered overall community composition due to the addition of insect taxa, large branchiopod community composition remained unchanged. However, we did see entirely different large branchiopod assemblages between spring and summer. Additionally, we observed significant population-level effects on large branchiopods, with differences in body size, abundance, and reproductive maturity between treatments that varied through time and by taxon. Our results suggest that while climate-induced reductions in insect colonization may not immediately alter resistant taxa richness or composition, colonization reduction could fundamentally change resistant taxa populations. These population-level changes may have important implications for ecosystem function and food webs, including impacts on migratory birds that depend on ephemeral wetland productivity. As climate change continues to alter wetland hydrology globally, understanding shifts in cryptic but functionally important ephemeral wetlands becomes increasingly critical for conservation and management.

id #135515

Anticipating ecological responses to extreme climatic events in rivers

Jonathan D Tonkin¹

1. *University of Canterbury, Christchurch, CANTERBURY, New Zealand*

Climate-driven extreme events are fundamentally reshaping riverine ecosystems, with floods, droughts, and heatwaves increasing in severity and frequency. River ecosystems, structured by their connected dendritic networks, are highly vulnerable to the propagation of localised disturbances. Both isolated and compounding extreme climatic events affect biodiversity across scales, from the erosion of genetic diversity to the loss of ecosystem functioning, and these impacts are often amplified by other underlying stressors. However, extreme events are rare by their very nature, making them challenging to understand due to limited opportunities to learn. This presents a conundrum for predictive modelling of their impacts. Faced with insufficient information about how ecosystems might respond to such events over both short and long timescales, managers may be forced to make decisions that lack scientific credibility. Traditional ecohydrological models, often based on empirical correlations within historical ranges of variability and assumptions of hydroclimatic stationarity, may be limited when anticipating the impacts of ECEs. First, extremes are rare events located in the tails of statistical distributions, thereby challenging conventional methods that focus on central tendencies, and providing questionable insight into, and opportunities to learn from, the dynamics of extreme values. Second, the effects of extremes can propagate across levels of ecological organisation and among components of a species' life history. Third, anomalous events are often treated as statistical anomalies that are scrubbed from the data prior to analyses. In light of these and other challenges, I discuss promising avenues for modelling frameworks that can be employed to predict the impacts of ECEs ranging from near-term forecasts to mechanistic approaches particularly suited to long-term scenario-based projections such as community-wide matrix population models. Distributional regression approaches, such as Generalised Additive Models for Location, Scale and Shape offer a particularly useful set of tools to understand the full range of ecological responses to extremes. Increasing risks of extreme events in freshwater ecosystems means new approaches are needed to transform the field from restoration thinking to resilience thinking.

id #136027

Hydrologic transitions drive divergent controls on dissolved oxygen behavior in non-perennial grassland streams

Erin C Seybold¹, Alexi Sommerville¹, Connor Brown¹, Amy Burgin², Sam Zipper¹, Kristen Underwood³, Donna Rizzo³

1. *Kansas Geological Survey/Department of Geology, University of Kansas, KS, United States*

2. *Ecology & Evolutionary Biology, Iowa State University, Ames, Iowa, United States*

3. *University of Vermont, Burlington, VT*

Dissolved oxygen (DO) is a master variable in aquatic ecosystems, dictating biogeochemical reactions and shaping biological communities. DO is influenced by a complex suite of biological and physical/hydrologic processes that control variation in DO over diel to seasonal time scales. In non-perennial streams that experience cycles of wetting and drying, this dynamic flow regime (including shifts between wet and dry, and flowing and pooled states) can exert strong controls on DO behavior, leading to distinct DO regimes with differing controls under divergent flow conditions.

We combine a unique dataset of high spatio-temporal resolution data from a non-perennial stream network in Kansas, USA with a novel machine learning approach (Tandem Evolutionary Algorithm) to identify the unique combinations of hydrometeorological and biological drivers that are associated with high and low DO concentration states in an intermittent river system, and assess how these drivers shift across varying flow states. Our analysis revealed equifinality in the drivers of DO concentrations, indicating that no single set of hydrometeorological or biological variables are associated with high or low DO concentrations at our study site. Rather, both high and low DO concentrations are associated with multiple distinct yet recurrent pathways. Taken together, these results show that dynamic hydrologic conditions, specifically changes in flow magnitude, velocity, groundwater contributions, and network connectivity, play a central role in reorganizing the mechanisms that drive high or low DO concentrations across a wide range of flow conditions.

id #134748

Impacts of a forested state park and weather on stream and reservoir nutrient concentrations in a Midwestern agricultural watershed

Bartosz Grudzinski¹, Owen Larson¹, Tessa Farthing¹, Eileen Rintsch¹, Thomas Fisher¹, Natalie Teale¹

1. Miami University (OH), OH, United States

Agricultural nonpoint source pollution from fertilizer runoff contributes to eutrophication in Midwestern freshwater systems. This study: 1) examines how a forested state park (Hueston Woods State Park) containing forested streams and a hypereutrophic reservoir influence nutrient concentrations in an agricultural watershed in southwest Ohio, and 2) examines relationships between changes water quality and weather (temperature and precipitation). Over a two-year period (2019–2021), biweekly water samples were collected from nine sites: four at agricultural-forested stream boundaries, four at stream-reservoir confluences, and one at the reservoir outflow. Samples were analyzed for total nitrogen (TN), nitrate (NO_3^-), total phosphorus (TP), and orthophosphate (PO_4^{3-}). Stream discharge was calibrated using stage-discharge curves and pressure transducers. Weather data for the watershed was attained from the PRISM Climate Group. Multivariate statistical analyses were performed from streamflow-weighted event mean concentrations to assess changes in nutrient concentrations within the full study area and the individual stream and reservoir sections. Multivariate analyses also revealed relationships between water quality changes and weather. Full study area results showed NO_3^- decreased by 29%, while TP increased by 60%. High TN:TP ratios throughout indicated the system was largely P limited. Nutrient changes were more strongly correlated with reservoir changes than with stream inputs, suggesting the reservoir's longer residence time and increased biogeochemical processes had a dominant influence on nutrient dynamics. Changes in water quality were related to both temperature and precipitation, but varied by water quality parameter. These findings highlight the need to consider both forested streams and reservoirs in water quality management across mixed-use landscapes.

id #136028

Does shredder size matter? Interspecific vs intraspecific effects on aquatic leaf litter decomposition

Mourine Jesire Yegon^{2,1}, Pratiksha Acharya^{1,3}, Katrin Attermeyer^{1,3}, Wolfram Graf², Simon Vitecek², Mourine Jesire Yegon^{2,1}

1. WasserCluster Lunz Biological station, Lunz am See, Lower Austria, Austria

2. Institute of Hydrobiology and Aquatic Ecosystem Management, University of Natural Resources and Life Sciences, Vienna, Austria

3. Department of Functional and Evolutionary Ecology, University of Vienna, Vienna, Austria

Biodiversity is a key driver of ecosystem functioning, with increased diversity often linked to greater ecosystem efficiency. However, the strength and direction of such relationships can vary, as intraspecific variation, including differences across ontogenetic stages, may modulate a taxon's contribution to ecosystem functioning. To explore these effects, we conducted laboratory experiments examining the influence of shredder identity, diversity, and larval stage on leaf litter decomposition and fine particulate organic matter (FPOM) production. Three caddisfly shredder taxa, *Allogamus*, *Potamophylax*, and *Sericostoma*, were collected in two different months and assigned to individual and mixed-species treatments using highly decomposable alder (*Alnus glutinosa*) leaves. The experiments incorporated distinct larval instars, reflecting natural variation in body size and ontogenetic stage, and were conducted in separate one-week incubations with a two-month interval between collection periods. We hypothesized that decomposition and FPOM production would differ between taxa and ontogenetic stages, with species diversity enhancing processing rates, and that younger, smaller larvae would exhibit higher processing rates due to elevated metabolic demands. The results showed that both shredder identity and ontogenetic stage significantly influenced leaf litter decomposition, FPOM production, and particle size distribution. Among taxa, the largest species, *Potamophylax*, exhibited the highest decomposition and FPOM production rates and produced the largest FPOM particles. In contrast, the smallest taxon, *Allogamus*, had the lowest rates and generated the smallest particles. Within each taxon, younger and smaller larval stages consistently processed leaves faster than their older conspecifics, highlighting the importance of ontogenetic variation. Mixed-species treatments showed intermediate patterns, suggesting that diversity effects were largely mediated by species identity rather than additive or synergistic interactions. These findings demonstrate that while biodiversity can enhance ecosystem functioning, the specific contributions of species and their ontogenetic stages play a critical role in shaping the magnitude and timing of leaf litter decomposition in aquatic ecosystems.

Two-station Models for Estimating Nitrate Uptake in Rivers

Robert Hall¹, Danielle Colling²

1. *University of Montana, Polson, MT, United States*

2. *University of North Carolina Chapel Hill, Chapel Hill, NC, USA*

Rivers can rapidly process nitrate via assimilatory and dissimilatory pathways. We typically measure these fluxes using experimental additions of nitrate, but the advent of high-frequency sensors allows rates to be estimated by fitting ambient nitrate dynamics to process models. Here, we couple high-frequency nitrate data collected with co-calibrated SUNAs deployed in a spring stream and in the Nyack reach of the Middle Fork Flathead River. We confronted these data with two different two-station models of uptake based on zero- and first-order kinetics, i.e., where nitrate uptake depends on its concentration. One-station models with first-order kinetics can satisfactorily estimate both assimilatory uptake via photoautotrophs and the overall rate of gross nitrate turnover, but they assume long upstream reaches. Two-station models account for reach heterogeneity. Models with zero-order kinetics showed no net change in nitrate flux across the reaches. Models with first-order kinetics showed that although there was no net change in nitrate flux, turnover was high, indicating high gross processing rates along each reach. Sensor data coupled with models allow estimation of nitrate dynamics in rivers that are far too large to easily conduct tracer experiments.

What is an algal bloom in a river? Diverse algal bloom types necessitate diverse research and management needs in flowing waters

Joanna R Blaszczyk¹, Jordan Zabrecky¹, Andrea Garcia Jimenez¹, Robert O Hall², Keith Bouma-Gregson³, John Gardner⁴, Riley Logan⁵, Laurel Genzoli⁶

1. *Natural Resources and Environmental Science, University of Nevada Reno, Reno, NV, USA*

2. *Flathead Lake Biological Station, University of Montana, Polson, MT, USA*

3. *U.S. Geological Survey, California Water Science Center, Sacramento, CA, USA*

4. *Earth, Marine and Environmental Sciences, University of North Carolina, Chapel Hill, Chapel Hill, NC, USA*

5. *Electrical & Computer Engineering Department, Montana State University, Bozeman, MT, USA*

6. *Independent Scholar, Missoula, MT, USA*

Algal blooms in freshwater ecosystems degrade critical ecosystem services provided by rivers, streams, and lakes, such as habitats that sustain biodiversity, irrigation and drinking water, and areas of significant cultural and recreational value. While planktonic algal blooms have been a major environmental issue studied in lakes, estuaries, and oceans for decades, algal blooms in rivers have only recently become an emergent environmental issue of broad scientific and management concern. Yet the meaning of the phrase "harmful algal bloom" in a river is often indistinct, making management and research priorities on this topic unclear. The complexity of river networks, which can include shallow waters, impoundments, and flow modifications, results in diverse bloom-like phenomena depending on the specific characteristics of sites within the network. In this talk, we define key types of algal blooms in rivers and discuss their differences and similarities, including benthic toxin-producing (i.e., harmful) cyanobacterial mats, nuisance benthic filamentous algal blooms, and planktonic blooms, either originating from within rivers or upstream lentic water bodies (i.e., reservoirs or lakes). Impacts from these different types of bloom events are on a spectrum from nuisance to harmful. We draw on recent literature to evaluate the potential of different tools to detect, quantify, predict, and manage these diverse bloom types (e.g., remote sensing, in-situ sensors). Finally, we outline key research priorities to better understand and predict the spatial extent, temporal dynamics, and ecological consequences of algal blooms in rivers.

Comparing the effects of climate and land use changes on stream communities

John R Olson¹, Lucas Raymond¹

1. *California State University Monterey Bay, Seaside, CA, United States*

Climate and land use changes have both disrupted freshwater ecosystems, and these changes will continue and potentially worsen into the future. These changes will favor some taxa and impair others, resulting in changes in richness and composition. However, the amount of impact due to changes in climate relative to changes in land use are not clear. Understanding which changes are likely to result in the greatest changes in composition and losses of biodiversity is necessary for effectively prioritizing actions and resource commitments. Therefore, we built models predicting richness and community composition for both fish and macroinvertebrates in US rivers and then applied forecasts of future climate and land use to these models to predict how richness and composition will change in response to environmental changes. We used fish and macroinvertebrate occurrence data collected as part of the EPA National River and Stream Assessment as response data for random forest models of assemblage richness and multivariate random forest models of assemblage composition. For predictors in our models, we used a combination of static environmental factors, like soils and topography, and dynamic factors like climate and land use. We derived values for dynamic factors from climate (MACA, CMIP5) and land use (FORE-SEE) models, using current projections for richness and composition for model training and then using mid-21st century projections to predict future richness and composition. The fish richness model explained 65% of the variation and the macroinvertebrate model explained 38%. For both models, temperature and forested land use were the most important dynamic predictors and had approximately equal influence. But maps of the

predicted changes indicated regional differences in richness change, with increasing fish and macroinvertebrate richness occurring in the western US and decreasing richness occurring in the Appalachians. Increased richness coincided with increased precipitation in the west, whereas decreased richness in the Appalachians was associated with increased temperatures. Improving our understanding of the relative impacts of climate and land use changes will help us make better decisions about priorities for management and mitigation of these threats.

id #135006

Influence of trash substrate on biofilm growth and metabolic activity in urban streams

Gianna Parrish¹, Megan Fork¹

1. West Chester University, West Chester, Pennsylvania

Trash accumulation in aquatic ecosystems is a growing environmental concern in part because of how chemical and physical effects of trash can harm primary producers such as biofilms. The food webs in aquatic ecosystems like streams are supported by biofilms, communities of algae and other microorganisms growing on substrates in the water. Biofilms play a crucial role in stream ecosystems by contributing to nutrient cycling and organic matter processing, yet their growth is dependent on access to nutrients and stable surfaces for attachment. Trash in urban streams can be a substrate for biofilms to grow when natural cobbles and wood are buried by sediment. This study investigated how biofilm growth and metabolism differ when stream biofilms colonize various types of trash. We found that biofilm growth and metabolic activity differed significantly among trash substrate types. Gross primary production (GPP) varied across substrates ($p = 0.0014$), with more textured materials, including avocado, fritted glass, and plastic bottles, supporting the highest GPP, while smooth plastic film supported the lowest. Community respiration also differed strongly by substrate ($p = 4.56 \times 10^{-6}$), with organic materials such as avocado and cardboard exhibiting higher oxygen consumption than plastic substrates. Biofilm biomass (AFDM) differed significantly among substrates ($p = 0.0001$), with avocado accumulating the greatest biomass and plastic film the least. Areal chlorophyll-a density showed weaker substrate effects ($p = 0.087$), though fritted glass supported higher chlorophyll-a than smoother plastics. Across substrates, GPP was positively related to AFDM, though this relationship was not statistically significant ($p = 0.1338$). Overall, textured and organic substrates supported greater biofilm biomass and metabolic activity than smooth plastics, indicating that surface properties and material composition strongly influence biofilm colonization and function in urban streams.

id #136030

Quantifying stream metabolic stability to storm disturbance using integrated resistance and resilience metrics

Michael C Beall¹, Scott C Brooks², Jesus D Gomez-Velez³, Erin R Hotchkiss¹

1. Biological Sciences, Virginia Tech, Blacksburg, Virginia, United States

2. Oak Ridge National Lab, Oak Ridge, Tennessee, United States

3. Civil and Environmental Engineering, University of Iowa, Iowa City, Iowa, United States

Storms are a dominant driver of variability in stream metabolism, yet comparing responses across events is challenging because disturbances differ widely in magnitude, duration, and recovery dynamics. We developed a disturbance–response framework that integrates resistance (impact magnitude) and resilience (recovery efficiency) to quantify whole-ecosystem metabolic stability during storm events. Storm impacts were calculated as cumulative departures of gross primary production (GPP) and ecosystem respiration (ER) from antecedent conditions and normalized by disturbance duration, and these components were combined into a unitless stability index (Ψ).

We applied this framework to >10 years of high-frequency dissolved oxygen data from East Fork Poplar Creek, Tennessee, focusing on the downstream monitoring station (efk5.4). A total of 1,560 storm events were delineated using the DMCA-ESR method, of which 115 groups met criteria for metabolic analysis and were classified as isolated or multi-storm (successional) disturbances. Preliminary results indicate that metabolic departures during multi-storm periods were comparable to or slightly greater than those during isolated storms. Instead, stability differences were driven by faster recovery. Median recovery factors increased by ~50% for GPP and ~10–15% for ER, resulting in ~20–30% higher overall stability (Ψ) during multi-storm periods. Across events, Ψ closely tracked recovery metrics rather than resistance, indicating that resilience dynamics, rather than reduced disturbance magnitude, primarily governed metabolic stability.

These results suggest that prior disturbances condition ecosystem responses to subsequent storms and demonstrate that recovery processes, rather than reduced disturbance magnitude, dominate whole-stream metabolic stability. This framework provides a process-based and transferable approach for comparing metabolic stability across heterogeneous hydrologic regimes.

id #134495

Thawing ground and changing streams: rusting rivers in Arctic Alaska

Lauren Yancy^{2,1}, Jeffrey Muehlbauer^{2,3}

1. Alaska Department of Fish and Game, Fairbanks, AK

2. University Alaska Fairbanks, AK, United States

3. Alaska Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Fairbanks, AK

The Arctic is experiencing rapid warming that is contributing to permafrost thaw. This increases hydrologic connectivity to previously sequestered metals in the Earth's active layer, mobilizing these metals and into streams located in remote, undeveloped ecosystems. The resulting chemical processes produce a visually striking rusting phenomenon in the Arctic's waterways. Despite increasing reports of rusting rivers across the Brooks Range, the ecological consequences of these metal-rich waters for stream biota remain poorly quantified. This study evaluates spatial and temporal patterns in stream biota across

a gradient of rusting influence in the Wulik River drainage of northwest Alaska. Using long-term biomonitoring data, streams were categorized by rusting influence based on field water quality metrics and visual indicators. We examined responses of periphyton biomass, benthic macroinvertebrate abundance and richness, and macroinvertebrate community composition across rusted and control sites. Multivariate analyses were used to assess shifts in community composition at multiple taxonomic resolutions, and mixed-effects models were used to evaluate biological responses across space and time. Preliminary results indicate declines in periphyton biomass and benthic macroinvertebrate abundance and richness at sites influenced by rusting seeps relative to control sites. Early analyses suggest that biological metrics vary consistently along a gradient of rusting influence, with stronger responses observed at sites experiencing persistent rusting conditions. Community composition analyses are ongoing, but initial ordination results suggest differences among rusted and non-rusted sites that warrant further investigation. Together, these results suggest that permafrost thaw and associated metal mobilization, linked to a changing cryosphere, influence biological conditions in Arctic streams.

id #135519

Characterizing variable inundation in non-perennial streams with a terrestrial model and commercial satellite imagery

Etienne Fluet-Chouinard¹, Sundar Niroula¹, Thiago Tayer², Peter Regier¹, Xingyuan Chen¹, James Stegen¹

1. Pacific Northwest National Laboratory (PNNL), Richland, WASHINGTON, United States

2. The University of Western Australia, Perth

Non-perennial streams are among the largest emitters of carbon fluxes per unit land area, yet their global contribution remains uncertain due to the challenges of characterizing their hydrological conditions, including inundation duration, controlling rates of sediment respiration. Observation of intermittent hydrology in narrow channels can benefit from the finer spatial and temporal resolution of recent commercial satellite imagery. Yet, characterizing intermittent headwater hydrology at the catchment scale remains a challenge to benchmark regional land surface models and upscale biogeochemical rates.

Here, we combine commercial satellite imagery and in situ wildlife cameras with clustering and segmentation models to generate near-daily water cover over the intermittent channels in the Satus Creek watershed located within the Yakima River Basin. To construct a training set of water presence, we tasked Umbra satellites to acquire sub-meter X-band Synthetic Aperture Radar (SAR) image resolution over wildlife camera locations targeting seasonal wet-dry transition periods across the watershed. Umbra images and manually interpreted water masks were paired with near-daily multispectral imagery from PlanetScope (3-5m resolution) to estimate water cover fraction per stream reach using the WaterDetect algorithm.

The resulting near-daily water masks over 2022-2025 were compared to channel ponded depth simulated with the Advanced Terrestrial Simulator (ATS) process-based model for individual 100m-long reach. We found variable agreement between the data sources across the catchment, with persistent errors over narrow headwater channels, overhanging vegetation canopy cover, shadows cast over the channel as well as over braided channels. We also evaluated the catchment-wide connectivity of the river network from the model and imagery to characterize how intermittency affects flow of water and matter in the catchment. This work represents an initial assessment of the lower detection limit of meter-scale imagery over narrow stream channels, and further steps will explore methods fusing optical and SAR commercial imagery to refine the water detection.

id #135775

Targeting biodiversity and standardization with a modified aquatic macroinvertebrate sampling method

Jacob Lott^{1,2}, Dean Fletcher², Stacey Lance², Dan Peach², Raven Bier²

1. Odum School of Ecology, University of Georgia, Athens, GA, USA

2. Savannah River Ecology Lab, University of Georgia, Aiken, SC, USA

Aquatic macroinvertebrates are vital to stream ecosystem function, including as food sources for higher trophic levels and are used as indicators of stream condition due to sensitivities to water quality and habitat disturbance. This study aims to synthesize several aquatic macroinvertebrate bioassessment methodologies into a single protocol that better captures biodiversity and reproducibility across sites and time. Incorporating sampling protocols from U.S. Environmental Protection Agency (USEPA), Georgia Environmental Protection Division (GAEPD), and South Carolina Department of Environmental Services (SCDES), to create a revised sampling protocol with the standardization and habitat-based sampling approach of USEPA/GAEPD, while sampling large areas and multiple habitat types to increase captured biodiversity, found in the SCDES protocol. To determine the number of samples within a stream reach needed to increase the biodiversity sampled while remaining standardized, we conducted a preliminary study using four different streams at the Savannah River Site (Aiken, SC, USA) across a gradient of size and degradation. We took samples from six leaf packs, six snags, five bank samples, and nine sand samples (three from pools, runs, and flats each). Specimens were identified to the Family level. To understand how biodiversity varied with sample number and habitat type, species accumulation curves and Venn diagrams were created. Preliminary analysis showed that 80% of the Family level aquatic macroinvertebrate biodiversity was found within three samples for leaf packs, snags, and banks, and six samples for sand habitats (two pools, flats, and runs each). Our preliminary study found 37, 50, 55, and 62 families at our four sites, compared to 15, 25, 14, and 30 families respectively at the same sites found during a 2022 SCDES protocol sampling event. Leaf packs largely contained families found within other habitats: only 0-5.4% of families were unique to leaf packs. These results allowed us to reduce the number of samples taken in each habitat in our follow up sampling comparing this methodology, SCDES, and GAEPD in the same streams, while also revealing the biodiversity that can be missed with non-standardized, less intensive sampling efforts.

id #136032

Unraveling drivers of variation in riparian vegetation recovery following wildfire

Leah Simantel¹, Kellie J Carim², Sean A Parks², Lisa A Eby¹

1. *Ecosystem & Conservation Science, University of Montana, Missoula, MT*

2. *Aldo Leopold Wilderness Research Institute, Missoula, MT*

The effects of fire on stream ecosystem processes and aquatic fauna are highly variable. This variation is, in part, explained by the extent and severity of the burn, as well as the time lag of the sampling and the fire event¹. Despite extensive research on fire effects on vegetation and post-fire recovery in terrestrial landscapes^{2,3}, comparatively little is known about how riparian vegetation responds to fire and what factors influence recovery. A better understanding of processes influencing riparian recovery is needed to link the effects of wildfire to stream ecosystems.

This study examined topographic, climatic, and wildfire-related drivers of riparian recovery in watersheds of the Bob Marshall Wilderness Area (Montana) using Landsat Normalized Difference Vegetation Index (NDVI) data spanning 1984-2024. We quantified changes in vegetation as the difference between pre-fire and post-fire greenness (Δ NDVI) for individual Landsat pixels in riparian areas up to 20 years post-fire. Fire severity strongly influenced riparian Δ NDVI during the three years following fire, with moderate and high-severity sites exhibiting significantly reduced greenness compared to low-severity and unburned controls. Even though differences in Δ NDVI among fire severity groups diminished over time, Δ NDVI remained significantly lower in sites that burned at high-severity relative to unburned sites 20 years post-fire. Approximately 90% of high-severity pixels achieved pre-fire levels after 19 years, compared to 16 and 3 years for moderate- and low-severity pixels, respectively.

We used random forest models to assess the influence of fire severity, site specific indices for vegetation type, soil moisture, immediate post-fire drought conditions, and annual seasonal climate variables (snowpack, summertime temperature and precipitation) on riparian recovery. Though all covariates were influential, fire severity was the strongest driver. Seasonal climatic variables influenced Δ NDVI at all time points, but their effect sizes and directional relationship varied with time since fire. These results improve our understanding of fire impacts on riparian zones and the multi-decadal influences on variation in riparian recovery, highlighting the importance of long-term studies to understand legacy impacts associated with wildfire disturbance.

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2. Hao, B., Xu, X., Wu, F., & Tan, L. (2022). Long-Term Effects of Fire Severity and Climatic Factors on Post-Forest-Fire Vegetation Recovery. *Forests*, 13(6), 883.
3. Kurbanov, E., Vorobev, O., Lezhnin, S., Sha, J., Wang, J., Li, X., ... & Wang, Y. (2022). Remote sensing of forest burnt area, burn severity, and post-fire recovery: A review. *Remote Sensing*, 14(19), 4714.

id #136033

Insights from long-term zooplankton and phytoplankton monitoring in a large, regulated, oligotrophic river

Mehzabeen Mannan^{1,2}, Kurt Chowanski^{1,2}, Genny Hoyle³, Lisa Kunza^{1,2}

1. *Center for Sustainable Solutions, South Dakota School of Mines and Technology, Rapid City, SD, USA , Rapid City, SD, USA*

2. *Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines and Technology, Rapid City, SD, USA , Rapid City, SD, USA*

3. *Environmental Department, Kootenai Tribe of Idaho, Bonners Ferry, ID, USA, Bonners Ferry, ID, USA*

Plankton dynamics in large rivers are shaped by strong hydrologic regulation and pronounced seasonal variability, yet long-term patterns remain unclear. We examined phytoplankton and zooplankton abundance from 2009 to 2021 across 13 mainstem and 4 tributary sites in the Kootenai River basin. Analysis of the annual April-June data showed that phytoplankton communities were generally dominated by Bacillariophyta, while zooplankton assemblages across all sites were consistently dominated by Rotifera. However, during June 2019-2021, phytoplankton communities across the river network exhibited a pronounced shift away from diatom dominance toward increased contributions from microflagellates and picoplankton, coinciding with elevated zooplankton abundance. Further analysis of extended seasonal records in two mainstem sites (from early spring through late summer) revealed that phytoplankton abundance exhibited consistent spring-early summer peaks while zooplankton's abundance displayed a delayed response compared to phytoplankton, suggesting bottom-up control during the early growing season. Moreover, several mainstem sites exhibited recurrent periods of elevated cyanobacteria abundance during mid to late summer (June-September). In contrast, tributary sites displayed more episodic and short-lived cyanobacteria increases, with sharp peaks occurring in some years where cyanobacteria temporarily equaled or exceeded Bacillariophyta abundance. Relationships between zooplankton abundance and environmental drivers were seasonally variable, with temperature effects strongest in spring and discharge dominating in summer. This multi-year analysis advances understanding of phytoplankton and zooplankton dynamics at the base of the food web, with direct relevance for adaptive management decisions to support higher trophic levels in oligotrophic river systems.

id #133986

Longitudinal patterns in hyporheic stygobiont distribution in a karst spring-fed river

Ben Hutchins¹, Kenneth Sparks², Safra Altman³, Kathryn Perez⁴, Zoey Chanin¹, Caroline Mierzejewski¹, Benjamin Schwartz¹

1. *Texas State University, San Marcos, TX, United States*

2. *City of Austin, Austin, TX, United States*

3. *U.S. Army Corps of Engineers, Vicksburg, MS, United States*

4. University of Texas Rio Grande Valley, Edinburg, TX, United States

Groundwater-obligate taxa (stygobionts) exhibit variable degrees of habitat specialization, ranging from micro-habitat specialists to generalists occurring across multiple habitat types. The karstic Edwards Aquifer is the most biodiverse groundwater ecosystem in North America, and among the most thoroughly studied. However, little is known about the distribution of stygobionts in the hyporheic zone of spring-fed rivers downstream from the aquifer. The goals of this research were to 1) assess the potential of the hyporheic zone as habitat for 'karst-adapted' stygobionts off-karst, and 2) to determine how stygobiont diversity and abundance are related to spatial (i.e., distance from karst) and environmental effects in the hyporheic zone of a spring-dominated river system. We enumerated stygobionts and quantified environmental characteristics for 55 hyporheic samples from 11 sites in the San Marcos River between three and 140 river kilometers downstream from the rivers' karstic Edwards Aquifer spring source. Linear mixed effects models and variance partitioning were used to investigate relationships between community metrics (richness, abundance, and diversity) and spatial/ environmental parameters. Eight stygobiontic taxa were recovered in hyporheic samples, but both stygobiont richness and diversity decreased with increasing distance from karst. Stygobiont richness was positively correlated with proximity to karst and hydraulic conductivity, which covaried and explained similar amounts of model variance. Stygobiont abundance was positively correlated with hydraulic conductivity and dissolved oxygen, which also explained similar amounts of model variance. Stygobiont diversity was positively correlated with proximity to karst and dissolved oxygen and negatively correlated with temperature, all of which covaried. Stygobiont communities in the hyporheic zone can be structured by both local environmental conditions, via habitat filtering, and by spatial effects (i.e., proximity to adjacent aquifers). The Edwards Aquifer appears to exert a major influence on community structure in the downstream hyporheic zone directly (as a source population for some stygobionts) and indirectly (influencing environmental conditions like hydraulic conductivity and temperature). Habitat specificity is poorly characterized for most rare stygobiont taxa, inhibiting effective conservation planning and implementation, but studies like this expand our understanding of occupied habitats. Maintaining connectivity between rivers and adjacent groundwater aquifers is critical for groundwater-obligate hyporheic residents.

id #136035

Investigating Seasonal Changes in Microplastic Pollution in the Davidson River, NC

Ariel Podlogar¹

1. Brevard College, North Augusta, SOUTH CAROLINA, United States

Microplastics are a contaminant of emerging concern, but very little is known about their presence and movement in freshwater ecosystems, and how seasonal changes can affect that movement, especially in the Appalachian Mountains. With the Appalachian region being the headwaters of many rivers, this experiment was conducted to begin filling the gaps in our understanding of microplastics in freshwater across the changes in seasons. By examining if there is a significant amount of microplastics being added to the Davidson River, which has its headwaters in the Pisgah National Forest, two sample sites were selected based on proximity to the headwaters and the mouth of the river. Three water samples were taken from each location, as well as a control blank sample, each size fractioned at 125 µm, 250 µm, and 500 µm. The samples were then separated first by density with CaCl₂, then by digestion of organic materials, before being filtered to be counted under a microscope. Seasonal changes affect water flow, which could impact how microplastics disperse and the overall impact they have on freshwater ecosystems. The findings of this study will provide insights into the effects of changes in climate on the distribution and dynamics microplastics pollution in river ecosystems.

id #136036

Understanding collaborative care of an urban river: An application of the Stewardship Mapping and Assessment Project and Environmental Stewardship Framework

Lauren Emer¹, Aaliyah Saintil¹, Natalia Piland², Elizabeth P Anderson¹

1. Florida International University, Miami, FL

2. The Center for Environment, Community, & Equity, Minneapolis

Governmental organizations, nonprofits, universities, neighborhood groups, and independent individuals engage in stewardship behaviors to protect and care for the Little River Watershed in Miami, Florida. The large number of stewards makes it challenging to keep track of everyone's stewardship efforts. As a result, organizations may engage in redundant or conflicting activities, miss out on relevant collaborations, or otherwise engage in stewardship that could be more effective if better organized. To support stewards in performing stewardship as efficiently and effectively as possible, we surveyed and interviewed organizations and individuals that care for the Little River Watershed. We used thematic coding, spatial analysis, and social network analysis to identify trends in a) stewardship locations, b) activities, and c) patterns of collaboration. We found that stewardship of the watershed is focused on the eastern portion of the river in public access parks. Stewardship functions ranged from community engagement (> 75% of surveyed stewards) to monitoring (50% of surveyed stewards). Organizations were not always connected with groups with similar activities, stewardship locations, or mission statements. Ocean conservation groups were the most aligned in stewardship locations and activities. Capacity in terms of time and resources emerged as a key theme in our analysis of stewardship barriers. These findings highlight areas of improvement for Little River stewardship including connecting potential collaborators, initiating stewardship in the western reaches of the river, and identifying resources for capacity building. Overall, this research strategy generated useful insights and action areas for supporting efficient care of the Little River and collaboration of diverse organizations. Researchers and managers should consider this stewardship mapping and assessment method to organize and better understand the watersheds in which they work.

id #134757

Spring-fed tributaries function as seasonal refuges for juvenile masu salmon

Masaru Sakai¹, Junjiro N Negishi², Kohma Arai³, Itsuro Koizumi², Kenta Iwasaki⁴

1. National Institute for Environmental Studies, Miharu, FUKUSHIMA, Japan

2. Hokkaido University, Sapporo, Japan

3. University of California, Davis, Davis, USA

4. Forestry and Forest Products Research Institute, Tsukuba, Japan

There is a growing interest in interactions between spring-fed and runoff rivers for conserving the health of river ecosystems. In particular, the stable temperature regime and slow current velocities in spring-fed streams are expected to provide refuges for fishes from flow and temperature extremes. However, how fishes seasonally utilize combinations of spring-fed and runoff rivers as habitat remains poorly known. Here, we investigated seasonal changes in population density and muscle carbon and nitrogen stable isotope ratios of juvenile masu salmon (*Oncorhynchus masou*) in spring-fed tributaries and adjacent runoff mainstems in the Shubuto River system, northern Japan. We also collected benthic macroinvertebrates as potential prey items for fish at the study site. The biomass of benthic macroinvertebrates was one-order of magnitude greater in spring-fed tributaries than in runoff rivers throughout the season, associated with higher fine sediment cover and greater standing stock of detritus on the streambed. Juvenile masu salmon were especially aggregated in the spring-fed tributaries in winter compared to other seasons. Among seasons, the stable isotope analysis revealed that individuals that presumably had recently inhabited mainstems were most frequently caught in spring-fed tributaries in February. These results suggest that warmer and slower current velocities of spring-fed tributaries can provide attractive wintering habitat for juvenile masu salmon. Our findings highlight the important ecosystem functions of spring-fed tributaries in structuring animal communities within river ecosystems.

id #133990

Drying history, not connectivity, controls nitrogen processing in nonperennial streams

Sarah M Flynn^{2,1}, Erin C Seybold¹, Amy J. Burgin³

1. Kansas Geological Survey, University of Kansas, Lawrence, KS, United States

2. Ecology and Evolutionary Biology, University of Kansas, Lawrence, KS, United States

3. Ecology, Evolution, and Organismal Biology, Iowa State University, Ames, IA

Nonperennial streams make up 80% of stream miles in Kansas, yet we still struggle to predict how drying and rewetting shapes biogeochemical processing. Hydrologic variation alters sediment redox conditions, organic matter quantity and quality, and shifts microbial community compositions. Most of our understanding of biogeochemical processes comes from perennially flowing systems. The studies that consider nonperennial streams do so when surface water is present, leaving a gap in how drying regulates biogeochemical processes. To address this, we performed monthly samplings and incubations across three watersheds. We used high-frequency water presence/absence data to characterize two network-scale and four local-scale drying metrics. We utilized structural equation modeling to understand the effects of drying on organic matter content and microbial N processes. Biogeochemical responses were driven more by site-level drying history than by network-scale connectivity. Dry sites accumulated more sediment organic matter, and metrics of local drying strongly predicted organic matter content ($\beta = 0.45$). Organic matter enhanced both DEA ($\beta = 0.45$) and nitrification ($\beta = 0.18$), but long dry durations suppressed the positive effect of organic matter on DEA ($\beta = -0.21$) (slope = 0.66 with low drying frequency and slope = 0.24 with high drying frequency). Nitrification increased with longer wet durations ($\beta = 0.28$), but intense wet-up events reversed the stimulatory effect of organic matter ($\beta = -0.30$). Simple-slope analysis showed that organic matter content stimulated nitrification under network-scale contraction conditions (slope = 0.48) and became slightly negative during network-scale wet up (slope = -0.12). NO_3^- concentrations decreased with increasing frequency of drying and increased with increasing nitrification potential. NH_4^+ concentrations increased with more frequent drying and decreased with DEA. While drying promotes the accumulation of organic matter in sediment, thus promoting NO_3^- removal through denitrification, it also suppresses nitrification and promotes the accumulation of NH_4^+ . Increasing NH_4^+ concentrations can result in nutrient pollution, harmful algal blooms, and oxygen depletion. With the ubiquity of nonperennial streams increasing in Kansas and globally, understanding the impacts of drying frequency and duration on N removal processes is imperative to inform sustainable management practices.

id #135014

20 Years and Counting: Long-term Monitoring of the Spokane Valley - Rathdrum Prairie (SVRP) Aquifer in Washington

Amy Sumner¹

1. Spokane County, Spokane, WA, United States

The Spokane Valley - Rathdrum Prairie (SVRP) Aquifer underlies about 370 square miles of the relatively flat valley spanning Washington and Idaho. Spokane County's SVRP Aquifer Long Term Water Quality Monitoring Program is a voluntary effort to monitor the quality and quantity of this important groundwater resource. Through this effort, consistent data has been collected since 1998 at monitoring locations across the aquifer. Data includes 17 inorganic chemicals, field parameters, and groundwater levels. A thorough analysis of the County's data from 1999 - 2019 was published in 2023, providing insights into the processes and human activities influencing groundwater quality and water levels. This presentation will provide background on the monitoring program, highlight the major findings of 20-year analysis, and discuss the next steps for evolving the monitoring program to address future concerns.

1. Sumner, A. (2023). Spokane Valley - Rathdrum Prairie Aquifer Spokane County, WA Long-term Monitoring Program 20-year Analysis (1999 - 2019). Spokane County Public Works, Water Resources.

id #135782

Relationships between benthic and hyporheic insect communities across environmental gradients in Texas rivers

Erika E Frandsen¹, Benjamin F Schwartz^{1,2}, Ben Hutchins², Weston Nowlin¹, Miranda Sams¹, Kierra Determan¹, Kirby Wright¹, Jessica Kittridge¹, Catherine Cline¹, Brook D Herman³, Erika E Frandsen⁴

1. *Biology, Texas State University, San Marcos, TX*
2. *Texas State University Edwards Aquifer Research and Data Center, San Marcos, TX*
3. *U.S. Army Engineer Research and Development Center, Vicksburg, MS*
4. *Biology, Texas State University, San Marcos, TX, United States*

Aquatic insects occur in both benthic and hyporheic habitats of lotic ecosystems. Although aquatic insects are typically assumed to inhabit benthic environments, they also utilize the hyporheic zone as a refuge from extreme conditions, predation, or drying. However, most studies examining insects in the hyporheic zone have focused either on single mesohabitat units (i.e., riffles) or relatively short reach-based approaches, with few investigations at larger scales. This highlights the need to understand relationships between benthic and hyporheic insect communities and to determine how those assemblages differ under specific environmental conditions, especially at large spatial scales (i.e., whole watersheds).

From 2021 to 2022, we sampled aquatic insects in the benthos and hyporheos at 64 riffle sites across 9 rivers across Texas, USA, along with a suite of site-specific environmental and hydrologic variables (e.g., dissolved oxygen, water temperature, conductivity, dissolved nutrients, discharge, hydraulic conductivity). Sites span wide hydrologic, geologic, land use, climatic, and ecoregion gradients, from sandy mainstem rivers in wet east and dry north Texas to rocky groundwater-dominated streams in sub-tropical central and semi-arid western Edwards Plateau. Using this dataset of paired benthic – hyporheic samples, we investigated relationships between benthic and hyporheic insect communities, asking whether the hyporheic insect community comprises consistent taxonomic or functional subsets of the benthic community across all river basins, or if the two communities vary independently. We also investigated relationships between environmental conditions and abundance and diversity of insects in the hyporheic zone.

The study is the first to investigate relationships between insect communities in the benthic and hyporheic zones across multiple large watersheds, and to quantify relationships between the two communities across large spatial, hydrologic (e.g., river position), and environmental gradients. Our results will support models that seek to understand how aquatic ecosystems will change under future environmental conditions, thus supporting more effective management and restoration of aquatic ecosystems.

id #134759

Flow regime shifts alter microbial extracellular enzyme production in non-perennial river systems

Claire R Utzman¹, Andrielle Larissa Kemajou Tchamba², Sarah Flynn³, Lydia H Zeglin¹

1. *Kansas State University, Manhattan, KS, United States*
2. *Department of Biology, University of Mississippi, University, MS, United States*
3. *The University of Kansas, Lawrence, KS*

Non-perennial systems experience repeated drying-rewetting cycles, which regulate microbial activity and nutrient processing, yet the functional consequences of dynamic flow patterns remain poorly resolved. To understand the effects of variable surface water flow on lotic microbial communities, we conducted a flow manipulation experiment on Shane's Creek at the Konza Prairie Biological Station in northeastern KS, USA, and conducted continued sampling through a natural dry-down period. We evaluated how microbial populations and decomposition enzyme activities in contrasting microhabitats (benthic sediment vs. epilithon) and geomorphological features (pool vs. riffle) responded to flow alteration. We quantified sediment and epilithon DNA concentrations, bacterial and fungal populations, extracellular enzyme activities (EEA, β -glucosidase (BG), N-acetylglucosaminidase (NAG), and alkaline phosphatase (AP)) across high-flow, drying, dry, and rewetting timepoints. We predicted that population size and enzyme production would increase with hydrologic connectivity due to higher nutrient delivery, with habitat-specific differences such as increased sensitivity to drying in epilithic biofilms and greater resistance to drying in sediment habitats.

Changes in surface water flow affected decomposition enzyme activities differentially among habitats. In riffles, epilithon NAG activity increased upon rewetting ($p = 0.001$), while sediment NAG activity increased during drying but decreased following rewetting ($p < 0.001$). In pools, sediment NAG activity was comparatively stable through changing flows ($p = 0.012$). BG and AP in sediments and epilithon also differed consistently between pools and riffles ($p < 0.03$). All BG:NAG activity ratios declined during the natural dry-down, indicating increasing N limitation, but showed no flow manipulation response. DNA yield, a proxy for microbial biomass, was more stable in epilithon than sediments, in that riffle DNA yield was low during high-flow periods, and increased during the natural drying period ($p < 0.001$). Thus, results fit our predictions, though with greater sensitivity in riffle sediments than expected. Collectively, the study demonstrates how drying-rewetting cycles affect microbiota in a non-perennial stream, with epilithon function and riffle sediment biomass exhibiting heightened sensitivity to dynamic surface flows. Further analysis of integrated decomposition rates and fungal and bacterial gene diversity will be used to assess microbial composition and function in response to flow modifications.

id #136039

Successional controls on trophic transfer in an oligotrophic river

Augustine Sitati¹, Michael Zampini^{2,3}, Michael Wulf², Saeed Kariunga², Jane C. Marks², Mary E. Power³, Steve A. Thomas¹

1. *Department of Biological Sciences, University of Alabama, Tuscaloosa, Alabama, USA*

2. Department of Biological Sciences and Center for Ecosystem Science and Society, Northern Arizona University, Flagstaff, Arizona, USA

3. Department of Integrative Biology, University of California, Berkeley, California, USA

Variation in the availability and quality of food sources for primary and secondary consumers shapes how nutrients enter and move through river food webs. In the nitrogen-limited South Fork Eel River (SFER), *Cladophora glomerata* dominates primary producer biomass during the summer growing season and, through colonization by epiphytic diatoms, becomes a key energetic and nutritional resource for consumers. In this study, we assessed microbial community composition associated with three nutritionally distinct successional stages of *Cladophora* (early green, middle yellow, and late red) in the SFER using 16S and 18S rRNA amplicon sequencing. We then used stable isotope tracing (^{13}C & ^{15}N) to measure primary productivity and subsequent transfer of fixed Carbon (C) and nitrogen (N) to turf-weaving midges (*Pseudochironomus richardsonii*; a dominant grazer) and onward to aeshnid larvae (*Aeshna walkeri*; a common predatory macroinvertebrate) across the three stages of *Cladophora* differing in epiphyte load. Algae, midges, and predators were incubated in flow through buckets in a shallow open river reach for eight days with six replicates per successional stage. C- and N-fixation were primarily controlled by *Cladophora* successional stage. The red stage exhibited higher mass-specific N-fixation rates than earlier stages (green < yellow < red), whereas mass-specific C-fixation declined from green to red stage *Cladophora* (green > yellow > red). C and N transfer to midges and aeshnids was also dependent on *Cladophora* successional stage and was greatest for the late stage *Cladophora*. Sequencing results suggest that these differences were driven by microbial variation, with red-stage *Cladophora* supporting a higher diversity and abundance of diazotrophic epiphytic diatoms (e.g., *Epithemia*). Together, these findings suggest that *Cladophora* stages differentially contribute to upward trophic transfer, with later stages pushing more C and N up the food chain because of *Cladophora*'s association with microbial partners. This study highlights the central role of seasonal algal variation and algal-diatom associations in regulating algal quality, nutrient cycling and energy transfer to consumers in stream food webs.

id #136040

Age of the two-stage: time since floodplain restoration mediates the hydrologic signature of storms in an agricultural watershed

Caroline G.T. Anscombe¹, Jennifer L. Tank¹, Ursula H. Mahl¹, Mohamed A. Aboelnour², Emma M. Thrift-Cahall¹, Todd V. Royer³, Caroline Anscombe¹

1. Department of Biological Sciences, University of Notre Dame, Notre Dame, IN, United States

2. Environmental Change Initiative, University of Notre Dame, Notre Dame, IN, United States

3. Paul H. O'Neill School of Public and Environmental Affairs, Indiana University, Bloomington, IN, United States

Stream channelization and subsurface tile drainage in agricultural watersheds optimizes runoff but also increases discharge variability, creating flashy hydrological networks. The two-stage ditch practice, which restores lateral floodplain benches adjacent to the stream channel, has emerged as a conservation strategy that can improve water quality by increasing water residence times and bioreactive surface area. However, limited research has quantified the efficacy of two-stage ditches in reducing storm-driven flow variability of agricultural streams or how the influence of constructed floodplains may change over time. We explored how floodplain restoration has altered flow regimes in a small agricultural watershed in Indiana. We used a 17 year dataset to compare flow characteristics before (2008-2017) and after (2018-2025) the phased implementation of 6.4 km of floodplain in Shatto Ditch (Kosciusko Co, IN). We delineated 24 storms, 12 before and 12 after, to examine the impact of floodplain implementation. We analyzed the runoff coefficient (RC) for each storm, which is the proportion of rainfall that becomes streamflow, and show a trend of lower RCs after floodplain restoration, although this trend was not significant (Pre: 0.25 ± 0.05 , Post: 0.19 ± 0.03 ; ANOVA, $p > 0.05$). Interestingly, in the first two years after construction, RCs were similar to pre-construction values (0.26 ± 0.03). In contrast, 3-4 yrs post-construction, we documented a significant decrease in RC's (0.14 ± 0.03 ; ANOVA, $p < 0.05$), indicating that more precipitation is retained as floodplains age. Similarly, we documented a significant decrease in precipitation-normalized peak discharge ($\text{L mm}^{-1} \text{ s}^{-1}$) for storms 3-4 yrs after floodplain restoration compared to before or soon after construction (Pre: 33.45 ± 2.57 , 1-2 yrs post: 47.45 ± 9.77 , 3-4 yrs post: 14.0 ± 2.39 ; ANOVA, $p < 0.05$). As constructed floodplains age, and vegetative cover is established, their ability to increase water residence times likely increases. Our results suggest that the two-stage ditch is an effective restoration practice for "naturalizing" storm-mediated flow variation in highly-modified agricultural streams, and that the efficacy of constructed floodplains increases over time.

id #133737

Navigating uncharted waters: establishing practical and cost-effective methodologies for environmental RNA (eRNA)

Elise D Snyder¹, Teya C Broyles¹, Sarah A Molinaro¹, Mark A Davis², Elise Snyder¹

1. Illinois Natural History Survey, University of Illinois at Urbana-Champaign, Champaign, Illinois, United States

2. University of Illinois at Urbana-Champaign, Champaign, ILLINOIS, United States

Nucleic acids (DNA and RNA) store and express the genetic code for all living organisms, and sampling these molecules after they are shed into the environment allows us to efficiently explore the distribution of aquatic species. While uncertainties remain in interpreting environmental DNA (eDNA) data, even less is known about environmental RNA (eRNA). Researchers initially assumed eRNA was too unstable to persist in water samples, but recent studies suggest eRNA can be abundant in aquatic environments. Further, the inherent instability of RNA molecules may yield more precise information from eRNA samples than eDNA samples. However, the volatility of eRNA also brings unique methodological challenges, making it unclear whether eDNA protocols are suitable for eRNA samples. To address this gap, we compared various methods for preserving and extracting eRNA from water samples using naturally-occurring eRNA and eDNA from Largemouth Bass (*Micropterus salmoides*) as well as DNA and RNA extracted from Bluegill (*Lepomis megalotis*) tissues. While working with eRNA presented both expected and unexpected challenges, these difficulties ultimately generated a deeper understanding of the methodology necessary to advance eRNA applications. We found that methods developed for eDNA samples can be readily repurposed and successfully applied to preserve and process eRNA with minimal modifications. For example, we found refrigeration of water samples to be an effective

method for short-term preservation of eRNA in the field, with water samples on ice in a cooler showing minimal eRNA degradation over 25 hours. We also found that a low-cost chloroform extraction protocol yielded comparable eRNA quantities relative to much more expensive commercial kits, in some cases even outperforming these methods. Overall, this work demonstrates the practical feasibility of eRNA sampling and provides foundational best practices for future work on eRNA in natural ecosystems.

id #134761

Time-varying drivers of phenological change in an estuarine food web

Robert Fournier¹, Tyler Marino¹, Stephanie Carlson¹, Albert Ruhi Vidal¹

1. *University of California Berkeley, CA, -*

Biota are shifting their phenologies in response to changing climates. However, the magnitude and direction of these shifts can differ among trophic groups, potentially decoupling food webs when patterns diverge. Moreover, phenological responses to environmental cues are often assumed to be static, but this assumption may misrepresent future dynamics as phenological plasticity can vary over time and responses may change under novel climatic conditions. Here, we assessed the time-varying drivers of phenological change for fishes and their zooplankton prey in the San Francisco Estuary. We found that phenological shifts were common across the community (22.4% of taxa), but trends diverged between trophic guilds, with zooplankton peaking earlier in the year (-0.95 ± 1.29 days per year) while fishes often peaked later ($.046 \pm .9$). Additionally, multivariate dynamic linear models indicate that phenology/environment relationships were non-static, with responses to temperature, salinity, and secchi changing in magnitude and, in some cases, direction throughout the time series. When we compared historical sensitivities to patterns within the last decade, we found that the environment is increasingly driving phenological change with recent phenological sensitivities being higher in 38.1% of fish time series and 37.9% of zooplankton time series. Taken together, our findings suggest that phenological responses are accelerating, with time-varying sensitivities and differential shifts between trophic groups amplifying the risk of trophic mismatches. Thus, accounting for temporally variable responses is critical to anticipating how climate change will affect species interactions and ecosystem function.

id #135785

Aquatic-terrestrial linkages and biodiversity spillover in forest streams

Francis J Burdon¹, Jasmine Brown¹, Mael Marguet¹, Martin Sarkezi¹, Stephen Pohe², Andrew Barnes¹

1. *Te Aka Mātuatua - School of Science, University of Waikato, Hamilton, New Zealand*

2. *Pohe Environmental Ltd, Whangarei, New Zealand*

Biodiversity is changing globally, with disproportionate losses in freshwater ecosystems. While the impact of biodiversity loss on ecological functions and stability within a single ecosystem is well understood, its influence across ecosystem boundaries remains less clear. Forested streams and their adjacent riparian zones offer a model "meta-ecosystem" to investigate these spill-over effects. Forest stream invertebrates heavily depend on terrestrial leaf litter for energy, and emerging aquatic insects are a crucial food source for riparian consumers like birds, bats, and spiders. To test how donor ecosystem biodiversity might influence recipient ecosystem functioning and create reciprocal feedback loops by facilitating prey diversity, we have begun a field experiment manipulating terrestrial litter inputs at the reach-scale in two forested streams with markedly different vegetation. One stream is dominated by pine (*Pinus radiata*), while the other stream features diverse native forest. Our experiment involves a BACI (Before, After, Control, Impact) design, where terrestrial litter inputs are being manipulated in two 30-m reaches of each stream for one year and compared with before data alongside control reaches. The two litter input treatments involve a labile litter type from a single tree species (monoculture) and a polyculture of litter from three tree species. Our sampling has included benthic and emergent aquatic insects, organic matter processing, and eDNA. Preliminary results based on eDNA show the natural variability among study reaches, with whole-community indicators suggesting little variation within each stream, but subtle differences between the pine and native forest sites. These responses will be complemented with food-web analyses using functional traits, biotracers, and gut-content metabarcoding. By using these innovative techniques in an ecosystem manipulation at the reach scale, we can better understand the importance of biodiversity within a landscape context. Addressing this frontier in ecology will improve our ability to conserve, restore, and manage biodiversity and ecosystems across spatial scales.

id #134506

Periphyton biomass in tropical montane streams of Puerto Rico: biotic and abiotic controls

Adriana Villanueva-Cruz¹, Vamery González Hernández^{1,2}, Alonso Ramirez^{1,2}, Pablo E. Gutierrez-Fonseca^{1,3}

1. *Applied Ecology, North Carolina State University, Rio Grande, PR, USA*

2. *Applied Ecology, North Carolina State University, Raleigh, North Carolina, USA*

3. *Biology, University of Puerto Rico Mayaguez Campus, Mayaguez, PR, USA*

Studies on factors controlling periphyton biomass are important for understanding food web structure and ecosystem functioning in tropical streams. Multiple factors potentially influence periphyton in streams, including floods, periods of low flow, nutrients, light availability, and grazing. Although periphyton are important components that promote secondary production, studies concerning their biomass in heterotrophic streams are scarce. Our goal is to assess the main biotic and abiotic controls that influence periphyton biomass production in an unimpacted tropical stream at the Luquillo Experimental Forest, Puerto Rico. We predict that periphyton biomass will be significantly influenced by canopy cover, discharge, and shrimp populations. Our study is part of the Stream Flow Reduction Experiment (*StreamFRE*), which addresses multiple ecosystem responses to prolonged low-flow conditions, including food web dynamics. The main drivers that will be examined are water discharge, nutrients (P and N), canopy cover, and shrimp populations, while benthic organic matter (i.e., AFDM) and chlorophyll will serve as biomass indicators. Preliminary analyses indicate that hydrology is a dominant driver of benthic organic matter and chlorophyll dynamics over the study period. In contrast, canopy cover showed no relation to periphyton biomass even following hurricane defoliation of riparian

vegetation. Experimental manipulation of shrimp grazing highlighted their critical role in periphyton biomass in our study stream. Overall, our research shows that periphyton biomass is controlled by both biotic and abiotic factors, with top-down control by shrimp playing a key role in its dynamics. Even responses to major disturbances like droughts and hurricanes are mediated by shrimp activity.

id #134762

Differentiating Recharge Zones and Pathways in the Rathdrum Prairie Aquifer

Jeff Langman¹, Seth Oliver², Isaac Hall¹, Tim Link^{3, 4}, Eric Aston¹

1. *University of Idaho, Earth and Spatial Sciences, Moscow, ID, United States*

2. *Idaho Department of Environmental Quality, Coeur d'Alene, Idaho*

3. *Forest, Rangeland, and Fire Sciences, University of Idaho, Moscow, Idaho*

4. *Chemical and Biological Engineering, University of Idaho, Moscow, Idaho*

The productivity and water quality of the Rathdrum Prairie Aquifer are inextricably linked to land-surface influences because of its unique geological composition derived from the Missoula floods, which allows for the quick entry of recharge into the aquifer. The aquifer's thick, unconsolidated Quaternary sediments provide a horizontal hydraulic conductivity that can range from 100 m/d to 3.5 km/d. The northern portion of the aquifer is primarily recharged through lake seepage, and the movement of groundwater into the southern portion of the aquifer is controlled by subsurface channels formed by bedrock ridges. To evaluate the distribution and mixing of recharge sources and the primary flowpaths within the northern Rathdrum Prairie Aquifer, an age-dating study was implemented through analysis of 40 groundwater samples for chlorofluorocarbon compounds, sulfur hexafluoride, per- and polyfluoroalkyl substances, and the stable isotopes of water. The chlorofluorocarbon compounds and sulfur hexafluoride are being used as a traditional, young water age-dating technique to evaluate dominant recharge sources and mixing of these sources as the groundwater moves into the channel connections. The per- and polyfluoroalkyl substances are being evaluated as a possible age-dating technique because of their environmental persistence and temporal differences of entry into the hydrologic cycle. The stable isotopes of water are assisting with the discrimination of recharge sources and their mixing patterns. The goal of the study is to refine our understanding of aquifer dynamics for the northern portion of the aquifer and its contribution to the larger Spokane Valley-Rathdrum Prairie Aquifer. Our improved understanding of the aquifer will assist with the new groundwater modeling effort that is being undertaken to improve wellhead protection and analysis of future water demands on this sole-source aquifer.

id #136043

Functional and Community Responses to Climate-Driven Drying in Human-Made Pond Ecosystems

Checo Colon Gaud¹, Kelsey A Wilbanks¹, Victoria T. Baglin¹

1. *Georgia Southern University, Statesboro, GEORGIA, United States*

Human-created freshwater ecosystems are increasingly common across altered landscapes, yet their biogeochemical dynamics and responses to climate-driven hydrologic change remain understudied. We used a decommissioned U.S. Fish and Wildlife Service hatchery in southeast Georgia, now a network of long-abandoned artificial ponds, to examine how hydrologic regime influences organic matter decomposition and benthic macroinvertebrate communities. Twelve experimental ponds were assigned to one of three hydroperiod treatments representing climate-change-driven drought scenarios: permanent ($n = 4$), temporary ($n = 4$), and dry ($n = 4$). Leaf packs were deployed and retrieved at 30, 60, and 90 days to quantify decomposition rates and associated macroinvertebrate assemblages. Decomposition proceeded more rapidly in permanent ponds, while temporary ponds exhibited reduced rates as drying progressed. Macroinvertebrate assemblages were initially similar across hydroperiods but diverged by 90 days, with temporary ponds showing lower diversity, density, and biomass as drying intensified. Our findings demonstrate that hydrologic alteration in human-made wetlands can significantly modify biogeochemical processing and benthic community structure, with implications for the resilience and functional stability of artificial freshwater ecosystems under future climate scenarios.

id #135788

Effective treatment of wastewater for microbial removal

Hemant Sagar¹, Subodh Sharma¹, Rajat Pant¹

1. *Indian Institute of Technology, Roorkee, Roorkee, UTTARAKHAND, India*

Cavitation bubble collapse induces a pressure wave, microjet, and temperature in the surrounding media. The intensities of the pressure wave and microjet is severe as per many experimental and numerical investigations. Very few reported the use of shock wave propagation and its implication to disinfect the water. This work mainly focuses on analysing the wastewater for microbial estimation before and after treatment using cavitation produced by ultrasound. For this purpose, we collected the water sample from various sources used for different purposes. Samples collected were divided into six subsamples, of which one was kept as control and remaining were exposed to cavitation for various duration. We observed that the samples exposed to cavitation after 75 minutes showed 99% removal of *Total coliform* and *E. coli* from the samples, while the control sample showed significant bacterial colony formation. Hence, our study proposes cavitation as an effective method for physical disinfection of water before releasing it to the freshwater ecosystem.

Causal Inference, Adaptive Management and Environmental Flows: a Cool Mix

Charles Yackulic¹, Drew Eppehimer¹, Kate Behn¹, Bridget Deemer¹, Kimberly Dibble¹, Maria Dzul¹, Lindsay Hansen¹, Brian Healy¹, Benjamin Miller¹, Josh Korman², Jeff Arnold³, Conor Clancy³, Dave Rogowski⁴, Kurt Shollenberger⁵, Laura Tennant⁵

1. US Geological Survey, AZ, -
2. Ecometric Research, Vancouver, BC, Canada
3. Glen Canyon National Recreation Area, National Park Service, Page, AZ
4. Research Branch, Arizona Game and Fish Department, Flagstaff, AZ
5. Grand Canyon National Park, National Park Service, Flagstaff, AZ

Uncertainty in the expected responses of aquatic ecosystems can be a barrier to environmental flow implementation. Robust assessments of flows when they occur can reduce this uncertainty and inform future flow implementation. True control river segments are usually not available, and many assessments rely on before-after designs that can lead to erroneous inference if replication is low, important environmental drivers cannot be controlled, or experimental flows are not prescribed randomly. Model-based approaches provide an alternative to design-based approaches to analysis. Specifically, models that consider both controlled and uncontrolled factors can be used to forecast responses under both the observed conditions and the expected conditions in the absence of flows (i.e., a counterfactual).

Here we assess the impacts of “cool mix” flows released from Lake Powell through Glen Canyon Dam into the Grand Canyon segment of the Colorado River during 2024 using counterfactual forecasting analyses. Cool mix flows involved releasing a portion of the daily flow through river outlet works located 100 feet deeper in the reservoir than the hydropower generating tubes from which water is typically released. Cool mix was designed to minimize spawning and somatic growth of Smallmouth Bass (*Micropterus dolomieu*: SMB). SMB were first observed reproducing below Glen Canyon Dam in 2022 coincident with the lowest reservoir elevations and warmest release temperatures in five decades. Cool mix was expected to increase phosphorous in addition to cooling water temperatures. Cool mix was expected to increase the somatic growth of Rainbow Trout (*Oncorhynchus mykiss*) as increasing phosphorous should increase tailwater ecosystem productivity and cooling temperatures favored this cold-water species. Cool mix was also expected to decrease the growth of Humpback Chub (*Gila cypha*) as they prefer warmer temperatures. Observed water quality, SMB catch, and somatic growth for all species was consistent with forecasts under a variety of assumptions, however uncertainty in flow effects varied.

High Sodium Lifestyle: The physiology and habitat preferences of a rare diving beetle endemic to an extreme environment.

Audrey Lindsteadt¹, Lusha Tronstad²

1. Zoology and Physiology, University of Wyoming, Laramie, Wyoming, United States
2. Wyoming Natural Diversity Database, University of Wyoming, Laramie, Wyoming, United States

Rare animals are difficult to study, but understanding their habitat selection and comparative physiology is essential for conserving populations and promoting biodiversity. Extreme environments often have high proportions of rare, endemic species, but we challenge the assumption that they thrive there because they are physiologically adapted to the harsh conditions. The Narrow-foot Hygrotus Diving beetle (*Hygrotus diversipes*) is a rare beetle that occupies eight intermittent streams that experience the harsh environmental conditions of arid basins, including desiccation, flash floods, and extreme specific conductivity. This beetle is only known from Wyoming and has been repeatedly petitioned for protection under the US Endangered Species Act. We hypothesize *H. diversipes* are physiologically stressed in these streams, but they select them due to a competitive disadvantage in less harsh ecosystems. Our lab experiments found 70% of the beetles kept in 30mS/cm died after just five days, and ecological death occurred when individuals remained in 16mS/cm for 30 days. Despite this, *Hygrotus diversipes* have been detected in conductivities as high as 64mS/cm, and the highest conductivity we measured at a known site was 110mS/cm! Additionally, our models using projected precipitation and temperature values predicted an increase in salinity at known *H. diversipes* sites. Currently, we are investigating the species' swimming and flying capabilities to understand the degree to which these beetles can leave unsuitable habitat. If this beetle is already living near its physiological limits, increases in conductivity due to anthropogenic water use and environmental change could lead to extirpation from known sites. Our results suggest that some rare species are not specialized to extreme conditions and they are instead living at the edge of their tolerance, making them especially vulnerable to habitat change.

Restoring Our Waters

Dave Upthegrove¹

1. Dept of Natural Resources, Olympia, WA, United States

From dams to culverts, fish passage barriers have slowed and stunted the pace of salmon recovery across our state. Removing these barriers is an essential part of restoring salmon populations - and restoring the health of Washington's waters. In this presentation, Commissioner Upthegrove will discuss the work of the Department of Natural Resources to remove the last fish passage barriers in our public waters - and the ways in which the agency is partnering with organizations across the state on nature-based solutions to restoring the health of Washington's rivers, lakes, streams, and sounds.

Local environmental factors outweigh spatial factors in shaping beta-diversity patterns of tropical island fish assemblages

Kai Chen¹, Xingwei Cai², Robert M. Hughes^{3,4}, Julian D. Olden⁵

1. *Nanjing Forestry University, Nanjing, JIANGSU, China*

2. *Hainan Academy of Ocean and Fisheries Sciences, Haikou, Hainan, China*

3. *Amnis Opes Institute, Corvallis, Oregon, USA*

4. *Oregon State University, Corvallis, Oregon, USA*

5. *University of Washington, Seattle*

Identifying the patterns and ecological determinants of riverine community composition remains a key goal to understand community assembly, biological conservation, and environmental management. However, the patterns and potential causes of beta diversity variation and species replacement in riverine fish metacommunities are largely scarce at tropical islands. In this study, we investigated the environmental and spatial factors influencing beta diversity of local fish assemblages across several hydrological units of tropical rivers in the Hainan island, China. By collating 189 standardized fish samples during the low-water season across eight hydrological units, we tested the hypothesis that environmental factors influence tropical fish assemblages beta-diversity more than spatial factors. We found marked consistency in high beta-diversity across hydrological units that influenced by species turnover rather than nestedness. Aligning with our hypothesis, environmental parameters alone explained a higher fraction of variation in beta-diversity than did spatial factors alone or shared environmental and spatial factors, implying that environmental filters constrain species distributions and abundances. In conclusion, our study suggested that conservation of fish diversity in tropical rivers requires maintenance of both local-scale habitat condition and riverine connectivity.

id #136047

TealWaters: Bridging disciplines and communities to operationalize wetland mapping and function

Sarah Fancher¹, Meghan Halabisky¹

1. *TealWaters, Seattle, Washington, United States*

Scientists from different disciplines often work in parallel rather than together, producing insights that are difficult to translate into tools and practices that endure beyond a single project or funding cycle. In freshwater and wetland management, this disconnect can limit the ability of managers and communities to respond to rapidly changing conditions while also addressing long-standing challenges. Bridging disciplinary, organizational, and community boundaries is therefore essential for producing science that is both actionable and durable.

TealWaters is a collaboration among multidisciplinary scientists, data engineers, and community partners focused on operationalizing science for wetland and watershed management. Our work emphasizes co-production, ensuring that scientific advances are shaped by real management needs and can be applied directly on the ground. Our flagship effort, the Wetland Intrinsic Potential (WIP) Tool, is a remote sensing-based product with an intuitive web viewer that improves both the accuracy and spatial extent of wetland mapping. The WIP addresses a deceptively simple but persistent question—where are the wetlands?—while allowing users to tailor supplementary estimates of wetland functions to their specific objectives.

Born from a National Science Foundation Convergence Accelerator Grant, TealWaters brings together collaborators from academia alongside community partners, including the Tulalip and Klamath Tribes. We present a case study from the Skykomish Watershed, recently affected by flooding from an atmospheric river precipitation event, to illustrate how this collaborative approach supports management-relevant outcomes. In addition to mapping wetlands, we estimate the proportion of previously unmapped wetlands, classify wetlands by hydrogeomorphic setting, identify catchments with high water storage potential, and estimate carbon storage. This example highlights both the opportunities and challenges of cross-disciplinary, cross-organizational collaboration and demonstrates how co-produced science can better support freshwater management in a changing climate.

id #138864

Marking 25 Years Since *Idaho v. U.S.*: Reflections and Impact on the Health of Coeur d'Alene Lake

Laura Laumatia¹, Caj Matherson¹

1. *Coeur d'Alene Tribe, Plummer, ID, United States*

Coeur d'Alene Lake has been the center of the Coeur d'Alene Tribe's homeland since Time Immemorial. In 2001, the U.S. Supreme Court affirmed the Tribe's rightful ownership of the beds and banks of its southern waters, supporting the Tribe's governance and its ability to address over a century of pollution from mining and development across the Basin. Presenters will discuss the cultural, political, social, and environmental relationship of the Tribe to the lake and its tributaries, as well as the limitations of Western environmental law in addressing lake protection and clean-up.

id #134513

Cultivating resilience in science and science communication

Faith Kearns¹

1. *Arizona State University, Rimrock, ARIZONA, United States*

Resilient freshwater systems and the resilience of the people who study them, communicate about them, and care for them go hand in hand. Yet science is often presented as objective and detached, while the humans who do science and communicate it are subjective and deeply connected. How do scientists and science communicators navigate the muddy waters of public engagement—especially around emotional and contentious topics—while remaining resilient and grounded?

Drawing on experiences of scientists and science communicators across disciplines and career stages, this talk explores concrete ways to forge forward in challenging times. Rather than framing resilience as individual toughness or emotional distance, it offers a more sustainable approach rooted in care, connection, and shared purpose.

At its core, this talk argues that staying human is not a liability in science or science communication—it is the point.

id #136049

Evaluating organic matter decomposition in headwater streams across a mining-induced salinity gradient throughout the Central Appalachian Coalfields

Megan Underwood¹, Lisa Tabor¹, Teresa Brown², Erin Hotchkiss³, Daniel McLaughlin⁴, Caleigh Meehan³, Stephen Schoenholtz⁴, Kelley Sinning¹, Carl Zipper⁴, Sally Entekin¹

1. Department of Entomology, Virginia Tech - Entekin Aquatic Entomology Lab, Blacksburg, Virginia, United States

2. Department of Natural Sciences, University of Virginia's College at Wise, Wise, Virginia, United States

3. Department of Biological Sciences, Virginia Tech, Blacksburg, Virginia, United States

4. Virginia Water Resources Research Center, Virginia Tech, Blacksburg, Virginia, United States

Coal-mining practices in Appalachia expose rocks to accelerated weathering, resulting in the release of minerals and associated ions into nearby streams. This ion runoff increases specific conductance (SC) and can alter microbial and macroinvertebrate communities, impacting such ecological processes as production, respiration, and decomposition. We assessed how a mining-induced salinity gradient affects organic matter decomposition by measuring changes in rates of tensile strength and ash-free dry mass (AFDM) loss, and measured change in microbial and algal activity on the organic matter. We predicted 1) decomposition would follow a subsidy-stress response, where rates increase alongside moderately elevated SC due to an increase in microbial and algal activity from ion subsidy, then decrease after a certain point due to ion stress, 2) woody organic matter decomposition would occur linearly with SC compared to cotton strips due to the carbon being slower to breakdown via hydrolysis, and 3) decomposition would occur at a faster rate in the streams than in the riparia due to an increase in microbial activity and physical abrasion created by streamflow. From October 2023 to February 2026, we deployed and retrieved wood veneers and standardized cotton strip assays seasonally in nine Virginia and West Virginia streams with a mining-induced salinity gradient of ~20 $\mu\text{S}/\text{cm}$ - ~1,185 $\mu\text{S}/\text{cm}$. We measured decomposition through tensile strength of cotton strips and AFDM of wood veneers, and measured microbial and algal activity by conducting chlorophyll-a and respiration analysis on wood veneers. Preliminary data show seasonal variation in decomposition on both substrates, with May and August 2025 having the fastest decomposition, and cotton strips decomposing faster in streams than in the riparia. Chlorophyll-a increased on the wood veneers with salinity, while respiration exhibited a subsidy-stress response to salts, with microbial activity decreasing after 400-450 $\mu\text{S}/\text{cm}$. Overall, elevated salinity does seem to have season-specific effects on organic matter decomposition and algal activity, though it doesn't appear to strictly follow a subsidy-stress response, while the microbial activity does seem to show a subsidy-stress relationship with salinity. These findings contribute to understanding salinity's impacts on organic matter decomposition and help guide future research into potential environmental remediation.

id #134003

Insect decline and ecosystem function in tropical streams in Costa Rica: A long-term comparison

Vamery González Hernández¹, Alex Shelton², Marcelo Ardon³, Alonso Ramirez¹

1. Applied Ecology, North Carolina State University, Raleigh, North Carolina, United States

2. Biological Sciences Department, North Carolina State University, Raleigh, North Carolina, United States

3. Forestry and Environmental Resources, North Carolina State University, Raleigh, North Carolina, United States

Declines in insect abundance and richness are now documented even from protected tropical forest ecosystems. Long-term studies from our lab, in lowland rainforest streams in Costa Rica, indicate that insect richness is decreasing. A decrease in insects could affect the rate of ecosystem processes, unless other organisms compensate for their decline. Fortunately, we have information on leaf litter decomposition and insect colonization from over 20-30 years ago that serves as a reference line to evaluate the effects on the decomposition process. The main objective of this study is to assess whether the observed regional decline in aquatic insects results in changes in insect colonization and decomposition rates of leaf litter. Our work was conducted at La Selva Research Station in Costa Rica. We ran a leaf decomposition study in seven streams along a natural phosphorus gradient. We placed replicated fine and coarse-mesh bags of *Ficus insipida* and *Castilla elastica* leaves, both used in studies during the 1990s and 2000s. We determined mass loss, decomposition rates, and quantified insect richness, abundance, and biomass. Although there has been an observed decline in aquatic insects at La Selva, decomposition rates have not changed from the 1990s or 2000s to our study results in 2025. The response to the P gradient was also similar to the one described in previous studies. For example, *Ficus insipida* leaves in coarse-mesh bags had decomposition rates 33.18% faster under high P, similar to previous reports. Insect assemblages colonizing leaves had lower richness than reports from the 1990s and 2000s and were dominated by larval Chironomidae. In contrast, previous studies report more diverse assemblages, dominated by chironomids, caddisflies, and other taxa. Our study provides critical insights into the consequences of insect diversity loss on a major, understudied ecosystem process in these globally significant habitats.

id #136051

Determining periods of beaver impoundment using floodplain softwood stress

Kendal Massey¹, Abigail Hullihen¹, Emily R Arsenault¹

1. Environmental Biology, SUNY ESF, Syracuse, New York, United States

The North American Beaver (*Castor canadensis*) changes its landscape by felling mainly hardwood trees, thereby impounding moving water to create flooded, low-energy systems that cause stress to impacted floodplain trees. These impacts are episodic and cyclical, with dams typically standing for one or more decades. Therefore, reconstructing the history of beaver influence in a system over time can be important for understanding long-term ecosystem dynamics. Commonly used methods to age beaver impoundments involve removing and aging wood at the base of dam structures, which is destructive and not possible in sensitive systems. Here, we propose and test a non-invasive way to estimate timing of flood events by assessing standing softwood stress using methods in dendrochronology. In an Adirondack hardwood forest, we sampled three beaver ponds over an age gradient, from intact dams to breached dams, to historically dammed reaches. At 150-meter increments along the streams, transects (n=45) were surveyed to obtain tree species and crown class information. In riparian areas of each site, softwoods were cored upstream and downstream of impounded areas (i.e., low-energy and high-energy sites, respectively). Tree cores were analyzed using skeleton plotting to determine years of stress (i.e., decreased annual growth) over time. Overall, we observed a significant difference in the proportion of softwood trees between low energy (71% softwood) and high-energy (37% softwood) reaches (p=0.004). In addition, we observed a clear pattern of softwood stress: we saw evidence of increased stress in low-energy segments and decreased stress in high-energy segments. This pattern likely indicates a flooding event while similar stress levels in adjacent high- and low-energy segments indicates a major impoundment breach. These findings agreed with estimated flood timings found using satellite imagery. Preliminary results indicate a new potential method for determining the hydrological history of streams and their riparian zones in systems engineered by beavers.

id #135796

Natural vs. Engineered Ponds: Divergent Water-Quality Outcomes in Urban Watersheds

Sandra Clinton¹, Emma Lacy¹, Safiyah James¹

1. University of North Carolina Charlotte, Charlotte, NORTH CAROLINA, United States

Urbanization has well known quantifiable impacts on stream ecosystems including altered hydrology, increased nutrients, and decreased biodiversity. Stormwater ponds are widely used to mitigate these impacts by retaining urban runoff and promoting sedimentation and nutrient processing before slowly releasing the water back to the stream. Over time however, these ponds accumulate excess nitrogen and phosphorus leading to nuisance algal blooms that may need to be managed to maintain water quality. Beaver populations are increasing across the United States, including in cities, where they alter the landscape by increasing the number of ponds and wetlands in urban watersheds. These ponds also retain sediment and nutrients and can function as a natural solution for treating stormwater runoff. The overall goal of our research is to evaluate the role of beaver ecosystems on urban water quality and compare beaver and stormwater pond potential for eutrophication. We quantified nutrients (nitrogen and phosphorus), algal biomass and composition, and total suspended solids (TSS) in three beaver and three stormwater ponds in Charlotte, NC, USA. Nutrient and TSS concentrations of input water are similar across stormwater and beaver ponds. Pond and output water however, varies with pond type and constituent. TSS is higher in stormwater output water compared to beaver ponds. Total phosphorus is higher and nitrate is lower in beaver ponds compared to stormwater ponds. Algal biomass was higher in stormwater ponds, which also received more light and had higher temperatures, compared to beaver ponds. Overall desmids and blue-green algae were more common in stormwater ponds compared to beaver ponds. Overall beaver ponds can function as a natural solution and improve urban water quality; however, more work is needed to evaluate the role of larger urban beaver complexes (as opposed to single ponds) and to integrate these systems into watershed scale nutrient models.

id #135029

Exotic or Endangered: The Conservation Paradox of Translocated *Astacopsis gouldi* Populations

Caitlin Silberberg¹, Chris Burrige¹, Jia-Huan Liew¹, Leon Barnuta¹

1. University of Tasmania, Seymour, TAS, Australia

Astacopsis gouldi is the largest freshwater invertebrate in the world and is restricted to northern Tasmania, Australia. It is threatened, with habitat degradation and poaching causing a decline in populations. Four covertly translocated populations have been established illegally outside their native range, representing a potential management paradox, as they may offer conservation benefits for the species, but threaten other native taxa, but their conservation value and hence future management is yet to be assessed.

This study aims to determine the long-term viability of translocated *A. gouldi* populations and assesses their conservation value. Seven locations — four translocated and three native — were surveyed using baited trapping and hand searching. Each survey searched 200–300 m of stream and deployed 12 baited traps. Environmental characteristics were measured at each location.

Astacopsis gouldi is persisting at two of the translocated locations. When *A. gouldi* was present at a translocated location, the environmental conditions were comparable to a native location. No females were observed at translocated locations, and a paucity of juveniles was observed, but the latter was also the case at native populations.

Translocated populations of *A. gouldi* found outside their native range may not be viable long-term without intervention, such as the supplementation of additional genetic material or introduction of females. Whether translocated *A. gouldi* are impacting other species in recipient ecosystems is yet to be evaluated.

id #135797

Building Interdisciplinary Capacity in Urban Science Through the REMUS REU Program

Sandra Clinton¹, Casey Davenport¹

1. *University of North Carolina Charlotte, Charlotte, NORTH CAROLINA, United States*

Over half (56%) of the global population currently lives in urban areas, with that proportion expected to increase to 68% by 2050. In the United States, an even larger share of the population lives in cities (83%), which is also expected to grow to 89% by 2050. Growth in urban populations, combined with the multisystem and multiscale effects of climate change, is anticipated to significantly stress urban areas through pressures on resources such as food, energy, housing, and water. Interdisciplinary groups of researchers are needed to advance our understanding of urban science, provide solutions to the many complex and interacting problems, as well as to develop and maintain sustainable cities of the future. Undergraduate students in earth and environmental disciplines are well positioned to contribute to this emerging field and develop innovative solutions.

The Research Experiences and Mentorship in Urban Systems (REMUS) program is an innovative undergraduate summer research experience designed to build interdisciplinary capacity in urban science. Hosted at the University of North Carolina at Charlotte, the program immerses students in the study of urban systems within a rapidly growing metropolitan region in the southeastern United States. REMUS engages diverse student cohorts in mentor-guided research projects that examine urban environments from multiple disciplinary perspectives. Faculty mentors represent urban ecology, environmental engineering, earth and atmospheric sciences, Geographic Information Science, and environmental justice, and are drawn from several departments and colleges. Through REMUS, the program: 1) enables students to better comprehend interdisciplinary interactions in a dynamic social-ecological-technological system; 2) immerses students in complex, real-world sustainability and environmental justice issues; 3) allows students to gain proficiency in data collection, management, and analysis; and 4) facilitates a peer mentoring and networking community within and across participant cohorts. This presentation will provide an overview of the REMUS program, including how it is uniquely positioned as an NSF REU site, the disciplinary make-up of the cohorts and faculty mentors, recent research projects, as well as the range of professional development activities including incorporation of python coding workshops.

id #136053

Incorporating angler perspectives and ecological evidence to inform management of non-native fish interactions with native Yellowstone Cutthroat Trout in the Teton River Basin, Idaho

Lucas Ellingson-Cosenza¹, Colden Baxter¹, Morey Burnham¹

1. *Idaho State University, Pocatello, IDAHO, United States*

Management actions involving non-native species, including, in some cases, their removal, can create tension between fisheries managers and anglers, which poses challenges in fostering collaborative efforts and leads to a lack of angler buy-in to new management. These are circumstances within which both ecological and social science may be needed to address these challenges, and may require viewing management as a collaborative effort, where angler's attitudes and anecdotal experiences are considered in decision-making. One such context occurs in the Teton River drainage (Idaho, USA), where Yellowstone Cutthroat Trout (YCT) (*Oncorhynchus clarkii bouvieri*) are a native trout species imperiled (reduced to half their historical range) due to habitat degradation and negative interactions with invasive, nonnative trout. In particular, Rainbow Trout (RBT) (*Oncorhynchus mykiss*) are considered a major threat due to their hybridization with YCT which decreases genetic diversity and quantity of pure YCT populations, but competitive interactions with nonnative Brook Trout (*Salvelinus fontinalis*) are also a concern. To address the status of YCT, we are integrating social research methods with an ecological field study. I conducted semi-structured interviews with Teton River anglers and fisheries managers to explore their values and perceptions of YCT conservation and management approaches to mitigate invasive species impacts, as well as the "tension" that I've observed in initial interviews. My interviews were coded into network mental models of both group's varying perspectives. The ecological component of my project focused on Badger Creek, a tributary of Teton River that received a rotenone treatment in fall 2025 to remove its high RBT numbers. I evaluated ecological assumptions underpinning this removal action and that may help in predicting its ecological outcomes. I also drew on a drainage-wide, 20-year time series of e-fishing monitoring and a species distribution snapshot (based on eDNA) to determine if there is evidence for additional locations that might be priorities for future removal efforts. Managing for native YCT against non-native species creates tensions with anglers, therefore considering angler opinion along with ecological research creates a more comprehensive management approach to preserving native YCT populations within the Teton River drainage.

id #136309

Do invasive crayfish have a metabolic advantage over native crayfish?

Adele Crabtree¹, Kaelyn Fogelman², Jim Stoeckel³, Zanethia Barnett⁴, Brian Helms¹

1. *Department of Biological and Environmental Sciences, Troy University, Troy, AL, USA*

2. *School of Renewable Natural Resources, Louisiana State University, Baton Rouge, LA, USA*

3. *School of Fisheries, Aquaculture and Aquatic Sciences, Auburn University, Auburn, AL, USA*

4. *Southern Research Station, Center for Bottomlands Hardwoods Research, US Forest Service, Clemson, SC, USA*

Invasive crayfish are a significant threat to freshwater biodiversity across the globe. Invasive crayfish often have plastic traits that enable them to persist in new environments— one such plastic trait may be metabolic rate. Estimating the standard metabolic rate (SMR) and routine metabolic rate (RMR) of invasives compared to natives allows for direct comparison of energy expenditure for survival at rest and under basic activity, respectively. Alabama has the highest diversity of crayfish species in the world, and includes some invasive crayfish. In central Alabama, the Cahaba River watershed has been invaded by *Faxonius virilis*, and in southern Alabama, the Conecuh River watershed has been invaded by *Procambarus clarkii*. Here we determined if there are differences in SMR between invasive and native crayfishes in both systems. We estimated SMR and RMR for two native and one invasive species from the Cahaba and in the Conecuh using intermittent-flow respirometry equipped with fiber optic dissolved oxygen (DO) sensors and automated data loggers. We compared *F. virilis* to *Cambarus coosae* and *Faxonius erichsonianus*.

within the Cahaba watershed and compared *P. clarkii* to *P. acutissimus* and *P. versutus* within the Conecuh watershed. All crayfish were kept in individual artificial freshwater holding tanks under a 12-hr light/dark cycle for a minimum of two weeks. DO readings from respirometry trials were converted to rates of consumption (MO₂ values), corrected for background respiration. SMR was estimated for each individual by determining the 20th percentile MO₂ during a 10-hour period of lowest respiration. RMR was measured as the median MO₂ during light cycle and dark cycle for each individual. We used 8-10 individuals of each species to estimate SMR and RMR. Preliminary data suggest that SMR and RMR are lower for invasive as compared to native crayfish, suggesting a lower metabolic cost for invasive crayfish. Lower SMR and RMR, and presumably metabolic needs, may be a species trait that can predict successful invasion, particularly in association with other demonstrated traits of successful invaders, such as elevated growth, fecundity and aggression.

id #137845

Long-term resilience, regime shifts and recovery of aquatic communities in metal-contaminated watersheds

William Clements¹

1. Colorado State University, CO, United States

The likelihood that degraded watersheds can recover after elimination of an anthropogenic stressor and the time required for these systems to return to pre-disturbance conditions (defined here as ecological resilience) are critical questions in applied ecology. Despite significant investments in restoration projects globally, quantifying ecological resilience and predicting the success of these programs remains challenging. Inadequate study designs, ineffective pre- and post-restoration monitoring and failure to consider ecological theory significantly impede our ability to learn from these management "experiments." Ecological resilience of streams is determined by numerous factors, including the specific nature of the disturbance (e.g., chemical versus physical), watershed characteristics, natural community variation and the proximity of sources of recolonization. Compound perturbations and the establishment of novel communities in disturbed ecosystems often impede recovery, even after complete elimination of a stressor. A better understanding of factors that influence ecological resilience is essential for predicting the success of watershed restoration projects. Using long-term (20-37 years) data from several western streams previously impacted by historic mining, I examined the relative importance of biotic and abiotic factors that affected ecological resilience. The distance from upstream sources of recolonization and the establishment of novel, metal-tolerant communities were especially important for determining the rate of recovery. Long-term shifts in community composition resulting from alterations in stream hydrology, water temperature and other physicochemical characteristics associated with climate change also affected recovery trajectories. Finally, because responses to restoration are endpoint-specific, our definition of what constitutes recovery is often influenced by conflicting ecological and socioeconomic considerations. Although full recovery of some watersheds is elusive, even highly damaged systems can eventually approach reference conditions following the removal of a stressor. Maintaining public support for ecological restoration requires that we objectively document the success of these projects but not overstate the potential for a complete return to pre-disturbance conditions.

id #135030

Trialling Whakaweku for Invertebrate Sampling in Tasmanian Streams

Caitlin Silberberg¹, Chris Burridge¹, Jia-Huan Liew¹, Leon Barmuta¹

1. University of Tasmania, Seymour, TAS, Australia

Invertebrates are an integral part of freshwater ecosystems and provide researchers with knowledge regarding the health and productivity of streams and lakes. However, freshwater invertebrates are declining across the globe due to factors such as climate change and habitat degradation. It is pertinent to ensure that freshwater invertebrate communities are frequently sampled to monitor changes in freshwater environments, such as changes in community composition and movement of invasive species.

There are a multitude of sampling methods available for freshwater invertebrates, with methods varying in deployment time, cost, manual labour and equipment. However, most existing sampling methods for freshwater invertebrates are labour intensive or often require large amounts of equipment. This can incur additional costs to research, such as hiring of extra hands or purchasing new equipment when required.

This project saw the novel trialling of Whakaweku traps, a traditional Maori method of capturing crayfish, as an invertebrate surveying method in Tasmania. Whakaweku are constructed from ≈10 individual *Pteridium esculentum* fern fronds, which are shaped into a bundle and deployed individually or on a line as a group. Whakaweku have proven to be successful in monitoring populations of native New Zealand fish and crayfish, and some research suggests Whakaweku provide accurate representations of macroinvertebrate populations.

In this study we trialled Whakaweku in Tasmania, primarily to passively survey juveniles of the giant freshwater crayfish *Astacopsis gouldi* while simultaneously surveyed the broader instream invertebrate fauna.

Preliminary results suggest that Whakaweku can be used to collect a large diversity of species and include specimens smaller than 1 mm in size. Results also suggest that Whakaweku samples can be used to reliably compare productivity and pollution levels between different Tasmanian streams. However, no juvenile freshwater crayfish were surveyed, and possible reasons for this will be discussed. Otherwise, Whakaweku provided a low-cost, passive invertebrate surveying method without sacrificing sample quality for ease of use.

id #135798

Synthesis of native, freshwater mussel survey protocols in North America

Shay S. Keretz^{1,2}, Rachel H. Toczydlowski², John P. Wares¹, Wendell R. Haag³

1. *Odum School of Ecology, University of Georgia, Athens, Georgia, United States*

2. *USDA Forest Service, Northern Research Station, Rhinelander, Wisconsin, United States*

3. *USDA Forest Service, Southern Research Station, Frankfort, Kentucky, United States*

Native, freshwater mussels (Bivalvia: Unionidae) are a highly imperiled faunal group in North America. However, unionids are often under-surveyed, particularly in remote regions, which can hinder conservation efforts. For example, National Forests in the Great Lakes Basin represent about 5% of the land base but to date little is known about unionids on these lands. Furthermore, survey protocols for unionids are highly variable and little guidance exists for deciding which protocol to use. To facilitate decision-making that benefits unionids on National Forest land, we are developing a standardized survey protocol for the National Forests in the Great Lakes basin to assess unionid assemblages and health. This protocol will also serve to provide data that relates to a broader Great Lakes basin conservation strategy that includes other federal, state, and tribal efforts. To guide development of this protocol, we are synthesizing a review of unionid survey protocols used throughout North America by academia, federal agencies, state agencies, nonprofit organizations, and industry. We categorized protocols by methodology (*i.e.*, time-search, transect, quadrat, or a mix) and regional basin. We also noted project objectives, site selection methods, survey results, and river size. We will use the results of this comprehensive literature review to create a decision tree of how existing survey protocols have been previously selected based on region, unionid abundance, river size, and project goals. Our synthesis currently includes >150 papers. Presently, our literature synthesis has revealed that time-search surveys are the most common survey type due to their ease of implementation and ability to cover large areas; however, person-hour effort is widely variable among studies. Transect, quadrat, and mixed survey types are relatively equally represented in the literature. Our project aims to facilitate the collection of comparable baseline unionid assemblage data by providing standardized guidelines and tools for both survey site selection and collecting and archiving detection data. This standardization will increase knowledge-sharing among partners and agencies in ways that support monitoring, protecting, and restoring native freshwater mussels, particularly within the Great Lakes region.

id #134775

Three Decades of Water Purveyors Advancing the Science and Stewardship of the Sole-Source Groundwater Supply for the Spokane/Coeur d'Alene Metropolitan Region

John J Porcello¹

1. *GSI Water Solutions, Portland, OREGON, United States*

The Spokane Valley-Rathdrum Prairie (SVRP) Aquifer in northeastern Washington and northern Idaho is the sole source of drinking water supply for the rapidly growing Spokane/Coeur d'Alene metropolitan region. The SVRP Aquifer is one of the world's most prolific aquifers, consisting of catastrophic glacial flood deposits of sands, gravels, cobbles, and boulders. Perimeter lakes and the upper reaches of the Spokane River provide substantial recharge to the aquifer, and most natural groundwater discharge occurs as baseflow to downstream reaches of the Spokane River and the Little Spokane River. The Spokane River interacts closely with the aquifer and is a highly valued natural, cultural, and recreational resource for the region. As a result, local water purveyors conduct their groundwater supply planning and management activities with the river's health in mind. This presentation will discuss how purveyor-led field and modeling studies during the past three decades have (1) dramatically improved the understanding and current state of knowledge of the region's hydrogeology and river-aquifer interactions and (2) allowed science-based planning work to occur that accounts for how future possible changes in land use, water demands, and climate conditions in the aquifer and its many contributing watersheds could affect the health of the SVRP Aquifer and the Spokane River.

id #135031

Assessing the potential for evolutionary rescue in imperiled Southern Plains fish species

Kasey Pruett¹, Edward Mager¹, David Hoeinghaus¹

1. *University of North Texas, Denton, TEXAS, United States*

Since 2000, global mean annual temperatures have risen by 0.750C and the Southern Plains region in north Texas has experienced record-breaking drought events. Intense drought exposes fishes to extreme water temperatures and hypoxic conditions, especially when streams are fragmented into isolated pools. Given that both the frequency and intensity of drought events is expected to continue to increase, futures for freshwater fishes are uncertain. One important unknown is the extent to which adaptive capacity may enable evolutionary rescue in fish populations impacted by increasingly extreme droughts. For four imperiled fish species, we have the unique opportunity to assess adaptive responses to changing environmental conditions over the past quarter century by comparing results from contemporary research to those from research conducted on the same populations 25 years ago. In this study, we quantified thermal and hypoxia tolerance for four imperiled fish species native to the Upper Brazos River – federally endangered Sharpnose Shiner (*Notropis buccula*), federally endangered Smalleye Shiner (*Notropis oxyrinchus*), state threatened Red River Pupfish (*Cyprinodon rubrofluviatilis*) and state SGCN Plains Minnow (*Hybognathus placitus*). Thermal tolerance (CTmax) and hypoxia tolerance (hypoxia LOE and Pcrit) metrics were quantified for 36 individuals of each species to determine underlying systemic stress responses. Sequential CTmax experiments were conducted using two heating rates: 0.30C/min to align with current convention, and 0.50C/min to allow for direct comparisons to prior research. A Loligo 4 chamber static respirometry system was used for hypoxia tolerance testing. We assessed interspecific and intraspecific variation among individual results for CTmax, Pcrit, and hypoxia LOE using mixed models, and compared our species-level results to those of Ostrand and Wilde (2001) using t-tests. Our study provides novel information regarding the potential for evolutionary rescue in these four imperiled species and can be used to inform conservation planning and management actions.

Non-perennial flow enhances impacts of sediment texture on hydro-biogeochemical parameterization

Yunxiang Chen¹, Jie Bao¹, Lupita Renteria¹, Xingyuan Chen¹, James Stegen^{1,2}

1. Pacific Northwest National Laboratory, Richland, WA, United States

2. School of the Environment, Washington State University, Pullman, WA, USA

The riverbed is a physical interface that connects stream water and subsurface sediments and microbial communities. Complex hydro-biogeochemical interactions and strong gradients occur at the interface driven by interactions among shear stress, pressure gradients, and turbulent mixing of solutes mediated by microbial reactions. Among those interactions, sediment size heterogeneity and flow intermittency play a fundamental role as defining geological and hydrological control factors. Despite widely recognized importance, the role of sediment heterogeneity and flow intermittency in hydro-biogeochemical parameterization is rarely studied due to the scarcity of sediment size data across sediment to river reach scales. We shed lights on this issue by first introducing AI4GSD, an end-to-end modular AI Python toolkit, capable of continually mapping sediment sizes from millimeters to a hundred meters scales along a river shoreline and then evaluating how this sediment size heterogeneity affects streamed hydrobiogeochemical parameterization using water depth and velocity simulated from a large-scale computational fluid dynamics modeling along the same shoreline. Our results showed that fine scale sediment texture contributes to 15-52% mean absolute percent biases in hydro-biogeochemical parameters compared to values estimated using reach-average grain sizes and such impact is further enhanced by up to several folds when flow shifts from perennial to intermittent stage.

Tide-driven soil respiration: Impacts of Tidally-Induced Water Table

Yinuo Yang^{2,1}, Chuanhui Gu¹, Emily Bernhardt²

1. Environmental Research Center, Duke Kunshan University, Kunshan, China

2. Duke University, Durham, NC STATE, United States

Riparian zones act as critical interfaces for carbon cycling, yet quantifying their greenhouse gas (GHG) emissions is challenging due to the mismatch between continuous soil respiration and episodic methane pulses. Traditional models often fail to reconcile these distinct temporal dynamics. This study proposes a Point-Event Dual-Scale Framework based on high-frequency monitoring of tidal riparian soils to address this gap.

For CO₂, we leveraged a comprehensive seasonal dataset to construct a continuous point-scale model. Results confirmed a robust water-limited aeration mechanism: groundwater drawdown significantly enhances aerobic respiration by increasing the vadose zone thickness. This seasonal model demonstrated high explanatory power ($R^2=0.48$) and strong generalizability, successfully capturing the dominant biological response to water level and temperature variations throughout the year.

For CH₄, given the stochastic nature of ebullition, we focused specifically on identifying the triggering mechanisms for moments of intense gas release. We revealed a hierarchical Two-Stage control: while biological methanogenesis initiates at a deep threshold of -1.3m, explosive emissions are only triggered when the water table breaches a shallow physical threshold of -0.91m. Crucially, these high-intensity events manifested as instantaneous pulses driven by hydrostatic unloading (physical shock) rather than cumulative biological production.

The proposed framework successfully decouples these processes: using a continuous approach for the stable CO₂ baseline and an event-based approach for sporadic CH₄ outbursts. We identify -0.91 m as a critical Safety Line for eco-hydrological regulation, providing a scientific basis for optimized sluice scheduling to mitigate carbon pulses.

Trait-based Prediction of Fish Assemblage Response to Hydrologic Alteration

Mark B. Lueders^{1,2}, Ryan A. McManamay², **Mark B. Lueders^{1,2}**

1. Minot State University, Minot, NORTH DAKOTA, United States

2. Environmental Science, Baylor University, Waco, Texas, United States

Environmental flow management seeks to balance human water use with the ecological needs of riverine systems, yet predictive tools are often limited to a subset of well-studied or economically important species. Trait-based approaches offer a mechanistic and transferable framework for anticipating fish assemblage responses to hydrologic alteration by linking flow regimes to functional characteristics shared across species. We present a holistic, multi-trait framework that integrates existing trait databases and synthesized flow–ecology relationships to predict fish species and assemblage responses to altered flow conditions.

Trait information for North American freshwater fishes was compiled from established databases and supplemented using phylogenetic surrogacy. Published flow–ecology relationships were translated into directional flow–trait linkages and applied across a broad suite of hydrologic metrics to estimate relative risk for individual species within local assemblages. Rather than isolating single traits or life history strategies, this approach incorporates a wide range of trophic, life history, habitat, and tolerance traits to better capture the multifaceted nature of ecological response to flow alteration.

The framework was evaluated using contrasting case studies representing short-term natural disturbance and long-term anthropogenic hydrologic modification. Predicted responses more closely aligned with observed assemblage changes under anthropogenic alteration, while responses to natural climatic events were more variable, highlighting differences in predictability related to magnitude and duration of flow change.

By emphasizing trait integration over taxonomic specificity, this approach supports environmental flow decision-making in data-limited systems and enables extrapolation to unstudied species. The results underscore the value of holistic, trait-based tools for assessing community-level vulnerability and informing adaptive environmental flow management.

id #137848

Beyond recovery: embracing the multiple dimensions of river resilience

Angus McIntosh¹

1. *University of Canterbury, Christchurch, New Zealand*

Resilience is a widely invoked concept in stream ecology, yet its multiple dimensions—ecological and human—are often underappreciated. Moreover, ecological resilience can be either beneficial or obstructive, depending on the context. Drawing on research from Aotearoa New Zealand, I explore these dimensions in detail.

First, while stream ecologists have emphasized dispersal and stochasticity in recovery from disturbance, density dependence remains a crucial, yet overlooked, mechanism. This is evident in fish and invertebrate responses to floods, drying, and fishing.

Second, recolonization rarely begins from a blank slate. The existing community can strongly influence assembly outcomes. This relates to what Sam Lake termed negative resilience, where, for example, a pre-existing community of tolerant taxa can inhibit the return of desired species, and our recent review of river restoration literature suggests that the linked biotic interactions are often ignored in stream restoration.

Third, key characteristics underpinning resilience operate at large spatial scales. Riverscape heterogeneity—variation in composition and asynchronous population dynamics across patches—supports landscape-level resilience via spatial and species-based insurance effects. In Aotearoa's flood-prone braided rivers these effects are especially important for sustaining this resilience.

Finally, experimental perturbations across rivers of varying sizes suggest that small rivers may be more vulnerable yet easier to restore, due to lower resistance and resilience. However, a purely ecological lens overlooks vital human dimensions. Indigenous communities in Aotearoa maintain deep, intergenerational connections to specific river places—especially large rivers—requiring ecologists to engage with these cultural contexts.

Embracing the full spectrum of riverine resilience—ecological and human—will enhance outcomes for both river ecosystems and the communities that depend on them.

id #136057

Plant Matter at the Finest Scales: Importance of Phylogenetic History, Hybridization, and Dioecy in Riparian Systems

Carri J. LeRoy^{1,2}, Iris J. Garthwaite^{1,2}, Walton M. Andrews^{1,3}, Joy M. Ramstack Hobbs⁴, Shannon M. Claeson⁵

1. *The Evergreen State College, WA, -*

2. *U.S. Geological Survey, Reno, NV, USA*

3. *San Juan Islands Conservation District, Friday Harbor, WA, USA*

4. *St. Croix Watershed Research Station, Science Museum of Minnesota, Marine on St. Croix, MN, USA*

5. *Pacific Northwest Research Station, USDA Forest Service, Wenatchee, WA, USA*

Organic matter processing is a key component of the global carbon cycle, governed by environmental conditions, the constitutive qualities of organic material, and their interactions. Important variation in litter quality is introduced into ecosystems through species diversity, genetic diversity, hybridization, dioecy, and tissue diversity. An overwhelming focus has been on the first two sources of variation, with less emphasis focused on finer scales of diversity. Here we will present evidence for important influences of plant matter at the finest scales on ecosystem processes and aquatic communities. Using data from several large-scale leaf litter decomposition studies, we will synthesize overall results to place this variation into a broader context. First, we will examine how leaf litter decomposition can be explained at a global scale using phylogenetic relatedness. We found a statistically significant phylogenetic signal for leaf litter decomposition rate ($k \text{ day}^{-1}$) when examining hundreds of results from published studies. Next, we will examine how the influence of hybridization between two *Populus* species can have similar effects on in-stream communities as species-level litter mixtures, showing that a genetic combination of two species is similar to a physical mixture of leaf litter from the same species. Third, we will demonstrate the influence of riparian plant dioecy on phytochemistry, where female willows (*Salix sitchensis*) have lower %N and lower C:N than males, and how female willows are more susceptible to herbivory by stem-boring herbivores that induce summer litter fall. These changes to litter chemistry and litter fall timing significantly accelerate leaf litter decomposition, and alter litter-associated macroinvertebrate and microbial communities. Finally, we will demonstrate that variation within individuals in terms of tissue types (*Salix sitchensis* leaves vs. catkins) can lead to increased abundances and diversity of litter-associated macroinvertebrates. Given the ubiquity of hybridization in plants and the relatively high frequency of dioecious trees in riparian zones, as well as the global distribution of *Salix* and *Populus* species, these fine-scale influences on organic matter processing in linked terrestrial-aquatic ecosystems may be widespread.

id #135802

Opportunities for accelerating flow regime restoration

Seth Wenger¹, Charles van Rees¹, Phillip Bumpers¹, Eric Walther¹, Aurora Fowler¹, Laura -¹, Mary Freeman¹, Edward Stowe¹

1. *University of Georgia, GA, -*

Meeting freshwater objectives under the Kunming-Montreal Global Biodiversity Framework will require an acceleration in the pace of environmental flow restoration (i.e., restoring some or all of the aspects of the natural flow regime in altered lotic systems). Here we highlight three flow restoration approaches with opportunities for growth, and note research needs associated with each. (1) Multipurpose floodplain reconnection. There is substantial scope to expand the use of nature-based solutions (e.g., "levee setbacks") that reconnect rivers with floodplains to reduce flood risk, restore river-floodplain flow connectivity, and provide other services. A key research need is to better characterize tradeoffs in the location and design of such projects to maximize desired benefits. (2) Water source-switching to maintain flows during droughts. In the Flint River Basin of Georgia, farmers are installing deeper wells to access alternative groundwater sources during drought periods to reduce pressure on surface waters to protect imperiled mussels. A key research need for guiding policy is to understand population and ecosystem responses to low flows, which generally requires long-term monitoring. (3) Converting from hydropeaking to quasi-run-of-river hydropower operation. Historically, electric utilities have used hydropeaking reservoirs as giant batteries that can be switched on during peak demand periods. However, expansion of solar power and improvement in battery technology means that alternative sources of peak power are increasingly available. Converting dam operations to run-of-river could provide extensive habitat restoration opportunities, but currently there are limited incentives for power companies to make this change. A key research need is to develop the science and policy for biodiversity credit programs, including biodiversity accounting schemes that are science-based, broadly applicable, and minimize unintended consequences.

id #136058

Two Extremes, One Park: A first look at the aquatic insects in Yellowstone's glaciers and hot springs.

Audrey Lindstead¹, Lusha Tronstad², Todd Koel³

1. *Zoology and Physiology, University of Wyoming, Laramie, Wyoming, United States*

2. *Wyoming Natural Diversity Database, University of Wyoming, Laramie, Wyoming, United States*

3. *Fisheries Management, Yellowstone National Park, Mammoth, Wyoming, United States*

Extreme environments often support rare and endemic species, but the adaptations that allow these organisms to persist there remain poorly understood. Because rare species are particularly vulnerable to environmental change, understanding their habitat preferences and physiological constraints is critical for effective management. We hypothesize that rare species inhabiting extreme thermal environments are experiencing physiological stress, but select for these habitats to avoid the greater biological diversity of less extreme ecosystems. Yellowstone National Park, the world's first national park, hosts a variety of unique and extreme habitats. Our research focuses on the aquatic macroinvertebrate communities in the park's 58 rock glaciers and >10,000 thermal features. To our knowledge, the ice-cold headwater streams draining rock glaciers have never been systemically surveyed, representing a major gap in our understanding of Yellowstone's biodiversity. More studies have been conducted on Yellowstone's thermal features, but none focused on aquatic insects. In 2025, we surveyed 33 sites near 14 rock glaciers to quantify used and available habitat across a broad temperature gradient. We also confirmed the status of streams draining rock glaciers as flowing or dry. Our focal species for these surveys is the Western Glacier Stonefly (*Zapada glacier*), a federally threatened insect in Wyoming and Montana. *Zapada glacier* likely occurs in Yellowstone based on proximity to other known populations, but they have never been surveyed there. Our samples confirm the presence of *Zapada*, but species-level identification requires pending DNA barcoding. Additionally, we investigated ~800 thermal features and conducted aquatic macroinvertebrate surveys at 51 suitable sites. We evaluated how community composition changed along a linear thermal gradient within each thermal outflow stream (i.e., distance from pool source). Future work will involve surveying additional rock glaciers and thermal features in Yellowstone. We will conduct *in situ* physiological experiments to test a mismatch between the fundamental and realized niche in each of these extreme environments. Our research will provide foundational data on the aquatic insect communities at both ends of Yellowstone's thermal spectrum, ensuring the park's biological diversity will be better understood and preserved.

id #134011

Linking oxygen and carbon dynamics to characterize dominant biogeochemical processes in headwater wetlands

Carla López Lloreda¹, James Maze², Emily Mulcahy³, Nicholas Corline⁴, Katherine Wardinski², Daniel McLaughlin⁵, Nate Jones⁶, Durelle Scott⁷, Margaret Palmer⁸, Erin R Hotchkiss¹

1. *Biological Sciences, Virginia Tech, Blacksburg, VA, USA*

2. *Water Institute, University of Florida, Gainesville, FL, USA*

3. *Biology, University of Louisiana, Lafayette, LA, USA*

4. *Wildlife, Fish, and Conservation Biology, University of California, Davis, CA, USA*

5. *Forestry and Environmental Conservation, Virginia Tech, Blacksburg, VA, USA*

6. *Biological Sciences, University of Alabama, Tuscaloosa, AL, USA*

7. *Biological and Agricultural Engineering, North Carolina State University, Raleigh, NC, USA*

8. *Entomology, University of Maryland, College Park, MD, USA*

Wetlands are important control points for carbon cycling from local to global scales. Carbon dioxide (CO₂) in wetlands is produced and consumed by primary producers and microbes, while groundwater acts as an additional major contributor of CO₂. In addition, physical processes also alter CO₂ concentrations in wetlands. And although the individual biogeochemical processes influencing CO₂ dynamics are relatively well understood, how these processes change with dynamic environmental conditions, particularly in small, hydrologically dynamic wetlands, requires a more integrated ecosystem-scale approach. Further, while wetlands are generally framed as carbon sinks, recent work highlights their potential role as carbon sources due to their significant carbon emissions.

We used a dissolved oxygen (DO) and CO₂ framework to characterize dominant biogeochemical processes across sites and seasons in three depressional forested wetlands that varied in size and inundation regime. This framework evaluates departures of DO and CO₂ from an expected theoretical metabolic 1:-1 line, where deviations indicate processes beyond photosynthesis and respiration (e.g., anaerobic metabolism, groundwater inputs, or physical controls). We deployed DO and CO₂ sensors, monitored wetland water levels, and collected quarterly surface water for water chemistry. To test how site and seasonal differences influenced biogeochemical processes, we calculated the offset for each paired DO-CO₂ observation from the expected 1:-1 metabolic line.

At the ecosystem scale, wetlands were substantially more depleted in oxygen (range = -406.3-207.7µM) and more enriched in CO₂ (range = -4.5-1331.9µM) than lakes and rivers. The largest, hydrologically stable wetland had lower offsets compared to the smaller wetlands, suggesting stronger controls of aerobic metabolism relative to other controls, likely due to greater light availability. Offsets during the drier summer and fall were higher, pointing to potentially increasing groundwater influence. There were periods of undersaturation, particularly in the large, hydrologically stable wetland, that point to potential periods of CO₂ sink behavior. Together, these results highlight the variability in processes influencing CO₂ dynamics and how these are mediated by hydrology, seasonality, and size. Understanding these controls will be critical to understanding the vulnerability of the wetland carbon cycle to future climatic changes, such as warming temperatures, storms, and droughts.

id #136059

Persistence of crayfish populations on coastal barrier islands of the southeastern USA

Nicholas A Macias¹, Nico Salazar², Reginald Turner³, Raymond P Kidder II³, Checo Colon Gaud³

1. *University of Georgia Marine Institute, Sapelo Island, GA*

2. *Florida Atlantic University, Boca Raton, FL*

3. *Georgia Southern University, Statesboro, GA*

Holocene hummock islands are a characteristic geological feature of the Southeastern US Atlantic coastline. These islands commonly contain upland maritime forest habitat, in which permanent and ephemeral wetlands support a variety of terrestrial and aquatic diversity, including crayfish of the genus *Procambarus*. In Georgia, historical surveys have documented seven native species across the state's fourteen major barrier islands. Recent studies have discovered additional, previously undocumented crayfish populations across a number of these islands, signifying the need for additional contemporary research. These spatially and genetically isolated populations face a host of anthropogenic stressors, including enhanced habitat connectivity, land use change, and an increased risk of habitat loss via salinization. Here, we examine the persistence of these populations with a focus on the environmental and ecological drivers shaping crayfish community composition across Georgia's barrier islands. Despite limited prior information, we identify key stressors influencing coastal crayfish populations, including saltwater intrusion, climatic deviation, indirect competition from invasive species, and urbanization. Together, these findings highlight how geographically isolated coastal crayfish persist under a suite of interacting environmental and anthropogenic pressures, revealing the complex ecological context shaping their distribution across Georgia's barrier islands. Our goal in this research is to provide insight and direction to guide the future of crayfish research in the Southeastern US and beyond; because the stressors impacting native crayfish in this region are not unique, elucidating their relative importance can help inform the management of crayfishes in an ever-changing landscape.

id #135804

Back to the future with environmental DNA: repurposing previously collected samples to study the history of a Texas freshwater jellyfish invasion

Matthew A Barnes¹, Morgan Jennison¹, Kaitlin Matheus¹, Vera Ye¹

1. *Texas Tech University, Lubbock, TX, United States*

Non-native, invasive species represent one of the greatest threats to freshwater biodiversity. For example, the freshwater jellyfish *Craspedacusta sowerbii* is native to the Yangtze River valley in China but has established populations in multiple Texas reservoirs, where its presence may disrupt food webs through voracious zooplankton grazing. In Texas, this species was first detected at Medina Lake near San Antonio and has since spread to other reservoirs near Dallas to the north and Houston to the east. The jellyfish spreads through connected river systems and human activity such as recreational boating. However, limited understanding exists about the historical spread and establishment of this jellyfish, and their distribution remains understudied. Therefore, we conducted an environmental DNA (eDNA) survey to assess the spatial distribution of *C. sowerbii* in reservoirs of central Texas. Specifically, we collected surface water at publicly accessible boat ramps and conducted on-site filtration before returning to the lab for quantification of jellyfish eDNA via qPCR. Moreover, we took advantage of the fact that our research team had previously collected samples at our study sites (years = 2014, 2020), repurposing old samples to examine invasion chronology through time. Occurrence data from our study will support state and local agencies in developing more effective management strategies to protect Texas freshwater ecosystems from invasive species and enhance conservation by identifying invasion hotspots.

id #136061

Quantifying Basal Resources of Aquatic Insect Assemblages in Beaver-Engineered Streams Using Two Stable Isotope Methods

Abigail Hullihen¹, Emily Arsenault¹

1. SUNY ESF, Syracuse, United States

North American beaver (*Castor canadensis*) create impoundments that alter the physical structure of a landscape on decadal scales due to alternating dam assembly and breach events. Habitat heterogeneity of a beaver-engineered landscape through the "lifecycle" of dam establishment, failure, and succession has been recognized as source of discontinuity in headwater systems, yet formal testing of how aquatic insect basal resources vary across this temporal gradient has not been explored. Here, we investigate this gap and directly compare results from two stable isotope methods: bulk-tissue stable isotope analysis of carbon and nitrogen (BT-SIA) and compound specific stable isotope analysis of amino acids (CSIA-AA). In freshwater food web studies, essential amino acid carbon stable isotope ($\delta^{13}\text{C}_{\text{EAA}}$) profiles have enabled more accurate estimates of basal resource use because such profiles are generally consistent among basal resources that share broad taxonomic classification (e.g., algae, bacteria, fungi, and terrestrial plants). Therefore, we compared CSIA-AA with BT-SIA to evaluate consistency in basal resource estimates. Aquatic insects were sampled using active and passive techniques across high-energy reaches bordered by forested riparian areas ($n = 24$) and low-energy reaches flowing through beaver meadows ($n = 21$) in six headwater streams representing the temporal gradient of the beaver dam "lifecycle" and analyzed using both BT-SIA and CSIA-AA. BT-SIA results indicated that periphytic algae was the dominant basal resource assimilated to aquatic insect biomass across all site types and temporal gradients for predators (median of 58.3–95.8%), shredders (median of 83.2–94.1%), and collectors (median of 51.4–85.6%). Preliminary results from linear discriminant analysis of $\delta^{13}\text{C}_{\text{EAA}}$ profiles show clustering of all aquatic insects with aquatic basal resources with the exception of one individual shredder, suggesting general agreement between the two methods. Results indicate CSIA-AA provides complementary validation and added resolution for detecting mixed resource pathways. For example, CSIA-AA clarified resource use of a shredder individual with multimodal BT-SIA results. Next, we will formalize these comparisons using Bayesian mixing models. This observed consistency in techniques further supports the importance of aquatic basal resources in headwater streams and suggests that this importance is consistent despite apparent beaver-driven habitat heterogeneity.

id #135806

Tolerant taxa and hydrologic isolation lead to few landscape effects on water chemistry and aquatic invertebrate communities in restored agricultural wetlands in central Wisconsin.

Jered Studinski¹, Jeff Lim¹, Macayla Greider²

1. University of Wisconsin - Stevens Point, WI, United States

2. Ducks Unlimited, Bismarck, ND, USA

Restored agricultural wetlands provide numerous ecosystem services and increase local biodiversity. Aquatic invertebrate communities are a diverse group of wetland biota and understanding the factors that affect their abundance and diversity may aid wetland restoration efforts and inform invertebrate-based wetland assessment tools.

We investigated the effects of landscape level factors such as land use and proximity to other open water habitats on water chemistry and aquatic invertebrate communities in 62 ponds on 41 conservation easements in central Wisconsin. Sample sites were limited to ponds that were permanent, restored via simple excavations (i.e. scrapes), isolated from other surface waters, and lacking populations of large benthivore fishes. Invertebrates were sampled from ponds in spring and again in late summer using activity traps and sweep nets. Additionally, macrophyte and fish communities were broadly assessed, as they may affect aquatic invertebrate communities.

Initial results suggest that annual variation in rainfall overwhelms the effects of landscape factors and water chemistry on aquatic invertebrates. However, diversity appears to decrease with increasing proportions of nearby developed land and with increasing isolation from other waterbodies. Linkages between land use and water chemistry are weak in hydrologically isolated ponds, suggesting a nuanced approach may be needed when selecting sites and interpreting data. Additional analyses will provide insights into sampling methodologies and effort and may inform sample site selection.

id #136062

An evaluation of localized habitat manipulation effects on native and nonnative trout in springbrooks of the Snake River-floodplain, Idaho, USA

Kaitlyn Warner¹, Colden Baxter¹, Zachery Wadsworth², Hunter Osborne²

1. Idaho State University, Pocatello, IDAHO, United States

2. Shoshone-Bannock Tribes Fish & Wildlife Department, Fort Hall, Idaho, United States

Human driven alterations to the flow of the Snake River and stocking of nonnative rainbow trout (*Oncorhynchus mykiss*) have compromised natural ecosystem processes and native species metapopulations. The Fort Hall Bottoms within the Fort Hall Indian Reservation of the Shoshone-Bannock Tribes (Idaho, USA), is a testament to these challenges. This culturally significant landscape is one of the few physically intact river-floodplain ecosystems remaining in the Snake River Basin. For over a decade, the Fort Hall Bottoms has seen a major reduction in the floodplain's natural flood regime due to the manipulations of the rivers outside reservation boundary lines. This reduction has led to a decline in suitable habitat for spawning of local species such as the Yellowstone cutthroat trout (*Oncorhynchus clarkii bouvieri*) due to the lack of natural riverbed scouring. To address this issue, the Shoshone-Bannock Tribal Fish and Wildlife Department implemented localized habitat manipulations to enhance stream habitats to increase fish abundance. We evaluated the before and after trends of trout abundance at two treated and one control site, by boat electrofishing and snorkel surveys. Preliminary results suggest there may not be detectable effects of habitat manipulation on localized trout abundance, but our analyses are ongoing. Across all sites, fish assemblages are dominated by rainbow trout and rainbow trout x Yellowstone cutthroat trout hybrids. Another dimension of this study is to investigate the proportions of the rainbow trout present at these sites that were stocked in nearby waters but have crossed reservation boundaries, and how this may be mediating any prospective responses by native Yellowstone cutthroat trout. We are also using population modeling to evaluate potential impacts of rainbow trout on Yellowstone cutthroat trout. Our preliminary analyses suggest that localized habitat manipulations alone may not be adequate to meet the management goals of the Tribes. Rather,

more challenging efforts that address issues beyond reservation boundaries, such as negotiating changes in the regulation of Snake River flows or alterations in stocking practices in the river and downstream reservoir, may be required. The outcomes of our investigations are intended to help the Tribal community identify management priorities for the coming years.

id #134527

Positive biotic interactions homogenize metacommunities

Akira Terui¹, Seoghyun Kim², Christine Dolph³

1. *University of North Carolina at Greensboro, Greensboro, NC, United States*

2. *Kangwon National University, Gangwon*

3. *University of Minnesota, St. Paul*

Biodiversity is regulated by a wide array of processes; among the most widely recognized drivers at broad spatial scales are environmental filtering and negative biotic interactions. In contrast, although ecologists have increasingly recognized the importance of positive biotic interactions—such as facilitation and mutualism—their influence on regional biodiversity patterns remains controversial, largely due to limited empirical evidence. Here, we examine the role of positive interactions in shaping diversity at the metacommunity level. Using extensive freshwater fish community data from North America, we focus on nest association—a form of reproductive facilitation between host and beneficiary species—as a model system to test whether and how the geographic range of host species (i.e., metapopulation occupancy) regulates regional biodiversity. We found that greater host occupancy increased local species richness (α diversity) but decreased β diversity, indicating that positive interactions can promote biotic homogenization. As a result, γ diversity remained unchanged because opposing patterns in α and β diversity offset one another at the regional scale. These patterns were supported by strong local associations between host and beneficiary species, demonstrating that positive interactions influence both local richness and spatial variation in community structure. Together, our results provide robust empirical evidence that positive biotic interactions can have far-reaching effects on biodiversity across spatial scales and represent an underappreciated yet important component of metacommunity dynamics with implications for large-scale biodiversity conservation.

id #135039

Particulate organic nitrogen losses partially offset nitrogen retention in a plant-dominated river

Sarah S Roley¹, John P Buster¹, Aaron C Pelly¹

1. *Washington State University, WA, -*

In most streams, assimilatory uptake dominates nitrogen (N) uptake. Assimilatory uptake is a temporary removal, and so understanding the fate of assimilated N is important to an overall understanding of stream N cycling patterns. In plant-dominated river reaches, N assimilation rates tend to be high, and the assimilated N is typically retained for longer than N assimilated by biofilm and other compartments. However, plants typically senesce at the end of the growing season and much of the biomass drifts downstream, potentially releasing a pulse of assimilated N as particulate organic N (PON). We hypothesized that a plant-dominated river reach would have N assimilation and inorganic N retention rates that exceeded typical rates in reaches dominated by biofilm, but that PON losses would offset much of the total assimilation. We measured assimilation, retention, and particulate N losses in the lower Yakima River, in Washington state, USA, where nearly the full river width contains water stargrass (*Heteranthera dubia*). We estimated gross assimilatory uptake as plant biomass from quadrat samples multiplied by N content. Next, we estimated annual inorganic N retention by doing a nitrate mass balance of a 35 km reach with no tributaries or groundwater inputs. To complete the nitrate mass balance, we used continuous nitrate and discharge data from USGS sensors. Inorganic N retention calculations do not include PON, and so we measured PON losses by deploying a drift net during plant senescence and measuring biomass and N content of drifting plant material. The annual gross plant N assimilatory uptake rate was $20 \text{ g N m}^{-2} \text{ yr}^{-1}$, with half subsequently released as nitrate, for an N retention rate of $10 \text{ g N m}^{-2} \text{ yr}^{-1}$. During senescence and drift, 10–20% of the retained N was released as PON, suggesting that PON losses are a meaningful portion of N loss, but even so, this plant-dominated river reach retained more N than a typical biofilm-dominated reach.

id #135807

Testing thermal niche adaptations on energy demands between high- and low-elevation fishes within benthic and pelagic habitat guilds

Karmann Kessler¹, Xingli Giam², Matthew Troia¹

1. *University of Texas at San Antonio, San Antonio, TX, United States*

2. *University of Tennessee, Knoxville, TN, United States*

Individual energy demands constrain survival and growth that determine population level performance ultimately driving geographic range limits. Differences in species traits like behavior and thermal physiology affect energy demands and therefore drive interspecific differences in the aforementioned performance metrics. We tested two hypotheses for four fishes in the Appalachian Mountains of the Southeastern USA: 1) pelagic species have higher energy demands and therefore higher consumption rates relative to benthic species and 2) high-elevation species exhibit a distinct thermal niche adaptation compared to low-elevation species. This distinction could either be (a) thermal adaptation, resulting in high-elevation species having lower thermal optimum than low-elevation species or (b) countergradient variation, resulting in high-elevation species having higher consumption rates across temperatures due to increased thermal sensitivity in comparison to low-elevation species. We performed consumption experiments in the laboratory at temperatures from 8 to 32°C to address these hypotheses. We found lower energy demands for benthic versus pelagic species across all temperatures. As example, pelagic species consumed mean $2,308 \text{ J/g/d}$ at 28°C whereas benthic species consumed mean 625 J/g/d at the same temperature, supporting Hypothesis 1. High and low-elevation pelagic species had the same optimum temperature for consumption (28°C) while no clear optimum was

present for benthic species, not supporting Hypothesis 2a. However, high-elevation species had lower consumption rates at warm temperatures for both habitat guilds, supporting Hypothesis 2a and refuting Hypothesis 2b. These findings provide evidence for thermal adaptation and emphasize distinctions in energy demands between habitat guilds and elevational ranges to inform conservation planning.

id #136063

Continental Mapping of Mesic Ecosystem Using Machine Learning: Effects of Training Dataset Size and Spatial Autocorrelation

JUAN CAMILO Rojas-Lucero¹, Trevor Caughlin¹, Jodi Brandt¹

1. Boise State University, BOISE, ID, United States

Mesic ecosystems occupy a very small portion of semi-arid landscapes, yet they function as ecological and hydrological keystones supporting biodiversity, wildlife, and moisture-dependent services across the sagebrush biome. Despite their importance, monitoring mesic ecosystems at broad spatial and temporal scales remains challenging because these habitats are rare, spatially fragmented, and highly variable across environmental gradients. Existing satellite products face trade-offs between spatial resolution (Sentinel-2) and temporal depth (Landsat), complicating efforts to generate long-term, high-resolution maps. Although machine learning offers new opportunities for continental-scale vegetation monitoring, performance is often constrained by limited training data and strong spatial autocorrelation. Addressing these challenges is essential for building reliable, generalizable models. This study developed a Random Forest regression framework integrating Landsat imagery, Sentinel-2 training products, and environmental covariates to generate growing-season mesic vegetation and surface-water fractional cover from 1984 to present at 30-m resolution across the sagebrush biome. We evaluated (1) how training dataset size affects model accuracy, (2) how errors vary across ecoregions, and (3) whether incorporating spatial structure improves performance. Approximately 250,000 ecoregion-stratified training samples were derived from Sentinel-2 fractional cover data (2017–2020). Five subsets (56k–250k points) were used to test sensitivity to training size. Two model configurations were compared: a spectral-only baseline and a spatially informed model including ecoregions and geographic coordinates. Independent validation samples were stratified by Level II ecoregions.

Results show that training dataset size strongly controls prediction accuracy. MAE exceeded 0.42 with 56k samples, declined substantially at 116k and 168k, and stabilized near 224k samples (MAE < 0.28), indicating a saturation point for continental-scale mapping. Spatial predictors produced modest improvements in overall error but enhanced ecological consistency across regions.

This study provides a scalable workflow for long-term mesic ecosystem monitoring. Using sufficiently large, ecologically representative training datasets and incorporating spatial structure are critical for accurate large-area mapping. The resulting 40-year products support conservation planning, hydrological monitoring, and long-term ecosystem assessment across the sagebrush biome.

id #135808

From Closed Doors to Open Windows: Reflections on “Failure”, Interdisciplinarity, and Opportunities in Academia and Non-Academia

Ryan McManamay¹

1. Baylor University, Waco, TEXAS, United States

Contemporary science not only requires, but often celebrates, interdisciplinarity or its derivatives (multi- or trans-disciplinarity). Yet interdisciplinarity is most often envisioned as individuals of different disciplinary expertise working together under various configurations. Very rarely, however, are interdisciplinarity individuals, i.e. those that cross multiple, often disparate, disciplinary lines in research, held in the same regard in modern academia. In part, this stems from strong linkage between degree programs, professional identity, scientific societal reputation, and building singular disciplinary expertise; this is only reinforced during cycles of academic hiring, tenure, and promotion. This modern construct poorly reflects our scientific predecessors and their contributions to, and perhaps, transformations of, society. Unfortunately, interdisciplinarity comes with increased risk of research and career failures. Furthermore, this viewpoint continues to thwart interdisciplinarians from the academic sector. In this presentation, I'll contrast modern views of interdisciplinarity to historical scientists and non-scientists and discuss how academia has and could pivot to meet growing societal and cultural needs. I'll reflect on my own journey as an ecologist through two diverse graduate degree programs, eight years at a National Laboratory, and the past seven years at a university that shaped 20 years of research spanning the disciplines of stream ecology, fisheries, environmental flows, hydrology, geography, environmental science, and civil engineering. I'll provide examples of how early failures in my dissertation and career failures led to new opportunities, illustrating how necessity is the mother of invention (or just simple luck) and how closed doors lead to open windows with even better, unanticipated outcomes. Hopefully, in reflecting on my own experiences and failures, I can give some perspective and advice to early career interdisciplinarity scientists, and those considering careers in corporate research or academia, especially in freshwater science.

id #136064

Impact of Climate Change on Macroinvertebrate Functional Traits in the Pacific Northwest

David Wooster¹, Chad Larson², Edward Krynak³

1. Oregon State University, OR, United States

2. *Washington State Department of Ecology, Olympia, Washington, United States*

3. *Washington State Department of Ecology, Olympia, Washington*

Climate change is expected to profoundly affect freshwater ecosystems by reshaping the distribution and prevalence of aquatic taxa, with cascading consequences for community structure and ecosystem function. One increasingly important impact of climate change is rising stream intermittency, which alters hydrologic regimes and imposes strong environmental filters on stream communities. To evaluate how projected increases in temperature and changes in precipitation may influence stream macroinvertebrates under increasing intermittency, we applied ensemble species distribution models to data from more than 1,000 stream sites across Washington State, with a focus on functional traits related to flow sensitivity and disturbance tolerance.

We compared downscaled CMIP6 climate projections for 2081–2100 under four greenhouse gas emission scenarios with historical climate conditions for 391 macroinvertebrate taxa. Analyses emphasized functional traits expected to respond predictably to intermittency, including voltinism, rheophily, and functional feeding groups. Our objectives were to (1) identify the bioclimatic variables to which macroinvertebrate taxa are most sensitive and (2) quantify projected changes in the prevalence and spatial distribution of functional trait modalities. We hypothesized declines in semi-voltine taxa and increases in multivoltine taxa, along with losses of erosional taxa and increased prevalence of depositional taxa.

Annual precipitation emerged as the most influential predictor for the majority of taxa, followed by annual mean temperature and precipitation seasonality. Projections indicated increases in the prevalence of multivoltine taxa and taxa associated with depositional habitats across emission scenarios, with stronger responses under higher carbon scenarios. In contrast, semi-voltine taxa and taxa associated with erosional habitats consistently declined in prevalence. Functional feeding groups also exhibited pronounced shifts, with declines in shredders and collector-filterers and increases in herbivores and parasites across scenarios.

Together, these results suggest that increasing stream intermittency may substantially restructure stream communities and alter ecosystem function by favoring fast-reproducing taxa and shifting the prevalence of functional feeding groups.

id #135043

Ionic contributions to specific conductivity dynamics in an urbanized stream network

Katherine X. Pérez Rivera¹, Alejandra L. Flota¹, Emma A. Lucier¹, Erin R. Hotchkiss¹

1. *Virginia Polytechnic Institute and State University, Blacksburg, VIRGINIA, United States*

Human activities are increasing dissolved salt concentrations in freshwater ecosystems, contributing to freshwater salinization. In urban environments, research on salinization has become a focal point regarding the sources of salt inputs to streams and rivers. Urban development alters stream structure through channelization, the construction of retention ponds, and the burial of streams, thereby modifying natural flow paths. However, the influence of stream flow modification on dissolved salts is not well addressed in the context of freshwater salinization. Here, we evaluated how urban heterogeneity influences dissolved salts within an urbanized stream network. To do this, we deployed high-frequency water level, temperature, and conductivity sensors at the two inlets and the outlet of a retention pond within Stroubles Creek in Blacksburg, VA, and collected surface water samples to analyze for major ions from May 2023 to December 2024. We used specific conductivity (SpC) as an integrative metric of salinization to compare sites, identify which dissolved salts contributed the most to total SpC, and evaluate the influence of water level on SpC dynamics. Our preliminary results identified Cl^- , Ca^{2+} , and Mg^{2+} as the major contributors to SpC among sites (> 20%), while reactive salts such as NO_3^- and PO_4^{2-} contributed < 5 %. The retention pond buffered SpC's response to water level at the outlet and increases in water level generally diluted SpC among all sites. Major ion contributions of Cl^- , Ca^{2+} , and Mg^{2+} are associated with impervious surfaces and road salts. Analyses of dissolved salts among sites found that NO_3^- , PO_4^{2-} , and Mg^{2+} were different ($p < 0.05$) and lower at the outlet compared to either inlet, suggesting net removal within the retention pond, potentially due to biological uptake and/or water-sediment interactions. Future work will focus on evaluating the effects of seasonality and water level on ionic contributions to SpC to assess how dissolved salts in our system may be changing, and its implications for water quality and ecosystem function.

id #136067

Linking Wildfire Disturbance to Patterns in Aquatic Macroinvertebrate Communities Using Spatially Diverse Biomonitoring Data

Brian Bush¹, Matthew Trentman¹, Kaitlyn Thomas¹, Ian Hellman¹

1. *University of Montana, Missoula, MT, United States*

Wildfires are a significant disturbance affecting aquatic ecosystems in the western United States, with an estimated 11% of streams impacted within the past 30 years (1). Despite extensive research on wildfire effects, the specific mechanisms by which fires influence aquatic systems, such as increased sedimentation and reduced riparian cover, remain highly system-dependent and poorly understood. As wildfire size and severity are projected to increase under anthropogenic climate change (2), improving mechanistic understanding of fire effects on aquatic ecosystems is increasingly important. Aquatic macroinvertebrates play a key role in stream ecosystem function and often respond to wildfire in predictable ways, making them an effective indicator of wildfire disturbance (3).

To address this knowledge gap, we combined macroinvertebrate data from the Bureau of Land Management's (BLM) Assessment, Inventory, and Monitoring (AIM) program with spatial fire datasets to construct a large ($N > 600$) dataset of stream reaches across BLM managed land in Idaho, including standardized community data, local environmental variables, and fire disturbance metrics. We used this dataset to explore relationships between wildfire disturbance and macroinvertebrate community composition.

Because our ultimate goal was to develop a Structural Equation Model (SEM) describing causal pathways of fire effects, we first identified broad-scale drivers of community structure independent of fire using multivariate analyses. PERMANOVA indicated that geographic location, represented by field office, explained approximately 15% of variation in community composition ($p < 0.001$). Distance-based redundancy analysis (dbRDA) then identified percent overhead cover and stream temperature as the strongest predictors of community composition. These variables will be incorporated into an SEM to evaluate whether higher fire

disturbance (characterized by greater percent of watershed burned, closer proximity to fire, and more recent fires) indirectly influenced macroinvertebrate communities through changes in overhead cover and sedimentation, measured as a decrease in median particle size (D50). Given the complex and indirect pathways through which wildfires affect aquatic ecosystems, this study highlights the value of spatially extensive, long-term monitoring datasets for disentangling fire-driven mechanisms and guiding more targeted hypothesis development.

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id #134532

Freshwater Science with the City of Austin

Abel Porras¹

1. *City of Austin Watershed Protection Department, TX, United States*

The City of Austin Watershed Protection (AWP) Department has long supported efforts in protecting and improving our streams. Watershed ordinances were first implemented in Austin in 1986 and a citizen sponsored ordinance protecting Barton Springs and its watersheds passed in 1998. To that end, since the early 1990's AWP has employed scientists, engineers, planners, and environmental coordinators in its mission to protect lives and properties from flooding, erosion, and pollution. Specifically, AWP engages in freshwater science to monitor the state of its water bodies and to evaluate interventions (such as policies, programs, and projects) that mitigate the impacts of urban development.

I am one in a long line of applied mathematicians/statisticians that have supported this work. Through consultations on design of experiments and statistical analyses, I have collaborated with other environmental scientists to assess various interventions and to document our results. These assessments range from impacts of wastewater discharges on our creeks to analyses of benthic macroinvertebrate community metrics to estimates of sediment quality.

As I grew within the department, I was lucky enough to become a section manager supervising a group of environmental modelers and programmers. As part of this group, we launched a project to simulate every watershed in Austin with the goal of quantifying hydrology, estimating sediment and pollution transport, and predicting bioassessments every kilometer along the creeks. Under my guidance we've expanded our skill set to incorporate modern statistical algorithms, such as Functional Data Analysis and Deep Learning methods, to find patterns in relationships between hydrology and riparian ecology. These efforts culminated in a published paper in *Freshwater Science* in December 2025. All of this work was done not as an academic exercise, rather to support decision making on what areas need prioritization and what intervention is needed.

I would be honored to speak about either my experiences and career growth in AWP or the various freshwater science opportunities available in AWP and its long-storied past in providing improved water quality to our freshwater streams.

id #136068

Nutrient Load Scaling: Deviations in Nested Relationships

Jazzmyn Luna¹, Alex Webster¹

1. *University of New Mexico, Albuquerque, NM, United States*

Monsoonal storm events in semi-arid montane watersheds generate short-lived but disproportionately large nutrient and suspended sediment load pulses with important downstream ecological consequences. Rapid increases in discharge during storm events mobilize and transport nutrients and sediment from hillslopes, soils, and floodplains into stream networks, often accounting for a substantial fraction of annual material export. These flood-driven expansion and contraction dynamics create critical hydrologic connections that regulate when, where, and how nutrients and sediment enter the aquatic system and are delivered to downstream waters. Understanding the controls on nutrient and sediment loading during storm events is therefore essential for predicting watershed-scale nutrient export under a changing climate.

We employed a nested watershed approach to quantify storm-driven nutrient dynamics across spatial scales. In situ sensors were deployed throughout the stream network to continuously measure flow, nitrate, dissolved organic carbon, and total suspended solids at 15-minute intervals, enabling capture of rapid changes in concentration and discharge during monsoonal storm events. Storm events were defined as periods when discharge exceeded two times base flow and were detected at a minimum of two locations within the watershed. Identified storms were then used to calculate nutrient loads by integrating solute concentrations over the duration of each event, quantified as the area under the concentration–time curve.

We hypothesized that nutrient loads would scale linearly with contributing drainage area. We found this to be true of dissolved organic carbon and total suspended solids, where there is a linear increase in load with contributing area. However, nitrate loads followed a unimodal pattern, suggesting non-conservative behavior during storm events. This talk explores the mechanisms underlying this deviation by integrating storm-based metrics, including hysteresis indices and flushing indices, to evaluate how event-scale transport, source activation, and in-stream processing shape the predictability and scaling of nitrate loads across stream networks.

id #134533

Storm-driven nitrate loading varies with land use along a headwater stream continuum

Joseph Baldus¹, Lara Munro¹, Wilfred Wollheim¹

1. University of New Hampshire, Durham, NH, United States

Excess nitrogen loading is a major threat to freshwater and downstream coastal ecosystems, particularly in human-impacted watersheds dominated by agricultural and urban land use. Nitrogen “impairment” can trigger a cascade of harmful ecological effects: eutrophication, biodiversity loss, and acidification, as well as broader negative impacts on atmospheric conditions, air quality, and human health. Storm events often contribute a disproportionately large fraction of annual nutrient exports, with pulses of mobilized nutrients overwhelming ecosystem function at higher flows. Yet storm-scale nitrogen dynamics remain poorly understood due to historical limitations in temporal data. As storm frequency and intensity increase under climate change, understanding what controls patterns of storm-driven nitrogen loading is increasingly important.

This study investigated how land use and hydroclimatic conditions jointly influence nitrate sources, transport pathways, and export during storm events along a mixed land-use stream continuum. We deployed high-frequency (15-minute) in situ sensors measuring nitrate, discharge, and specific conductance at three nested sites along College Brook (New Hampshire, USA), spanning agricultural, forested, and urban reaches. Storm event nitrate loading was quantified using concentration–discharge (C–Q) relationships, hysteresis metrics, and event-scale nitrogen fluxes.

We hypothesize that agricultural and urban reaches provide readily mobilized nitrate sources and minimal riparian buffering, which will generate flushing (increased concentrations) and positive hysteresis, whereas forested reaches exhibit dilution (decreased concentrations) due to source limitation and riparian buffering. We also hypothesize that longer antecedent dry periods between storms enable nitrate accumulation in soils via mineralization, resulting in stronger initial flushing and storm-scale nitrogen fluxes during subsequent storms.

We present comparisons of storm event nitrate loading across land-use reaches and evaluate how these responses vary with precipitation characteristics, seasonality, and antecedent moisture conditions. Preliminary results show greater flushing, fluxes, and positive hysteresis linked agricultural land use, before marked dilution by forested land continuing downstream. By capturing nutrient dynamics at multiple points along a stream continuum, this work disentangles effects of fine-scale land use that are often obscured in single downstream monitoring locations within complex, mixed land-use watersheds. These findings will inform stream restoration efforts in College Brook and help watershed managers identify high-priority locations for mitigation.

id #135814

Coupled watershed and stream modeling to evaluate wastewater management scenarios in an urban stream

Angel Santiago¹, Abel Porras¹, Dana Thomas¹, Bianca Perez¹

1. City of Austin, Austin, TEXAS, United States

Urban streams in rapidly developing regions increasingly receive treated wastewater effluent as a significant component of baseflow, altering hydrology and water quality during dry-weather conditions. Although wastewater discharges operate in compliance with permitting frameworks, these permits may not include sufficiently protective limits for nutrients such as nitrogen and phosphorus, which can drive eutrophication, excessive algal growth, and dissolved oxygen depletion. As a result, evaluating how different treatment levels and effluent quality influence nutrient dynamics and benthic algae growth remains a challenge, particularly under variable hydrologic conditions that control dilution and transport.

We present a coupled modeling approach that integrates watershed-scale hydrology with in-stream water quality processes to evaluate wastewater impacts and management scenarios in an urban watershed. The Gridded Surface Subsurface Hydrologic Analysis (GSSHA) model was used to simulate watershed hydrology and generate flow inputs to the receiving stream. These flows were then applied as boundary conditions in the QUAL2K model to simulate in-stream water quality responses, including nutrient dynamics, dissolved oxygen, and benthic algae growth.

The approach was applied to an urban watershed in Austin, Texas, where wastewater discharges influence stream conditions. QUAL2K was calibrated using observed data and subsequently used to evaluate alternative wastewater management scenarios, including variations in treatment performance and effluent quality. Scenario analyses quantify changes in nutrient concentrations and benthic algae growth along the stream network under different discharge conditions.

Results demonstrate that integrating watershed hydrology with stream water quality modeling provides a robust framework for isolating wastewater contributions and evaluating the effectiveness of treatment and management strategies. This approach supports decision-making by enabling comparison of regulatory and operational scenarios and identifying conditions under which wastewater discharges may contribute to water quality impairment or improvement.

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id #134535

Spatiotemporal variation in algal and bacterial assemblages associated with toxigenic benthic cyanobacteria *Microcoleus* and *Anabaena* in northern Californian rivers

Jordan Zabrecky¹, Taryn Elliott¹, Meghan Hickey¹, Raina Fitzpatrick², Rosalina Stancheva³, Laurel Genzoli¹, Grant Johnson⁴, Shadman Kaiser⁵, Abeer Sohrab⁵, Ramesh Goel⁵, Joanna Blaszcak¹

1. University of Nevada, Reno, Reno, NV, United States
2. Northern Arizona University, Flagstaff, AZ, United States
3. George Mason University, Fairfax, VA, United States
4. Karuk Tribe Water Quality Program, Happy Camp, CA, United States
5. University of Utah, Salt Lake City, UT, United States

Proliferations of benthic cyanobacteria associated with the production of neurotoxic cyanotoxins known as anatoxins have been increasingly observed in rivers worldwide. However, controls on their anatoxin production are poorly understood, as previous studies have found weak relationships between anatoxin concentrations and environmental factors such as water quality. This suggests that perhaps biotic factors, such as species interactions among co-occurring taxa, may influence anatoxin production. In this study, we investigated how algal and microbial assemblages within benthic mats dominated by one of two cyanobacterial taxa associated with anatoxins, *Microcoleus* and *Anabaena*, varied through time and among rivers and with variation in anatoxin concentrations. From June to September in 2022, we collected *Microcoleus* and *Anabaena* mats biweekly from the South Fork Eel, Russian, and Salmon Rivers in northern California. We quantified the relative abundances of algal groups with microscopy and bacteria with 16s rRNA, and analyzed samples for anatoxin concentrations using LC-MS/MS. Both algal and bacterial assemblages associated with *Microcoleus* significantly differed among rivers while assemblages associated with *Anabaena* were similar among rivers. *Microcoleus* samples from the South Fork Eel River were associated with higher relative abundances of nitrogen-fixing diatoms (*Epithemia*) and cyanobacteria (e.g., *Anabaena*, *Nostoc*) and had a higher median bacterial diversity than that of *Microcoleus* from the Salmon River as quantified by the Shannon Diversity Index (5.7 vs. 4.9). *Anabaena* samples from all three rivers had high relative abundances of diatoms (including *Epithemia*), coccoid cyanobacteria, and other anatoxin-producing cyanobacteria (especially *Geitlerinema*) but had similar bacterial diversity among rivers (5.5-5.6). Preliminary results suggest that algal and bacterial assemblages associated with *Microcoleus* and *Anabaena* changed with increasing anatoxin concentrations within the South Fork Eel River where anatoxin concentrations reached a maximum of over 100 µg anatoxins g⁻¹ organic matter, but not for *Anabaena* in the Russian River where anatoxin concentrations only reached up to 5 µg anatoxins g⁻¹ organic matter. Future work will identify taxa responsible for composition differences and will incorporate theoretical functional groups of the bacterial assemblages as determined by PICRUST2.

id #135815

Parallel patterns or divergent responses? Investigating species-genetic diversity correlations among macroinvertebrates in streams across the southern United States

Meryl C Mims¹, Samuel C Silknetter², Michael T Bogan³, Albert Ruhi Vidal³, Daryl R Trumbo⁴, Brian A Gill³, Kyle Leathers⁵, Zacchaeus G Compson⁶, Travis M Apgar⁷, Carla L Atkinson⁸, Jacob Dorris³, Shang Gao³, Kierstyn T Higgins⁹, Kelsey D Hollien³, Megan Malish¹⁰, Tom Neeson¹¹, Ariel Shogren⁸, Chelsea Smith⁸, Daniel C Allen⁹

1. Virginia Tech, Blacksburg, VA, United States
2. Susquehanna University, PA
3. School of Natural Resources and the Environment, University of Arizona, Tucson, Arizona, USA
4. Colorado State University Pueblo, Pueblo, CO, USA
5. California Water Science Center, U.S. Geological Survey, Sacramento, CA, USA
6. Biological Sciences, University of North Texas, Denton, Texas, USA
7. Water Branch, California Department of Fish and Wildlife, Sacramento, CA, USA
8. Biological Sciences, University of Alabama, Tuscaloosa, Alabama, USA
9. Department of Ecosystem Science and Management, Penn State, University Park, PA, USA
10. Department of Fisheries and Wildlife, Michigan State University, East Lansing, MI, USA
11. Department of Geography and Environmental Sustainability, University of Oklahoma, Norman, OK, 73019

Biodiversity across levels of biological organization, from genes to communities to ecosystems, is responding to rapid environmental change globally. Yet the degree to which diversity at different biological scales responds to environmental drivers in parallel or in divergent ways remains poorly understood and difficult to predict. This is true for benthic organisms affected by stream drying, a phenomenon increasing in frequency and severity globally. Despite advances in understanding how stream drying independently influences species diversity and genetic diversity of aquatic organisms, whether species and genetic diversity are responding in parallel ways to stream drying dynamics remains unclear. For example, does high species diversity always correlate with high genetic diversity? Or, does the relationship between species and genetic diversity depend on environmental or biological factors? As part of the StreamCLIMES Project, we tested species-genetic diversity correlations (SGDCs) for multiple macroinvertebrate taxa in five stream basins across the southern United States. We generated population genetic datasets (COI and double-digest restriction-site associated DNA) for focal mayfly taxa (Baetidae/Caenidae) from each stream basin for comparison with macroinvertebrate community surveys. We hypothesized that the strength and direction of SGDCs vary among basins and taxa, and that a combination of biological (trait-based) and environmental (landscape, flow permanence) factors explain the strength and direction of observed SGDCs. We present findings from a weighted, information-based decomposition framework to compare levels of allelic and species diversity. Overall, our study helps inform when positive SGDCs can be expected in stream ecosystems and the role that stream drying plays. This is particularly vital when conservation and research resources prohibit genetic analysis for all species of interest, in which cases community-level data may serve as a proxy for population-level biodiversity.

id #136071

Finding a Fluorinated Fingerprint? Examining spatial variation in and correlations between different PFAS compounds in New Hampshire groundwater

William G McDowell¹, Sophia Beland¹

1. Merrimack College, NH, United States

Per- and polyfluoroalkyl substances (PFAS) are an emerging contaminant in freshwater ecosystems. Due to their persistence in the environment, this broadly used class of chemicals have been dubbed “forever chemicals” and can have negative human health consequences even in extremely low concentrations, with safe exposure rates in the parts per quadrillion range for PFOA and PFOS. Following the discovery of large PFAS contamination surrounding the Saint Gobain plastics production plant in Merrimack, New Hampshire in 2016, the state Department of Environmental Services began an aggressive testing campaign for both municipal and private water sources. Private water sources represent an intriguing data source, as residences that do not have municipal water rely on groundwater sources from private wells. We examined state data for over 17,000 well samples that tested for five different PFAS compounds (PFOS, PFOA, PFNA, PFBS, PFHXS) from 2016-2025. Using ArcGIS, we interpolated between sampling points to create a statewide map of PFAS concentrations in groundwater. We found PFAS “hotspots” throughout the state, particularly surrounding current and former landfill sites, the former location of the plastics plant, and former military bases. We are currently examining using a non-metric multidimensional scaling (NMDS) analysis to determine if sites adjacent to different types of contamination sources have a distinct “fingerprint” based on concentrations of the five different PFAS compounds.

id #135560

Eat at your own risk: PFAS in Lake Michigan prey fish

Daniele Miranda¹, Alison Zachritz¹, Heather Whitehead¹, Graham Peaslee¹, Gary Lamberti¹

1. University of Notre Dame, Notre Dame, IN, United States

A multimillion-dollar salmonid fishery in Lake Michigan is enabled by abundant prey fish that serve as a key link between basal resources and top predators. This role also makes prey fish an important conduit for per- and polyfluoroalkyl substances (PFAS) in the food web. We quantified 21 PFAS in six prey fish species (alewife, bloater, round goby, deepwater sculpin, slimy sculpin, and gizzard shad) to understand how habitat use, feeding guild, and ontogeny influence PFAS accumulation in the critical middle of the Lake Michigan food web. Fish were collected from seven harbors encircling the lake during the summers of 2020 and 2021. PFAS were detected in all fish ($N = 301$), with species-specific concentrations ranging from 3.22 ± 0.8 (gizzard shad; mean $\pm$ SE) to 100.9 ± 11 ng/g wet weight (slimy sculpin); upper-end concentrations were notably high relative to top predators studied previously in the same region (Lake Michigan, Great Lakes, North America). Clustered isotopically ($\delta^{15}\text{N}$ and $\delta^{13}\text{C}$), benthic species associated with fine substrates (deepwater and slimy sculpins; 45.7 ± 2.63) had higher PFAS than those found on coarse substrates (round goby; 3.71 ± 0.35), suggesting that fine sediments are an important PFAS reservoir and exposure pathway for organisms. Pelagic zooplanktivores combined (alewife and bloater; 6.18 ± 0.07) had intermediate PFAS concentrations, whereas the filter-feeding planktivore gizzard shad exhibited the lowest overall PFAS levels. Carbon stable isotopes indicated narrow dietary niches for most species, with broader niches for gizzard shad and round goby. However, broader diets were not associated with higher PFAS loads in these species. Ontogenetic patterns appeared to influence PFAS uptake; PFOS, the most frequently detected PFAS, increased with age in deepwater sculpin, decreased with age in bloater, and showed no pattern in alewife, likely reflecting species-specific habitat shifts and benthic–pelagic coupling. Overall, PFAS accumulation was strongly influenced by species ecology, highlighting how ecological traits in the food web can influence contaminant transfer to higher trophic levels.

id #136072

PFAS contamination in South Dakota: Insights from a statewide soil and water survey

Mehzabeen Mannan^{1,2}, Bret Lingwall^{1,2}, Kurt Chowanski^{1,2}, Heidi L. Sieverding^{1,3}, Daniel K. Owens^{1,2}, David M. Waterman^{1,3,4}, Lisa Kunza^{1,2}

1. Center for Sustainable Solutions, South Dakota School of Mines and Technology, Rapid City, SD, USA , Rapid City, SD, USA

2. Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines and Technology, Rapid City, SD, USA , Rapid City, SD, USA

3. Department of Civil and Environmental Engineering, South Dakota School of Mines and Technology, Rapid City, SD, USA, Rapid City, SD, USA

4. Deceased , Rapid City

Per- and polyfluoroalkyl substances (PFAS) are persistent contaminants of increasing concern for environmental and human health, yet state-level data remain sparse for many regions of the United States. This study presents a statewide survey of PFAS occurrence in South Dakota (SD) and a few nearby locations based on analysis of samples collected (431 samples) and compiled through 2023, with emphasis on environmental matrices relevant to agriculture and human exposure, including surface water, groundwater, drinking water, and soil. Samples were analyzed for up to 30 PFAS compounds using LC-MS/MS. Overall, non-detects dominated the dataset; however, detection frequency was strongly influenced by the number of PFAS compounds analyzed per sample. Among samples in which six or fewer PFAS were assessed ($n = 284$), only 1.7% exhibited detectable PFAS concentrations. In contrast, samples analyzed for 19 or more PFAS compounds ($n = 147$, all from South Dakota) showed detections in 53.7% of cases. Detected PFAS were associated with localized point and land-use related sources. Elevated PFAS concentrations in surface waters were consistently observed downstream of wastewater treatment plant effluent, although concentrations generally remained below 10 ng/L. Groundwater contamination was rare and confined almost exclusively the vicinity of a fire training area with historic use of Aqueous Film-Forming Foam (AFFF) and the surrounding area, where all groundwater samples within 3 km of the fire training area contained detectable PFAS. In contrast, groundwater samples outside this region were uniformly non-detected. Soil contamination reflected multiple contributing pathways, including irrigation with contaminated groundwater, historical biosolids application, intensive fertilizer and pesticide use, and areas of heavy vehicle

activity. Several soils without biosolids application exceeded 200 ng/kg PFAS, while sites with extensive biosolids use or groundwater irrigation exceeded 1,000 ng/kg. PFOS and PFHxS were the most frequently detected compounds, followed by PFHxA, PFOA, PFBS, and PFBA. PFAS concentrations in soils and waters from high-elevation and remote areas were uniformly non-detect, suggesting negligible atmospheric deposition. Results from this study establish a baseline PFAS dataset for SD and provide critical evidence to support targeted monitoring, source control, and risk-informed management strategies for protecting agricultural systems and rural water resources.

id #134537

Arctic rusting streams and rivers show spatial and temporal variability in dissolved metal concentrations from acid rock drainage

Abagael N Pruitt¹, Taylor G Evinger¹, Jonathan A O'Donnell², Joshua C Koch³, Michael P Carey³, Carson A Baughman³, Kenneth Hill⁴, Brett A Poulin¹

1. *University of California, Davis, Oakland, CA, United States*
2. *Arctic Inventory and Monitoring Network, U.S. National Parks Service, Anchorage, Alaska*
3. *Alaska Science Center, U.S. Geological Survey, Anchorage, Alaska*
4. *Arctic and Central Alaska Networks, U.S. National Parks Service, Fairbanks, Alaska*

Streams and rivers in Arctic Alaska are turning orange due to acid rock drainage that mobilizes metals from sulfide minerals below thawing permafrost. These “rusting” streams are often characterized by lower pH and elevated sulfate (SO_4^{2-}) and metal concentrations (e.g., iron (Fe), aluminum (Al), and zinc (Zn)) with potential consequences for fisheries, biodiversity, and human health. However, the spatial and temporal variability of chemical signatures from acid rock drainage, and the relative contributions of point versus diffuse sources remain poorly understood. Here, we performed repeat synoptic samplings of hillside seeps ($n=4$), tributaries ($n=6$), and mainstem sites ($n=7$) in the Nakolikurok Creek watershed (817 km^2) in the Brooks Range, Alaska, USA, during June, July, and twice in August 2024 to understand how acid rock drainage chemical signatures varied spatially and temporally. Across the whole watershed, the pH ranged from 2.8–8.5, and was lower in the seeps and impaired tributaries compared to mainstem sites (Kruskal-Wallis $p<0.001$). Seeps and impaired tributaries showed distinct chemistry from unimpaired tributary and mainstem sites. Hillside seep chemistry was highly temporally variable; for example, SO_4^{2-} concentrations in a single seep varied from 13,700 mg/L in July, to 5960 mg/L in August, and climbed to 13,200 mg/L the following week. A subwatershed leverage (relative chemical influence) analysis indicated that tributaries (impaired and unimpaired combined) contributed -13-6465% of dissolved (filter-passing) Fe and -8-151% total Fe to the outlet of Nakolikurok Creek, suggesting that some tributaries dilute Fe while others disproportionately enrich downstream concentrations. Tributaries sampled June-August showed variable spatial stability (temporal persistence of spatial patterns) in both dissolved and total Fe (40% stable, 40% unstable), Al (66% unstable) and Zn (33% unstable), likely due to episodic inputs from impaired tributaries. Mainstem sites were predominantly moderately stable. Sulfate, a conservative tracer of acid rock drainage, was strongly related to continuous specific conductance ($R^2=0.96$), with specific conductivity and sulfate concentration peaking in late July. These findings demonstrate that acid rock drainage influences Arctic river chemistry through highly heterogeneous and dynamic source contributions, complicating efforts to predict downstream metal exposure under continued permafrost thaw.

id #135049

Holistic approach to lake restoration

Johnny S Richardson¹

1. *Leon County Public Works, Tallahassee, FL, United States*

Lake Munson is an approximately 288-acre, cypress-rimmed lake located south of the City of Tallahassee, FL. Through the mid-1900s, decades of development in the watershed, wastewater treatment facilities discharging to the tributary system, and drainage activities focused on flood reduction contributed to high nutrient loads entering Lake Munson, resulting in severe water quality and ecological problems. In the early 1980s, Lake Munson was ranked one of the most polluted lakes in Florida. Since then, efforts to restore this valuable resource, improve the water quality, and reduce the nutrient loading in Lake Munson have been ongoing. The assessment of data, analyses, and scientific findings is consistently conducted to determine optimal strategies and emerging possibilities aimed at enhancing the water quality of Lake Munson. The primary objective is to achieve significant improvements in lake water quality while minimizing adverse effects, and this effort remains ongoing. This presentation will explore the current path to restoration.

id #136075

Scaling Nitrogen Fixation in a Coastal California River

Steve Thomas¹, Augustine Sitati¹, Michael Zampini², Michael Wulf², Saeed Kariunga², Frank Akamagwuna¹, Adrian Sun³, Mary Power³, Jane Marks²

1. *University of Alabama, Tuscaloosa, AL, United States*
2. *Northern Arizona University, Flagstaff, Arizona, United States*
3. *University of California - Berkeley, Berkeley, California, United States*

In nitrogen-limited ecosystems, biological nitrogen (N) fixation can provide an important subsidy to ecosystem N budgets. In streams, however, N-fixation is most commonly measured at the scale of individual substrates or patches and has rarely been scaled to reaches or river segments. Here, we combined substrate-specific N-fixation rates with high-resolution spatial data to generate reach-scale estimates of N-fixation in the South Fork of the Eel River (SFER), California. We used a $^{15}\text{N}_2$ tracer approach to quantify mass-specific N-fixation rates and paired these measurements with areal standing stock estimates and percent cover

derived from drone imagery. Five integrated reach-scale (~100 m) assessments were conducted across early, mid, and late summer during 2024–2025. A random, stratified sampling design was used to quantify N-fixation in (1) floating *Cladophora glomerata* mats and associated epiphytes, (2) benthic *Cladophora* turfs, (3) epilithic algae, and (4) moss.

Across both years, *Cladophora* and its epiphytes represented the largest biomass pool, whereas epilithic algae had the greatest spatial coverage. The diatom *Epithemia* spp., which hosts an N-fixing endosymbiont, dominated both epilithic and *Cladophora* epiphytic assemblages by mid- to late summer, comprising >80% of diatom abundance in *Cladophora* epiphytes and ~60% in epilithon. Preliminary 2024 estimates indicate high reach-scale N-fixation rates (~20 mg N m⁻² d⁻¹), with *Cladophora* epiphytes and epilithic assemblages contributing 45% and 54% of total activity, respectively. To place these rates in context, we estimated an annual dissolved N flux of 102.3 kg N yr⁻¹ into the study reach using discharge and repeated measurements of DIN and DON. Assuming N-fixation persists for 45 days annually, approximately 15 km of stream length would fix an amount of N equivalent to the annual upstream dissolved N input. Despite this, downstream increases in dissolved N were not observed, suggesting rapid biological uptake. If efficiently transferred through the food web, observed N-fixation rates could support secondary production of ~5.6–8.4 g dry mass m⁻² yr⁻¹. Combined, these measurements illustrate the ecosystem-scale importance of *Epithemia* spp. driven N-fixation in the SFER and the potential ecological consequences of a possible shift to N-fixing cyanobacteria if conditions change.

id #135052

Modeling multi-species assemblages of *Lestes* damselflies across a hydroperiod gradient

Peter O.L. Martin¹, Phoebe L. Zarnetske², Alisha A. Shah¹

1. Department of Integrative Biology, W.K. Kellogg Biological Station, Michigan State University, Hickory Corners, MI, United States

2. Department of Integrative Biology, Michigan State University, East Lansing, MI, United States

Freshwater systems are impacted by multiple threats such as pollution, warming, land cover change, and deoxygenation. Monitoring efforts for these systems should utilize bioindicator species, whose prevalence and abundance can inform assessments of habitat quality. In temperate regions, lentic waterbodies commonly fall along a hydroperiod gradient, from ephemeral pools to permanent lakes. *Lestes* damselflies (Zygoptera: Lestidae) segregate along this gradient, and they are effective bioindicators. As active predators that are aquatic in their juvenile stages but terrestrial as adults, *Lestes* damselflies connect the aquatic and terrestrial portions of their habitats, and their low dispersal ability links population persistence to local environmental conditions. There are however key obstacles to monitoring their populations, including the spatial patchiness of these damselflies' distributions, the known detection biases in their presence records, and the lack of prior knowledge on which environmental drivers affect their geographic distributions. As a result, it is difficult to identify probable "hotspots" for a given *Lestes* species. To address these issues, we applied multi-species distribution models, fit to presence data for ten *Lestes* species in the state of Michigan. We leveraged three separate frameworks: 1) a stacked species distribution model, 2) hierarchical modeling of species communities (HMSC), which can quantify co-occurrence patterns among species, and 3) a spatiotemporal joint species distribution model within a flexible site-occupancy framework, which can account for imperfect detection by observers. We compared the predictive performance of these three frameworks, used model output to describe the environmental, geographic, and seasonal drivers of each *Lestes* species' distribution, and, through a model ensemble approach, identified hotspots in the state of Michigan where these damselflies are more likely to be found.

id #135564

Exploring the relationship between sediment size and macroinvertebrate individual size distribution in North American freshwater streams

Morgan Hrivnak¹, Jeff Wesner¹

1. University of South Dakota, Vermillion, SD, United States

Individual size distribution (ISD) is the scaling relationship between body mass and frequency of individuals within a community. Sediment can impact the physical environment where macroinvertebrates live and can carry potentially harmful chemicals and pollutants. Previous studies have examined the relationship between the macroinvertebrate communities and chemical properties associated with sediment, yet the physical size of sediment has received less attention. This project aimed to explore the relationship between the sediment size of the substrate and the ISD of macroinvertebrates over 21 National Ecological Observation Network (NEON) Wadeable Streams across North America from Alaska to Puerto Rico. We estimated the particle size distribution of each pebble count survey from 2017 to 2021 and the corresponding ISD values of macroinvertebrates. Preliminary results show that sediment size does not impact the ISD of macroinvertebrates on a continental level. These results support the robust community-level property of ISD that remains stable across a wide gradient of sediment size.

id #135820

Evaluating the Trojan Y chromosome strategy for the removal of invasive Sacramento Pikeminnow from the Eel River, California

Alexander Juan¹, Andre Buchheister²

1. University of California, Berkeley, Berkeley, CA, United States

2. California State Polytechnic University, Humboldt, Arcata, CA, United States

The introduction and spread of Sacramento Pikeminnow (*Ptychocheilus grandis*) in northern California's Eel River basin represents a significant threat and impediment to the recovery of several threatened native fish species. This study was

undertaken to evaluate the Trojan Y chromosome (TYC) strategy for the extirpation of pikeminnow from the basin. The TYC strategy is a biocontrol method that relies on the production and stocking of fish with YY sex chromosomes, which may be phenotypically male (YY male or Myy) or female (YY female or Fyy). These YY fish produce all-male offspring when mating with their wild conspecifics, potentially leading to the extirpation of wild-type females over time. To evaluate the strategy, we constructed a mechanistic, individual-based population model to simulate a TYC Sacramento Pikeminnow stocking program. The specific study objectives were to (1) estimate the amount and time scale of YY Sacramento Pikeminnow stocking needed to achieve elimination; (2) evaluate how a TYC program would be affected by the rate of manual pikeminnow suppression, the relative survival of YY pikeminnow, and the phenotype of stocked YY fish (i.e., Fyy or Myy); (3) evaluate the mathematical feasibility of a TYC program for population reduction in an open system where elimination would not be possible due to influx from outside sources (e.g., a river tributary); and (4) identify key biological uncertainties for future investigation.

id #134541

Charades, association games, improvisation, and videography: Fun, interactive, and effective activities for ice breakers, science communication, and outreach

Juliana D'Andrilli^{1,2}, Kelly Basinger¹, Brandon Gaesser^{3,1}, Jeffrey A. Hawkes⁴, Kaleb Heinrich⁵

1. *Advanced Environmental Research Institute, Denton, Texas, USA*

2. *Biological Sciences, University of North Texas, Denton, Texas, USA*

3. *Media Arts, University of North Texas, Denton, Texas, USA*

4. *Uppsala University, Uppsala, Sweden*

5. *New Mexico State University, Las Cruces, New Mexico, USA*

Science communication and outreach should be fun, just like learning should be fun. After all, who wants to endure a boring lecture, activity, or workshop? Exactly. No one. Think back to your own undergraduate days: didn't we all gravitate towards professors and opportunities that sounded fun? And we learned that fun doesn't necessarily mean just one thing. It is not limited to physical movement, being loud and/or eccentric, or something wild and crazy. Fun is about building experiences that spark curiosity and connection and makes lasting impacts. In our careers, we embraced this philosophy by exploring diverse, creative ways to make science communication and outreach enjoyable and inclusive. When we design events, activities, or broader impacts, we aim for variety. We do this because just as not everyone learns the same way, not everyone likes to have fun the same way. In this presentation, we will share practical examples of science charades, association and connection games, improvisation exercises, and videography formats that can be used to engage audience members in person and online, worldwide. Each example will be accompanied by metrics of evaluation and feedback because we grow the most by reflecting on previous experiences, both individually and from community input. These approaches work in classrooms, workshops, webinars, social media posts, community events, and even casual gatherings with friends and family. Our science communication examples come from real projects with real goals including freshwater ecology research, professional development workshops, aquatic organic matter characterization trainings, biological and chemistry courses, and previous SFS meetings and international conferences. We aim to inspire others to build a toolkit of creative strategies that make science communication memorable, meaningful, and fun for diverse audiences.

id #135309

High-frequency sensor evidence for widespread, condition-dependent photodegradation and recharging of aromatic DOM

Alex Collins¹, Paris Collingsworth¹, Tingyu Hou¹, Jake Hosen¹

1. *Purdue University, Lafayette, INDIANA, United States*

Photochemical processing of riverine dissolved organic matter (DOM) strongly influences its composition, bioavailability, and downstream fate. Photobleaching and transformation of aromatic DOM can affect in-stream metabolism and shift DOM composition toward more microbially reactive forms. Despite its importance, photodegradation within natural waters remains poorly quantified. Most estimates are derived from lab experiments or a limited number of site-specific environmental case studies. Aromatic DOM is abundant in many freshwater systems and is highly susceptible to photodegradation. This aromatic DOM can be tracked using light absorbance of water at 254 nm (a₂₅₄), a reliable estimate of aromatic and conjugated DOM. High frequency, in situ observations diel hysteresis of a₂₅₄ levels can be used to estimate photodegradation rates at a scale not possible with other methods.

To quantify photodegradation dynamics, we analyzed high frequency sensor records from 23 NEON sites spanning diverse climatic and hydrologic regimes, complemented by NOAA buoy observations in Lake Erie. We estimated photodegradation rates using diel hysteresis between a₂₅₄ and photosynthetically active radiation (PAR) across sites and interpreted these patterns alongside temperature, turbidity, and discharge.

Across all sites, we detected ~13,000 diel cycles, of which nearly 9,000 passed quality-control criteria and were retained as hysteresis events. We found both clockwise hysteresis loops, indicating net a₂₅₄ photodegradation, and counterclockwise hysteresis loops that indicate "recharging" of aromatic DOM via in situ transformation or external inputs. Net decreases in a₂₅₄ displayed strong seasonality, with the largest declines occurring during periods of greater PAR. The extent of photodegradation increased with lower latitudes, consistent with greater UV exposure. Lastly, many sites showed diel increases in a₂₅₄, demonstrating that aromatic DOM can be replenished on short timescales by hydrologic connectivity or in stream processing.

id #135565

Tradeoffs between nutrient retention and greenhouse gas emissions in managed agricultural wetlands

Amanda K Suchy¹, Roderick Lammers¹, Aloff Maureen¹, Brown Autumn¹, Jankowski Brendan¹, Rassette Rachel¹, Reser Ainsley¹, Shablin Julia¹, Wheelock Bridget¹

1. Central Michigan University, Mount Pleasant, MI, United States

The Michigan Department of Natural Resources (DNR) manages numerous wetland areas for the primary purpose of bird habitat and waterfowl hunting opportunities. These managed wetland units fall into three broad types: (1) marshes, which have wetland vegetation, are inundated year-round, and typically have a less intensively managed hydroperiod, (2) moist soil units, which are partially cultivated but also support wet-meadow vegetation and have managed inundation from mid-fall through spring; and (3) flooded agricultural units, which are actively cultivated for row crops (corn, soybean, buckwheat) during the growing season and also have managed inundation from mid-fall through spring.

Our aim was to examine what services (nutrient retention) and potential disservices (greenhouse gas emissions) these wetland management regimes provide in addition to their primary purpose of supporting bird habitat. To that end, we collected soil cores from three units in spring, summer and fall of 2024 and 2025. We measured soil nitrogen and phosphorus, net mineralization, and potential denitrification rates. In addition, we quantified limiting substrates of denitrification using factorial additions of NO_3^- and carbon. Greenhouse gas emissions were quantified using static chambers deployed during the same sampling periods as soil core collection.

Preliminary results suggest that moist soil and flooded agricultural units had higher soil nitrogen and phosphorus concentrations than marshes, reflecting the addition of fertilizers for growing crops. Denitrification potential (denitrification rates when both NO_3^- and carbon are provided) did not differ among units, but moist soil and flooded agricultural units were carbon limited, while marsh denitrification was nitrogen limited, suggesting more capacity for NO_3^- removal in marsh units. Additionally, moist soil and flooded agricultural units emitted more CO_2 and N_2O gases than marshes, but marshes had high CH_4 emissions and thus higher overall greenhouse gas production (as a CO_2 -equivalent).

Overall, these preliminary results suggest nutrient cycling and retention differs among unit types and there may be tradeoffs in management strategies (e.g. more potential NO_3^- removal but higher greenhouse gas emissions in marshes). Continued work will provide more actionable information that can help the Michigan DNR balance these tradeoffs to maximize the services these systems provide.

id #134542

It's grow time! Seasonal and annual variability in juvenile salmonid growth within a glacierized watershed

Lindsey C McCulloch¹, Ryan Bellmore², Jason B Fellman³, Megan V McPhee¹

1. University of Alaska Fairbanks, AK, United States

2. Pacific Northwest Research Station, USDA Forest Service, Juneau, Alaska, United States

3. Alaska Coastal Rainforest Center and Program on the Environment, University of Alaska Southeast, Juneau, AK, United States

Glacial watersheds can provide diverse foraging and growth opportunities to aquatic consumers through the juxtaposition of cold, meltwater mainstems with warmer, rain-fed tributaries and floodplain channels. Adjacent habitats often display stark contrasts in food availability, temperature, and other environmental parameters on small spatial scales. While streams fed by meltwater are classically considered unproductive, these harsh environments provide habitat for robust salmon populations in Alaskan watersheds. Meltwater streams may benefit fish by providing foraging opportunities when food availability in rain-driven streams is low, generating spatiotemporal asynchronies in growth potential. Our study investigates seasonal and interannual differences in juvenile salmonid growth between meltwater and rain-driven streams within a glacial watershed in Southeast Alaska. We collected cohort and individual growth histories of PIT-tagged juvenile Coho Salmon and Dolly Varden through intensive mark-recapture at three sites consisting of a glacial mainstem and two distinct rain-fed tributaries across two growing seasons, and generated growth curves that describe how growth rates change through time at each site. We also measured a diverse set of environmental parameters that characterize the distinct seasonal cycles of our meltwater and rain-driven streams. We found that the meltwater mainstem displayed significant differences in the seasonal timing and magnitude of fish growth when compared to rain-driven tributaries, creating asynchronies in growth potential between adjacent habitats. We also observed larger interannual variation in the magnitude and timing of peak growth within the meltwater mainstem when compared to the rain-driven tributaries. Spatial asynchrony and annual variation in growth were explained by differences in environmental conditions between streams. Our results suggest that the dynamic and variable conditions offered by adjacent meltwater and rain-driven streams provide a portfolio of growth opportunities that may support the resilience of fish populations. However, decreases in meltwater runoff is eroding environmental heterogeneity in watersheds undergoing rapid glacial melt, which is likely to temporally synchronize and simplify the growth landscape through the loss of unique growth trajectories associated with disappearing meltwater streams.

id #135310

Urbanization restructures but does not consistently reduce freshwater pond biodiversity

Julia Portmann¹, David Skelly¹

1. Yale University, Hamden, CT, United States

Urbanization is a significant and increasing driver of landscape change. It has been linked to habitat fragmentation and degradation, particularly in freshwater systems. Small, seasonal ponds are a particular system of interest; they support high biodiversity but are often overlooked. Many studies focus on vertebrates and plants in lentic systems, but macroinvertebrates are an essential and abundant component of every pond. We monitored macroinvertebrate predator communities annually in thirty-two ponds spanning an urban gradient in northeastern USA (New Haven County, CT) from 2019 to 2024. We quantified

'urbanicity'—how urban the setting is in which a given pond is located—using a PCA of land cover, night time lights, and road density, measured pond surface area, water depth, water temperature, pH, conductivity, and canopy cover. We did not find clear patterns of reduced biodiversity due to urbanization, although we did detect shifts in community composition. Rather, pH, water temperature, and canopy cover were the most important predictors for overall macroinvertebrate richness and diversity. At the order level, we found that Hemiptera abundance increased with urbanicity, Trichoptera abundance declined with urbanicity, and Odonata and Coleoptera abundances were not predicted by urbanicity. At the genus/family level, only three taxa were predicted by urbanicity: *Aeshna* presence was negatively predicted while Corixidae and *Anax* presence were positively predicted. Across a robust dataset, we found minimal evidence that urbanization reduces lentic macroinvertebrate biodiversity, but that it instead shifts community composition. Urban ponds are important habitat for native biodiversity and our results suggest that loss of urban ponds is a greater threat than the conditions within or surrounding them.

id #136079

Risk assessment for invasive species in the Laurentian Great Lakes Region

Reuben Keller¹

1. *Loyola University Chicago, IL, -*

The Laurentian Great Lakes are one of the most highly invaded freshwater ecosystems in the world. Non-native species have been introduced through a variety of vectors, including ballast water, and the bait, live food, aquarium and watergarden trades. Substantial federal and international efforts have been put towards reducing the introduction of species through ballast water. In contrast, the trades in live organisms are growing in overall size and the number of species that they import. Because non-native species in these trades support commerce there is reason to allow the species if they are unlikely to become invasive. At the same time, preventing the introduction of species likely to become harmful is the best way to reduce the overall impacts of invasive species. This has led to the development and application of risk assessment tools across the Great Lakes Basin. These tools take the history of invasions and search for common traits among species that have become invasive. The assumption is that traits that have been strongly associated with invasion in the past can be used to predict which species are likely to become invasive in the future. Local, state/provincial, and federal jurisdictions across the region have independently developed risk assessment tools. These range from expert opinion to machine learning algorithms, and it is not surprising that individual species can be assessed as having very different risks depending on the tools used. Many species are now banned from import in some jurisdictions but allowed in others. The large number of risk assessment tools in use, and the long period over which they have been developed, makes the Great Lakes a particularly good case-study to investigate the promise and pitfalls of efforts to prevent new invasions in freshwater ecosystems. There are few regions in the world where so many jurisdictions are working to address invasive species in the same ecosystem. I will review the development of risk assessment tools in the Great Lakes, how and why different approaches have become popular, and how progress can be made to ensure that high risk species are prevented from introduction across the region.

id #134288

Inundation history controls sediment oxygen consumption more strongly than pyrogenic organic matter additions in parafluvial systems

Vanessa Garayburu-Caruso¹, Robert Danczak¹, Xia Gillespie¹, Alyssa Radakovich¹, Lupita Renteria¹, James Stegen¹, Nick Ward², Jianqiu Zheng³, Allison Myers-Pigg²

1. *Pacific Northwest National Laboratory, Richland, WA, United States*

2. *Pacific Northwest National Laboratory, Sequim, WA, United States*

3. *Pacific Northwest National Laboratory, Seattle, WA, United States*

River corridors are impacted by natural and anthropogenic disturbances that can interact in complex ways. Hydrologic disturbances such as drought or prolonged dry periods cause flow cessation and sediment drying in nearly all river systems, particularly within parafluvial zones that experience intermittent drying and rewetting. These variably inundated systems commonly occur in arid regions where wildfire risk is elevated, creating opportunities for interactions between hydrologic and fire disturbances. Wildfires alter river corridor processes through changes in hydrologic connectivity and pyrogenic organic matter (PyOM) inputs, yet how variable inundation and PyOM interact to influence sediment biogeochemistry remains unknown. We hypothesized that dry sediments would exhibit slower oxygen consumption rates than wet sediments, and that PyOM and organic matter additions would have greater effects on respiration in dry versus wet sediments. To test this hypothesis, we conducted a full factorial laboratory experiment using sediments from wet (continuously inundated) and dry (intermittently inundated) locations within the Columbia River parafluvial zone. Following a 7-day pre-incubation, sediments were exposed to three treatments: synthetic river water (control), unburned Douglas fir litter leachate (9 mg C/L), and high-severity burned Douglas fir litter leachate (9 mg C/L). During the incubation, we measured dissolved oxygen consumption rates continuously using optical sensors, and post-incubation we analyzed dissolved organic carbon, CO₂, and nutrients. Our results showed that wet sediments exhibited significantly faster O₂ consumption rates than dry sediments regardless of treatment, supporting our first hypothesis. However, contrary to our second hypothesis, PyOM and unburned organic matter additions did not significantly affect respiration rates in either sediment type. Dry sediments released substantially higher water-extractable organic carbon concentrations compared to wet sediments, potentially masking treatment effects. We used process-based models to distinguish biotic versus abiotic contributions to oxygen consumption. Abiotic vs biotic rate ratios ranged from 0.08-0.2, indicating predominantly biotic control of oxygen consumption. Wet sediments showed slightly higher abiotic contributions

than dry sediments suggesting greater initial chemical oxygen demand from reduced species accumulated during sustained inundation. These findings suggest that inundation history impacted sediment biogeochemical responses during rewetting more strongly than PyOM additions.

id #135312

Rapid GPU-Accelerated Fish Track Detection from Split-Beam Hydroacoustics for Riverine Lake Sturgeon Monitoring

Eamonn Powers¹, Eric Snyder¹, Stephanie Ogren ², Marty Holtgren ³, Bill Flanagan ⁴

1. GVSU, Grand Rapids, MICHIGAN, United States

2. Grand Rapids Public Museum , Grand Rapids, Michigan, US

3. Encompass Socio-ecological Consulting, Manistee , Michigan , United States

4. John Ball Zoo , Grand Rapids, Michigan, US

The purpose of this study is to generate a population estimate for Lake Sturgeon in the Grand River, Michigan, a threatened population, especially given the presence of a dam in downtown Grand Rapids. Hydroacoustic surveys monitoring adult sturgeon were coupled with juvenile pit tagging/genetic surveys. Both surveys allow for a greater understanding of the threatened Lake Sturgeon population in the Grand River. Genetic parentage analysis indicates a small population of spawning adults, a conclusion in the process of being evaluated by fixed hydroacoustic surveys.

Hydroacoustic surveys are a valuable, non-invasive method of monitoring fish in larger rivers, yet split-beam datasets can be incredibly slow to turn into usable information when processing and interpretation requires a substantial amount of manual effort. Our Lake Sturgeon research team presents a significantly simpler workflow that speeds up the path between raw split-beam recordings, to summarize fish tracks that are ready for review, export, and further analysis.

We used an HTI, Hydroacoustic Technology Inc. 241 split beam echosounder with a $2.5 \times 10^\circ$ elliptical beam transducer. The workflow we have developed is designed to support routine use during the field season by producing track candidates quickly, as well as organizing them in a fashion that makes it easy to confirm positive observations.

The outputs we generate are formatted for relatively common monitoring questions tied to Lake Sturgeon monitoring, such as where and when fish are encountered and how movement patterns vary by both season and environmental conditions. Overall, this approach demonstrates an advanced path towards making non-invasive split-beam hydroacoustics scalable for riverine monitoring of fishes.

id #135313

Mercury Dynamics in Beaver-Dominated Mountain Streams

Clifford Adamchak^{2,1}, Keira Johnson², John Christensen³, Katherine B Lininger⁴, Eve-Lyn S Hinckley^{2,1}

1. Department of Ecology and Evolutionary Biology, University of Colorado, Boulder, Boulder , Colorado, USA

2. University of Colorado, Boulder/CIRES, Boulder, COLORADO, United States

3. Lawrence Berkeley National Labs, Berkeley, CA, USA

4. Department of Geography, University of Colorado, Boulder, Boulder, CO, USA

Atmospheric deposition of mercury (Hg) is increasing in some regions of the Western U.S. When transported to anoxic environments, inorganic Hg can be transformed into methylmercury (MeHg), a bioaccumulating neurotoxin. Simultaneously, there is evidence that American beaver (*Castor canadensis*) populations are increasing, and their modifications of ecosystems (e.g., construction of dams, lodges, food caches, and burrowing in the floodplain) expand anoxic conditions, highlighting a need to understand how beaver activity alter MeHg production, storage, and transport to downstream water bodies.

In this work we investigate the seasonal variation in Hg concentration, speciation, and dissolved versus particulate fractions across two stream reaches (~1.5 km), one beaver-dominated and one unimpounded. We hypothesize that, compared to the unimpounded reach, the beaver-dominated reach will 1) exhibit higher dissolved MeHg transport due to enhanced particulate settling and microbial methylation in anoxic pond sediments and 2) have the greatest MeHg enrichment during midsummer, when prolonged water residence time, warm temperatures, and anoxia stimulate high rates of microbial reduction. To test these hypotheses, we collected surface, ground, and porewater and paired stream discharge throughout the snow-free 2025 season in Coal Creek watershed near Crested Butte, Colorado. We collected samples at the inlet, outlet, and on the floodplain in paired beaver-dominated and impounded reaches and analyzed the water samples for dissolved and particulate total Hg/MeHg along with sulfate, dissolved organic carbon, and physical water quality parameters. Preliminary results suggest that hydrologic connectivity may be a stronger control on Hg cycling than beaver activity, with similar changes across the hydrograph observed in both the control and beaver dominated reach. The outcomes of this work will provide valuable information to land and water managers and restoration practitioners who must consider the expanding presence of beaver activity and its impacts on water quality and the bioavailability of MeHg in the river corridor.

id #136081

Hyporheic Microbial Communities at Two Central Texas Rivers

Eryl Austin-Bingamon¹, Benjamin F Schwartz¹, Annette S Engel², Audrey T Paterson², Ben T Hutchins³, Brook D Herman⁴

1. Texas State University, TX, United States

2. University of Tennessee, Knoxville, Tennessee, United States

3. *Texas State University Edwards Aquifer Research and Data Center, San Marcos, TX, United States*

4. *US Army Engineer Research and Development Center, Vicksburg, Mississippi, United States*

The wellbeing of freshwater ecosystems relies on critical biogeochemical processes facilitated by aquatic microbial communities. In a stream, many of these processes may occur in the hyporheic zone, an ecotone located beneath the stream bed where surface water and groundwater mix. Microbial communities in the hyporheic zone attenuate contaminants, facilitate nutrient turnover in a stream, and drive up to 95% of whole-river metabolism. Despite this, our understanding of hyporheic microbial communities is generally limited, and their significance is underemphasized in river research and conservation. In this study, we sampled water from the hyporheic zone for its geochemical properties and microbial diversity at sites along the Llano and Concho Rivers, tributaries of the Colorado River in central Texas (USA) that had contrasting flow regimes (high vs. low flow, respectively) and sediment properties (cobble vs. sand/silt, respectively). Microbial diversity was characterized from 16S rRNA gene sequence analysis and their ecologically relevant traits using FAPROTAX, a tool used to predict metabolic functions. Results from the rivers were compared to assess the effects of geochemistry, flow, and sediment properties on community diversity and functionality to advance our understanding of hyporheic microbial processes, and improve our ability to predict how future ecological changes may alter microbial communities and how that, ultimately, will impact river functionality.

id #136082

Context matters: how riparian forest restoration and landscape resilience shape water quality in headwater streams

João Miguel Mercês Bega¹, Giulia Domingues Pedro¹, Bruno Moreira Felipe¹, Paula Caroline dos Reis Oliveira¹, Ricardo Taniwaki², Solange Filoso³, Davi Gasparini Fernandes Cunha⁴, Sílvia Frosini de Barros Ferraz¹

1. *Forest Sciences, University of São Paulo, Piracicaba, São Paulo, Brazil*

2. *Center for Engineering, Modeling and Applied Social Sciences, Federal University of ABC, São Paulo, Brazil*

3. *University of Maryland Center for Environmental Science, Cambridge, Maryland, United States*

4. *Hydraulic Engineering and Sanitation, University of São Paulo, São Carlos, São Paulo, Brazil*

Forest restoration has been widely promoted as a nature-based solution to protect water quality by reducing nutrient and sediment inputs to streams. Nevertheless, the magnitude over which restored riparian forest functions as an effective buffer, resulting in lower in-stream concentrations and reduced pollutant exports, remain poorly constrained, particularly across landscapes with contrasting disturbance histories. Here, we assessed how different stages of riparian vegetation influence stream water quality and in-stream pollutant exports across headwater catchments spanning a gradient of ecological conditions in two contrasting environments: one with low ecological resilience shaped by intensive agricultural practices, and another with high resilience, characterized by lower historical disturbance and greater landscape connectivity. Across these environments, we sampled a total of fifty-seven catchments (approximately thirty per environment) during the dry season of 2025 and the rainy season of 2026. Catchments were classified into four forest-cover classes according to the type and extent of vegetation cover: (i) catchments dominated by native forest remnants; (ii) pasture-dominated catchments with well-developed riparian buffers; (iii) pasture-dominated catchments with narrow riparian thicket buffers; and (iv) catchments largely covered by pastures. Stream water was sampled to determine concentrations of total dissolved carbon, dissolved organic carbon, dissolved inorganic carbon, total suspended solids, fixed and volatile suspended solids, total nitrogen, nitrite, nitrate, total phosphorus, and orthophosphate. Landscape metrics, including vegetation indices and land-use/land-cover composition, were derived using GIS at reach, riparian, and catchment scales. Stream discharge was measured using the pulsed release of a conservative solute, enabling the calculation of mass loads and the estimation of pollutant exports across multiple temporal scales. Our results provide novel, process-based insights into how riparian forest restoration stage and landscape context jointly regulate in-stream water quality and pollutant export in headwater systems. These findings offer actionable guidance for the design and prioritization of riparian restoration strategies aimed at improving water quality and reducing downstream pollutant delivery.

id #135315

Potential Fate, Transport, and Impact of Enhanced Rock Weathering Products in Freshwater Ecosystems

Emily Bernhardt¹, Jonathan Gilman¹, Heili Lowman¹, Katryna Niva¹, Shannon Plunkett¹, Spencer Rhea¹, Veronica Slevin², Yuyang Wang¹

1. *Duke University, DURHAM, NC, United States*

2. *University of North Carolina, Chapel Hill, NC, USA*

The Earth's carbonate-silicate cycle is an important regulator of atmospheric carbon dioxide, which has stabilized Earth's climate over geologic time. New geoengineering approaches aim to enhance the rate of silicate rock weathering by mining, milling and distributing silicate minerals on land to facilitate heightened rates of CO₂ consumption in weathering reactions. In order to be effective as a climate mitigation approach, the bicarbonate ions and cations produced by enhanced rock weathering (ERW) must ultimately be transported to the deep ocean. While a great deal of recent research has focused on the efficacy of this geoengineering approach on land, far less attention has been paid to the fate, transport and impact of these ERW solutes as they are routed through freshwater ecosystems. Here we attempt to synthesize what is already known about the fate, transport and impact of bicarbonate and cations in rivers and suggests priorities for the freshwater science community for helping maximize the efficacy and minimize the unintended consequences of this important new climate mitigation strategy.

id #135572

Organic matter dynamics in a coastal plain watershed: insights from nested synoptic sampling of dissolved and particulate optical properties in the Southeastern United States.

Laura M Phillips¹, Ariel J Shogren², Andrew Ali², Zacharie Loveless²

1. Biological Sciences, University of Alabama W2W RaMP Program, Tuscaloosa, AL, United States

2. Biological Sciences, University of Alabama, Tuscaloosa, AL, United States

Headwater stream networks are significant aquatic ecosystems, as they constitute the majority of river miles and are prominent carriers of biogeochemical signals from the surrounding terrestrial environment. Despite their importance, organic matter dynamics remain understudied in humid, low-relief headwater networks in the Gulf Coastal Plain of the Southeastern United States (US). To explore organic matter dynamics in an understudied region, we intensively sampled 20 subcatchments nested within the forested and actively managed South Sandy Watershed (125 km²) in the Talladega National Forest of central Alabama, US. Using samples collected monthly from September 2025 to January 2026, we evaluated the properties of dissolved organic matter (DOM) and particulate organic matter (POM) pools at each site. We characterized samples using optical properties, including specific ultraviolet absorbance at 254 (SUVA₂₅₄), Fluorescence Index (FI), Humification Index (HI), and Biological Index (BIX). Collectively, these indices reflect the source, quality, and complexity of organic matter in each water sample. Notably, our seasonal sampling period encompassed organic matter dynamics before, during, and after leaf-off. We hypothesize that: 1) POM will experience a sharper increase in complexity during leaf-off and decrease afterward relative to DOM, 2) Both DOM and POM will have a strong allochthonous signal during leaf-off, followed by a more autochthonous signal, and 3) DOM and POM properties of each subcatchment will be correlated with vegetation cover. Results from our study will improve understanding of the contributions of headwater networks to the global carbon cycle, which is crucial for sustainable land management.

id #136084

Nutrient enrichment alters stream fungal communities: evidence from microscopy and next-generation sequencing

Vladislav Gulis¹, Jennifer Bogdanoff^{1,2}, Megan Cevasco¹, Jonathan P. Benstead³, Amy D. Rosemond⁴

1. Coastal Carolina University, Conway, SC, United States

2. Craven Community College, New Bern, NC, United States

3. University of Alabama, Tuscaloosa, AL, United States

4. University of Georgia, Athens, GA, United States

Litter-associated fungi are key intermediaries in carbon flow and nutrient cycling in streams. They can obtain nitrogen (N) and phosphorus (P) from both substrates and the water column. We used streamside channels at the Coweeta Hydrologic Laboratory (NC, USA) to test the effects of dissolved inorganic N (DIN) and soluble reactive P (SRP) on fungal reproduction (sporulation rate of aquatic hyphomycetes) and community structure associated with plant litter (maple and rhododendron leaves and wood). To evaluate sporulation and fungal communities, we combined traditional microscopy with next-generation sequencing (NGS, Illumina) targeting fungal ITS2 rDNA region. Our bioinformatics pipeline included QIIME2 coupled with the UNITE database supplemented with >100 novel sequences from our fungal culture collection. Sporulation rates and cumulative spore production of aquatic hyphomycetes showed significant relationships with DIN rather than SRP, with the greatest effects of nutrients found on wood. Cumulative spore production was strongly correlated with plant litter decomposition rates (R^2 0.68-0.91). Dominant species of aquatic hyphomycetes were similar as assessed by microscopy and Illumina sequencing. Dissolved nutrients affected fungal community structure based on microscopy (ANOSIM, $p=0.001$), while the effect of DIN was less pronounced based on Illumina data (PERMANOVA, $p=0.02$) and not significant for SRP. The effect of substrate (even type of leaf litter when wood was omitted) was greater than the effect of dissolved nutrients in shaping fungal communities. Microscopy appeared to be more sensitive in detecting changes in fungal community structure than NGS. This could be due to (i) microscopy assessing the effects of nutrients on reproductive output of aquatic hyphomycetes while NGS relied on DNA from vegetative fungal biomass embedded in plant litter, and (ii) microscopy focusing on highly active, truly aquatic fungi while NGS also detected DNA from less active, dormant or dead terrestrial fungi initially present on autumn-shed leaves that produced interfering background signals. Due to differences in activity and elemental stoichiometry of fungal species, changes in fungal communities may affect plant litter decomposition and carbon, nutrient, and energy flows in streams.

id #135061

Establishing a Functional Flows Framework for Rivers and Wetlands in the Great Salt Lake Basin

Noelle Patterson¹, Lane Belize¹, Sarah Null¹, Farah Nusrat¹, Jeffrey Ostermiller², Bethany Neilson¹, Melissa Stamp¹, Michelle Baker¹

1. Utah State University, Logan, Utah, United States

2. Utah Department of Environmental Quality, Utah Division of Water Quality, Salt Lake City, Utah, United States

Terminal basins are often highly sensitive to streamflow contributions, making them especially vulnerable to freshwater ecosystem degradation when streamflow is diverted or altered. The Great Salt Lake (GSL) terminal basin in Utah supports a unique ecosystem including the saline lake, peripheral wetlands, and snowmelt-driven streams, all which face high stress from competition for freshwater resources. Recent efforts to increase streamflow to GSL and reverse lake decline underscore the need to simultaneously consider instream flow requirements for contributing rivers and delivery needs for the GSL peripheral wetlands, so water dedicated and delivered to the GSL can achieve ecological benefits to the basin as a whole. Here we present a functional flows framework for rivers and managed wetlands draining to the GSL. Functional flows are components of the annual hydrograph that support riverine functions and processes critical to aquatic ecosystem health. Functional flow components specific to the GSL basin describe aspects of the region's snowmelt-dominated annual hydrograph including winter low flow, spring ascension flow, peak flow, recession flow, and summer low flow, which are characterized by metrics quantifying the magnitude, timing, duration, frequency, and rate of change of flow. These metrics were generated for 180 reference-condition streamgages within the GSL basin and the Middle Rockies ecoregion. The functional flow metrics were then paired with basin characteristics and climate data to predict water year-specific functional flow target ranges at all ungaged streams in the GSL basin, and results are

being made publicly available through map-based visualization. These predicted flows can be used as initial targets to inform decisions on when, where, and how much water to deliver to streams, managed wetlands, and the GSL.

id #135062

From Streams to Landscapes: how the Tennessee Aquarium is advancing watershed conservation

Helaina Gomez¹

1. Tennessee Aquarium Conservation Institute, TN, United States

The Tennessee Aquarium's mission is to connect people with nature and support informed decision-making about water and wildlife. The Aquarium's team of conservation professionals advances this mission by contributing essential applied freshwater research and ecosystem management expertise. Working at the intersection of science, restoration, and public engagement, the Watershed Conservation Coordinator integrates field research, social-ecological theory, restoration practice, and stakeholder engagement to improve watershed health across southeast Tennessee.

The Aquarium's conservation staff lead long-term, field-based programs that yield measurable outcomes, including a 25-year lake sturgeon restoration effort, a region-wide land management initiative exceeding \$10 million in investment, and multiple research and monitoring projects that guide the recovery of threatened and endangered species. Partnerships with agencies, NGOs, landowners, and community groups are central to the work of the Watershed Conservation Coordinator and demonstrate the importance of cross-sector collaboration in applied conservation science.

This integrative approach positions ecology as both a conceptual framework and a practical toolkit for addressing real-world conservation challenges. Through interdisciplinary work, the Watershed Conservation Coordinator translates academic research into effective strategies while generating insights that contribute to the broader scientific community. This model highlights impactful, community-embedded career pathways available to students and early career scientists beyond traditional academic settings.

id #135574

Assessing Bergmann's rule in freshwater macroinvertebrates using NEON data

Reece Bollinger¹, Jeff Wesner², James Junker³, Justin Pomeranz¹

1. Colorado Mesa University, Grand Junction

2. University of South Dakota, Vermillion

3. University of North Texas, Denton

Bergmann's rule is based on the observation that closely related species within a genus, or populations within a species, tend to be smaller on average at high temperatures and larger at cold temperatures. The distribution of body sizes has important implications for food web structure and the flow of energy to higher trophic levels. Bergmann's rule has wide support in the literature for endothermic animals, but support for ectotherms, such as insects, is equivocal. Therefore, we analyzed macroinvertebrate body size data from the NEON stream sites to assess if average body size of closely related taxa changes consistently across a 26°C gradient of environmental temperature. We hypothesize that average body size will decrease in related taxa with increasing environmental temperature. The findings of this study will provide important implications for understanding how aquatic macroinvertebrates adapt to their local thermal conditions at the continental scale, and may provide insight into how body sizes at the population- and community-scale may change in response to a rapidly warming climate. This information will help us understand how climate change is affecting the structure and function of aquatic communities.

id #136086

Community body-size distributions inform ecological integrity in aquatic systems

Brent Murry¹

1. West Virginia University, WV, United States

Taxa-based indices of biotic integrity (IBIs) are commonly used to evaluate ecological well-being of freshwater systems including streams, rivers, and wetlands. The occurrence and prevalence of taxa and functional groups are correlated in multivariate space to disturbance gradients where ultimately the relative abundance of sensitive to tolerant taxa provide an indication of ecosystem integrity or impairment. Because IBIs are taxa-based they are inherently sensitive to regional differences in biodiversity, therefore IBIs need to be constructed regionally and for specific habitat types (e.g. streams vs wetlands). Community body size distributions, on the other hand, are ataxic and as such more universal (i.e. all communities have body sizes). Community body size distributions represent an emergent property of aquatic communities and have been used to infer food web function. It is hypothesized that ataxic community body size distributions can provide insightful metrics of ecosystem integrity complementary to traditional taxa-based metrics. Community size spectra (CSS) is a theoretical approach for examining patterns in community (or assemblage) – wide body size distribution. CSS quantifies the relative abundance of individuals, regardless of taxonomic identity, across a range of body sizes. There is a near universal decline in abundance with increasing body size in aquatic systems directly related to size-related metabolism and gape-limited foraging. The rate of decrease, often referred to as the CSS slope, represents food web efficiency or degree of biomass/energy in the smaller size classes (lower trophic positions) passed to successively larger (often higher trophic position) organisms. The CSS slope responds in a predictable fashion to internal food web changes in species relative abundance as well as external watershed and habitat characteristics. In general, however, the CSS is broadly stable around a long-term mean and deviations from that mean (outside typical temporal variability) are indicative of ecosystem change. In this presentation I will share a series of published and unpublished case studies to demonstrate the

responsiveness of CSS and make a case for the use of ataxic community body size distributions as complementary measures ecosystem integrity.

id #135063

The Trees Speak: Examining the Health of a Managed River through Riparian Growth

Noelle Patterson¹, Lane Belize¹, Xiaoli Dong², Adam Csank³, Stewart Rood⁴, Samuel Sandoval-Solis⁵

1. *Utah State University, Logan, Utah, United States*

2. *Department of Environmental Science and Policy, University of California, Davis, Davis, California, United States*

3. *Department of Geography, University of Nevada, Reno, Reno, Nevada, United States*

4. *Department of Biological Sciences, University of Lethbridge, Lethbridge, Alberta, Canada*

5. *Department of Land, Air, and Water Resources, University of California, Davis, Davis, California, United States*

The Truckee River is Nevada's largest river system, and supports a unique haven of river, riparian, and terminal lake ecosystems in the arid Great Basin. Decades of intense diversions from the Truckee River caused terminal lake levels at the river's outlet to drop severely, nearly eliminating endemic fish species and causing widespread reductions in riparian forest cover. Following this ecological collapse, increased attention to the decline of the river's health led to implementation of new streamflow management policies, called Environmental Flows, to better protect the river and its dependent ecosystems. This field-based research project uses tree-ring growth to explore how decades of river management, including periods of both heavy diversion and subsequent environmental management, have impacted the health of the riparian environment through the lens of cottonwood tree growth.

id #135575

Skewed Power Laws: Biased Data Bends λ and How to Bend it Back

Justin Pomeranz¹, Jeff Wesner², James Junker³

1. *Colorado Mesa University, CO, United States*

2. *University of South Dakota, Vermillion, SD*

3. *University of North Texas, Denton, TX*

Individual size distributions (ISD; also known as size spectra, community biomass distributions) are a powerful framework for understanding biological communities by incorporating processes from the individual- to the population-level. ISD relationships are described by a power law: $N \sim M^\lambda$, where N is abundance and M is individual body mass. Because M is a continuous measure, the relationship simplifies to a frequency distribution in the form of $f(M) \sim M^\lambda$. Determining the value of λ from a vector of individual body sizes, m , is a frequent goal of ecologists. However, typically have biases associated with them, particularly with under-sampling of the smaller values. These empirical data biases result in only some portion of the right tail of the data fitting a power law distribution well. Although the potential issues of estimating λ from a biased empirical sample have been previously discussed, and solutions (typically "cutting" the data below some arbitrary size) have been offered, to our knowledge no formal assessment of the effects that biased data has on estimates of λ have been performed. Here, we used a numerical simulation study design to assess how biases (specifically, under-sampling of small values) affect estimates of λ , as well as testing a solution proposed by Clauset et al. (2009) for determining the best point to "cull" the data to maximize the fit to a power law distribution. We found that estimates of λ became less accurate as the degree of under-sampling increased. When the under-sampled data was culled, λ estimates universally improved, although the improvement was not as great with the extremely under-sampled data. Based on our results, we recommend that all future studies of individual size distributions pre-process their data to remove biases associated with under representation of the smaller values. This will improve estimates of λ and will aid in understanding the remarkable consistency of λ across biological communities, and, importantly, to better understand when and why λ does vary.

id #135576

Groundwater-surface water exchange influences on the longitudinal spatial distribution of toxigenic cyanobacterial proliferations in the South Fork Eel River, California, USA

Andrea Garcia Jimenez¹, Janette L Davidson¹, Robert Padan¹, Donovan Davis¹, Danielle Hare², Mark B Hausner³, Christopher B Kratt⁴, Sydney Brown⁵, Jacob K Mormando⁵, Ramon C Naranjo⁶, Gregory L Boyer⁷, Zac Triumph⁷, Rosalina Christova⁵, Ramesh Goel⁸, Joanna Blaszcak¹

1. *Department of Natural Resources, University of Nevada, Reno, NV, Washoe*

2. *Earth, Geographic, and Climate Sciences, University of Massachusetts, Amherst, MA*

3. *Division of Hydrologic Sciences, Desert Research Institute, Reno, NV, Washoe*

4. *Center for Transformative Environmental Monitoring Programs, University of Nevada, Reno, NV, Washoe*

5. *Environmental Science and Policy, George Mason University, Woodbridge, VA*

6. *Nevada Water Science Center, US Geological Survey, Carson City, NV*

7. *Department of Chemistry, SUNY- College of Environmental Science and Forestry, Syracuse, NY*

8. *Civil and Environmental Engineering, University of Utah, Salt Lake City, UT*

Benthic algal assemblages, including toxin-producing cyanobacteria, are increasingly detected in rivers globally and threaten aquatic ecosystems and public health. Anatoxins are a neurotoxic cyanotoxin produced by the cyanobacterium *Microcoleus*, which can dominate benthic habitats with low dissolved reactive phosphorus and low dissolved inorganic nitrogen, indicating that water-column nutrients alone cannot explain bloom formation or toxicity. Other environmental conditions commonly associated with *Microcoleus* proliferation include elevated water temperature, stable summer discharge, and moderate to high specific

conductance, some of which are typically indicative of groundwater influence in rivers. Groundwater-surface water exchanges in rivers vary longitudinally and can influence nutrient and dissolved ion concentrations, water temperature, and benthic algal assemblages. We hypothesized that longitudinal variation in groundwater-surface water exchanges along a river influences the distribution of *Microcoleus* growth and anatoxin production through its effects on nutrient availability and water temperature. To test this, we investigated the role of groundwater influences on *Microcoleus* proliferation and anatoxin production in the South Fork Eel River, northern California, where recurrent proliferations have been observed for over a decade. Groundwater or hyporheic inflows were identified using fiber-optic distributed temperature sensing (FO-DTS) and continuous temperature loggers. We installed monitoring wells to characterize the groundwater chemistry and temperature. We estimated percent cover along 57 transects within a 2 km reach and collected surface water velocity, substrate size, and *Microcoleus* mats, which we analyzed for anatoxins using liquid chromatography with tandem mass spectrometry (LC-MS/MS). Preliminary field observations indicate that side channels with warmer temperatures had a more extensive *Microcoleus* mat cover, which appeared to be at a more advanced successional stage than the main channel. This suggests that channel complexity and groundwater-surface water exchange may influence the longitudinal spatial distribution of *Microcoleus* by creating warmer water temperatures. Our study provides insights into how groundwater processes may influence the ecology and spatial distribution of toxigenic benthic cyanobacteria in rivers.

id #135832

Effects of beaver and beaver dam analogs on post-fire recovery of stream and riparian ecosystems

Camille McNeely¹, Rebecca Brown¹, Katelin Killoy², Sawyer Nagle¹, Joe Weirich³, Alexa Whipple³

1. Eastern Washington University, WA, -

2. Montana Fish, Wildlife, and Parks, Dillon, MT, USA

3. Methow and Okanogan Beaver Project, Twisp, WA, USA

Channel incision is a global problem for watersheds - one that is growing in the American west, where wildfire exacerbates channel erosion. Incision lowers water tables, draining aquifers and wetlands. Beaver activity can reduce incision, improving riparian habitat and retaining water, sediment, and nutrients. Unfortunately, it can be difficult to restore beaver to degraded areas. Consequently, man-made Beaver Dam Analogs (BDAs) have become a popular restoration technique. We report results of a Before-After Control-Impact study comparing 5 fire-impacted BDA restoration sites with 5 paired controls and 4 reaches with beaver activity in the Methow and Okanogan watersheds (Washington, USA).

Pre-restoration data show substantial effects of beaver on water storage and riparian condition. Beaver sites had width-to-depth ratios over twice as large as pre-BDA sites, but not significantly different from control sites ($p=0.0033$). Beaver sites had floodplain widths over twice as large as both pre-BDA sites and control sites ($p=0.0002$). Beaver ponds had a higher frequency of fine sediment sizes, whereas non-beaver sites had a wider distribution of sediment sizes (Kruskal-Wallis, $df=2$, $p < 0.01$). We observed dramatically slower water travel times through beaver reaches compared to control and planned restoration reaches, during both low and high flow conditions ($p < 0.05$).

BDA installation was completed for 3 reaches during 2022 and the remaining 2 in 2023. In post-installation data, BDAs were effective in slowing water during both low-flow conditions and high-flow conditions (paired t-test of differences between pre and post restoration travel times, $p < 0.05$). However, beaver impoundments slowed water much more, under both low and high flow conditions. BDAs were very effective in capturing sediment, resulting in increased frequency of fine sediment sizes (Mann-Whitney test, $p < 0.001$). We are currently analyzing data on floodplain topography and riparian vegetation post-restoration. Our data indicate BDAs are effective in slowing water flow, and storing water and sediment, but that beaver are more effective in storing water. Protection of beaver populations is an important goal for protecting water quality in the semi-arid western US.

id #136088

Intraseasonal covariation in autotrophic and heterotrophic microbial function in a non-perennial polar stream

Lydia H Zeglin¹, Michaela K Halsey¹

1. Kansas State University, KS, United States

Seasonally ephemeral streams are common globally, and are inhabited by organisms that can survive long dry periods, then capitalize on resources mobilized as flow resumes. In the McMurdo Dry Valleys of Antarctica, a polar desert, ephemeral streams contain diverse food webs supported by microbial photoautotrophy. Here, benthic microbial mats dominated by nitrogen (N)-fixing *Nostoc* spp., or "black" mats, provide N for primary production in neighboring Oscillatoriacean dominated, or "orange" mats, and for heterotrophy and N-cycling activity within the mats and underlying hyporheic sediments. This study sought to understand how these communities change upon re-wetting early in the flow season, and as flows decrease through the season's end. We collected samples from one stable-hydrology stream and two flashier systems, between late-December 2024 and late-January 2025, spanning before and after flow resumed and three more occasions. We quantified organic matter (OM) content of sediments and mats, measured DNA and chl-a (biomass proxies), and extracellular enzyme activities (EEA), which assess microbial investment to access carbon (C) and N. We predicted that EEA would increase upon re-wet, then C availability would increase but N availability decrease as the season progressed.

Interestingly, sediment OM declined as the season progressed, more quickly in black than orange mats, with some recovery at the end of the season ($P < 0.05$). Sediment EEA was not temporally dynamic ($P > 0.05$); however, sediment EEA was higher ($P < 0.05$) and expressed lower N-limitation ($P < 0.01$) under black (N-fixing) mats than under orange mats. In the mats, C-limitation was lower than sediments, and lowest in the stable stream and in black mats. N-limitation was lowest in black mats from the stable stream, and showed contrasting shifts through time in the flashy streams. Overall, patterns suggest that C demand by sediment heterotrophs is strong, could contribute to OM consumption through the flow season, and that N availability may accelerate this process. Concurrently, C is more available in mats, likely due to primary production there, and is promoted by N-fixation and flow stability. Work is in progress to finalize autotroph and heterotroph biomass estimates which will inform ideas about growth-driven nutrient C and N supply and demand.

Hydraulic controls on microplastic transport and deposition across riverine flow habitats

Enahoro Kennedy Owowenu¹, Chika Felicitas Nnadozie¹, Nelson Odume¹

1. *Institute for Water Research, Rhodes University, Grahamstown/Makhanda, Eastern Cape Province, South Africa*

Hydrodynamic conditions play a crucial role in governing microplastic (MP) transport, deposition, and resuspension in riverine habitats. However, their specific effects across distinct flow zones remain poorly understood. This study investigated how two key hydraulic indices, the Froude number (Fr) and Reynolds number (Re), influence the spatial and seasonal distribution of MPs in pools/backwaters and runs/riffles of three subtropical urban rivers in the Eastern Cape Province, South Africa. Microplastics were sampled from suspended fractions in surface water and settled fractions from the riverbed during dry/cold and wet/hot seasons. Identification was performed using chemical digestion, density separation, stereomicroscopy, and FTIR-ATR spectroscopy. Results revealed that settled MPs were strongly influenced by hydraulic conditions, with Fr being the most consistent predictor across habitats. In run/riffle zones during the wet/hot season, higher Fr values were associated with significantly lower settled MP concentrations ($p < 0.001$), suggesting enhanced flushing and reduced retention under more energetic flow conditions. Additionally, Fr was the only hydraulic index linked to changes in the size distribution of settled MPs (multivariate $p = 0.009$), primarily driven by the 1-2 mm size class ($p = 0.007$). In contrast, suspended MPs exhibited weaker and more variable relationships with both Fr and Re, suggesting that once MPs are entrained in the water column, turbulence and mixing processes dominate over hydraulic sorting. Elevated Re was occasionally linked to increased resuspension in deeper pool habitats, but these patterns were inconsistent across seasons. These findings demonstrate that MP-hydraulic interactions are habitat-specific and seasonally dynamic. Understanding how Fr and Re influence MP retention and mobility offers valuable insights for refining monitoring strategies and predictive models. This knowledge is crucial for enhancing our ability to forecast MP fate and inform targeted mitigation efforts in fluvial systems.

Use of Thermal Refugia by Brown Trout on the North Branch Au Sable River, Michigan

Logan D. Clark¹, M. Woller-Skar², Mark R. Luttenton¹

1. *Grand Valley State University Annis Water Resource Institute, Allendale, MI, USA*

2. *Grand Valley State University, Allendale, MI, USA*

Brown Trout (*Salmo trutta*) response to warming stream temperatures has become an important consideration for fishery managers. To evaluate Brown Trout response to warm stream temperatures we tracked 12 adults (29-50 cm) implanted with temperature-sensitive radio tags (ATS #F1150) from July 1 to August 19 on the North Branch Au Sable River (Michigan). We were particularly interested in determining if fish would actively seek thermal refuge. Tracking occurred on 34 dates, multiple days a week and up to four relocations per day. Every 15 minutes water temperature and light were recorded at multiple locations within the study reach and at one "air" location using Onset MX2201 and MX2202 loggers. Water temperature was also recorded adjacent to relocated fish. Additional environmental data included water depth at one fixed site and barometric pressure. Out of 482 relocations, fish were found at their homesites 87% of the time. Only two fish had internal temperatures cooler (4.3% of relocations, 21 observations) than ambient stream temperatures. One fish (17 observations) occupied a homesite apparently setting over a groundwater discharge. A second fish initially occupied a cool water homesite (4 observations) but abandoned the site. When stream temperatures exceeded 20°C (58.8% of relocations, 284 observations) or 25°C (12.6% of relocations, 61 observations), except as noted above, there were no differences between body and stream temperatures. When Brown Trout left homesites at stream temperatures >21°C (24 observation) or >25°C (5 observations), body temperature remained similar to ambient. Most movement occurred during low-light conditions and increased stream depth regardless of temperature. Overall, Brown Trout exhibited limited use of thermal refugia and were not observed actively seeking refuge during warmer stream temperatures (23-26°C). Limited use of thermal refugia may reflect weak thermal cues due to diffuse groundwater inputs, homesite fidelity, or adaptations to intermittent thermal stress.

Environmental filtering plays a larger role than resource stoichiometry in macroinvertebrate community stoichiometry across North America.

Nathan Tomczyk¹, Chelsea Smith², Mark Nessel³, Kauan Fonseca⁴, Viviana Bravo Ortiz⁵, Jessica Corman⁶, Eric Moody⁷, Hal Halvorson⁸

1. *New Mexico Highlands University, NM, United States*

2. *University of Alabama, Tuscaloosa*

3. *Oklahoma State University, Stillwater*

4. *Universidade do Estado do Rio De Janeiro, Rio de Janeiro*

5. *University of Georgia, Athens*

6. *University of Nebraska-Lincoln, Lincoln*

7. *Middlebury College, Middlebury*

8. *University of Central Arkansas, Conway*

Anthropogenic nutrient enrichment has altered the relative supply of nitrogen (N) and phosphorus (P) in freshwater ecosystems, with implications for organismal growth, nutrient recycling, and ecosystem functioning. Ecological stoichiometry predicts that alignment between consumer nutrient demand and resource supply should reduce metabolic costs, yet most aquatic macroinvertebrates exhibit strong stoichiometric homeostasis at the individual level. Whether communities of macroinvertebrates

can achieve stoichiometric flexibility through species sorting or intraspecific variation remains unclear, particularly at broad spatial scales.

We evaluated community-level stoichiometric flexibility of stream macroinvertebrates across North America using data from 27 National Ecological Observatory Network (NEON) stream sites spanning wide gradients in basal resource stoichiometry. We combined long-term macroinvertebrate abundance and biomass data with taxon-specific body stoichiometry from the STOICH database to estimate community-weighted C:N:P for three functional feeding groups: herbivores, collector-gatherers, and collector-filterers. We quantified apparent stoichiometric flexibility as the relationship between resource stoichiometry (periphyton or seston) and community body stoichiometry using the inverse homeostasis coefficient (H^{-1}). To identify mechanisms and constraints, we compared observed communities with counterfactual scenarios that isolated the effects of intraspecific variation, global trait availability, and local species pools.

Across nearly all nutrient combinations and feeding groups, observed macroinvertebrate communities were effectively homeostatic ($H^{-1} \approx 0$), showing little alignment with resource stoichiometry despite large variation in basal resource C:N:P. Intraspecific variation in body stoichiometry did not increase apparent flexibility. In contrast, theoretical communities assembled from the global species pool exhibited substantial flexibility, indicating that sufficient trait variation exists at the continental scale. However, communities constrained by local species pools generally lacked the stoichiometric range necessary to track resource gradients, demonstrating that environmental filtering and dispersal limitation overwhelm resource-driven species sorting.

Our results suggest that resource stoichiometry plays a limited role in structuring macroinvertebrate communities across landscapes, leading to persistent mismatches between consumer demand and resource supply. As nutrient imbalances intensify with human activity, the apparent homeostasis of stream communities may have important consequences for nutrient retention, recycling, and downstream ecosystem function.

id #134042

Spatial and Temporal Variability of Nitrogen Cycling in a Non-Perennial Stream

Lien Tran¹, Sarah Flynn², Michelle Busch³, Kaci Zarek⁴, Erin Seybold^{5, 6}, Lydia Zeglin⁷, Amy Burgin¹

1. Ecology, Evolution, and Organismal Biology, Iowa State University, Ames, IA, USA

2. Ecology and Evolutionary Biology, Kansas University, Lawrence, KS, USA

3. Michigan Natural Features Inventory, Michigan State University, East Lansing, MI, USA

4. Ecology and Evolutionary Biology, Cornell University, Ithaca, NY, USA

5. Kansas Geological Survey, Kansas University, Lawrence, KS, USA

6. Geology, Kansas University, Lawrence, KS, USA

7. Biology, Kansas State University, Manhattan, KS, USA

Non-perennial streams have a distinct hydrology governed by alternating drying and rewetting cycles that strongly shape biogeochemical cycling. However, the effects of shifting hydrological regimes on nitrogen cycling processes (e.g., N_2 fixation and denitrification) remain understudied. Our study addresses this knowledge gap by examining the spatial and temporal variability of nitrogen dynamics within the South Fork of King's Creek in the Konza Prairie Biological Station (Kansas, USA). We conducted watershed-scale synoptic sampling of surface water at 50 headwater sites (1-3rd order), 42 of which had water present for sampling. To capture seasonal variability, seven of the sites were subsequently sampled across spring, summer, and fall in 2022 and 2023. We used membrane inlet mass spectrometry (MIMS) to quantify N_2 excess, which can indicate net N_2 -fixing or denitrifying conditions. Additional water samples were collected to measure physicochemical variables and characterize microbial community composition (16S rRNA). N_2 excess signatures varied spatially, with most sites indicating net N_2 fixation conditions (30 sites; -0.04 ± 0.02) and fewer showing net denitrification conditions (12 sites; 0.15 ± 0.21). In contrast, all seven seasonally sampled sites exhibited N_2 excess signatures indicative of net N_2 fixation (-0.29 ± 0.10). No physicochemical variable was significantly correlated with N_2 excess, and differential abundance analysis determined similar microbial community composition between net N_2 fixation and denitrification conditions sites. Antecedent drying conditions, along with other hydrological metrics such as network connectivity and topographic wetness index, will be examined in future analyses to assess their potential effects on nitrogen dynamics. Ultimately, our study provides greater insight into the spatial and temporal variability of nitrogen cycling processes in a non-perennial stream.

id #135322

Nutrient history and plankton succession can influence large river harmful algal bloom formation.

Justin Murdock¹, Jingjing Li¹, Anna Linhoss², Bill Ford³, Jimmy Fox³, Prem Parajuli⁴, Mindy Yeager-Armstead⁵

1. Tennessee Tech University, TN, United States

2. Auburn University, Auburn, AL

3. University of Kentucky, Lexington, KY

4. Mississippi State University, Starkville, MS

5. Marshall University, Huntington, WV

Harmful algal blooms (HABs) in large rivers are an emerging global concern, yet the mechanisms driving bloom initiation, composition and toxin production remain poorly understood compared to lakes. The Ohio River, USA has experienced several extreme blooms, with the largest in 2015 covering over 1100 river km and lasting over two months. Our group is developing a hydrodynamic-based river HABs prediction model that integrates river water quality, plankton ecology and physiology, and channel and watershed hydrology. This presentation highlights how nutrient, light, and temperature relationships were developed to parameterize the model using 40-day laboratory incubations of Ohio River plankton communities. Cyanobacteria dominated ending communities at extreme temperatures and light, with *Pseudanabaena* sp. dominant at low light and temperature, and *Microcystis* sp. at high light and temperature. As expected, nutrient concentrations regulated cyanobacteria biomass, but nitrogen

(N) and phosphorus (P) ratios also regulated composition, as low N:P favored heterocyte containing *Dolichospermum* sp., and high N promoted non-heterocyte containing *Microcystis* sp. and *Pseudanabaena* sp. Unexpectedly, maximum cyanobacteria and total algal biomass were better predicted by nutrient conditions ~35 days prior to blooms than ambient total N and P at peak biomass, including treatments with high N-fixation rates. This laboratory result parallels the prolonged (> 30 day) low-flow conditions observed in the Ohio River prior to blooms with presumably limited external nutrient input, and introduces the concept of “nutrient history” for bloom forecasting models. These findings add mechanistic insight into large river bloom ecology, suggesting initial succession community composition and competition may interact with chemical and physical environmental controls to influence bloom composition and biomass end points.

id #135834

Seasonal and spatial patterns of fish community composition in Tonle Sap Lake revealed by environmental DNA

YI YU¹, Tsuyoshi Kinouchi¹, Sobot Soth², Seyha Hok², Rei Itsukushima³, Kazuki Kanno⁴, Davin Tes¹, Kong Heng²

1. Institute of Science Tokyo, School of Environment and Society, Tokyo, Japan

2. Inland Fisheries Research and Development Institute (IFReDI), Phnom Penh, Cambodia

3. Kyushu University, Japan

4. KenKan Consultants Co., Ltd., Japan

Tonle Sap Lake is the largest freshwater lake in Southeast Asia. As a key ecological system of the Mekong Basin, it sustains highly productive fisheries and rich fish biodiversity across the surrounding floodplain through seasonal flood-pulse dynamics. However, most existing knowledge of fish biodiversity in Tonle Sap Lake is based on traditional capture-based surveys with limited spatial coverage. Despite its ecological and socio-economic importance, lake-wide patterns of fish biodiversity remain insufficiently understood.

In this study, we investigated lake-wide spatio-temporal patterns of fish communities with environmental DNA (eDNA) metabarcoding, which allowed us to explore fish community composition across the entire lake–floodplain system at broader spatial coverage and higher resolution. eDNA samples were collected from more than 200 sampling sites across the lake–floodplain system, including the Tonle Sap River (TSR) and Mekong River (MR). A total of seven surveys were conducted between 2024 and 2025, covering both dry and wet seasons.

Spatial and seasonal patterns in fish communities are analyzed based on species richness and community composition dissimilarity. Hierarchical modeling of species communities was also combined with eDNA-derived presence–absence data for analyzing species-specific responses to seasonality effect and area. We found there is pronounced spatial heterogeneity in fish community composition across the lake–floodplain system. In particular, fish diversity in floodplain habitats were higher than those in the open lake, and stronger community difference was observed at the northern and southern ends of the lake compared to the central region. Seasonal patterns of species richness and assemblage composition were comparatively stable at the community level, whereas species-level analyses indicated that individual fish species exhibited distinct seasonal variation.

Overall, our results revealed that environmental DNA can effectively capture fish community composition in Tonle Sap Lake, complementing traditional monitoring approaches and providing new support for ecological modeling for the whole system.

id #136090

Operationalizing Wetland Intrinsic Potential (WIP) as a Foundational Layer for Mapping Wetlands and Their Functions

Meghan Halabisky¹, Sarah Fancher¹, Dan Miller¹, Steven Pestana¹, Maureen Ryan¹, Erin Donley Marineau¹

1. TealWaters, Seattle, WA, United States

Wetlands are critical components of freshwater systems, yet they remain difficult to map and integrate into operational hydrologic and ecological analyses. In the United States, the National Wetlands Inventory (NWI) remains the most widely used wetland dataset, but it is limited by high errors of omission—particularly for small, vegetated, and intermittently inundated wetlands—and by spatial misalignment resulting from its legacy, photo-interpreted origins. These limitations hinder integration with other spatial datasets, including remote sensing imagery, hydrologic models, terrain analyses, and stream network data, constraining the ability to characterize wetland functions at management-relevant scales.

We describe the Wetland Intrinsic Potential (WIP) tool, a scalable, process-informed remote sensing and machine learning framework designed to serve as a foundational spatial layer for wetland characterization. Rather than producing a static wetland classification, WIP represents wetland presence as a continuous probability surface that reflects the intrinsic capacity of landscapes to support wetlands. The model is implemented using a random forest approach that integrates multi-scale terrain metrics derived from digital elevation models, hydrologic indices, soils, land cover, and climate variables. By explicitly incorporating geomorphic and hydrologic controls on water flux, WIP captures cryptic, ephemeral, and degraded wetlands that are systematically missed by spectral classification approaches and legacy inventories.

WIP is designed to enable integration across datasets and disciplines. As a spatially continuous, topographically aligned surface, it provides a common spatial framework for linking wetlands with stream networks, hydrologic connectivity models, time-series remote sensing products, and biogeochemical datasets. Building on this foundation, we describe modular applications that characterize wetland functions, including estimates of water storage and hydrologic connectivity, and discuss how the same framework supports extension to additional function-based analyses.

All workflows are implemented in a repeatable, cloud-based architecture that enables high-resolution lidar-based analyses at the watershed scale and scalable processing at 10–30 m resolution across state and national extents. We illustrate applications in the Pacific Northwest, at national scale, and through partnerships with Ministries of Environment in several African countries via Digital Earth Africa. These examples demonstrate how WIP operationalizes wetland science as durable, reusable infrastructure for freshwater analysis rather than a one-off mapping product.

Clearing Blackwaters: Saltwater Intrusion Changes the Color of Southeastern Coastal Rivers.

Spencer Rhea¹, Lauren Krohmer¹, Ryan Emanuel¹, Emily Bernhardt¹

1. Duke University, Durham, NORTH CAROLINA, United States

Saltwater intrusion, the landward movement of marine salts, is becoming an increasing problem in coastal areas due to sea level rise, storm surge, drought, and hydrologic modification of freshwaters. Saltwater intrusion has clearly documented impacts on terrestrial ecosystems such as killing coastal trees, changing greenhouse gas emissions from wetlands, and reducing the productivity of farmland. But how this change in salinity affects river environments is less clear. To evaluate the impact of saltwater intrusion on coastal river ecosystems, we assessed changes in river color, an integrative metric of many important ecosystem parameters. We used the Harmonized Landsat-Sentinel and an integrated Landsat dataset to extract river reflectance values from National Hydrography Dataset river reaches wider than 60m in the U.S. southeast. We looked at long-term trends in river color (1985-present) and seasonal patterns along the river environments. In a preliminary analysis of North Carolina rivers, we found most river reaches have darkened (78%) since 1985. We also found a positive correlation between drought and river greenness in 15% of river reaches, most of which occurred near the outlet of the river. Although remote sensing is a powerful tool that allows exploration across large scale, in this case it lacks the ability to identify the mechanisms behind observed changes. We speculate changes in river brightness could either be caused by reductions in suspended sediment or increases in dissolved organic matter, both of which have been observed across many U.S. rivers. The relationship between drought and river greenness we suspect is caused by flocculation of dissolved organic matter by divalent cations in sea salt. This phenomenon is known to occur in organic rich waters and could lead to clearer water, which has higher green reflectance, or could increase light availability and allow growth of algae. In our analysis, we observe widespread changes in southeastern rivers but lack the ability to definitively attribute these changes to any one mechanism. Future work should explore the cause of change in river brightness and the impact of saltwater intrusion on coastal river ecosystems.

Occupancy drivers of Midwestern headwater stream fishes in a northeastern Indiana watershed

Darren J Shoemaker¹, Peter C Smiley Jr.², Krista M Ruppert³, Robert B. Gillespie⁴

1. University at Buffalo, Amherst, NY, United States

2. USDA Agricultural Research Service, Columbus, Ohio, United States

3. Department of Biological Sciences, University of North Texas, Denton, Texas, United States

4. Department of Biology, Purdue University - Fort Wayne, Fort Wayne, Indiana, United States

Fishes are among the most imperiled classes of vertebrate animals, with approximately 39% of North American fish species exhibiting population declines and extinction risk. Headwater stream fish assemblages are comprised of unique assemblages of headwater specialists, adaptable generalists, and imperiled species. These assemblages are of particular interest in the Midwestern United States given the high rates of modification and degradation of headwater streams in that region. In addition to supporting their own diverse communities, headwater streams are also ecologically important due to their strong influence on downstream water quality. Many studies have previously examined how hydrology, water chemistry, landscape modification, and climate affect stream fish communities in the Midwest. However, relatively few studies have examined relationships between those environmental metrics and individual species. To address this knowledge gap, we performed occupancy modeling using fish captures and instream habitat surveys from 14 years of electrofishing surveys in the Cedar Creek watershed, Indiana and East Branch of the Saint Joseph River, Michigan. Specifically, we sought to broadly describe occupancy trends, quantify occupancy drivers for each species, and identify differences in drivers between species and the overall community. We identified over 64,000 individual fish and 39 unique species, retaining 31 species for modeling. Latent occupancy probability increased for 13 species, decreased for six species, and was roughly stable for nine species. We examined four sets of environmental covariates: water chemistry, hydrology, substrate, and instream vegetative cover. Stream depth and width were the most important drivers of occupancy for most species, although water temperature and conductivity were also important for some species. A few species also showed strong distinctions between their occupancy drivers and those of the overall community, such as conductivity, dissolved oxygen, and temperature for blacknose dace and silverjaw minnow. Our results emphasize the importance of considering the needs of individual species when making conservation decisions and suggest a one-size-fits-all management plans are likely inadequate to maintain diverse headwater stream fish assemblages.

Using the National Ecological Observatory Network (NEON) aquatic sites to explore the drivers of stream particulate quantity and quality across the United States and Puerto Rico

Zacharie Loveless¹, Ariel Shogren¹, Jonathan Benstead¹, David Manning²

1. The University of Alabama, AL, United States

2. The University of Nebraska Omaha, Omaha, NE, United States

River networks are dynamic conveyors and processors of particulate material, commonly quantified as total suspended solids (TSS) and further characterized by particulate carbon (C), nitrogen (N), and phosphorus (P) content. The transport and processing of these materials play key roles in in-stream biological and biogeochemical processes. However, understanding of the drivers of particulate material content and transport is limited, especially at large geographical scales. Here, we expand our understanding of drivers of particulate material at the continental scale by using long-term data collected from the National Ecological Observatory Network (NEON). The NEON dataset includes regularly collected particulate chemistry data from 26 established NEON aquatic river and stream sites across the United States and Puerto Rico. At each site, we first examined relationships

between TSS and concentrations of particulate C, N, and P using high-frequency turbidity data. We then leveraged additional site data, including land use and land cover, high-frequency sensor data, and continuous discharge to examine and enhance our models to ascertain the main drivers of particulate material composition and transport across the US. Our initial results demonstrate that TSS and all three elemental concentrations have significant relationships with turbidity at roughly half of the sampling sites. These initial results enhance our understanding of the driving forces of particulate material in river networks and highlight the use of large-scale sensor and sampling networks to advance the study of riverine biogeochemistry.

id #136092

Influence of land use and hydrology on plastic pollution in the riparian zone of a Mediterranean river

Eugenia Martí¹, DELFINA CORNEJO¹, Berta Bonet¹, Henar Margenat¹, Joan Blàvia¹, Sara Castelar¹, Miquel Ribot¹, David Calvo Mora¹, Wilfred M. Wollheim², Timothy J. Hoellein³, Helena Guasch¹

1. Centre d'Estudis Avançats de Blanes (CEAB-CSIC), Blanes, Spain

2. Earth Systems Research Center, University of New Hampshire, Durham, NH, USA

3. Department of Biology, Loyola University Chicago, Chicago, IL, USA

Riparian zones are located at the interface between terrestrial and fluvial ecosystems; and thus, they are particularly vulnerable to plastic pollution. Mismanaged plastic can reach riparian zones from local inputs of adjacent urban, industrial or agricultural areas. In addition, during high flow conditions or episodic floods, plastic can be flushed from riparian zones into the stream and transported downstream, where in turn, it can be trapped into riparian zones, especially in the presence of vegetation. Therefore, abundance of plastic in a riparian zone can result from the interplay between local human activity and the net effect of stream hydrologic dynamics. In this study, we assessed the degree of plastic pollution (i.e., abundance and category type of the fraction >5 mm, macroplastic-MacP) in the riparian zones of a Mediterranean stream, and examined how it is affected by the catchment land use and the stream hydrological regime. Thirteen sites spanning forested, agricultural, and urban areas were surveyed for MacP abundance on a single date in the riparian zone and stream channel. Additionally, five sites were sampled bimonthly to assess temporal variation of MacP abundance. MacP was mostly found in riparian zones (67%) rather than in the stream channel (<15%). Indicating that riparian zones are hot spots of plastic accumulation. Abundance of MacP was higher in riparian zones nearby urban areas than nearby forested and agricultural areas. This evidences the different influence of human activities on plastic pollution in riparian zones. Floods cause a distinct effect on riparian MacP abundance among riparian zones, which was mostly associated with the vegetative structure of these zones than with their location along the fluvial network. Together, these results shed light about some factors and mechanisms that explain how plastic pollution in catchments ultimately reaches marine ecosystems. It also provides insights on potential management strategies to help reducing plastic pollution in stream-riparian ecosystems.

id #135069

Longitudinal patterns of dissolved organic matter following floodplain restoration in a Northwestern forested river system

Arya Mohanan¹, Emily Coronado¹, James Junker¹, Marc Peipoch², Robert Payn³, Michael DeGrandpre⁴, Juliana D'Andrilli¹

1. Department of Biological Sciences and the Advanced Environmental Research Institute, University of North Texas, Denton 76201, Texas, USA

2. Stroud Water Research Center, Avondale, Pennsylvania 19311, USA

3. Department of Land Resources and Environmental Sciences, Montana State University, Bozeman, Montana 59717, USA

4. Department Of Chemistry and Biochemistry, University of Montana, Missoula, Montana 59812, USA

The Upper Clark Fork River (UCFR) has a long history of contamination, including a flood in 1908 where heavy metal mine tailings were distributed across the channel and floodplain. Our study investigated how restoration due to legacy contamination influences the DOM quality and quantity over a five-year period to better understand patterns of aquatic health following restoration using a carbon cycling perspective. The study will showcase how aquatic ecosystems are influenced by the re-establishment of a 'flood pulse' by restoring the river's connection to its floodplain and what carbon quality signals may serve as healthy ecosystem markers after metal contamination restoration. Our results showed DOC concentrations exhibited distinct annual cycles, with increases during high discharge events. Average DOC concentrations were largest in spring and gradually declined to winter. Excitation Emission Matrix-Parallel Factor (PARAFAC) analysis modeled five fluorescence components, describing humic- and protein-like DOM of varying sizes, reactivities, and origin. More labile, autochthonous microbial-like DOM had more spatial and temporal variation, as expected during the spring summer seasons. From 2017-2022, DOM character increased abruptly in aromatic and more plant/soil "humic-like" nature, determined from specific ultraviolet absorbance at 254 nm (SUVA₂₅₄) and PARAFAC analysis reflecting more terrestrial contributions to the river. Fluorescent humic-like terrestrial signals were dominant from upstream to downstream sites seasonally and annually. Fluorescent DOM temporal variability dominated over spatial variability. Humic-like fluorescence showed synchronous trends during high hydrological connectivity. Interestingly, during the Fall 2018, despite lower DOC concentration, the dominant chemical character was large humic-like DOM. Protein-like fluorescence increased during baseflow as expected, reflecting in-stream production and processing. Fluorescence index (FI) decreased longitudinally and humification index (HIX) increased temporally and spatially, indicating that DOM character was more terrestrial and humified as water moves downstream over time. Furthermore, tributaries exerted a strong control on DOM composition throughout the river by acting as a source and/or diluter.

id #135325

Meltwater Source as a Driver of Mountain Stream Biogeochemistry in the Western United States

Anna Shampain¹, Scott Hotaling¹, Janice Brahney¹

1. Utah State University, Logan, UTAH, United States

Climate change is accelerating the decline of mountain ice worldwide. Over the past 35 years, North American glaciers have lost substantial mass, alongside more variable precipitation and declining seasonal snowpack. Subsurface ice features like rock glaciers—mixtures of rock and ice insulated by debris—are expected to persist even as surface glaciers disappear.

Meltwater sources strongly influence biogeochemical and physical processes in headwater streams. Surface and rock glaciers sustain streamflow during the melt season, regulate temperatures, supply key nutrients (nitrogen, phosphorus, carbon), and release trace elements into aquatic ecosystems. Further, headwater streams are the source of 75% of the drinking water in the western United States and serve as important habitat for sensitive mountain species. However, meltwater from glaciers can also contain elevated trace elements that may occur naturally in bedrock or be added to the system through atmospheric deposition. Our research aims to understand how meltwater sources, including seasonal snow, groundwater, surface glaciers, and rock glaciers, influence key physical and biogeochemical processes in mountain streams. In the summers of 2024 and 2025, we surveyed mountain streams along a latitudinal gradient in the western U.S. from Glacier National Park (MT) to the Teton Range (WY) and the Wasatch (UT).

We sampled 18 streams fed by varying meltwater sources and visited each stream three times during the summer (early-, mid-, and late-season) to capture intra-seasonal variations. We collected water samples for analysis of key nutrients (nitrogen, phosphorus, and carbon), major ions, and trace elements, and measured physical site conditions. Our results demonstrate key differences in stream physical and chemical properties across locations as a product of the meltwater source. These findings will inform conservation efforts for sensitive aquatic species and water management strategies (i.e., downstream water quality treatment and flow management) to address the impacts of changing meltwater sources.

id #135581

Restoration in riverscapes: using spatial food-web modelling to inform restoration planning

Holly Harris¹, Ryan Bellmore², Joe Benjamin³, Aimee H. Fullerton⁴, Casey Justice⁵, Matt Kaylor⁵, Joe Lemanski⁶, Les Naylor⁷, Adam Price⁸, Gabriel Rossi⁹, Brian Staab⁸, Sean Welch¹⁰, Michael Wisdom⁸

1. Juneau Forestry Sciences Lab, Pacific Northwest Research Station, Juneau, Alaska, United States

2. Pacific Northwest Research Station, Juneau, -, United States

3. Forest and Rangeland Ecosystem Science Center, USGS, Boise, Idaho, United States

4. Northwest Fisheries Science Center, NOAA, Seattle, Washington, United States

5. Columbia River Inter-Tribal Fisheries Commission, Portland, Oregon, United States

6. Oregon Department of Fish and Wildlife, Salem, Oregon

7. Confederated Tribes of the Umatilla Indian Reservation, La Grande, Oregon

8. Pacific Northwest Research Station, USFS, Portland, Oregon

9. University of California at Berkley, Berkley, California

10. Bonneville Power Administration, Portland, Oregon

River networks comprise spatially-structured food webs across headwaters, mainstems, and floodplains that provide diverse foraging opportunities for mobile consumers. These so-called foodscapes are theorized to promote the productivity and stability of aquatic consumers, such as stream fishes, but such theory is rarely incorporated into watershed management. Here, we develop a spatially-explicit food web simulation model to predict salmon and trout responses to restoration alternatives in the Meadow Creek watershed in northeastern Oregon, USA. The model tracks consumer-resource biomass dynamics over time in different reaches of the river network that are linked by spatial flows of basal resources, invertebrates, and fish. We parameterized the model with in-stream and riparian vegetation conditions collated from previous studies, and simulated patterns of periphyton, invertebrate, and fish abundance across the watershed. This spatially-explicit model gives insight into the potential for effects of local restoration actions to carry-over into different parts of the watershed, which can feed into the adaptive management process. This model is a step towards greater understanding of how watershed management influences foraging patterns and growth potential critical to supporting mobile consumers in river foodscapes.

id #135837

Urbanization stressors on avian invertivore assemblages mediated by aquatic macroinvertebrate diversity: Implications for multi-scale riparian restoration in an increasingly urbanized future

Giordano Jacuzzi¹, Julian D Olden¹

1. University of Washington, Seattle, United States

Freshwater and riparian ecosystems are tightly coupled by the flow of resource subsidies across aquatic and terrestrial boundaries, yet highly vulnerable to the environmental stressors associated with urban expansion. Ongoing declines in insectivorous and migratory bird populations are attributed in part to parallel reductions in riparian habitat and the availability of aquatic and terrestrial prey. However, our understanding of how anthropogenic stressors alter aquatic resource subsidies, and how these changes propagate to avian consumers, remains limited. Leveraging benthic macroinvertebrate surveys and avian acoustic monitoring data from streams spanning an urbanization gradient in the Seattle metropolitan area of Washington, USA, we use multi-species occupancy and structural equation modeling to disentangle the effects of biotic and abiotic factors on the

diversity of birds and their aquatic insect prey during the breeding season. We found that a benthic index of biotic integrity (B-IBI) strongly influenced both the occupancy and richness of invertivorous predator and migrant guilds, more so than measurements of impervious surface (ISC) and tree canopy cover (TCC) within the stream reach or catchment. Indirect effects of urbanization pressures on avian richness operated through two pathways at multiple spatial scales: reach-scale ISC reduced local TCC, and catchment-scale ISC reduced B-IBI. Therefore, aquatic prey subsidies and riparian habitat conditions can mediate the influence of anthropogenic stressors on avian assemblages. By applying our model to projections of ISC through the year 2100 we show that restoration at multiple spatial scales is necessary to maintain present-day richness levels in the face of continued urban expansion and its effect on aquatic subsidies. These findings indicate that the future conservation of terrestrial and aquatic biodiversity in urban environments is tightly intertwined and dependent on co-management of the landscape.

id #136093

Redlined water bodies are smaller and more polluted

Anika R Bratt¹, Alexandra Jabbarpour^{1,2}, CJ Denney^{1,3}, Ahmed Abdallah Ahmed¹, Georgia Akins¹, Ollie Branch¹, Mara Pirone¹, Lucia Zuvela¹, Sophie Carpenter¹

1. Macalester College, Saint Paul, MN, United States

2. Friends of the Mississippi River, Minneapolis, Minnesota, US

3. Yale University, New Haven, Connecticut, USA

Redlining was a racist, discriminatory practice developed by the Home Owners Loan Corporation (HOLC) in the mid-1930s that assigned grades to residential neighborhoods based on their perceived "mortgage security". Neighborhoods were graded A ("best"), B ("still desirable"), C ("definitely declining"), or D ("hazardous"), with minoritized neighborhoods rated C or D. D neighborhoods were colored red on maps, thus the term redlining. While this policy ended in the mid-1950s, numerous studies have connected redlining to disparities in temperature, air quality, and green space between graded neighborhoods. There is, however, limited research exploring the connections between surface water and redlining. This study investigated how water quality and water body size varied across HOLC grades in Saint Paul and Minneapolis, MN. Stream data from the United States Geological Survey (USGS) was also used to assess stream flashiness across graded neighborhoods throughout the US. We show that water bodies in redlined neighborhoods are significantly smaller, but also had significantly cooler water temperatures. Further, our data demonstrate that this translates into less of a cooling effect in redlined neighborhoods during summer months. Long term records of water quality show that phosphorous was also significantly higher in C and D neighborhoods. USGS data suggest that streams and rivers in A and B ranked neighborhoods exhibit flashier discharge patterns. Overall these results show important disparities in water quality and ecosystem services in urban landscapes.

id #134302

Habitat variability along a harsh-mild gradient to help determine potential climate refugia for cold-adapted taxa in alpine streams

Deb Finn¹, Scott Hotaling², Lusha Tronstad³

1. Missouri State University, Springfield, MISSOURI, United States

2. Utah State University, Logan, Utah, USA

3. University of Wyoming, Laramie, Wyoming, USA

One approach to mitigating biodiversity loss associated with climate change is identifying and protecting local habitats that can act as refugia for vulnerable taxa. Alpine streams are habitat to many climate-vulnerable cold stenotherms and are highly heterogeneous from one to the next, driven largely by variability in hydrological source. In the Rocky Mountains (USA), major alpine streamflow sources include snowmelt, surface-glacier runoff, and runoff from ice locked in cold rocky landforms (CRLs) such as rock glaciers. CRL ice is relatively buffered from synoptic-scale climate, hence streams sourced by CRL ice ("icy seeps") have potential as climate refugia for cold-adapted taxa. However, substantial variability also exists among icy seeps. Our objective was to quantify abiotic and biotic variability among alpine streams representing a range of habitat characteristics, with an emphasis on icy seeps. During a 2-week mid-summer period in two consecutive years, we sampled 7 icy seeps, plus 5 snowmelt-fed and 4 glacier-fed streams in the Teton Range, Wyoming. We asked if abiotic habitat variables hypothesized to occur along a harsh-mild gradient could predict aspects of benthic macroinvertebrate community structure. Habitat characteristics included streambed stability, suspended solids, water temperature, conductivity, flow intermittency, elevation, aspect, and slope. An abiotic glaciality index clearly differentiated icy seeps from the other stream types. Macroinvertebrate communities varied somewhat among stream types, but there was also substantial variability among communities within stream type, particularly for icy seeps. Of 43 total taxa, few were indicative of a single stream type. For example, a threatened stonefly (*Lednia tetonica*) occurred in both glacier-fed streams and icy seeps. A multivariate habitat "mildness" metric was strongly correlated with community structure. Icy seeps with greater mildness had greater taxa richness, including cold stenotherms. Variables increasing habitat mildness included higher streambed stability, lower diel temperature range, lower suspended solids, and combinations of elevation, slope, aspect that decrease duration of snow cover. These results suggest that abiotic variability among icy seeps might predict the potential of individual streams to act as climate refugia for vulnerable taxa. Identifying readily measurable habitat variables indicative of greatest refugial potential will be an important next step for conservation.

id #136094

Using remotely sensed data to quantify the influence of beaver dams on surface water in small prairie streams.

Matt Webster¹, Colleen Piper¹, Lisa Eby¹, Blake Hossack^{2,1}, Patrick Donnelly^{3,1}, Thomas Riecke¹

1. *University of Montana; W.A. Franke College of Forestry and Conservation, Missoula, Montana, United States*

2. *U.S. Geological Survey; Northern Rocky Mountain Science Center, Missoula, Montana, United States*

3. *University of Montana; Numerical Terradynamic Simulation Group (NTSG), Missoula, Montana, United States*

Water limits productivity in semi-arid and arid regions. Beaver-based restoration is increasingly being used to conserve and restore streams and riparian areas and to promote ecosystem resilience to drought. However, the influence of beaver dams on surface water has not been examined in small prairie streams. Watersheds in the northern Great Plains are characterized by erosive soils and sparse woody vegetation, often resulting in deeply incised channels with confined stream flow. This region is also known for variable and unpredictable summertime flows across intermittent and permanent streams. Although data from satellite imagery has dramatically increased our ability to monitor beaver dams and surface water change across landscapes, assessing the effects of beaver in prairie watersheds has been hindered by many factors, including stream size, flow intermittence, and deeply incised stream banks. To quantify the effects of beaver in prairie streams in eastern Montana, we created a random forest-based land cover classifier that uses fine scale, freely available satellite imagery to identify surface water across landscapes. We then coupled this surface water data with drainage area, topographic confinement and stream flow data to quantify the relative influence of beaver dams on surface water, relative to modeled stream flow. Validation results show that our random forest land cover classifier performs well, with an overall accuracy of 0.86 for all land cover types (water, vegetation, bare ground), and an average consumer's accuracy of 0.95 for surface water. Despite considerable variation in the magnitude of effect among watersheds, topographic ponding capacity and stream flow were clear drivers of surface water area. Beaver dams increased surface water area within a limited window of stream flow conditions. Notably, this window tends to correspond with mid- to late summer stream flow levels, which has important implications for late season mesic productivity. Our research offers a framework by which to explore the hydrologic and topographic conditions that moderate beavers' influence on surface water in prairie streams, which can help target areas for beaver-based restoration, and inform expectations for when dams may increase surface water availability.

id #136350

Crayfish effects on salmonids in fresh waters: A synthesis of current knowledge and research priorities

Julian Olden¹

1. *School of Aquatic and Fishery Sciences, University of Washington, Seattle, Washington, USA*

Large-scale assessments of threats to salmonids have increased in recent decades; yet the role of crayfish—both native and nonnative—has received comparatively limited attention, and existing knowledge remains fragmented across the literature. This study addresses this gap by synthesizing current evidence of crayfish effects on salmonids across ontogenetic stages and identifying critical knowledge gaps to guide future research. The state of published knowledge on the impact of crayfish on salmonids is relatively disparate across taxonomic groups, geographic regions, and environmental contexts. Our review identified 35 articles (comprising 60 crayfish-salmonid study outcomes) published over the past half-century, involving 10 crayfish species and 12 salmonid species. Half of all study outcomes explored the salmonid egg stage (51%), alevin (8%), fry (10%), juveniles (14%), adults (12%), and post-mortality carcasses (5%). Salmonids experience substantial mortality during early life stages, particularly from egg deposition through fry emergence. Salmonid eggs are vulnerable to predation by both native and nonnative crayfish, which act as active foragers as well as scavengers. Although laboratory studies have provided valuable insights, their applicability to natural systems remains limited. Key knowledge gaps persist, particularly regarding the functional feeding responses of crayfish (i.e., egg consumption as a function of availability) and the interactions between environmental and biological factors that shape per capita effects on salmonids. Crayfish-salmonid interactions extend to fry and juvenile stages through both direct predation and competition for shelter and invertebrate prey, which may constrain salmonid growth. Despite these plausible mechanisms, empirical studies are very scarce. Evidence for effects on adult salmonids is even more limited, with correlative studies yielding mixed results, and the role of crayfish in nutrient cycling via carcass scavenging remains poorly understood. By illuminating the complex, context-dependent nature of crayfish-salmonid interactions, we propose 10 priority research directions to inform future management and conservation actions.

id #133791

When floodplain channels desynchronize: consequences of lateral connectivity dynamics on ecosystem metabolism

Basile BC Cousin¹, Sylvain SD Dolédec¹, Jean-Michel JMO Olivier¹, Camille CL Lebrun¹, Mathieu MT Tocaben¹, Fanny FC Colas¹

1. *UMR LEHNA - Laboratoire d'Ecologie des Hydrosystèmes Naturels et Anthropisés, VILLEURBANNE, France*

The lateral dimension of fluvial hydrosystems encompasses the interactions between main and floodplain channels, creating diverse environmental conditions that shape ecosystem metabolism in space and time. Most studies on the metabolism of fluvial hydrosystems (i.e., defined by the balance between Gross Primary Production (GPP) and Respiration (ER)) have focused on main channels, overlooking the lateral floodplain dimension. We developed a novel approach to quantify temporal dynamics of lateral channel connectivity and we further assess their influence on ecosystem metabolism. To this end, we used hourly water table and temperature data over one year in nine floodplain channels and two main channels of the French Upper Rhône to compute a Synchronism Index (SI) describing the degree of hydrological coupling between floodplain and main channels. In addition, concurrent continuous dissolved oxygen measurements enabled metabolism quantification of each floodplain channel. Our results show (i) the relevance of SI in characterizing lateral connectivity dynamics, and (ii) a consistent increase in GPP and ER in desynchronized channels. This suggests that reduced hydraulic constraints favor higher primary production and organic matter accumulation, leading to enhanced respiration. This study seems pioneering in revealing how variability in lateral connectivity shapes metabolism in fluvial hydrosystems and provides initial knowledge for management in a context of intensified disconnection of floodplain ecosystems.

How does drought alter the synthesis and aquatic-terrestrial transfer of omega-3 fatty acids?

Madeline Hannappel¹, Robert Fournier¹, Adrienne Smits², Lauren Esposito³, Albert Ruhi Vidal¹

1. *Environmental Science, Policy, and management, UC Berkeley, Berkeley, CA, United States*

2. *Dept. of Environmental Science and Policy, University of California, Davis, Davis, CA*

3. *Islands and Seas, San Francisco, CA, USA*

Emerging insects from freshwater ecosystems can be key prey to riparian predators, but research on these cross-ecosystem subsidies has typically focused on biomass fluxes rather than on nutritional aspects. Omega-3 long-chain polyunsaturated fatty acids (ω 3 LC-PUFAs) synthesized by microalgae are rare in the terrestrial environment but critical to consumer growth, immune function, and reproduction. Here, we reviewed drought impacts on the production and transfer of ω 3 LC-PUFAs from riverine to terrestrial ecosystems. Putative mechanisms include reductions in surface water and habitat connectivity, shifts in algal communities toward LC-PUFA-poor taxa (e.g. cyanobacteria rather than diatoms), shortened hydroperiods that constrain aquatic insect life cycles, and altered emergence phenology that may decouple timing of subsidies from riparian consumer demand. To explore these mechanisms, we conducted a pilot study comparing ω 3 LC-PUFA profiles of aquatic insects from temporary and perennial streams in Pinnacles National Park, CA. Fatty acid profiles differed significantly among insect taxa and between hydrology types, although ω 3 LC-PUFA concentrations at the community level did not differ between temporary and perennial streams. We then compared 2 abundant taxa that differ in life histories: water striders *Aquarius* sp. (Gerridae), short-lived insects that survive drought by migrating to wet habitat (i.e., resilient strategist); and fishflies *Neohermes* sp. (Corydalidae), long-lived insects that survive drought locally via physiological and life-history adaptations, by entering a dormant stage (i.e., resistance strategist). We found significant effects of hydrology on ω 3 LC-PUFA concentrations in drought-adapted *Neohermes*, with concentrations in temporary sites increasing with insect body mass. Because compound-specific stable isotopes suggest *Neohermes* did not change diet with intermittency or body size, we propose that these insects bioaccumulate ω 3 LC-PUFA over the repeated dormancy-growth cycles - becoming very rich in ω 3 prior to metamorphosis. These results highlight the complexity of interpreting fatty acid data, and the importance of natural history (e.g., resilient v.s. resistant strategies) in influencing fatty acid profiles and responses to environmental variation. We contend that intensifying droughts may alter the amount and form of ω 3 fatty acids transferred to the riparian environment, 'silently' disrupting a key linkage between rivers and their watersheds.

Unpredictable waters: lentic community responses to experimental hydroperiod variability

Courtney Hendrickson¹, Betsy Bancroft², Annika Rose-Person², Tiffany Garcia¹

1. *Oregon State University, Corvallis, OR, United States*

2. *Gonzaga University, Spokane, WA, United States*

Climate change is increasing the frequency, intensity, and unpredictability of extreme events experienced by freshwater communities. These impacts, which include unseasonal floods and droughts, can disrupt historically predictable hydroperiod patterns. Therefore, understanding how communities with differing natural disturbance histories respond to hydrological stochasticity is critical for predicting future pond resilience under ecosystem change. We explored shifts in pond community structure challenged with altered hydroperiod regimes using frameworks of species diversity (richness) and phenotypic flexibility (body size variability). Communities that experience regular disturbances, such as seasonal pond drying, are expected to support populations with flexible life-history strategies and higher body size variability, whereas communities from perennial ponds may exhibit higher species diversity but lower flexibility in body size response. Thus, we tested whether communities emerging under stochastic hydroperiod conditions are characterized primarily by species diversity, trait flexibility, or a combination of both, and whether these mechanisms differ among communities with contrasting disturbance histories. We conducted a fully factorial mesocosm experiment using two community types—sourced from ephemeral and perennial ponds—and three hydroperiod regimes—stable, drying, and stochastic. We used stovepipe samples to stock mesocosms with zooplankton, macroinvertebrates, amphibian, and fish species in naturally relevant densities from these two pond types. We observed pronounced differences in community composition, body size structure, and developmental patterns across hydroperiod treatments, and particularly under stochastic drying regimes. Community responses varied depending on disturbance history, with taxa exhibiting differential responses in species and body size diversity. These patterns suggest that increasing hydroperiod variability can restructure freshwater communities through multiple, potentially interacting mechanisms with implications for predicting community organization under future climate scenarios.

Environmental consulting and the adventures of a fish biologist

Christina Linkem¹

1. *Stillwater Sciences, Arcata, California, United States*

Environmental consulting is a common yet mystifying career path for freshwater scientists. In this talk, I will share my career trajectory as a fish biologist working in environmental consulting, highlighting how academic training in aquatic ecology translates into applied science, regulatory processes, and on-the-ground conservation outcomes. Drawing on experiences from salmonid habitat restoration, flow augmentation studies, endangered species consultation, and field-intensive monitoring programs, I will describe the diversity of projects, roles, and day-to-day work that define my consulting career.

The presentation will explore key skills that are critical for success outside academia, including project management, interdisciplinary collaboration, regulatory navigation, and effective communication with agencies, clients, and stakeholders. I will also touch on how consultants and academic researchers can work together to address complex restoration questions, before, during, and after project implementation.

I will conclude by discussing the trade-offs and rewards of consulting work, ranging from seasonal field “adventures” to report writing and long-term project development, and how applied careers can offer meaningful opportunities to contribute to both ecological understanding and real-world conservation outcomes. This talk is intended for students and early-career scientists exploring non-academic career options, as well as for those interested in applied pathways within freshwater science. I also welcome discussions from academic and non-academic professionals alike on how we can work together to achieve habitat restoration objectives for the betterment of our shared natural resources.

id #135585

Contaminant spiraling as a unifying conceptual model for predicting fate and effects of contaminants on aquatic-terrestrial meta-ecosystems

Mirco Bundschuh¹, Bridger M Creel², Christopher Kotalik³, Johanna M Kraus³, Jeffrey D Muehlbauer⁴, Brittany Perrotta³, Freya Rowland³, Ralf Schulz¹, David Walters³, Jeff Wesner⁵

1. *Institute for Environmental Sciences, University of Kaiserslautern-Landau, Landau, Germany*

2. *Division of Biological Sciences, University of Montana, Missoula, MT, USA*

3. *U.S. Geological Survey, Columbia, MO, United States*

4. *Alaska Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Fairbanks, Alaska, USA*

5. *Department of Biology, University of South Dakota, Vermillion, South Dakota, USA*

Freshwater streams receive nutrients, contaminants, and organic material from their catchment, which are then transported downstream. Here we develop contaminant spiraling as a unifying conceptual model within the study of the fate and effects of contaminants in aquatic-terrestrial meta-ecosystems throughout stream networks. We start by taking an expansive view of nutrient spiraling, adding concepts of ecotoxicology, food web theory, ecological subsidy theory, the River Continuum Concept, and contaminant fate and transport modeling to characterize a multidimensional contaminant spiral with components of spiraling length, width, and transport velocity. To provide a broad context for the types of biological responses that might drive contaminant spiraling, our conceptual model distinguishes between contaminants considered non-toxic and toxic to the base of the food web at environmentally relevant concentrations. For example, if microbial communities are not directly affected by non-toxic contaminants, then uptake in those biofilms and emerging aquatic insects should increase with a contaminant's propensity to bioaccumulate, leading to wider, shorter spirals. In contrast, toxic contaminants can reduce biofilms, affecting the contaminants' transport within aquatic-terrestrial meta-ecosystems. At high concentrations, toxic contaminants are expected to increase spiral length, reduce spiraling width, and reduce insect-mediated contaminant flux to land. Position in the stream network will also affect contaminant spiraling by influencing the probability of exposure to these contaminants as well as the productivity and basis of production of the stream and riparian ecosystems. Testing these and further hypotheses derived from our conceptual model is instrumental to understanding the spatio-temporal dynamics of contaminants' fate and effects within aquatic-terrestrial meta-ecosystems, ultimately directing respective policy decision making and management.

id #136097

Looking back and looking forward: river conservation in a human-dominated world

N LeRoy Poff¹

1. *Colorado State University, Fort Collins, COLORADO, United States*

In this 30-minute wrap-up presentation, I will give a personal perspective on contributions of myself, students and collaborators on the last 25 years to the broad field of stream and river ecology and conservation. I will highlight several challenges that we as a scientific community have made significant progress on, as well as offer up several additional challenges that face us now and in the future. The second half of the presentation will be oriented toward audience interaction and brainstorming about the future of riverine conservation.

id #135330

Nitrogen flows in two arctic stream food webs of contrasting disturbance regime

Tori A Hebert¹, Alexander D Huryn¹, Jonathan P Benstead¹, Stephanie M Parker², Diane M Sanzone³, William B Bowden⁴, Bruce J. Peterson⁵

1. *The University of Alabama, Tuscaloosa, AL, United States*

2. *National Ecological Observatory Network, Boulder, CO, USA*

3. *Ramboll, Portland, ME, USA*

4. *University of Vermont, Burlington, VT, USA*

5. *The Ecosystems Center, Marine Biological Laboratory, Woods Hole, MA, USA*

Here, we present the results of a LINX 1-style ¹⁵N tracer addition conducted in the summer of 2002 in two adjacent headwater tributaries of the Ivishak River in the Arctic National Wildlife Refuge, Alaska. Our objective was to assess how contrasting disturbance regimes drive differences in stream N-uptake dynamics. The Ivishak spring-stream is groundwater-fed and maintained relatively constant discharge during the ¹⁵N addition (110-180 L/s). The Ivishak mountain stream is fed by precipitation and experienced more variable discharge through the addition period (60-2200 L/s). The isotope tracer (¹⁵NH₄Cl) was added

simultaneously to both reaches for 4 weeks to quantify N-uptake dynamics and food-web interactions. Preliminary results show that ^{15}N uptake was rapid in both streams. Uptake lengths of $^{15}\text{N-NH}_4$ (S_w) in the mountain stream were high at the beginning of the tracer release (7 km) due to elevated discharge but decreased significantly thereafter (200–400 m). Uptake lengths and their temporal variability in the spring stream were consistently lower (80–150 m). Most of the added $^{15}\text{N-NH}_4$ in the spring-stream was taken up by bryophytes, whereas uptake of N in the mountain stream was dominated by filamentous algae. Chironomids appeared to be feeding primarily on epilithon and filamentous algae, and simuliids fed on seston in both streams. Additionally, the spring-stream contained a predatory stonefly (*Isoperla petersoni*), which became enriched midway through the addition. We are currently analyzing this unpublished dataset with the Bayesian hidden Markov model in the R package 'IsoTracer' to quantitatively compare N-dynamics in streams with differing disturbance regimes.

id #135586

beyond phytoplankton and phosphorus: assessing trophic state in freshwater benthic ecosystems

Paige M Kleindl¹, Carolina Candelario¹, Evelyn E Gaiser¹

1. *Institute of Environment, Department of Biological Studies, Florida International University, Miami, Florida, United States*

Phytoplankton biomass and water column total phosphorus (TP) concentrations are common measures of lake trophic status. However, these commonly used trophic state indicators often lose their effectiveness in P-limited lentic systems dominated by benthic microbial and macrophyte producers, including both algae and cyanobacteria. We undertook a literature review to document the ranges and forms of the relationship between TP and phytoplankton, benthic algal, and macrophyte biomass across P-limited lentic benthic aquatic ecosystems. While phytoplankton and benthic algal biomass responded positively to water TP in littoral zones and shallow lakes and ponds as expected, benthic algal biomass was weakly and negatively related to water TP in wetlands. To better understand this negative relationship, we examined benthic microbial and macrophyte biomass relative to TP standing stocks across a nutrient gradient in two karstic wetlands and found strong positive associations between biomass and TP stock for both producers. Some wetland trophic assessments acknowledge that planktonic TP and chlorophyll-*a* fail to accurately represent trophic status. Incorporating a standing stock approach could enhance the effectiveness and reach of producer and TP relationship indicators for lentic systems at large.

id #135843

Detecting spreading impacts of a persistent invasive predator across a wetland landscape.

Flavien Garcia¹, Nathan J Dorn¹

1. *Florida International University, Miami, FL, United States*

Impacts of invasive species are most often evaluated locally and shortly after establishment, yet far fewer studies examine how impacts spread and vary as invaders persist and expand across space and time. The non-native Asian swamp eel (*Monopterus albus/javanensis*) has been established in the relatively small (<250 km²) Taylor Slough drainage of the freshwater Everglades (FL) for 12–16 years, where it has reduced or collapsed (>90%) several fish and crayfish populations. The eel has now spread throughout much of the remaining 6000 km² ecosystem, including larger drainages, seasonally drying marshes, and wetlands with less depth fluctuation. We used long-term monitoring datasets (1996–present) of small native marsh fishes and decapods to quantify invasive eel impacts in three additional regions with contrasting hydrologic regimes and species compositions. For more permanent slough regions, we fitted baseline species–hydrology models for dominant fishes and decapods using pre-invasion data (Shark River Slough < 2021; Water Conservation Area < 2017). Invasion impacts were assessed using Monte Carlo randomization to test whether post-invasion residual deviations exceeded those expected from pre-invasion variability. Prey responses were spatially heterogeneous and weaker in the region with least hydrologic variability. Most, but not all native fish species exhibited substantial density declines relative to expectations in one or both slough regions (mean loss = 46.4%), ranging from moderate reductions for bluefin killifish (*L. goodei*; –16.6%) to local extirpations of flagfish (*J. floridae*). Slough crayfish (*P. fallax*) declined by an average of 59.8%, whereas grass shrimp (*P. paludosus*) occasionally exceeded expected densities (+87.1% in Shark River Slough). Using a Before-After-Control-Impact approach and a reference region far west of the invasion, we even found evidence for 41% reductions in crayfish densities (*P. alleni*) in shallower seasonally drying marshes (Impact*Post = –0.46, *p* = 0.039). Several species responses were consistent with patterns reported from the region of first invasion, while magnitudes were somewhat smaller. Our results demonstrate that invasion impacts can emerge as diffuse spreading processes that progressively erode prey populations with some variability among systems. Accounting for continued invasion dynamics is therefore essential for understanding freshwater ecosystem resilience and anticipating long-term consequences of persistent non-native species.

id #136099

Using Hydroacoustics to Analyze Microcystis and Fish Distribution During Harmful Algal Blooms in Lake Erie

Ali Shakoore¹, Donna Kashian¹

1. *Wayne State University, MI, -*

In Western Lake Erie (WLE), Harmful Algal Blooms (HABs) are occurring with increased frequency and duration. Microcystis aeruginosa, the primary HAB-forming cyanobacterium, increases turbidity, disrupts food webs, and produces toxins that can

harm humans and animals. While most research focuses on toxin production and bloom formation, less is known about HABs impacts on WLE's food web and fisheries. This study aimed to determine subsurface *Microcystis* concentrations and fish community distribution inside and outside of HABs events in WLE using multi-frequency hydroacoustics. Partnering with NOAA-GLERL, data was collected using a Biosonics hydroacoustics system, utilizing 70, 120, 200, and 420 kHz transducers towed along a transect covering HABs and non-HABs areas. Data was collected over two days, during daytime and nighttime cruises, during August of 2022. Fixed station water quality data consisting of Chl-a, *Microcystis* concentrations, microcystin concentrations, temperature and dissolved oxygen were also collected. Preliminary data indicates *Microcystis* is distributed throughout the water column, likely from wind-mixing in shallow WLE. Fish in HABs zones appeared constrained between surface blooms and possible toxic bottom conditions. These spatial shifts in fish distributions during HABs events may affect fish abundance indices used in surveys. Improved understanding of these dynamics can help fisheries managers refine regulations and ensure sustainable resources use.

id #136100

Spatial and temporal variation in the composition of Grand Canyon parasite communities, surveyed through 18s metabarcoding.

Isaac J Schuman¹, Sanders Justin¹, Anna Jolles¹, Theodore Kennedy², Kate Behn², David Lytle¹

1. Oregon State University, Corvallis, OREGON, United States

2. Grand Canyon Monitoring and Research Center, Flagstaff, AZ, United States

The diversity of biological communities is often described without considering parasites. Instead, parasites are often studied alongside bacterial and viral pathogens using the tools of disease ecology. Parasites become obvious in situations such as zoonotic spillover and outbreaks among host populations. In contrast, the loss of parasite biodiversity can go unnoticed. This project uses DNA metabarcoding as a method to survey the parasite communities of fish in the Colorado River in Grand Canyon, focusing on protists, or unicellular eukaryotes. Parasitic protists are often neglected because they are difficult to locate and identify, making them good subjects for a molecular approach. Parasites have pervasive effects on behavior and mortality in host populations, and they create unique trophic linkages between intermediate and definitive hosts. In a heavily monitored and managed ecosystem like Grand Canyon, understanding parasite communities deepens our understanding of effective management and emerging threats.

DNA was extracted from samples collected between 2022 and 2024 throughout the length of the canyon, including filtered water samples, as well as fecal samples from over 250 *Catostomus latipinnis* (flannelmouth sucker), 20 *Oncorhynchus mykiss* (rainbow trout), and 77 federally threatened *Gila cypha* (humpback chub). Eukaryotic 18s sequences were amplified and Illumina NGS sequenced to generate taxon lists which were then filtered to parasitic taxa. In addition, data on fish size and health was collected in the field.

In our updated multi-year results we detected over 200 parasites of fish and invertebrates, originating in samples sourced from both fish and water. NMDS ordination further revealed distinct community fingerprints for each fish species and sampling year. Water from tributary streams was especially diverse, supporting current ecological theories of dam-driven environmental homogenization and degradation in the mainstem. Key parasites of fishes we detect include the introduced Asian fish tapeworm *Bothriocephalus acheilognathi*, as well as myxozoan DNA present in over half the samples. We observe distinct instances of temporal patterning in parasite occurrence, such as a four-fold spike in microsporidian DNA detections in 2023. These results provide a baseline for monitoring parasite biodiversity and prevalence, and inform the rapidly evolving management situation in Grand Canyon.

id #134309

Emergent aquatic insects link freshwater integrity to riparian bird communities at continental scales

Christian Schuerings¹, Julian Olden¹

1. University of Washington, Seattle, WASHINGTON, United States

Freshwater ecosystems are increasingly degraded by land use change, hydrologic alteration, and climate stress, yet their consequences for adjacent terrestrial food webs remain insufficiently quantified at large spatial scales. Aquatic insects, particularly mayflies, stoneflies, and caddisflies (EPT), provide a critical energetic and nutritional subsidy to riparian predators through adult emergence and are widely used as indicators of freshwater ecosystem integrity. Here, we test whether variation in EPT richness predicts riparian bird communities across continental scales. We integrated benthic macroinvertebrate monitoring data with over two million bird observations from eBird across 14,177 river sites in the contiguous United States. Using hierarchical hurdle models we show that EPT richness strongly predicts the prevalence of aerial insectivorous birds, with more than threefold higher prevalence at high- versus low-EPT sites. This association was consistent across hydrological, thermal, and geomorphological river types, including intermittent rivers, but was attenuated by urban land use and, to a lesser extent, intensive agriculture. Non-aerial insectivores and herbivores showed substantially weaker or inconsistent associations, supporting a trophic mechanism linked to emergent aquatic prey. Our findings provide continental-scale evidence that freshwater integrity sustains riparian bird assemblages through aquatic insect subsidies, highlighting that protecting rivers yields co-benefits that extend beyond the aquatic realm.

id #135333

Islands in the stream: Aquatic bryospheres are sites of high nitrogen-fixation in oligotrophic headwaters

Heili Lowman¹, Brooke Hassett¹, Blanka Aguero¹, Christopher T Solomon², Emily Bernhardt¹

1. Duke University, Durham, NC, United States

2. Cary Institute of Ecosystem Studies, Millbrook, NY, United States

Aquatic bryophytes commonly form mats in small, headwater streams that trap sediment and nutrients and provide substrate for epiphytic algae. However, aquatic bryophytes are rarely included in our current conceptual models of stream ecosystem function or biogeochemical cycling. Here, we examine the potential for the “bryosphere” - inclusive of live and senescent bryophyte tissue, trapped particulates, and associated organisms - to contribute to in-stream biogeochemical function. We measured nitrogen (N) fixation rates from samples of two species of aquatic bryophytes - a liverwort (*Marsupella emarginata*) and a moss (*Codriophorus aduncoides*) - collected from Hubbard Brook Experimental Forest (New Hampshire, USA) and examined how N fixation rates vary by horizon (live/senescent tissue) and mat thickness. All incubated samples demonstrated measurable N fixation (mean $\pm$ s.d. = $70 \pm 26 \mu\text{g N m}^{-2} \text{ hr}^{-1}$). We found scant evidence of epiphytic cyanobacteria, but nearly all samples had $\delta^{15}\text{N}$ signatures indicative of N fixation (1 to -3‰), suggesting potential contributions from endophytic communities. When comparing between live and senescent tissues, N fixation was greater in the live, green horizon, and in samples collected from shallower mats (<5 cm thick), likely due to shallow mats consisting primarily of live tissues. The streams from which these aquatic bryophytes were collected are very oligotrophic, but our findings suggest aquatic bryophytes are nevertheless able to access a “new” source of nitrogen. We propose that estimates of bryophyte coverage and nitrogen fixation should be included in stream N budgets, even when stream water nutrient concentrations remain low.

id #136101

Strategies and Challenges in the Data Stewardship Practices of a Highly Collaborative Wetland Monitoring Program

Raissa Mendonca¹, Ishfaq Rahman², Connor Gluck¹, Olivia Schloegel¹, Lauren E Kinsman-Costello¹

1. Kent State University, OH, United States

2. University of Toledo, Lake Erie Center, Oregon, OH, USA

Environmental monitoring programs can generate actionable data that informs management and policy, but their long-term success requires a high level of multi-institutional and interdisciplinary collaboration. Data co-produced under a shared organizational umbrella can strain quality management throughout the data lifecycle. From a technical perspective, challenges include varying technological aptitudes among personnel, decentralized or nonstandard data collection, lack of version control, and/or incomplete metadata. More broadly, the design of shared data infrastructures must consider the timeline of decision-making activities and primary audiences they serve. Careful consideration of potential pain points and plans for effective solutions are critical to ensure robust data-to-action systems, and those strategies are often best forged through thoughtful engagement with a variety of stakeholder perspectives. The H2Ohio Wetland Monitoring Program (WMP) was established in 2020 to evaluate the nutrient removal effectiveness of varied wetland restoration projects implemented by the Ohio Department of Natural Resources (ODNR) as part of a state-wide water quality improvement program. The WMP is implemented by principal investigators and technicians from multiple Ohio universities who frequently interact with agency personnel, land managers, and wetland practitioners in a variety of capacities. The information created by the WMP informs future wetland restoration approaches and management activities by providing a community of practice with transparent and trusted science. To do so, the data stewardship and governance policies for the WMP have focused on strategies and tools that ensure data preservation, guarantee scientific rigor, and optimize efficiencies from data collection to dissemination by adopting FAIR principles. Our data infrastructure has been built in a stepwise iterative process, adapting and improving systems using on-the-ground feedback from data collectors. We have used custom digital surveys to centralize data entry and continuous integration systems to automate and document critical data lifecycle steps. We built standardized quality checks to produce verified and integrated datasets. These approaches ensure robust data toward actionable results and repeatable analyses, while also promoting the usability of these datasets in other applications.

id #135590

Evaluating the effects of warming on the resistance and resilience of stream communities to drought

Allison Sutcliffe¹, Wyatt Cross¹, Nate Heili¹, Mark Ledger², Kieran Khamis², Mia Berwick², Jonathan Benstead³, Arial Shogren³, Alexander Huryn³

1. Department of Ecology, Montana State University, Bozeman, MT, U.S

2. University of Birmingham, Birmingham, U.K

3. The University of Alabama, Tuscaloosa, Alabama

Climate change is altering the structure and function of stream ecosystems by increasing temperature and intensifying the frequency and duration of droughts. While the individual effects of warming or drought have received substantial attention, little is known about how chronic warming will influence stream ecosystem responses to acute disturbances such as drought. We conducted an artificial stream flume experiment at the University of Birmingham (U.K.) EcoLaboratory and quantified the effects of chronic warming on structural (invertebrate communities) and functional (leaf decomposition) responses to drought. Replicate mesocosms ($n = 16$) were assigned to either ambient or warmed ($+5^\circ\text{C}$) treatments, and invertebrate fauna were acclimated to these treatments for eight months prior to a simulated three-week drought event. We also deployed coarse and fine leaf packs monthly before and after the drought. We hypothesized that chronic warming would favor drought-resistant and resilient taxa, resulting in compensatory structural dynamics that buffer effects of drought on ecosystem functions, including leaf decomposition. Preliminary results show that mean total invertebrate abundance (individuals/m²) was roughly two times higher in the warmed flumes than in ambient flumes prior to the drought, despite similar mean taxon richness in both treatments. Higher abundance in the warmed flumes was driven by high apparent fecundity and survival of gastropods (e.g. *Potamopyrgus antipodarum*, *Planorbidae* *Gyraulus*, *Radix*, *Physa*). Warming increased the rate of leaf litter decomposition by $\sim 30\%$, likely driven by increased microbial activity and resilience of shredder taxa under warmer temperatures, whereas drought substantially reduced

decomposition in ambient flumes. Our results aid in understanding how chronic warming may influence how stream communities and ecosystems respond to additional acute stressors.

id #135335

The ripples of fire: sediment, carbon, nutrients and lake metabolism responses to fire in the northern Rocky Mountains

Sage Fletcher¹, Ashley Ballantyne¹, Robert O Hall², Jessica R Corman³

1. *University of Montana, Missoula, MONTANA, United States*

2. *Flathead Lake Biological Station, University of Montana, Polson, Montana*

3. *University of Nebraska-Lincoln, Lincoln, Nebraska*

Wildfires are increasing in severity and areal extent across the western United States, which can alter carbon and nutrient budgets of aquatic ecosystems in burned watersheds. Given the potential impacts that wildfires have on physical and chemical properties in watersheds, it is crucial to assess how wildfires affect elemental loading and biological processes in lakes. We monitored five lakes (three in Idaho and two in Montana) that experienced recent wildfires in their watersheds (burned lakes) and compared them with five unburned reference lakes. Burned lakes had higher levels of total phosphorus (TP), total dissolved phosphorus (TDP), total dissolved nitrogen (TDN), total nitrogen (TN), dissolved organic carbon (DOC), and total organic carbon (TOC) when compared to the reference lakes. Across all ten lakes sampled in 2024-2025, burned lakes had 47 percent higher total suspended solids (TSS) than reference lakes with peak concentrations in July. Of the five lakes in Idaho, three of the lakes had prior baseline data, which also showed relative increases in nutrient concentrations (TDP, TDN, TN, and in some cases soluble reactive phosphorus (SRP) and TP) in 2025. Our results of increased mass loading of nutrients and carbon in burned lakes over time and with respect to reference lakes are consistent with previous research. Strong negative excursions in dissolved oxygen in burned lakes also indicate enhanced ecosystem respiration.

id #136103

Comprehensive Toolkit for Assessing, Restoring, and Protecting River-Estuarine Ecosystems

Benjamin Block¹, Jennifer Stamp¹

1. *Tetra Tech, VT, United States*

Over the past 6 years, the U.S. EPA has partnered with the Saugus River Watershed Council, the State of Massachusetts, the MassBays National Estuary Program, and Northeastern University to develop a suite of indicators and tools to: (1) set robust and attainable goals to protect and restore aquatic resources, select appropriate management actions, and track progress towards achieving those goals along the Saugus River; and, (2) clearly communicate to the public and stakeholders the environmental, social, and economic consequences of management decisions. The tools will be accessible via an online application that provides user-friendly access and data visualization to allow managers and the public to readily understand the community and regional aquatic resource goals, the rationale behind these goals, and to monitor progress toward those goals. An essential feature of this online forum is the capability to visualize the information at user-selected scales, whether river segment, embayment, region, or baywide, so that the data and information can be applied at the scale relevant for taking action. These tools and indicators have been developed by U.S. EPA and its partners to support states, Tribes and territories develop and manage their water quality management programs consistent with CWA sections 101, 303, 304, and 320.

id #135336

Harnessing airborne thermal infrared survey data of stream temperature to inform restoration across the Willamette River basin, Oregon from 1999 to 2024.

Sylvia R Gholson¹, Christian E Torgersen¹, Matthew I Barker², Caelan E Simeone³, Russell N Faux⁴, Francine H Mejia¹, Will B Long⁵, Brandon T Overstreet², Martin A Briggs⁶

1. *U.S. Geological Survey, Seattle, WA, United States*

2. *U.S. Geological Survey, Corvallis, OR*

3. *U.S. Geological Survey, Missoula, MT*

4. *Applied River Analytics, Corvallis, OR*

5. *Oregon Water Resources Department, Portland, OR*

6. *U.S. Geological Survey, Storrs Mansfield, CT*

Airborne thermal infrared (TIR) remote sensing offers unique insights into drivers and ecological effects of riverine thermal diversity across the 'thermalscape'. Like the riverscape, the thermalscape represents a complete and continuous picture of thermal patterns across multiple temporal and spatial scales. Integrating TIR data with empirical and process-based models and in-situ temperature data helps identify the context of hydrogeological processes in the thermalscape, with the goals of identifying cold-water patches critical to aquatic biota and understanding and restoring thermal heterogeneity in riverine ecosystems. The USGS Next Generation Water Observing System (NGWOS) and Integrated Water Availability Assessments (IWAAs) programs

are currently compiling and updating over 1,600 km of existing TIR datasets from intermittent surveys conducted in 1999 to 2024 and continuous in-situ water temperature records across 788 locations spanning 2011 to 2024 in the Willamette River basin, Oregon to inform resource managers of historical stream temperature conditions and identify stream reaches of importance for repeat surveys. These datasets represent records from multiple federal, state, and local agencies, non-profit organizations, and private industry partners and are currently being archived and published in accessible formats in the federal repository, Science Base. This effort also seeks to provide outreach to local, state, federal, and non-governmental organizations to develop and improve understanding of TIR remote sensing techniques and post-processing methods and to explore new applications of TIR data.

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id #135592

Towards understanding the combined effects of warming and drought on stream food web stability

Nate M Heili¹, James R Junker², Jonathan P Benstead³, Ariel Shogren³, Alexander D Hury³, Mark Ledger⁴, Kieran Khamis⁴, Wyatt F Cross¹

1. Montana State University, Bozeman, Montana, USA

2. University of North Texas, Denton, TX, USA

3. The University of Alabama, Tuscaloosa, Alabama, USA

4. University of Birmingham, Birmingham, UK

Environmental warming modifies consumer-resource interactions and, in turn, alters food web structure, energy flow, and ecosystem stability. Despite the widespread expectation that warming destabilizes ecological networks, there is surprisingly little empirical evidence that warmed food webs are less able to resist or recover from additional, acute disturbances such as drought. Here, we combine empirical food web data from geothermal streams in Iceland spanning a wide thermal gradient (~5 to 27°C) with bioenergetic models to quantify energy fluxes and evaluate how assemblage structure and pathways of energy flow influence food web stability. Building on previous work showing that warming increases total energy demand and shifts energy flux toward small-bodied, fast-turnover taxa, we apply Jacobian matrices of interaction strengths at seasonal and annual time scales to assess stability across temperature regimes. We further simulate drought across the thermal gradient through trait-based, non-random species removals to test whether long-term warming amplifies or compensates for the destabilizing effects of flow reduction. As global change intensifies stressors on streams and rivers, this approach links temperature-driven changes in community composition and energy flux to predictions of food web stability under increasing hydrologic disturbance, offering a roadmap for future empirical studies.

id #135848

Population genetic structure and connectivity of *Allocapnia rickeri*

Camryn A Larson¹, Ashley Dowling², Clay Prater¹, Michelle A Evans-White¹

1. Biological Sciences, University of Arkansas, Fayetteville, AR, United States

2. Entomology and Plant Pathology, University of Arkansas, Fayetteville, AR, United States

Despite being important to freshwater ecosystems, stoneflies (Order: Plecoptera) are imperiled worldwide. Examining landscape genetics can provide information on the genetic diversity and population connectivity of these organisms. This data would aid their conservation by providing a framework for their suitable habitat and dispersal potential and allowing for more informed management decisions. The field of landscape genetics hypothesizes that organisms with a high dispersal ability should have low genetic distances across all geographic distances. Applying this hypothesis to riverscapes has proven challenging as factors like direction of flow, mode of dispersal, and the high dynamism of the environment all have to be considered. We are assessing the riverscape genetics patterns for a winter stonefly, *Allocapnia rickeri*, across the White River catchment in Northwest Arkansas to address two main objectives: 1. Determine their main mode of dispersal, and 2. Correlate riverscape or landscape covariates associated with any genetic differentiation or barriers to dispersal observed. We hypothesize that *A. rickeri* will show genetic patterns that indicate strong connectivity between sites that are in close physical proximity, but these patterns will be stronger within separate watersheds. While this species is common in the region, knowing this information about a common species can have practical applications to rarer *Allocapnia* species that are prioritized for conservation need in the state by allowing us to infer their primary dispersal mode and identifying any major barriers to dispersal these species may have in this region.

id #136104

Stronger together? Exploring synergistic potential in mussel-macroinvertebrate nutrient dynamics

Rowan Scott¹, chester zarnoch², Brendan Foster³, Denise Bruesewitz⁴, Jess Jones⁵, Sally Entekin¹, Timothy Hoellein⁶

1. Virginia Polytechnic Institute and State University, Blacksburg, VIRGINIA, United States

2. Natural Sciences, City University of New York, Baruch College, New York, United States

3. United States Geological Survey, Richmond, Virginia, United States

4. Colby College, Waterville, Maine, United States

5. Department of Fish and Wildlife Conservation, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, United States

6. Biology, Loyola University Chicago, Chicago, Illinois, United States

Urbanization and high nutrient loading from anthropogenic activities can saturate biological demand in streams, leading to eutrophication. Freshwater mussels (FWM) can assimilate, transform, and remove excess nutrients via biofiltration, biological storage, and enhancement of sediment microbial activity. FWM cohabit with macroinvertebrates, who also influence nutrient fluxes through functional feeding (FFG) and bioturbation, making them attractive organisms for best management practices. While mussels and macroinvertebrates independently influence stream chemistry, there is potential for synergistic interactions. The nutrient-rich biodeposition and gas exchange conditions of mussels beds could enhance the capacity for burrowing detritivores to influence environmental conditions which promote microbial nutrient transformations like denitrification. This study aims to evaluate how macroinvertebrate and FWM interactions alter dissolved nutrient and gas fluxes and if there is a synergistic effect on nitrogen removal. We hypothesized that 1) mussel-macroinvertebrate cores will exhibit greater dissolved oxygen (DO) loss due to combined respiration of organisms and bioturbation-enhanced O₂ respiration in sediment, 2) combined mussels and macroinvertebrates will show greater net effluxes of dissolved nitrogen and phosphorus solutes via bioturbation-facilitated mineralization, and 3) synergistic mussel-macroinvertebrate interactions will increase denitrification. To test these hypotheses, we conducted an in-situ core experiment using 10 cores across three treatments: macroinvertebrates only, mussels and macroinvertebrates, and a sediment-only control. Utilizing *Elliptio complanata* and a representative macroinvertebrate assemblage, we measured changes in dissolved nutrients (NO₃⁻, NO₂⁻, NH₄⁺, SRP) and gases (O₂, N₂) before and after 50% DO reduction was reached. Preliminary analyses indicated that macroinvertebrates influence SRP and ammonia (NH₃) exchange, with invertebrate cores increased to ~0.005 mg/L SRP more and ~0.025 less N/L as NH₃ than the control, supporting the need to investigate their role in multi-species nutrient dynamics. Invertebrate cores had ~41% ammonia and ~72% nitrate concentration of the mussel and macroinvertebrate cores. Knowing that insect macroinvertebrates have a significant signal, we will conduct further laboratory experiments with a mussel-only group. Results from this project will quantify whether potential synergistic effects between mussels and macroinvertebrates on nutrient cycling, informing their capacity to improve water quality in FWM streams.

id #134313

Persist, persist, and if necessary, pivot – lessons from establishing model ecosystems in the lab

Angus Webb¹, Yiwen Xu¹, Wim Bovill¹

1. University of Melbourne, Carlton, Vic, Australia

Laboratory experiments are an attractive option for testing theoretical predictions not amenable to field-based investigations. The lesson from our work is to not underestimate the time, effort, and repeated failure required to establish a novel laboratory experimental system; and to be ready to pivot to a 'safe' option when continued failure may compromise a project.

We established laboratory systems to test theoretical predictions regarding the effects of habitat network structure on population outcomes. Previous empirical work in this space used protist systems, mostly with 'migration' supplied by pipetting between habitat patches (jars). Our systems increased the scale of organism, using *Daphnia* sp. as the model organism, and allowed self-directed movement by connecting habitat patches with aquarium tubing.

Early attempts were beset by high mortality and extinctions. Literally months were required to balance feeding rate and water replacement, enabling habitat networks that supported populations over months. Successful experiments then tested the effects of network structure on equilibrium population outcomes and the effects of disrupting connectivity of established populations.

However, attempts to introduce directional bias to movement between habitat patches were unsuccessful. Movement bias is common in systems like rivers, or habitat networks subject to strong prevailing winds. Despite promising pilot results, attempts to introduce directional flow were unsuccessful at full experimental scale. Similarly, using a light gradient with the phototactic *Daphnia* appeared promising but could not be scaled up.

At that stage, we pivoted to model-based simulations to assess the effect of movement bias and rate. However, to inform the efforts of future laboratory researchers, we parameterised the models using data from the successful experiments and used a reasonable number of replicates that one might see in a well-resourced laboratory. Simulation-based power analyses identified potentially observable effects in a laboratory setting.

By remaining flexible and planning for contingency the laboratory program was robust to the many failures along the way and the broader project was robust to the ultimate failure of some laboratory tests. Planning for contingencies, learning from failure, and documenting the trials and tribulations inherent in establishing this type of system can help future researchers to go further, faster.

id #134825

Dissolved inorganic carbon inputs to streams are shaped by soil respiration, riparian groundwater respiration, and subsurface weathering

Ashif Hasan Abir¹, Erin Hotchkiss², Jeremy Jones³, David Butman⁴, Kaelin Cawley⁵, Keli Goodman⁵, Frances Iannucci², Kristin Kraus³, Hannah Conroy⁴, Wilfred Wollheim¹

1. Natural Resources and Environment, University of New Hampshire Main Campus, Durham, New Hampshire, United States

2. Biological Sciences, Virginia Polytechnic Institute and State University, Blacksburg, VA, United States

3. Department of Biology and Wildlife, University of Alaska Fairbanks, Fairbanks, AK, United States

4. Civil & Environmental Engineering, University of Washington, Seattle, WA, United States

5. National Ecological Observatory Network, Boulder, CO, United States

Across the terrestrial-aquatic continuum, dissolved inorganic carbon (DIC) and its component species (H_2CO_3 , HCO_3^- , CO_3^{2-}) vary across upland soil water, groundwater and surface water. The variation can be attributed to different dominant processes across these pools. Soil respiration produces CO_2 in the rooting zone of soil, forming carbonic acid as it dissolves in soil porewater. Once infiltrated into groundwater, this acidic water drives the weathering of carbonate and non-carbonate (e.g., silicate) minerals, releases alkalinity in the form of carbonates and bicarbonates (HCO_3^- , CO_3^{2-}), that enters aquatic systems. In undisturbed ecosystems, both respiration and weathering convert biologically produced free CO_2 into dissolved alkalinity, acting as a long-term sink for atmospheric CO_2 . However, the macro-scale controls on observed variation in DIC concentration and speciation across watersheds remain poorly understood.

Using standardized measurements from NEON at 25 co-located terrestrial-aquatic sites, this study asks: (a) How and why do DIC concentration and its component species vary across soil-groundwater-surface water pools over different ecoregions? (b) What proportion of stream DIC originates from soil respiration vs. subsurface weathering? Results suggest that DIC concentrations are consistently higher in riparian groundwater compared to upland soils across sites ($\text{DIC}_{\text{groundwater}}:\text{DIC}_{\text{soil}} > 1$). This difference can be attributed to complex roles of deep subsurface weathering and riparian contribution of respired CO_2 . DIC concentration decreases from groundwater to streams ($\text{DIC}_{\text{groundwater}}:\text{DIC}_{\text{stream}} > 1$), indicating evasion of free CO_2 from streams into the atmosphere. We further aim to uncover the watershed attributes (e.g., geology, meteorology, channel morphology) that control variation in DIC concentration and speciation across upland soil - groundwater- surface water pools over different ecoregions of the United States.

id #135849

Effects of temperature on *Cladophora*'s microbial assemblage, photosynthesis, and nitrogen fixation in a nitrogen limited river

Michael Wulf¹, Michael Zampini¹, Augustine Sitati², Saeed Kariunga¹, Steve Thomas², Mary Power³, Jane Marks¹

1. Northern Arizona University, Flagstaff, AZ, United States

2. University of Alabama, Tuscaloosa, AL, United States

3. University of California, Berkeley, CA, United States

Microbial communities associated with benthic macroalgae play a central role in river ecosystem metabolism by mediating carbon and nutrient cycling, regulating nutrient availability, organic matter processing, and the transfer of energy to higher trophic levels. We studied how temperature affects microbial community composition, photosynthesis, and nitrogen fixation associated with the dominant filamentous green macroalga *Cladophora glomerata* in the South Fork Eel River, California. *Cladophora* often grows attached to the scoured bedrock and harbors a diverse epiphytic microbiome consisting of prokaryotic and eukaryotic taxa. In this nitrogen-limited river, the epiphyte assemblage often becomes dominated by *Epithemia* spp., diatoms with nitrogen-fixing endosymbionts that play a crucial role in the N budget of the river. To determine how temperature alters the microbial communities controlling nutrient cycling in *Cladophora* mats, we manipulated temperature in artificial flow-through stream mesocosms, maintaining three temperature treatments (ambient, +4 °C, +8 °C). We used stable isotopes of carbon (^{13}C) and nitrogen (^{15}N) to measure photosynthesis and nitrogen fixation. We analyzed changes in the microbial community using 18S and 16S rRNA amplicon sequencing. Results indicate that increasing temperature significantly altered both eukaryotic and prokaryotic communities, with a shift toward taxa commonly associated with decomposition, suggesting that warming may favor microbial assemblages linked to the breakdown of *Cladophora* biomass. Together, these results indicate that microbial assemblages associated with *Cladophora* respond to increasing temperature, while photosynthesis and nitrogen fixation rates remain unaffected across treatments.

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id #135338

Fire effects on nutrient stores and transport in aridland watersheds

Heili Lowman¹, Ashley Cale², Xiaoli Dong³, Stevan Earl⁴, Nancy Grimm⁴, Shanthini Ode⁵, Alex Webster⁶, Ara Winter⁶, Tamara Harms⁵

1. Duke University, Durham, NC, United States

2. University of Nevada Reno, Reno, NV, United States

3. University of California Davis, Davis, CA, United States

4. Arizona State University, Phoenix, AZ, United States

5. University of California Riverside, Riverside, CA, United States

6. University of New Mexico, Albuquerque, NM, United States

Fires in the western U.S. are increasing in both size and severity and may significantly impact streams and rivers through the addition of burned materials and the alteration of upland flowpaths. Research on the effects of fire on freshwaters has been largely focused in mesic regions, though arid regions are at risk of lagged or more prolonged responses to fire due to highly variable precipitation regimes. Flashy precipitation and discharge regimes have contributed to uncertainty in predicting the effects of fire on watershed biogeochemistry and its implications for water quality in aridlands. We propose that stream chemistry (i.e., concentration) information should be paired with discharge records to discern whether changes in water quality are due to the input of burned material or altered hydrology. To examine the effects of fire on both solute stores and transport, we assembled

long-term observations of stream chemistry and discharge for >45 watersheds across arid ecoregions of the western U.S. We used multi-level Bayesian regression models to quantify changes in concentration-discharge (C-Q) relationships for three years before and after fire. We interpret changing C-Q slopes to reflect the effects of fire on solute transport routes and post-fire change in intercepts to indicate changes in solute stores. We then examined how fire size, catchment characteristics (i.e., elevation, slope), vegetation, hydrology, and adaptation to fire influence post-fire changes in C-Q relationships. Initial results suggest that C-Q slopes for nitrate (NO₃⁻) became more positive following fire (i.e., stronger flushing), though the magnitude of effects varied significantly across ecoregions. We also found that the percent of the watershed burned had a positive effect on post-fire NO₃⁻-C-Q intercepts, suggesting an increase in NO₃⁻ availability. Examining the magnitude and controls of post-fire stream chemistry change is critical so that we may identify watersheds in arid regions that are most vulnerable to large and prolonged changes in water quality.

id #135850

Exploring the interaction between stream warming and benthic biofilms on the fate and transport of antimicrobial resistance genes (ARGs) using experimental mesocosms

Grace C Krane¹, Jennifer L Tank¹, Mitchell J Liddick¹, Emma M Thrift-Cahall¹, Caroline G.T. Anscombe¹, Kyle Bibby²

1. *Environmental Biology, University of Notre Dame, Notre Dame, Indiana, United States*

2. *Civil and Environmental Engineering and Earth Sciences, University of Notre Dame, Notre Dame, Indiana, United States*

The widespread use of antibiotics in animal agriculture has increased the environmental prevalence of antimicrobial resistance genes (ARGs) which poses a growing public health concern. Streams can facilitate the downstream dispersal of ARGs while benthic biofilms can remove ARGs from the water column, influencing persistence and propagation. Despite a growing body of literature on environmental ARGs, the relative influence of water temperature and benthic biofilms on the removal of ARGs from the water column remains understudied. We predict that stream warming will increase biofilm biomass thereby altering the fate of ARGs through water column removal. Using n=24 temperature-controlled recirculating mesocosms at the Notre Dame Experimental Mesocosm Facility (ND-EMF), we tested the interacting effects of water temperature (20° C vs. 26° C), biofilm presence (colonized vs. bare), and the presence of autotrophic vs. heterotrophic constituents using inorganic (rock) and organic (cellulose) substrata on ARG removal rates. We conducted pulse additions of a ARG-rich manure slurry, and collected sequential water samples from each mesocosm at five time points (0.5, 2, 4, 10, and 24hrs) to quantify ARG removal rates (as *k*; hr⁻¹). For each filtered water sample, using digital PCR, we measured concentrations of ARGs for tetracycline (*tetW*) and erythromycin resistance (*ermB*), as well as a ruminant fecal indicator (*bacR*), and a bacterial marker gene (16s rRNA). We hypothesized that warmer temperatures would increase removal rates through augmented biofilm biomass, which would promote the physical removal of ARGs. Additionally, we hypothesized that the algal growth on the organic substrata would not degrade ARG-containing particles, while heterotrophic biofilm communities colonized on the inorganic substrata would. Previous research has demonstrated that warmer water temperatures can increase the water column removal of other molecular constituents (e.g., environmental DNA), but warming has not been examined specifically for ARGs. While the molecular detection results of this experiment are forthcoming, determining how water temperature influences ARG transport in streams will help mitigate the spread of antibiotic resistance in aquatic ecosystems.

id #135595

Mining effects on stream physical properties from minutes to millennia

Charles M Shobe¹

1. *U.S. Forest Service Rocky Mountain Research Station, Fort Collins, COLORADO, United States*

Mining can affect the physical structure of streambed habitat by changing the amount and size of sediment supplied to streams. Resulting changes in stream channel shape and bed grain size influence the viability and composition of aquatic ecosystems. Here I review mining effects on stream sediment dynamics as a function of the timescale over which they appear, with the goal of identifying knowledge gaps that present opportunities to quantify and forecast mining effects on aquatic ecosystems.

Short-term (minutes to years) effects are observed where mining activities directly interact with the riparian corridor. Examples include hydraulic gold placer mining in the Sierra Nevada, burial of headwater streams by mountaintop removal/valley fill coal mining in Appalachia, and modern river sand mining for construction materials. These effects commonly include elevated turbidity, changes in river-bed grain size, and rapid changes in channel form. Medium-term effects occur wherever near-channel mining landforms can source sediment to the river channel over years to centuries and include many small-scale, off-channel forms of mining, such as Appalachian bench-and-highwall coal mining and early gold mining in the Colorado Rockies. This intermediate timescale also encompasses corollary effects of mining, such as deforestation, that themselves influence stream processes by triggering pulses of water and/or coarse sediment. Long-term (centuries, millennia, and longer) effects occur primarily where mining has caused extensive and dramatic changes to the topography such that natural erosion trajectories are supplanted by novel landscape evolution pathways. Topographic effects of modern, large-scale surface mining fall into this category because altered topography causes streams in mined watersheds to experience perturbations to their water and sediment regimes over timescales longer than most human planning efforts.

Important knowledge gaps arise from the immense variability in 1) how mining affects the land surface as a function of the target deposit, physiographic setting, ecosystem, and regulatory environment, and 2) how land-surface processes filter the expression of mining-driven alterations in streambed habitat structure. Progress along these lines may enhance our ability to effectively balance society's needs for minerals, energy, and functioning aquatic ecosystems.

id #135851

Scaling up: How fish ontogeny and evolution shape ecosystem nutrient flux

Amina Mohamed¹, Gracie Brehm¹, Brittany Rakhmonova¹, Swanne Gordon¹, Andres Lopez-Sepulcre¹

1. Cornell University, NY, United States

Nutrient cycling is key to ecosystem functioning and maintaining balance across trophic levels. Consumers are central to this process, selectively taking up, storing, and releasing nutrients in ways that shape primary productivity and biogeochemical processes. Freshwater fish, for example, continuously exchange water through their gills, releasing limiting nutrients like phosphorus that primary producers can uptake. Despite the importance of consumer-driven nutrient recycling (CNR), we lack a mechanistic understanding of how ontogeny and demographic structure influence nutrient fluxes, especially in species undergoing rapid evolutionary change. Consumer traits like nutrient consumption, assimilation, allocation, excretion, and egestion change significantly as they grow, so linking these shifts to demographic structure is critical for predicting population-level contributions to nutrient recycling. This project investigates how nutrient storage and recycling shift across ontogenetic stages in *Poecilia reticulata*, a globally invasive fish species with well-documented variation in growth rates. We reared guppies adapted to high- and low-predation environments on different diets and across ontogenetic stages to quantify how dietary phosphorus is assimilated into the body and allocated to bone for storage and to RNA for growth. We also measured excretion rates to determine how much phosphorus is returned to the environment. Excretion analysis is underway, and the body nutrient, bone, and RNA analyses will begin in February 2026. Studying invasive species like guppies is especially important in a changing world, as their rapid life histories and adaptability allow them to alter nutrient dynamics and adversely affect native biodiversity and ecosystem resilience.

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id #134572

Fish community structure, diet composition, and gut microbiome of the two dominant fish species in an African savanna river

Joshua Benjamin¹, Christopher Dutton¹, Tavis Goldwire¹, Liam O'Connor², Ava Gabrys¹, Therese Frauendorf³, Joseph Gathua⁴, Edward Njagi⁴, Ray Schmidt⁵, David Post⁶, Amanda Subalusk¹

1. University of Florida (UF), FL, United States

2. School of Geography and the Environment, University of Oxford, Oxford, Oxfordshire, England

3. Biology, Colgate University, Hamilton, New York, USA

4. Entomology, National Museum of Kenya, Nairobi, Kenya

5. Biology, Randolph-Macon College, Ashland, Virginia, USA

6. Department of Ecology and Evolutionary Biology, Yale University, New Haven, Connecticut, United States

Studies on spatial and interannual variation in fish community structure, diet composition, and gut microbiome remain limited in Afrotropical river systems. As a result, little is known about the diet composition and gut microbiomes of *Labeobarbus altianalis* and *Labeo victorinus*, two dominant and ecologically important species in the Mara River. This study addresses this gap by examining the relationships among fish community structure, diet composition, and gut microbiome of the two dominant fish species along a longitudinal gradient in the Mara River Basin in East Africa.

We compared historical (2013–2016) and recent (2021–2022) datasets and found no significant differences in species diversity and fish biomass between the two sampling periods. Stable isotope values ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) revealed clear spatial and species-specific patterns across sites. Where both species co-occurred (M4, M5, M6, M7, and M9), *Labeobarbus altianalis* consistently occupied a slightly higher trophic position than *Labeo victorinus*, as indicated by enriched $\delta^{15}\text{N}$ values. Gut content analyses showed that both species were primarily supported by amorphous detritus and non-algal biofilm. However, the relative contribution of these resources varied markedly among sites, particularly at M4, M7, and M9. PERMANOVA indicated no significant main effect of species on overall gut composition ($p = 0.468$), suggesting broadly similar diets when averaged across the river system. A significant species-site interaction ($p = 0.015$) further showed that the two species did not respond uniformly to these spatial differences; instead, each species altered its diet in distinct ways across sites. The two fish species were dominated by Clostridia, Gammaproteobacteria, Bacteroidia, Bacilli, Fusobacteria, and Planctomycetes. Overall, *Labeo victorinus* exhibited higher microbiome diversity. Consequently, microbiome diversity declined along the longitudinal gradient across the three sites (M4, M7, M9).

Collectively, these findings demonstrate that while fish community structure in the Mara River has remained relatively stable over time, underlying trophic pathways, feeding strategies, and host-associated microbiomes vary strongly across space. This study highlights the importance of integrating multiple ecological compartments to better understand how tropical river ecosystems function and respond to environmental gradients, and provides a valuable baseline for future assessments of ecological change in Afrotropical rivers.

id #134828

Essential fatty acid and methylmercury fluxes from emerging aquatic insects: implications for wetland insectivores

Alan J Mock¹, Joel C Trexler², Nathan J Dorn¹

1. *Florida International University, FL, United States*

2. *Coastal and Marine Laboratory, Florida State University, St. Teresa, FL, United States*

Emerging aquatic insects can transport beneficial nutrients and harmful contaminants accumulated during aquatic life stages to terrestrial food webs. Contrasting the beneficial "bright" and harmful "dark" sides of aquatic insect fluxes between habitats of different productivity can improve understanding of how aquatic ecosystem stressors such as eutrophication may affect their net benefits and costs to recipient consumers. We quantified the seasonal (wet and dry) fluxes of the essential fatty acid eicosapentaenoic acid (EPA; 20:5n-3) and the neurotoxin methylmercury (MeHg) from emerging aquatic insects in an oligotrophic and a eutrophic freshwater marsh of the subtropical Florida Everglades. To assess potential consequences for terrestrial insectivores, we also quantified EPA and MeHg availability in above-water invertebrates inhabiting emergent vegetation. EPA flux was 241X greater in the eutrophic habitat across both seasons but varied seasonally within each habitat. In the eutrophic habitat, the wet season EPA flux was 2.3X greater than in the dry season and in the oligotrophic habitat, the dry season EPA flux was 13.7X greater than in the wet season. The MeHg flux did not differ between habitats but was 4.7X greater in the dry season. Patterns of EPA and MeHg availability in above-water invertebrates mirrored emerging insect fluxes, indicating that aquatic export directly translated into availability for terrestrial insectivores. Spiders (Araneidae, Tetragnathidae, Philodromidae, Pisauridae, and Salticidae) and adult damselflies (Zygoptera) dominated the MeHg and EPA availability in both habitats. These findings suggest that wetland trophic status is linked to the quantity and quality of emerging aquatic insect subsidies, potentially enhancing EPA availability for terrestrial insectivores without increasing MeHg exposure. This study also illustrates that variation in aquatic ecosystem productivity can decouple EPA and MeHg dynamics in linked aquatic-terrestrial food webs.

id #135340

Multi-marker eDNA reveals contrasting biodiversity responses to watershed land use

Kara Andres¹, Jenna Benson¹, Gwen Church¹, Katein Meek¹

1. *Illinois State University, Normal, IL, United States*

Environmental DNA (eDNA) metabarcoding offers an efficient approach for assessing biodiversity across space and time. A key benefit of eDNA approaches is the ability to target multiple taxonomic groups using different genetic markers, providing a more comprehensive view of biodiversity across trophic levels within ecosystems. Here, we applied a multi-marker eDNA framework to characterize biodiversity patterns across a gradient of land use in an agriculturally dominated watershed. To do so, we analyzed three loci targeting different taxonomic groups: 18S rRNA (broad eukaryotic diversity), COI (macroinvertebrates), and 12S rRNA (fish). We compared within- (α -diversity) and among-site (β -diversity) biodiversity patterns across sites spanning a gradient of land use intensity, as indicated by the percent agriculture and urban cover within the upstream catchment of each sampling site. We found divergent responses among target species groups, with sites with higher land use intensity exhibiting higher overall eukaryote diversity, while fish and macroinvertebrate diversity was lower at more impacted sites. These contrasting results suggest that land use impacts may reduce diversity in traditional bioindicator species groups while increasing overall eukaryotic diversity, potentially due to shifts toward disturbance-tolerant microeukaryotic taxa. Despite these differences in α -diversity, all species groups showed high β -diversity across sites, indicating strong turnover in community composition across the land use gradient. By combining fish and macroinvertebrate metabarcoding with broad eukaryotic diversity metrics, this approach provides a more complete picture of how watershed land use impacts freshwater biodiversity across multiple trophic levels.

id #135596

Bioenergetic plasticity of larval Chironomidae across an Icelandic stream thermal gradient

Sam J Larkin¹, Jonathan P Benstead², Alexander D Huryn², Ariel Shogren², Wyatt F Cross¹

1. *Montana State University, Bozeman, MT, United States*

2. *University of Alabama, Tuscaloosa, AL, United States*

Anthropogenic warming of high-latitude stream ecosystems is influencing the energy budgets of aquatic organisms, with important implications for the movement and storage of energy and materials in food webs. Although acute exposure to higher temperatures often leads to reduced growth efficiencies (i.e., the proportion of ingested or assimilated resources that contribute to biomass production), bioenergetic plasticity in response to long-term warming may allow partial or complete metabolic compensation, thus allowing some populations to persist at higher temperatures. We conducted a reciprocal transplant experiment of *Eukiefferiella* minor (Diptera: Chironomidae) across a natural stream geothermal gradient (5°C - 22°C) in southwestern Iceland to test the hypothesis that organisms acclimated or adapted to chronic warming will exhibit higher energetic efficiencies than organisms chronically exposed to lower temperatures, via an elevated growth to respiration ratio. Our preliminary results show that individuals from warmer streams had higher mass-specific growth rates and down-regulated respiration rates compared to individuals from colder streams across the range of temperatures, which indicates partial compensation of *E. minor* in response to chronic warming. These results, together with complementary efforts at our study site, imply that bioenergetic plasticity is a potentially significant factor in determining how aquatic organisms respond to warming, and can influence trophic transfers of energy in aquatic food webs.

id #135852

Between a rock and a cold place: Effects of rapid environmental warming on alpine stream communities

Katrina Heide¹

1. *Colorado Mesa University, Grand Junction, COLORADO, United States*

Individual size distributions (ISD; also known as size spectra), are an empirical relationship describing the decline in relative abundance from small to large individuals. ISD relationships are incredibly consistent across ecosystems and habitats and have been proposed as a simple, universal metric for assessing ecological status. Increased environmental temperatures are expected to favor small individuals leading to steeper ISDs supporting fewer large individuals. The Teton Alpine Stream Research (TASR) project has been monitoring alpine streams since 2015. Streams are grouped based on water sources: glacier, rock glacier, and snowfield. Monitoring includes hourly stream temperature measurements and annual collection of quantitative benthic macroinvertebrate samples. Temperature analyses have indicated summer temperatures in glacier and rock glacier streams are consistent across years, but snowfield streams are warming rapidly. This system presents a unique opportunity to test if ISD relationships change or are consistent in response to increased environmental temperatures.

ISD relationships were estimated from quantitative macroinvertebrate samples from two time points (first and last sample) from the three stream categories, and will be analyzed with a two-way ANOVA to test for effects of stream group, time point, and their interactive effects.

We hypothesize that the ISD relationships will remain consistent through time in both the glacier and rock glacier streams. However, we predict steeper ISD relationships in the snowfield stream due to a higher relative abundance of small individuals.

The long-term and repeatable data collection offers a unique opportunity to assess ISD pattern changes in response to rapidly rising temperatures. The information from this research is widely applicable across mountain and alpine environments which are experiencing rapid climate warming.

id #135342

The forgotten utility: Managing stormwater in the built environment

Ariana Dionisio¹

1. *Public Works, City of Mukilteo, Mukilteo, WA, Snohomish*

Adjacent to drinking water and wastewater, stormwater is often forgotten utility. However, in the region of western Washington where periods of heavy and consistent rain are common occurrences the stormwater utility increases in importance. This talk will go over how the stormwater team at the City of Mukilteo manages a natural system (e.g. rain) within an unnatural environment (e.g. urban) and how the team strives to make data informed management decisions. Mukilteo, WA is a growing city that is primarily urban landscape, contains 13 first order streams, steep slopes, and abuts the Puget Sound. These characteristics develop a unique challenge for stormwater management. Primary goals of the stormwater team are to 1) Ensure proper drainage to prevent flooding 2) Mitigate pollutants that enter waters of the state and 3) Educate the public on best management practices. This talk will go over how the stormwater team fulfills these goals and the next steps the program will take to better serve a growing city.

id #136110

Rapid phosphorus uptake and storage by aquatic hyphomycete fungi

Lydia P McGregor Bravo¹, Jonathan P Benstead¹, Kevin A Kuehn², Steve A Thomas¹

1. *The University of Alabama, Tuscaloosa, AL, United States*

2. *The University of Southern Mississippi, Hattiesburg, MS, United States*

Classic studies in the 1980s using ³²P demonstrated the importance of leaf litter and associated microorganisms for dissolved phosphorus (P) uptake. However, the functional response of aquatic hyphomycete fungi – dominant microbial litter decomposers in forest streams – to elevated dissolved P concentrations remains poorly understood. Whole-stream nutrient-addition experiments at Coweeta Hydrologic Laboratory (North Carolina, USA) have showed that, as the percent P content of litter increases, fungal biomass increases at a slower rate, indicating 'luxury' storage of P. In this study, we investigated whether litter-associated fungi store excess P by conducting P-enrichment experiments that tested short-term effects of elevated dissolved P concentration on fungal biomass and litter P content. After incubating two leaf species, fast-decomposing tulip poplar (*Liriodendron tulipifera*) and slow-decomposing white oak (*Quercus alba*), in a second-order stream in the Coweeta basin, we returned at early- (day 50), mid- (day 100), and late-decay (day 150) stages and added 200 µg/L of P over one-week periods, collecting litter after 0, 4, 8, 24, 72, and 168 h of P enrichment. Litter samples were analyzed for C, N, and P content and fungal biomass (measured as ergosterol). Fungal biomass did not increase proportionally with rising fungal P content, suggesting that fungi stored excess P. To investigate this mechanism further, nuclear magnetic resonance (NMR) was used to distinguish among P-containing compounds in the litter, and estimate the mass of polyphosphate, the primary storage form of P in fungi. Polyphosphate mass was significantly higher in enriched compared to non-enriched reaches across all white oak decay stages and the early decay stage of tulip poplar. This supported the hypothesis that aquatic hyphomycetes store excess P as polyphosphate under elevated dissolved P conditions.

id #134320

Ecological risk assessment of flow alteration: A regional, multi-species framework for flow management

Kris Taniguchi-Quan¹, Lane Belize², Jacob Morgan³, Anzy Lee², Katie Irving⁴, Eric Stein⁴, Lara Jansen⁴, Ted Grantham⁵, Gabe Rossi⁵

1. *Southern California Coastal Water Research Project, Costa Mesa, CA, USA*

2. *Utah State University, Logan, UT, USA*

3. *Ulteig, Nashville, TN, USA*

4. *Southern California Coastal Water Research Project, CA, United States*

5. *University of California, Berkeley, Berkeley, CA*

Streamflow alteration can degrade aquatic ecosystem health. However, determining acceptable levels of alteration is challenging when evaluating ecological risks across multiple species, life stages, and a diversity of stream types within a region. Natural resource managers need tools that can be applied at the watershed or regional scale to evaluate potential ecological consequences of flow alteration and to identify protective ranges of alteration. Risk-based approaches provide an advantage over traditional approaches of establishing environmental flow targets by providing the flexibility to adjust management objectives based on risk tolerance, the needs of different species, and trade-offs with other water demands. In this study, we developed an ecological risk assessment framework for identifying protective ranges of flow alteration, leveraging existing studies and datasets to account for varying risk profiles of specific species and life-stages of interest. We evaluated changes in regional ecological risks associated with incremental deviations from unimpaired flow for different species life-stage needs and under different annual climate conditions and physical channel settings. The framework is designed to protect the habitat needs of salmonids, such as Chinook salmon and steelhead trout, alongside other ecological endpoints including amphibians, benthic macroinvertebrates, and algal communities. This multi-species approach enables assessment of how flow alteration may affect both species of primary management concern, and the broader ecological functions supported by the natural hydrograph. We applied this framework in a coastal watershed in northern California, USA, synthesizing ecological risk curves to reveal inflection points that suggest levels at which diversion rate limits could be established. These insights can help inform regional flow management policies that balance human water use with aquatic ecosystem protection.

id #134576

Fish community response after dam removal: eDNA metabarcoding reveals diversity recovery and short-and-long term impacts

Stephanie Burnette¹, James Skelton²

1. *College of William & Mary, Williamsburg, VA, United States*

2. *Plant Pathology, University of Florida, Gainesville, Florida, United States*

Dams threaten freshwater fish populations by blocking migratory species from accessing habitat that is essential to their life cycle, by altering the physical landscape, by promoting invasive species, and by homogenization of fish communities. There are thousands of aging and deteriorating dams across the United States, and dam removal is gaining popularity as a mitigation strategy to reconnect aquatic habitats. However, restoration after dam removal is not instantaneous or guaranteed – decades of sediment may be released throughout the catchment area, water flow may dramatically increase, and invasive species may be able to expand their ranges. All of these changes impact the fish that live above and below a dam removal. This study uses a replicated design to evaluate how fish communities respond after removal. Using environmental DNA metabarcoding, we surveyed eight sites where dams have been removed. We selected dam removals spanning two decades (2004-2024) to create a time series. For every removal, two reference sites were selected: a dammed site with an intact dam, and an undammed site where no human-made dam has ever existed. Every site was split into two sub-sites above and below a fixed point: the removed dam at the removal sites, the existing dam at the dammed sites, or a location chosen based on accessibility at the undammed sites. We compared patterns of alpha and beta diversity, as well as re-colonization time for native migratory, resident species, and invasive species. The stratified design allowed for comparison at both the site and regional levels. The results I will discuss can be used to inform dam mitigation strategies, biodiversity monitoring, fisheries management, and future restoration projects.

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id #134321

Shell Shocked: How Toxins and Salt Stir Up Oyster Microbiomes

Champagne Cunningham¹, Rachel E Diner¹

1. *Department of Biological Sciences, University of Memphis, Memphis, Tennessee, United States*

Microcystin is commonly measured in freshwater environments. However, during high rainfall events *Microcystis* and the corresponding toxin, microcystin, can be introduced through drainage to brackish or marine environments. The high rainfall events also introduce freshwater into marine environments, decreasing salinity in these systems. Oysters in the systems are then exposed simultaneously to microcystin and low salinity. Potential increases in rainfall place the health of wild and farm raised oysters in danger. Oysters are sessile, benthic filter feeders. Their feeding methods directly expose them to microcystin in the water column. Microcystin accumulates in the tissues of oysters, which could also impact their microbiomes. Oysters are highly sensitive to changes in salinity; however, the effects of low salinity in combination with microcystin on oyster microbiomes and oyster health have not been examined. This study will measure potential changes in the microbiomes of oysters after exposure to microcystin and low salinity. *Crassostrea virginica* will be experimentally tested for microbiomes changes after microcystin exposure (3ug and 8ug) in low (10 ppt) and regular salinity (20 ppt) measurements. Oysters will be exposed to high or low salinity and co-exposed to low or high

microcystin concentrations. Knowledge of the response of eastern oyster microbiomes to low salinity and microcystin will further our understanding of the physiology and microbiome of eastern oysters.

id #135345

Exploring relationships between river flows and multiple trophic levels in the San Francisco Estuary (California)

Angelika L. Kurthen¹, Sam Bashevkin², Laura Twardochleb², Albert Ruhi Vidal¹, Angelika L. Kurthen¹

1. *Environmental Science, Policy, and Management, University of California Berkeley, Berkeley, California, US*

2. *California State Water Resources Control Board, Sacramento, CA, USA*

River flows drive seasonal and interannual changes in spatial environmental gradients in estuarine ecosystems, creating a mosaic of conditions that largely controls food-web structure and responses. In the San Francisco Bay-Delta, freshwater inputs and Delta outflows are known to influence zooplankton, benthic invertebrates, and fish populations. However, there are still uncertainties in the specific mechanisms connecting interannual variation in freshwater flows to fish population abundances. At least two reasons may have contributed to this challenge. First, many flow-ecology models in the past have not explicitly considered biotic interactions, potentially confounding responses of focal taxa to flow with flow-mediated responses of competitors and predators. Second, in a non-stationary environment, some flow-ecology relationships could be more accurately modeled with time-variant methods. We assess the value of information provided by increasing biotic complexity (i.e., inclusion of multi-trophic levels of biotic interactions, from phytoplankton to fish) and abiotic complexity (i.e., allowing for a time-varying strength of flow-ecology relationships) to flow-ecology models for fish in the San Francisco Bay-Delta. To this end, we developed single-taxon and multi-taxon models for zooplankton and fish assemblages. For each level of biotic complexity, we fitted a multivariate autoregressive state-space or MARSS model, which assumes a static relationship between flow and the taxa, and a dynamic linear model, which allows for a time-varying relationship between flow and taxa. All models were fed the same data, combining Delta outflows and spatially-replicated ecological time series data collected by the California Dept. of Fish and Wildlife and the California Department of Water Resources (Bay Study, Fall Midwater Trawl, 20mm Survey, and Environmental Monitoring Program). We compared these four model structures representing different levels of ecological complexity to evaluate the value of including biotic interactions and time-varying relationships in models of flow-ecology relationships. Including biotic interactions and time-varying abiotic processes in models could help academics and practitioners seeking to understand and forecast estuarine dynamics.

id #136113

Meta-food webs in dendritic systems: decreased resistance and increased resilience to disturbance

Sarah F Sorensen^{2, 1}, Dan Magoulick^{2, 1, 3}

1. *Arkansas Cooperative Fish and Wildlife Research Unit, Fayetteville, Arkansas, USA*

2. *University of Arkansas, Fayetteville, ARKANSAS, United States*

3. *US Geological Survey, Fayetteville*

Ecological communities are being impacted by climate change at increasing rates, including an increase in the intensity of disturbances, impacting communities through the destruction and isolation of biota. Meta-communities allow dispersal between communities and habitat patches and are associated with increased population resilience. Recently, meta-community research has expanded to incorporate a food web component. Meta-food webs allow an exploration of how spatial connectivity and dispersal contribute to food web stability. Examining how network topology interacts with food webs is an important step, especially in streams which have dendritic networks that are known to directly influence community structure. We modeled food webs in single patches and in multi-patch linear, lattice, and dendritic networks with and without disturbance. Disturbance models were run for ramp and pulse disturbances at mild and severe intensities for each network topology. Network robustness and mean patch robustness were calculated at each timestep. Multi-patch topologies had a greater network robustness and lower patch robustness than single patches. Disturbance caused a greater decrease in patch robustness and network robustness in the dendritic topology than linear or lattice topologies. However, after the disturbance was over, patch robustness recovered most in dendritic topologies. Network topology and connectivity play a critical role in the resistance and resilience of communities during and after disturbance. Our results highlight the potential capacity for streams to recover from disturbance, which will become more important in freshwater systems as hydrologic disturbance increases. These results have substantial implications for metacommunity, meta-food web, and stream research, as dendritic topologies are not equivalent to the more commonly studied lattice and linear topologies and are likely to have different relationships with dispersal, disturbance, and community structure.

id #134834

Characterizing the distribution of macrophytes on the Kootenai River using unmanned aerial vehicles

Patrick K Kozak^{2, 1}, Nathan Jensen³, Brandon Diller³, Michael Beason⁴, Christopher Brooks², Lisa A Kunza^{2, 1}

1. *Center for Sustainable Solutions, South Dakota School of Mines, Rapid City, SD, USA*

2. *Department of Chemistry, Biology, and Health Sciences, South Dakota School of Mines, Rapid City, South Dakota, USA*

3. *Fish and Wildlife Department, Kootenai Tribe of Idaho, Bonners Ferry, Idaho, USA*

4. *South Dakota Wing, Civil Air Patrol, Rapid City, South Dakota, USA*

Emergent and submersed aquatic vegetation provide hotspots for food resources and their distribution, which are essential elements of large, oligotrophic riverine systems. A complete understanding of riverine system dynamics must include submersed aquatic vegetation (SAV) diversity and distribution to assess faunal habitat, water flow, sediment stabilization, water quality, and SAV contribution to biogeochemical cycling for food resources. However, spatial data are typically compiled from riverine transects and/or point-intercept surveys, which can be time-consuming, logistically challenging, and costly, particularly for large river systems. The Kootenai River, which originates in British Columbia, Canada, flows through Montana and Idaho before returning to British Columbia and entering Kootenay Lake. The Kootenai River provides an excellent opportunity to apply a remotely sensed macrophyte characterization approach to a riverine system where current research is ongoing. We plan to fly the 75-kilometer stretch of the Kootenai River in Idaho, from just west of Bonners Ferry to the Canadian border, which includes the main deep, slow-moving meander section, islands, and engineered structures. Our analysis will be conducted using a Wingtra Gen II fixed-wing UAV with the Micasense RedEdge-P multispectral imager at 2–4 m spatial resolution in the visible and near-infrared bands. The planned data acquisition will be conducted during a low-flow period, thereby maximizing the penetration depth of the imaging sensor into the water. Initial analysis of the USGS National Agricultural Imagery Program (NAIP) 2023 VNIR imagery demonstrated that a remote-sensing approach can detect SAV along the Kootenai River at 1 m spatial resolution. SAV was found in areas where water movement was reduced, or the depth became shallower, such as near islands, point bars, backwaters, and man-made structures. The initial NAIP classification using 1-meter imagery will enable us to optimize the Wingtra Drone flight plan, reducing collection, processing, and analysis time and costs. Food resource availability in large oligotrophic rivers is crucial for supporting higher trophic levels, as these environments are inherently nutrient-poor and biomass-limited. Mapping the distribution of macrophyte assemblages at this scale provides cost-effective and timely information to support river management decisions.

id #136114

The Impact of Non-perennial Flow Regimes and Urban Land Use on Headwater Stream Microbial Community Composition and Extracellular Enzyme Activities.

David Roberts¹, Michaela Halsey¹, Roberto Maldonado Flores¹, Lydia Zeglin¹

1. KSU, Manhattan, KS, United States

Stream ecosystems are increasingly under pressure from urban land development and altered hydrological regimes. Microorganisms play vital roles in stream nutrient cycling, transforming and taking up dissolved carbon and nitrogen while also generating biomass for the basal food web. With the expansion of urban centers and the increased drying of perennial streams, understanding how these factors affect microbial communities will help us understand integrated stream ecosystem health and metabolism. This study investigates how land use and flow permanence (perennial vs. non-perennial) shape microbial community composition and function, measured using prokaryotic and protistan gene sequencing and decomposition extracellular enzyme activity (EEA) potentials, respectively. We predicted that community composition would differ between flow regimes and that urban land use would exacerbate that dissimilarity, and that urban land use would suppress EEA due to potential increases in nutrient concentrations. We collected benthic sediment, epilithic biofilm, leaf litter, and water column samples from headwater tributaries (n=24) to East Fork Poplar Creek in Oak Ridge, TN, at locations including perennial and non-perennial headwaters along an urban-to-forested gradient. Bacterial community composition was determined by 16S rRNA gene metabarcoding and high-throughput sequencing, and EEAs were measured using fluorescence assays.

Preliminary analysis reveals that both sediment ($p = 0.002$) and epilithic biofilm ($p = 0.04$) community compositions are influenced by the percentage of urban land development. Further, sediment communities were dissimilar between flow regimes ($p = 0.01$), whereas epilithic communities were not. Functional patterns were weaker, as EEAs in epilithon did not vary across the study gradients, and benthic sediment nitrogen-acquiring enzymatic activity negatively correlated with urban development ($p = 0.08$, $\rho = -0.24$) but did not vary with flow permanence. Additional results from the water column and leaf litter communities are forthcoming, as is protistan community composition, and comparisons between summer, fall, and winter seasons. Our initial findings suggest that the microbial community response differs across microhabitats, with the sediment community more sensitive to land-use and flow-regime factors. Understanding how microorganisms across stream microhabitats are affected by anthropogenic changes will give us a more comprehensive picture of the health and metabolism of our stream ecosystems.

id #133811

Salinization of lakes in west Michigan: how problematic is it?

Alan Steinman¹, Sean Woznicki¹, Michael Hassett¹, Katie Tyrrell¹

1. Annis Water Resources Institute-GVSU, MI, United States

Salinization of freshwater ecosystems is a growing concern, especially in north temperate regions due to runoff from de-icing salts. Studies have identified a positive association between lakes with elevated chloride (Cl^-) concentrations and proximity to major highways but it is less clear if elevated Cl^- is found in more rural regions. This knowledge gap inspired a lake survey in 5 west Michigan counties to determine the extent of chloride contamination. Our results indicated an absence of elevated Cl^- with the exception of previously studied urbanized lakes. Land use affected both Cl^- and conductivity, with forest land cover negatively correlated and developed land positively correlated to both parameters. In addition, there was a moderately negative correlation between Cl^- and distance to major roads and a moderately positive correlation between chloride and estimated salt application. The epilimnion and hypolimnion chloride measurements were compared with chloride predictions from previously published models; Cl^- levels fell within model prediction intervals, although Cl^- levels compared to a second model suggested our sampled lakes had not reached a Cl^- equilibrium. We developed a Michigan-specific epilimnion and hypolimnion chloride model using our samples and data from the U.S. Water Quality Portal, which predicted greater Cl^- concentrations in lakes with urbanized watersheds. Although the lakes surveyed in this study currently are not being impacted by road salt runoff, the correlation data suggest they may be at risk in the future; increasing air temperatures may offset this risk if less road salt is needed to protect the roads.

Wildfire impacts on stream organic matter chemistry depend on hillslope-stream connectivity

Allison Myers-Pigg¹, Peter Regier¹, Jake Cavaiani¹, Andrew Burgess¹, Xingyuan Chen², Brieanne Forbes², Vanessa Garayburu-Caruso², Xia Gillespie², Amy Goldman², Sophia McKeever², Sundar Niroula², Lupita Renteria², Yi Xiao², Faria Zahura², James Stegen²

1. *Pacific Northwest National Laboratory, Sequim, WASHINGTON, United States*

2. *Pacific Northwest National Laboratory, Richland, Washington, United States*

Wildfires impact watersheds in varied ways, with complex relationships between burn and watershed characteristics influencing downstream aquatic biogeochemistry. In particular, wildfire alterations to the terrestrial landscape influence water delivery to stream networks which can have hard-to-predict impacts on organic matter (OM) chemistry and related biogeochemical processes in streams. Here we present an ongoing case study exploring wildfire impacts on hydrologic connectivity across a small watershed that was entirely burned, and the resulting impacts on aquatic biogeochemistry. In November 2024, four months after fire ignition, we established five sites across the Oak Creek watershed in the Yakima River Basin in Washington, USA. Each site is equipped with YSI EXO2 sondes to measure temporally resolved proxies for groundwater inputs (specific conductivity) and fluorescent dissolved OM (fDOM) along with basic water quality (including dissolved oxygen) and depth. We also collect end-member samples from precipitation, the hyporheic zone, soil leachates, and deep groundwater across the watershed to assess end-member contributions to the streams. Previous work has suggested that groundwater contributions may mediate the relationship between OM chemistry and burn severity in streams draining burned watersheds. Therefore, we explored the relationship between stream OM chemistry and subsurface contributions to streams across the watershed. We found spatiotemporal differences in the contributions of groundwater across our sites, based on specific conductance and hydrologic tracer simulations. This corresponded to shifts in OM signatures over the study period and predicted hillslope contributions of OM to the river corridor. We find that spatial and temporal variability in wildfire impacts on OM quality in streams post-fire is dependent on sub-watershed variations in hydrologic connectivity.

Being a successful role model (organism): Insights into diversity and distribution of apple snails (*Pomacea*) gained through field and molecular ecology

Romi L Burks¹

1. *Southwestern University, Georgetown, TX*

Ecologically and economically devastating non-native invasive species, including, but not limited to, rusty crayfish, zebra mussels, and Asian carp, feature prominently in the career of retiring ecologist Dr. David Lodge. Studying species traits and invasive pathways occupied theses and dissertations of dozens of Lodge students. Moreover, Dr. Lodge's focus on placing organismal biology within a larger community context fostered critical consideration of what it takes to establish a successful model organism for study. Collective insight gained about what makes a successful invasive species shaped policy and regulatory decisions, starting in the Great Lakes and extending nationally. Taking into account some lessons learned from Lodge's collective works, this talk reviews past, present and future aspects of managing invasive apple snails (i.e., genus *Pomacea*). These species pose growing economic and ecological threats to the southeastern US and represent an emergent model of successful invasive species. Historical lessons include considering pathways of spread (e.g., aquarium and bait industries), understanding the central position of omnivorous species in food webs, analyzing reproductive potential and quantifying size-selective pressures on competitive outcomes. Apple snails initially established in Texas and Florida from aquarium dumps, voraciously consume plants, but also recycle organic matter, produce large clutches over a long reproductive season, and exhibit superior competitive abilities due to their large size. More recent lessons stress the importance of confirming species identification and untangling taxonomic uncertainty to help streamline regulations and management efforts. Multiple non-native apple snails species exist and show extensive phenotypic plasticity, but they can exhibit different environmental tolerances making range expansions challenging to predict. The future of combating apple snails lies in combining field work with molecular approaches, including environmental DNA, to confirm identity, detect establishment as early as possible, and verify success of local eradications. Depositing their clutches above the water's surface, apple snails possess both a lung and a gill and lay and also serve as a model for aquatic-terrestrial linkages. Overall, this talk summarizes how strong role models and successful model invasive species share certain key traits: resilience, adaptability, and the ability to make a broader impact beyond anyone's expectations.

Effects of repeated alum treatments on phytoplankton and zooplankton community structures in a managed urban lake.

Trajan EL Bitner¹, Yangdong Pan¹, Mark Rosenkranz²

1. *Portland State University, Portland, OR, United States*

2. *Aquatic Insight, Portland, OR, US*

Urban lakes provide a myriad of benefits to users and residents but face high nutrient enrichment from anthropogenic sources leading to eutrophication. Eutrophication is of concern due to the resulting harmful algal blooms (HABs), which are potentially toxic to humans and wildlife and decrease the aesthetic value of the lake in addition to other deleterious effects. Aluminum sulfate (alum) is a common chemical management tool for both nutrient reduction and HAB treatment. Although there are decades of research on alum's effectiveness and ecological impact, few studies have considered how repeated alum applications affect planktonic communities in lakes over time. To measure these effects, I examined the changes in zooplankton and phytoplankton communities over twelve years in a eutrophic and HAB impaired urban lake in western Oregon that used alum as its primary

methods of HAB management. Phytoplankton and zooplankton samples were collected at least monthly from May through October over three management periods; (I) 2002-2004 before alum treatments began, (II) 2005-2010 during annual alum surface applications, and (III) 2011-2014 after alum surface applications ended. I used PerMANOVA tests to evaluate changes in phytoplankton and zooplankton communities across the three management periods and Pairwise Wilcoxon tests to identify changes in major phytoplankton and zooplankton groups across the periods. The phytoplankton communities were observed to be significantly different between periods I and II ($p=0.004$) and I and III ($p=0.001$) while no statistically significant difference was observed in the zooplankton communities. The proportion of diatoms ($p=0.041$) and cyanobacteria ($p=0.009$) significantly increased and decreased, respectively, between periods I and III. The proportion of rotifers ($p<0.05$) decreased significantly after each period, with more copepods ($p=0.003$) during period II and more cladocerans ($p=0.009$) during period III. This analysis shows that planktonic communities respond differently to ongoing alum applications than to the resulting change of environmental conditions within the lake following a single application. Further analysis into the relationship between factors associated with individual applications and changes in the planktonic populations will help inform managers planning alum applications to be more ecologically sound.

id #135605

Investigating optimal detection strategies for and ecological impacts of invasive European Frog-bit (*Hydrocharis morsus-ranae*) in Michigan inland lakes

Joshua Tellier¹

1. Michigan Department of Environment, Great Lakes, and Energy, Lansing, MICHIGAN, United States

European Frog-bit (*H. morsus-ranae*), or EFB, is an aquatic invasive plant in North America primarily found in sheltered, shallow water. First found in Michigan in 1996, it remained primarily along Great Lake coastlines. Despite being a prohibited species in the state, rapid spread to inland lakes began occurring in 2016. Several populations are now known to exist in inland lakes throughout the state. Where present, EFB can form dense mats, smothering native vegetation and substantially altering habitat in important nursery areas for aquatic animals. Recognizing the potential for EFB to impact Michigan's abundant freshwater resources, the Michigan Department of Environment, Great Lakes, and Energy (EGLE) set out to identify optimal detection strategies for EFB and to characterize the ecological impacts of this invader. Beginning in 2022, and in collaboration with several other agencies and university groups, Michigan EGLE piloted a study comparing detection frequency and the aquatic plant communities of lakes known to contain EFB against those known to be free of EFB. Twenty lakes (ten containing EFB, ten without) of similar relative size, habitat characteristics, and trophic state were sampled using multiple methods (e.g., point-intercept, targeted shoreline meander) to identify the most efficient method for detecting EFB, balancing detection probability with field staff effort. During the two-year sampling period, eighty-seven other aquatic plant species were identified across the twenty study lakes. This robust dataset allowed EGLE to compare aquatic plant communities in lakes with and without EFB to inform management decisions and invasion response. Several statistical methods were used to investigate whether EFB alters the littoral plant community, most notably cluster analysis applied to dissimilarity matrices among lakes. Lakes naturally clustered into groups of similar composition, but EFB did not appear to be a determining factor for any cluster in-group.

id #135861

Agricultural Impacts on Invertebrate Diversity in the Spokane River Watershed

Kaitlin Abell¹, Camille McNeely¹

1. Eastern Washington University, Spokane Valley, WASHINGTON, United States

Changes in land use, such as agricultural expansion, have impacted stream systems by modifying discharge, substrate diversity, channel morphology, and nutrient inputs. The Spokane River watershed is heavily impacted by human activity, including urban and agricultural development. Rehabilitation efforts are being organized to reintroduce Salmonidae species in the Spokane River Watershed. The first step of this reintroduction is to perform habitat studies to inform allocation of restoration efforts. As part of this effort, I performed benthic macroinvertebrate surveys throughout the watershed. Fifty sites were sampled during the summers of 2024 and 2025. In each reach, 4 subsamples, each comprising 0.19² m, were collected from the riffles, using a 500 µm mesh D-frame net then combined for preservation. I am analyzing the invertebrate samples for taxonomic and trait composition. Macroinvertebrate traits are increasingly used as a bioassessment tool, as they are directly impacted by abiotic factors within their aquatic environment. Traits for this study include habit (how invertebrates move around their habitat), feeding group (how invertebrates obtain nutrients), voltinism (number of generations per year), and ventilation mechanism (how invertebrates receive oxygen). The community data is being used to compare α and β diversity for taxonomic and functional composition of the invertebrates throughout the watershed. I am using GIS to quantify watershed land use and analyze the relationship between agricultural land use and invertebrate diversity. I hypothesize there will be higher taxonomic and functional diversity at low to moderate intensity of agricultural land use, and low diversity at high intensity of agricultural land use. Preliminary data shows a strong difference in taxonomic composition between stream reaches, with several streams dominated by Diptera populations. This data will provide awareness of the factors impacting invertebrate diversity in our region's waterways, and support habitat restoration efforts for native salmonids. Additionally, these results will provide insight into the shape of the relationship between agricultural land use and invertebrate diversity.

id #136117

Cryptic chemoautotrophic and methanotrophic processes complicate the use of carbon stable isotopes in rivers

Amanda G DeVecchia¹

1. UNC Chapel Hill, NC, United States

Over the past decade, increasing evidence has shown a prevalence of anoxic conditions and contributions of methane-derived carbon to freshwater food webs, highlighting the potential for methanotrophy and chemoautotrophy to occur. However, $\delta^{13}\text{C}$ values are still commonly interpreted under the assumption that all potential basal carbon sources are photosynthetic. Methanotrophy and chemoautotrophy can provide non-photosynthetic carbon sources that are sometimes cryptic, or present at such low concentrations that they might not provide an obvious carbon source, especially in rivers which are typically conceptualized as well-oxygenated. At most, chemoautotrophy and methanotrophy might be important, yet overlooked, carbon sources, and at least, even a small amount of these isotopically lighter contributions could complicate assessments of basal carbon sources to river food webs. Even a 1% contribution of methane-derived carbon to consumer biomass could cause 30% error in estimates of allochthony. Stream consumer biomass has included substantial contributions of non-photosynthetic carbon in habitats such as chalk streams, hyporheic zones, backwater pools, and even floodplains of glacial rivers. Herein I review the spatiotemporal distribution of stream conditions conducive to the production and potential consumption of cryptic carbon resources. I argue for a nuanced interpretation of carbon isotope values that explicitly considers three assumptions: representation of dietary end members using bulk sampling, estimation of fractionation rates associated with microbial oxidation reactions, and the decision of whether or not to include proxies for the heavily fractionated biomass of chemoautotrophs and methanotrophs. I suggest that these assumptions be explicitly considered in food web studies conducted in low-gradient, warmer, organic matter-rich, or spring-fed streams which are those which most commonly have the highest methane concentrations, and thus high probabilities of cryptic carbon production. In accounting for cryptic carbon sources, stream food web studies could consider the biogeochemical complexity that is possible in these heterogeneous environments, and obtain a more comprehensive understanding of the energy sources that sustain stream consumers.

id #135862

Methane-derived carbon subsidizes freshwater food webs in glacier-fed Alaska watersheds

Ryan Bellmore¹, Jason Fellman², Amanda DelVecchia³, Lindsey McCulloch⁴, Amy Holt², Holly Harris¹, Emily Whitney⁵, Matthew Dunkle⁴

1. US Forest Service, Juneau, ALASKA, United States

2. University of Alaska Southeast, Juneau, Alaska, United States

3. University of North Carolina, Chapel Hill

4. University of Alaska Fairbanks, Juneau

5. Alaska Coastal Rainforest Center, University of Alaska Southeast, Juneau, Alaska, United States

Methane-derived carbon has been shown to support stoneflies in hyporheic zones and midges in lake sediments. But evidence for widespread aquatic food web reliance on these non-photosynthetic energy pathways is limited. Here, we examined methane-derived carbon contributions to glacier-fed food webs in southeast Alaska coastal watersheds. These rivers are highly oligotrophic, especially in winter when low light and temperatures limit photosynthesis. We conducted seasonal sampling in three streams to measure (1) stable isotopes ($\delta^{13}\text{C}$) of fish and aquatic invertebrate biomass, and (2) dissolved methane concentrations from surface and well-water. Winter methane concentrations were high in both surface and well water, ranging to 10 μM in the surface and 30 μM subsurface. Correspondingly, we found strong seasonal methane incorporation into river food webs evidenced by isotopic values. In the winter, aquatic invertebrates were highly depleted in ^{13}C ($\delta^{13}\text{C}$ of -65 to -45) relative to summer months ($\delta^{13}\text{C}$ of -35 to -25), and isotope mixing model analyses suggested that methane was an important energy source fueling river food webs in the winter. We combined this information with a previously constructed quantitative food web for one of our study sites to estimate the proportion of secondary invertebrate and fish production supported by methanotrophy. These seasonal methanotrophic pathways fueled an estimated 20% of annual secondary invertebrate production at this study site. Together, these findings suggest that methane could be a quantitatively significant energy source in these coastal Alaska watersheds, seasonally subsidizing river food webs in the winter when availability of autotrophic resources are limited.

id #134583

Pathogen accumulation and the collapse of the invasive rusty crayfish population in Trout Lake

Lindsey S Reisinger¹, Cheyenne Stratton^{1,2}, Jamie Bojko³, Donald Behringer¹

1. University of Florida, Gainesville, FL, United States

2. Missouri Department of Conservation, Columbia, MO

3. Teesside University, Middlesbrough, England

David Lodge's thorough documentation of the rusty crayfish invasion in north temperate lakes over the past few decades provides important insight into how invasion dynamics change over time. Invasive species are thought to initially dominate biological communities because they have fewer natural enemies than native species. Over long timescales, as enemies such as pathogens accumulate in an invasive species, the density and impacts of the invader should decline. This process has rarely been documented, since few invasions are investigated over long timescales. Rusty crayfish populations have recently declined in many north temperate lakes, but the cause of these declines is uncertain. We examined how pathogens affect rusty crayfish fecundity, mortality, behavior, and ecological impacts in Trout Lake, Wisconsin, using controlled mesocosm experiments and field

data from 12 sites. Rusty crayfish abundance data in Trout Lake has been collected periodically since the early 1980s, and our field sampling was conducted from 2021 – 2023. During our study, there was an outbreak of a microsporidian pathogen (*Nosema rusticus*) in the invasive crayfish population and a substantial decline in rusty crayfish abundance. We found that this pathogen reduced rusty crayfish fecundity and survival and was more prevalent at high density sites. We also found that infected crayfish were more lethargic, and that infected individuals had reduced ecological impacts due to trait-mediated effects of the pathogen. Crayfish in Trout Lake were also often infected with trematode parasites (*Microphallus* sp.). Our previous work on trematodes indicates that these parasites do not cause direct mortality but are associated with an increase in crayfish boldness. Rusty crayfish infected with trematodes have greater ecological impacts because they spend less time hiding. Finally, we observed an increase in the abundance of native virile crayfish in Trout Lake. Virile crayfish were often infected with trematodes, but none were infected with *N. rusticus*. Overall, this research demonstrates that pathogens can substantially alter the impacts of invasive species by changing their behavior and abundance, and in some cases, this can promote the recovery of the native ecological community.

id #136119

Bugs care how many hippos are there

Therese Frauendorf¹, Amanda Subalusky², Christopher Dutton², James Junker³, Laban Njoroge⁴, Joshua Benjamin², Emma Rosi⁵, David Post⁶

1. Colgate University, NY, United States
2. University of Florida, Gainesville, FL
3. University of North Texas, Denton, Texas
4. National Museums of Kenya, Nairobi, Kenya
5. Cary Institute of Ecosystem Services, Millbrook, NY
6. Yale University, New Haven, CT

Animal subsidies have important effects on aquatic food webs, yet we know little about how these effects vary with the magnitude of subsidies. The Mara River in Kenya has over 4000 hippopotamuses who forage on savanna grasslands during the night and rest in the river during the day. We selected six sites along the hippo abundance gradient (from 0 to 4300 hippos) in the Mara River and measured invertebrate biomass, production, and diversity at each site. Hippo dung increased from 0 to 70 g of dry mass/m², which resulted in a > 10-fold growth in invertebrate biomass. Invertebrate production increased exponentially from 840 mg/m²/year to 7080 mg/m²/year at the site with the highest hippo dung. These changes in production were strongly driven by increases in scrapers like the Tricorythid mayflies, while other functional feeding groups increased to a lesser extent. The number of invertebrate species doubled at sites with hippo dung, but they did not change with the magnitude of the subsidy. Lastly, the density of invertebrates within hippo dung itself did not vary across the gradient, although invertebrate density in hippo dung did increase during high discharge periods. These patterns indicate that it is not just the occurrence, but the magnitude of animal subsidies that play an important role in structuring freshwater macroinvertebrate communities.

id #134328

From swamps to stormwater ponds: Ecosystem functions lost (and gained?) due to reorganization of the waterscape

AJ Reisinger¹, Audrey Goeckner¹, Meredith Holgerson², Jacqueline Gerson², Anika Bratt³

1. University of Florida, Gainesville, FLORIDA, United States
2. Cornell University, Ithaca, NY, United States
3. Macalester College, Saint Paul, MN, United States

Urban expansion alters surface runoff due to increased impervious surfaces across the landscape. To address this urban hydrological issue, many regulatory agencies now require developers to reduce the impacts of urbanization on watershed hydrology. Florida regulations, for example, require developments to not exceed the pre-development runoff levels. Due to their well-established hydrological efficacy, stormwater ponds (SWPs) are one of the most common stormwater management approaches for new development. These de novo ecosystems are particularly common in Florida, where there are >75,000 SWPs statewide. The construction of SWPs doesn't happen in a vacuum, though, as the expansion of urban areas has come at the cost of a variety of terrestrial and aquatic ecosystems, including wetlands. For example, there was a 33% loss of wetland area in the Tampa Bay watershed between 1950 and 2007 while the construction of artificial aquatic ecosystems offset 75% of this wetland loss. While SWPs are clearly not identical to natural wetlands, they are capable of providing some similar functioning, including nutrient processing, carbon sequestration, and habitat for wildlife. It is unclear, however, how this reorganization of the waterscape from natural wetlands to de novo ecosystems affects the functions being provided. Quantifying the functions that SWPs (and other de novo ecosystems) provide is important to improve the design and management of these human-derived ecosystems, maximizing their benefits to society. But it is also important to balance these functional gains with those lost due to wetland loss. By combining empirical field measurements and literature values with historical remote sensing imagery, we can estimate the magnitude of wetland-SWP transitions from an areal extent and quantify the effect of these transitions on regional biogeochemical dynamics. This approach will provide an understanding of how urban development has influenced downstream water quality over previous decades. Combining this approach with future land use projections, we can identify the potential impacts of future development and develop strategies to protect ecosystem functions while still allowing for responsible urban expansion.

id #136120

The space-time discontinuum: correcting FLIR-derived temperature profiles to account for diel temperature variation.

Geoffrey C Poole¹, Marty King², Bethy Rogers-Pachico², Scott J O'Daniel²

1. *Montana State University, Bozeman, MONTANA, United States*

2. *Confederated Tribes of the Umatilla Indian Reservation, Pendleton, OR, USA*

Forward-looking infrared (FLIR) surveys conducted from helicopters provide spatially continuous temperature profiles across tens to hundreds of river kilometers, but the extended duration of these surveys means that temperature readings at different locations are collected at different times of day—conflating spatial patterns with diel thermal dynamics. We developed and validated a statistical method for correcting FLIR-derived temperature profiles to account for the confounding effects of diel temperature variation during airborne surveys. We hypothesized that FLIR observations, combined with information about the time of data collection, could accurately predict temperature metrics such as daily maximum temperature at any surveyed location -- even where temperature readings were collected hours before or after the daily temperature peak. We deployed stationary temperature loggers at multiple sites along the Umatilla River (Oregon, USA) during FLIR surveys and used these data to build a multiple regression model predicting daily maximum water temperature from three variables: the FLIR-derived water temperature (extracted via quantile spline regression), the hour of day when each FLIR observation was collected, and a categorical year term to account for annual temperature variation and other factors, such as sensor calibration differences for surveys conducted a decade apart. The regression model explained 98% of the variance in observed daily maximum temperatures ($R^2 = 0.98$, RMSE = 0.49°C), with all predictor variables highly significant ($p < 0.001$). The hour-of-day coefficient confirmed that FLIR observations collected earlier in the day required larger upward adjustments to estimate daily maximum temperature, consistent with expected diel warming patterns. Our approach provides a template for transforming FLIR surveys from point-in-time temperature readings collected across both time and space into estimates of temperature metrics that are more meaningful for river research and management. Our approach is transferable to other FLIR applications and provides a framework for maximizing the inferential value of airborne thermal surveys in stream temperature monitoring programs.

id #135865

Analysis of Aquatic Invertebrate Diversity and Body Length Along a Sodium Gradient

Connor Foley¹, Sally Entekin², Michelle Evans-White³, Clay Prater³, Samuel Dias³, Natalie Clay¹, Maggie Herrmann¹, Beck O'Neill¹

1. *Entomology, University of Arkansas, Fayetteville, AR, United States*

2. *Entomology, Virginia Tech, Blacksburg, VA, USA*

3. *Biology, University of Arkansas, Fayetteville, AR, United States*

Aquatic invertebrates are highly sensitive to changes in the concentrations of sodium in their environment, increasingly struggling to maintain osmotic and ionic balance in their bodies with greater departure of environmental sodium concentrations from their biological needs. Nonetheless, they require sodium for innumerable cellular functions. The Sodium Subsidy-Stress (SSS) hypothesis predicts that organisms are subsidized by sodium addition to the water when availabilities are lower than an organism's needs, are healthiest at an "optimum point" of sodium availability near the concentration in hemolymph, and then experience stress when water ion concentrations exceed internal levels. Individual aquatic invertebrates (i.e., body mass) and, by extension, invertebrate communities (i.e., diversity) are expected to display SSS along this scarcity-optimum-overabundance curve. To test SSS, a paired terrestrial-aquatic mesocosm experiment was used to analyze the response of aquatic invertebrate diversity and body mass to a sodium addition gradient with six weekly treatments: 0 (controls), 100, 200, 300, 400, 500 mg/L NaCl addition ($n=5$ per treatment). Ten additional mesocosms received 300mg/L NaCl addition to streams ($n=5$) or to terrestrial soils ($n=5$). Aquatic invertebrates were sampled monthly using benthic cores. All the organisms from a subset of 12 samples have been identified to order, and body length, a proxy for biomass, was measured. So far, 18 orders have been identified. Oligochaeta was most frequently encountered, followed by Ostracoda. Classes Insecta and Clitellata, as well as the subphylum Crustacea accounted for ~20%, ~40%, and ~25% of individuals, respectively. Typically sensitive taxa such as Ephemeroptera and Plecoptera were infrequently collected in sampled replicates. Notably, a number of terrestrial (e.g., Acari: Oribatida) or amphibious (e.g., Clitellata) taxa were collected in experimental streams and provide a measure of terrestrial subsidies to stream systems. From the organisms most represented in preliminary samples, Oligochaeta had the greatest body sizes in control mesocosms, whereas Ostracoda have slightly longer body lengths in relatively elevated [NaCl]. These unique, paired, and experimental riparian mesocosms will enable us to examine links between systems and how communities change in response to both terrestrial and aquatic salinization.

id #134587

Fifteen miles: Partnerships and management of the Rattlesnake Creek watershed from wilderness to downtown Missoula, MT.

Matt Trentman¹

1. *University of Montana-O' Connor Center for the Rocky Mountain West, MT, United States*

The Rattlesnake Creek watershed is located just north of Missoula, Montana and is valued for its ecological, cultural, and recreational abundance. The watershed is in the historic homelands of the modern-day Confederated Salish and Kootenai Tribes where the original Peoples were sustained by the prolific bull trout (*Salvelinus confluentus*) in the creek. In recent history, European settlers, and modern day Missoulians have relied on the creek as a direct municipal water source through the 1980's. The Rattlesnake Creek watershed offers an abundance of recreation opportunities and is beloved by Missoulians for its year-round access to a pristine, federally recognized, Wilderness and National Recreation Area within a 20-minute drive from downtown Missoula. The Rattlesnake Creek Watershed Group (RCWG) was established in 2008 based on a grassroots response

to invasive plants encroaching on habitat in the lower Rattlesnake Creek neighborhood. Over the past 15 years, the group has facilitated between stakeholders in the watershed, actively engaged in watershed planning, and being a voice of the watershed to the public. In recent years new threats have emerged to the health of the watershed. For example, the water rights of the creek, and the management of multiple wilderness dams that have exceeded their useful life, have shifted from private to public ownership with the City of Missoula. Decisions must be made on whether to rehabilitate these dams, and potentially use them to augment stream flows, or to decommission the dams and return the lakes to their natural state. This talk will focus on a new collaboration between RCWG and the O'Connor Center for the Rocky Mountain West at the University of Montana to build partner capacity and conduct restoration planning via a Bureau of Reclamation WaterSMART grant. I will discuss (1) our approach to make the RCWG a clearinghouse of ecological and cultural information in the watershed, (2) how we are engaging with the many city, county, federal, tribal, and non-profit stakeholders in the watershed to prioritize specific needs, and (3) initial baseline data collection, research, and watershed restoration planning.

id #134843

ENVIRONMENTAL FLOWS AND THE LAW

David Allan¹, Don Blankenau²

1. *University of Michigan, MI, United States*

2. *Blankenau, Wilmoth & Jarecke, LLP, Lincoln , NE, USA*

The natural flow regime has become the dominant framework for environmental flow management, appearing in numerous technical and guidance documents, and showing how scientific concepts can profoundly influence policy implementation. Owing to the scarcity of water resources in many regions and growing demand, the need for science-based protection of environmental flows will become increasingly urgent. To protect flows for ecological uses, science and law are necessarily intertwined but the law is not uniform. It is location and case specific, which requires that the science be designed to address the applicable law. Protection of interstate water resources requires a precise showing that harm occurs to species of value when flows are diminished beyond a certain amount and for a certain duration. Protecting flows is multi-disciplinary. Hydrologists must show that flow declines are directly caused by human activity. Economists/sociologists must show that the ecological harm to the particular species causes a human hardship. Using a case study of a failed effort to apportion flows between Georgia and Florida, we illustrate the challenges in demonstrating harm and remedy in a highly contested dispute over interstate water use. State law often provides much greater clarity to obtain flow protection but similarly requires scientific support before such protection will be granted by the state water agency. At least 13 states allow for some level of instream flow protection, but generally differ in evidence required for an application to be granted. Other states may lack specific statutory guidance and hence provide less clarity on how an application for instream flows can be argued. Few instream flow applications have been granted without litigation, and for that reason, the scientific and economic evidence needs to be focused on the specific criteria for granting a permit. Among the exceptions have been the Loup, Elkhorn, and Niobrara Rivers in Nebraska – a state with some of the most detailed statutory criteria. These examples illustrate some common challenges with establishing instream flow rights.

id #135867

Assessing long-term impacts of mining in a subarctic stream

Thomas House^{1,2}, Jeffrey Muehlbauer^{1,2}

1. *University of Alaska Fairbanks, Fairbanks, AK, United States*

2. *Alaska Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, Fairbanks, AK, USA*

Our presentation reviews recent research to assess energetic budgets of juvenile salmonids in a mining impacted sub-arctic stream. Our study site is located at Coal Creek, a tributary of the Yukon River in Yukon-Charley Rivers National Park and Preserve in Alaska. This work is especially important in the context of the current collapse in Yukon River Chinook salmon (*Oncorhynchus tshawytscha*) populations. This collapse has had not only devastating effects on ecosystem health, but also an outsized impact on the local Indigenous people who rely on salmon for subsistence. In this research we demonstrate how mining-induced alterations to habitat impact growth potential for juvenile salmonids – principally by quantifying and characterizing macroinvertebrate prey communities and using this information, along with temperature to inform a growth model. Because salmon are limited in their access to prey by gape size it is critical that they grow rapidly to access larger, more energetically dense prey. We hypothesize that the impacted conditions in Coal Creek, principally redistribution of the stream into a novel channel dominated by mine tailings, is associated with slower growth rates in juvenile salmonids. Subsequently, we predict that total production of the macroinvertebrate community will be lower in Coal Creek than in adjacent, unmined reference streams, with concomitant impacts on salmonid growth rates between these streams.

id #136123

STUDIES ON THE MAJOR CHARACTERISTICS OF THE KARATOA RIVER AT SHAHJADPUR AND ITS FISHERIES

Bijoy Kumar Ghosh¹

1. *Ecohydrology Lab Department of Civil and Environmental Engineering Clarkson University 226 Rowley, Box 5710, 8 Clarkson Avenue Potsdam, NY 13699, Potsdam, NY 13699, NY, United States*

Studies on the major characteristics of the Karatoa river and its fisheries at Shahjadpur upazilla of Sirajganj district, was carried out during a period from October 1999 to September 2000. Physical, chemical and biological parameters were observed. Fishing methods, marketing system and socio-economic condition of the people those who depends upon the resources of the river were studied.

Soil analysis revealed that sandy-silty soil of the study area endowed with organic materials. Maximum and minimum air temperature were recorded as 29.25°C (June) and 18.80°C (January), water temperature were found to vary from 1 to 3°C less than air temperature. Maximum rainfall was 150.7 mm (May) and in the months of November, December and March there were trace rainfall. Relative humidity varied from 88.5% (September) and 66% (March). Maximum and minimum values of water depth were 6.75 m (October) and 1.95 m (March), water area were 93.91 m² (October) and 10.05 m² (February), turbidity were 397.98 mg/l (June) and 250.83 mg/l (September), water velocity were 0.533 m/sec (October) and 0.071 m/sec (February), pH were 7.89 (October) and 6.81 (December), range of DO was 4.0 to 9.0 mg/l, CO₂ were 8.57 mg/l (August) and 5.0 mg/l (January). The families Chlorophyceae, Bacillariophyceae and Myxophyceae were dominant among the phytoplanktons throughout the year. Moreover, some common aquatic plants were also found. Among the zooplanktons copepods, cladocerans, ostracods, rotifers, aquatic insect and crustacean larvae were common with copepods as dominant group.

A total of 104 different species of fishes of 55 genera, 24 families and 10 orders were found of which 17 species are rare. Six different types of crafts, 5 nets and a number of traps and other fishing gears were found to be used. Marketing system at the fish markets were found to be traditional one. Socio-economic condition of the fishers and fish traders were found as of very poor standard. The fishers only get 55.50 to 58.30% and 55.57 to 65.50% of the consumer's price for the larger and smaller fish species respectively.

Key word: Socio-economic condition, Sandy-silty soil, Relative humidity, Turbidity, Phytoplankton, Zooplanktons, Dominant group, Consumer's price.

id #135868

Can thermal performance curves predict fish distribution across a thermally diverse watershed?

Benjamin Zdasiuk¹

1. Simon Fraser University, Port Moody, BRITISH COLUMBIA, Canada

As coupled climate change and human land use transform the hydrologic regimes of watersheds in Northwestern North America, endemic fishes may sometimes encounter novel temperature and flow conditions. Thermal performance curves offer a mechanistic description of the temperature-performance response function and might be able to predict individual-level responses to temperature conditions outside of historical norms. However, the degree to which novel temperature responses affect populations and communities is complicated by ecological contexts including physical habitat characteristics, phenology, and spatial population structure, among others. Here, we describe a coupled study which fit thermal performance curves through an exhaustive meta-analysis to predict thermal habitat suitability and coupled snorkel surveys to investigate whether thermal habitat suitability predictions translate to occupancy or density outcomes in a thermally diverse watershed. Collectively, we find that thermal performance might influence occupancy and density trends only once critical habitat characteristics and spatial structure have been accounted for. We conclude by detailing how our findings relate to persistence and resilience of populations facing thermal challenges associated with climate change.

id #134845

Mussel-mediated nutrient fluxes in a coastal plains river

Madison P. Kuczek¹, Michael D. Kaller², Kaelyn J. Fogelman², Mysha Ahmed³, Michael P. Hayes³, Garrett W. Hopper²

1. School of Renewable Natural Resources, Louisiana State University, Baton Rouge, LA, United States

2. School of Renewable Natural Resources, Louisiana State University, Baton Rouge, LA, United States

3. School of Plant, Environmental and Soil Sciences, Louisiana State University, Baton Rouge, LA, United States

Animal-driven nutrient cycling processes can be important to ecosystem nutrient dynamics and are influenced by habitat size, ambient nutrient concentrations, and species-specific traits such as body size and nutrient composition. Nutrients released by animals contribute valuable ecosystem services and promote biological diversity by acting as sources of bioavailable energy and nutrient subsidies. Freshwater mussels (Order: Unionoida) are a diverse, threatened group of sedentary, filter-feeding bivalves that predominantly occur in dense multispecies aggregations and can create hotspots of biogeochemical activity. Although occurring within the same feeding guild, interspecific differences in body size and tissue stoichiometric ratios can result in interspecific differences in nutrients assimilated and released. Thus, differences in mussel assemblage structure should affect site level nutrient availability. We quantified nitrogen (N) and phosphorus (P) excretion, particulate organic matter (POM) egestion, and tissue carbon (C), N, and P for dominant species spanning nine sites in the Bogue Chitto Watershed, Louisiana and Mississippi, and scaled individual rates to the assemblage level. Egestion, excretion, and storage varied interspecifically, with the greatest differences occurring for P excretion rates and storage, which consequently led to interspecific differences in N:P ratios. High density sites composed of large-bodied individuals had the highest areal excretion, egestion, and stored nutrients. Differences in assemblage composition altered site level stoichiometric ratios, producing differences in assemblage scale N:P excretion. We estimated temporal changes in mussel contributions throughout the seasons with volumetric excretion, such that at lower stream discharge mussels have a greater contribution to ambient concentrations. This study provides novel information about how the distribution of mussel biomass and assemblage structure may influence nutrient cycling in a biodiverse coastal plain river.

Severe storm response in Central Appalachian streams: focusing on benthic macroinvertebrate community dynamics

Quinn Weigel¹, Kelley Sinning¹, Amaryllis Adey¹, Caleigh Meehan², Megan Underwood¹, Lisa M Tabor¹, Teresa Brown³, Stephen Schoenholtz⁴, Erin R Hotchkiss², Daniel McLaughlin⁴, Carl Zipper⁴, Sally Entrekin¹

1. *Virginia Tech Aquatic Entomology Lab, Blacksburg, United States*

2. *Department of Biology, Freshwater Ecology & Biogeochemistry Lab, Blacksburg, VA, USA*

3. *Department of Natural Sciences, University of Virginia's College at Wise, Wise, VA, USA*

4. *Virginia Tech Water Resources Research Center, Blacksburg, VA, USA*

Though headwater ecosystems in central Appalachia are distinct for their high biodiversity, including aquatic insects, intense seasonal weather patterns including sustained storm systems could pose a threat. Tropical hurricanes, flooding, and icemelts have caused an increasing appearance in freshwater stream integrity declines, coinciding with greater ecological damage. Macroinvertebrate communities act as complex bioassessment indicators for these systems, and factors such as richness, biomass, and taxa dominance can indicate quality of taxa refugia. After preliminary analysis of Hurricane Helene confirmed immediate declines in richness and abundance, our hypothesis that amplified storm events reduce an ecosystem's macroinvertebrate community resilience can be further applied to another storm event, and expanded to include how intensity of stream flow surges correlate with specific assemblage abnormalities. Predictions include increased dominance of sediment and salinity tolerant taxa, community disruption in later months post impact, and significant biomass differences among salinity classes. Hurricane Helene of September 2024 and a historic freeze in mid February 2025 each caused multiple days of streamflow output beyond the 90th percentile of normalcy, with both events distinct in their increased precipitation. Hydrological and macroinvertebrate data from nine Virginian and West Virginian headwater streams included in or adjacent to the affected storm event regions were collected from September 2023 to August 2025, with macroinvertebrates identified to the highest taxonomic resolution possible (e.g., genus). Study streams were located in multiple watersheds across a salinity gradient where specific conductivity ranged from 19–1,185 $\mu\text{S}\cdot\text{cm}^{-1}$, with intersecting effects of salinity and high flow events to be evaluated. A full year of pre-hurricane collection data acted as a comparative baseline for measuring the two storm events' disturbance, wherein significant decreases in overall taxa richness due to the hurricane were found. Data will highlight both fluctuation in community biomass and dominance of subjects such as *Leuctra* stoneflies. Through this combined monitoring, population dynamics can be understood with higher fidelity, especially when considering other similar disturbance events that interact with salinity. Creating a profile of ecosystem behavior after significant disturbances could help assist in forming recovery expectations and disturbance predictability in a world with consistently record breaking storm events.

From strategy to structure: decomposition dynamics of unionid mussel shells

Lauren M Morris¹, Garrett W Hopper², Taylor M Michael¹, Jonathan W Lopez³

1. *Biological Sciences, University of Alabama, Tuscaloosa, Alabama, United States*

2. *Renewable Natural Resources, Louisiana State University, Baton Rouge, Louisiana, United States*

3. *Department of Biology, University of Mississippi, Oxford, Mississippi, United States*

Freshwater mussels are a diverse taxonomic group that can influence nutrient dynamics in rivers through the storage of carbon (C), nitrogen (N), and phosphorus (P) in their soft-tissues and shells. Following mortality, nutrients stored in soft tissues are rapidly released, whereas nutrients in shells are released over months to decades, allowing shells to act as long-term reservoirs of nutrients. Mussel life history strategies reflect trade-offs among processes like growth and structural investment, which generate predictable patterns in shell morphology with opportunistic and periodic species generally producing thinner shells than equilibrium species. Previous studies of shell stoichiometry suggest that shell morphology (e.g., shell thickness, surface area, etc.) and life history strategy, may influence shell nutrient composition and decomposition rates. However, little is known regarding how specific morphological traits, such as shell thickness and surface area, regulate shell decomposition or how rates of nutrient leaching from shell material varies among C, N, and P. We hypothesized that thinner-shelled species and shells with greater surface areas will have faster decomposition rates than thicker-shelled species and shells with lower surface area because they are more vulnerable to mechanical breakdown, with opportunistic and periodic species decomposing faster than equilibrium species. To test this, we deployed coarse mesh bags containing shells of seven freshwater mussel species with varying life history traits (e.g., shell thickness, surface area) along three sites in the Sipsey River, Alabama, USA that were collected periodically over four years to estimate decomposition rates. As predicted, thicker shells decomposed slower than thinner shells; however, increasing shell surface area tended to result in slower decomposition for some species. Ongoing work aims to evaluate how shell nutrient release changes throughout decomposition and how decomposition rates differ among elements. Understanding how shell traits and decomposition vary among species improves our ability to predict mussel-mediated nutrient provisioning and the long-term influence of mussels shells on ecosystem services in rivers.

From streamlines to storylines: A Poff Lab-er's journey from science to freshwater filmmaking

Jeremy Monroe¹

1. *Freshwaters Illustrated, OR, -*

LeRoy Poff's career studying rivers modeled a meandering channel through basic organismal ecology, applied management-focused science, and adaptive conservation strategies. Like many of his students, I proudly followed a similar nonlinear path... one that took me right out of academic science and into the world of educational filmmaking. Through my lens as a visual

storyteller, I will share some examples of how my work carries the spirit of discovery and intricacy that were part of what LeRoy inspired in students, along with lessons and fond memories from my time in the Poff Lab.

id #134591

The assessment of riverine wetland conditions, macroinvertebrate assemblages, and physical habitat conditions of streams on Santa Rosa Island, Channel Islands National Park.

Daniel Pickard¹, Parsa Saffarinia²

1. Aquatic Bioassessment Lab (ABL) - Chico, Chico, CA, United States

2. Biological Sciences, CSU Chico, Chico, CA, United States

Santa Rosa Island is an 82 sq. mi. island approximately 26 mi. off the southern California coast, located within Channel Islands National Park. Thirty long term monitoring sites were assessed during two sampling events for riparian wetland health with CRAM (California Rapid Assessment Method). Bioassessment (including physical habitat assessment) was done on ten of these sites. CRAM scores averaged 76 (range 60-91), with nine (14%) sites scoring impaired for overall CRAM score. Thirty-two (50%) sites scored impaired for the BIO/PHYS CRAM submetrics. California Stream Condition Index (CSCI) scores averaged 0.73 (range 0.58-0.95) and showed impairment at 17 (77%) sites. Physical habitat metrics indicated grazing impacts. All but one site showed impairment for specific conductivity. NMDS analyses revealed a grouping of communities by watershed suggesting that community structure is driven by local watershed conditions rather than island-wide effects.

id #135359

Long-term monitoring of alpine streams in the Teton Range: biotic and abiotic trajectories vary among hydrologic sources

Lusha M Tronstad¹, Deb S Finn², Scott Hotelling³

1. University of Wyoming, WY, -

2. Missouri State University, Springfield, MO, USA

3. Utah State University, Logan, Utah, USA

Water stored as snow and ice in mountain ecosystems is changing globally due to rising air temperatures. Glaciers and perennial snowfields are receding, and seasonal snowpack is declining; however, far less is known about the trajectory of cold rocky landforms such as rock glaciers. Rock glaciers are large masses of debris-covered ice that flow downhill and are predicted to persist longer than ice located on the surface. In 2015, we formed the Teton Alpine Stream Research project to measure the degree to which climate change is altering alpine biodiversity in streams fed by each of the dominant alpine water sources in the middle Rocky Mountains (USA): glaciers, rock glaciers, and perennial snow. Each August, we collect a suite of abiotic (e.g., temperature) and biotic (e.g., macroinvertebrate community structure, algae) data in 12 alpine streams in the Teton Range, Wyoming. Our first 11 years of monitoring showed that summer stream temperatures have increased in snowmelt-fed streams. Algal biomass, and invertebrate densities and biomass were highest in rock glacier-fed streams ($p < 0.05$). The federally protected stonefly, *Zapada glacier*, primarily lives in rock glacier-fed streams, as do several other cold stenotherms. Our results highlight the power of long-term monitoring for quantifying changes in mountain ecosystems and the potential for rock glaciers to act as key climate refugia into the future.

id #134592

Physiological Responses of *Pteronarcys californica* to Simulated Wildfire Ash Leachates

Jacob D McArtor^{2,1}, Benjamin P Colman¹, Rachel L Malison²

1. University of Montana, Missoula, MT, United States

2. Flathead Lake Biological Station, Polson, MT, United States

Wildfires are natural disturbances, but their frequency and severity are increasing across the western United States due to climate and land-use change. Post-fire runoff can substantially alter stream chemistry by delivering nutrients, sediments, organic matter, and metals from burned landscapes. Although the effects of wildfire on freshwater communities are well documented, the mechanisms linking post-fire water chemistry to biological responses remain relatively unexplored. In particular, there are limited data on the effects of metals from ash on aquatic macroinvertebrates post-fire, though leached ash can release alkaline water with potentially toxic concentrations of a range of metals including aluminum, copper, and selenium. Subsequent dilution and neutralization during mixing with streamwater may alter metal speciation, solubility, biological availability, and toxicity. We conducted a laboratory microcosm experiment to evaluate the immediate post-fire effects of metallic and ionic constituents derived from wildfire ash on the common and ecologically important stream macroinvertebrate, *Pteronarcys californica*. Solutions were prepared using laboratory-grade reagents to match the dominant metallic and ionic composition of ash leachates. These leachates were generated in the laboratory from ash collected in the Seeley-Swan Valley, Montana. Treatment solution pH was adjusted to match that of the source streamwater used in the experiment. Wild-caught *P. californica* nymphs were exposed to a gradient of concentrations for three weeks, after which we quantified survival, growth, critical thermal maximum (CT_{MAX}), and metabolic rates. Survival, growth, and metabolic rates did not differ detectably among treatments, despite aluminum concentrations in our treatment solutions as high as 30,000 ppb, and copper concentrations as high as 60 ppb. In contrast, CT_{MAX} showed a modest but clear effect of treatment, with higher leachate concentrations associated with increased CT_{MAX} , which was

contrary to our expectation. Overall, the muted physiological responses across treatments suggest that chemical transformations associated with neutralization of ash leachates mediates metal bioavailability and biological effects. This work highlights the importance of chemical context in understanding post-fire stressor impacts and helps inform a more mechanistic understanding of how wildfire-altered water chemistry influences aquatic organisms in streams.

id #135872

Implementing freshwater restoration under real-world constraints: European lessons for building resilient watersheds

Yaron HersHKovitz^{2,1}, Sebastian Birk¹

1. *Aquatic Ecology, University of Duisburg-Essen, Essen, Germany*

2. *Steinhardt Museum of Natural History, Tel Aviv University, Tel Aviv, Israel*

Building resilient watersheds and communities requires not only ambitious restoration goals, but also a clear understanding of how restoration measures are implemented in practice under real-world constraints. This presentation contributes to the transatlantic dialogue on freshwater restoration by synthesising European experiences from the EU-funded MERLIN project, with a focus on implementation processes rather than ecological outcomes alone.

Across diverse river, wetland, and floodplain restoration case studies, MERLIN examined how planning and implementation unfold within complex governance settings shaped by permitting regimes, land ownership patterns, stakeholder interests, funding rules, and technical uncertainty. Drawing on qualitative case narratives and practitioner reflections, the synthesis highlights how restoration delivery is rarely linear: measures are frequently adapted, redesigned, or sequenced differently than originally planned in response to social, institutional, and ecological constraints. Importantly, such adaptations often reflect learning and responsiveness rather than implementation failure.

The analysis identifies recurring implementation patterns, including the importance of early institutional alignment, sustained stakeholder engagement, and the need for long pre- and post-project phases to support durable outcomes. These insights resonate strongly with North American watershed restoration experiences, where adaptive management and community engagement are likewise central to success. By foregrounding implementation as a negotiated and adaptive process, this contribution offers transferable lessons for designing restoration programmes that are robust to complexity and capable of strengthening both ecological and social resilience across freshwater landscapes.

id #136128

How surface water modification alters carbon storage and flux in a protected savanna landscape

Audrey Goeckner¹, Tatenda Dalu², Christopher Dutton¹, Dumisani Khosa³, Robert McCleery⁴, Linton Munyai², Amanda Subalusky¹

1. *Department of Biology, University of Florida, Gainesville, FL, United States*

2. *School of Biology and Environmental Sciences, University of Mpumalanga, Mbombela, Mpumalanga, South Africa*

3. *South African National Parks, Skukuza, Mpumalanga, South Africa*

4. *Department of Wildlife Ecology and Conservation, University of Florida, Gainesville, FL, United States*

Human modification of freshwater flows is widely associated with shifts in carbon pools and fluxes, most commonly studied in urban and developed landscapes. However, flow modification also occurs in otherwise pristine ecosystems for unique human-wildlife management purposes. Kruger National Park (South Africa) is a 19,000 km² savanna landscape that contains more than 40 dammed impoundments constructed along intermittent streams to provide dry-season water resources for wildlife. These impoundments aggregate large herbivores (e.g., hippopotami, elephants, giraffes), which can further influence aquatic carbon biogeochemistry through physical disturbance, grazing, and subsidies of organic matter and nutrients. We investigated how artificial impoundments and variation in large herbivore density and diversity influence carbon cycling in a savanna grassland landscape by comparing dammed impoundments to unaltered intermittent streams. Across 11 watersheds, we selected paired sites consisting of one intermittent stream and one dammed stream (hereafter, impoundment). Impoundments feature a gradient of low to high large herbivore density. Impoundments spanned a gradient of low to high large herbivore density. At each site, we sampled three positions along an elevation gradient spanning the terrestrial-aquatic interface: upland (rarely inundated), transitional, and bank (seasonally or persistently inundated). At each position, we measured carbon dioxide and methane fluxes and collected 30-cm deep soil and sediment cores to quantify carbon stocks and bacterial abundance. Preliminary dry season gas flux comparisons between ephemeral stream and impoundment transects show that uplands are similar in both carbon dioxide and methane fluxes, and that ecosystem types become increasingly divergent in transitional and bank zones. Specifically, carbon dioxide is greater, and methane is lower in impoundments compared to intermittent streams in transitional and bank elevations. We further expect that impoundment carbon stocks and bacterial abundance will increase from upland to bank zones, and that transitional and bank zones in impoundments will be greater than those of intermittent streams due to longer hydroperiods. This study provides a unique opportunity to study how human modification of surface waters influences biogeochemical pools and fluxes in undisturbed savanna landscapes, and how large herbivorous wildlife mediate these functional properties.

id #135617

Stream warming drives biofilm metabolism recovery following drought

Annel R Schreiber¹, Ariel J Shogren¹, Jacqueline E Todd¹, Nina Keck¹, Mia Berwick², Kieran Khamis², Mark Ledger², Alexander D Hurn¹, Jonathan P Benstead¹, Wyatt F Cross³

1. *The University of Alabama, Tuscaloosa, AL, United States*
2. *University of Birmingham, Birmingham, United Kingdom*
3. *Montana State University, Bozeman, Montana*

Headwater streams play an essential role in biogeochemical cycling via the carbon metabolism and nitrogen transformations mediated by benthic biofilms. As climate change progresses, higher temperatures will co-occur with more frequent and intense drought disturbances, with unknown consequences for stream ecosystem function. We tested the effect of warming on stream biofilm recovery from drought in an outdoor mesocosm experiment by measuring two key biofilm processes: metabolism and dinitrogen (N₂) flux. We inoculated the flumes with biofilms from nearby streams and maintained ambient and warmed (+5°C) treatments for one year before subjecting all flumes to a 1-month drought. We measured biofilm metabolism before and after drought by incubating biofilm-covered stones in recirculating chambers and estimated rates gross primary production (GPP) and community respiration (CR) as the change in dissolved oxygen over time in light and dark conditions, respectively. To measure potential N₂ flux, we collected dissolved N₂ gas samples at the start and end of the incubations. We measured biofilm chlorophyll a concentration and ash-free dry mass from each set of stones. Our data show increased GPP (g O₂/g Chl-a/hr) in the warmed flumes after drought, and a decrease in ambient flumes. Additionally, CR (g O₂/g AFDM/hr) was similar between temperature treatments and was lower following drought. Preliminary N₂ flux results show evidence of N₂ fixation by both ambient and warmed biofilms. Our data provide evidence that long-term warming may affect the recovery of benthic biofilm metabolism after drought, suggesting that metabolic responses to disturbance are influenced by temperature regime.

id #136130

Building resilience in large rivers of the northwest: Linking Science, Management, and Collaboration

Lisa Kunza¹, Genny Hoyle², Brandon Diller³, Kurt Chowanski¹, Nathan Jensen³, Mark Elliston³, Campbell Olson³, Brooke Long-Fox¹, Mehzaheen Mannan¹, Aisling Hall¹, Christopher Brooks¹, Gregory Hoffman⁴

1. *Center for Sustainable Solutions; Chemistry, Biology, and Health Sciences, South Dakota Mines, Rapid City, South Dakota, United States*
2. *Environmental Department, Kootenai Tribe of Idaho, Bonners Ferry, Idaho, United States*
3. *Fish and Wildlife Department, Kootenai Tribe of Idaho, Bonners Ferry, Idaho, United States*
4. *US Army Corps of Engineers, Libby, MT, United States*

Large, regulated rivers of the Northwest anchor biodiversity, Tribal lifeways, and regional economies. Yet hydropower, land-use change, and climate stressors have reshaped flow and thermal regimes, sediment and nutrient dynamics, floodplain connectivity, and food webs. Building resilience in these systems requires cross-disciplinary science and sustained collaboration among communities, Tribes, agencies, and universities. This opening presentation frames the special session by highlighting the monitoring, management, and restoration approaches needed to support resilience in large rivers. We invite practitioners and scientists to share research and programs that directly inform resilient watershed management, including work linking community-based monitoring and integrative indicators (e.g., whole-river metabolism, production, food-web quality, thermal refugia, eDNA/bioassessment, and sensor and remote networks) to management actions, including Indigenous-informed decisions. We begin the session with examples from the Kootenai River collaborative group to illustrate how integrative science and partnership-based approaches can connect ecosystem processes to restoration effectiveness in floodplains, side channels, and riparian areas, and to provide a common context for the talks that follow.

id #134851

Protecting aquatic habitats in Minnesota: Science, standards, and stakeholders

Will Bouchard¹

1. *Minnesota Pollution Control Agency, MN, United States*

The Minnesota Pollution Control Agency's (MPCA) goal is to ensure that Minnesotans have healthy air, sustainable lands, clean water, and a better climate through the prevention and reduction of air, land, and water pollution. To achieve this objective, the MPCA employs dedicated scientists, technicians, attorneys, and support staff and collaborates with other Minnesota agencies, local partners, academics, other states, and federal agencies. I will discuss my career as an aquatic entomologist, water quality standards rule coordinator, and scientist employed by the agency responsible for implementing the Clean Water Act in Minnesota. The focus of my work at the MPCA is the development and adoption of water quality standards and their implementation within the state's aquatic resource management programs (e.g., permitting, total maximum daily loads). I will provide an overview of some projects in which I have been involved including the development and adoption of biological criteria, Tiered Aquatic Life Uses, eutrophication standards, and site-specific standards. In addition to research, my position at the MPCA also includes writing rule language and navigating a maze of administrative procedures to promulgate water quality standards into Minnesota rules. This includes collaboration and input from state, federal, local, and academic partners which complicates but also strengthens the final rules. The administrative work can be challenging for a scientist, but this step translates the science to implementable standards that directly protect and restore water quality. My presentation will discuss the challenges and rewards of working in a non-academic career that includes technical, scientific research and collaboration with diverse stakeholders to develop protective water quality standards that directly benefit the quality of aquatic resources and the lives of the public.

Diatoms.org: training, transforming, transcending

Sarah Spaulding¹

1. INSTAAR, University of Colorado, CO, -

Diatoms play essential roles in aquatic ecosystems; they produce an estimated 20–30 percent of Earth's atmospheric oxygen, regulate major biogeochemical cycles, and provide key information used in water quality assessment and environmental monitoring programs. Accurate identification and ecological understanding of diatoms are therefore fundamental to biodiversity conservation, public health protection, and natural resource management.

However, substantial portions of diatom diversity and ecological function remain undocumented. Taxonomic research, species descriptions, and the maintenance of high-quality identification resources are chronically underfunded, and governmental agencies generally do not prioritize support for taxonomy or foundational ecological work. This lack of stable funding has led to significant gaps in scientific knowledge, limited training opportunities, and insufficient infrastructure for managing diatom biodiversity data.

The diatoms.org project includes expert taxonomists, students, early-career scientists, and analysts who contribute peer-reviewed "species pages", biogeographic records, and high-resolution light and scanning electron micrograph imagery. The platform supports environmental monitoring programs, academic researchers, state and federal agencies, and the public by providing tools for species identification, comparative analysis, distribution modeling, and taxonomic verification. We reach the public through simple and compelling leads into science through imagery and art projects, and in the process, develop collaborations with designers, students, and artists. Diatoms.org plays a key role in training and professional development through the Diatom Web Academy, the Society for Freshwater Science Taxonomic Certification Program and field station courses. While most current content focuses on freshwater diatoms from North America, diatoms.org is laying the groundwork to expand the platform to more complete freshwater and marine species, extending its relevance from watersheds to coastal systems.

Monitoring benthic macroinvertebrates in urban stream restorations

Chris Ruck¹, Jonathan Witt¹

1. Fairfax County DPWES Watershed Assessment Branch, VA, United States

Does urban stream restoration improve biological function? Fairfax County monitored benthic macroinvertebrates at municipal restoration projects implemented from 2010 to present. This presentation discusses the evolution of a project-specific monitoring program, its status, and "early" results. Overhauled in 2015-16, the current monitoring framework needs at least seven years before assessing the post-restoration biological condition of sites. Supplemented by the county's annual monitoring, this data set has over 75 proposed or constructed stream projects with restoration benthic monitoring data; and over 25 streams with pre- and post-restoration information. These data are compared to countywide averages, and pre- vs. post- analyses of individual projects. This presentation discusses trends and changes to macroinvertebrate assemblages that may be attributed to urban stream restoration. Further, this presentation discusses optimal experimental designs needed for restoration monitoring, factors to be applied to the data, and how these data may help direct restoration implementation.

Carbon reactivity in freshwaters across the continental US was related to climate and organic matter composition

Jake Hosen¹, Tingyu Hou¹, Joanna Blaszcak², Alexander Collins¹, Robert Hensley³

1. Purdue University, Lafayette, INDIANA, United States

2. Natural Resources & Environmental Science, University of Nevada, Reno, Nevada, USA

3. NEON, Battelle, Boulder, Colorado, USA

The fate of organic matter in natural waters influences microbial activity, food webs, water clarity, and aquatic photosynthesis, among other ecosystem impacts. Organic matter quantity and composition change in response to ultraviolet photodegradation, photosynthetically active radiation (PAR), and microbial metabolism individually. Yet, very few studies have examined how water samples respond to all of these processes individually.

To address this knowledge gap, we obtained water samples from 19 NEON lake, wetland, stream, and river aquatic sites spanning the continental United States. We subjected each sample to the following treatments: 1) incubated in the dark, filtered to 0.2 μm , 2) incubated in the dark, unfiltered, 3) incubated in a solar simulator, filtered to 0.2 μm , 4) incubated in a solar simulator, unfiltered, 5) incubated under PAR without ultraviolet light. This allowed us to estimate the impact of respiration, photodegradation, and photosynthesis + respiration individually for each sample processed. We used the change in total dissolved organic carbon (DOC) to measure net change in carbon production. We used ultraviolet-visible (UV-Vis) absorbance and fluorescence to characterize dissolved organic matter (DOM) composition changes.

We found that the amount of DOC degraded due to respiration increased with the amount of protein-like, low aromatic DOM present within a sample. Photodegradation was conversely positively correlated with the aromatic DOM content in a sample as measured by metrics including specific ultraviolet absorbance at 254 nm (SUVA₂₅₄). Climatological factors were also important predictors of DOC transformation. Increased solar insolation was related to decreased photodegradation and biodegradation of DOC. Higher rates of photosynthesis were correlated with sites with high forest and grassland watershed land cover. Overall, our results show that DOM composition is more responsive to photosynthesis, respiration, and photodegradation than is DOC quantity.

Urban stream reflections: What can we learn from long-term waterway management and research?

Brian Murphy¹, Mateo Scoggins², Christopher Herrington³, Rhys Coleman⁴

1. *River Network, Denver, CO, United States*

2. *City of Bellingham, Bellingham, WA*

3. *Department of Public Works and Environmental Services, County of Fairfax, Fairfax, VA*

4. *Melbourne Water, Melbourne, Australia*

Urban waterways are under increasing pressure from a range of anthropogenic threats, including urbanization and climate change, resulting in rapid loss of freshwater biodiversity in comparison to other habitats. In response, there has been an increasing investment globally to protect and improve the health of urban waterways. Historically waterway management has had a narrow focus, most typically on drinking water supply, flood protection, or water extraction to support agricultural activities. In recent decades, in many parts of the world, this scope has expanded to the management of waterway health more holistically, including pollution reduction, riparian and instream habitat improvement, and the provisioning of instream connectivity for animal migration. Management considerations also incorporate social values, such as amenity, recreation, community connection, and cultural values. In the face of the expanding scope of waterway managers, urgency to act, and the need to understand the complexity of social-ecological context in which to waterway managers typically operate, it is more important than ever for waterway managers to be able to understand drivers, opportunities, and priorities.

Inspired by discussions prior and during the SUSE6 in Brisbane, Australia, this presentation will describe approaches and lessons from practitioners and researchers from the USA, Australia, and France that each have had decades of experience in waterway management, and research. These lessons are captured through the use of case studies. Covering core themes across a waterway management cycle, this presentation will also outline a strategic waterway management planning framework that incorporates input from community and researchers, insights from long-term collaborative research in waterway management, and spatially-explicit data and models for improved decision-making, engagement, and monitoring.

Biogeomorphic feedbacks along the Middle Green River, Utah: Implications of flow management on vegetation encroachment and channel narrowing

Catherine M. McClure¹, Christopher L. Holmquist-Johnson², Jonathan M. Friedman², Dusty W. Perkins³, Luke Gommerman³, Melissa Trammell⁴, Ryan R. Morrison¹, Catherine M. McClure¹

1. *Colorado State University, CO, United States*

2. *U.S. Geological Survey, Fort Collins Science Center, Fort Collins, Colorado, USA*

3. *National Park Service, Northern Colorado Plateau Network, Moab, Utah, USA*

4. *National Park Service, Grand Junction, Colorado, USA*

In the Colorado River watershed, decreases in peak flows have coincided with channel narrowing and degradation of habitat for native fishes. Along the middle Green River, decreases in flow magnitude and duration have promoted proliferation of riparian vegetation resulting in sediment deposition and channel narrowing. We analyzed five years (2020-2024) of riparian plant data coupled with plot inundation-duration and elevation data for approximately 2,000 1m-X-1m plots at six river sites on the middle Green River. Mean percent vegetative cover of plots near the channel increased during moderately dry and average flow years and decreased in the one moderately wet year. Additionally, we observed an increase in elevation of more densely vegetated plots compared to sparsely vegetated plots in wetter years. Plots that increased in elevation were dominated by a mix of annual herbs and grasses as well as perennial woody species, particularly coyote willow (*Salix exigua*) with lesser abundance of Fremont cottonwood (*Populus fremontii*) and tamarisk (*Tamarix* sp.). While the annuals have little opportunity to influence sediment deposition, the stems of the perennial woody species trap sediment and the roots reduce erosion. These results suggest that woody plants established in dry years subsequently trap sediment in wetter years. This process may explain why multi-year droughts have been strongly associated with narrowing along the Green River. Annual drone imagery of the same 20km reach of the Green River confirms this relationship between reduced flow and channel narrowing. Findings from this study suggest that flow peaks with high magnitude and duration can promote wide, complex channels which ultimately improve habitat for threatened and endangered fishes.

Together, we find a way: Conserving freshwater resilience at The Nature Conservancy

Peter S Levi¹

1. *The Nature Conservancy, Ashland, WI, United States*

The Nature Conservancy (TNC) has developed bold, ambitious goals to advance meaningful progress towards mitigating the global crises of climate change and biodiversity loss. Our 2030 Goals provide a pathway for sustaining and, where necessary, restoring our natural systems based on ecological and social science. Regarding freshwater ecosystems, TNC seeks to conserve 1 million kilometers of rivers and 30 million hectares of wetlands and lakes by 2030. We have achieved 17.3% and 51.1% of those goals to date, respectively. Most of our conservation goals are achieved through (1) direct action by TNC, (2) collaboration with diverse partners, from academics and aligned non-profit organizations to federal and state agencies, and (3) indirect influence via policy change. Our on-the-ground work touches down in 40 countries and we have partnerships in an additional 41 countries, which provides us with the opportunity to make meaningful global change through local action. As a science-based organization, we prioritize our work across watersheds using a myriad of western and traditional ecological knowledge and

geospatial data. For example, the “Resilient River Explorer” identifies the watersheds, streams, and rivers to protect or restore to sustain freshwater biodiversity given the threats posed by global environmental changes. In this moment, the threats to our natural systems are great. But, together, we find a way.

id #134854

Elevated carbon gases in beaver ponds: spatial and temporal trends across southern New England

Lev Becker¹, Cassandra Hryniewicz¹, Skylar Ross¹, Ashley Helton^{2, 1}, Beth Lawrence^{2, 1}

1. Department of Natural Resources and the Environment, University of Connecticut, Storrs, Connecticut, USA

2. Center for Environmental Science and Engineering, University of Connecticut, Storrs, Connecticut, USA

Through their activity as ecosystem engineers, North American beavers (*Castor canadensis*) are anticipated to play an increasingly important role in landscape carbon cycles as their populations expand. However, data on carbon gas dynamics from beaver-modified ecosystems is sparse, and broader trends across these spatially heterogeneous and seasonally variable ecosystems are poorly resolved. To examine how concentrations of dissolved carbon gases change as water flows through beaver-modified habitat, we sampled thirty beaver-modified ecosystems across southern New England, USA in late summer 2024 and chose a subset of five sites for twice-monthly temporal sampling from spring to early fall of 2025. At each beaver pond and associated inlet and outlet streams, we measured dissolved methane and carbon dioxide concentrations from surface water, alongside in situ surface water quality (temperature, pH, specific conductivity, and dissolved oxygen). Across all 30 sites, methane and carbon dioxide were highly variable, (ranging 5 and 3 orders of magnitude, respectively). Methane was six times higher in ponds than inlets and slightly elevated in outlets. Likewise, carbon dioxide was twice as high in ponds than in inlets or outlets. Water temperature tended to be higher in ponds and outlets, and dissolved oxygen was lower in ponds than inlets or outlets. Time series data show methane and carbon dioxide increasing as water temperature rises and as dissolved oxygen drops over the course of the growing season in beaver ponds. These results indicate that southern New England beaver ponds are active environments for anaerobic processes associated with carbon gas emissions, and next steps include investigating the underlying biotic processes involved in the carbon cycle within these ecosystems.

id #135878

Emergent concentration-discharge (C-Q) relationship across nested subcatchments in a coastal plain watershed.

Andrew A Ali¹, Arial J Shogren¹

1. Department of Biology, The University of Alabama, Tuscaloosa, Tuscaloosa, AL, United States

Watersheds are dynamic systems in which carbon and nutrient exports vary with hydrologic conditions and subcatchment characteristics, reflecting shifting connectivity between terrestrial sources and stream networks. Concentration-discharge (C-Q) relationships are widely used to examine these dynamics across spatial and temporal scales and infer whether solute export is controlled by transport/source limitation or hydrologic mixing. However, most C-Q frameworks are derived from event-focused studies, leaving uncertainty about how seasonal hydroclimate and landscape position regulate solute behaviour, particularly in low-gradient, subtropical Coastal Plain watersheds. Here, we examined seasonal and spatial variability in C-Q relationships for dissolved organic carbon (DOC), total dissolved nitrogen (TDN), and phosphate (PO_4^{3-}) across three nested subcatchments in the South Sandy watershed (~125 km²), in central Alabama, USA. We collected biweekly water samples paired with concurrent discharge measurements (June 2024-July 2025). We then fitted log-log C-Q regressions by site and seasonality to quantify slope magnitude and direction, enabling comparison across subcatchments and seasons. We found that DOC consistently enriched at high flows (positive C-Q slope), particularly during wet periods in spring, indicating transport-limited mobilization from organic-rich soil. In contrast, TDN and PO_4^{3-} showed chemostatic (near-zero slopes) to negative values, reflecting source-limited behavior in higher flows. Differences among site pairs reflected landscape position, with headwater subcatchments showing stronger seasonal responses than the outlet, where hydrologic mixing neutralized C-Q signals. Results from our study will provide insight into solute source dynamics in low-relief Coastal Plain watersheds.

id #136134

Catching what we miss: what eDNA metabarcoding adds to endangered shiner monitoring in the Brazos River watershed

Zacchaeus G. Compson¹, Kaley Cave¹, Lindsey Davis¹, Michael Curtis¹, Chase Nimee², Krista Ruppert¹, Tobin J. Davidson¹, Grace Deckard¹, Angel Montenegro¹, Pedro Rodriguez¹, Carmen G. Montaña², David J. Hoeinghaus¹

1. Department of Biological Sciences, Advanced Environmental Research Institute, University of North Texas, Denton, Texas, United States

2. Department of Biology, Stephen F. Austin State University, Nacogdoches, Texas, United States

Sharpnose Shiner (*Notropis oxyrhynchus*) and Smalleye Shiner (*N. buccula*) are Brazos River endemics that have undergone range contractions and were federally listed as endangered in 2014, motivating an urgent need for robust, long-term monitoring that can track populations, habitat associations, and responses to changing flow and other stressors. Here, we detail a multi-year project that integrates standardized traditional fish sampling with novel, DNA-based approaches, including environmental DNA (eDNA) metabarcoding and the use of loop-mediated amplification (LAMP) assays to (1) optimize survey design for these focal minnows and their co-occurring fish assemblages, (2) support data-driven recommendations to agencies and stakeholders, and (3) develop transferable field and analytical tools (e.g., occupancy-detection modeling frameworks) that can be adopted for long-term, repeatable monitoring. We present project highlights with an emphasis on what we are learning by pairing capture-based

surveys with genomics-enabled detection. Specifically, we synthesize emerging insights from (i) cross-method comparisons of taxonomic richness and community recovery, (ii) spatial and temporal patterns in fish assemblage composition across sites and seasons, as well as the environmental drivers of these patterns, and (iii) the degree of concordance (and mismatch) between traditional sampling and eDNA-based approaches for monitoring focal endangered shiners. Collectively, these findings reinforce the value of an integrated monitoring strategy: genomics tools, like eDNA metabarcoding, add sensitive, non-invasive, repeatable, community-level detection that strengthens occupancy inference and can complement and, in some cases, improve upon capture methods—especially for endangered species where handling risk, low density, and imperfect detection complicate conventional monitoring.

id #135879

Delivery of carbon gases by preferential groundwater discharge across a river network

Ashley M Helton¹, Alaina M Bisson², Eric M Moore¹

1. *University of Connecticut, CT, -*

2. *Harvard University, Cambridge, MA, United States*

Streams play an important role in the global carbon cycle, in part by emitting carbon gases (CO₂ and CH₄) to the atmosphere. Though we know that substantial amounts of carbon gases emitted from streams are derived from groundwater, we understand less about the spatial variability of groundwater carbon delivery to streams. Thus, our objective was to investigate the spatial patterns of groundwater carbon gas concentrations discharged to streams by linking spatially extensive surveys of groundwater discharge with measurements of groundwater discharge carbon gas concentrations. We used handheld thermal infrared imagery to conduct surveys of groundwater discharging directly into streams from exposed streambanks, based on the thermal contrast between groundwater and surface water. Groundwater discharge samples were collected (393 individual samples collected from 313 locations between 2019 and 2021) across >60 km of stream length surveyed throughout the Farmington River watershed in Connecticut and Massachusetts, USA. We found that CO₂ and CH₄ concentrations in groundwater discharging to streams are spatially heterogeneous, varying by 1.9 and 4.6 orders of magnitude, respectively, across the stream network. We linked sampling locations to modelled locations of groundwater discharge and used existing groundwater flow path models to link our sampling location with contributing groundwater characteristics and land cover at the location of groundwater infiltration. Best fit linear mixed effects models for CO₂ and CH₄ explained 31% and 49% of variation, respectively, and included groundwater discharge chemistry (dissolved organic carbon, dissolved oxygen, and nitrate), percent riparian forest land cover, percent infiltrating land cover (forest for CO₂ and wetland for CH₄), and modelled groundwater flow path characteristics (residence time for CO₂ and groundwater discharge rate for CH₄). This study highlights the spatial variability of groundwater carbon gas concentrations in a mixed land-use watershed, and underscores the importance of understanding the multi-scale spatial controls on groundwater discharge carbon gas dynamics.

id #136135

Influence of Reservoir Structure on Fish Community, Development, and Trophic Interactions in Puerto Rico

Angel Sebastian Estruche Santos¹, Omar Perez-Reyes¹

1. *University of Puerto Rico Rio Piedras Campus, San Juan, PUERTO RICO, Puerto Rico*

Across the 20th century, Puerto Rico faced significant development of freshwater systems through channelization, embankment, and dam projects. The damming of rivers create a novel environment that affects river continuity and the affinity of species to aquatic environments above, in, and below the dam. This study focuses on the influence of reservoir structure on species abundance, growth, and trophic interactions within the Curias Reservoir and Isabella Retention Pond, a spillway-embankment and a retention pond, respectively. The reservoirs differ in use, structure, age, and connection to freshwater ecosystems. The methods consist of long-term censuses conducted across each reservoir: the littoral zone, the transitional zone, the inlet, and the outlet. The captured fish and decapods were measured, weighed, photographed, and dissected to quantify stomach contents. Both reservoirs were dominated by introduced cichlid species and other large-bodied, low-trophic-level fish, such as tilapia cichlids, midas cichlids, and armored catfish. The Curias Reservoir has a greater diversity of fish and decapods, and it supports native shrimp species. The species assemblage within the Isabella Retention Pond was dominated by non-native fish and decapod species, including the Australian redclaw crayfish. The Curias Reservoir connection to the main river channel, the natural littoral zones (vegetation and sediment), and a funnel-shaped spillway structure allow colonization by native decapod species. The fish show intra- and interspecific variation in weight, length, and shape, including snout projection and body depth, between both locations. The fish from the Curias Reservoir had significantly greater weights at equal lengths than those in Isabella Retention Pond. River embankment cichlids were more robust, showed slower growth scaling, and had distinctive reproductive traits. Moreover, the fish assemblages within Isabella Retention Pond showed greater constraints on functional diversity and traits than those in the Curias Reservoir. The Curias Reservoir fish stomach contents demonstrated greater niche partitioning and trophic levels, with representation of piscivore, herbivore, and detritivore species; however, both reservoir types were dominated by generalist and benthic foraging species. These results indicate how reservoir structure, stability, and connectivity strongly mediate species colonization, growth, functional diversity, and trophic interactions within fish communities.

id #135880

Prioritizing watersheds to support aquatic biodiversity under changing climates

Allison Roy¹, Graziella DiRenzo¹, Rebecca O'Brien², Jennifer Rogers³, Jason Carmignani⁴, Rebecca Quiñones⁴, Todd Richards⁴

1. U.S. Geological Survey, Massachusetts Cooperative Fish and Wildlife Research Unit, Amherst, MA, United States
2. Pennsylvania Cooperative Fish and Wildlife Research Unit, Pennsylvania State University, State College, Pennsylvania, United States
3. Narragansett Bay Estuary Program, Roger Williams University, Providence, Rhode Island, United States
4. Massachusetts Division of Fisheries and Wildlife, Westborough, MA, United States

Freshwater ecosystems are facing a biodiversity crisis due in part to a suite of anthropogenic threats and exacerbated by changing climates. To mitigate biodiversity loss, conservation managers are tasked with distributing limited resources. Limited resources suggest the importance of geographic prioritization of conservation-related activities. Understanding biotic distributions under current and future climates can aid in identifying areas for protecting biodiversity. We assessed the distribution of freshwater mussel (12), fish (53), and turtle (1) species across six Northeastern states (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont). For each of the three taxonomic groups, we created habitat suitability maps using survey data and corresponding geographic, climatic, and land use data (or, in the case of wood turtle, used published maps), and we identified groups of species that were most vulnerable to changing temperature and streamflow. For freshwater mussels, we identified watersheds as high priority for land protection based on greater overall species richness, higher percent forest and wetland cover in the watershed, and higher climate resilience. For fishes, the novel biodiversity groups had different distributions than traditionally-used species guilds (temperature guilds, habitat guilds) and may offer a novel approach for identifying vulnerable species. Forest cover throughout the watershed was critical for sustaining freshwater fish and mussel species that are vulnerable to climate change. In general, watersheds with highest richness of sensitive mussels, proportional abundance of native fishes, and prevalence of wood turtles were non-overlapping, suggesting that efforts to support one group may not be sufficient to benefit all taxa, limiting the usefulness of umbrella species in this region. The broad-scale species distribution maps can be used by managers to identify watersheds for developing more focused, site-specific management actions.

id #135881

Wading in to wetland monitoring together: How cross-sector collaboration co-creates knowledge for better water quality

Lauren E Kinsman-Costello¹, Olivia Schloegel¹, Raissa Marques Mendonca¹, Larrick Siena², Klukow Mary Ellen¹, Kenneth Anderson¹, Kerns Janice³, Saas Eric³, Winslow Christopher⁴, Fussell Kristen⁴

1. Kent State University, Kent, OH, United States
2. Bridgewater State University, Bridgewater, MA, United States
3. Ohio Department of Natural Resources, Huron, OH, United States
4. Ohio State University, Columbus, OH, United States

The H2Ohio Wetland Monitoring Program (WMP) was established in 2020 to evaluate the effectiveness of wetland construction, restoration, and enhancement projects supported by the Ohio Department of Natural Resources as part of a state-wide water quality improvement program, the "H2Ohio Initiative." The primary focus of the water quality initiative is to minimize recurring harmful algal blooms in Lake Erie and other freshwater ecosystems by mitigating excess nutrient loading. The WMP is led by a multi-university team affiliated with the Lake Erie and Aquatic Research Network (LEARN). Since its inception, the WMP has collected data from over 45 of the hundreds of H2Ohio wetland projects and produced results that directly inform state-wide investments and local strategies in wetland restoration and management to mitigate nutrient loads. Our researchers have engaged with policy makers and practitioners at multiple decision-making levels, from agency directors and members of the governor's office, to state agency staff and local conservation partners and managers. We have collaborated with geographically distributed partners in most of the major watersheds of Ohio in both the Lake Erie and Ohio River basins. Our distributed engagement within a harmonized framework has strengthened the capacity for more effective water quality management by building trust among our stakeholders, investing in data management and documentation standards, and co-creating meaningful communication products. We will outline the strategies employed, tactics implemented, and practical lessons learned in the first five years of the WMP. The WMP is at an inflection point after experiencing a major shift in funding, and we will also preview our map for a future that continues to cultivate monitoring, research, and management. Our experiences demonstrate the necessity of cross-sector collaboration to effectively produce robust, relevant, and actionable freshwater science.

id #134858

Moving mud: Ecosystem engineering by an invasive crayfish alters stream bank stability and water condition regardless of disease expression

Anna C French¹, Samuel F Fritz¹, Hayley C Oakland¹, Stacy Schmidt¹, Ainsley Nystrom¹, Geoffrey C Poole¹, Lindsey K Albertson¹

1. Montana State University, Bozeman, MONTANA, United States

Invasive species and disease are among the greatest contemporary threats to aquatic biodiversity and ecosystem functioning. The virile crayfish (*Faxonius virilis*) is an invasive ecosystem engineer that exhibits behaviors including burrowing and sediment transport and is experiencing outbreaks of crayfish plague. Diseased individuals can develop external lesions that may influence ecosystem engineering ability. We elucidated how crayfish engineering behaviors may impact stream biogeochemistry and bank stability using crayfish with or without visible injury. Male, virile crayfish were collected outside their native range in southwest Montana for use in a 24-hour study with mesocosms containing manually formed clay banks and stream water. We monitored burrow size, sediment movement distance, turbidity, and concentrations of stream nutrients including nitrate, nitrite, ammonium, and phosphate. We found that crayfish substantially disturbed the clay and reduced water clarity. All crayfish in this study exhibited burrowing behaviors and full bank collapses due to bioturbation occurred within 24 hours in over 90% of the crayfish trials. Crayfish with lesions burrowed as frequently and to a similar extent as crayfish without lesions. Instead, the magnitude of sediment changes was positively correlated to crayfish body size. The movement of stream bank sediment into main channels and weakening of bank structure has the potential to strongly influence concentrations of dissolved nutrients in streams experiencing

high crayfish activity. As virile crayfish continue to invade new freshwater habitats and show novel patterns of disease expression, improved understanding of how engineering activities modulate abiotic stream conditions can improve predictions of the effects of future invasions on stream health, community structure, and ecosystem function.

id #135882

Organic Matter Composition and Turnover in Ephemeraally Wetted Soils and Hyporheic Sediments of the McMurdo Dry Valleys, Antarctica

Sophia M. Gomez¹, John E. Barrett¹, Warren S. Vidar², Rachel E.B. Reid³, Tyler J. Kohler⁴

1. *Biology, Virginia Tech, Blacksburg, Virginia, United States*
2. *Chemistry and Biochemistry, University of North Carolina Greensboro, Greensboro, North Carolina, United States*
3. *Geosciences, Virginia Tech, Blacksburg, Virginia, United States*
4. *Science Faculty, Charles University in Prague, Prague, Czech Republic*

Terrestrial organic matter content and composition is crucial to quantify and characterize, as soils act as an essential carbon sink but, due to mineralization, can also act as a carbon source to ecosystems and the atmosphere. The McMurdo Dry Valleys (MDV) of Antarctica are a polar desert ecosystem, characterized by extreme cold and dry conditions for the majority of the year. Yet, each summer, melt connects and reactivates desiccated communities of heterotrophic and autotrophic organisms concentrated in microbial mats. Hyporheic microbial mats in the MDV fix carbon and nitrogen essential to downstream productivity, and hyporheic soils represent an area of relatively high soil biological activity due to enhanced access to water. Hyporheic and ephemeraally saturated sediments could become increasingly significant as critical zones of productivity in the MDV, as enhanced melt leads to an expanded spatial distribution and longer duration of hyporheic saturation for microbial mat communities. Thus, it is crucial to characterize the composition and rate of turnover of organic matter in hyporheic and ephemeraally hydrated soils in the MDV to understand the controls on and dynamics of biogeochemical cycling within this ecotone. Since this region of Antarctica is devoid of vascular plants, all primary productivity and decomposition pathways are within microbial mats. External influences on the soil ecosystem in the MDV are minimal and well constrained. Therefore, MDV are an ideal system to describe and assess the biotic influence on the processing and composition of organic matter within soils that are ephemeraally wetted from snow melt or hyporheic processes. Using untargeted metabolomics on a Ultraperformance High-Pressure Liquid Chromatography Mass Spectrometry (UPLC-MS) and isotopic characterization (¹³C, ¹⁴C, and ¹⁵N) analyses, I aim to explore the influence of microbial mats on soil biogeochemical cycling. Through the identification of metabolites and biomarkers within the soils, I seek to elucidate and describe the chemical composition of the organic matter deposited into the soils, facilitating predictions on the fate and processing of organic matter in both soils and adjacent stream ecosystems.

id #134347

The Presence of Regulated Crayfish Species in Aquarium Stores in the Great Lakes States

William Ota¹, Rachel Feagley¹, Brian Roth¹

1. *Michigan State University, MI, United States*

The aquarium trade is a known pathway for a variety of non-native and invasive species, and it is of increasing concern to managers and researchers. Non-native crayfish are increasingly available to individuals through retail markets as a part of the aquarium trade. Due to concerns about invasion by non-native and invasive crayfish species, states in the Great Lakes region regulate the possession and sale of some of these species. We conducted aquarium and pet store visits across six Great Lakes states (IL, IN, MI, NY OH, & PA) to survey the species present and assess employees' knowledge of crayfish regulations and identification. At each location, the species present were identified, and employees were offered a survey to complete. We found that ~30% of surveyed aquarium stores sold crayfish. Eight different species were present across all surveys, with Everglades crayfish (*Procambarus alleni*) being the most common. Two species of regulated crayfish (red swamp crayfish and marbled crayfish) were found during retailer visits. The frequency of regulated species and crayfish varied across states from 0% of stores selling crayfish in Pennsylvania to over 40% of stores in Michigan. Retail employee surveys revealed several valuable insights into their level of knowledge surrounding crayfish and their regulations. Forty percent of retailers surveyed indicated that they received unrequested (hitchhiker) crayfish, and the majority of these retailers did not attempt to identify or simply gave away hitchhikers to patrons. Our survey indicated that employees rarely identified crayfish species due to a lack of time and knowledge to classify morphologically similar species. This project demonstrates that despite recent regulations prohibiting invasive crayfish, they are both present in the retail trade and stores may require additional training to assist in the implementation of these regulations.

id #134859

Investigating spatial and temporal dynamics in the “fire pulse” across 70 years of monitoring freshwater ecosystems

Jeremy M Brooks^{1,2}, Colden V Baxter³, Kellie Carim⁴, Mark Coleman⁵, Lisa Eby¹, Becca Jonswold⁵, Dan Preston²

1. *Wildlife Biology, University of Montana, Missoula, MT, United States*
2. *Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, CO, United States*
3. *Idaho State University, Pocatello, ID, United States*
4. *US Forest Service, Rocky Mountain Research Station, Aldo Leopold Wilderness Research Institute, Missoula, MT*
5. *Northern Water, Berthoud, CO*

Wildfires can be either a threat or a boon to freshwater biodiversity and ecosystem function, and understanding the complex drivers leading to these contradictory outcomes is needed to support more resilient ecosystems. One prominent way fire affects freshwater ecosystems is the “fire pulse,” or a post-fire increase in bottom-up productivity driven by increased nutrients and light availability. Decades of research show that the fire pulse can be spatially and temporally variable and inconsistently propagate through food webs (e.g., periphyton-invertebrates-fish or phytoplankton-zooplankton). This has been linked to variation in 1) fire severity; 2) changes in light following the loss and regeneration of riparian vegetation; 3) nutrient and light-limiting ash inputs; and 4) community traits and interactions that direct energy flow. Spatial and temporal variation in the fire pulse raises the possibility that contradictory responses among patches may contribute to a dynamic portfolio of ecological responses across the riverscape. Theory suggests that such a portfolio may increase the resiliency of affected ecosystems.

We evaluated this possibility by leveraging three studies with 70 years of monitoring from the Bitterroot River watershed (MT; 15 years post-fire), Big Creek watershed (ID; 37 years encompassing multiple fires), and Three Lakes reservoir-complex (CO; 18 years with 5 post-fire). The stream studies evaluated changes in habitat, aquatic invertebrates, fish-centric food webs, and aquatic-terrestrial linkages, and the lake study considered nutrient inputs and responses by plankton.

Collectively, the fire pulse was spatially and temporally variable, and the affected foodweb pathways and ecosystem functions exhibited diverse responses. In the Bitterroot River watershed, fire was associated with elevated invertebrate drift and caloric content of fish gut contents for most sites, but experienced limited increases in aquatic insect emergence. Across the Big Creek watershed, chlorophyll-*a* and benthic invertebrate biomass were elevated post-fire, but the length of this pulse ranged from a few years to over a decade. The Three Lakes reservoir-complex experienced nutrient fluxes, but the pulse inconsistently propagated to phytoplankton and zooplankton. Spatial and temporal dynamics in the fire pulse contributed to a portfolio of post-fire responses, which was associated with the maintenance of biodiversity and ecosystem function across the riverscapes.

id #136139

Wood, Water, and Wildlife: Avalanche-Delivered Wood Accumulations Support Elevated Fish Abundance and Riparian Bird Species Richness

Anna Reside¹, Colden Baxter¹

1. Idaho State University, Pocatello, IDAHO, United States

Large woody debris (LWD) and large wood accumulations (logjams) play a critical role in creating and maintaining habitat complexity in river corridors. However, the relationship between large wood, habitat complexity, and biodiversity is poorly resolved. Here, we assess how logjams influence aquatic and riparian vertebrate diversity in Big Creek, a 6th-order wilderness river in central Idaho, USA. Big Creek is one of the few remaining montane rivers with both an intact disturbance regime and full faunal assemblage. It therefore represents a rare opportunity to examine how disturbance-driven processes shape ecological communities in the relative absence of anthropogenic forces. Following a 2014 avalanche, thousands of standing dead trees were deposited into the upper reaches of Big Creek. Subsequently, that wood has been mobilized downstream, accumulating in large logjams and forming localized side channel, backwater, and riparian habitats that are used by a variety of organisms. Here, we map patterns of wood recruitment, mobility, and retention throughout the river network using aerial imagery (2013–2023) and pair these data with an intensive biodiversity assessment conducted at twelve logjam habitat complex- and three reference- sites during summer 2025. Using a combination of snorkel surveys and point counts, we quantified diversity and abundance of birds and fishes at each site. Our preliminary results indicate that logjam habitat complexes support 2.9 times higher overall fish density and 16 times higher juvenile fish density than reference reaches. Although mean fish species richness did not differ between site types, logjam complexes supported a greater number of unique species–age class combinations than reference sites (20.7 vs. 14.3), reflecting the increased juvenile density. Riparian bird abundance did not differ between site types; however, bird species richness was nearly two times higher at logjam sites, which supported an average of 11.6 species compared to 6.3 species at reference sites. Together, these data indicate that logjams and their associated habitats are important determinants of both aquatic and riparian vertebrate community structure. We are expanding upon this research by exploring the impacts of logjams and logjam-associated habitats on invertebrate communities as well as the structure of community interaction webs.

id #136907

Migratory Shorebird Stopover Habitat Management Enhances Wetland Biogeochemical Function in the Mississippi Alluvial Plain Agroecosystem

Jason Taylor¹, Matthew T Moore², Frank E Johnson II², Amit Chatterjee³, Martin A Locke⁴, Jason D Hoeksema¹

1. Center for Water and Wetlands Resources, Department of Biology, University of Mississippi, University, MS, USA

2. U.S. Department of Agriculture, Agricultural Research Service, National Sedimentation Laboratory, Water Quality & Ecology Research Unit, Oxford, MS, USA

3. U.S. Department of Agriculture, Agricultural Research Service, National Laboratory for Agriculture and the Environment, Soil, Water & Air Research Unit, Ames, IA, USA

4. U.S. Department of Agriculture, Agricultural Research Service, National Sedimentation Laboratory, Oxford, MS, USA

The historical ecological state of the Mississippi Alluvial Plain (MAP) was a vast bottomland hardwood forest with seasonal widespread inundation from the Mississippi River. Today the MAP is an extensive, highly productive, row-crop agricultural region that is protected from flooding by significant hydromodification. The MAP and other similar modern agroecosystems represent coerced complex adaptive systems with uncertain ecological resilience due to their relatively new ecological condition. While quantitative measures of resilience for highly modified agroecosystems are elusive, finding ways to insert biogeochemical function provided by wetlands back into agroecosystems is key to maintaining dynamic ecological landscapes that enhance availability of clean water and provide habitat for wildlife, all while maintaining productive agricultural activities. We are working with farmers to evaluate how post-harvest flooding for migratory shorebird habitat influences nutrient biogeochemistry in row-crop fields across

a portion of the MAP. Our observations demonstrate that flooding fields in the fall support the rapid buildup of benthic macroinvertebrate densities that provide a stopover food resource for migratory shorebirds. Flooding after corn harvest resulted in net denitrification rates that were sustained throughout the inundation period and culminated in an estimated 33-45 kg ha⁻¹ of excess N removed. Lower denitrification benefits were associated with flooding soybean fields compared to corn fields. Flooding reduces soil nitrate (NO₃-N) while increasing soil ammonium (NO₄-N) over time in corn and soybean fields. Flooding also led to reductions in mean storm event runoff concentrations for suspended solids, NO₃-N, phosphate (PO₄-P), and total phosphorus (P). Comparisons of soil nutrients pre- and post-flooding indicated that flooding led to lower plant available P in soils, but the degree of this reduction was relatively minor, and was only observed in fields flooded for the entire fall-winter sequence. No yield impacts attributed to managed flooding for migratory shorebird habitat have been reported by 10 different participating farmers across 19 locations. Our results suggest that novel timing of shallow water habitat management provides temporary wetland structure and biogeochemical function that can provide critical shorebird habitat and support downstream water quality benefits, without impacting crop yields within MAP agroecosystems.

id #134348

Interannual flow variability and its effects on isotopic niche overlap in a regulated, high-elevation river

Kelley A Sinning^{1,2}, Helen Acosta², Brien Rose³, Yoichiro Kanno^{1,2}

1. Department of Ecosystem Conservation and Sciences, University of Montana, Missoula, Montana, United States

2. Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, Colorado, United States

3. Blue Valley Ranch, Kremmling, Colorado, United States

In arid and high-elevation streams of Colorado, dams are necessary for water supply. However, they can alter ecological structure and function with regulated flow regimes and anthropogenic oligotrophy. *Didymosphenia geminata* ("Didymo") is a single-cell, stalked diatom species that becomes abundant in phosphorus-limited stream conditions, often below dams. In response to low nutrients, thick extracellular stalks form from *Didymo* cells and can grow rapidly in attempts to acquire nutrients from the water column. It has been documented that *Didymo* mats can reduce habitat for macroinvertebrates and disrupt recreation activities. Additionally, through field and laboratory research it has been identified that *Didymo* is resistant to high-shear environments but grows preferentially in stable flow conditions. However, knowledge gaps exist about long-term food web responses to *Didymo*-dominated algal communities. The Lower Blue River in Kremmling, CO below the Dillon Dam and Green Mountain Dam is one such example where nuisance *Didymo* blooms have formed. From 2021 to present, algae, macroinvertebrate, and fish data have been collected every year in May, August, and October. More frequent algae data have been collected since 2023 with paired Benthotorch and velocity measurements. Using this dataset, we evaluated the differences in stable isotopes in macroinvertebrates and fishes across different flow events and seasons and infer the dominance of *Didymo* associated with other biological data. The study period encompassed drought and wetter-than-average years. We hypothesized that interannual flow variability may predict niche overlap between Brown Trout (*Salmo trutta*) and Mottled Sculpin (*Cottus bairdii*) because of significant negative relationships that exist between *Didymo* and high velocities and discharge which may enhance feeding opportunities for fishes. Preliminary results found that high interannual variability in seasonal patterns, not necessarily predicted by interannual flow patterns, did not inform stable isotope relationships between Brown Trout and Mottled Sculpin. Flow management on regulated rivers may encourage scouring of nuisance *Didymo* blooms while not affecting higher trophic level patterns, of which knowledge is limited. Yet, future further analysis of fish diets, macroinvertebrate assemblage, and fatty acid content will inform long-term ecological patterns on the regulated Lower Blue River.

id #134860

Macroinvertebrate and habitat responses to low-tech process-based stream restoration in northeastern Oregon

Tucker D Hoffman¹, David E Wooster¹, Sandra J DeBano¹

1. Fisheries, Wildlife, and Conservation Science, Oregon State University, Corvallis, Oregon, USA

Low-tech process-based restoration (LTPBR) techniques like beaver dam analogs (BDAs) are increasingly popular as a means for restoring degraded streams. One expected outcome of these methods is increased habitat heterogeneity, and a commensurate biotic response (e.g., increased macroinvertebrate richness). To test whether habitat and biotic communities respond to LTPBR as expected, we sampled macroinvertebrates and habitat variables at four stream restoration and paired unrestored control sites in northeastern Oregon, USA. These sites were restored with BDAs that were designed to benefit native salmonids by increasing habitat and water availability. We used targeted riffle (TR) and multi-habitat transect (MHT) sampling to collect macroinvertebrates and sampled early and late summer 2023 and 2024. We investigated whether restored reaches (1) had greater habitat heterogeneity than control reaches, (2) had greater macroinvertebrate richness/diversity than control reaches, and (3) whether there was a detectable relationship between habitat heterogeneity and macroinvertebrate richness/diversity. We predicted that habitat heterogeneity and macroinvertebrate richness/diversity would be greater in restored than control reaches. Additionally, we expected MHT samples to have higher habitat heterogeneity and macroinvertebrate richness/diversity than TR samples. Finally, we expected to find a relationship between macroinvertebrate richness/diversity and habitat heterogeneity. Contrary to our predictions, we found variability in velocity, cover, and substrate diversity were significantly lower in restored versus control reaches for MHT and we detected no differences in habitat responses for TR. Consistent with our predictions, macroinvertebrate richness and diversity were both higher in restored reaches than control reaches. However, this was only observed in TR samples, and not in MHT samples. We found that variation in velocity was significantly and positively related to macroinvertebrate richness and diversity. Thus, the lack of invertebrate response to restoration in MHT habitats may be because restoration decreased variation in velocity, rather than increasing it as predicted. These results indicate that restoration may not necessarily lead to the predicted changes in habitat variables that are expected to increase macroinvertebrate richness/diversity. They also suggest that macroinvertebrate communities in riffles may be responding to habitat variables that we did not measure. This study illustrates the complex nature of monitoring ecological stream restoration projects.

Diving in headfirst: Investigating inundation dynamics and nutrient cycling at a wetland restored on agricultural land

Hana Esber¹, Kenneth Anderson¹, Michael Back¹, Eileen Acosta¹, Jackson Miller¹, Lauren E Kinsman-Costello¹

1. Kent State University, Kent, Ohio

Wetlands are being constructed for ecological functions including wastewater treatment, flood abatement, habitat, and nutrient and pollutant removal within a landscape. Nutrients are stored within the sediment and vegetation of newly constructed wetlands, potentially removing excess nitrogen (N) and phosphorus (P) running off the landscape into freshwater bodies. Topography in constructed wetlands leads to variation in water level and soil moisture, a common control on wetland biogeochemical cycling. Under inundated and reducing conditions, denitrification can drive N removal from sediment, however there is potential for P release through the dissolution of iron oxides. Despite the differences between wetlands and their designs, water level remains a common control on the biogeochemical cycling that occurs, which makes understanding how man-made systems like constructed wetlands experience nutrient cycling between topographic features under varying hydrologic conditions crucial. The main questions for this study are: 1) Does nutrient flux in constructed wetland pools increase under a gradient of inundation status? 2) Will soils exhibiting a lower inundation status experience a release of ammonia and dissolved reactive phosphorus when rapidly inundated? To explore how future fluctuating hydrologic conditions may affect wetland nutrient cycling, we conducted an intact sediment core incubation at an H2Ohio wetland in northeast Ohio. The Trumbull Creek wetland is a former agricultural field, recently restored with diverse topographic features that experience variable levels of inundation throughout the year. Across inundation regimes, sediment and surface water exchange rates of NO_3^- ranged from $-2\text{--}3\text{ mg/m}^2/\text{day}$, NH_4^+ from $-500\text{--}1,000\text{ mg/m}^2/\text{day}$ and dissolved reactive phosphorus (DRP) from $-4\text{--}2\text{ mg/m}^2/\text{day}$. We observed that inorganic nutrient flux between sediments and surface waters was highly variable across inundation regimes and throughout time. In some instances, sediments switched from being a sink of DRP to a source during the second and third day of the incubation period. As climate change continues to affect inundation regimes in regions across the globe, exploring how more frequent and intense dry/rewetting periods change the duration of overlying surface water and nutrient cycling in wetlands will become increasingly important.

Aquatic breeding in a dry land: determining drivers of amphibian water site use and reproductive success in the Sonoran Desert

Sadie Roth¹, Matthew A Barnes¹, Kerry L Griffis-Kyle¹

1. Texas Tech University, Lubbock, TX, United States

Amphibians present a particularly complex conservation challenge due to their biphasic lifecycle (i.e., aquatic larvae and terrestrial adults), in which successful reproduction depends on the quality of multiple habitats that must co-occur in time and space. In a highly dynamic, arid system such as the Sonoran Desert, the alignment of terrestrial habitat conducive to adult activity and aquatic habitat suitable for larvae may occur only within a small window. Here, the water sites required for larval development are rare islands on a dry landscape and the monsoonal rains that provide moisture for terrestrial adult activity are extremely stochastic both spatially and temporally. Furthermore, as the climate changes, the timing and volume of monsoonal rains are expected to shift, and with it the timing and availability of breeding habitat for amphibians. Further altering their breeding resources, constructed waters have been built throughout the region to increase water availability for wildlife; however, these sites differ from natural sites in their structure and water chemistry and may not always provide quality larval habitat. As the impacts of climate change and water management practices alter the quality and timing of breeding resources for amphibians in the Sonoran Desert, identifying important habitat characteristics for adult breeding effort and tadpole occurrence may be vital to future conservation of populations. To investigate features of desert waters related to patterns of adult occurrence, breeding effort, and breeding success, we sampled waters in southwestern Arizona for amphibians in five years using visual, acoustic, and dip-net survey methods and implemented a multi-state occupancy model. We found higher probabilities of adult occurrence and breeding effort at natural water sites compared to constructed waters. Breeding effort was also more likely to occur with more precipitation, and the probability of breeding success increased with higher breeding effort. Thus, actions that target and protect natural water sites may be vital to support adult breeding effort, and thereby breeding success, to bolster amphibian conservation in arid lands through the impacts of climate change.

Long-term water quality trends in the headwaters of the Snake and Green River Basins

Samantha Dilworth¹, Ginger Paige¹, David Williams², Fabian Nippgen¹, Zhufeng Fang¹

1. Ecosystem Sciences and Management, University of Wyoming, Laramie, WY, United States

2. Botany, University of Wyoming, Laramie, WY, United States

Temperate mountain systems capture and store winter precipitation as snow and ice and slowly release it to downstream ecosystems and human communities during the warm season. Climate warming in these regions is shifting precipitation from snow to rain and shortening the duration and increasing the variability and magnitudes of spring runoff. These changes impact the chemical and physical properties of stream water in these systems, although knowledge of spatial and temporal patterns at the watershed and basin scales is limited. Our study seeks to address this knowledge gap by focusing on headwater streams of the Snake and Green Rivers in the Greater Yellowstone Area (GYA), a snow-dominated temperate mountain landscape where more than half of annual precipitation falls as snow. On average, temperatures in the GYA for the last two decades are higher than during any other period over the last 20,000 years. In the Upper Green River Basin in Wyoming, temperatures observed between 1950 and 2018 have increased by $1.7\text{ }^\circ\text{C}$, contributing to a 44% reduction in snowfall while total precipitation remained largely the same (Hostetler et al. 2021). We combine multi-decadal water quality observations (including pH, conductivity,

dissolved oxygen, and turbidity) with measured and modeled stream temperature and discharge to assess how climate driven changes in stream flow have altered the physical and chemical characteristics of water exported from the mountain headwaters of the Upper Green and Snake River Basins of Western Wyoming. By quantifying long-term changes in the lessening meltwater-driven streamflow and water quality, this work advances understanding of how cryospheric change is reshaping freshwater ecosystems in temperate mountain environments. These insights have broad relevance for aquatic habitat, downstream water quality, and water resource management, and contribute to a growing, cross-regional effort to understand the cascading effects of a changing cryosphere on freshwater ecosystems.

1. Hostettler, S., C. Whitlock, B. Shuman, D. Liefert, C. W. Drimal, and S. Bischke. 2021. Greater Yellowstone climate assessment: Past, present, and future climate change in greater Yellowstone watersheds. Montana State University, Institute on Ecosystems. (Available from: <https://scholarworks.montana.edu/xmlui/handle/1/16361>)

id #134863

Does size matter? Quantifying crayfish ecosystem engineering behavior across body and sediment sizes under global change

Lydia J L Bushey¹, Lindsey K Albertson¹, Stacy A Schmidt^{1,2}, Susan B Adams³, David A Schmetterling⁴

1. Montana State University, MT, United States

2. Montana Fish, Wildlife & Parks, Helena, MT, United States

3. USDA Forest Service, Southern Research Station, Oxford, MS, United States

4. Montana Fish, Wildlife & Parks, Missoula, MT, United States

Biological invasions are one of the most pressing threats to ecosystem stability and biodiversity in freshwater systems. Virile crayfish (*Faxonius virilis*) are an invader of increasing concern across Montana, USA where expanding distributions and ecosystem engineering behaviors (sediment sorting/burrowing) have raised questions about their ecological impacts. Crayfish burrows have the potential to buffer against climate stresses for other freshwater species by providing refuge in low flow conditions and facilitating biogeochemical cycling in the hyporheic zone, but can also augment channel erosion. These effects on sediment movement may intensify under global change if behaviors are altered or rates of activity are increased at higher water temperatures. To investigate how warming waters may impact *F. virilis*' burrowing behavior we ran a randomized fully-blocked laboratory experiment in insulated mesocosms at 15°C, 23°C, and 30°C (n = 27 per temperature, 81 total). Each contained either colored sands (D₅₀ = 1 mm; mimicking a stream bank) or gravels (D₅₀ = 7.5 mm; representing channel streambed surfaces) and one crayfish. Individuals were acclimated at their assigned water temperature for 24 hours, prior to exposure to the experimental substrates. Over just 24-48 hours, we found a strong relationship between body size and the mass of sediment an individual moved for both sands and gravels, but no clear effect of water temperature. Larger individuals selected larger rocks to excavate under, with many creating multiple burrows regardless of warm temperatures that exceed natural maximum summer temperatures by 1.35 times. Our findings demonstrate that *F. virilis*' sediment sorting behavior is driven by body size, and resistant to short periods of exposure to warming waters. This resistance to short-term stress could have implications for the future role of invasive crayfish in freshwater systems. As waters warm, if crayfish activity and burrowing behavior remains constant, they have the potential to act as either a strong stabilizing or destabilizing force on an ecosystem, regarding sediment movement. Future research could consider how this relationship shifts over longer periods.

id #136143

Whole-stream isotope additions reveal light and predator controls on nitrogen fluxes in Tropical stream food webs

Frank C Akamagwuna¹, Steve Thomas¹, Matthew Dunkle², Matthieu Bruneaux³, Andrés López-Sepulcre⁴, Sarah Collins⁵

1. University of Alabama, Tuscaloosa, Alabama, United States

2. Institute of Arctic Biology, University of Alaska, Fairbanks, Alaska, United States

3. Jyväskylä, Finland

4. Department of Ecology and Evolutionary Biology, Cornell University, Ithaca, New York, United States

5. Department of Zoology and Physiology, University of Wyoming, Laramie, Wyoming, United States

Whole-ecosystem isotope tracer additions provide a powerful tool for quantifying nitrogen cycling and food-web dynamics in streams. We conducted whole-stream ¹⁵N tracer additions in tropical headwater streams in Trinidad to test how predator introduction and canopy thinning shape nitrogen fluxes through stream food webs. We combined tracer addition with Bayesian hierarchical modeling to examine how guppy introduction (*Poecilia reticulata*) and canopy thinning affect nitrogen fluxes among ecosystem compartments, including inorganic nitrogen pools, basal resources, detrital compartments, invertebrate consumers, and fishes (*P. reticulata* and *Anablepsoides hartii*). Experimental methods consisted of a 10-day constant-rate addition of ¹⁵NH₄⁺ (target enrichment ~20,000‰) in two reaches (control and + guppy) in two parallel streams (ambient and thinned-canopy). Identical experiments were conducted in 2010 and 2019. All food web compartments were sampled before tracer addition, repeatedly during enrichment (days 1, 3, 5, 7, and 10), and five times over 30 days following the enrichment period. Tracer data were modeled using the *IsoTracer* Bayesian hidden Markov model to simultaneously estimate nitrogen fluxes between food web compartments and to compute posterior contrasts that account for propagated uncertainty. Our results demonstrated that both guppy introduction and canopy thinning altered nitrogen cycling, and that these effects persisted for nearly a decade as guppies adapted to novel predator conditions. However, the relative effects were largely context dependent, with both thinning and guppy presence increasing nitrogen flux from basal resources to invertebrate consumers. Posterior contrasts indicated that total nitrogen flux from basal resources to consumers increased in thinned canopy reaches (Pr(Δ > 0) = 0.62). The introduction of guppies also had a positive effect on N flux to consumers (Pr(Δ > 0) = 0.76). Quantitative comparisons between 2010 and 2019 are ongoing, but early results suggest that the influence of canopy thinning and guppy introduction has remained consistent through time. By

integrating ecosystem-scale manipulations with Bayesian tracer modeling, this study provides a mechanistic framework for understanding how light and consumer community traits regulate nutrient cycling in streams.

id #134352

Trash Talk: River flow dynamics and bridges impact anthropogenic litter patterns

Dana C Garcia¹, Rae McNeish¹

1. Biology, California State University Bakersfield, Bakersfield, California, United States

Anthropogenic litter (AL; *i.e.*, human trash) in freshwater environments can cause serious threats to wildlife through injury and exposure to persistent organic pollutants, impacts expand to include marine ecosystems as AL travels across freshwater-marine interfaces. Freshwater systems are also under extreme pressures from anthropogenic activities that result in water scarcity via overextractions and diversions. An important river habitat supported by river flows are debris dams, which historically were comprised solely of woody debris and other organic matter, but are currently also comprised of AL. Our study aims to understand how anthropogenic activities linked with river water availability and road bridges impacts the distribution and abundance of AL in the Lower Kern River (LKR), which historically was perennial-like but currently ecologically functions as an ephemeral river through Bakersfield, CA USA. Three bridge sites were surveyed for AL at four river habitats: upstream and downstream of bridges ($n = 3$ transects/habitat) and underneath bridges at debris dam (DD) and non-DD bridge piers ($n = 5$ pier pairs). Sampling occurred in the LKR during dry, summer 2025 season (no flows) and again in Fall 2025 when the LKR had flowing water for the first time since summer 2025. While fall sample processing is still underway, preliminary results from the dry, summer season suggested 1) greater accumulation of AL under bridges within DD habitats, 2) plastic AL dominated DDs (80%), whereas the non-DDs had less plastic but contained more AL variation ($X^2_{[50]} = 113.64$; $P < 0.0001$), and 3) river habitat significantly impacted AL distribution, with more less dense items more abundant under road bridges ($F_{[5]} = 19.823$; $P < 0.0001$). Shifts in AL distribution during the dry season may be linked to wind transportation, human visitors (*e.g.*, unhoused, recreational), and/or legacy effects from the last time the river had flowing water. These results highlight the importance of studying AL similar to any material that accumulates and transports through ecosystems, since AL materials are now a permanent material in the environment. Therefore, understanding variables that impact the abundance, distribution, and fate of AL materials will be necessary to develop and implement AL reduction management strategies and policies.

id #136144

Spatial Drivers of Soil Organic Carbon and Carbon Sequestration Rates within estuarine marsh plains along the California Coast.

Jessica Turner¹, Adina Paytan², Kevin O'Connor³, John Olson¹

1. CSU Monterey Bay, Marina, CALIFORNIA, United States

2. UC Santa Cruz, Santa Cruz

3. Central Coast Wetlands Group, Moss Landing

Coastal wetlands provide many ecosystem services including storm and erosion control, water quality benefits, and important wildlife habitats, but among their most important benefits is their role in carbon sequestration. Despite only covering 0.3% of land, coastal wetlands are responsible for 1%-2% of carbon sequestered within the U.S.. While wetland Soil Organic Carbon (SOC) data are available in California, the California Air Resources Board (CARB) reports an uncertainty range of at least $\pm 90\%$ in existing SOC estimates. These large uncertainties are due to a lack of estimation of carbon storage, export or sequestration rates within California that are specific to each site including species specific vegetation. The CARB recommends improved vegetation classification and mapping where samples are taken to increase the accuracy of SOC estimates. Carbon concentration can vary within a single wetland type as microclimates cause changes in precipitation and temperature leading to differences in vegetation composition and geomorphology, all of which are important when predicting carbon storage potential. We collected sediment cores at 13 sites along the California coast that were within Estuarine Marine Protected Area Monitoring Program areas. We selected sites with marsh plains and additional sites to fill longitudinal gaps within existing data. These cores were analyzed in 1cm increments for SOC at all sites. ^{210}Pb dating was used to estimate C sequestration rate in sites that have not had earth moving. Our measurements of carbon content will establish baseline carbon stocks for future monitoring. In addition, we will combine our SOC data with existing data from the Coastal Carbon Network to model the spatial patterns of SOC concentration and C sequestration rates. The carbon content and sequestration rate of these coastal wetlands was modeled with potential spatial drivers such as, land use, temperature, precipitation, vegetation species, and mean high tides. Using the Random Forest machine learning algorithm, we identified which spatial drivers of SOC and C sequestration rates are the most influential. Understanding the spatial patterns of C sequestration and what environmental factors increase rates will inform decisions for wetland management and help prioritize restoration and conservation of carbon sinks.

id #134353

The impacts of an invasive aquatic plant on freshwater ecosystems: A study of *Ludwigia peploides* in ponds

Isaac Owens¹, Rae E McNeish¹

1. California State University Bakersfield, CA, United States

Pond ecosystems can have substantial impacts on water quality, support diverse flora and fauna, and provide essential resources for human consumption. However, their small size makes them vulnerable to biological invasions. Floating primrose-willow (*Ludwigia peploides* [(Kunth) P.H. Raven]; hereafter FPW) is an aquatic plant known to in the littoral areas of pond systems and grows in dense mats horizontally and vertically similarly to the aquatic plant common water hyacinth (*Eichhornia crassipes* [Emmerik]). To understand how FPW influences pond environmental conditions and aquatic macroinvertebrate communities, we

compared littoral habitats dominated by FPW to adjacent non-vegetated habitats within an invaded pond at the Kern Water Bank, CA. We predicted that nutrient concentrations would be greater and light availability and water clarity will be reduced in habitats with FPW compared to habitats without aquatic vegetation due to the dense vegetation mats that limit sunlight reaching the water column. We also predicted that macroinvertebrate density and diversity would be highest in FPW habitats and that macroinvertebrate communities would differ taxonomically and functionally, with collector-gatherers, detritivores, and predators dominating areas beneath FPW mats, and grazers and predators dominating non-vegetated habitats. Colorimetric methods were used to determine nutrient concentrations, background environmental data were measured across habitat types, and macroinvertebrates were collected using D-frame net sweeps within 1.0 m² plots ($n = 10$). Organisms were identified to family level and assigned to functional feeding groups to evaluate taxonomic and functional differences between habitats. Preliminary results suggest that FPW presence produces measurable differences in environmental conditions and macroinvertebrate communities. Littoral areas with FPW supported greater macroinvertebrate abundance and higher diversity compared to non-vegetated habitats. Macroinvertebrate communities beneath FPW mats were dominated by Chironomidae and copepods, whereas *Daphnia* spp. dominated non-vegetated zones. Understanding how invasive aquatic plants reshape pond habitats will aid in effective management and restoration efforts in freshwater systems.

id #134865

Oxygen dynamics regulate sediment-water phosphorus exchange in agricultural headwater streams

Grace A Watson¹, Jay F Martin², Jim M Hood¹

1. *Evolution, Ecology, and Organismal Biology, Ohio State University, Columbus, Ohio, United States*

2. *Department of Food, Agricultural and Biological Engineering, Ohio State University, Columbus, Ohio, United States*

Agricultural headwater streams are critical pathways for phosphorus (P) export from tile-drained landscapes, yet the mechanisms controlling sediment-water P exchange in these systems remain poorly constrained. Shallow, hydrologically modified channels experience strong seasonal and diel variability in dissolved oxygen (DO), temperature, and biological activity, which can alter redox-sensitive P mobilization from sediments. Here we examined spatial and temporal patterns of DO and evaluated their influence on sediment porewater soluble reactive phosphorus (SRP) flux in three agricultural drainage-ditch streams within the Shallow Run sub-watershed of the Maumee River basin, Ohio, during the 2025 growing season.

Monthly reach-scale surveys were conducted from April through September to characterize water chemistry, physical habitat, and biological conditions, while continuous optical sensors quantified high-frequency DO dynamics at the bottom of the reach. In situ diffusive gradients in thin films (DGT) probes were deployed concurrently to provide time-integrated, depth-resolved estimates of labile SRP flux across the sediment-water interface (SWI). Continuous monitoring revealed pronounced seasonal declines in DO and strong diel oscillations, with extended hypoxic and anoxic conditions during mid-summer, particularly at the most nutrient-enriched site.

Sediment porewater SRP concentrations measured by DGT (cDGT) exhibited substantial temporal variability, with the highest porewater SRP concentrations occurring during periods of elevated temperature and reduced DO. Depth-resolved cDGT profiles indicated seasonal shifts in the depth of maximum SRP mobilization within sediments, suggesting dynamic redox controls on P release.

Preliminary data analysis suggests that porewater SRP concentrations are higher under warm conditions, in the presence of vegetation, and when sediment P saturated is high. Porewater SRP concentrations declined with stream water DO concentrations, presumably because this facilitated the release of porewater P into the water column. Stream water SRP concentrations were highest when DO concentrations were low.

These results demonstrate that low-oxygen conditions in agricultural headwater streams can substantially enhance sediment-derived P flux during the growing season. By integrating continuous oxygen measurements with in-situ porewater flux estimates, this study provides mechanistic insight into how redox dynamics regulate internal P loading in intensively managed agricultural stream networks, with important implications for downstream nutrient export and water-quality management.

id #136145

Toward Regional Thermal Tolerance Indices for the Pacific Northwest: Evaluating the Weighted Averaging Optimum Framework for Benthic Macroinvertebrates using State-wide Biomonitoring Datasets.

Garrison S Ferone¹, Chris C Caudill¹

1. *College of Natural Resources, Department of Fish and Wildlife Sciences, University of Idaho, Moscow, ID, United States*

Quantifying macroinvertebrate thermal niches supports the development and refinement of temperature regulatory thresholds in Pacific Northwest streams. The Macroinvertebrate Thermal Tolerance Index (MTTI; Hubler et al. 2024) was proposed as an estimate of local thermal suitability using taxa-specific thermal tolerances (weighted average optima, WAO) and local assemblage structure. Taxon-specific WAO values are estimated using a geographically defined distribution dataset and thus application of WAO estimates across regions may reveal either consistent estimates or biases caused by biological and analytical factors. In this study, we compared WAO and MTTI estimates from ~7,200 Idaho BURP observations (1993–2021) with estimates calculated from the combined Oregon-Washington dataset used to develop the MTTI. We aimed to: (1) quantify patterns in thermal optima

and breadth for macroinvertebrate taxa between regions (ID vs. OR-WA); (2) evaluate the utility and transferability of the MTTI across regional versus sub-regional scales; and (3) identify potential methodological refinements.

We estimated the WAO of 221 shared operational taxonomic units (OTUs) observed in ~1,200 BMI assemblage samples. OTUs were categorized by thermal response curve shape (e.g., cold-water/warm-water specialists vs. generalists) and site-specific MTTI scores were compared to predicted stream temperatures. Model fit declined when WAO from one region was applied to another, as expected. While most Idaho taxa exhibited concordant response curves with WA-OR, OTUs were more frequently classified cold-water specialists in the ID dataset. Pairwise comparison of WAOs calculated for taxa occurring in ID and WA-OR revealed lower WAO in ID than WA-OR, suggesting sampling frames imposed by state-wide monitoring programs may bias WAO estimation. Specifically, we speculate that, on average, ID observations capture the cold portion of each OTU's range, causing underestimation of WAO. Our results indicate that thermal tolerance indices are constrained by the range of stream temperatures included in sampling and thus suggest robust MTTI estimates require sampling a fuller thermal gradient of BMI taxa present to reduce biases associated with sampling the edges of a given taxon's thermal range. Furthermore, taxonomic uncertainty may have contributed to patterns, and we suggest integrating genetic barcoding with traditional morphological approaches to improve taxonomic resolution and comparability across assessment programs.

1. Hubler, S., Stamp, J., Sullivan, S. P., Fernandez, M., Larson, C., Macneale, K., ... & Bierwagen, B. Improved Thermal Preferences and a Stressor-Specific Index Derived from Modeled Stream Temperatures and Regional Taxonomic Standards for Freshwater Macroinvertebrates of the Pacific Northwest, USA. DOI: <https://doi.org/10.1016/j.ecolind.2024.111869>

id #134866

Mining, selenium, and the conundrum of fish-tissue based criteria in the Kootenai River, Idaho and Montana

Erin M Murray¹, Christopher A Mebane¹, Lauren M Zinsser¹, Vicki S Blazer², James L Dunnigan³

1. USGS, Boise, ID, United States

2. USGS, Kearneysville, WV

3. Montana Fish, Wildlife, and Parks, Libby, MT

Risk interpretation and regulatory criteria to manage selenium risks to aquatic organisms are often based on fish-tissue concentrations. Because selenium toxicity in early-life stage fish is thought to stem from maternal transfer to embryos, selenium levels in egg and ovary tissues are considered the most relevant indicator of risk to fish populations. However, establishing a regulatory water-quality criterion expressed as a chemical concentration in fish egg and ovary tissues poses practical problems for monitoring. Capturing gravid fish may be difficult, particularly for species that leave the water body under assessment to spawn in tributaries. Small-bodied fish such as sculpin may be more readily catchable, but laboratory protocols may not accommodate analysis of small-mass tissues. The USEPA's aquatic life criterion for selenium document includes extensive analyses and assumptions for relating selenium concentrations in egg and ovary tissues to concentrations in whole-body or muscle fillet tissues, which may be easier to obtain. In the Kootenai River basin (Idaho and Montana, USA), we have extensively sampled fish tissues to examine selenium concentration patterns across different species, seasons, and water concentrations from 2019 to 2024. We compare concentrations in ovary, liver, muscle, and reconstructed whole-body tissues for many species, including *Prosopium williamsoni* (Mountain Whitefish), *Oncorhynchus mykiss* (Rainbow Trout), *Richardsonius balteatus* (Redside Shiner), *Cottus cognatus* (Slimy Sculpin), and *Ptychocheilus oregonensis* (Northern Pikeminnow). Our robust paired sampling datasets allow us to explore selenium in these fish tissues throughout the spawning cycle, comparing observed concentrations to reproductive factors such as egg development stage (assessed visually or via histology), gonadosomatic index (GSI), and collection date. We observed positive correlations between liver and muscle selenium concentrations and ovary selenium concentrations within the same individual fish, though the strength of the relationship varied by species. Some species exhibited significant negative correlations between selenium concentrations in ovary, muscle, or liver tissues and GSI. Results from this study will inform how tissue relations of selenium in Kootenai River fish compare to the relation outlined in the USEPA criterion, particularly regarding the suitability of muscle or whole-body concentrations as surrogates for egg and ovary concentrations.

id #136146

Watershed land use influences the prevalence of antibiotic resistance genes (ARGs) in streams and rivers

Mitchell J. Liddick¹, Jennifer L. Tank¹, Abagael N. Pruitt^{2,1}, Kyle Bibby³

1. Biological Sciences, University of Notre Dame, Notre Dame, IN, United States

2. University of California, Davis, Davis, CA, United States

3. Civil and Environmental Engineering and Earth Sciences, University of Notre Dame, Notre Dame, IN, United States

Antimicrobial resistance (AMR) threatens public health by undermining the effectiveness of antibiotics. However, the influence of land use on the occurrence of antimicrobial resistance genes (ARGs) along river networks remains poorly understood. National monitoring assessments show streams with degraded habitat have more ARGs, but lack of spatial resolution limits attribution to specific land use drivers across watersheds. To understand how ARG signatures vary temporally between streams and rivers of varying land use, we collected water samples during Fall 2023 and Summer 2024 from 105 sites (72 tributary and 33 mainstem sites) across four Midwestern watersheds. The watersheds included the forest-dominated Manistee (56% forest), the mixed but largely forested Muskegon (39% forest), the agriculturally and urban influenced St. Joseph (48% agriculture, 16% urban), and the agriculture-dominated Tippecanoe River (72% agriculture). For each sampling location, we analyzed water samples for four

ARGs (*tetW*, *tetQ*, *ermB*, *sul1*), two fecal indicators (*bacR*, PMMoV), and a 16S rRNA bacterial marker. We hypothesized that ARG concentrations across watersheds would be driven by land use intensity, with higher ARG concentrations in urban influenced systems associated with human wastewater inputs and in agricultural systems associated with fertilizer inputs, relative to forested systems. We found that *tetW* and *tetQ* concentrations (gene copies mL⁻¹) significantly differed among watersheds (generalized linear mixed model, $p < 0.05$), with the highest estimated marginal means (EMM) observed in Muskegon and St. Joseph watersheds. In contrast, the lowest *tetW* and *tetQ* concentrations occurred in the Tippecanoe and Manistee watersheds, which were not significantly different (EMM, $p > 0.05$). This pattern was unexpected, as the Tippecanoe watershed has the highest % agricultural land use, whereas the Manistee watershed is the most forested. We also found that ARG concentrations were positively correlated to % urban ($p = 0.31$ to 0.34) and % agricultural ($p = 0.17$ to 0.23) land use, while concentrations were negatively correlated with % forest ($p = -0.13$ to -0.21). Together, these findings indicate that ARG distributions along Midwestern river networks are shaped by complex interactions among land use types, rather than by agricultural intensity alone, highlighting the need to account for agricultural type (livestock versus row-crop production) when evaluating environmental ARG patterns.

id #134868

Dynamic variation in the impact of Artificial Light at Night (ALAN) on aquatic-terrestrial insect fluxes

Elizabeth M Parkinson^{1,2}, Nicole D Wagner¹, Scott D Tiegs¹

1. Oakland University, MI, -

2. Cranbrook Institute of Science, Bloomfield Hills, MI, United States

Aquatic and terrestrial ecosystems are connected by resource subsidies, often in the form of emergent aquatic insects that enter terrestrial environments after metamorphosis. Emergence frequently occurs in discrete pulses, periods when large numbers of individuals move from freshwaters into terrestrial environments and become food for consumers. This resource transfer is impacted by anthropogenic factors, including the pollutant artificial light at night (ALAN). The widespread adoption of LED technology has increased shoreline lighting and shifted the spectral composition of ALAN towards peaks in short wavelengths. A unique characteristic of LEDs is their tunability to emit specific wavelengths, creating an opportunity to minimize impacts, but how different LED wavelengths influence emergent insects across taxa and periods of high and low emergence remains understudied. To determine whether the effects of quasi-monochromatic LEDs on emergent flux varied through time, we installed replicate treatments of four spectral compositions (violet [410 nm], green [530 nm], red [630 nm], broad-spectrum white [4000K]) and a dark control in littoral and riparian zones of a Michigan (USA) lake. We collected insects on seven nights (June through September) using pan traps, encompassing periods of both high and low emergence. We evaluated differences in insect abundance, biomass, individual body size, and diversity among light treatments, habitats, and dates using three-way repeated-measures ANOVAs. We calculated differences in the effect size of wavelength and habitat on date over time, and community composition using non-metric multidimensional scaling (NMDS) and adonis permutation procedures. Insect abundance and biomass were greatest in violet treatments, though the magnitude of this effect strongly depended on sampling date, ranging from 352–5133% greater abundance and 389–6824% greater biomass relative to dark controls. Individual insect size was also greater under violet light but varied by date. These patterns were driven by taxa-specific responses, with emergent aquatic insects being more strongly attracted to short-wavelength light than insects of terrestrial origin, particularly during peak emergence of large-bodied caddisflies. Overall, our results demonstrate that ALAN alters aquatic–terrestrial fluxes, but effects depend on wavelength and sampling date, highlighting the dynamic nature of these impacts.

id #134869

Macroinvertebrates and microplastics: understudied ecological communities in Appalachian freshwater springs

Benjamin E Jackson¹, Emma Hall¹, Dorothy J Vesper¹, Brent A Murry¹

1. West Virginia University, Morgantown, WV, United States

Freshwater springs represent unique ecological systems that support diverse biological communities and provide critical ecosystem services with direct human benefits. Because of their intersectional role, it is essential to understand the factors affecting their biogeochemical integrity. Central Appalachia was chosen as the focal area for this study because of its abundance of freshwater springs and their common use for critical water resources.

Across all freshwater systems, microplastics represent a potential threat to both human and environmental health. Previous research has shown the sheer abundance and extensive spread of these pollutants across Appalachia, in surface waters. Springs may be more insulated from microplastic contamination; however, flow from groundwater may still contribute to contamination. While geochemical characteristics of Appalachian springs have been documented extensively, biological data remain scarce, leaving critical gaps in understanding community composition and ecosystem function. This lack of biological data limits our ability to assess how spring communities contribute to ecosystem services and respond to emerging contaminants such as microplastics. We sampled the ecological communities of Appalachian springs across a gradient of carbonate - shale - sandstone

lithologies to evaluate the degree of microplastic contamination in spring macroinvertebrate communities and determine if specific geochemical traits predict microplastic presence.

We sampled springs throughout central Pennsylvania and West Virginia and specifically collected water to analyze the concentration of microplastics and macroinvertebrates to develop a greater understanding of community composition and biomagnification of pollutants within spring ecosystems. We hope to compare this data with simultaneously collected and historical geochemical data in order to better understand the movement of microplastics into and through springs, as well as potential health risks that microplastics pose. We predict that springs across Appalachia will have moderate concentrations of microplastics, which will in turn affect macroinvertebrate communities additively with geologic and chemical factors. Our results not only supplement the current lack of biological record across Appalachian springs for future research but also provide insight into the role microplastics play within springs, a vital resource for both human and environmental health.

id #135125

When temperatures change, nutrition follows: Seasonality of insect emergence in Australian alpine headwaters in changing climates

Sithara Sirimanna¹, Sally Maxwell², Aleicia Holland¹, Michael Shackleton¹

1. *Department of Ecological, Plant and Animal Sciences, La Trobe University, Wodonga, Victoria, Australia*

2. *Center for freshwater ecosystems, La Trobe University, Wodonga, Victoria, Australia*

Alpine ecosystems are particularly vulnerable to climate change due to their pronounced seasonal temperature fluctuations. Future projected atmospheric temperature increases may alter the phenology of insects in the Australian alps, imposing nutrient limitations on alpine food webs.

Aquatic insects are one of the main biological components in Australian alpine ecosystems and have a prominent role in cross-ecosystem nutrient fluxes between aquatic and terrestrial environments. Aquatic insects are particularly high in poly-unsaturated fatty acids (PUFA) that are essential but limited in terrestrial ecosystems. Cross-ecosystem fluxes, driven by emerging aquatic insects, transport these scarce nutrients to land, where they are consumed by terrestrial animals. As an initial step toward understanding nutrient dynamics in these systems, we conducted a study to quantify temporal and spatial variation in the abundance and diversity of alpine insects and the transfer of aquatic insect fatty acids into the terrestrial environment.

Insects were sampled monthly at Falls Creek, Victoria, Australia from November 2024 until May 2025 using emergence and malaise traps. Temperature loggers were deployed over the same time frame to collect water and air temperature data. The abundance and biomass of insect orders were determined, and fatty acids were quantified to explore seasonal variation in aquatic and terrestrial insects.

Insect emergence rates and biomass varied significantly across seasons peaking in summer and declining into autumn. Preliminary findings revealed significant differences in PUFA levels between aquatic and terrestrial insects. These results suggest seasonal variation in the cross-ecosystem transfer of insect-derived fatty acids in alpine ecosystems. Further analyses for the study are currently in progress. Future focus will explore seasonal variation through development of models that link antecedent temperatures with biomass and fatty acid loads. Moreover, these models will be used to infer how nutrient landscapes may shift with changing climate under future climate scenarios.

id #135893

Salinization impacts on terrestrial and aquatic decomposition, soil respiration, and photosynthesis

Natalie A Clay¹, Maggie Herrmann¹, Sally Entrekin², Samuel Dias¹, Connor Foley¹, Clay Prater¹, Vien Tran³, Beck O'Neill¹, Michelle Evans-White¹

1. *University of Arkansas, Fayetteville, ARKANSAS, United States*

2. *Virginia Tech University, Blacksburg, Virginia*

3. *University of North Texas, Denton, Texas*

Global salinization is impacting both freshwater and terrestrial systems, and impact from increased salts can change processes within systems and energy and matter exchange between systems. Yet these systems are often studied in isolation. To determine how terrestrial and freshwater salinization changes ecosystem function in riparian nutrient cycling, we initiated a greenhouse mesocosm salinization experiment with paired and connected terrestrial and stream systems, where NaCl was added as 100, 200, 300, 400, 500 mg/L or none as controls (n=5 per treatment) or just streams or just soils at 300 mg/L. Preliminary results on a subset of mesocosms demonstrated that terrestrial (after ~4 months) and stream (after 26 days) decomposition averaged ~49% and ~57% mass loss in open bags and ~43% and ~58% mass loss in closed bags, respectively. For streams, closed and open bags had similar decomposition rates, but NaCl addition up to 400 mg/L tended to slightly increase decomposition. For open bags, the addition of NaCl up to 300mg/L tended to slightly increase decomposition rate relative to controls in terrestrial systems, but had no effect >300mg/L NaCl, whereas any NaCl addition tended to slightly decrease decomposition rate in closed bags relative to controls. The difference in decomposition rate between closed and open bags was greatest when NaCl was added up to 300mg/L. Similarly, soil respiration decreased relative to controls in all NaCl addition treatments. This suggest that soil salinization stimulates detritivores at low levels but suppresses microbial activity. Photosynthesis and ecosystem respiration increased with NaCl additions up to 300mg/L relative to controls resulting in controls having the greatest net ecosystem productivity. Although preliminary, these results suggest a slight sodium subsidy-stress response in both systems.

id #133846

River Stories: The Kalamazoo River in Watercolor

Bonnie F Lord¹

1. Albion College, Albion, MI, United States

The Kalamazoo River is a sixth order stream in southwest Michigan that drains to lake Michigan. With a complex history including one of the largest inland oil spills in U.S. history, pollution resulting in several Superfund sites, and socially and economically enforced segregation, the Kalamazoo River has been at the center of several statewide issues in Michigan. In Albion, Michigan, this importance increases tenfold. The north and south branches of the headwaters meet in Albion, and recreation, drinking water quality, and research have all stemmed from its presence in the city's landscape. However, without a career dedicated to the history, science or maintenance of the Kalamazoo River, understanding something everyone in the Albion community seems to have a relationship with becomes a relatively intimidating task. In a series of written features centering the stories of the Kalamazoo River's greatest advocates and Albion's most dedicated citizens, I have captured an inter-generational, intersectional and deeply human-centered snapshot of the river's history. Accompanying these in-depth pieces, I have created a large-scale watercolor illustration of the stories I've heard across the landscape they center, culminating in a unique piece of artwork to display the relationship between the Kalamazoo River and Albion's community. Each character, represented as a creature native to the river's local ecosystem, inspires curiosity leading the viewer to interact with the digitally published written work documenting their stories.

id #135894

Spatial distribution and hydrology as controls on aquatic insect beta diversity across geographically isolated wetlands

Elizabeth Sicking¹, Kier Klepzig¹, Steve Golladay¹, Daniel McLaughlin², Sally Entrekin²

1. The Jones Center at Ichauway, Newton, GA

2. Virginia Tech, Blacksburg, Virginia, United States

The alteration of wetlands and creation of man-made wetlands is leading to landscape fragmentation, intensifying the need to understand what factors control the dispersal of organisms among sites. Wetlands are both sources and sinks of water, nutrients, and organisms, and act as important breeding habitats for amphibians, migrating birds, and dispersing aquatic insects. In geographically isolated wetlands (GIWs), long-term hydrological patterns are one of the most important determinants of emergent insect timing and community composition. However, the extent to which distance between sites, hydrological diversity, and seasonal timing control individual wetland aquatic insect diversity across GIWs is unknown, making it difficult to predict how changes to these factors may influence regional diversity in wetlands. To address this gap, we assessed emergent aquatic insect communities across a hydroperiod in 11 wetlands. We sampled five cypress swamps and six sedge marshes in the southeastern Coastal Plain of Georgia, USA during 2023, and measured total beta diversity, nestedness, and turnover between sites using distance, hydrology, wetland type, and time as predictors. We hypothesized that communities would be most similar in closely located GIWs with similar hydrology, late in the inundated period when emergence rates were greatest. We also predicted that communities would be most different between marshes and swamps, due to differences in habitat. Preliminary results suggest that wetlands were most similar when they were closer together, but there is less dissimilarity between communities with different hydrological regimes and between wetland types than expected. Our analysis will continue diving into these patterns by adding a temporal perspective and incorporating isolation as an additional predictor to address our objective.

id #136150

Movement strategies dictate the spatial dynamics of consumer nutrient and energy fluxes

Mack White^{1,2}, W. Ryan James^{3,4}, Justin S Lesser³, Rolando O Santos^{3,4}, Jennifer S Rehage^{2,4}

1. Department of Evolution, Ecology, and Organismal Biology, The Ohio State University, Columbus, OH, United States

2. Department of Earth and Environment, Florida International University, Miami, FL, US

3. Department of Biological Sciences, Florida International University, Miami, FL, US

4. Institute of Environment, Florida International University, Miami, FL, US

Animal movement – from daily excursions that span a few meters to annual migrations that take place over thousands of kilometers – is both a cause and consequence of spatiotemporal variation in resource availability. Though a large body of research has highlighted the role of mobile consumers in facilitating ecosystem processes, few studies have linked empirical movement data to the functional roles of animals – defined as their influence on nutrient and energy fluxes. Further, we know little about how individual variation in movement behavior governs the spatial dynamics of consumer function. Using a suite of simple movement metrics calculated from 13 years of acoustic telemetry data, we identified recurrent patterns of intraspecific movement behavior for a highly mobile mesopredator, Common Snook (*Centropomus undecimalis*), in the Shark River, Everglades National Park. We paired these data with elemental flux-, home range-, and stable isotope mixing models to investigate the ecological consequences of individual movement strategies. Specifically, we estimated the magnitude (mg d^{-1}), spatial distribution ($\text{mg m}^{-2} \text{d}^{-1}$), and connectivity (g month^{-1}) of the three key functions (i.e., consumption, excretion, and secondary production) for 174 individuals and their associated movement strategies. The functional roles of residents were spatially concentrated due to their high site fidelity and minimal displacement. In contrast, nomads exhibited the greatest functional connectivity, driven by low site fidelity and high displacement that integrated resources across a gradient from freshwater to marine prey. Notably, the functional roles of migrants exhibited intermediate effects on both the spatial distribution and connectivity of their functions, highlighting that

migratory animals exert intermittent control on ecosystem-level processes across multiple spatial and temporal scales. As global change continues to alter animal movement patterns, untangling the threads that link animal behavior to ecosystem processes will be needed to predict and mitigate changes in ecosystem function.

id #134871

Mercury does not stop at the shoreline: terrestrial insects accumulate mercury in High Arctic pond food webs.

Camryn Middlebrooks¹, Matt Chumchal¹, Benjamin Strang¹, Kurt Burnham², Aleah Appel¹, Sommerlyn Babineau¹, Ethan Cary¹, Titus Crawford¹, Sage Dale², Charlie Duethman¹, Aidan Duffield¹, Piper Dumont², Skylar Dunn¹, Madeline Hannappel³, Sydney Hill¹, Ramsey Jennings¹, Ben Katzenmeyer¹, Chidi Mbagwu¹, David Peebles¹, Denice Rodriguez¹, Emma Sullivan¹, Kyle Trevor¹, Tristan Williams¹, Tyler Williams¹, Lance Viscioni-Wilson¹, Kimberlee Whitmore¹, David Wright¹

1. Texas Christian University, Fort Worth, TX, United States

2. High Arctic Institute, Orion, Illinois, United States

3. California Academy of Sciences, San Francisco, California, United States

Rapid environmental change in the Arctic has the potential to alter mercury (Hg) cycling and shape methylmercury (MeHg) transport across connected freshwater and terrestrial food webs. Freshwater ecosystems are important sources of MeHg, a potent neurotoxin, to terrestrial consumers, yet the degree to which this subsidy elevates Hg concentrations in terrestrial predators remains unclear in High Arctic landscapes. We quantified Hg concentrations in spiders and their potential prey at three distances from pond shorelines (0m, 10m, 35m) in six freshwater ponds in High Arctic Greenland to evaluate cross-ecosystem transfer of Hg and the role of distance from pond shorelines and trophic position in structuring exposure. Across all ponds, spiders consistently exhibited Hg concentrations elevated relative to their prey; however, these concentrations were not related to distance from shorelines, suggesting that spiders may be feeding on high-Hg prey far from the pond shorelines. Contrary to expectations that terrestrial insects would exhibit lower Hg concentrations than aquatic taxa, aquatic and terrestrial insects did not differ in overall Hg. Instead, Hg exposure within terrestrial insects was strongly structured by feeding guild, increasing with trophic position from neotivores to predators. Predatory terrestrial insects accumulated Hg concentrations approaching those of spiders, indicating substantial biomagnification occurring within the terrestrial invertebrate community. These results suggest that trophic position, rather than habitat (aquatic versus terrestrial), may be the primary determinant of Hg exposure in this High Arctic food web. Elevated Hg concentrations in terrestrial insects reveal an important pathway for contaminant transfer to terrestrial predators, with implications for consumers such as insectivorous birds that rely heavily on spiders and aquatic and terrestrial insects during the Arctic breeding season.

id #135895

Alpha, beta, and functional diversities of benthic macroinvertebrates within three habitats of a large floodplain -river corridor

Kelsey Wilbanks¹

1. Georgia Southern University, Aiken, SC, United States

River corridors are interspersed with a mosaic of habitats that promotes biodiversity. Flow is important in shaping physical habitats, driving abiotic processes and maintaining biotic communities, but climate change and the associated hydrological drying threaten these systems. This study aimed to understand role of lateral river corridor habitats and hydrological drying in structuring the diversity of macroinvertebrates. Aquatic invertebrate assemblages were sampled during a descending limb of the hydrograph at six sites on of the Savannah River, southeastern, USA. Floodplain, oxbow lake and the river channel habitats were sampled within six sites for a total of 113 samples. We found differences among alpha diversity of macroinvertebrates, including Shannon, inverse Simpson and rarefied richness between habitats using linear mixed model. Oxbow lakes had the highest alpha diversity, with the highest number of genera. We also found differences in beta diversity, where Bray-Curtis dissimilarity accounted for 43% of the variation in community composition, where habitats were the dominant driver of variation (26.5%) and drought stage accounted for a smaller portion (4.8%) of variation. However, a significant interaction revealed that drought did not shift communities in a similar patterns among habitats and multivariate dispersion showed that mid-drought had a higher dispersion than early or late. Understanding the influence drought will have on aquatic invertebrates in these ecosystems will be critical to anticipate changes and develop strategies to protect freshwater ecosystems. The loss or alteration of river corridor habitats will have consequences for biodiversity and ecosystem services. A mosaic of habitat types with differing levels of connectivity and multiple transition zones between habitats will provide maximum biodiversity and allow for recovery of taxa after hydrological extremes.

id #134105

Do quantitative macroinvertebrate samples from riffles and depositional habitats provide accurate estimates of stream condition across a gradient of agricultural land cover?

Brice Crum¹, Sergio Sabat-Bonilla¹, Matthew Cashman², Kelly Maloney³, Daniel McLaughlin⁴, Tristan Mohs⁵, Greg Noe⁶, Sally Entrek¹, Paul Angermeier⁷

1. Department of Entomology, College of Agriculture and Life Sciences, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, United States

2. U.S. Geological Survey, Water Mission Area, Reston, Virginia, United States

3. U.S. Geological Survey, USGS Eastern Ecological Science Center, Kearneysville, West Virginia, United States
4. College of Forest Resources and Environmental Conservation, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, United States
5. U.S. Geological Survey, Maryland-Delaware-DC Water Science Center, Catonsville, Maryland, United States
6. U.S. Geological Survey, Florence Bascom Geoscience Center, Dover, Delaware, United States
7. U.S. Geological Survey, Virginia Cooperative Fish and Wildlife Research Unit, Virginia Polytechnic Institute and State University, Blacksburg, VA

Historical modification of land cover has resulted in stream macroinvertebrate communities shifting to generalist taxa in forested, grassland, and agricultural watersheds. Conversion from undeveloped to agricultural land cover alters water quality, organic matter inputs, retention, and habitat heterogeneity in streams. Erosional and depositional habitat heterogeneity is essential for supporting unique macroinvertebrate niches, although sampling protocols designed to sample riffle-only habitats could miss taxa exclusive to depositional habitats. The United States Environmental Protection Agency (USEPA) multihabitat Rapid Bioassessment Protocol (RBP) collects invertebrates from multiple habitats but aggregates all samples from a single site, so communities are not associated with habitats. We will quantify macroinvertebrate assemblages from both erosional and depositional habitats to better understand the biotic communities in each habitat type and how they may influence measures of stream condition. We will assess five quantitative biomass-weighted macroinvertebrate samples in depositional and riffle habitats across 24 streams along a gradient of agricultural land cover. Our study will provide information on often-undersampled habitats in macroinvertebrate bioassessment. We hypothesize that depositional habitat types (i.e., debris accumulations, undercut banks) will support more taxonomically and functionally diverse communities than riffles. Due to depositional habitat diversity providing niches for macroinvertebrates, we also predict and will present data on whether streams with greater depositional habitat diversity have greater species richness. Additionally, we predict and will present data on whether depositional habitats will support more biomass-weighted clinger and shredder taxa than riffles. Mixed pasture-row crop land-cover within the watershed ranged from 10% to 76%, and Shannon habitat diversity ranged from 0 (one habitat type) to 1.3 (different proportions of all five habitat types). Mean richness across 7 preliminary sites indicated that riffles (mean = 44, standard deviation = 8.50) and depositional habitats (mean = 43, standard deviation = 12.49) contained similar amounts of taxa. Our study will determine the unique taxonomic and functional assemblages that riffle and depositional habitats provide to stream bioassessments, and how these assemblages change along a gradient of agricultural land cover and depositional habitat heterogeneity.

id #134873

Variation in basal resource isotope values and their contributions to a keystone wetland piscivore

Matthew S. Foli¹, Ben J. Spitz¹, John M. Farrell¹, Emily R. Arsenault¹

1. SUNY College of Environmental Science and Forestry, Syracuse, NEW YORK, United States

Basal resources form the foundation of aquatic food webs, transferring energy from primary producers to higher trophic levels. Understanding how these resource pathways change over space and time to support consumers improves our understanding of food-web function. However, resource use in wetland habitats can be complex, given a wide diversity of available basal sources and environmental conditions. We investigated how differences in geomorphology across eight wetland sites influence the relative contribution of basal resources, including phytoplankton, periphyton, macrophytes, and detritus, to the energetic pathways supporting young-of-year (YOY) Northern Pike (*Esox lucius*) an important wetland predator. Given their generalist feeding behavior, known habitat use over ontogeny, and role as apex predators, YOY Northern Pike serve as an ideal organism for evaluating how basal resource contributions differ across wetland types. Eight nursery sites were selected: seven in the Thousand Islands region of the Upper St. Lawrence River and one in Cranberry Lake, located in Adirondack Park (both New York, USA). These sites span a gradient of wetland geomorphic types (protected and open coastal embayment, drowned river mouth, and palustrine) and aquatic plant invasion impact. We used stable isotope analysis of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and Bayesian mixing models to identify energetic pathways providing dominant support for YOY Northern Pike. Preliminary results suggest variation in the isotopic composition of basal resources across wetland types, with resources in coastal embayments being more enriched in ^{13}C compared to those of drowned river mouth wetlands (mean $\pm$ SD $\delta^{13}\text{C}$ = $-20.8 \pm 7.39\text{‰}$ vs. $-28.6 \pm 2.96\text{‰}$). Additionally, YOY Northern Pike trophic niche size was higher in drowned river mouth wetlands when compared to coastal embayments (total area = 9.6 vs 3.2), suggesting a diversified use of resources within these habitats. Investigating basal resource variation among nursery habitats provides insight into how energy moves through wetland food webs, how trophic pathways differ among habitat types, and how basal production ultimately supports higher trophic levels. Additionally, incorporating YOY Northern Pike introduces a tertiary consumer into nursery food-web analyses, linking basal resources to the energetic support of a top predator during a critical early life stage.

id #135897

Controls on nutrient export moving downstream an intermittent stream network

Karessa G De La Paz¹, Shannon L Speir¹, Kathleen J Cutting¹

1. University of Arkansas, Fayetteville, AR, United States

Intermittency impacts approximately 60% of the world's streams and rivers, and their extent is expected to increase with future climate change. Despite their ubiquitous nature, there is a lack of understanding surrounding the impact of intermittency on nutrient loss in streams. More specifically, the propagation of nutrient export moving downstream is not well characterized across a range of hydrologic connectivity conditions. To address this gap, we studied Brush Creek, an intermittent stream in Northwest Arkansas, sampling four nested sites along the mainstem of the watershed for nitrate and soluble reactive phosphorus (SRP). Using the *loadflex* model, we used continuous discharge measurements and grab sample data to estimate daily nutrient concentrations and loads at each site from October 2024 to November 2025. Using metrics of synchrony, we quantified the effect of drying on the coupling between upstream and downstream nutrient dynamics, specifically measuring the extent to which

nutrient patterns propagate through the stream network on signal propagation downstream. Preliminary results indicate that drying reduces synchronicity in nutrient dynamics across nested sites. This suggests that surface hydrology exerts significant control on nutrient exports in intermittent networks, as loss of surface flow disrupts the downstream propagation of nutrient signals. Our results will help inform how hydrologic connectivity is impacting nutrient loading as stream intermittency becomes more prevalent in our changing climate.

id #135898

Wetland Methane Fluxes: Sensor Development and Environmental Drivers

Clark Jackson¹, Jake Hosen¹, Ayobami Oni¹, Zaven Arra²

1. *Purdue University, Lafayette, IN, United States*

2. *River Restoration Intelligence and Verification, Oakland, CA, USA*

In recent decades, anthropogenic and natural sources of greenhouse gases (GHGs) have been broadly studied to quantify their impact on the earth's changing climate. Natural sources constitute 40-50% of global methane (CH₄) emissions with wetlands contributing upwards of 40% of those natural sources. Despite being a major source of natural CH₄ emissions, researchers' ability to predict emission rates from individual wetlands is poor. This knowledge gap exists largely due to the high cost of existing flux measurement systems as well as the variability of wetland CH₄ emissions over space and time. These factors unite to prevent comprehensive quantification of CH₄ emissions from wetlands. Given that CH₄ emissions have a global warming potential (GWP) over 25 times that of carbon dioxide (CO₂), quantifying environmental drivers and flux rates is crucial for filling the knowledge gap. To address this need, we have developed a low-cost CH₄ flux chamber that can autonomously collect data in the field using a semiconductor gas sensor. While these sensors are sensitive to CH₄, limits to manufacturing processes mean that each device must be calibrated individually to incorporate the influence of temperature and humidity on sensor outputs. We evaluated regression and machine learning approaches including generalized additive models (GAMs), random forests, and support vector machines to create calibration protocols and equations that will reduce the time and expertise required to calibrate each sensor. GAMs were the most accurate method for generating a reliable calibration model for this specific sensor, particularly when CH₄ concentrations were high. We applied this flux protocol approach to examine the impact of environmental drivers on fine scale variation of depressional wetlands in Midwestern, USA. Specifically, we evaluated the impact of microtopography and freeze/thaw cycles, which are key factors related to wetland CH₄ emissions. Additionally, we found high spatial variation related to microtopography and water depth. Saturated soils had the highest CH₄ fluxes whereas dry soil had the lowest. Increased water depth slightly attenuated CH₄ fluxes and increased the temporal variation of fluxes, likely due to ebullition. Rapid thawing events in the winter resulted in substantial CH₄ flux events, especially saturated zones with ice cover.

id #135899

That's a mouthful: availability, feeding ecology, and life history variation of stream fishes during periodical cicada emergences

Rich H Walker¹, Olivia Daugherty¹, Tyler Davis¹, Drew Wildman¹, Levi Wright¹, Darrin Hunt², Bryan M Maitland³

1. *University of Tennessee at Chattanooga, Chattanooga, United States*

2. *Department of Environmental and Sustainability, Shenandoah University, Winchester, Virginia, United States*

3. *Rocky Mountain Research Station, USDA Forest Service, Boise, Idaho, United States*

Aquatic and terrestrial food webs are closely linked through the exchange of allochthonous resources that cross ecosystem boundaries. While many subsidies occur annually, rarer, pulsed subsidies may have disproportionate long-term effects on recipient communities. Periodical cicadas (*Magicicada* spp.; hereafter cicadas) represent one of the largest natural resource pulses in North America, emerging every 13 or 17 years depending on species and delivering large quantities of biomass and nutrients to forest and stream ecosystems. Optimal foraging theory predicts that as the density of a high-value resource increases, consumers will preferentially exploit resources while ignoring lower-value alternatives. Despite their magnitude, the ecological effects of cicada emergences on stream fishes remain poorly understood. Cicadas are relatively large and may impose morphological constraints on consumption by gape-limited fishes. Fishes also rely on multiple sensory modalities (e.g., visual, olfactory, and auditory cues) to detect and exploit prey, suggesting that both physical and behavioral traits may influence access to cicadas. Our objectives were to: (1) quantify cicada drift density and biomass using drift nets (26 × 45 cm), (2) assess cicada occurrence in fish diets, and (3) examine relationships between foraging strategy, life history, and gape size. We leveraged the emergence of Broods XIX and XII in 2024 and XIV in 2025, sampling drift from 88 streams across Arkansas, Missouri, Illinois, Iowa, Tennessee, Kentucky, and Ohio, and collected fishes from a subset of 39 streams. We focused on a variety of fish species, including Green Sunfish (*Lepomis cyanellus*), Longear Sunfish (*Lepomis megalotis*), Bluegill (*Lepomis macrochirus*), Creek Chub (*Semotilus atromaculatus*), Bullhead Catfish (*Ameiurus* Spp.), Madtoms (*Noturus* Spp.), Sculpins (*Cottus* spp.), and Topminnows (*Fundulus* spp.). Using these data, we assess whether cicada consumption varies among surface, pelagic, and benthic foragers, and how traits such as gape size, feeding mode, and habitat preference mediate access to this pulsed subsidy. By integrating drift dynamics with consumer traits, this study provides insight into how infrequent, large-scale resource pulses influence fish feeding ecology, trophic interactions, and energy flow in stream ecosystems, contributing to a broader understanding of consumer–resource dynamics under rare but ecologically significant phenomenon.

id #134876

SVRP Aquifer Protection Efforts by Panhandle Health District in Kootenai County, Idaho

Jenny Gray¹

1. *Panhandle Health District, Hayden, IDAHO, United States*

This talk will provide an overview of prevention-based aquifer protection efforts by Panhandle Health District (PHD) in Kootenai County, Idaho. Historical efforts dating back to the late 1970s through present-day will be discussed. The main focus will be on PHD's Critical Materials Program, which works with commercial and industrial facilities over or adjacent to the Rathdrum Prairie Aquifer (Idaho side of the greater SVRPA) to ensure that chemicals and non-domestic wastewater are properly stored and disposed of. Additional topics will include an overview of PHD's aquifer sampling program, which has provided over 50 years of groundwater quality data; and PHD's education and outreach efforts.

id #135900

Connecting in-stream toxicity testing with community-level responses in an impacted stream

Louise Stevenson¹, Paul Matson¹, Peyton Marcum¹, Scott Brooks¹, Trystan Bordeau¹, Allison Fortner¹, Natalie Griffiths¹, Trent Jett¹, Michael Jones¹, Nikki Jones¹, Peijia Ku¹, Teresa Mathews¹

1. Oak Ridge National Lab, Oak Ridge, TENNESSEE, United States

Laboratory toxicity tests are a standard tool for evaluating the potential impairment of freshwater systems by anthropogenic activities. Although these controlled assays yield valuable information about individual stressors, it remains uncertain whether they accurately reflect community-level impacts on aquatic organisms in natural environments. We present a unique, decades-long dataset combining in-stream toxicity testing and macroinvertebrate community surveys from sites along Bear Creek on the Oak Ridge Reservation (Tennessee, USA), downstream of the Y-12 National Security Complex, allowing us to assess the predictive power of toxicity tests conducted in the lab using stream water for community-level ecological effects. The toxicity testing data set consists of biannual three-brood *Ceriodaphnia dubia* tests from 2005 to 2025, and significant impacts of the stream water on *C. dubia* reproduction have been consistently observed at certain sites through time. Analysis of water quality data in conjunction with these responses has revealed that water quality parameters, specific conductance to the greatest extent, have a strong correlation with observed sublethal toxicity to *C. dubia* (higher conductivity values are correlated with greater toxicity). We identified trends in macroinvertebrate communities, both community-level diversity metrics and family-level presence/absence and densities, that significantly correlated with conductivity and sublethal toxicity responses of *C. dubia* to Bear Creek water samples collected at the same sites during the same period. We found that Ephemeroptera, Plecoptera, Trichoptera (EPT), and non-EPT species density and richness had significant relationships with *C. dubia* reproduction as percent of control (greater toxicity observed in the lab correlated with decreased invertebrate diversity), indicating that toxicity signals measured in the lab have a significant relationship with instream communities. At the family level, we found that many pollution-tolerant families were present and at higher densities during periods of sublethal toxicity, as measured by percent control reproduction of *C. dubia* in the lab, while some sensitive families were absent and/or at lower densities during these periods. These extensive, multi-decadal data sets allow us to investigate the potential for laboratory toxicity tests to detect changes in water quality that impact the diversity of aquatic communities in the field.

id #134622

Think global, teach local: Engaging undergraduates in long-term monitoring of temperature, stream flow, and biological indicators in local headwater streams.

Shannon J O'Leary¹

1. Saint Anselm College, Manchester, NH, United States

Rising air temperatures, altered precipitation regimes, reduced snowfall, and more frequent extreme hydrologic events are expected to increase stream temperatures and modify flow regimes, with cascading effects on aquatic communities. Headwater streams play a key role in maintaining watershed integrity and biodiversity and can function as climate refugia. This presentation describes and undergraduate-centered, inquiry-based teaching and research framework designed to characterize physical and biological responses to environmental change in headwater streams of the Piscataquog River in southern New Hampshire and identify possible climate refugia. This project uses temperature and water-level loggers complemented by imaging approaches, to quantify thermal regimes and streamflow dynamics. Biological responses, habitat use and quality are assessed using environmental DNA and traditional sampling of brook trout and macroinvertebrates. This research is intentionally embedded within the curriculum of a small liberal arts college, where long-term data collection is sustained through student involvement. Students from biology, environmental science, and other majors engage with the project through scaffolded, inquiry-based teaching modules, course-based research experiences, and directed research. Students receive training in lab and fieldwork including experimental design, eDNA methods, data analysis and management, microscopy, and species identification, enabling them to contribute meaningfully to ongoing research questions. Additionally, this project contributes to the Ecological Research as Education Network (EREN) initiative INVURTS (Investigations with NEON: Variability of Macroinvertebrates versus Urban and Rural Temperature Dynamics in Streams), allowing students to integrate local findings with regional and national datasets and to compare patterns across diverse stream systems using NEON-linked data. Student learning is supported and assessed through analysis of research products, data interpretation, and reflective synthesis connecting local watershed observations to broader ecological and climatic patterns. This framework helps students to directly apply course content and place their local watershed in a broader ecological and climatic context, while reinforcing the value of long-term monitoring. This presentation will demonstrate how undergraduate-driven research can simultaneously advance freshwater science, conservation, and education by sharing teaching and laboratory modules, implementation strategies, lessons learned, and pedagogical challenges, as well as pitfalls to avoid when embedding long-term freshwater monitoring into undergraduate curricula.

id #135134

Quantifying the impact of water abstraction on macroinvertebrate communities in Austrian alpine streams

Alexander Auhser¹, Patrick Leitner¹, Kevin Merl¹, Wolfram Graf¹, Franz Greimel¹

1. Department of Ecosystem Management, Climate and Biodiversity, BOKU University, Vienna, Austria

Publish consent withheld

id #136158

Seasonality modulates urban influences on dissolved organic carbon bioavailability

Mary Munt¹, Jacob C. Rudolph¹, Jennifer L. Morse¹, Rebecca Hale²

1. Portland State University, OR, United States

2. Smithsonian Environmental Research Center, Edgewater, MD

Urbanization impacts aquatic ecosystems in a heterogeneous manner, and the driving factors fluctuate both spatially and temporally. One area which is still not well understood is how urbanization impacts the bioavailability of dissolved organic carbon. Bioavailable dissolved organic carbon (BDOC) is a measure of how readily organic carbon is utilized by heterotrophic microorganisms and is dependent on the size and molecular composition of the dissolved organic carbon. We hypothesized that BDOC would peak in the dry season and decrease through the wet season, and that urban factors would significantly influence BDOC. Seasonal surface water sampling occurred in eight watersheds within the Portland, OR metropolitan area between July 2023 and May 2024. The samples were filtered, inoculated with mixed unfiltered water, then incubated in the dark at 20°C for 28 days. Temporal and spatial difference were evaluated using mixed-model ANOVA and the relationship between BDOC and urbanization was modeled using stepwise regressions per season, and redundancy analysis (RDA) for the whole year. BDOC levels changed significantly seasonally as well as between watersheds. Four watersheds saw BDOC decrease from summer peaks through Winter and then increase again in Spring. The other four trended differently, some increasing in Fall and Winter, or decreasing in the Spring. The regressions modeled that correlation with BDOC fluctuates between urban characteristics during the dry season and in-stream processes during the wet season. BDOC was consistently associated with stream metabolism, and the direction of the seasonal correlations suggest that BDOC reflects both autotrophic production and microbial uptake. In the dry season, BDOC shared significant relationships with urban factors (Summer R^2 0.84), and in the wet season BDOC was associated with in-stream processes (Spring R^2 0.60). The RDA identified relationships between the BDOC and factors such as infrastructure age, urban density, canopy cover, and precipitation. The model also revealed that, while urban factors do influence BDOC, seasonal fluctuations govern these urban stream systems, further evidenced by the variable correlations identified by the regression models. These results contribute to further understanding heterogeneous urban impacts coupled with seasonality and how these factors influence the freshwater carbon cycle.

id #135647

EVALUATING CHEMICAL SIGNATURES IN CONSTRUCTED WETLANDS TREATING RECLAIMED WATER: A COMPARATIVE STUDY WITH NATURAL WETLANDS

Chamoda P.D.M. Dissanayake¹, AJ Reisinger², Ashley Hennessey², Michael Mantone², Basil Iannone¹

1. School of Forrest, Fisheries, and Geomatics Sciences, University of Florida, Gainesville, Florida, United States

2. Department of Soil, Water, and Ecosystem Sciences, University of Florida, Gainesville, Florida, United States

As demand for clean water rises, water reuse has become an important strategy. Reclaimed water (RW), treated residential and municipal wastewater, has emerged as a potential solution for irrigation, industrial applications, and groundwater recharge. Despite these potential benefits, RW can contain elevated levels of contaminants such as nutrients, heavy metals, pharmaceuticals, and microplastics, which pose environmental risks when discharged into aquatic ecosystems. Wastewater treatment plants often incorporate constructed wetlands (CWs) to treat RW by mimicking natural wetland functions (i.e., microbial activity, plant uptake, and soil remediation). However, there is limited understanding of how the chemical signatures of CWs differ from natural wetlands, and how these signatures vary across different CW designs (single vs. hybrid connected systems). Our study addresses this gap by quantifying chemical signatures of natural and engineered wetlands and identifying the chemical parameters driving these variations. We hypothesize that (1) CWs differ chemically from natural wetlands, and (2) Hybrid CWs more closely resemble natural wetlands than single CWs, with downstream cells in hybrid systems exhibiting more natural chemical signatures. To test these hypotheses, we sampled 7 natural wetlands, 6 single CWs, and 1 hybrid CW system with 9 wetland cells spanning four treatment stages during the wet season across north-central Florida, USA. At each site, we measured in situ physical parameters (temperature, DO, pH, and conductivity) and collected water samples to characterize various forms of nitrogen, phosphorus, and carbon. Multivariate analyses revealed clear differentiation in chemical signatures between natural wetlands and CWs. CWs were associated with higher nitrogen, phosphorus, and conductivity, while natural wetlands showed higher dissolved oxygen and temperature. Chemical signatures also varied among CW designs. In hybrid systems, chemical signatures shifted progressively as RW moved through treatment stages, with downstream cells exhibiting lower nutrient concentrations and distinct physical characteristics. Our findings suggest that engineered wetlands differ chemically from natural wetlands, and that CW design and management influence their chemical signatures. Our research has the potential to improve the design and implementation of CWs for RW treatment, reducing environmental risks, which is particularly important as RW is increasingly used worldwide to support water conservation efforts.

id #136159

Legacy Mercury, Modern Risks: Identifying Methylmercury Sources & Biogeochemical Processes in an Impaired California Reservoir

Virginia M. Krause¹, Benjamin D. Peterson², Tyler A. Hewett¹, Brett A. Poulin¹

1. Department of Environmental Toxicology, University of California, Davis, Davis, CA, USA

2. School of Freshwater Sciences, University of Wisconsin, Milwaukee, Milwaukee, WI, USA

Mercury contamination poses a significant threat to aquatic ecosystems, with reservoirs in arid regions providing ideal conditions for microbial transformation of inorganic mercury into neurotoxic methylmercury. This transformation, driven by redox-sensitive biogeochemical processes, leads to bioaccumulation and biomagnification of methylmercury in aquatic food webs. Historic mercury mines act as sources of inorganic mercury to local watersheds, particularly in areas impacted by mining such as the California Coastal Range. Presently, 131 reservoirs across the state of California have been listed as mercury-impaired and in need of remediation. Guadalupe Reservoir, in the New Almaden Mining District of Santa Clara County, is among these impaired reservoirs and exhibits record-high fish mercury concentrations statewide. Remediation strategies, including use of sorbent amendments, are under exploration to mitigate methylmercury formation in the system. However, critical knowledge gaps regarding the biogeochemical controls and locations of methylmercury production hinder the effective remediation of mercury concentrations in fish. This study aims to integrate biogeochemical measurements and microbial metagenomics to identify the primary locations of methylmercury production within Guadalupe Reservoir and determine the key biogeochemical and microbial processes responsible for its production. In this assessment, we conducted in-situ incubations in the water column and sediments throughout the stratification cycle along with monthly biogeochemical monitoring to understand mercury methylation processes in varying redox conditions. These experiments will be complemented by metagenomic analyses to identify the microbial clades responsible for mercury methylation under specific conditions in the water column. Ultimately, this research aims to identify the locations and rates of mercury methylation in Guadalupe Reservoir, which in turn will assist water managers in targeted remediation strategies.

id #135905

From Riparia to Stream and Everything in Between: How Salinization Alters Decomposition Rates in Terrestrial and Aquatic Systems

Giisella DePiazza¹, Natalie Clay², Lisa Tabor¹, Megan Underwood¹, Amaryllis Adey¹, Michelle Evans-White³, Sally Entrekin¹

1. Entomology, Virginia Tech, Blacksburg, Virginia, United States

2. Entomology and Plant Pathology, University of Arkansas, Fayetteville, Arkansas, United States

3. Biological Sciences, University of Arkansas, Fayetteville, Arkansas, United States

Terrestrial and aquatic ecosystems are linked through reciprocal exchanges of nutrients, energy, and organisms. Salinization is a large-scale global change impacting both aquatic and terrestrial ecosystems, yet how it affects the linkages between these systems remains poorly understood. Understanding how salinization alters ecosystem processes across the aquatic-terrestrial interface is critical for predicting responses to ongoing environmental change and informing conservation strategies. In this study, we tested the Sodium Subsidy-Stress (SSS) hypothesis, which predicts that moderate salinization subsidizes ecosystem processes while high salinization causes stress, resulting in a hump-shaped response. This talk will focus on organic matter decomposition to quantify the amount and rate that salinization alters decomposition in streams and adjacent riparian zones across an agriculturally and urban induced salinity gradient. We predict that the stress threshold will exist at a lower sodium (Na) level in streams than terrestrial riparia thus displaying a hump-shaped response across the two ecosystems. Using 30 stream sites across the central Shenandoah Valley in Virginia and West Virginia, spanning a gradient from reference conditions to highly salinized (50 to 900 uS/cm specific conductivity) streams. At each site, we deployed paired decomposition assays in the stream and terrestrial riparian zone consisting of red maple leaves and cotton fabric strips to quantify microbial and invertebrate-mediated decomposition. We also measured stream water chemistry and riparian soil ion concentrations to relate decomposition rates. We will present how decomposition responses to salinization differ between aquatic and terrestrial habitats, with potential decoupling of these linked processes under high salinity stress. Understanding how anthropogenic salinization affects carbon cycling and energy flow at ecosystem boundaries will allow managers to predict ecosystem responses and develop strategies to protect these important aquatic-terrestrial linkages in impacted watersheds.

id #135650

Utilizing environmental DNA analysis for monitoring invasive species from lakes and rivers

Kristy Deiner¹, Elvira Maechler¹, Christopher Hemple¹

1. SimplexDNA, Winterthur, Switzerland

Environmental DNA (eDNA) methods are transforming invasive species monitoring by enabling sensitive, cost-effective detection and community-level insights that often outperform traditional field surveys. The quagga mussel (*Dreissena rostriformis bugensis*), a prolific freshwater invader originating from the Black Sea, has rapidly spread through Swiss perialpine lakes and rivers since its first detection in 2014, with profound ecological and socio-economic impacts on water quality, native biodiversity, and hydropower infrastructure in affected cantons.

We present a comprehensive eDNA-based monitoring framework developed by SimplexDNA for cantonal authorities in Zurich and Aargau, integrating quantitative PCR (qPCR) assays with high-throughput metabarcoding to detect and quantify quagga mussel DNA in water samples. Our qPCR assays target species-specific genetic markers to achieve high sensitivity for early invasion fronts and low-density populations, while metabarcoding datasets provide broader context on co-occurring biota and the ecosystem response. These dual approaches support robust presence/absence and relative abundance assessments that inform adaptive management strategies across diverse freshwater systems.

This work aligns with the career-long research trajectory of Dr. David Lodge, whose impactful studies in invasive species ecology and ecological forecasting have emphasized the need for early-detection tools to support decision making and risk assessment.

Lodge's advocacy for bridging academic research with practical management including pioneering eDNA as a decision-support tool underpins the conceptual framework of this project.

Our findings demonstrate that integrating qPCR and metabarcoding eDNA methods can enhance surveillance efficiency, provide actionable data for government authorities, and contribute to evidence-based management of high-priority aquatic invaders such as the quagga mussel.

id #134372

Functional Structure of Native Community Explains Variation in Invasibility

Arif Jan¹, Brooke Penaluna², Laura Hauck², Tiffany Garcia¹, Mark Raggon³, Stacy Crumley¹

1. Oregon State University, Corvallis, OREGON, United States

2. 2Pacific Northwest Research Station, U.S. Department of Agriculture Forest Service, 3200 SW Jefferson Way, Corvallis, OR 97331, USA, Oregon State University, Corvallis, Oregon, United States

3. Bureau of Land Management and USDA Forest Service Interagency Watershed Monitoring Pacific Northwest Region, Aquatic and Riparian Effectiveness Monitoring Program, Bureau of Land Management and USDA, Corvallis, Oregon, United States

Biological invasions are reshaping communities worldwide, yet their mechanistic links to their functional organization have rarely been quantified at broad spatial scales. Classical theories invoke biotic resistance and biotic acceptance. However, empirical understandings that integrate functional structure and invasion dynamics across spatial scales still lacks a unified framework thereby hindering biodiversity conservation strategies. Here, we integrate multi-gene eDNA metabarcoding data from 139 watersheds across the western United States with a harmonized trait database for fishes, amphibians, mammals, and invertebrates to quantify how functional organization of communities governs invasibility by nonnative species. We characterized multiple dimensions of functional diversity including richness, redundancy, evenness, originality, and trait-space vacancy of communities, and related them to watershed-level invasion indices. Across watersheds, the proportion of unoccupied functional trait space i.e., functional vacancy, emerged as the strongest predictor of invasibility ($R^2 = 0.51$, $P < 0.01$). Communities exhibiting high functional redundancy and compact trait distributions resisted invasion, whereas those with greater vacancy and reduced overlap among species were more susceptible. Environmental gradients, particularly stream temperature, stream gradient, and habitat complexity, indirectly shaped invasion risk by altering community functional architecture that in turn affected the opportunity for nonnative species to establish and persist. Together, these results unify classical invasion hypotheses by mechanistically linking functional trait-space structure to invasion opportunity within a quantitative, trait-based framework from eDNA data, showing that invasion opportunity arises from measurable gaps in community trait space.

Keywords: Functional diversity, trait-space vacancy, biological invasions, eDNA, watershed, ecosystems, community invasibility

id #136164

Examining the roles of beavers and time in promoting recovery following stream channel restoration

Ben Colman¹, Alyssum Ahler-Mull¹, Daylen Egger¹, Klemensas Krasaitis¹, Danielle Novotny¹, Eamon Peterson¹, Claire Shady¹, Kory Shoja-Chaghorvand¹, Shawnalee Voyles¹, Paul Parson²

1. Franke College of Forestry and Conservation, University of Montana, Missoula, MT, 59812

2. Trout Unlimited, Missoula, MT, USA

Historical placer gold mining had profound impacts on stream ecosystems, leaving straightened streams that were moved to the margins of their valleys and channelized between 10m tall banks. These physical changes transformed these ecosystems' hydrology, sediment transport, and ecology. Channel reconfiguration may be necessary to restore stream geomorphology and hydrology, and replanting vegetation can help start the process of revegetating riparian zones. However, revegetation can be slow, taking decades to restore stream reaches to the point where they begin to resemble a reference condition. In North America, American Beavers are increasingly recognized for the beneficial effects of their dam-building in some settings, which can accelerate the recovery of floodplain soils and vegetation. To examine the combined effects of time since restoration and colonization by beavers on soil properties and vegetation cover following channel reconfiguration, we studied Ninemile Creek in western Montana, USA. The river was placer-mined from the late 1800s into the early 1900s, and we studied three reaches that were restored in 2014, 2016, and 2018. We examined vegetation cover, soil characteristics, and the number of beaver structures across all restored reaches in both 2019 and 2021. In 2019, beaver had started recolonizing the 2014 reach, which is also the most upstream reach, and had built seven dams and one lodge. By 2021, they had expanded their dams to 36 and built an additional lodge. By 2021, soil moisture and organic matter were both three-fold higher in the 2014-restored reach than in the downstream reaches, overlapping with values from the reference. Additionally, the mean woody vegetation cover was ~5-fold higher in the 2014 reach than in the 2016 or 2018 reaches. These results build on literature suggesting that partnering with beavers either intentionally or through happenstance can greatly accelerate the pace of recovery of restored streams in North America.

id #135909

Nitrous oxide emissions from tailwater recovery systems

Auldyn L. Faulk¹, Shannon L. Speir¹, Nellie I. Little¹, Frank E. Johnson III², Mike B. Daniels¹

1. University of Arkansas, Fayetteville, ARKANSAS, United States

2. U.S. Department of Agriculture, Agricultural Research Service, Oxford, Mississippi, United States

Though agriculture is a cornerstone of the Arkansas economy, it is also a significant contributor to gaseous nutrient losses and the biggest driver of water consumption, accounting for about 70% of freshwater withdrawal globally. Additionally, nitrous oxide (N₂O) is a potent reactive gas in the atmosphere. While irrigation tailwater recovery systems (TWRS) have gained traction in Eastern Arkansas for their ability to conserve groundwater, recycle irrigation water, and reduce nutrient losses, there is limited evidence regarding their role in gaseous nutrient losses. This gap in knowledge has prevented such systems from being recognized as a resilient conservation practice. As such, TWRS may either mitigate or contribute to emissions depending on their management. Without clear data, their potential tradeoffs remain poorly understood. To address this gap, we conducted year-round monitoring of N₂O emissions within three different TWRS located on Arkansas Discovery Farm sites across Eastern Arkansas. To quantify N₂O emissions in TWRS, we sampled emissions utilizing floating chambers placed at three distributed locations in each TWRS, quantifying emissions via gas chromatography. Preliminary results show that N₂O emissions vary seasonally across sites, with the spring season having the highest emissions. Additionally, emissions are driven by nitrate concentrations in the TWRS, indicating that increased fertilizer runoff will enhance N₂O loss to the atmosphere. Additionally, preliminary data indicates that tailwater recovery systems are typically net sources of N₂O to the atmosphere. These findings emphasize the potential tradeoff between water conservation and greenhouse gas emissions from TWRS, suggesting that net environmental benefits depends on the implantation of targeted strategies to also reduce nitrate-driven N₂O losses.

id #135398

Monitoring Starry Stonewort (*Nitellopsis obtusa*) Invasion in Chicago's Lake Michigan Harbors Via SCUBA Surveys and the Development of an Environmental DNA Assay

Megan M. Cunnison¹, Reuben P. Keller¹

1. Loyola University Chicago, Chicago, ILLINOIS, United States

Starry stonewort (*Nitellopsis obtusa*) is a macroalgae causing ecological and economic impacts as it spreads in the US Northeast and Midwest. Monitoring the species' distribution is essential for understanding and controlling its spread. Starry stonewort is established within the Great Lakes basin and was first recorded in Chicago's Lake Michigan harbors in August 2023. Because boats are a known vector for the species and often travel among Chicago's harbors, we suspected the species was present in additional harbors. Prior work showed that SCUBA surveys are substantially more effective for detecting starry stonewort than standard rake toss methods. We performed SCUBA surveys in six of Chicago's harbors to establish a baseline of starry stonewort presence and growth form. SCUBA surveys consisted of two divers searching for five minutes at 10 sites within each harbor. We identified starry stonewort in multiple forms in every harbor and at 75% of all sites surveyed. There were areas where we could not perform SCUBA surveys due to lack of access, low visibility, or heavy boat traffic. To address these shortcomings, we are developing a species-specific environmental DNA (eDNA) assay for starry stonewort. We capture eDNA present in the harbor by pumping water through a filter, extract DNA from the filter, and perform qPCR using primers that amplify starry stonewort DNA. We are able to detect starry stonewort from harbor samples using low annealing temperatures and we can prevent amplification of closely related species using high annealing temperatures. We are testing adjusted extraction and qPCR protocols to increase the sensitivity of the assay while maintaining the specificity. Once the assay is optimized, we will test filters from nine of Chicago's harbors. Our distribution and growth form data from the SCUBA surveys will aid in the validation of our eDNA results. SCUBA surveys offered high resolution monitoring but required extensive time and trained divers, in addition to having limits associated with access and conditions. Ultimately, we aim to produce a reliable and deployable methodology for detecting starry stonewort eDNA so that more frequent and widespread monitoring for the species will be possible.

id #135910

Frequent hypoxia increases the potential for benthic sediment P release within the Maumee River, OH

Sophie E. Dziekan¹, Mike Kulasa¹, Lydia K. Schiopota¹, Teige O'Brien², Casey Godwin², Jim M. Hood¹

1. The Ohio State University, Columbus, OHIO, United States

2. Cooperative Institute for Great Lakes Research, University of Michigan, Ann Arbor, Michigan, United States of America

The accumulation of excess phosphorus (P) in freshwater ecosystems can increase eutrophication, resulting in harmful cyanobacteria blooms that impair ecosystem services and human health. In many large rivers, sediment bound-P accumulates in slow-flowing areas, creating large sediment beds which can contribute to eutrophication under certain conditions. The potential for P exchange between sediment and dissolved pools is shaped by river physicochemistry. Low soluble reactive phosphorus (SRP) and dissolved oxygen (DO) concentrations can enhance sediment P release, particularly when P is bound with redox-sensitive iron and aluminum compounds. Since sediment P release can undermine expensive efforts to limit P losses from agricultural or urban areas, it is important to understand the potential for river sediment to contribute to eutrophication. We examined the potential for sediment P release in a slow-flowing section of the Maumee River between Defiance, OH and Grand Rapids and Providence Dams. Maumee River P exports are the primary cause of harmful cyanobacteria blooms in Lake Erie. At four sites, we monitored DO using a sensor positioned near the sediment-water interface (SWI) and 0.5 m below the surface; we complemented those high-frequency measurements with spatial DO surveys. Monthly, between May and September, we measured benthic sediment characteristics including P fractions, P saturation, sediment P release rates, and other aspects of sediment composition. At all sites, sediment was saturated with P increasing the risk of P release. While the majority of P was associated with a tightly-bound residual fraction, the second most common P fraction was P bound with iron and aluminum which could be released under low dissolved oxygen concentrations. During the mid-summer, SRP concentrations declined while algal biomass increased; hypoxia was common near the SWI. Thus, our results suggest that the risk of sediment P release within this section of the Maumee River is high. Sediment P release would increase the potential for eutrophication within the Maumee River and, potentially, the export of bioavailable P downstream to Lake Erie where it could enhance or sustain harmful cyanobacteria blooms.

Flow-driven plant removal can reduce river ecosystem metabolism

Aaron C Pelly¹, Sarah S Roley¹

1. Washington State University, WA, United States

In aquatic plant-dominated stream and river reaches, ecosystem metabolism can be much higher than in other rivers, but few metabolism studies exist for medium and large rivers like these. In the lower Yakima River (Washington State, USA), dampened spring peak flows have led to the proliferation of water stargrass (*Heteranthera dubia*). This native aquatic plant dominates the ecosystem, often growing across the full width of the river. Large spring peak flows can remove water stargrass biomass, potentially reducing river metabolism. We tested the effect of spring flow by comparing metabolism during the growing season between a high-flow, low-water stargrass year and 5 years with typical low flows and high water stargrass biomass. We estimated metabolism with the 1-station Bayesian inverse modeling approach, using dissolved oxygen (DO) data from a U.S. Geological Survey monitoring location and from PME miniDOT DO sensors. In the low-flow years, mean summer gross primary production (GPP) was $36.5 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$ (range: $23.7 - 41.3 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$) and mean summer ecosystem respiration (ER) was $-25.4 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$ (range: $-15.9 - -28.4 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$). The high flow year with low plant abundance had lower metabolic values: mean summer GPP was $22.3 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$ and mean summer ER was $-15.1 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$. By the fall, GPP and ER were similar between the years. In the low flow years, mean summer metabolism was strongly net autotrophic but became less autotrophic over the season and into the fall. Conversely, the high flow year began slightly net heterotrophic and became strongly net autotrophic over the year. Compared to other plant-dominated streams, in the low flow years, the lower Yakima River had far higher GPP rates but similar or slightly higher ER rates. In the high flow year, GPP was higher, but less so, and ER was comparable. Overall, our results demonstrate that plants can have a strong effect on river ecosystem metabolism and that flow-driven plant removal can mediate this effect.

Assessing structural barriers as a potential method of containing the spread of an invasive crayfish

Jacob B. Watson¹, Pete Cavalli², Kevin R. Gelwicks², Annika W. Walters³, Eric R. Larson¹

1. Department of Natural Resources and Environmental Sciences, University of Illinois Urbana-Champaign, Urbana, Illinois, USA

2. Wyoming Game and Fish Department, Cheyenne, Wyoming, USA

3. U.S. Geological Survey, Wyoming Cooperative Fish and Wildlife Research Unit, Department of Zoology and Physiology and Program in Ecology and Evolution, University of Wyoming, Laramie, Wyoming, USA

Invasive crayfish pose a significant threat to native crayfish, making the identification of effective management strategies a priority. One option is to use existing structures that fragment streams (e.g., culverts and weirs) to exclude invasive crayfish from habitats occupied by native crayfish. We assessed whether existing structural barriers are containing the upstream spread of the invasive virile crayfish (*Faxonius virilis*), which is displacing the native pilose crayfish (*Pacifastacus gambelii*) in the upper Bear River watershed (Wyoming, Idaho, and Utah). In the summer of 2025, we sampled 82 sites and surveyed 18 prospective crayfish barriers throughout the watershed, aiming to identify *F. virilis* leading edges and establish its distribution. We assessed the passability of prospective barriers by the presence or absence of *F. virilis* upstream and downstream of the prospective barriers. Structure attributes (water velocity, drop height, and bank conditions) were measured to identify traits of successful barriers responsible for containing *F. virilis* spread. No structures were found to be successful barriers, likely due to attributes that allowed crayfish to climb structures and the absence of sufficient drop heights. Given that structural barriers do not appear to be limiting *F. virilis* spread in this watershed, barrier removal or modification for connectivity management of native fishes in the watershed is not in conflict with *P. gambelii* conservation. Alternate mechanisms driving the distribution of *F. virilis* in the watershed are discussed, as are future directions for conserving *P. gambelii*, a rare crayfish species, and assessing the stability of its populations in the upper Bear River watershed.

Using molecular and physiological approaches to support freshwater mussel conservation decisions in the Clinton River Watershed

Brittany Perrotta¹, Jason Magnuson¹, Brandon Sansom², James Kunz¹, Allison Sieja¹, Megan Bradley³, Jessica Pruden⁴, Mandy Annis⁴, Jeffery Steevens¹

1. US Geological Survey, MO, United States

2. University of Missouri, Columbia, MO

3. U.S. Fish and Wildlife Service, Genoa, WI

4. U.S. Fish and Wildlife Service, East Lansing, MI

Freshwater mussels are sensitive to contaminants, particularly in early life stages which has contributed to widespread population declines across North America. Historically, the Clinton River Watershed, had a thriving and diverse mussel population. In past decades, chronic anthropogenic disturbances and contaminants in the agricultural/urban watershed have reduced mussel populations. Recent remediation efforts have substantially increased water quality, and mussel augmentation efforts began in 2024 to re-introduce laboratory-reared Fatmucket (*Lampsilis siliquoidea*). Juvenile mussels were measured and deployed into flow-through mussel silos at five locations for three months to assess site suitability across the watershed. Sondes were deployed to continuously measure dissolved oxygen, water temperature, conductivity, and depth. At the end of the deployment period, juvenile mussels were measured, and their digestive glands were dissected for microbiome and transcriptome analysis. DNA and

RNA were co-extracted and sequenced. Sonde measurements were used to estimate stream metabolism over the mussel deployment period. We found that all the sites were suitable for mussel re-introduction as we observed minimal juvenile mussel mortality in the silos and adequate water quality during the deployment period. We found some evidence that mussel growth was correlated with stream metabolism, where the fastest mussel growth rates were measured at the sites with the highest estimated gross primary productivity. Within the mussel microbiome, we determined that longitudinal connectivity was an important factor in shaping the microbiome community, but that all sites except the two closest sites (<1 mile) had distinct microbiome communities. Interestingly, we found overall microbiome function was similar among all assessed sites. We are currently analyzing the mussel transcriptomes to gain further insight into juvenile mussel gene expression at each site during deployment in the silos. This work provides insight into mussel health, including stressors, during deployment and demonstrates endpoints to consider during site assessment for augmentation.

id #135400

The importance of small streams

Walter Dodds¹

1. Kansas State University, Manhattan, KS, United States

Small streams are key providers of ecosystem services including water quantity and quality, biodiversity, and cultural values. Small streams (low order) are the predominant interface between terrestrial and aquatic habitat. Given the immense pressures these streams face globally, and the hesitation in the US to regulate them to achieve environmental protection goals, I review progress in establishing the importance of these streams and try to distill simple talking points that can be used to emphasize how essential it is to protect and research small streams. 1) Most stream length and bottom surface area is made up of low-order streams. 2) Adventitious streams are most commonly low order streams, even those adventitious to rivers. 3) Most days of flow in most streams are driven by groundwater inputs and small streams are an essential interface with groundwater. 4) Low order streams and their riparian zones control downstream pollution, even when low order streams are intermittent. 5) Low order streams mediate water yield, and this can be even more important in areas with ephemeral and intermittent channels. 6) Small streams and their riparian zones are key support areas for terrestrial and aquatic biodiversity. and 7) pristine small streams are culturally valuable.

id #135912

Trait-based ecology and LeRoy's quest for generality over contingency

Julian D Olden¹

1. School of Aquatic and Fishery Sciences, University of Washington, Seattle, Washington, USA

Historical roots in natural history provided the foundation for over a century of ecological research focused predominantly from a taxonomic perspective. Although recognized a century ago, it was only in recent decades that scientific thought branched out to include ecological theories and frameworks built on traits rather than taxonomy. The notion is that species traits serve as a universal currency for studying the patterns and drivers of biodiversity across diverse taxonomies and geographies, enabling ecological generalization. In this presentation, I trace how LeRoy Poff's foundational research in the 1990s sparked the emergence of trait-based approaches in freshwater ecology.

id #134633

Patterns in freshwater mussel community reassembly and composition relative to disturbance

Tyler Schartel¹, Brian Metzke²

1. Illinois Natural History Survey, Springfield, ILLINOIS, United States

2. Illinois Department of Natural Resources, Springfield, IL, USA

Post-disturbance patterns in freshwater community reassembly can be affected by a multitude of factors including regional species pools, time since disturbance, local habitat, anthropogenic stressors, and species' life histories. However, community reassembly patterns and the time scales over which they occur remain poorly understood for freshwater mussels. Here we utilize mussel occurrence and abundance data from 20 disturbed and 47 undisturbed (i.e., reference) sites to characterize temporal dynamics in community composition and species abundances relative to site disturbance. Changes in community composition, as well as rates in composition change, were site-specific and varied substantially across sites relative to disturbance status and time between sampling events. Community trajectory analyses revealed idiosyncratic site trajectories with respect to both community composition and abundance. Disturbed site trajectories indicated these sites were unlikely to return to either pre-disturbance community composition or abundances in the timeframes considered here. Ecological state assessment developed reference envelopes that can be leveraged to determine if sampling events or entire trajectories in disturbed sites differ exceptionally in mussel community compositions and abundances. Multiple sampling events in disturbed sites were found to fall outside this reference envelope. Ultimately, results of this investigation will aid managers in defining expectations for community reassembly and recovery, identifying conservation actions necessary to reach conservation goals, and developing tools for identifying whether disturbed sites are beginning to return to baseline or reference expectations with respect to mussel community composition or abundances. To this end, new perspectives about what constitutes mussel community recovery, and how recovery is assessed, will likely need to be adopted.

id #134889

Heterogeneous Freshwater and Solute Export from a Mountainous Headwater Stream Network

William F. Mejia-Garcia¹, Alex J. Webster¹

1. University of New Mexico, Albuquerque, NEW MEXICO, United States

Mountainous headwater stream networks are key drivers of freshwater and solute export to downstream waters and exert a strong influence on riverine chemistry due to their high surface-area-to-volume ratios and terrestrial-aquatic connectivity. Varied flow paths across heterogeneous terrestrial subcatchments result in distinct assemblages of geogenic and biogenic solutes delivered to streams, where they may serve as indicators of ecosystem function. Understanding how landscape characteristics influence freshwater and solute export is particularly important as high-elevation watersheds undergo climate-driven shifts in precipitation patterns and vegetation communities, affecting water resource resilience worldwide. Since 2023, we have conducted monthly catchment-wide synoptic sampling campaigns of streamflow and water chemistry across gradients of elevation, aridity, vegetation composition, topography, and hydrological connectivity within a major municipal watershed of the southwestern United States. We characterized the magnitude and variability of subcatchment contributions to total watershed export through a residual analysis and inferred stream network function via mass-balance-based ecohydrological metrics. Our residual analysis suggests that high-elevation subcatchments, which receive greater mean annual precipitation, export higher-than-expected and less variable amounts of freshwater, nitrate, phosphate, and dissolved organic carbon (DOC), relative to lower-elevation, drier subcatchments. These patterns reflect hydrological controls on solute export and likely arise from shallower, more dynamic flow paths associated with biogenic solute transport. However, several sites exhibit behavior opposite to that of adjacent subcatchments with similar terrestrial characteristics, suggesting heterogeneous terrestrial nitrate pools, contrasting source types, or localized land disturbance. Furthermore, ecohydrological metrics reveal seasonal DOC enrichment and nitrogen uptake at lower elevation stream reaches, highlighting differences across the catchment. Through these analyses, we identify subcatchments and stream reaches that disproportionately contribute to catchment-wide export and exert influence on stream network function. These findings provide insights to inform watershed management strategies aimed at maximizing the reliability of freshwater export, improving water quality, and supporting watershed resilience under changing climatic conditions.

id #135913

Examining influences of submerged macrophytes on fluvial CH₄ and CO₂ drivers and concentrations.

Jaden Nguyen¹, Emily H Stanley¹

1. Center for Limnology, University of Wisconsin, Madison, Wisconsin, USA

Fluvial systems are disproportionately large sources of greenhouse gases to the atmosphere, but demonstrate considerable variability in both the amounts and drivers of their contributions. Submerged aquatic vegetation is a prominent feature within many freshwater systems and is a strong driver of greenhouse gas dynamics in wetlands and lakes. However, its influence remains largely unexamined in rivers and streams. Submerged macrophytes may alter carbon cycling, contributing to dissolved greenhouse gas concentrations by creating conditions favorable to their production or conducive to their consumption, oxidation, or removal. During summer 2025, we conducted a short-term, small-scale (600 sq. meter) removal of macrophytes in a river in south-central Wisconsin. Dissolved greenhouse gases, river sediment, and water chemistry measurements were collected sub-weekly over a period of eight weeks. Following macrophyte removal, we observed a decrease in the percentage of sediment organic matter (1.86% to 1.47%) and a decrease in methanogenic potential (4.013 $\mu\text{mol L}^{-1} \text{ day}^{-1}$). These findings were coupled with an unexpected 20% increase (0.013 $\mu\text{mol/L}$) in daytime dissolved CH₄ and a 12.5% (0.762 $\mu\text{mol/L}$) increase in daytime dissolved CO₂ concentrations compared to the adjacent reference area. These results suggest that macrophytes increase methane oxidation and/or removal of CH₄ and CO₂ and raises the possibility that they may not necessarily enhance gas emissions in these ecosystems.

id #134634

Microbial organic matter decomposition along a P-gradient in tropical streams

Alex Shelton¹, Vamery González Hernández¹, Alonso Ramirez¹

1. North Carolina State University, Raleigh, NC, United States

With phosphorus (P) pollution rapidly increasing over the past decade, it is becoming increasingly important to understand the impacts that P can have on biodiversity and ecosystem function. The geothermally modified groundwater at La Selva Research Station in Costa Rica offers a unique opportunity to study the impacts of P on organic matter processes across a naturally occurring P-gradient. While studies have been conducted at La Selva to determine the impacts of P on leaf litter decay, deploying a cotton strip assay will remove the effects of physical decomposition from flow and macroinvertebrate presence, further isolating the decomposition effects of P and the resulting microbial activity. Therefore this project aims to determine the impact of microbial activity on organic matter decomposition rates across a naturally occurring P gradient. A standardized cotton strip assay was deployed across 13 streams that form a natural P gradient. Strips were collected between 10 and 19 days, with collection dates chosen according to estimated decomposition rates of each stream to ensure > 50% loss of tensile strength. Mean tensile strength loss across all streams was $3.66 \pm 1.97\%$ per day, which aligns with the predicted cotton strip decomposition rate at La Selva of 3.90% per day, based on models developed from leaf litter studies. Microbial respiration rates ranged from 0.11 ± 0.01 to $0.34 \pm 0.01 \text{ mgO}_2\text{gDM}^{-1}\text{hr}^{-1}$. Tensile strength loss and microbial respiration rates were regressed against Soluble Reactive Phosphorus (SRP). The relationship between these variables and the SRP were non-linear, therefore, a Michaelis-Menten model was used, following previous studies. These results indicate, as predicted, both decomposition and microbial respiration rates are high and P limited at low P concentrations ($<10.15 \mu\text{g SRP L}^{-1}$). Identifying the role microbes play in decomposition is becoming increasingly more important as biodiversity declines worldwide. They may be able to compensate for the loss of biodiversity and maintain ecosystem function, especially in the case of decomposition. The extent to which microbes can help mitigate changes in stream ecosystem processes could provide valuable insights into the resilience of tropical streams to global change.

id #135402

The Puzzle of “Broken” Streams: Linking Barrier Assessment to Fish Community Structure in Intermittent Streams of the Intermountain West

Stephanie Webster¹, Christopher Cheek¹

1. South Dakota State University, Brookings, SOUTH DAKOTA, United States

Stream fragmentation caused by dams and road-stream crossings is pervasive across the landscape, yet many of these structures remain unassessed in the Intermountain West. This study inventoried dams and road-stream crossings across Montana and Wyoming to evaluate their impacts on aquatic connectivity and fish communities. We utilized the North Atlantic Aquatic Connectivity Collaborative (NACC) Rapid Assessment protocol to assess road-stream crossing structures and we will investigate how the placement of road crossings and dams influences fish community assemblage along the longitudinal gradient in highly disturbed intermittent streams during the dry phase by sampling fish and environmental variables in wetted pools and applying trait-based multivariate community analysis.

id #135914

Are positive effects associated with ecosystem engineers detectable at large scales: Beyond mesocosms and mechanisms

Samuel F Fritz¹, Hayley C Oakland², Anna C French¹, Geoffrey C Poole², Lindsey K Albertson¹

1. Ecology, Montana State University, Bozeman, Montana, United States

2. Land Resources and Environmental Sciences, Montana State University, Bozeman, Montana, United States

Disentangling the influence of biotic and abiotic drivers on the distribution of individual species, or the composition of communities, is a fundamental challenge in ecology. Ecosystem engineers exist at an important nexus between biotic and abiotic forces, shaping the physical environment and often facilitating other species in the process. In freshwater ecosystems, net-spinning caddisflies (Trichoptera: Hydropsychidae) are abundant, widely distributed ecosystem engineers. Caddisfly silk nets and retreats have well documented positive effects on stream macroinvertebrates and microbial communities. However, because caddisfly distribution itself is also driven by the abiotic environment, their importance to biotic interactions is difficult to predict at the landscape scale. We sampled caddisflies and other macroinvertebrates across a 1.5km reach of Cherry Creek, MT to evaluate whether high resolution, georeferenced data on substrate type, substrate size, geomorphic units, current velocity, water depth, temperature, light, and dissolved oxygen or the presence of caddisflies best predicts macroinvertebrate community structure. We found that net-spinning caddisflies ranged widely in density (0-1200/m²) and abundance was highest in large substrates. We discuss how to use hierarchical modeling of species communities (HMSC) to explore the abiotic factors driving both the occurrence of an engineer of interest, such as net-spinning caddisflies, and associated macroinvertebrates across stream reaches. We then review the evidence for positive associations between caddisflies and their potential beneficiaries after accounting for the influence of the abiotic environment. Our work informs ideas about the spatially patchy importance of ecosystem engineering and advances the goal of understanding the interplay between biotic and abiotic drivers at scale.

id #135915

Shifting Communities: How Salinization Alters Riparian Plant Diversity, Phytochemistry, and Litterfall into Streams

Maggie C Herrmann¹, Michelle Evans-White¹, Sally Entekin², Samuel Diaz¹, Clay Prater¹, Beck O'neil¹, Misty Goings¹, Cale Prosen¹, Vien Tran¹, Anna Ortegón¹, Connor Foley¹, Lilia Stemet³, Natalie Clay¹

1. University of Arkansas, Fayetteville, ARKANSAS, United States

2. Biology, Virginia Tech University, Blacksburg, Virginia

3. Biology, University of Arkansas, Monticello, Arkansas

Riparian ecosystems are highly productive habitats that support a diverse assemblage of vegetation which buffers freshwater bodies, traps sediment, and delivers organic matter to adjacent aquatic systems. These systems are increasingly threatened by soil and freshwater salinization, which can alter plant phytochemistry and reduce species diversity, richness, and evenness and thus the quantity and quality of stream inputs and services. Despite their ecological importance, the effects of salinization across sodium gradients on the phytochemistry and community composition of inland riparian plants remain poorly understood. To address this gap, we conducted a year-long greenhouse paired riparian-stream mesocosm experiment where both systems simultaneously received one of six NaCl addition levels: 100, 200, 300, 400, 500 mg/L NaCl or just water as controls. An additional two treatments were added where just the riparian soil or stream received salt additions (n = 5 replicates per treatment; total N = 40 mesocosms). Salt treatments were added to each mesocosm once weekly. Additionally, we measured soil conductivity, litterfall, plant diversity and height, and leaf chemistry. After ~4 months soil conductivity averaged 73, 78, 149, 117, 192, and 199 uS/cm for controls through 500 mg/L NaCl addition, respectively. Mesocosms had ~56 plant species across all treatments with an average of ~17 per mesocosm. Control treatments had the highest diversity (H'=1.73) and richness (~8) and the 500mg/L treatments had the lowest (H'=1.41, ~7 respectively). Overall, control treatments were 19% more diverse and ~10% richer than 500 mg/L treatments. Litterfall averaged .4994 mg/m² across treatments and was highest in the 100mg/L treatments (8520 mg/m²) and lowest in the 400mg/L treatments (2330 mg/m²). Litterfall chemistry will be analyzed and discussed. Together these results suggest that salinization may reduce riparian plant diversity and richness. Lower plant diversity combined with alterations to soil conductivity and declines in litterfall rate as salinization increases are likely to alter the quality and quantity of organic matter reaching adjacent streams.

id #136172

Biodiversity of Adult Odonata Communities of South Central Virginia

Scott M Starr¹, Joe Newcomer¹, Victoria Fenton¹, Cullyn Cary¹, Will Gardner¹

1. Hampden-Sydney College, Farmville, VA, United States

Aquatic habitats and biodiversity are in peril around the world. Anthropogenic impacts on these systems are immense and complex due to habitat degradation and water pollution. Aquatic macroinvertebrates are used as bioindicators in-order to study current aquatic habitat health. Order Odonata, commonly known as dragonflies and damselflies and are found across the globe. Order Odonata is a very understudied group in many regions of the United States. The current citizen science database focused on the distribution, biogeography, biodiversity, and identification of order Odonata is called Odonata Central (www.odonatacentral.org). The south central region of Virginia, including Prince Edward County and its neighboring 7 counties (Amelia, Appomattox, Buckingham, Charlotte, Cumberland, Lunenburg, and Nottoway) are one of these understudied regions for Odonate biodiversity. When examining the surrounding counties, only 174 records have been recorded between 2002 and 2021 on Odonata Central. In total, 80 unique species had been identified during that time period in those counties with lowest county diversity consisting of 3 species and the highest county diversity having 42 identified species. Six of the eight counties have less than 20 species identified.

This presentation will focus on data collected in Prince Edward County. Over the past four years (2022-2025), field surveys were conducted from early April to late October. Streams, rivers, wetlands, ponds, lakes, meadows, and other habitats favored by adult odonates were surveyed. Aerial nets and photography were used to capture physical specimens and detailed images that could be used for identification. Specimen vouchers (male and female) were preserved for a reference collection. When surveys first started, only 18 species had been documented in Prince Edward County. Current species richness now stands at 68 species. This large spike in known diversity wasn't just due to the field surveys that were conducted. Citizen science observations that were posted to Odonata Central and iNaturalist were key in helping to find rarer species. This study demonstrates how scientific survey along with citizen science platforms can be vital in documenting the biodiversity of specific groups of organisms, especially in regions where they are understudied.

id #134637

Acid Rock Drainage in a Warming Arctic: Abrupt Onset and Persistent Downstream Effects in Rivers Across Alaska's Brooks Range

Taylor G Evinger¹, Jonathan A O'Donnell², Michael P Carey³, Joshua C Koch³, Carson A Baughman³, Kenneth R Hill⁴, Rebecca Frei¹, Brett A Poulin¹

1. University of California, Davis, Davis, CA, United States

2. Arctic Network, National Park Service, Anchorage, Alaska, United States

3. Alaska Science Center, U.S. Geological Survey, Anchorage, Alaska, United States

4. Arctic and Central Alaska Networks, National Park Service, Fairbanks, Alaska, United States

Streams and rivers in Alaska's Arctic provide essential ecological and societal benefits but are increasingly vulnerable to degradation from permafrost thaw and the release of toxic elements. Here, we evaluated the water chemistry of six watersheds across Alaska's Brooks Range (2022-2024) to quantify the effect of thaw-driven acid rock drainage on river chemistry. As a control, we sampled pristine mainstem rivers absent of major point-source acid rock drainage inputs and found low metal concentrations (median; sum of 23 metals: 99 µg/L) in conjunction with alkaline pH (pH: 8.3). In contrast, point source acid rock drainage from hillside seeps (locations of acidic groundwater upwelling on hillsides) exhibited low pH (pH: 3.2), high sulfate (SO_4^{2-} : 3,025 mg/L), and higher major and trace element concentrations compared to alkaline streams (sum: 451 mg/L), consistent with metal-rich sulfide mineral sources. Similarly, tributaries impaired by seeps exhibited lower pH (pH: 6.9), higher sulfate (SO_4^{2-} : 390 mg/L), and elevated metals (sum: 9.2 mg/L) than rivers from control, unimpaired watersheds. Across the six watersheds, point-source acid rock drainage inputs from seeps and tributaries increased mainstem concentrations of SO_4^{2-} up to 962% and many other elements (Fe, Al, Mn, Zn, Ni, Co, Cd, Cu, and rare earth elements (REE)). Further, a temporal analysis (2015-2024) of SO_4^{2-} concentration revealed an abrupt increase of 174-263% in 2019 and 2020 across two watersheds following record warm and snowy years. This timing is consistent with the onset of orange staining in multiple watersheds identified by remote sensing observations. In the Salmon River watershed, impaired tributaries significantly increased filtered and particulate constituent concentrations (SO_4^{2-} , Fe, Zn, Ni, and Cd) that remained elevated >100 km downstream, highlighting the potential for long-range transport. The low pH and high metal concentration of seeps and tributaries impaired by seeps observed across watersheds reveal the high potential for acid generation and metal release across watersheds spanning the western Brooks Range. Collectively, these results document a high potential for acid rock drainage to affect mainstem river chemistry across the western Brooks Range that persists downstream, with implications for biogeochemical cycles and ecological and human health.

id #135405

Macroinvertebrate diversity in climate-stressed ponds: when will communities recover from drying?

Joshua Rihs¹, Kaitlin Haase¹, Rebecca Best¹

1. School of Earth & Sustainability, Northern Arizona University, Flagstaff, AZ, USA

Aquatic diversity in the arid Southwest is sustained by thousands of small ponds and wetlands, many of which are ephemeral. The maintenance of taxonomic and functional diversity across this system therefore depends on how well taxa can either survive drying events or disperse across the landscape to reestablish their populations. To track changing ephemerality, connectivity, and biodiversity through >5 years of drought conditions our lab has combined remote sensing and community sampling in nearly 50 ponds. Remote sensing using Sentinel-2 imagery revealed a consistent decrease in habitat availability in the early summer period between snowmelt and the monsoon season. Despite this frequent drying, many ephemeral and permanent ponds support equivalently diverse communities, each with distinct specialists, such as fairy shrimp in ephemeral ponds and amphipods in

permanent ponds. Drying events in ponds without a history of disturbance can shift communities into a lower diversity state dominated by generalists, but recolonization by generalists is remarkably fast, with no evidence for dispersal limitation. In the absence of specialists, diversity appears to depend not on connectivity or hydroperiod but on habitat quality indicators such as submerged aquatic vegetation and turbidity, which can be impacted by invasive species and land management. Together, these studies show that climate effects on macroinvertebrate diversity depend on the traits of the original species pool as well as interactions with other global change stressors.

id #135917

DISTANCE: How do particle size and type interact with seasonal dynamics to impact the fate of DNA in a prairie stream?

Tobin J Davidson¹, Leonardo A Medellin², Hannah L Braslau³, Jacquie Molloseau³, Krista M Ruppert¹, Pedro FP Brandão-Dias⁴, Daniel C Allen³, Ariel J Shogren², Zacchaeus G Compson¹

1. Department of Biological Sciences, Advanced Environmental Research Institute, University of North Texas, Denton, TX, USA

2. Department of Biological Sciences, The University of Alabama, Tuscaloosa, AL, USA

3. Department of Ecosystem Science and Management, Penn State University, University Park, PA, USA

4. School of Marine and Environmental Affairs, University of Washington, Seattle, WA, USA

Environmental DNA (eDNA) is increasingly used as a cost-effective, non-invasive alternative or complement to traditional sampling methods, with applications ranging from targeted species detection to long-term biodiversity monitoring. The NSF-funded project, "Collaborative Research: MRA: Determining InStream Transport of eDNA across NEON sites at the Continental Scale (DISTANCE)," examines how abiotic factors influence the transport, persistence, and fate of eDNA in stream ecosystems. We will conduct repeated, constant-rate releases of Atlantic Herring (*Clupea harengus*) DNA (in the form of a homogenized tissue slurry) to quantify downstream eDNA transport dynamics across National Ecological Observatory Network (NEON) stream sites spanning multiple ecoregions in the United States. These nutrient-style additions will elucidate how spiraling dynamics vary across a wide range of streams with different biogeochemical conditions. In addition to Atlantic Herring releases, we will conduct seasonal releases to characterize temporal DNA flux patterns from a suite of locally significant species with differing tissue types in North Texas within Pringle Creek (NEON domain 11). This will allow for a comparative analysis of how biological source material and particle size affect the spatio-temporal distribution of DNA under fluctuating seasonal parameters. We hypothesize that fine-scale eDNA particles originating from labile tissues will exhibit increased transport distances and accelerated degradation rates relative to larger particles. Preliminary results from the DISTANCE project will advance mechanistic understanding of eDNA dynamics and inform the application of eDNA methodologies in lotic ecosystems nationwide.

id #136173

Flawed theory or lack of biological understanding?: Responses of benthic macroinvertebrate taxa to environmental alteration in western USA streams are neither strongly nor predictably associated with biological traits

Charles P Hawkins¹, Andrew M Caudillo¹, Jennifer L Courtwright¹

1. Watershed Sciences, Ecology Center, and National Aquatic Monitoring Center, Utah State University, Logan, Utah, United States

Trait-based analyses are promoted as a way to better understand the forces that drive patterns of adaptive radiation in evolutionary time and influence the assembly of communities in ecological time. Another potential advantage of trait-based analyses is that they are potentially more transferable across regions, which typically differ in species composition, than taxonomic-based analyses. We used a western-wide multivariate species distribution model derived from 659 reference-quality sites to estimate the predicted probabilities of capture (pc) of 267 taxa at both reference and 350 probabilistically selected sites sampled between 2018 and 2022 throughout the western USA. We then estimated both expected (F_e) and observed (F_o) frequencies of occurrence for each taxon as $F_e = \text{Sum}(pc)$ and $F_o = \text{Sum}(\text{presences})$ for non-reference and reference sites, respectively. We also assigned 86 biological trait states to each taxon to assess which trait states were most strongly associated with whether taxa were increasers or decreasers. Dominant environmental stressors in western USA streams include alterations associated with dams, urbanization, agriculture, and livestock grazing. We found that 98 taxa were strong decreasers ($F_o/F_e < 0.8$) and 91 taxa were strong increasers ($F_o/F_e > 1.2$). F_o of common taxa decreased more often than that of rare taxa, which tended to increase in F_o more often than common taxa. However, only 5 trait states (slow development, rare occurrence in drift, synchronous emergence, short lifespan, and free-range attachment) were associated with differences in taxa composition between reference and non-reference sites. Despite 21-27% of these streams having lost > 28% of their local taxa, these trait states were only weakly correlated with differences in taxa composition between reference and non-reference sites (Mantel r values = 0.30-0.39). It was also difficult to interpret how shifts in these trait states were mechanistically related to either natural environmental gradients or variation in land use. Ecology has struggled as a discipline to develop strong, predictive theories. Moreover, the ecological literature is full of poorly designed studies or studies whose conclusions suffer from confirmation bias. True progress will come only when we become more rigorous in designing, interpreting, and evaluating the research we conduct.

id #135918

Climate and hydrologic environmental thresholds for stream fish species and assemblages across flow regimes

Dan Magoulick¹, Ana Meza Salazar¹, Chloe Moore Voorhees¹, Tyler Fox¹, Sarah Sorensen¹

1. Arkansas Cooperative Fish and Wildlife Research Unit, University of Arkansas, Fayetteville, Arkansas, USA

Identifying thresholds in fish-environment relationships is a goal of many studies and decision making. We compiled landscape-scale, geo-referenced species occurrence datasets to examine and compare patterns of fish assemblage and species-level turnover across stream flow regimes. A gradient forest machine learning approach was used to quantify multi-species threshold responses along climate and hydrologic disturbance gradients in groundwater, runoff, and intermittent streams in the Interior Highlands, USA. Spatial, climate, soils and lithology variables were most important in explaining fish assemblage structure, but their importance varied by flow regime. Thresholds were common but species and assemblage dependent. Assemblage level responses typically showed fewer thresholds than species level responses. Climate and hydrology importance were flow regime and species dependent. This approach can be used to inform conservation efforts for multiple species, individual species of conservation concern and whole assemblages. Results of this study also provide understanding of complex nonlinear environmental and disturbance threshold relationships driving patterns in fish assemblages and species in streams.

id #135151

A Saltier Future for Europe's Rivers: Continental-Scale Evidence of Freshwater Salinization

Sandra Mingarelli¹, Julie Crabot², David Cunillera-Montcusi³, Gary Free¹, Daniel Hernandez-Carrasco⁴, Martyn Kelly⁵, Alvaro Moyano², Ralf Schaefer⁶, Miguel Canedo-Arguelles²

1. *European Commission Joint Research Centre (JRC), Ispra, Italy*

2. *Institute of Environmental Assessment and Water Research (IDAEA), Barcelona, Spain*

3. *Institute of Aquatic Ecology, HUN-REN Centre for Ecological Research, Budapest, Hungary*

4. *School of Biological Sciences, University of Canterbury, Christchurch, New Zealand*

5. *Bowburn Consultancy, Durham, UK*

6. *Research Centre One Health Ruhr and Faculty of Biology, Ecotoxicology, University of Duisburg-Essen, Essen, Germany*

Freshwater salinization is an increasingly widespread environmental challenge, driven by climate change, hydrological alteration, land-use intensification, and growing human demands on water resources. Rising salt concentrations are reported across rivers and streams worldwide, with significant consequences for aquatic ecosystems, water quality, and the sustainability of freshwater services. In Europe, salinization is of particular concern given extensive river regulation, high population density, and the connectivity of transboundary river networks. Both the European Water Framework Directive (WFD) and the United Nations Sustainable Development Goal (SDG) indicator 6.3.2 explicitly require the assessment of salinity and the establishment of criteria for good ecological status. However, few European countries have developed salinity thresholds, and their ecological rationale is often unclear, highlighting the urgent need for a continent-scale understanding of the extent, drivers, and impacts of freshwater salinization to inform robust criteria.

Using a pan-European modelling approach, we show that freshwater salinization is already pervasive. Of the 35 countries with available data, 28 contain severely salinized rivers or streams, and an estimated 14–26% of the European river network exhibits some degree of salinization. We confirm several well-known hotspots—including the Ebro basin in Spain, the Meurthe in France, Alpine streams in Italy, and coastal rivers in Belgium, the Netherlands, and Denmark—while also identifying previously unreported regions with elevated salinity. Observed levels are sufficiently high to restructure aquatic communities, reduce biodiversity, disrupt ecosystem functioning, and increase the likelihood of harmful algal blooms and fish mortality. In many locations, salinity exceeds recommended limits for drinking water, irrigation, and industrial use.

Our results indicate that vulnerability to salinization is shaped by natural factors such as soil composition, geology, and rock weathering, but human pressures—including urbanization and agriculture—are strong predictors of elevated salt concentrations. Ion ratios further suggest that evaporation is an important driver, particularly in western Europe. As freshwater salinization intensifies under future climate and land-use change, these findings underscore the need for coordinated, transboundary management and the development of scientifically robust, ecologically meaningful salinization criteria to support WFD implementation and progress towards SDG 6.3.2, preventing future ecological and societal impacts.

id #135407

Invertebrate community dynamics along a rock glacier outflow in the alpine zone of the Sierra Nevada (California) in years of high and low snowpack

David Herbst¹, David Herbst¹

1. *Sierra Nevada Aquatic Research Laboratory, Nevada City, CALIFORNIA, United States*

Rock glaciers (RGs) are unusual high elevation cold rocky areas where melt from buried glaciers form emerging springbrook habitats whose biology has not been well explored. As climate warms, surface glaciers disappear, and drought threaten headwaters, these kinds of environments may become important refugia. Besides documenting the benthic invertebrate fauna of RGs, this study asks whether community patterns are local and stable with downstream distance from the source over years of varied snowpack. Midges prevail in these underground outflows from within rock jumbles and always dominated in the areas nearest the RG. EPT (mayfly, stonefly, caddisfly taxa) were most common and diverse downstream where RG waters mixed with surface tributary sources. These sites are all within ~1 km of each other. In years of high snowpack, with later melt and colder temperatures, the community across sites sustain the greatest population densities of midges including *Diamesa*, the ice fly, along with *Paratrichocladius* and trichad flatworms. In years of low snowpack drought conditions with earlier snowmelt and warmer conditions, *Hydrobaenus* and *Diplocladius* came to dominate and the others all but disappeared. The golden algae *Hydrurus foetidus* also was most common upstream in cold years and diminished under drought flows. If droughts and warming become more prevalent in alpine zones, some species of RG midges seem likely to become displaced while others become more prominent. These taxa along with a possibly endemic caddis and stonefly are being barcoded.

id #135919

The effects of in-stream habitat complexity on translocated bluebreast, Tippecanoe, and variegate darter populations

Julia Glandorff¹, Lauren Pintor¹, Brian Zimmerman¹

1. The Ohio State University, Columbus, OH, United States

Although the passage of the Clean Water Act spurred significant recovery of native fish populations, habitat degradation continues to limit species recovery into their historic ranges. Conservation translocation – reintroducing fish into their former native range – is a common management strategy used to overcome these barriers to natural recovery, but not all fish species respond similarly to translocation efforts. Thus, there is a critical need to learn from translocation successes and failures to maximize the chance of recovery success. The bluebreast darter (*Nothotus camurus*), variegate darter (*Etheostoma variatum*), and Tippecanoe darter (*N. tippecanoe*) were formerly endangered in Ohio and have been reintroduced to their native ranges in six Ohio rivers via yearly translocation events that began in 2016, 2020, and 2018, respectively. There has been wide variation in species re-establishment following translocation. One proposed explanation for this variation is differences in habitat quality. Habitat complexity, or the spatial arrangement of objects such as rocks and logs in a riverine habitat, is an understudied component of habitat quality that may influence this variation. High levels of benthic habitat complexity provide shelter and surface area for foraging and reproduction, therefore encouraging population growth following translocation. However, siltation lowers benthic habitat complexity by occluding benthic interstitial space. To determine the influence of benthic habitat complexity on translocated darter populations, we employed a transect-based method to measure substrate size (including silt) and rugosity, or surface roughness, at each translocation site. We predict that 1) sites with higher rates of siltation and 2) less diverse substrate assemblages will be associated with lower darter population sizes due to less physical space to forage, reproduce, and shelter in. This study will provide critical knowledge natural resource managers can reference to perform more effective conservation translocations, as well as generally further the field of conservation science.

id #136175

Decadal trends in native and non-native fish assemblages in the Willamette River in Oregon

Stan Gregory¹, Brooke Penaluna¹, Randy Wildman¹

1. Oregon State University, Corvallis, OR, United States

Major declines in biodiversity are occurring globally, and data for native and non-native fish assemblages in freshwaters, especially large rivers, are limited. We assessed assemblages of native and non-native fishes of the mainstem Willamette River, Oregon in 2011-2013 and 2021-2023, as well as three major tributaries in 2015-2017, using consistent boat and backpack electrofishing methods. The Willamette River basin contains 34 species of native fish and 30 non-native species; the mainstem river contains 31 native and 25 non-native species. Abundance and species richness of native fish in both decades decreased longitudinally downstream, and abundance and richness of non-native fish increased downstream. In the mainstem Willamette in 2011-2013, 93% of the individual fish captured were native species. In monitoring of three major river tributaries in 2015-2017, 97% of the individual fish captured were native species. In the period of 2021-2023, abundance of non-native fish in the mainstem Willamette was significantly greater as compared to the previous decade, and the increase was greater in the downstream reaches of the river. Unlike the earlier mainstem monitoring, only 75% of the individual fish captured were native fish. Sixty percent of individual native fishes showed evidence of a decline in abundance or occupancy between decades, but absolute differences were relatively small for most species. Regional temperatures were exceptionally warm in 2015 and 2021. Decadal trends revealed that the greatest decline for native fishes of small cold streams and cool channels was in the upper mainstem, and the greatest decline for native backwater fishes occurred downriver. Non-native fishes, which are classified as warm-water tolerant, increased in abundance over the decade. This study demonstrates that a decade may be sufficient to detect changes in large river fish assemblages. Accelerated restoration efforts in rivers throughout the world are essential to protect and recover native fish assemblages. Restoration programs focused on correcting problems caused by past environmental degradation likely will be challenged by warming trends in many rivers. Success in restoring rivers and recovering native fish assemblages may be limited if we do not understand the processes that are changing and fail to anticipate the future river.

id #135408

Evaluating benthic macroinvertebrate communities in streamside management zones in private, working forests.

Haley Vaglianti¹, Checo Colon Gaud¹, Raymond Kidder¹, Zanethia Barnett², Nicholas Macias³, Daniel Hanks⁴, Brent Hawks⁴, Angela Larsen-Gray⁵

1. Georgia Southern University, Statesboro, GA, United States

2. Southern Research Station, Forest Service, Clemson, South Carolina

3. Marine Institute, University of Georgia, Sapelo Island, Georgia

4. Weyerhaeuser Company, Seattle, Washington

5. National Council for Air and Stream Improvement, Blacksburg, Virginia

Timber harvesting can increase sediment and nutrient inputs to streams, affecting water quality and aquatic biota. Forestry best management practices (BMPs), including streamside management zones (SMZs), are designed to mitigate these effects by reducing sediment delivery and moderating temperature, flow, and nutrient regimes. We assessed ecological integrity at 24 intermittent stream sites with SMZ in private, working forests across the Piedmont and Coastal Plain of Georgia. At each site, we conducted physical habitat evaluations and characterized benthic macroinvertebrate assemblages. Sites represented a gradient of SMZ widths, enabling comparisons of biological and physical indicators across buffer conditions. To evaluate how well our sampled SMZ sites approximate regional reference conditions, we compared study sites to ecoregion-specific reference streams drawn from a standardized regional database. Preliminary analyses suggested that physical habitat conditions differed between

our sampled sites and the regional references, with variation evident in select habitat components rather than overall habitat scores. In the Piedmont region, initial results suggested that macroinvertebrate community composition varies across SMZ width categories, but not across adjacent pine age groups. EPT percentages were higher in wider (>69-120m) than narrower (32-69m) SMZ. Further statistical analyses will focus on stream condition among SMZ widths and quantifying the degree to which our sampled SMZ sites resemble regional reference conditions. This work provides an updated assessment of SMZ performance in non-perennial systems and will inform forest managers about capacity and potential limitations of current BMPs to maintain ecological integrity in working forests.

id #135920

Influence of beaver engineering on fatty acid content and contaminant load of predatory stream insects

Emily R Arsenault¹, Evie S Brahmstedt², Roxanne Razavi¹, Abigail Hullihen¹, Sujan Fernando³, Lisa Cleckner⁴

1. State University of New York College of Environmental Science and Forestry, Syracuse, NY, United States

2. New York State Water Resources Institute, Cornell University, Ithaca, NY

3. Clarkson University, Potsdam, NY

4. Finger Lakes Institute, Hobart and William Smith Colleges, Geneva, NY

As ecosystem engineers, North American beaver (*Castor canadensis*) punctuate stream continua over space and time through the construction of dams that retain water in open-meadow habitats. Such dams undergo natural assembly and disassembly with changes in intensity of beaver activity over time, which has the potential to alter nutrient and contaminant pathways in stream food webs. Here, we explored the nutritional quality (polyunsaturated fatty acid PUFA content) and mercury load (total mercury THg and methylmercury MeHg) of predatory aquatic insects in beaver-engineered stream habitats in the New York Adirondacks (USA). We hypothesized that (1) predatory insects inhabiting beaver-impounded sites would exhibit elevated Hg content, in line with the "reservoir effect," but that this effect would decline with impoundment age and successional stage, and (2) that predatory insect PUFA content would be highest above beaver dams, given favorable growing conditions for algae in open-canopy meadows. We collected predatory aquatic insects representing the orders Odonata, Megaloptera, and Plecoptera from stream transects located above and below beaver-impounded areas representing the following dam conditions: intact dam (East Creek), recently breached dam (Sucker Brook), and dam absent, having breached >10 years prior (Chair Rock Creek). Insect samples were freeze-dried, homogenized using a mortar and pestle, and analyzed to obtain THg, MeHg, and PUFA concentrations. Preliminary results from our case study suggest that aquatic insects collected from meadowed, beaver-impounded areas had overall higher %MeHg (mean $\pm$ SD = $92 \pm 8\%$) than aquatic insects collected in forested reaches immediately downstream ($76 \pm 17\%$). Conversely, PUFA concentrations of predatory insects were lower in beaver impounded sites (range = 0–50 ng/gdw) than in downstream forested reaches (range = 10–750 mg/gdw), with non-detectable levels of eicosapentaenoic acid (EPA) in most insects collected upstream of beaver impoundments. Together, these preliminary results postulate the role of beaver in altering not only stream habitats, but also the nutrient and contaminant dynamics of stream food webs, with further implications for terrestrial consumers of emergent aquatic insect prey. Overall, this research highlights the potential influence of ecosystem engineers on broader food web processes.

id #134641

Nutrient makeover: Effects of a wastewater treatment plant upgrade on a delta ecosystem

Kyle Leathers¹, Tamara Kraus¹, Keith Bouma-Gregson¹, Kyle Nakatsuka¹, Brian Bergamaschi¹

1. US Geological Survey California Water Science Center, Sacramento, CA, United States

Anthropogenic nutrient enrichment from wastewater treatment plants (WWTPs) drives water quality in estuaries globally, but the ecological impacts of effluent nutrient reductions from WWTP upgrades are uncertain, especially in hydrologically complex estuaries. In California's Sacramento-San Joaquin Delta (Delta), a major WWTP upgrade was completed in 2021 that reduced effluent ammonium and increased effluent nitrate concentrations, with an overall reduction in dissolved inorganic nitrogen. Some researchers predicted this change would bolster the pelagic food web by fostering a more abundant and nutritious phytoplankton community; however, the ecosystem response was uncertain. We evaluated the spatial extent of the WWTP effluent influence on nutrient concentrations across the Delta before and after the upgrade and assessed the phytoplankton response. We analyzed water quality data from 22 Delta sites over the four years preceding and three years following the upgrade. Generalized additive models showed that, prior to the upgrade, effluent nutrient (i.e., ammonium, nitrate, dissolved inorganic nitrogen, and phosphate) inputs affected Delta nutrient concentrations up to approximately 80 kilometers downstream. Following the upgrade, effluent ammonium concentrations declined to non-detectable levels. Furthermore, effluent dissolved inorganic nitrogen and phosphate affected Delta site concentrations in a smaller spatial area post-upgrade. Finally, we found that ammonium and nitrate concentrations downstream of the WWTP were primarily positively associated with chlorophyll concentration and the prevalence of diatoms. These initial findings suggest that WWTP upgrades can alter nutrient regimes in complex estuaries with implications for primary producers.

id #135666

Seasonal and hydrologic variation influences habitat and functional structure of stream fish assemblages

Joshua D Tivin¹, Dan D Magoulick^{1,2}

1. Arkansas Cooperative Fish and Wildlife Research Unit, Biological Sciences, University of Arkansas, Fayetteville, AR, United States

2. U. S. Geological Survey, Arkansas Cooperative Fish and Wildlife Research Unit, University of Arkansas, Fayetteville, AR, United States

Hydrologic variability is a key driver of ecological structure in lotic systems, shaping habitat conditions, taxonomic diversity, and the functional traits that mediate species' persistence and performance (e.g., reproductive success). While many studies examine taxonomic responses to variation in flows, few evaluate how spatiotemporal hydrologic variation influences the functional organization within stream fish communities. We quantified seasonal habitat structure and functional trait diversity of fish assemblages across six Ozark Plateau headwater streams representing two contrasting flow regimes: Groundwater Flashy and Runoff/Intermittent Flashy. Fish and habitat data were collected seasonally during a dry year (2002) and a wet year (2003). Functional space was constructed using PCoA of morphological, ecological, and life-history traits, and functional diversity was measured using community weighted means (CWMs), functional richness (FRic), functional evenness (FEve), and functional divergence (FDiv). We found that habitat structure differed strongly by flow regime and season, with Runoff/Intermittent streams exhibiting pronounced reductions in depth, area, and velocity, while groundwater streams remained structurally stable. Functional identity of assemblages was similar across flow regimes, dominated by benthic, hydrodynamic taxa with opportunistic and periodic life-history strategies. However, functional structure differed significantly: FEve and FDiv were consistently lower in Runoff/Intermittent Flashy streams in both years, indicating assemblage dominance of species with similar trait combinations and reduced trait partitioning under variable flow. FRic and taxonomic richness remained stable across seasons and flow regimes, suggesting high functional redundancy despite species turnover. Together, results show that flow regime mediates both habitat structural stability and functional organization. As climatic warming and extreme drought increase hydrologic instability in headwaters, functional trait approaches provide a sensitive tool for detecting losses of functional roles that may not be evident by using taxonomic metrics alone.

id #135411

The interactive effects of hydrology and plant species on freshwater wetland CH₄ cycling.

Aileen Taylor¹, Margaret Palmer¹

1. University of Maryland, Silver Spring, MD, United States

Freshwater wetland systems can often display a mosaic of different vegetation types across the landscape. These "patches" are often characterized by functionally distinct vegetative groups that tend to display unique methane (CH₄) flux signatures. However, it is unclear whether these unique signatures are due to the differences in the vegetation present or some other variable in the system (like hydrology). Natural field conditions limit our ability to isolate the individual effects of cooccurring factors on the stages of CH₄ cycling as many of these factors interact. For example, vegetation and hydrology both have the potential to alter the redox state in soil, which will directly affect microbial activity levels. However, a wetland's hydroperiod will also affect the location and growth of plant species. To investigate the relative impact of hydrologic stage vs. plant species type on CH₄ fluxes, we designed a mesocosm experiment using various combinations of hydrologic stage (standing water, saturated, and low saturation) and vegetative states [emergent sedges (*Carex striata*), emergent grass (*Panicum hemitomon*), low growing forbs (*Persicaria hydropiperoides*/*Proserpinica pectinata*), submerged vegetation (*Juncus repens*/*Proserpinica pectinata*), and no vegetation] across 60 mesocosms. In the standing water treatments, we found significant differences in the CH₄ fluxes between vegetation treatments, which matched CH₄ fluxes in field conditions with comparable water levels. We also found that there was a definitive interactive effect between vegetation and hydrology on CH₄ fluxes, and that fluxes from emergent types of vegetation displayed an opposite pattern of response to changing water level compared to submerged groups. Whether these results are due to differences in how vegetation affects CH₄ transport, production, or oxidation will require extended study.

id #135923

Insect Emergence and Riparian Predator Abundance in Urban Ponds Across a Land Cover and Chemical Gradient

Jennoa E Fleming¹, Jackson A Bates¹, Jeremy M Brooks¹, Michelle L Hladik², Johanna M Kraus³, Dan L Preston¹

1. Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, Colorado, United States

2. California Water Science Center, U.S. Geological Survey, Sacramento, California, United States

3. Columbia Environmental Research Center, U.S. Geological Survey, Columbia, Missouri, United States

Urbanization affects aquatic and terrestrial biodiversity by fragmenting habitats, introducing pollutants, and shifting species composition. The presence of ponds within urban areas may act as refugia for aquatic species, but it is unclear how ecosystem services shift in relation to surrounding land cover and concomitant changes in water chemistry. A significant ecosystem service provided by urban ponds is the emergence of aquatic insects which transfer energy into terrestrial environments. The aim of this study was to determine how the flux of aquatic insect emergence shifts in relation to urban land cover and water chemistry (including nutrients, pesticides, and trace metals), and how riparian predators may be impacted by these rates. Aquatic emergence traps were deployed at 16 ponds along a gradient of urbanization (measured by percent impermeable land surface) within Fort Collins, Colorado for one month in late summer. Bird and spider surveys were conducted at each site to test if emergence was correlated with riparian predator abundances. Ponds with reduced riparian habitat complexity and increased surrounding areas of impervious surfaces appear to have reduced biomass and reduced species richness of insect emergence. The ponds that were most impacted by urbanization had higher concentrations of pesticides and emergence comprised often exclusively of pollution tolerant midges (*Chironomidae*). Ponds with more complex riparian habitats, less surrounding areas of impervious surfaces, and lower concentrations of pesticides had more diverse taxa such as *Trichoptera*, *Odonata*, *Ephemeroptera*, and *Diptera* with higher richness and abundance. Preliminary data also shows that riparian spider and insectivorous bird abundance increase with riparian vegetation complexity. These findings indicate that urban ponds may be an important source for energetic subsidies for riparian environments as riparian predators prepare to overwinter or migrate, but water chemistry and local habitat features may regulate the taxonomic composition of subsidies.

Spatiotemporal trends in nutrients in two biodiverse rivers

Annie Blalock¹, Phillip Bumpers¹, Byron Freeman¹, Seth Wenger¹

1. River Basin Center, University of Georgia, Athens, GA, USA

Nutrient enrichment is a ubiquitous stressor to freshwater ecosystems. In order to effectively manage freshwater ecosystems, we need a better understanding of spatiotemporal dynamics of water chemistry at multiple spatial scales. We used a long-term dataset (27 years) of watershed-scale synoptic sampling of two rivers nationally recognized for their aquatic biodiversity, the Etowah and Conasauga Rivers. Water quality monitoring programs were initiated in both rivers in 1997, with synoptic sampling continuing through 2025 at frequencies ranging from one to ten times per year. We evaluated watershed controls on spatiotemporal patterns in nutrients and conductivity. Preliminary analyses revealed that nitrate and SRP concentrations declined modestly but significantly over time in both the Etowah ($\beta = -0.028 \pm 0.007 \text{ SE yr}^{-1}$, $p < 0.001$; $\beta = -0.069 \pm 0.009 \text{ SE yr}^{-1}$, $p < 0.001$) and the Conasauga ($\beta = -0.029 \pm 0.007 \text{ SE yr}^{-1}$, $p < 0.001$; $\beta = -0.051 \pm 0.007 \text{ SE yr}^{-1}$, $p < 0.001$). However, temporal stability at the watershed scale does not preclude substantial spatial heterogeneity within watersheds. Watershed-level analyses may obscure important variation occurring at the subcatchment scale. Spatially distributed sampling therefore provides a critical opportunity to test the spatial scales which may be controlling nutrient fluxes and to identify areas that exert disproportionate influence on watershed-scale nutrient dynamics. To address this, ongoing analyses integrate long-term water quality data with land-use change information and subcatchment leverage calculations to examine how nutrient sources and sinks within subcatchments influence flux at watershed outlets. Additionally, leverage analyses allow for the identification of specific subcatchments that function as biogeochemical control points. Comparing patterns between rivers with differing geology and land-use histories may further reveal common small-scale drivers that shape watershed-scale nutrient behavior. By identifying the location and spatial scale of these control points, this work provides spatially explicit frameworks for the focal watersheds which could improve the efficiency of monitoring programs and help inform targeted management and conservation strategies.

Characterizing spatiotemporal patterns in benthic algal assemblage responses to reduced flow in a Central Texas intermittent river

Ailish S Collins¹, Paula C Furey², Candice D Piercy³, Weston H Nowlin¹

1. Biology, Texas State University, San Marcos, TX, United States

2. Biology, St. Catherine University, Saint Paul, Minnesota, United States

3. Engineer Research and Development Center, US Army Corps of Engineers, Vicksburg, MS, United States

Intermittent rivers and ephemeral streams (IRES) comprise over half of the global river network. Research in IRES represents an opportunity to examine how spatiotemporal variation in environmental drivers can impact assemblage-level responses. Until recently, IRES have been under-studied, leading to a lack of protection despite their global prevalence and importance. Although IRES research has expanded, substantial questions remain about algal assemblages and their roles in community dynamics and ecosystem processes. Algae is the dominant primary producer in many IRES because of a lack of macrophytes and limited riparian vegetation in more arid regions. In this study, we collected benthic algae samples from riffles and pools at nine sites in the San Saba River (Texas, USA) throughout Summer 2024. The San Saba River has a hydrological gradient of flow intermittency and variability: the groundwater-fed upper reach has consistent perennial flow; the middle reach frequently undergoes drying and fragmentation; and the lower reach exhibits a more variable flow regime but typically maintains flow. We deployed unglazed ceramic tiles in riffle and pool mesohabitats at nine different study reaches and collected and redeployed tiles every three weeks. We additionally collected *in situ* limestone cobbles from mesohabitats at each site every three weeks. Algal biofilms were removed from tiles and rocks to estimate ash-free dry weight (AFDW), chlorophyll-a concentration, and algal assemblage composition. Results indicate that AFDW of cobbles was higher in pool mesohabitats and declined with increasing canopy cover. However, chlorophyll-a concentrations on tiles were higher in riffles than pools and declined with increasing canopy cover. Algal assemblage composition differed significantly between the different reaches of the river as well as the mesohabitats but did not vary significantly among sampling dates.

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Food Webs in Flow: Spatiotemporal Variation in Diversity and Food Web Structure in an Intermittent River

Jered Davenport¹, Weston Nowlin¹, Todd M Swannack²

1. Department of Biology, Texas State University, San Marcos, TX, United States

2. ECOMOD, US Army Corps of Engineers, San Marcos, TX, United States

Intermittent rivers and ephemeral streams are critical yet understudied ecosystems, particularly in semi-arid regions where hydrologic variability often governs community structure and ecosystem processes. This study examined how variation in hydrologic regimes affected the diversity and composition of fish and benthic invertebrate communities and food web

characteristics. We examined these responses in nine study sites in the San Saba River (TX, USA) in which sites had one of three hydrologic regimes (perennial and temporally stable flows, intermittent flows, and seasonally variable but perennial flows) at two seasonal periods (late Spring; growing season versus early Fall; dry season). Riffle and pool habitats at each site were sampled twice and water chemistry, benthic algae, terrestrial detritus, macroinvertebrates, and fish were sampled in both mesohabitat types. Diversity (taxonomic richness) of macroinvertebrates and fish were assessed within and across mesohabitats, sites, as well as across hydrological regime types. Stable isotopes of C and N were used to examine food chain length, basal resource dependence (allochthonous versus autochthonous) among mesohabitats, sites, and hydrologic regimes. Invertebrate taxonomic richness was greater in hydrologically consistent sites when compared to sites with more variable discharge regimes, but riffles habitats consistently had greater invertebrate richness than pools. Similarly, fish taxonomic richness was greater in perennially flowing sites, but the greatest richness was in pool habitats. The trophic position of blacktail shiner (*Cyprinella venusta*), a cosmopolitan fish species found at all sites, was one trophic level lower at intermittent flow sites when compared to the perennially flowing sites, indicating that variability in discharge and the number of days with zero flow influences the length of food chains. Cumulatively, these results indicate that the effects of flow regime variability on richness and food web characteristics depend on the specific group of study organisms which are examined and the mesohabitat unit that is sampled.

id #134901

Aquatic insect declines in protected tropical rainforest streams, Costa Rica

Alonso Ramirez¹, Pablo E. E. Gutiérrez-Fonseca², Marcelo Ardon¹

1. North Carolina State University, NC, United States

2. Department of Biology, University of Puerto Rico, Mayagüez, Mayagüez, Puerto Rico, USA

Biodiversity loss is a major global environmental challenge, driven primarily by human activities like land-use change and climate change. While tropical ecosystems were once thought to be relatively stable, emerging evidence shows significant declines even within protected tropical forests. Here, we used a 27-year monitoring program to: (1) quantify long-term changes in the richness and composition of aquatic insect assemblages in two tropical headwater streams in Costa Rica, and (2) evaluate how shifts in regional climate, particularly precipitation patterns, might be driving these changes. We hypothesized that taxonomic richness has been declining and that these declines are associated with a new disturbance regime due to changes in precipitation. Aquatic insects were sampled from 1997 to 2024 in two headwater streams at La Selva Research Station, Costa Rica. Samples were collected with a corer in run habitats by the same technician over the entire period. Insects were identified, mostly to genus level, and taxonomic richness and assemblage composition were assessed. Stream discharge, precipitation, and air temperature data were collected as potential environmental drivers. We found 87 and 96 taxa in each stream, with Diptera being the most diverse and abundant order. There was a significant exponential decline in richness in both streams, dropping to about one-third of the original value by 2016, before showing a minor increase in the 2024 sampling event. The density of nearly all dominant groups also declined by up to 90%. The assemblage composition became progressively more dissimilar over time, reaching maximum difference by 2016, although the 2024 assemblage showed a partial overlap with earlier years. Our 27-year study reveals a concerning and robust pattern of biodiversity loss in protected tropical headwater streams, with a consistent decline in richness and abundance across almost all taxa. While no taxa showed significant increases, the partial recovery observed in 2024 suggests that some resilience may exist. The observed declines and resulting changes in community composition are linked to climate change-related factors and long-term hydrological variability. The findings underscore that protected tropical forests are undergoing pervasive, multifaceted biodiversity loss, highlighting an urgent need for long-term research to identify ecosystem consequences.

id #135669

Cross-Sectoral Governance for Water-Linked Disease Risk Management in South Africa: A Qualitative Review of Constitutional, Legislative, and Policy Frameworks

Chika F Nnadozie¹, Nelson O Odume¹, Siyabonga M Mazibuko¹, Phindile Madikizela¹

1. Center for Environmental Water Quality, Institute for Water Research, Rhodes University, Grahamstown, Eastern Cape, South Africa

Water-related diseases remain a significant public health challenge globally, with effective governance critical for prevention. In South Africa, complex interactions between environmental, agricultural, veterinary, and health sectors create opportunities for controlling diseases linked to contaminated freshwater sources. Despite multiple overlapping mandates, there is limited understanding of how constitutional, legislative, and policy frameworks collectively support cross-sectoral governance for preventing these water-associated health risks. Existing governance analyses often overlook the systemic integration of environmental surveillance and communicable disease control, particularly regarding how statutory and policy instruments operationalize cooperation across sectors for waterborne disease risk management.

This study aims to assess how South Africa's constitutional, legislative, and policy frameworks define and support cross-sectoral governance structures, surveillance obligations, and reporting pathways relevant to preventing diseases associated with freshwater water contamination.

A qualitative, desktop-based research design was employed, combining doctrinal legal analysis with document analysis. Key national laws, regulations, and policy documents from health, water, agriculture, veterinary, and environmental sectors were systematically reviewed. Analytical focus was placed on identifying formal surveillance obligations, mandated reporting pathways, sectoral roles, and governance mechanisms facilitating integrated disease risk management.

The review revealed that while no single framework explicitly addresses waterborne disease prevention, multiple legislative instruments establish surveillance, environmental monitoring, and communicable disease control mandates that collectively underpin such efforts. These frameworks define hierarchical governance roles, formalize multi-level reporting channels, and promote coordination across sectors through mechanisms that can be aligned with One Health principles. However, operational gaps exist, particularly concerning detailed reporting procedures and integration of environmental data into disease surveillance. Incident-based reporting under water and disaster laws complements system-wide monitoring, while health and veterinary statutes mandate disease notification and environmental health investigations. Therefore, there is the need for enhanced policy coherence and the development of integrated operational protocols to translate legal mandates into effective cross-sectoral action against water-related disease risks. Strengthening collaboration, data sharing, across health, environmental, agricultural, and veterinary sectors will be essential to operationalize One Health governance in South Africa's water governance landscape. This study provides a foundational understanding for policymakers and practitioners aiming to advance holistic disease prevention strategies at the human-environmental water-animal interface.

id #135925

Interactive Effects of Invasive Riparian Litter and Rusty Crayfish on Detrital Processing and Nutrient Dynamics

Justin Sturtz¹, Christopher Cheek¹

1. South Dakota State University, SD, United States

Riparian ecosystems can be heavily influenced by inputs of allochthonous litter. Changes in the quality or quantity of litter inputs can alter organic matter and nutrients in aquatic habitats. Throughout the western United States, Russian Olive (*Elaeagnus angustifolia*) has invaded riparian habitats. Concurrent declines of native riparian trees such as Cottonwood (*Populus deltoides*) have replaced high-quality resource litter with a low-quality recalcitrant litter that has the potential to alter aquatic food webs. The invasion of Russian olive has the potential to facilitate subsequent aquatic invasions including common carp (*Cyprinus carpio*) and channel catfish (*Ictalurus punctatus*). However, few studies have considered the ability of Russian olive to facilitate the invasion by crayfish. Our study we compare the growth of an invasive Rusty crayfish (*Faxonius virilis*) in experimental stream mesocosms with native and invasive litter inputs. In addition, we evaluate the ability of crayfish to facilitate breakdown of recalcitrant litter compared to native litter. Our experimental design consisted of twenty-four tanks in which we simulated isolated pool-riffle ecosystems under low-flow conditions. We used a fully factorial design comprising of three litter types Cottonwood, Russian olive and control and crayfish presence/absence. Our study lasted 90 days in which we monitored water chemistry, decomposition rates, as well as the biotic responses of algae, invertebrates, zooplankton, and crayfish growth. We hypothesized that crayfish will experience less growth when provided with olive litter, crayfish will release more nutrients from the olive litter in the water column, and we will see no changes in invertebrates in the larval phase. Preliminary results indicate there is no assimilation of olive that contributes to stomatic growth of the crayfish. We see that there is no difference in nutrient concentrations in the system.

id #136181

Separating the wheat from the chaff: How sensitive are biodiversity surveys and ecological assessments to the bioinformatics pipelines used in eDNA metabarcoding

Leonce Ngirinsuti¹, Torrey Rodgers¹, Charles Hawkins¹

1. Utah State University, Logan, UTAH, United States

Metabarcoding of environmental DNA (eDNA) is increasingly used in both biodiversity surveys and ecological assessments. However, the metabarcoding process is susceptible to errors in assigning sequences to described species or identifying undescribed species (e.g., operational taxonomic units – OTUs). These errors are often caused by either (1) field and laboratory contamination, (2) errors during PCR amplification due to primer biases, or (3) formation of artifactual sequences during sequencing, which can result in both false positive and negative detections. Bioinformatics pipelines differ in how they remove sequence errors and the tradeoffs associated with retaining erroneous sequences and removing true biological sequences. To date, only a few studies have examined these tradeoffs and their subsequent effects on both biodiversity surveys and ecological assessments. In this study, we addressed two specific questions: (1) to what extent do the different noise filtering methods used in two different pipelines affect estimates of taxa richness and composition measured as operational taxonomic units (OTUs)? (2) does choice of pipeline influence the strength of OTU-environment relationships? To test these ideas, we used two bioinformatic pipelines to identify the OTUs present in 751 eDNA samples collected from streams in eastern Washington, USA: the Ribosomal Database Project (RDP) classifier and the High-Accuracy Pipeline for Processing deep metabarcoding data (HAPP). The RDP relies on the DADA2-denoising process, and the HAPP uses both DADA2 and additional filtering steps designed to remove low quality sequences and chimeras. The raw OTU output from both pipelines contained substantial errors which greatly inflated richness. We then removed additional likely errors by dropping singleton OTUs. The RDP identified more OTUs (345 OTUs) than HAPP (257 OTUs). However, RDP- and HAPP-based NMDS ordinations were similar to one another, and the direction and strength of longitude, latitude, and temperature vectors overlaid on OTU ordinations were nearly identical ($R^2 = 0.61, 0.74, \text{ and } 0.01$ for RDP and $0.6, 0.68, \text{ and } 0.01$ for HAPP). The metabarcoding community needs standardized methods for quantifying erroneous OTUs, and practitioners need to be both transparent in reporting metabarcoding errors and aware of how these errors can influence estimates of biodiversity and assessments of ecological condition.

Stream macroinvertebrate functional organization and biological integrity across headwaters in the Chesapeake Bay watershed.

Sergio A. Sabat-Bonilla¹, Kelly O. Maloney², Greg J. Pond³, Gregory B. Noe⁴, Sally E. Entrekin¹

1. Entomology Department, Virginia Tech, Blacksburg, VA, USA

2. Eastern Ecological Science Center, U.S. Geological Survey, Keareysville, WV, USA

3. Laboratory Services and Applied Science Division, U.S. EPA Region 3, Wheeling, WV, USA

4. Florence Bascom Geoscience Center, U.S. Geological Survey, Dover, DE, USA

Understanding how biological communities vary across regions with distinct environmental contexts is central to stream bioassessment, yet most frameworks emphasize taxonomic structure while giving more limited attention to functional organization. Using a coordinated regional dataset from 90 Wadeable streams across three regions of the Chesapeake Bay watershed (Valley & Ridge, Piedmont, and Coastal Plain), we evaluated how macroinvertebrate structural and functional indicators vary among regions and biological condition classes defined by the Chesapeake Basin-wide Benthic Index of Biotic Integrity (B-IBI). Structural organization was characterized using taxonomic composition and diversity metrics commonly incorporated into multimetric indices (taxonomic richness and Shannon diversity). Functional composition was quantified using density- and biomass-weighted community-weighted means (CWMs) of traits associated with environmental sensitivity (response traits), ecosystem functions and energy pathways (effect traits), and organism-habitat and resource interactions (linking traits), organized within a response-linking-effect (REL) framework. Functional organization beyond composition was evaluated using functional evenness (FEve), Rao's quadratic entropy (RaoQ) and functional redundancy (FR), which describe the distribution, dispersion and overlap of traits within communities. Because the B-IBI is grounded in taxonomic composition and tolerance, we evaluated whether structural indicators and response traits aligned with biological condition classes, and whether linking traits, effect traits, and functional organization exhibited stronger region- and weighting-dependent patterns. Relationships between macroinvertebrate indicators and B-IBI varied by region, weighting scheme and REL domain and were not consistently monotonic. Structural indicators showed limited or inconsistent correspondence with B-IBI across regions. However, FEve, RaoQ, FR and selected REL composites and trait CWMs related to habitat association, emergence timing and body size showed more consistent region-specific associations with B-IBI. Consequently, streams with similar B-IBI condition classes often differed in functional organization and energetic structure, while streams with different B-IBI scores also exhibited comparable functional responses. Together, these patterns indicate that biological condition classes capture important variation in community structure but do not correspond to a single, uniform trajectory of functional organization. Similar B-IBI scores can arise from distinct functional configurations that differ in trait composition, redundancy, and energetic structure, reflecting differences in how stream communities may buffer disturbance and reorganize following environmental change.

Do spongy moth mating disruption treatments alter headwater stream ecological processes?

David Speth¹, Ksenia Onufrieva¹, Andrea Hickman¹, Christina Navarro¹, Mehdi Ashraf-Khorassani¹, Sally Entrekin¹

1. Virginia Tech, Blacksburg, VA, United States

Headwater stream environments and processes have been undervalued historically, despite serving as vulnerable keystone habitats and supporting downstream waterbodies through organic matter processing, carbon sequestration, and nutrient cycling. SPLAT[®] GM-Organic, a wax-based mating disruption treatment used to control the spread of *Lymantria dispar dispar* (European spongy moth), is applied aerially above headwater streams with an unknown impact on streams. The spongy moth is a destructive folivore, native to Eurasia, that was introduced to the eastern United States in 1869. The Slow the Spread Program (STS), funded by the US Forest Service, became operational in 2000 and has successfully limited the spread of the spongy moth, primarily via the use of mating disruption treatments. SPLAT[®] GM-Organic was introduced by ISCA Technologies in 2010 and became the sole mating disruption treatment used by the STS in 2017. Once applied aerially, SPLAT[®] sticks to various substrates, including leaves, and steadily releases the pheromone disparlure. By preventing spongy moth-induced defoliation of riparian vegetation, SPLAT[®] treatments are understood to have indirect positive effects on stream health; however, our research aims to identify the previously uncharacterized direct effects of the treatment on stream biological diversity and processes. The deposition of allochthonous organic matter (i.e. leaves) is a driving factor in headwater nutrient and carbon cycling. We hypothesize that deposition of the SPLAT[®] matrix into the stream mainly occurs through leaf fall, and the matrix, once present in the stream, may limit oxygen diffusion and serve as a food source for macroinvertebrates. We selected five headwater streams in the Slate River Watershed of Appomattox, Virginia, to be sampled from June 2024 to July 2026. Each stream was divided into an upstream control reach, a buffer, and a downstream experimental reach which was treated with SPLAT[®] GM-Organic on June 11, 2025 by the STS. Using a Before-After-Control-Impact (BACI) experimental design, the short- and long-term effects of SPLAT[®] treatments can be identified. We are sampling macroinvertebrates, algae, organic matter, water chemistry, dissolved oxygen, and SPLAT[®]. Our research will evaluate whether current treatment practices alter the ecological processes of streams.

Ochéthi and Wóunspe: How Freemont Cottonwood Origin and Growth History Shape Microbial Response to Stream Warming

Amanda Rouillard¹, Jane Marks¹, Michaela Hayer¹, Egbert Shwartz¹, Bruce Hungate¹, Rebecca Best¹, David Pineda-Morante², Aaron Barnes¹

1. Northern Arizona University, Flagstaff, ARIZONA, United States

2. Integrative Freshwater Ecology Group, Centre d'Estudis Avançats de Blanes (CEAB-CSIC), Blanes, Girona, Spain

Headwater streams are predicted to warm with climate change, altering the microbial processes that control leaf litter decomposition. In this study, we integrate Dakota concepts of *Ochéthi* (origin/home camp, representing genetic inheritance) and *Wóuŋspe* (learning through lived experience, representing phenotypic plasticity) to examine how tree origin and growth history shape microbial responses to stream warming. We asked: How do the genetic origin, *Ochéthi*, and growth environment, *Wóuŋspe*, of Fremont cottonwood (*Populus fremontii*) influence microbial growth and leaf litter decomposition under simulated stream warming? We used cuttings from two genetically distinct provenances (warm and cold adapted) and grew them for five years in two common gardens (hot and cold), creating four leaf types. Leaves were then decomposed for 35 days in 48 flow-through stream mesocosms across three temperature treatments (ambient, +3°C, +6°C). We measured decomposition via mass loss and quantified taxon-specific microbial growth rates for fungi and bacteria using quantitative stable isotope probing (qSIP) and DNA sequencing. Tree population, *Ochéthi*, was the strongest predictor of microbial growth rates and litter mass loss, while a significant population × garden interaction showed that genetic effects depended on climate of growth, *Wóuŋspe*. This study bridges tree *Ochéthi* and *Wóuŋspe* with microbial processes, highlighting the connected roles of plants, microbes, and temperature in influencing litter decomposition and the stability of headwater streams in a warming climate.

id #135929

Neutral theory and the beaver: facilitation of fish communities by ecosystem engineering.

Niall Clancy¹, Annika Walters²

1. Salish Kootenai College, Pablo, MT, United States

2. U.S. Geological Survey, Wyoming Cooperative Fish & Wildlife Research Unit, Laramie, WY, United States

Ecosystem engineers, such as beavers, facilitate other species through habitat modification. Beaver conservation and mimicry is rapidly growing as a restoration tool, but their ability to facilitate other species is poorly understood in many systems. Further, it is unknown if beaver dams uniquely benefit some functional groups (species with similar traits) more than others. We therefore examined the impact of beaver dams on fish communities in the northwestern Great Plains of the United States through two related studies, one observational and one experimental. In a survey of 30 streams, each with a paired beaver and control site, we found total and native fish abundance was higher in beaver complexes, but introduced species' abundance was not. Functional group proportions within trophic, feeding, spawning, thermal, and habitat categories were not significantly different between beaver and control sites. We compared these results to those of an experimental, beaver-mimicry restoration in which 18 beaver-dam analogs were installed over three years on a prairie stream. Like the observational study, total fish abundance increased in the beaver-mimicry reach relative to a control reach. Functional group proportions changed slightly after initial restoration, but these differences were not sustained in the following year. Taken together, these studies suggest that beaver dams, by increasing wetted habitat, increase the total abundance of fish, but the proportional makeup of functional groups remains similar to the starting fish community. This serves as an example of how neutral theory can be applied to restoration and ecosystem engineering to better understand the mechanisms underlying assemblage-level responses.

id #135162

Quantifying the effects of interannual variation in flow and temperature on the abundance of fishes in two biodiverse southern Appalachian rivers

Phillip Bumpers¹, Mary Freeman¹, Byron Freeman¹, Seth Wenger¹

1. University of Georgia, GA, -

Discharge and thermal regimes are widely considered to be among the strongest drivers of ecosystem structure and function in rivers and streams. Both of these drivers are modified by climate change, land use change, and water resource management, creating persistent threats to riverine ecosystems. The field of flow ecology has greatly advanced our understanding of how animals respond to flow alteration, but our ability to predict responses is still limited, in part, due to a lack of understanding of how flow interacts with other variables such as temperature. Jointly considering flow and temperature together can greatly improve our predictions of how organisms respond to global change. We used long-term (26 yr) data of annual count surveys of shoal-dwelling fishes in two biodiverse rivers in the southern Appalachians in GA, USA to test hypotheses of how high and low flows and stream temperature could affect abundances of fishes. We evaluated specific hypotheses for seven species, including one species of high conservation concern, using Bayesian generalized mixed models. We found that high flow events decreased abundances in the current year for all species, but had mixed effects the following year, with three species showing positive responses and three species exhibiting negative responses. A possible mechanism for the lagged positive effect of high flows is improved habitat conditions from flushing of sediment. Low-flow extremes also generally had a negative effect on species abundances in the current year, but less of an effect the following year relative to high-flow events. Warmer water temperatures were negatively correlated with abundances for most species, but there were strong positive effects for one species (*Noturus leptacanthus*). Our results provide evidence that changes in flow and temperature regimes may impact native fishes in southern Appalachian rivers. Future analyses will include a greater suite of species (22) and consider if historical biogeography (i.e., evolutionary climates) improves predictions of species responses.

id #135674

Plastic pre-production pellets, “nurdles”, in rivers: Sampling methods, distribution, and implications for policy

Timothy J Hoellein¹, Olivia Schaul¹, Maya Worek¹, Alyssa Younus¹, Andrew Conover¹, Betsy Gates-Alford²

1. *Loyola University Chicago, Chicago, ILLINOIS, United States*

2. *Burke, Warren, MacKay & Serritella, P.C. , Chicago, Illinois, USA*

Microplastics (i.e., particles < 5 mm) are ubiquitous pollutants. Microplastics include a diverse suite of materials, which vary by polymer type, shape, and chemical additives. Sources of microplastics to rivers include wastewater, stormwater, atmospheric deposition, and industrial effluent. The latter includes 'nurdles' or pre-production pellets, commonly found as 2-3 mm disks. About 10 trillion nurdles are littered annually. In marine sites, nurdles are dispersed long distances, ingested by animals, and leach chemical additives. However, they are among the least well-studied types of microplastics in freshwater ecosystems. There are no standard protocols for quantifying their density in rivers and little is known about their distribution. Our objective was to locate potential nurdle sources and quantify their distribution in rivers throughout northeastern Illinois. We generated a map of sites by identifying plastic manufacturing facilities and their proximity to rivers. At several potential sources, we quantified nurdle abundance up- and downstream. We used four approaches, depending on nurdle density and environmental conditions: quadrats, nets, passive samplers, and visual surveys. We found nurdles at almost all study sites, regardless of proximity to potential sources. While density and assemblage were highly variable within and among study sites, nurdles were most abundant in shallow waters with organic matter accumulations (i.e., riparian vegetation and boat launches). Our results show nurdles are common throughout the region and likely widely dispersed from potential sources. These preliminary data will be integrated with additional measurements to inform standardized protocols for nurdle quantification in rivers. Results will inform emerging policies for nurdle pollution reduction.

id #135930

Drivers of hyporheic nitrogen concentrations across wetting, flowing, and drying periods in an intermittent stream

Angel Colón-Santiago¹, Alba Argerich^{1,2}

1. *University of Missouri, MO, United States*

2. *Missouri Water Center, Columbia, Missouri, US*

Intermittent streams comprise a large fraction of river networks, yet most mechanistic understanding of nitrogen dynamics derives from perennial systems. In intermittent streams, repeated transitions among wetting, flowing, and drying modify hydrologic connectivity and exchange between surface water, hyporheic sediments, and riparian zones, generating strong spatial and temporal variability in physicochemical conditions. Understanding how nitrogen responds to these connectivity shifts provides important context for predicting nutrient cycling under increasing hydrologic variability conditions.

We investigated drivers of dissolved nitrogen concentrations in surface water, hyporheic, and riparian compartments along a 5-km reach of Brushy Creek, a second-order intermittent stream in the Ozark Border region of south-central Missouri, USA. From October 2021 to June 2023, we collected monthly surface and subsurface water samples at four sites instrumented with six wells each (two hyporheic and four riparian) to characterize vertical and lateral connectivity among stream compartments. Hydrologic connectivity was quantified as the percentage of surface water contribution to well water using chloride as a conservative tracer.

Nitrogen concentrations varied across hydrologic periods and compartments, with the highest total dissolved nitrogen and nitrate concentrations observed in surface water during the drying period, followed by lower concentrations during wetting and flowing conditions. Dissolved inorganic nitrogen dominated during drying and flowing periods, whereas dissolved organic nitrogen contributed most strongly in the hyporheic zone during flowing conditions. Hydrologic connectivity was significantly associated with nitrate transformation in subsurface compartments, with higher surface water contributions corresponding to shifts in nitrate uptake and retention across hyporheic and riparian zones.

id #138746

FUTURE HABITAT SUITABILITY OF NATIVE, NON-NATIVE, AND GAME FISHES ACROSS FIVE LARGE LANDSCAPES

Taylor Woods¹, Ken Eng², Jared Smith², Matthew Cashman², Jaime Ashander¹, Allison Benscoter³, Wesley Daniel³, Michael Doddrell⁴, Corbin Hilling⁵, Joshua Hubbell⁶, Abigail Lynch⁷, Kelly Maloney¹, Daniel Wierich⁸, Michael Wiczorek⁹, Tanja Williamson¹⁰, Daniel Kowalski¹¹, Robert Zuellig¹¹, Sean Emmons¹², Anna Kaz¹³, James McKenna Jr.¹³, Mary Freeman¹³

1. *Eastern Ecological Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

2. *Water Mission Area, U.S. Geological Survey, Kearneysville, WV, USA*

3. *Wetland and Aquatics Research Center, U.S. Geological Survey, Kearneysville, WV, USA*

4. *Western Fisheries Research Center, U.S. Geological Survey, Kearneysville, WV, USA*

5. *Great Lakes Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

6. *Geological Survey of Alabama, Alabama, USA*

7. *National Climate Adaptation Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

8. *Core Systems Science, U.S. Geological Survey, Kearneysville, WV, USA*

9. *Maryland-Delaware-District of Columbia Water Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

10. *Ohio-Kentucky-Indiana Water Science Center, U.S. Geological Survey, Kearneysville, WV, USA*

11. *Colorado Parks and Wildlife, Denver, USA*

12. *Conservation Science Partners, USA*

13. *Unaffiliated, USA*

Freshwater ecosystems are particularly vulnerable to direct and indirect anthropogenic stressors, leading to shifts in fish communities in response to environmental changes. Traits that mediate fish community responses to altered environments by

influencing species' survival and adaptability can inform individual species and fish communities responses to changing environments. Here, we use a trait-based approach to predict how stream fish communities vary in response to environmental and land use change in five U.S. ecoregions: the Columbia, Upper Colorado River, and Mobile Basins, and the Great Lakes and Chesapeake Bay watersheds. We examine changes in habitat suitability for native, non-native, and game fishes from changing environments. By integrating trait-based models with landscape-scale data, we can refine predictions of ecosystem responses and better inform sustainable resource management.

id #134651

Do agricultural conservation practices promote biogeochemical resilience in a tile-drained and intensively fertilized landscape?

Lindsey M. Rasnake¹, Todd V. Royer¹, Ursula H. Mahl², Matt T. Trentman³, Shannon L. Speir⁴, Lienne R. Sethna⁵, Anna E.S. Vincent⁶, Abagael N. Pruitt⁷, Jennifer L. Tank²

1. *Indiana University, Bloomington, INDIANA, United States*

2. *University of Notre Dame, Notre Dame, Indiana*

3. *University of Montana, O'Connor Center for the Rocky Mountain West, Missoula, Montana, USA*

4. *University of Arkansas, Fayetteville, Arkansas, USA*

5. *Science Museum of Minnesota, St Paul, Minnesota*

6. *Northwestern University, Evanston, Illinois*

7. *University of California, Davis, California*

Agriculture increases nutrient loading to freshwater ecosystems, particularly in tile-drained landscapes. In these systems, increasing hydroclimatic variability raises questions about how agricultural conservation practices alter carbon and nitrogen availability and, in turn, regulate denitrification rates and biogeochemical resilience. While nitrate (NO_3^-) runoff is a primary target for conservation practices, dissolved organic carbon (DOC) fuels heterotrophic metabolism and nutrient processing, including denitrification. Accordingly, we focus primarily on patterns of DOC export across field and watershed scales, and use $\text{DOC}:\text{NO}_3^-$ ratios to assess responses to a common conservation practice. Winter cover crops (CC) can reduce NO_3^- losses, yet their influence on DOC export and $\text{DOC}:\text{NO}_3^-$ stoichiometry remains unclear. We measured DOC and NO_3^- concentrations from 2017–2022 across field and watershed scales in two northern Indiana watersheds (>80% row crops) with variable CC coverage. We quantified how hydrologic variability and CC interact to shape DOC export and $\text{DOC}:\text{NO}_3^-$ stoichiometry. Watershed-scale DOC yield was largely controlled by hydrology rather than CC, ranging from 5 to 36 kg C ha⁻¹ yr⁻¹ and peaking during the wettest years. One watershed exhibited episodic DOC export linked to antecedent rainfall, suggesting mobilization from near-surface soils, while the less flashy watershed showed lower variability and reduced sensitivity to high flows, likely reflecting groundwater contributions, soil texture, or management differences. Although CC sometimes reduced subsurface tile flow and DOC export, field-scale DOC responses to CC were inconsistent, and high CC years coincided with high precipitation and DOC yields, potentially masking CC effects at the watershed scale. In contrast, CC effects on NO_3^- consistently reduced field-scale NO_3^- losses (65% on average). Molar $\text{DOC}:\text{NO}_3^-$ ratios generally remained carbon-limited for denitrification (<2:1 C:N) at both field and watershed scales. Significant differences in field-scale $\text{DOC}:\text{NO}_3^-$ were observed in only one watershed, with higher ratios in tiles draining CC compared to conventional management, particularly during spring and summer. These results indicate that cover crops can transiently increase DOC availability, reduce NO_3^- loss, and modestly increase $\text{DOC}:\text{NO}_3^-$ ratios. However, DOC export remains hydrologically controlled and ratios suggest carbon-limitation for denitrification, potentially constraining gains in biogeochemical resilience in tile-drained agricultural systems.

id #135675

Acclimation in action: Recycled water boosts nutrient uptake over time

Ashley Hennessey¹, Chamoda PDM Dissanayake², Michael Mantone¹, Audrey Goeckner³, Ashley Smyth⁴, Lindsey Reisinger², AJ Reisinger¹

1. *Soil, Water, & Ecosystem Sciences, University of Florida, Gainesville, FL, United States*

2. *School of Forestry, Fisheries, & Geomatic Sciences, University of Florida, Gainesville, FL, United States*

3. *Department of Biology, University of Florida, Gainesville, FL, United States*

4. *Tropical Research and Education Center, Department of Soil, Water, and Ecosystem Sciences, University of Florida, Homestead, FL, United States*

Recycled water (RW), municipal wastewater that has been processed to remove impurities (including solids) and treated to reduce harmful substances, is an increasingly important component of freshwater systems. RW discharge, intentional or unintentional, represents a major pathway for excess nutrients, pharmaceuticals, antimicrobial resistance genes (ARGs), and other biologically active contaminants to enter wetlands and downstream ecosystems. Constructed wetlands are widely used as tertiary treatment systems to “polish” RW prior to discharge, yet the ecological consequences of RW remain poorly understood. To evaluate how exposure to RW influences the biogeochemistry of wetland sediments, we conducted controlled microcosm experiments to isolate the effects of RW exposure on ecosystem respiration, nutrient cycling, and microbial processes in sediments from six Florida wetlands (three natural and three constructed) across seasons. Sediments were incubated with either native site water or RW collected from local municipalities under acute (Day 0) and acclimated (Day 7) exposure conditions. We quantified sediment respiration, denitrification (net N_2 production), nutrient uptake (NH_4^+ , NO_x , PO_4^{3-}), and dissolved organic matter (DOM) composition, and characterized microbial communities to evaluate how RW exposure alters wetland biogeochemical processes and microbial structure. Preliminary wet-season (May–October) results show functional shifts following RW exposure. Natural wetland sediments initially released ammonium and phosphate under RW exposure, whereas constructed wetland sediments exhibited weaker or neutral responses. After seven days of RW acclimation, all RW-amended treatments shifted toward net NH_4^+ and PO_4^{3-} uptake (including both natural and constructed wetland sediments), indicating rapid microbial or geochemical adjustment to RW chemistry. In contrast, NO_x dynamics remained highly variable, suggesting complex interaction

among assimilatory and dissimilatory (e.g., nitrification and denitrification). Together, these results demonstrate that short-term RW acclimation can rapidly reorganize wetland biogeochemical function, with important implications for the influence of pharmaceutical-contaminated RW on ecosystem processes. Ongoing analyses of microbial community composition, denitrification genes, and ARG abundance will reveal mechanistic linkages between contaminant exposure, biological responses, and biogeochemical impacts, further resolving how prior exposure to RW mediates potential resistance or persistence to contaminants in wetland ecosystems.

id #135931

Long-term nutrient impacts of the afforestation of agricultural lands reveal a lag in dissolved phosphorus reduction

Kathleen Cutting¹, Shannon Speir¹, Rebecca Hale²

1. *University of Arkansas, AR, United States*

2. *Smithsonian Environmental Research Center, Edgewater, Maryland, United States*

Agriculture is associated with increased nutrient inputs to the landscape, which can drive the eutrophication of critical downstream systems, such as the Chesapeake Bay. To address these concerns, restoration techniques such as afforestation, the planting of trees on land that was not recently forested, can be utilized to decrease nutrient loss in impacted watersheds. Despite efforts to remediate the myriad of impacts from agriculture on our freshwater systems, we still lack an understanding of the long-term recovery trajectory for the restoration of croplands. Here, we used monthly composite water quality data from the greater Rhode River watershed (near Edgewater, Maryland), collected as a part of the Smithsonian Environmental Research Center Weir Network. We compared 44 years (1978-2023) of total and dissolved nitrogen (N) and phosphorus (P) from a continuously forested watershed and an agricultural watershed that was replanted with trees in 2013 as part of a tree diversity experiment, BiodiversiTREE. Preliminary results suggest that dissolved N concentrations decreased after fertilizer application ceased and afforestation occurred primarily due to decreases in nitrate ($p=0.05$). In contrast, TN and TP concentrations increased post-afforestation, likely resulting from increased organic matter. Interestingly, soluble reactive phosphorus (SRP) concentrations increased across the entire study period, with a greater increase post-afforestation ($p<0.01$). The observed increase in SRP underscores the potential influence of legacy P, which may be mobilized for many years after N and P inputs are reduced, delaying water quality recovery. Comparatively, nutrient concentrations remained constant through time at the fully forested watershed. Our results highlight the importance of utilizing restoration techniques to reduce nutrient export in impacted systems, while underscoring that managers must anticipate multi-decadal recovery periods driven by legacy nutrient mobilization.

id #135676

Stream restoration reorganizes riparian arthropod communities along lateral gradients

Katelyn P Driscoll^{1,2}, Nicole R Roberts¹, Taylor L Pearson¹, Laurel F Martinez^{1,2}, Thomas F Turner²

1. *Forest Service, NM, -*

2. *The University of New Mexico, Albuquerque, NM*

Riparian zones occur at the interface of aquatic and terrestrial ecosystems and are biologically diverse components of the landscape. Here, terrestrial arthropods support food webs as prey for birds, bats, reptiles, and other consumers, contribute to pollination and decomposition, and mediate energy and nutrient exchanges across ecosystem boundaries. Despite their key role, arthropods remain relatively understudied compared to vegetation and vertebrates, and restoration monitoring often emphasizes habitat structure or vegetation recovery while providing limited insight into ecosystem function. We evaluated how lateral position, vegetation structure, and stream restoration interact to shape riparian arthropod communities in montane streams of northern New Mexico. Using a before-after-control-impact (BACI) design, we sampled ground-dwelling arthropods across four distances from the stream (1-50 m) at sites restored using beaver dam analogs (BDAs) or plug-and-pond (P&P) treatments. Arthropods were identified to family and assigned to functional feeding guilds (spiders, non-spider predators, herbivores, detritivores). Sampling was paired with concurrent measurements of riparian vegetation structure and functional traits to evaluate habitat-mediated responses. Across sites, arthropod communities were strongly structured by lateral gradients. Total abundance, spiders, and detritivores declined with distance from the stream, whereas herbivores increased and non-spider predators peaked at the outer riparian margin. Vegetation structure—particularly canopy cover—explained additional variation beyond distance. Against this strong spatial template, BDA restoration produced clear responses, including increases in total abundance, family richness, spiders, non-spider predators, and detritivores, and declines in herbivores. In contrast, P&P restoration showed limited responses during a shorter post-treatment monitoring window. Where BDA effects occurred, they were not spatially uniform. Responses varied across the riparian corridor or depended on vegetation context, indicating reorganization across existing gradients rather than wholesale community reassembly. Riparian arthropods respond rapidly to restoration and provide sensitive, functionally informative indicators of riparian recovery. Yet, corridor-wide arthropod sampling is critical to fully capture restoration outcomes in coupled stream-riparian ecosystems.

id #134653

It's a Consultant's Life for Me: From Data to Decisions in Aquatic Ecosystem Management

Benjamin Block¹

1. *Tetra Tech, VT, United States*

Applied freshwater science is critical for turning research into actionable strategies that protect and restore aquatic ecosystems. This presentation highlights my career path and current work as a project manager, aquatic ecologist, and data analyst at Tetra Tech (a global consulting firm in water, environment, and sustainable infrastructure). My projects focus on supporting state agencies and the U.S. EPA on a variety of monitoring and assessment projects under the Clean Water Act, with a primary

emphasis on developing bioassessment models (i.e., multimetric indices). Other topics I will highlight include creating R packages to support automated water quality analyses, conducting ad-hoc research and literature reviews to inform policy and guidance decisions, and science communication using ArcGIS StoryMaps. A throughline of all these projects is the delivery of science-based recommendations that inform regulatory programs protecting and restoring waterbodies nationwide. Complementing these efforts, I will highlight the work of my colleagues in our Ecological Analytics and Modeling Group who advance projects in watershed modeling, remote sensing, climate change and ecosystem resilience assessments, risk analyses, causal assessment, and technical communication. This talk will explore how applied freshwater science supports policy development and ecosystem management, emphasizing the technical and collaborative aspects of consulting work. Beyond technical outcomes, these projects illustrate the importance of interdisciplinary approaches, which combine ecological theory, data analytics, and stakeholder engagement to achieve measurable improvements in aquatic health. Attendees will gain insight into the unique challenges and opportunities of non-academic career pathways, including how applied science complements academic research and drives real-world environmental solutions.

id #135678

A Knowledge-Guided Machine Learning Approach for Inferring Metabolism Regimes in Streams and Rivers

Peter Regier¹, Vanessa Garayburu-Caruso², Yi Xiao², Allison Myers-Pigg¹, James Stegen²

1. Pacific Northwest National Laboratory, Sequim, WA, United States

2. Pacific Northwest National Laboratory, Richland, WA, United States

Metabolism fundamentally shapes biogeochemical cycling in streams and rivers, yet estimating it remains challenging in many environments. Conventional reach-scale metabolism models require continuous measurements of dissolved oxygen, discharge, depth, and other physical parameters, and rely on assumptions about stationarity and mixing. These constraints limit the spatial and temporal extent of metabolism estimates, particularly in data-poor or physically complex systems (i.e., steep gradients, groundwater contributions). Because of these constraints, there is a need for approaches that can provide insight into whether oxygen dynamics are biologically or physically driven without detailed knowledge of reach-scale properties.

We present a knowledge-guided, data-driven, machine learning-enabled framework that infers dominant controls on stream oxygen dynamics using only high-frequency measurements of dissolved oxygen and temperature. Rather than explicitly estimating metabolism, our approach leverages characteristic patterns and covariation in these signals to identify whether oxygen variability at a site is primarily structured by biological processes or by physical drivers. To develop this framework, we explored a wide variety of approaches on large public datasets of coupled sensor measurements and metabolism estimates. Our framework has the potential to expand the range of systems where metabolism-relevant dynamics can be meaningfully interpreted, providing a rapid screening tool to guide when and where more intensive modeling efforts are warranted. We hope to offer a new lens for understanding how physical and biological processes jointly structure oxygen dynamics and associated biogeochemical functions in stream ecosystems.

id #136190

How can afforestation benefit or impair hydrological processes and aquatic ecosystems in tropical regions?

Silvio Ferraz¹, Nataly Ortega¹, Alessandro Silva¹, Maria Clara Marques¹, João Miguel Bega¹, Paula Oliveira Reis¹

1. University of São Paulo, Piracicaba, SP, Brazil

According to Brazilian restoration targets, more than one million hectares are expected to be restored within the Atlantic Forest biome. Multiple sectors of society place high expectations on the potential benefits of forest restoration for water provision, regulation and aquatic ecosystem, given the well-established association between native forest cover and the availability of high-quality water resources. Within this context, there is a pressing need to advance the understanding of how afforestation influences ecohydrological processes and, consequently, water resources and aquatic ecosystems in tropical regions.

This study employs field data from experimental catchments undergoing active and passive restoration with native vegetation, as well as from catchments afforested with *Pinus* and *Eucalyptus* plantations, to evaluate the effects of forest restoration in the Atlantic Forest region. The hydrological responses to land-use conversion to forest vary according to forest growth rates and the relative topographic position of new forest within the catchment. The establishment of forest cover is not necessarily accompanied by immediate recovery of hydrological processes; instead, the existence and duration of time lags depend on the ecological condition of the new forest, historical land-use legacies, and local physical characteristics.

Positive effects of forest restoration on the hydrological regime are expected primarily in the long term, when improvements in streamflow regulation offset the short-term reductions in water yield commonly observed after forest establishment. Key processes such as stormflow attenuation, the recovery of mature riparian forest functions, and baseflow regulation are essential for the maintenance of aquatic ecosystems; however, the thresholds governing these processes remain insufficiently understood. Finally, this study proposes and discusses a conceptual framework describing the effects of forest restoration on water resources at the catchment scale.

id #134655

Agricultural land use and hydrology controls patterns and persistence of multiple antimicrobial resistance genes (ARGs) in an agricultural stream

Emma M. Thrift-Cahall¹, Jennifer L. Tank¹, Daniel M.C. Hallack², Diogo Bolster^{3,1}, Mitchell J. Liddick¹, Ursula H. Mahl¹, Kyle Bibby³

1. Department of Biological Sciences, University of Notre Dame, Notre Dame, Indiana, United States

2. Department of Civil Engineering, McMaster University, Hamilton, Ontario, Canada

3. Civil and Environmental Engineering and Earth Sciences, University of Notre Dame, Notre Dame, Indiana, United States

Antibiotics are widely used in livestock production, and up to 90% are excreted unmetabolized, which maintains selective pressure on manure-associated bacteria. Application of manure as a soil amendment to farm fields can introduce antibiotic residues, resistant bacteria, and antimicrobial resistance genes (ARGs) into adjacent streams, serving as potential sources of resistance to humans and the environment. While previous studies have documented that ARGs are frequently found in agricultural streams, mechanisms controlling their fate and transport remain poorly understood. We hypothesized that animal feeding operation (AFO) inputs would elevate downstream ARG loads, with highest values during spring and fall manure-application periods. Over 24 months, we sampled upstream and at four sites downstream of an AFO and examined relationships between the concentrations of four ARGs, dissolved nutrients, and physiochemical parameters. Finally, we developed a simplified hydrological model to characterize reach-scale ARG transport patterns. The AFO input significantly increased stream ARG loads during fall, winter and spring (estimated marginal means [EMM], all $p < 0.05$). We also predicted that discharge and manure-associated drivers (e.g., ammonium, soluble reactive phosphorus [SRP], turbidity, and conductivity) would correlate with ARGs, and found that SRP and turbidity explained the greatest variation in ARG concentrations at both upstream and downstream sites, correlating positively with all four ARG targets at the downstream sites. For all seasons, ARG concentrations positively correlated with ammonium (Spearman's $\rho = 0.49-0.65$, $p < 0.05$), and SRP ($\rho = 0.63-0.84$, $p < 0.01$), and during winter, were positively correlated with discharge ($\rho = 0.45-0.63$, $p < 0.01$), suggesting that runoff due to increased precipitation drives manure-derived inputs. Summer ARG concentrations were negatively associated with dissolved oxygen ($\rho = -0.66$ to -0.78 , $p < 0.001$) and positively with temperature ($\rho = 0.49-0.68$, $p < 0.01$), potentially reflecting increased in-stream microbial activity. Across 60 reach-scale model fits, 31 indicated removal-dominated ARG dynamics with lateral inputs comparable to stream concentrations. An additional 26 runs supported a dilution+removal scenario with diffuse inputs, including 18 with net downstream enrichment and 8 where inputs were offset by removal. These results demonstrate that AFOs can substantially elevate stream ARG loads and that hydrology can influence downstream persistence. Optimizing manure application timing relative to seasonal hydrology could reduce ARG dissemination into streams.

id #135935

Tracking changes in dissolved N concentrations, retention, and export in a rapidly developing Mountain West community

Russell Conti¹, Wyatt Cross¹

1. Montana State University, Bozeman, MT, United States

Despite increasing concern about rapid land-use change and eutrophication of temperate mountain ecosystems, uncertainty persists about how stream ecosystems will retain, process, and export elevated concentrations of reactive nitrogen (N). Mountain watersheds that exhibit spatial heterogeneity in land use and large temporal variation in biotic demand for N can provide the opportunity to address these uncertainties, particularly where long-term data exist. We quantified spatial and temporal variation in dissolved N in tributary outlets that contribute to the West Fork of the Gallatin River, which drains a rapidly developing ski resort community in Big Sky, MT, USA. We collected bi-weekly stream grab samples and deployed continuous nitrate sensors at 3 sites from October 2024 to 2025, varying in development intensity. We then paired these data with comparable data from 2005 to estimate changes in concentration and N loading over a period during which housing development nearly doubled (~3300 to 5300 structures). Preliminary results suggest that rapid development has continued to elevate N concentrations and export, particularly in highly developed streams, suggesting early signs of N saturation driven by increased wastewater inputs. Streams draining high-development landscapes exhibit four-fold higher average N concentrations and a much higher magnitude of diel fluctuations relative to nearby pristine stream sites. Although enriched tributaries had higher algal biomass than pristine ones, elevated concentrations throughout the year suggest reduced ability of these ecosystems to retain excess N. Without further efforts to curtail N inputs, our data suggest that continued enrichment of downstream ecosystems is inevitable. By comparing water quality data across 20 years of development and leveraging continuous diel water quality signals from sites with varying levels of development intensity, our research helps to deepen our understanding of how temperate mountain streams respond to long-term land-use change.